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Exercise - I (Conceptual Questions) Build Up Your Understanding

ELECTRIC CHARGE & METHODS OF CHARGING

1. Which of the following charges cannot be present on an oil drop in Millikan's experiment :–
(1) 4.0×10^{-19} C (2) 6.0×10^{-19} C
(3) 10.0×10^{-19} C (4) all of them
PEL001

2. In nature, the electric charge of any system is always equal to :
(1) half integral multiple of the least amount of charge
(2) zero
(3) square of the least amount of charge
(4) integral multiple of the least amount of charge
PEL002

3. Consider a neutral conducting sphere. A positive point charge is placed outside the sphere. The net charge on the sphere is then :
(1) negative and distributed uniformly over its surface.
(2) negative and appears only at the point on the sphere closest to the point charge
(3) negative and distributed non-uniformly over its entire surface of the sphere
(4) zero
PEL003

COULOMB'S LAW AND EQUILIBRIUM OF CHARGE SYSTEM

4. Force between two identical spheres charged with same charge is F. If 50% charge of one sphere is transferred to the other sphere then the new force will be :–
(1) $\frac{3}{4}F$ (2) $\frac{3}{8}F$
(3) $\frac{3}{2}F$ (4) none of these
PEL004

5. Two point charges placed at a distance 'r' in air exert a force 'F'. The distance at which they exert same force when placed in a certain medium (dielectric constant K) is :–
(1) rK (2) r/K
(3) $r/\sqrt{K}$ (4) $r\sqrt{K}$
PEL005

6. Two charges are placed as shown in figure. Where should a third charge be placed so that it remains at rest?



(1) 30 cm from 9e (2) 40 cm from 16e
(3) 40 cm from 9e (4) (1) or (2)
PEL006

7. Two point charges +9q and +q are kept 16 cm apart. Where should a third charge Q be placed between them so that the system remains in equilibrium ?
(1) 24 cm from + 9q (2) 12 cm from + 9q
(3) 24 cm from + q (4) 12 cm from + q
PEL007

8. Two balls carrying charges +7 μC and –5 μC attract each other with a force F. If a charge –2 μC is added to both, the force between them will be :–
(1) F (2) $\frac{F}{2}$ (3) 2F (4) zero
PEL008

9. Two equal and like charges when placed 5 cm apart experience a repulsive force of 0.144 newtons. The magnitude of the charge in micro–coulomb will be
(1) 0.2 (2) 2 (3) 20 (4) 12
PEL009

10. Two point charges of + 2 μC and + 6 μC repel each other with a force of 12 N. If each is given an additional charge of – 4 μC, then force will become:–
(1) 4 N (attractive) (2) 60 N (attractive)
(3) 4 N (repulsive) (4) 12 N (attractive)
PEL010

11. What equal charges should to be placed on earth and moon to neutralize their gravitational attraction? (mass of earth = 10^{25} kg, mass of moon = 10^{23} kg)
(1) 8.6×10^{13} C (2) 6.8×10^{26} C
(3) 8.6×10^{3} C (4) 9×10^{6} C
PEL011

12. A point charge q_1 exerts a force F upon another point charge q_2. If a third charge q_3 be placed quite close to the charge q_2 then the force that charge q_1 exerts on the charge q_2 will be :-

(1) F (2) > F (3) < F (4) zero

PEL012

13. A charge Q is divided in two parts Q_1 and Q_2 and these charges are placed at a distance R. There will be maximum repulsion between them when :-

(1) $Q_1 = Q - q; Q_2 = q$ (2) $Q_1 = \frac{2Q}{3}, Q_2 = \frac{Q}{3}$

(3) $Q_1 = \frac{3Q}{4}, Q_2 = \frac{Q}{4}$ (4) $Q_1 = Q_2 = \frac{Q}{2}$

PEL013

14. The force of repulsion between two point charges is F, when these are 1 m apart. Now the point charges are replaced by conducting spheres of radii 5 cm having the charge same as that of point charges. The distance between their centres is 1 m, then the force of repulsion will :–

(1) increase (2) decrease

(3) remain same (4) become $\frac{10F}{9}$

PEL014

15. Two charges 4q and q are placed at a distance ℓ apart. A third charged particle Q is placed at the middle of them. If resultant force on q is zero then the value of Q is :–

(1) q (2) – q (3) 2q (4) – 2q

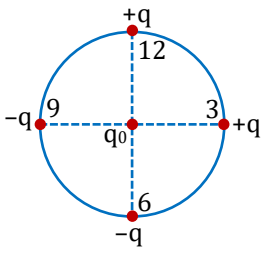
PEL015

16. Two similar spheres having +q and –q charges are kept at a certain separation. F force acts between them. If another similar sphere having +q charge is kept in the middle of them, it experiences a force in magnitude and direction as :–

(1) zero having no direction
(2) 8F towards +q charge
(3) 8F towards –q charge
(4) 4F towards +q charge

PEL016

17. Four charges are placed at the circumference of the dial of a clock as shown in figure. If the clock has only hour hand, then the resultant force on a positive charge q_0 placed at the centre, points in the direction which shows the time as :–



(1) 1:30 (2) 7:30 (3) 4:30 (4) 10:30

PEL017

18. Two small spheres each having a charge +Q are suspended by insulating threads of length L from a hook. This arrangement is taken to a space where there is no gravitational effect, then the angle between the two threads and the tension in each will be :-

(1) $180°, \frac{1}{4\pi\varepsilon_0}\frac{Q^2}{(2L)^2}$ (2) $90°, \frac{1}{4\pi\varepsilon_0}\frac{Q^2}{L^2}$

(3) $180°, \frac{1}{4\pi\varepsilon_0}\frac{Q^2}{2L^2}$ (4) $180°, \frac{1}{4\pi\varepsilon_0}\frac{Q^2}{L^2}$

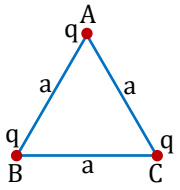
PEL018

19. Identify the wrong statement in the following :-
Coulomb's law correctly describes the electric force that.

(1) binds the electrons of an atom to it's nucleus
(2) binds the protons and neutrons in the nucleus of an atom
(3) binds the atoms together to form molecules
(4) binds the atoms and molecules together to form solids

PEL019

20. In the fig. force on charge at A in the direction normal to BC will be :–



(1) $-\frac{kq}{a^2}$ (2) $-\frac{kq^2}{2a^2}$ (3) $\frac{kq^2}{2a^2}$ (4) $\frac{\sqrt{3}kq^2}{a^2}$

PEL020

ELECTRIC FIELD INTENSITY

21. Two charges 9e and 3e are placed at a separation r. The distance of the point where the electric field intensity will be zero, is :–

(1) $\frac{r}{(1+\sqrt{3})}$ from 9e charge

(2) $\frac{\sqrt{3}r}{\sqrt{3}+1}$ from 9e charge

(3) $\frac{r}{(1-\sqrt{3})}$ from 3e charge

(4) $\frac{\sqrt{3}r}{1+\sqrt{3}}$ from 3e charge

PEL021

22. A ring of radius R is charged uniformly with a charge + Q. The electric field at a point on its axis at a distance r from any point on the ring will be :–

(1) $\frac{KQ}{(r^2-R^2)}$ (2) $\frac{KQ}{r^2}$

(3) $\frac{KQ}{r^3}\left(r^2-R^2\right)^{1/2}$ (4) $\frac{KQr}{R^3}a$

PEL022

23. Two positive charges of 1 μC and 2 μC are placed 1 metre apart. The value of electric field in N/C at the mid point of the two charges will be :–

(1) 10.8×10^4 (2) 3.6×10^4

(3) 1.8×10^4 (4) 5.4×10^4

PEL023

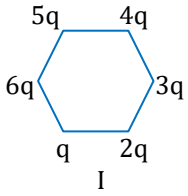
24. The electric field in a certain region is given by $\vec{E}=(\frac{K}{x^3})\hat{i}$. The dimensions of K are :–

(1) $MLT^{-3}A^{-1}$ (2) $ML^{-2}T^{-3}A^{-1}$

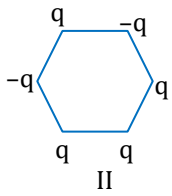
(3) $ML^4T^{-3}A^{-1}$ (4) $M^\circ L^\circ T^\circ A^\circ$

PEL024

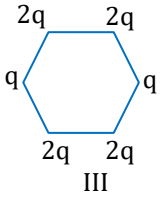
25. Figure below shows regular hexagon, with different charges placed at the vertices. In which of the following cases is the electric field at the centre zero ?



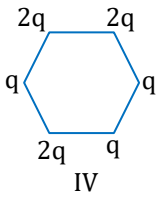
I



II



III

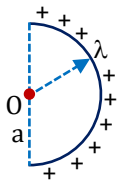


IV

(1) IV (2) III (3) I (4) II

PEL025

26. Electric field at the centre 'O' of a semicircle of radius 'a' having linear charge density λ is given as :-



(1) $\frac{2\lambda}{\epsilon_0 a}$ (2) $\frac{\lambda\pi}{\epsilon_0 a}$

(3) $\frac{\lambda}{2\pi\epsilon_0 a}$ (4) $\frac{\lambda}{\pi\epsilon_0 a}$

PEL026

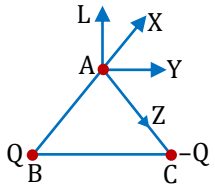
27. A semicircular ring of radius 0.5 m is uniformly charged with a total charge of 1.4×10^{-9} C. The electric field intensity at the centre of this ring is :–

(1) zero (2) 320 V/m.

(3) 64 V/m. (4) 32 V/m.

PEL027

28. For the given figure the direction of electric field at A will be :



(1) towards AL (2) towards AY

(3) towards AX (4) towards AZ

PEL028

29. -1×10^{-6} C charge is on a drop of water having mass 10^{-6} kg. What electric field should be applied on the drop so that it is in the balanced condition with its weight ?

(1) 10 V/m upward

(2) 10 V/m downward

(3) 0.1 V/m downward

(4) 0.1 V/m upward

PEL029

30. Two small identical spheres, each of mass 1 g and carrying same charge 10^{-9} C are suspended by threads of equal lengths. If the distance between the centres of the spheres is 0.3 cm in equilibrium then the inclination of the thread with the vertical will be :–

(1) $\tan^{-1}(0.1)$ (2) $\tan^{-1}(2)$

(3) $\tan^{-1}(1.5)$ (4) $\tan^{-1}(0.6)$

PEL030

ELECTRIC FIELD LINES, ELECTRIC FLUX & GAUS'S THEOREM

31. Two infinitely long parallel wires having linear charge densities λ_1 and λ_2 respectively are placed at a distance R. The force per unit length on either wire will be :–

(1) $\frac{k2\lambda_1\lambda_2}{R^2}$ (2) $\frac{k2\lambda_1\lambda_2}{R}$

(3) $\frac{k\lambda_1\lambda_2}{R^2}$ (4) $\frac{k\lambda_1\lambda_2}{R}$

PEL031

32. Choose the correct statement regarding electric lines of force :–

(1) They emerge from negative charge and terminate at positive charge

(2) The electric field in that region is weak where the density of electric lines of force are more

(3) They are in radial directions for a point charge

(4) They have a physical existence

PEL032

33. A solid sphere of radius R, is charged uniformly with a total charge Q. Then the correct expression for electric field is.

(r = distance from centre) :–

(1) $\frac{KQr}{R^3}$, where $r < R$

(2) $\frac{KQ}{r^2}$, where $r \geq R$

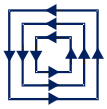
(3) it is zero, at all points

(4) (1) and (2) both

PEL033

34. Which one of the following pattern of electrostatic lines of force is not possible ?

(1)  (2) 

(3)  (4) 

PEL034

35. A sphere of radius R and charge Q is placed inside a concentric imaginary sphere of radius 2R. The flux associated with the imaginary sphere is :–

(1) $\frac{Q}{\epsilon_0}$ (2) $\frac{Q}{2\epsilon_0}$

(3) $\frac{4Q}{\epsilon_0}$ (4) $\frac{2Q}{\epsilon_0}$

PEL035

36. A nonconducting solid sphere of radius R is charged uniformly. The magnitude of the electric field due to the sphere at a distance r from its centre :–

(a) increases as r increases, for $r < R$

(b) decreases as r increases, for $0 < r < \infty$

(c) decreases as r increases, for $R < r < \infty$

(d) is discontinuous at $r = R$

(1) a, c (2) c, d (3) a, b (4) b, d

PEL036

37. 20 μC charge is placed inside a closed surface; then flux linked with the surface is ϕ. If 80 μC charge is put inside the surface then change in flux is :–

(1) 4ϕ (2) 5ϕ (3) ϕ (4) 8ϕ

PEL037

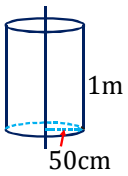
38. In a region of space the electric field is given by $\vec{E} = 8\hat{i} + 4\hat{j} + 3\hat{k}$. The electric flux through a surface of area of 100 units in the x–y plane is :–

(1) 800 units (2) 300 units

(3) 400 units (4) 1500 units

PEL038

39. Electric charge is uniformly distributed over a long straight wire of radius 1 mm. The charge per cm length of the wire is Q coulombs. A cylindrical surface of radius 50 cm and length 1m encloses the wire symmetrically as shown in fig.



The total flux passing through the cylindrical surface is :–

(1) $\frac{Q}{\epsilon_0}$ (2) $\frac{100Q}{\epsilon_0}$

(3) $\frac{10Q}{\pi\epsilon_0}$ (4) $\frac{100Q}{\pi\epsilon_0}$

PEL039

40. Gaus's law is given by $\epsilon_0 \oint_s \vec{E}.d\vec{s} = q$, if net charge enclosed by a Gaussian surface is zero then :–

(1) E must be zero on the surface

(2) Number of incoming and outgoing electric lines are equal

(3) there is a net incoming of electric lines

(4) none

PEL040

41. The electric field is 100 V/m, at a distance of 20 cm from the centre of a dielectric sphere of radius 10 cm. Then E at 3 cm distance from the centre of sphere is :–

(1) 100 V/m (2) 125 V/m

(3) 120 V/m (4) zero

PEL041

42. If the electric flux entering and leaving a closed surface is ϕ_1 and ϕ_2 respectively then electric charge inside the surface will be :-

(1) $(\phi_1 + \phi_2)\varepsilon_0$ (2) $(\phi_2 - \phi_1)\varepsilon_0$

(3) $\frac{\phi_1 + \phi_2}{\varepsilon_0}$ (4) $\frac{\phi_2 - \phi_1}{\varepsilon_0}$

PEL042

43. The electric field in a region of space is given by $E = (5\hat{i} + 2\hat{j})$ N/C. The electric flux through an area of 2 m^2 lying in the YZ plane, in S.I. units is:–

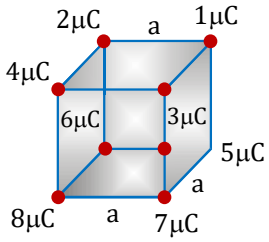
(1) 10 (2) 20

(3) $10\sqrt{2}$ (4) $2\sqrt{29}$

PEL043

44. The total flux associated with the given cube will be- where 'a' is side of the cube :–

$(\frac{1}{\epsilon_0} = 4\pi \times 9 \times 10^9$ SI units)



(1) $162\pi \times 10^{-3}$ Nm^2/C

(2) $162\pi \times 10^{3}$ Nm^2/C

(3) $162\pi \times 10^{-6}$ Nm^2/C

(4) $162\pi \times 10^{6}$ Nm^2/C

PEL044

45. A point charge is placed at a distance $\frac{a}{2}$ perpendicular to the plane and above the centre of a square of side a. The electric flux through the square is :–

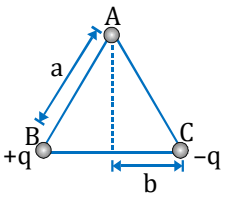
(1) $\frac{q}{\epsilon_0}$ (2) $\frac{q}{\pi\epsilon_0}$

(3) $\frac{q}{4\epsilon_0}$ (4) $\frac{q}{6\epsilon_0}$

PEL045

ELECTRIC POTENTIAL, EQUIPOTENTIAL SURFACE & RELATION BETWEEN FIELD & POTENTIAL

46. As shown in the fig. charges + q and – q are placed at the vertices B and C of an isosceles triangle. The potential at the vertex A is :-



(1) $\frac{1}{4\pi\epsilon_0}.\frac{2q}{\sqrt{a^2+b^2}}$ (2) zero

(3) $\frac{1}{4\pi\epsilon_0}.\frac{q}{\sqrt{a^2+b^2}}$ (4) $\frac{1}{4\pi\epsilon_0}.\frac{-q}{\sqrt{a^2+b^2}}$

PEL046

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47. At any point on the perpendicular bisector of the line joining two equal and opposite charges :-
(1) the electric field is zero
(2) the electric potential is zero
(3) the electric potential decreases with increasing distance from their mid point
(4) the electric field is perpendicular to the line joining the charges

PEL047

48. What is the electric potential at a distance 'x' from the centre, inside a conducting sphere having a charge Q and radius R ?

(1) $\frac{1}{4\pi\in_0}\frac{Q}{R}$ (2) $\frac{1}{4\pi\in_0}\frac{Q}{x}$

(3) $\frac{1}{4\pi\in_0}\cdot\frac{QX}{R^2}$ (4) zero

PEL048

49. Certain positive charge is given to a conductor. Then its potential :–
(1) is maximum at the surface
(2) is maximum at the centre
(3) remains same throughout the conductor
(4) is maximum somewhere between the surface and the centre

PEL049

50. Potential inside a charged spherical shell is :–
(1) uniform
(2) proportional to the distance from the centre
(3) inversely proportional to the distance
(4) inversely proportional to square of distance from the centre

PEL050

51. A solid conducting sphere having a charge Q is surrounded by an uncharged concentric conducting spherical shell. Let the potential difference between the surface of the solid sphere and the outer surface of the shell be V. If the shell is now given a charge -3Q the new potential difference between the same two surfaces is :-
(1) V (2) 2V (3) 4V (4) –2V

PEL051

52. Four charges 2C, – 3C, –4C and 5C respectively are placed at the four corners of a square. Which of the following statements is true for the point of intersection of the diagonals ?
(1) E = 0, V = 0 (2) E ≠ 0, V = 0
(3) E = 0, V ≠ 0 (4) E ≠ 0, V ≠ 0

PEL052

53. A non - conducting ring is of radius 0.5 m. 1.11×10^{-10} coulombs charge is non - uniformly distributed over the circumference of ring which produces electric field E around itself. If $\ell = 0$ is the centre of the ring, then the value of $\int_{\ell=\infty}^{\ell=0} -\vec{E}.d\ell$ is :–
(1) 2 V (2) – 2 V (3) –1 V (4) zero

PEL053

54. A hollow metal sphere of radius 5 cm is charged such that the potential on its surface is 10 volts. The electric field at the centre of the sphere will be :-
(1) 50 volt/meter (2) 10 volt/meter
(3) 5 volt/meter (4) zero

PEL054

55. The potential at a distance R/2 from the centre of a conducting sphere of radius R containing charge Q will be :-

(1) 0 (2) $\frac{Q}{8\pi\in_0 R}$

(3) $\frac{Q}{4\pi\in_0 R}$ (4) $\frac{Q}{2\pi\in_0 R}$

PEL055

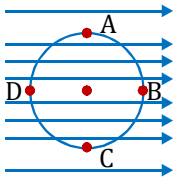
56. Four charges +Q, –Q, +Q and –Q are situated at the corners of a square; in a sequence then at the centre of the square :-
(1) E = 0, V=0 (2) E = 0, V ≠ 0
(3) E ≠ 0, V=0 (4) E = 0, V ≠ 0

PEL056

57. Electric field at a distance x from the origin is given as E = $\frac{100N-m^2/C}{x^2}$. Then potential difference between the points situated at x = 10 m and x = 20 m is :-
(1) 5 V (2) 10 V (3) 15 V (4) 4V

PEL057

58. A circle of radius R is drawn in a uniform electric field E as shown in the fig. V_A, V_B, V_C and V_D are respectively the potentials of points A, B, C and D on the circle then :-



(1) $V_A > V_C$, $V_B = V_D$
(2) $V_A < V_C$, $V_B = V_D$
(3) $V_A = V_C$, $V_B < V_D$
(4) $V_A = V_C$, $V_B > V_D$

PEL058

59. A uniform electric field pointing in positive x–direction exists in a region. Let A be the origin, B be the point on the x–axis at x = +1 cm and C be the point on the y–axis at y = +1 cm. Then the potentials at the points A, B and C satisfy :-

(1) $V_A < V_B$ (2) $V_A > V_B$
(3) $V_A < V_C$ (4) $V_A > V_C$

PEL059

60. The electric potential V is given as a function of distance x (metre) by $V = (5x^2 - 10x - 9)$ volts. The value of electric field at x=1 m is :–

(1) 20 V/m (2) 6 V/m (3) 11 V/m (4) zero

PEL060

61. The electric potential and electric field at a point due to a point charge are 600 V and 200 N/C respectively. Then magnitude of the point charge should be :-

(1) 3 μC (2) 30 μC
(3) 0.2 μC (4) 0.5 μC

PEL061

62. Two concentric spheres of radii R and r have similar charges with equal surface charge densities (σ). What is the electric potential at their common centre?

(1) σ/ε_0 (2) $\frac{\sigma}{\varepsilon_0}(R-r)$
(3) $\frac{\sigma}{\varepsilon_0}(R+r)$ (4) None of these

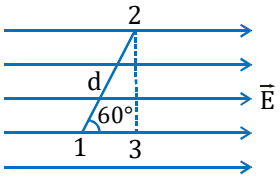
PEL062

63. Three charges 2q, -q, -q are located at the vertices of an equilateral triangle. At the centre of the triangle :-

(1) the field is zero but potential is non-zero
(2) the field is non-zero but potential is zero
(3) both field and potential are zero
(4) both field and potential are non-zero

PEL063

64. The electric field $\vec{E}$ is constant in both magnitude and direction. Consider a path of length d at an angle $\theta = 60°$ with respect to field lines as shown in figure. The potential difference between points 1 and 2 is :-



(1) $\frac{E}{d\sin 60°}$ (2) Ed cos 60°
(3) $\frac{Ed}{\cos 60°}$ (4) $\frac{E}{d}\sin 60°$

PEL064

65. The electric potential in a certain region is expressed by $V = 6x - 8xy^2 - 8y + 6yz - 4z^2$ volts. The magnitude of the force acting on a charge of 2 C situated at the origin will be :–

(1) 2 N (2) 6 N (3) 8 N (4) 20 N

PEL065

66. Which statement is true?

(i) A ring of radius R carries a uniformly distributed charge +Q. A point charge –q is placed on the axis of the ring at a distance 2R from its centre and released. The particle executes simple harmonic motion along the axis of the ring.
(ii) Electrons move from a region of higher potential to that of lower potential

(1) only (i) (2) only (ii)
(3) (i), (ii) (4) none of them

PEL066

67. Two conducting spheres of radii r_1 and r_2 have same electric field near their surfaces. The ratio of their electric potentials is :-

(1) r_1^2/r_2^2 (2) r_2^2/r_1^2
(3) r_1/r_2 (4) r_2/r_1

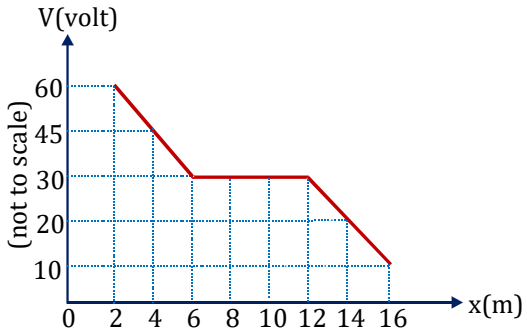
PEL067

68. A charged hollow metal sphere has a radius r. if the potential difference between its surface and a point at distance 3r from the centre is V, the electric intensity at a distance 3r from the centre is :-

(1) $\frac{V}{6r}$ (2) $\frac{V}{4r}$ (3) $\frac{V}{3r}$ (4) $\frac{V}{2r}$

PEL068

69. The variation of potential with distance x from a fixed point is shown in figure. The electric field at x =13 m is

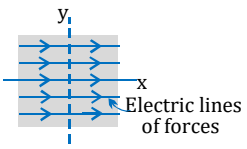


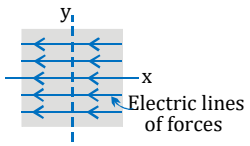
(1) 7.5 volt/meter (2) –7.5 volt/meter
(3) 5 volt/meter (4) –5 volt/meter

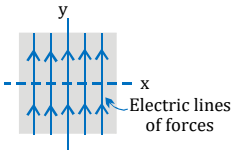
PEL069

70. List I gives certain situations in which electric field is represented by electric lines of forces in x-y plane. List II gives corresponding representation of equipotential lines in x-y plane. Match the figures in List I with the figures in List II and indicate your answer.

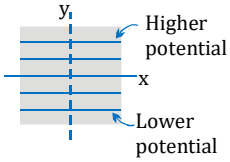
List - I

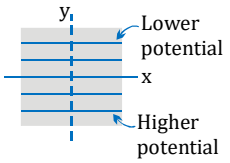
(P) 

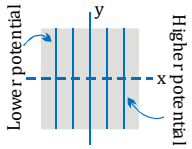
(Q) 

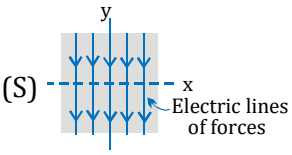
(R) 

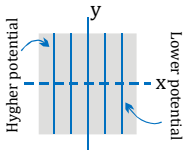
List - II

(1) 

(2) 

(3) 

(S) 

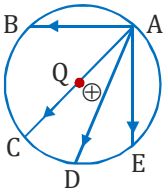
(4) 

Codes	P	Q	R	S
(1)	1	2	3	4
(2)	4	3	2	1
(3)	3	4	2	1
(4)	2	1	3	4

PEL070

ELECTRICAL POTENTIAL ENERGY

71. In the electric field of charge Q, another charge is carried from A to B, A to C, A to D and A to E, then work done will be :-



(1) minimum along the path AB
(2) minimum along the path AD
(3) minimum along the path AE
(4) zero along each path

PEL071

72. Choose the incorrect statement :–

(1) the potential energy per unit positive charge in an electric field at some point is called the electric potential.
(2) the work required to be done to move a point charge from one point to another in an electric field depends on the position of the points
(3) the potential energy of the system will increase if a positive charge is moved against the Coulombian force
(4) the value of fundamental charge is not equivalent to the electronic charge.

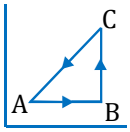
PEL072

73. A charge of 10 esu is placed at a distance of 2 cm from a charge of 40 esu and 4 cm from another charge of – 20 esu. The potential energy of the charge 10 esu is :– (in ergs)

(1) 87.5 (2) 112.5 (3) 150 (4) zero

PEL073

74. As shown in figure, on bringing a charge Q from point A to B and from B to C, the work done are 2 joules and – 3 joules respectively. The work done in bringing the charge from C to A will be



(1) – 1 joule (2) 1 joule
(3) 2 joules (4) 5 joules

PEL074

75. 15 joule of work has to be done against an existing electric field to take a charge of 0.01 C from A to B. Then the potential difference $(V_B - V_A)$ is :–
(1) 1500 volts (2) – 1500 volts
(3) 0.15 volts (4) none of these

PEL075

76. A 5 C charge experiences a force of 2000 N when moved between two points along the field separated by a distance of 2 cm in a uniform electric field. The potential difference between the two points is :–
(1) 8 volts (2) 80 volts
(3) 800 volts (4) 8000 volts

PEL076

77. When the separation between two charges is increased, the electric potential energy of the system of charges :-
(1) increases
(2) decreases
(3) remains the same
(4) may increase or decrease

PEL077

MOTION OF CHARGED PARTICLE IN ELECTRIC FIELD AND ELECTRIC DIPOLE

78. An electron enters an electric field with its velocity in the direction of the electric field lines then :–
(1) the path of the electron will be a circle
(2) the path of the electron will be a parabola
(3) the velocity of the electron will decrease just after the entry
(4) the velocity of the electron will increase just after the entry

PEL078

79. An electron and a proton are set free in a uniform electric field. The ratio of their accelerations is :–
(1) unity (2) zero (3) $\frac{m_p}{m_e}$ (4) $\frac{m_e}{m_p}$

PEL079

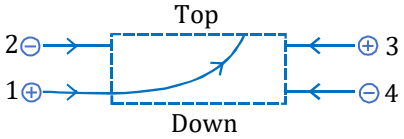
80. A particle of mass m and charge q is released from rest in an electric field E. Then the K.E. after time t will be :-
(1) $\frac{2E^2t^2}{mq}$ (2) $\frac{E^2q^2t^2}{2m}$
(3) $\frac{Eq^2m}{2t^2}$ (4) $\frac{Eqm}{2t}$

PEL080

81. A charge q is projected into a uniform electric field E; work done when it suffers a displacement Y along the field direction is :-
(1) qEY (2) $\frac{qY}{E}$ (3) $\frac{qE}{Y}$ (4) $\frac{Y}{qE}$

PEL081

82. The figure below shows the path of a positively charged particle 1 through a rectangular region of uniform electric field as shown in the figure. What are the direction movement of particles 2, 3 and 4?



(1) Down, top, down (2) Down, down, top
(3) Top, top, down (4) Top, down, down

PEL082

83. For a dipole, the value of each charge is 10^{-10} stat coulomb and their separation is 1Å, then its dipole moment is :–
(1) one debye (2) 2 debye
(3) 10^{-3} debye (4) 3×10^{-20} debye

PEL083

84. The electric potential and field at a point due to an electric dipole are proportional to :-
(1) r, r^{-1} (2) r^{-1}, r^{-2}
(3) r^{-2}, r^{-3} (4) r^{-2}, r^{-2}

PEL084

85. When an electric dipole $\vec{p}$ is kept in a uniform electric field $\vec{E}$ then for what value of angle between $\vec{p}$ and $\vec{E}$, will the torque be maximum ?

(1) 90° (2) 0° (3) 180° (4) 45°

PEL085

86. What will be the ratio of electric field at a point on the axis and an equidistant point on the equatorial line of a dipole :–

(1) 1 : 2 (2) 2 : 1 (3) 4 : 1 (4) 1 : 4

PEL086

87. For a dipole $q = 2 \times 10^{-6}$ C ; d = 0.01m; find the maximum torque on the dipole if $E = 5 \times 10^5$ N/C :–

(1) 1×10^{-3} Nm^{-1} (2) 10×10^{-3} Nm^{-1}

(3) 10×10^{-3} Nm (4) 1×10^{-4} Nm

PEL087

88. Two particles each of mass M is attached to the two ends of a massless rigid non-conducting rod of length L. The two particles carry charges +q and –q respectively. This arrangement is held in a region of uniform electric field E such that the rod makes a small angle θ (< 5°) with the field direction. The time period of rod is (rod oscillates about its centre of mass) :-

(1) $2\pi\sqrt{\frac{ML}{2qE}}$ (2) $\pi\sqrt{\frac{ML}{2qE}}$

(3) $\frac{\pi}{2}\sqrt{\frac{ML}{2qE}}$ (4) $4\pi\sqrt{\frac{ML}{2qE}}$

PEL088

89. The electric potential at a point due to an electric dipole will be :-

(1) $k\frac{\vec{p}.\vec{r}}{r^3}$ (2) $k\frac{\vec{p}.\vec{r}}{r^2}$

(3) $k\frac{\vec{p}\times\vec{r}}{r^3}$ (4) $k\frac{\vec{p}\times\vec{r}}{r^2}$

PEL089

90. The force on a charge situated on the axis of a dipole is F; if the charge is shifted to double the distance, the force acting will be :-

(1) zero (2) $\frac{F}{2}$ (3) $\frac{F}{4}$ (4) $\frac{F}{8}$

PEL090

91. A small electric dipole is of dipole moment p. The electric potential at a distance 'r' from its centre and making an angle θ from the axis of dipole will be :-

(1) $\frac{kp\sin\theta}{r^2}$ (2) $\frac{kp\cos\theta}{r^2}$

(3) $\frac{kp}{r^3}\sqrt{1+3\cos^2\theta}$ (4) $\frac{kp}{r^3}\sqrt{1+3\sin^2\theta}$

PEL091

92. If an electric dipole is placed in an electric field generated by a point charge then :–

(1) the net electric force on the dipole must be zero

(2) the net electric force on the dipole may be zero

(3) the torque on the dipole due to the field must be zero

(4) the torque on the dipole due to the field may be zero

PEL092

CONDUCTORS

93. Two conductors are of same shape and size. One of copper and the other of aluminium (less conducting) are placed in an uniform electric field. The charge induced in aluminium :-

(1) will be less than that in copper

(2) will be more than that in copper

(3) will be equal to that in copper

(4) cannot be compared with that of copper

PEL093

94. A big hollow metal sphere A is charged to 100 volts and another smaller hollow sphere B is charged to 50 volts. If B is put inside A and joined with a metallic wire, then the direction of charge flow :-

(1) is from A to B

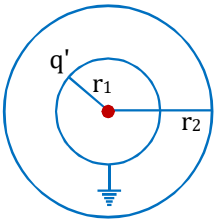
(2) is from B to A

(3) no charge flows

(4) depends on the radii of spheres

PEL094

95. Two concentric conducting spheres are of radii r_1 and r_2. The outer sphere is given a charge q. The charge q' on the inner sphere will be (inner sphere is grounded) :-



(1) q (2) –q

(3) $-q\frac{r_1}{r_2}$ (4) Zero

PEL095

96. A charge given to any conductor resides on its outer surface, because :-

(1) the free charge tends to be in its minimum potential energy state.

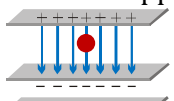
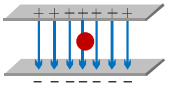
(2) the free charge tends to be in its minimum kinetic energy state.

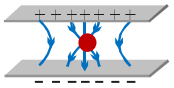
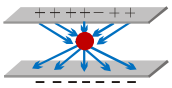
(3) the free charge tends to be in its maximum potential energy state.

(4) the free charge tends to be in its maximum kinetic energy state.

PEL096

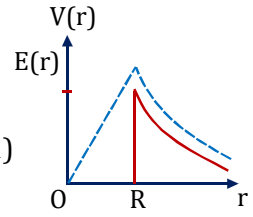
97. An uncharged sphere of metal is placed in between two charged plates as shown. The lines of force appear as :-

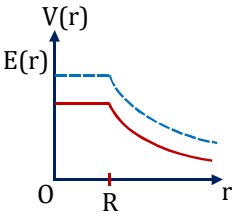
(1)  (2) 

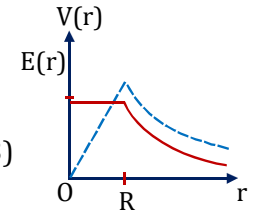
(3)  (4) 

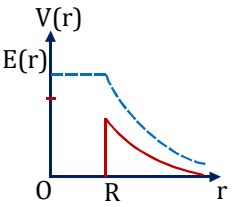
PEL097

98. Consider a conducting spherical shell of radius R with its centre at the origin, carrying uniform positive surface charge density. The variation of the magnitude of the electric field $|\vec{E}(r)|$ and the electric potential V(r) with the distance r from the centre, is best represented by the graph (Here dotted line represents potential curve and bold line represents electric field curve) :-

(1) 

(2) 

(3) 

(4) 

PEL098

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	4	1	3	4	2	1	1	1	1	1	4	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	2	1	2	4	2	3	2	3	2	3	4	2	2	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	3	4	3	1	1	1	2	2	2	3	2	1	2	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	2	1	3	1	1	2	1	4	3	1	1	3	2	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	3	3	2	2	4	4	3	1	3	2	4	4	3	2	1
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	1	4	3	3	2	1	1	1	3	1	2	3	1	1	4
Question	91	92	93	94	95	96	97	98							
Answer	2	4	3	2	3	1	3	4							

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. A conducting sphere of radius R is given a charge Q. The electric potential and the electric field at the centre of the sphere are respectively :-

(1) Zero and $\frac{Q}{4\pi\in_0 R^2}$

(2) $\frac{Q}{4\pi\in_0 R}$ and Zero

(3) $\frac{Q}{4\pi\in_0 R}$ and $\frac{Q}{4\pi\in_0 R^2}$

(4) Both are zero

PEL119

2. In a region, the potential is represented by V(x, y, z) = 6x – 8xy – 8y + 6yz, where V is in volts and x, y, z are in metres. The electric force experienced by a charge of 2 coulombs situated at the point (1, 1, 1) is :-

(1) $6\sqrt{5}$ N (2) 30 N

(3) 24 N (4) $4\sqrt{35}$ N

PEL120

AIPMT 2015

3. The electric field in a certain region is acting radially outward and is given by E = Ar. The charge contained in a sphere of radius 'a' centred at the origin of the field, will be given by :

(1) $A\,\varepsilon_0 a^2$ (2) $4\pi\varepsilon_0 Aa^3$

(3) $\varepsilon_0 Aa^3$ (4) $4\pi\varepsilon_0 Aa^2$

PEL121

Re-AIPMT 2015

4. If potential (in volts) in a region is expressed as V (x,y,z) = 6xy – y + 2yz, the electric field (in N/C) at point (1,1,0) is :

(1) $-(6\hat{i}+9\hat{j}+k)$ (2) $-(3\hat{i}+5\hat{j}+3k)$

(3) $-(6\hat{i}+5\hat{j}+2k)$ (4) $-(2\hat{i}+3\hat{j}+k)$

PEL122

NEET-II 2016

5. An electric dipole is placed at an angle of 30° with an electric field intensity 2×10^5 N/C. It experiences a torque equal to 4 Nm. The charge on the dipole, if the dipole length is 2 cm, is :-

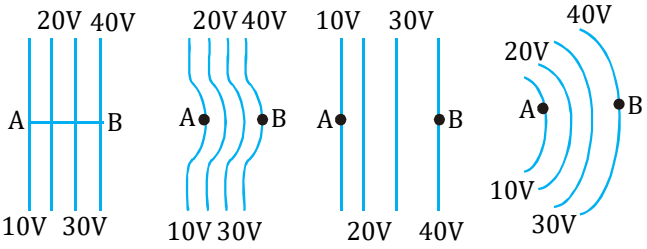
(1) 5 mC (2) 7 μC

(3) 8 mC (4) 2 mC

PEL123

NEET(UG)-2017

6. The diagrams below show regions of equipotentials :-



(a) (b) (c) (d)

A positive charge is moved from A to B in each diagram.

(1) In all the four cases the work done is the same

(2) Minimum work is required to move q in figure (a)

(3) Maximum work is required to move q in figure (b)

(4) Maximum work is required to move q in figure (c)

PEL124

7. Suppose the charge of a proton and an electron differ slightly. One of them is – e, the other is (e + Δe). If the net of electrostatic force and gravitational force between two hydrogen atoms placed at a distance d (much greater than atomic size) apart is zero, then Δe is of the order of [Given mass of hydrogen $m_h = 1.67 \times 10^{-27}$ kg]

(1) 10^{-23} C (2) 10^{-37} C

(3) 10^{-47} C (4) 10^{-20} C

PEL125

NEET(UG)-2018

8. An electron falls from rest through a vertical distance h in a uniform and vertically upward directed electric field E. The direction of electrical field is now reversed, keeping its magnitude the same. A proton is allowed to fall from rest in through the same vertical distance h. The time of fall of the electron, in comparison to the time of fall of the proton is :-

(1) smaller

(2) 5 times greater

(3) 10 times greater

(4) equal

PEL126

NEET(UG)-2019

9. A hollow metal sphere of radius R is uniformly charged. The electric field due to the sphere at a distance r from the centre :
(1) increases as r increases for r < R and for r > R
(2) zero as r increases for r < R, decreases as r increases for r > R
(3) zero as r increases for r < R, increases as r increases for r > R
(4) decreases as r increases for r < R and for r > R

PEL127

10. Two parallel infinite line charges with linear charge densities +λ C/m and –λ C/m are placed at a distance of 2R in free space. What is the electric field mid-way between the two line charges?

(1) zero
(2) $\frac{2\lambda}{\pi \in_0 R}$ N/C
(3) $\frac{\lambda}{\pi \in_0 R}$ N/C
(4) $\frac{\lambda}{2\pi \in_0 R}$ N/C

PEL128

11. Two point charges A and B, having charges +Q and –Q respectively, are placed at certain distance apart and force acting between them is F. If 25% charge of A is transferred to B, then force between the charges becomes:

(1) F
(2) $\frac{9F}{16}$
(3) $\frac{16F}{9}$
(4) $\frac{4F}{3}$

PEL129

NEET(UG)-2019 (Odisha)

12. A sphere encloses an electric dipole with charge ±3 × 10^{-6} C. What is the total electric flux across the sphere ?
(1) –3 × 10^{-6} Nm^2/C
(2) zero
(3) 3 × 10^{-6} Nm^2/C
(4) 6 × 10^{-6} Nm^2/C

PEL130

NEET(UG)-2020

13. A short electric dipole has a dipole moment of 16 × 10^{-9} C m. The electric potential due to the dipole at a point at a distance of 0.6 m from the centre of the dipole, situated on a line making an angle of 60° with the dipole axis is :

$\left(\frac{1}{4\pi \in_0} = 9\times10^9 \,N m^2/C^2\right)$

(1) zero
(2) 50 V
(3) 200 V
(4) 400 V

PEL131

14. In a certain region of space with volume 0.2 m^3 the electric potential is found to be 5 V throughout. The magnitude of electric field in this region is :
(1) 5 N/C
(2) Zero
(3) 0.5 N/C
(4) 1 N/C

PEL132

15. A spherical conductor of radius 10 cm has a charge of 3.2 × 10^{-7} C distributed uniformly. What is the magnitude of electric field at a point 15 cm from the centre of the sphere ?

$\left(\frac{1}{4\pi \in_0} = 9\times10^9 \,N m^2/C^2\right)$

(1) 1.28 × 10^7 N/C
(2) 1.28 × 10^4 N/C
(3) 1.28 × 10^5 N/C
(4) 1.28 × 10^6 N/C

PEL133

NEET(UG)-2020 (Covid-19)

16. The electric field at a point on the equatorial plane at a distance r from the centre of a dipole having dipole moment $\vec{p}$ is given by
(r >> separation of two charges forming the dipole, $\in_0$ - permittivity of free space)

(1) $\vec{E} = \frac{\vec{p}}{4\pi \in_0 r^3}$
(2) $\vec{E} = \frac{2\vec{p}}{4\pi \in_0 r^3}$
(3) $\vec{E} = -\frac{\vec{p}}{4\pi \in_0 r^2}$
(4) $\vec{E} = -\frac{\vec{p}}{4\pi \in_0 r^3}$

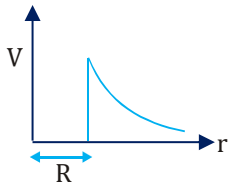
PEL134

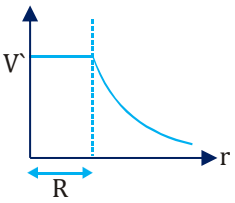
17. The acceleration of an electron due to the mutual attraction between the electron and a proton when they are 1.6 Å apart is,
($m_e \simeq$ 9 × 10^{-31} kg, e = 1.6 × 10^{-19} C)
(Take $\frac{1}{4\pi\varepsilon_0}$ = 9 × 10^9 Nm^2 C^{-2})

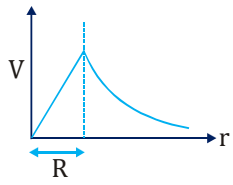
(1) 10^{24} m/s^2
(2) 10^{23} m/s^2
(3) 10^{22} m/s^2
(4) 10^{25} m/s^2

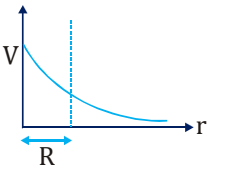
PEL135

18. The variation of electrostatic potential with radial distance *r* from the centre of a positively charged metallic thin shell of radius R is given by the graph

(1) 

(2) 

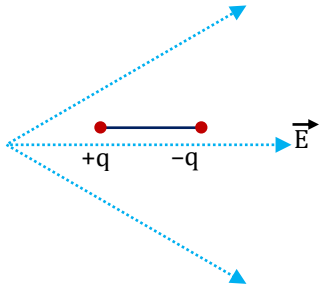
(3) 

(4) 

PEL136

NEET(UG)-2021

19. A dipole is placed in an electric field as shown. In which direction will it move ?



(1) towards the left as its potential energy will increase.
(2) towards the right as its potential energy will decrease.
(3) towards the left as its potential energy will decrease.
(4) towards the right as its potential energy will increase.

PEL137

20. Two charged spherical conductors of radius R_1 and R_2 are connected by a wire. Then the ratio of surface charge densities of the spheres (σ_1/σ_2) is:

(1) $\frac{R_1}{R_2}$ (2) $\frac{R_2}{R_1}$ (3) $\sqrt{\left(\frac{R_1}{R_2}\right)}$ (4) $\frac{R_1^2}{R_2^2}$

PEL138

21. Twenty seven drops of same size are charged at 220 V each. They combine to form a bigger drop. Calculate the potential of the bigger drop.

(1) 660 V (2) 1320 V
(3) 1520 V (4) 1980 V

PEL139

NEET(UG)-2022

22. The angle between the electric lines of force and the equipotential surface is :

(1) 45° (2) 90° (3) 180° (4) 0°

PEL140

23. Two point charges -q and +q are placed at a distance of L, as shown in the figure.



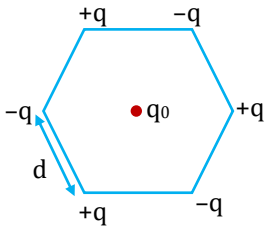
The magnitude of electric field intensity at a distance
R (R >>L) varies as :

(1) $\frac{1}{R^3}$ (2) $\frac{1}{R^4}$ (3) $\frac{1}{R^6}$ (4) $\frac{1}{R^2}$

PEL141

RE-NEET(UG)-2022

24. Six charges +q, -q, +q, -q, +q and -q are fixed at the corners of a hexagon of side d as shown in the figure. The work done in bringing a charge q_0 to the centre of the hexagon from infinity is : (ε_0 - permittivity of free space)



(1) Zero (2) $\frac{-q^2}{4\pi\varepsilon_0 d}$

(3) $\frac{-q^2}{4\pi\varepsilon_0 d}\left(3-\frac{1}{\sqrt{2}}\right)$ (4) $\frac{-q^2}{4\pi\varepsilon_0 d}\left(6-\frac{1}{\sqrt{2}}\right)$

PEL142

NEET(UG)-2023

25. An electric dipole is placed at an angle of 30° with an electric field of intensity 2×10^5 NC^{-1}. It experiences a torque equal to 4 Nm. Calculate the magnitude of charge on the dipole, if the dipole length is 2 cm.

(1) 6 mC (2) 4 mC (3) 2 mC (4) 8 mC

PEL170

26. If $\oint_s \vec{E}.\vec{dS} = 0$ over a surface, then:

(1) the magnitude of electric field on the surface is constant.

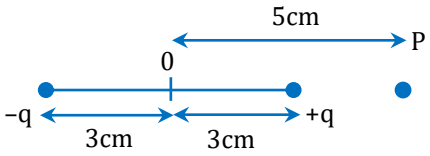
(2) all the charges must necessarily be inside the surface.

(3) the electric field inside the surface is necessarily uniform.

(4) the number of flux lines entering the surface must be equal to the number of flux lines leaving it.

PEL171

27. An electric dipole is placed as shown in the figure.



The electric potential (in 10^2 V) at point P due to the dipole is (ϵ_0 =permittivity of free space and $\frac{1}{4\pi\epsilon_0} = K$):

(1) $\left(\frac{5}{8}\right)qK$ (2) $\left(\frac{8}{5}\right)qK$

(3) $\left(\frac{8}{3}\right)qK$ (4) $\left(\frac{3}{8}\right)qK$

PEL172

RE-NEET(UG)-2023

28. According to Gauss law of electrostatics, electric flux through a closed surface depends on :

(1) the area of the surface

(2) the quantity of charges enclosed by the surface

(3) the shape of the surface

(4) the volume enclosed by the surface

PEL173

29. A charge Q μC is placed at the centre of a cube. The flux coming out from any one of its faces will be (in SI unit) :

(1) $\frac{Q}{\epsilon_0}\times10^{-6}$ (2) $\frac{2Q}{3\epsilon_0}\times10^{-3}$

(3) $\frac{Q}{6\epsilon_0}\times10^{-3}$ (4) $\frac{Q}{6\epsilon_0}\times10^{-6}$

PEL174

30. If a conducting sphere of radius R is charged. Then the electric field at a distance r(r > R) from the centre of the sphere would be, (V = potential on the surface of the sphere)

(1) $\frac{rV}{R^2}$ (2) $\frac{R^2V}{r^3}$ (3) $\frac{RV}{r^2}$ (4) $\frac{V}{r}$

PEL175

NEET(UG)-2024

31. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R.**

Assertion (A) :- The potential (V) at any axial point, at 2 m distance (r) from the centre of the dipole of dipole moment vector $\vec{P}$ of magnitude,

4×10^{-6} C m, is $\pm 9 \times 10^3$ V.

(Take $\frac{1}{4\pi\epsilon_0} = 9\times10^9$ SI Units)

Reason (R) :- $V = \pm\frac{2P}{4\pi\epsilon_0 r^2}$,, where r is the distance of any axial point, situated at 2 m from the centre of the dipole.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true and R is NOT the correct explanation of A.

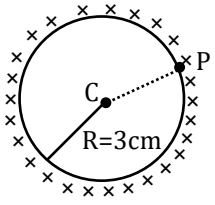
(3) A is true but R is false.

(4) A is false but R is true.

PEL176

32. A thin spherical shell is charged by some source. The potential difference between the two points C and P (in V) shown in the figure is :

(Take $\frac{1}{4\pi\epsilon_0} = 9\times10^9$ SI units)



q = 1μC

(1) 3×10^5 (2) 1×10^5

(3) 0.5×10^5 (4) zero

PEL177

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RE-NEET(UG)-2024

33. A metal cube of side 5 cm is charged with 6μC. The surface charge density on the cube is :

(1) 0.125×10^{-3} C m^{-2}

(2) 0.25×10^{-3} C m^{-2}

(3) 4×10^{-3} C m^{-2}

(4) 0.4×10^{-3} C m^{-2}

PEL178

34. The value of electric potential at a distance of 9 cm from the point charge 4×10^{-7} C is $\left(\text{Given } \frac{1}{4\pi\varepsilon_0} = 9\times10^9 \text{ Nm}^2\text{C}^{-2}\right)$:

(1) 4×10^2 V (2) 44.4 V

(3) 4.4×10^5 V (4) 4×10^4 V

PEL179

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	2	3	4	1	2	1	2	3	2	2	3	2	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	2	2	2	4	2	1	1	3	4	4	2	4	3
Question	31	32	33	34											
Answer	3	4	4	4											

Exercise - III (Analytical Questions) Master Your Understanding

1. A ring with a uniform charge distribution with a total charge Q and radius R is placed in the yz plane with its centre at the origin then which of the following is incorrect about this situation:

(1) The electric field at the origin is zero

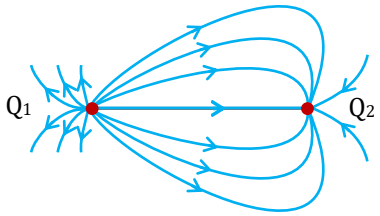
(2) The electric potential at the origin is $\frac{1}{4\pi\varepsilon_0}\frac{Q}{R}$

(3) The electric field at the point (x, 0, 0) is $\frac{1}{4\pi\varepsilon_0}\frac{Q}{x^2}$

(4) The electric field at the point (x, 0, 0) is $\frac{1}{4\pi\varepsilon_0}\frac{Qx}{\left(R^2+x^2\right)^{3/2}}$

PEL143

2. A few electric field lines for a system of two charges Q_1 and Q_2 fixed at two different points on their axis are shown in the figure. With the figure what information about charges and electric field is incorrect.



(1) $|Q_1| > |Q_2|$

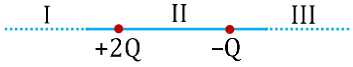
(2) $Q_1 \rightarrow$ +ve, $Q_2 \rightarrow$ –ve

(3) At a finite distance to the left of Q_1 the electric field is zero

(4) At a finite distance to the right of Q_2 the electric field is zero

PEL144

3. The figure shows, two point charges $q_1 = +2Q$ and $q_2 = -Q$. The charges divide the line joining them, in three parts I, II and III as shown in figure then which of the following statements is/are correct:



(1) Electric field may be zero in region III

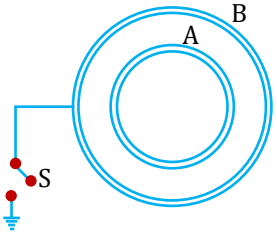
(2) Electric field may be zero in region I

(3) Electric field may be zero in region II

(4) Electric field can not be zero anywhere on this line

PEL145

4. Two concentric shells have radii R and 2R charges q_A and q_B and potentials 2V and (3/2)V respectively. Now shell B is earthed by closing the switch S and let charges on the shells be changed to q_A' and q_B'. Then incorrect option is:



(1) $|q_A|/|q_B| = 1$

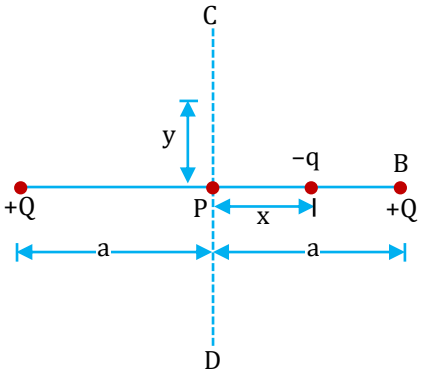
(2) $|q'_A|/|q'_B| = 1$

(3) Potential of B after earthing becomes zero.

(4) Potential difference between the shells depends on charge of inner sphere.

PEL146

5. For the arrangement shown in the Fig., the two positive charges, +Q each, are fixed. Mark out the correct statement regarding a third charged particle –q placed at mid point P that can be displaced along or perpendicular to the line connecting the charges,



(1) The particle will perform SHM for x << a.

(2) The particle will undergo oscillatory motion but not SHM about P for x comparable to a.

(3) The particle will perform SHM for y << a.

(4) The particle will undergo oscillatory motion but not SHM about P for y << a.

PEL147

6. Four charges Q_1, Q_2, Q_3 and Q_4 of same magnitude are fixed along the x axis at x = –2a, –a, +a and +2a, respectively. A positive charge q is placed on the positive y axis at a distance b > 0. Four options of the signs of these charges are given in List I. The direction of the forces on the charge q is given in List II. Match List I with List II and select the correct answer using the code given below the lists.

	List-I		List-II
P	Q_1, Q_2, Q_3, Q_4 all positive	1	+x
Q	Q_1, Q_2 positive, Q_3, Q_4 negative	2	–x
R	Q_1, Q_4 positive, Q_2, Q_3 negative	3	+y
S	Q_1, Q_3 positive, Q_2, Q_4 negative	4	–y

Codes:

(1) P-3, Q-1, R-4, S-2

(2) P-4, Q-2, R-3, S-1

(3) P-3, Q-1, R-2, S-4

(4) P-4, Q-2, R-1, S-3

PEL148

7. Match Column-I with Column-II

	Column-I		Column-II
I.	Electric field due to Infinite plane sheet of charge	a.	0
II.	Electric field due to Infinite conducting plane sheet of uniform thickness	b.	$\frac{\sigma}{2\varepsilon_0}$
III.	Electric field due to Non-conducting charged solid sphere at its surface	c.	$\frac{R\rho}{3\varepsilon_0}$
IV.	Electric field due to Non-conducting charged solid sphere at its centre	d.	$\frac{\sigma}{\varepsilon_0}$

(1) I-c, II-d, III-a, IV-b

(2) I-b, II-d, III-c, IV-a

(3) I-b, II-d, III-a, IV-c

(4) I-b, II-a, III-d, IV-c

PEL149

8. Match the facts given in Column I with the Column II :

	Column-I		Column-II
I.	Charge cannot exist	a.	without charge
II.	Mass can exist	b.	without mass
III.	Charge is	c.	not conserved
IV.	Mass is	d.	conserved

(1) I-b, II-a, III-d, IV-c (2) I-b, II-a, III-c, IV-d

(3) I-b, II-d, III-a, IV-c (4) I-b, II-c, III-d, IV-a

PEL150

9. Match Column-I with Column-II :-

	Column-I		Column-II
I.	ε_0	a.	$M^{\circ}L^{\circ}T^{\circ}A^{\circ}$
II.	ε_r	b.	$[M^1L^1T^{-2}]$
III.	E	c.	$[M^{-1}L^{-3}T^4A^2]$
IV.	F	d.	$[M^1L^1T^{-3}A^{-1}]$

(1) I-c, II-a, III-b, IV-d (2) I-a, II-c, III-b, IV-d

(3) I-b, II-c, III-a, IV-d (4) I-c, II-a, III-d, IV-b

PEL151

10. Electric potential on the surface of a uniformly charged solid sphere is V. Radius of the sphere is 1m. Match the following two columns.

	Column-I		Column-II
(a)	Electric potential at $r=\frac{R}{2}$	(p)	$\frac{V}{4}$
(b)	Electric potential at r = 2R	(q)	$\frac{V}{2}$
(c)	Electric field at $r=\frac{R}{2}$	(r)	$\frac{3V}{4}$
(d)	Electric field at r=2R	(s)	$\frac{11V}{8}$

(1) a-p, b-s, c-r, d-q (2) a-p, b-r, c-q, d-s

(3) a-s, b-q, c-q, d-p (4) a-q, b-q, c-r, d-s

PEL152

11. Given below are two statements :

Statement I : Two concentric conducting spherical shells are charged. The charge on the outer shell is varied keeping the charge on inner shell constant, as a result the electric potential difference between the two shells does not change.

Statement II : In the above situation, the electric field in the region between both the shells depends only on the charge existing on the inner shell.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both statement I and statement II are correct.

(2) Statement I is correct and statement II is incorrect.

(3) Statement I is incorrect and statement II is correct.

(4) Both statements I and statements II are incorrect.

PEL153

12. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : If a charge is released from rest in an electric field, it will always start to move along an electric field line.

Reason (R) : Force on a charged particle is always in the direction of electric field.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PEL154

13. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Electrostatic potential is a relative physical quantity.

Reason (R) : Electric potential difference is an absolute physical quantity.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

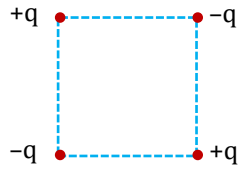
(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PEL155

14. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): The electric potential and the electric field intensity at the centre of a square having four fixed point charges at their vertices as shown in figure are zero.



Reason (R) : If electric potential at a point is zero then the magnitude of electric field at that point must be zero.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PEL156

15. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The tyres of aircrafts are slightly conducting.

Reason (R) : If a conductor is connected to ground, the extra charge induced on conductor will flow to ground.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PEL157

16. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Any arbitrary displacement of charges inside a conducting shell does not introduce any change in the electrostatic field of the outer space.

Reason (R) : A closed conducting shell divides the entire space into the inner and outer parts which are completely independent of one another in respect of electric fields.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PEL158

17. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : If a positive point charge is brought near an isolated neutral metal cube, the cube becomes negatively charged.

Reason (R) : If a positive point charge is brought near an isolated neutral metal cube, the cube remains neutral.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PEL159

18. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In a given situation of arrangement of charges, an additional charge is placed outside the Gaussian surface. In this situation, in the Gauss theorem $\oint \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\epsilon_0}$, q_{in} remains unchanged whereas electric field $\vec{E}$ is changed.

Reason (R) : Electric field $\vec{E}$ at any point on the Gaussian surface is due to inside charge only.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PEL160

19. If a body is charged by rubbing it with another body then weight of the system :–

(A) may increase slightly

(B) may decrease slightly

(C) must increase slightly

(D) remains precisely constant

(1) Only A (2) AB (3) D (4) C

PEL161

20. **Statement-I :** Coulombic force follows the principle of superposition.

Statement-II : Coulombic force is directly proportional to the charges on both particle.

(1) Statement–I is True, Statement–II is True; Statement–II is a correct explanation for Statement–I.

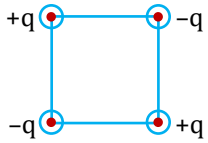
(2) Statement–I is True, Statement–II is True; Statement–II is not a correct explanation for Statement–I.

(3) Statement–I is True, Statement–II is False.

(4) Statement–I is False, Statement–II is True.

PEL162

21. **Statement-I :** The electric potential and the electric field intensity at the centre of a square having four point charges at their vertices (as shown) are zero.



Statement-II : Electric field is negative derivative of the potential.

(1) Statement-I is True, Statement-II is True; Statement-II **is** a correct explanation for Statement-I

(2) Statement-I is True, Statement-II is True; Statement-II **is NOT** a correct explanation for Statement-1

(3) Statement-I is True, Statement-II is False

(4) Statement-I is False, Statement-II is True.

PEL163

22. Two point charges 2 μC and 8 μC are placed at position x = 0 and x = 4 m respectively. The distance of the neutral point form the smaller charge is found to be at x= α/β. Find the value of $(\alpha + \beta)$.

(1) 7 (2) 5 (3) 4 (4) 3

PEL164

23. **Statement-I :** In a given situation of arrangement of charges, an extra charge is placed outside the Gaussion surface. In the Gauss Theorem

$$\oint \vec{E}.d\vec{s} = \frac{Q_{in}}{\epsilon_0}$$

remains unchanged where as electric field $\vec{E}$ at the site of the element is changed.

Statement-II : Electric field $\vec{E}$ at any point on the Gaussian surface is due to inside charge only.

(1) Statement-I is true, statement-II is true and statement-II is correct explanation for statement-I.

(2) Statement-I is true, statement-II is true and statement-II is NOT the correct explanation for statement-I.

(3) Statement-I is true, statement-II is false.

(4) Statement-I is false, statement-II is true.

PEL165

24. Two concentric spherical shells are given charges:-

(A) Outer shell will be at higher potential if the charges given to both shells are negative

(B) Inner shell will be at higher potential if the charges given to both shells are positive

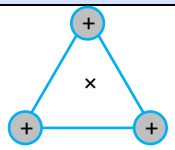
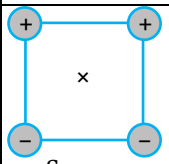
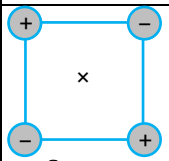
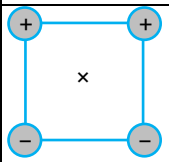
(C) Outer shell will always be at higher potential irrespective of the sign of charges given to two spheres

(D) No prediction can be made using given data.

(1) B (2) A,B (3) C (4) D

PEL166

25. In the following diagrams, all the charges have equal magnitude

Column-I		Column-II	
(A)	 Equilateral triangle	(P)	The potential is zero at the centre
(B)	 Square	(Q)	The electric field is zero at the centre
(C)	 Square		
(D)	 Rectangle		

(1) A → Q; B → P; C → P,Q; D → P

(2) A → P,Q; B → P; C → P,Q; D → Q

(3) A → Q; B → P,Q; C → P; D → P,Q

(4) A → P,Q; B → P; C → Q; D → P

PEL167

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26. **Statement-I :** Work done by electric field on moving a positive charge on an equipotential surface is always zero. **Statement-II :** Electric lines of forces are perpendicular to equipotential surface.

(1) Statement–I is True, Statement–II is True; Statement–II is a correct explanation for Statement–I.

(2) Statement–I is True, Statement–II is True; Statement–II is not a correct explanation for Statement–I.

(3) Statement–I is True, Statement–II is False.

(4) Statement–I is False, Statement–II is True.

PEL168

27. **Statement-I** : Electric field outside the uniformly charged spherical shell (due to shell) is the same as that of a point charge at its center.

Statement-II : The centre of mass of uniform shell is at its center.

(1) Statement-I is true, statement-II is true and statement-II is correct explanation for statement-I.

(2) Statement-I is true, statement-II is true and statement-II is NOT the correct explanation for statement-I.

(3) Statement-I is true, statement-II is false.

(4) Statement-I is false, statement-II is true.

PEL169

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	1	1	3	1	2	1	4	3	1	3	2	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27			
Answer	1	4	3	3	2	2	1	3	2	1	1	2			

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Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

ELECTRIC CURRENT & DRIFT VELOCITY

1. If 10^6 electrons/s are flowing through an area of cross section of 10^{-4} m^2 then the current will be :–
(1) 1.6×10^{-7} A (2) 1.6×10^{-13} A
(3) 1×10^{-6} A (4) 1×10^{2} A
PCE001

2. The current in a conductor varies with time t as $I = 2t + 3t^2$ A where I is amperes and t in seconds. Electric charge flowing through a section of the conductor during t = 2 s to t = 3 s is :-
(1) 10 C (2) 24 C (3) 33 C (4) 44 C
PCE002

3. 10,000 electrons are passing per minute through a tube of radius 1cm. The resulting current is :
(1) 10000 A (2) 0.25×10^{-16} A
(3) 10^{-9} A (4) 0.5×10^{-19} A
PCE003

4. There are 8.4×10^{22} free electrons per cm^3 in copper. The current in the wire is 0.21 A ($e = 1.6 \times 10^{-19}$ C). Then the drifts velocity of electrons in a copper wire of 1 mm^2 cross section, will be :–
(1) 2.12×10^{-5} m/s (2) 0.78×10^{-5} m/s
(3) 1.56×10^{-5} m/s (4) none of these
PCE004

5. There is a current of 40 amperes in a wire of 10^{-6} m^2 area of cross-section. If the number of free electrons per m^3 is 10^{29}, then the drift velocity will be
(1) 1.25×10^{3} m/s (2) 2.50×10^{-3} m/s
(3) 25.0×10^{-3} m/s (4) 250×10^{-3} m/s
PCE005

6. S.I. unit of current is :-
(1) C (2) A (3) A/s (4) N/s
PCE006

7. When no current flows through a conductor :-
(1) the free electrons do not move
(2) the average speed of a free electron over a large period of time is zero
(3) the average velocity of a free electron over a large period of time is zero
(4) the average of square of velocities of all the free electrons at an instant is zero
PCE007

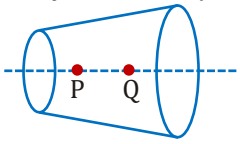
8. The number of free electrons per 10 mm of an ordinary copper wire is about 2×10^{21}. The average drift speed of the electrons is 0.25 mm/s The current flowing is :-
(1) 0.8 A (2) 8 A (3) 80 A (4) 5 A
PCE008

9. In a Neon discharge tube 2.9×10^{18} Ne^+ ions move to the right each second, while 1.2×10^{18} electrons move to the left per second; electron charge is 1.6×10^{-19} C. The current in the discharge tube is :-
(1) 1 A towards right
(2) 0.66 A towards right
(3) 0.66 A towards left
(4) zero
PCE009

10. Two wires each of radius of cross section r but of different materials are connected together end to end (in series). If the densities of charge carriers in the two wires are in the ratio 1:4, the drift velocity of electrons in the two wires will be in the ratio :
(1) 1 : 2 (2) 2 : 1 (3) 4 : 1 (4) 1 : 4
PCE010

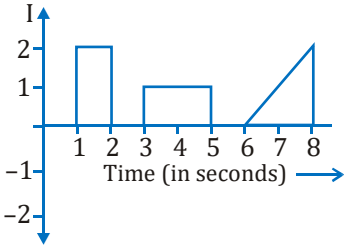
11. A current I flows through a uniform wire of diameter d when the electron drift velocity is v. The same current will flow through a wire of diameter d/2 made of the same material if the drift velocity of the electrons is
(1) v/4 (2) v/2 (3) 2v (4) 4v
PCE011

12. A wire has a non–uniform cross–section as shown in figure. A steady current flows through it. The drift speed of electrons at points P and Q is v_P and v_Q, then :-



(1) $v_P = v_Q$
(2) $v_P < v_Q$
(3) $v_P > v_Q$
(4) data is insufficient
PCE012

13. The plot represents the flow of current through a wire for different time intervals. The ratio of charges flowing through the wire corresponding to these time intervals is (see figure) :-



(1) 2 : 1 : 2 (2) 1 : 3 : 3
(3) 1 : 1 : 1 (4) 2 : 3 : 4

PCE013

OHM'S LAW & ELECTRICAL RESISTANCE

14. Three copper wires are there with lengths and cross-sectional areas as (ℓ, A); $\left(2\ell, \frac{A}{2}\right)$ and $\left(\frac{\ell}{2}, 2A\right)$. Resistance :-

(1) minimum for the wire of cross-sectional are $\frac{A}{2}$
(2) minimum for the wire of cross-sectional are A
(3) minimum for the wire of cross-sectional area 2A
(4) same for all the three cases.

PCE014

15. A wire of uniform cross-section A, length ℓ and resistance R is bent into a complete circle; the resistance between any two of diametrically opposite points will be :–

(1) $\frac{R}{2}$ (2) $\frac{R}{4}$ (3) $\frac{R}{8}$ (4) 4R

PCE015

16. The electric resistance of a certain wire of iron is R. If its length and radius both are doubled,
then :–
(1) the resistance will be halved and the specific resistance will remain unchanged
(2) the resistance will be halved and the specific resistance will be doubled
(3) the resistance and the specific resistance, will both remain unchanged
(4) the resistance will be doubled and the specific resistance will be halved.

PCE016

17. When a piece of aluminium wire of finite length is drawn to reduce its diameter to half its original value, its resistance will become :-

(1) two times (2) four times
(3) eight times (4) sixteen times

PCE017

18. As the temperature of a metallic resistor is increased, the product of resistivity and conductivity :-

(1) increases
(2) decreases
(3) may increase or decrease
(4) remains constant.

PCE018

19. If a wire is stretched, so that its length is 20% more than its initial length, the percentage increase in the resistance of the wire is :-

(1) 40% (2) 10%
(3) 44% (4) 25%

PCE019

20. The length of a given cylindrical wire is increased by 100%. Due to the consequent decrease in diameter the change in the resistance of the wire will be :–

(1) 300% (2) 200%
(3) 100% (4) 50%

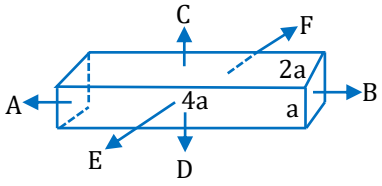
PCE020

21. On increasing the temperature, the specific resistance of a conductor and a semiconductor-

(1) both increase
(2) both decrease
(3) increases and decreases respectively
(4) decreases and increases respectively

PCE021

22. A conductor with rectangular cross section has dimensions (a × 2a × 4a) as shown in figure. Resistance across AB is x, across CD is y and across EF is z. Then



(1) x = y = z (2) x > y > z
(3) y > x > z (4) x > z > y

PCE022

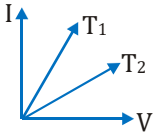
23. Specific resistance of a conductor increases with:–
(1) increase in temperature.
(2) increase in cross–sectional area
(3) increase in cross–sectional area and decrease in length.
(4) decrease in cross–sectional area.

PCE023

24. The temperature coefficient of resistance of a wire is 0.00125 per degree celcius. At 300 K its resistance is 1 ohm. The resistance of the wire will be 2 ohms at a temperature :–
(1) 1154 K (2) 1127 K
(3) 600 K (4) 1400 K

PCE024

25. The current voltage graph for a given metallic conductor at two different temperatures T_1 and T_2 are as shown in the figure. Then :–



(1) $T_1 > T_2$
(2) $T_1 = T_2$
(3) nothing can be said about T_1 and T_2
(4) $T_1 < T_2$

PCE025

26. The effective resistance is $\frac{6}{5}\ \Omega$, when two wires are joined in parallel. When one of the wire breaks, the effective resistance is 2 ohms. The resistance of the broken wire was :–
(1) $\frac{3}{5}\ \Omega$ (2) $2\ \Omega$ (3) $\frac{6}{5}\ \Omega$ (4) $3\ \Omega$

PCE026

27. At what temperature will the resistance of a copper wire become three times its value at 0° C? [Temperature coefficient of resistance for copper = 4×10^{-3} per °C] :–
(1) 400° C (2) 450° C
(3) 500° C (4) 600° C

PCE027

28. Copper and silicon are cooled from 300 K to 60 K; the specific resistance :–
(1) decreases in copper but increases in silicon
(2) increases in copper but decreases in silicon
(3) increases in both
(4) decreases in both

PCE028

29. Two resistances R_1 and R_2 are made of different materials. The temperature coefficient of the material of R_1 is α and that of the material of R_2 is $-\beta$. The resistance of the series combination of R_1 and R_2 does not change with temperature, then the ratio of resistances of the two wires at 0°C will be :

(1) $\frac{\alpha}{\beta}$ (2) $\frac{\alpha+\beta}{\alpha-\beta}$
(3) $\frac{\alpha^2+\beta^2}{\alpha\beta}$ (4) $\frac{\beta}{\alpha}$

PCE029

COMBINATION OF RESISTANCES & KIRCHHOFF'S LAW

30. A metal wire of resistance R is cut into three equal pieces which are then connected side by side to form a new wire, the length of which is equal to one third of the original length. The resistance of this new wire is :-

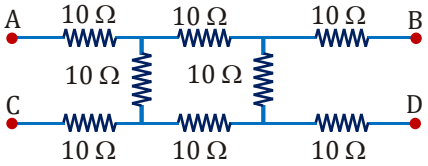
(1) R (2) 3R (3) $\frac{R}{9}$ (4) $\frac{R}{3}$

PCE030

31. Three resistances of values 2 Ω, 3 Ω and 6 Ω are to be connected to yield an effective resistance of 4 Ω. This can be done by connecting :
(1) 3 Ω resistance in series with a parallel combination of 2 Ω and 6 Ω
(2) 6 Ω resistance in series with a parallel combination of 2 Ω and 3 Ω
(3) 2 Ω resistance in series with a parallel combination of 3 Ω and 6 Ω
(4) 2 Ω resistance in parallel with a parallel combination of 3 Ω and 6 Ω

PCE031

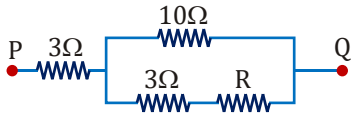
32. What will be the equivalent resistance between the points A and D?



(1) 10 Ω (2) 20 Ω
(3) 30 Ω (4) 40 Ω

PCE032

33. In the circuit shown here, what is the value of the unknown resistance R so that the total resistance of the circuit between points 'P' and 'Q' is also equal to R :–

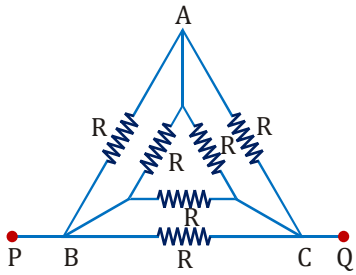


(1) 3 Ω (2) $\sqrt{39}\,\Omega$

(3) $\sqrt{69}\,\Omega$ (4) 10 Ω

PCE033

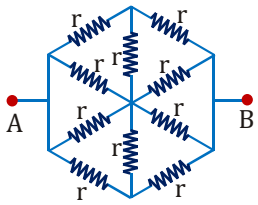
34. The resistance across P and Q in the given figure is



(1) $\frac{R}{3}$ (2) $\frac{R}{2}$ (3) 2R (4) 6R

PCE034

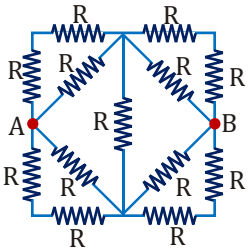
35. The resistance of the circuit between A and B is :



(1) r (2) 0.5r (3) 2r (4) 3r

PCE035

36. Thirteen resistances each of resistance R Ω are connected in the circuit as shown in the figure. The effective resistance between A and B is :-

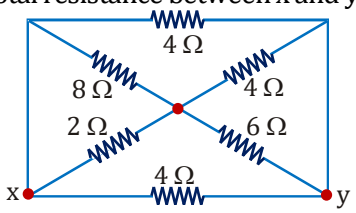


(1) $\frac{4R}{3}\Omega$ (2) 2RΩ

(3) R Ω (4) $\frac{2R}{3}\Omega$

PCE036

37. The total resistance between x and y in ohms is:-



(1) 1 Ω (2) 4 Ω

(3) $\frac{4}{3}$ Ω (4) $\frac{2}{3}$ Ω

PCE037

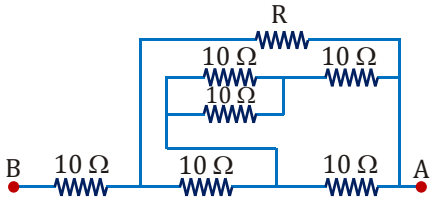
38. The resultant resistance of n wires each of resistance r ohms is R, when they are connected in parallel. When these n resistances are connected in series, the resultant resistance will be :-

(1) $\frac{R}{n}$ (2) $\frac{R}{n^2}$

(3) nR (4) n^2R

PCE038

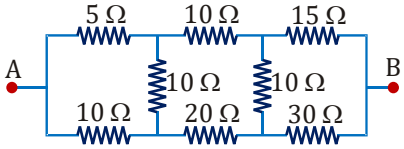
39. For the network of resistance shown in the fig. the equivalent resistance of the network between the points A and B is 18 Ω. The value of unknown resistance R is :-



(1) 8 Ω (2) 10 Ω (3) 16 Ω (4) 24 Ω

PCE039

40. In the arrangement of resistances shown below, the effective resistance between points A and B is

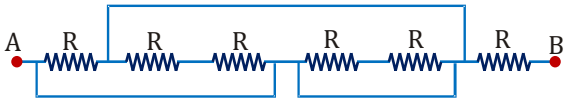


(1) 20 Ω (2) 30 Ω

(3) 90 Ω (4) 110 Ω

PCE040

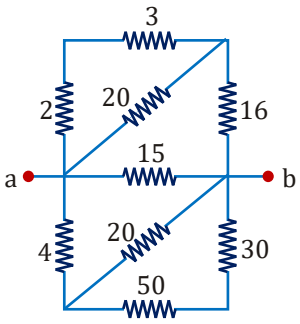
41. In the circuit shown the equivalent resistance between A and B is



(1) R (2) 2R/5 (3) R/3 (4) 2R

PCE041

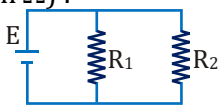
42. In the figure the numerical values denote resistances in SI units. The total resistance of the circuit between a & b will be:



(1) 12 ohms (2) 24 ohms
(3) 15 ohms (4) 6 ohms

PCE042

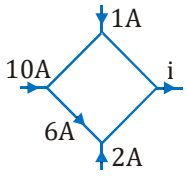
43. Resistors R_1 and R_2 have an equivalent resistance of 6 ohms when connected in the circuit shown below. The resistance of R_1 could be (in Ω) :-



(1) 1 (2) 5 (3) 8 (4) 4

PCE043

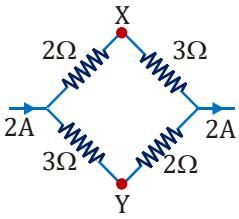
44. Value of current i in the following circuit is :-



(1) 13 A (2) 12 A
(3) 9 A (4) none of the above

PCE044

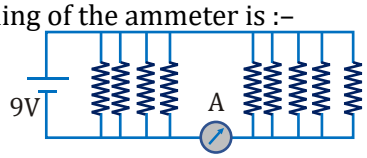
45. The potential difference between X and Y in volts is :-



(1) 1 (2) –1 (3) 2 (4) –2

PCE045

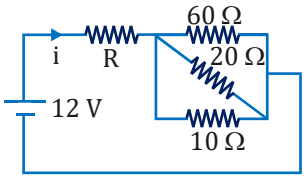
46. If each resistance in the fig. is of 9 Ω then reading of the ammeter is :–



(1) 5 A (2) 8 A (3) 2 A (4) 9 A

PCE046

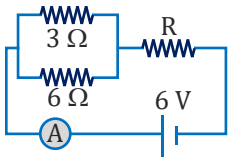
47. If i = 0.25 A in figure, then value of R is :–



(1) 48 Ω (2) 12 Ω (3) 120 Ω (4) 42 Ω

PCE047

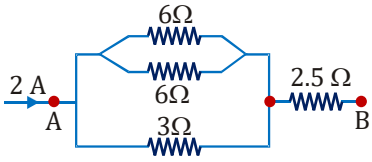
48. If the ammeter in the given circuit reads 2 A, the resistance R is :–



(1) 1 ohms (2) 2 ohms
(3) 3 ohms (4) 4 ohms

PCE048

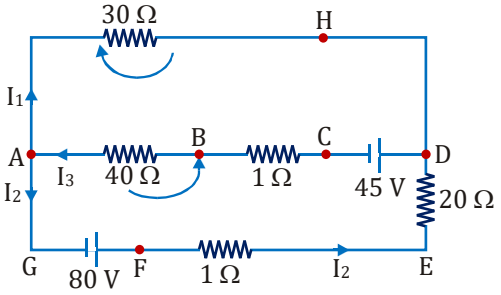
49. The equivalent resistance and potential difference between A and B for the circuit are respectively :–



(1) 4 Ω, 8 V (2) 8 Ω, 4 V
(3) 2 Ω, 2 V (4) 16 Ω, 8 V

PCE049

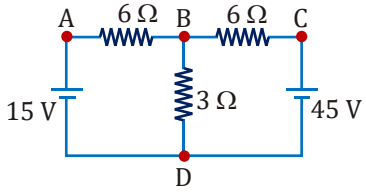
50. For the current loops shown in the figure, Kirchhoff's loop rule for the loops AHDCBA and AHDEFGA yields these equations respectively :-



(1) $-30\ I_1 - 41\ I_3 + 45 = 0$ and $-30\ I_1 + 21\ I_2 - 80 = 0$
(2) $30\ I_1 - 41\ I_3 + 45 = 0$ and $30\ I_1 - 21\ I_2 - 80 = 0$
(3) $30\ I_1 + 41\ I_3 - 45 = 0$ and $-30\ I_1 + 21\ I_2 + 80 = 0$
(4) $-30\ I_1 - 41\ I_3 - 45 = 0$ and $-30\ I_1 + 21\ I_2 - 80 = 0$

PCE050

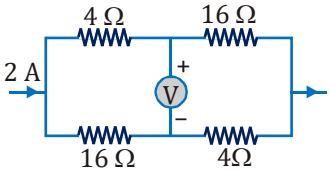
51. In the circuit shown in figure, find the current through the branch BD.



(1) 1 A (2) 5 A
(3) 3 A (4) 7 A

PCE051

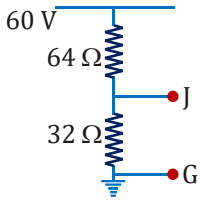
52. In the circuit shown below, the reading of the voltmeter V is :-



(1) 12 V (2) 8 V
(3) 20 V (4) 16 V

PCE052

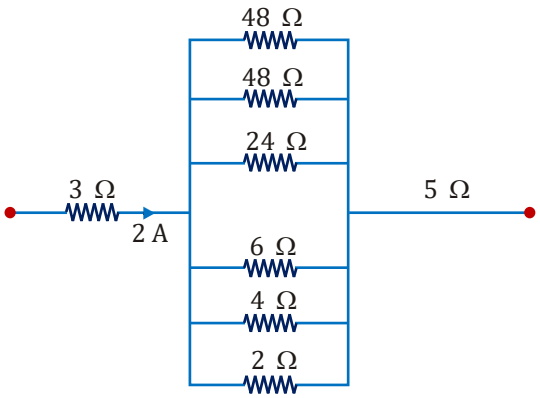
53. Find the potential of J with respect of G :-



(1) 40 V (2) 60 V
(3) 20 V (4) 30 V

PCE053

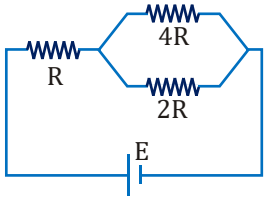
54. Find the potential difference across the 24 Ω :-



(1) 48 volts (2) 2 volts
(3) 4 volts (4) 1 volts

PCE054

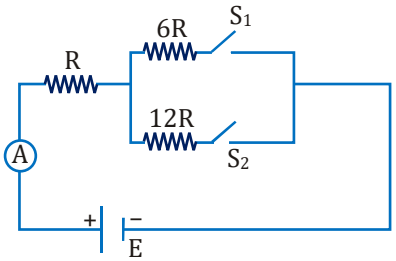
55. In a network as shown in the figure the potential difference across the resistance 2R is (the cell has an emf E with no internal resistance) :



(1) 2E (2) $\frac{4E}{7}$ (3) $\frac{E}{7}$ (4) E

PCE055

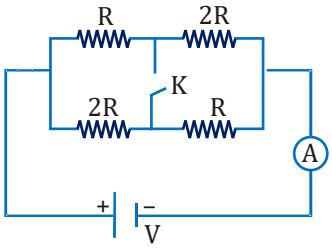
56. The reading of ammeter is I_1 when only S_1 is closed and I_2 when only S_2 is closed. The reading is I_3 when both S_1 & S_2 are closed simultaneously then :-



(1) $I_1 > I_2 > I_3$ (2) $I_2 > I_3 > I_1$
(3) $I_3 > I_2 > I_1$ (4) $I_3 > I_1 > I_2$

PCE056

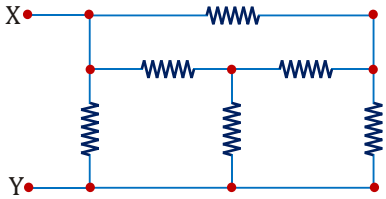
57. How will the reading of ammeter change if the key k is closed?



(1) Increase
(2) Decrease
(3) Remains same
(4) Information insufficient

PCE057

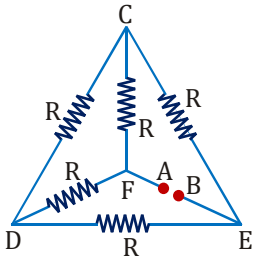
58. Six resistors each of 10 Ω are connected as shown. The equivalent resistance between points X and Y is :



(1) 20 Ω (2) 5 Ω (3) 25/3 Ω (4) 10 Ω

PCE058

59. Five equal resistances each of resistance R are connected as shown in the Figure. A battery of voltage V is connected between A and B. The current flowing in AFCEB will be

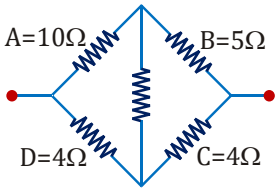


(1) $\frac{V}{R}$ (2) $\frac{V}{2R}$

(3) $\frac{2V}{R}$ (4) $\frac{3V}{R}$

PCE059

60. In a typical Wheatstone bridge the resistance in cyclic order are A = 10 Ω, B = 5 Ω, C = 4Ω and D = 4 Ω. For the bridge to be balanced :

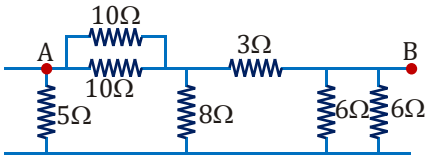


(a) 10 Ω should be connected in parallel with A
(b) 10 Ω should be connected in series with A
(c) 5 Ω should be connected in series with B
(d) 5 Ω should be connected in parallel with B

(1) a, b (2) b, c
(3) a, c (4) all

PCE060

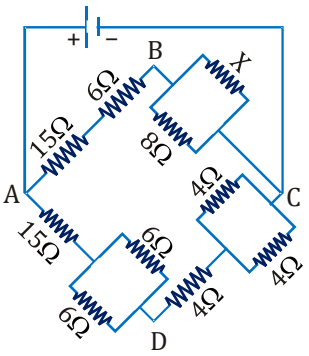
61. Seven resistances are connected as shown in the figure. The equivalent resistance between A and B is :–



(1) 3 Ω (2) 4 Ω
(3) 4.5 Ω (4) 5 Ω

PCE061

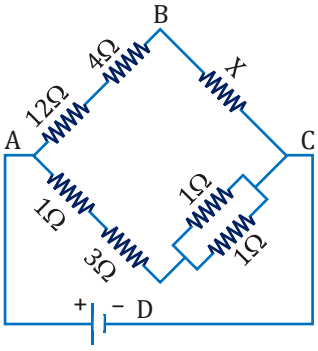
62. In the following circuit diagram the value of resistance X for the potential difference between B and D to be zero is :-



(1) 4 ohms (2) 6 ohms
(3) 56 ohms (4) 9 ohms

PCE062

63. In the arrangement of resistances shown in the circuit, the potential difference between points B and D will be zero, when the unknown resistance X is :-



(1) 4 Ω (2) 3 Ω (3) 2 Ω (4) 1 Ω

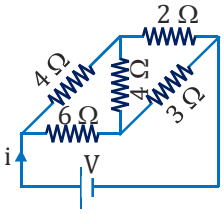
PCE063

64. The resistance of each arm of a Wheat stone bridge is 10 Ω. A resistance of 10 Ω is connected in series with the galvanometer then the equivalent resistance across the battery will be :–

(1) 10 Ω (2) 15 Ω (3) 20 Ω (4) 40 Ω

PCE064

65. For the network shown in the figure the value of the current i is :–

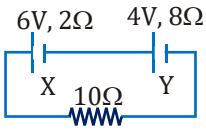


(1) $\frac{18V}{5}$ (2) $\frac{5V}{9}$ (3) $\frac{9V}{35}$ (4) $\frac{5V}{18}$

PCE065

CELLS & COMBINATION OF CELLS, ELECTRIC POWER & ENERGY

66. Two cells X and Y are connected to a resistance of 10 Ω as shown in the figure. The terminal voltage of cell Y is :-



(1) zero (2) 2 V (3) 4 V (4) 10 V

PCE066

67. A battery has e.m.f. 4 V and internal resistance 'r'. When this battery is connected to an external resistance of 2 ohms, a current of 1 A flows in the circuit. What current will flow if the terminals of the battery are connected directly?

(1) 1 A (2) 2 A
(3) 4 A (4) infinite

PCE067

68. Internal resistance of primary cell depends on :-

(1) the nature of electrolyte
(2) the area of plates immersed in the electrolyte
(3) the concentration of electrolyte and distance between the plates
(4) all the above

PCE068

69. The potential difference between the terminals of a cell is found to be 3 volts when it is connected to a resistance of value equal to its internal resistance. The e.m.f. of the cell is :-

(1) 3 V (2) 6 V (3) 1.5 V (4) 4.5 V

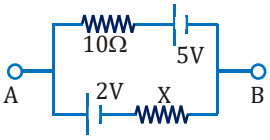
PCE069

70. A current of 2 A is flowing through a cell of e.m.f. 5 V and internal resistance 0.5 Ω from negative to positive electrode. If the potential of negative electrode is 10 V, the potential of positive electrode will be :-

(1) 5 V (2) 14 V (3) 15 V (4) 16 V

PCE070

71. In the following circuit if $V_A - V_B = 4$ V, then the value of resistance X in ohms will be :-



(1) 5 (2) 10 (3) 15 (4) 20

PCE071

72. Electromotive force of a cell is basically a

(1) force (2) power
(3) work (4) current capacity

PCE072

73. The terminal voltage is $\frac{E}{2}$ when a current of 2 A is flowing through 2 Ω resistance; the internal resistance of the cell is :–

(1) 1 Ω (2) 2 Ω (3) 3 Ω (4) 4 Ω

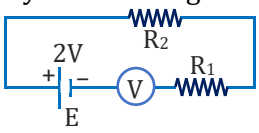
PCE073

74. When a resistance of 2 ohms is connected across the terminals of a cell, the current is 0.5 A. When the resistance is increased to 5 ohms, the current becomes 0.25 A. The e.m.f. of the cell is:-

(1) 1.0 V (2) 1.5 V (3) 2.0 V (4) 2.5 V

PCE074

75. A cell of e.m.f. 2 V and negligible internal resistance is connected to resistors R_1 and R_2 as shown in the figure. The resistance of the voltmeter, R_1 and R_2 are 80 Ω, 40 Ω and 80 Ω respectively. The reading of the voltmeter is :



(1) 1.78 V (2) 1.60 V
(3) 0.80 V (4) 1.33 V

PCE075

76. A cell supplies a current of 0.9 A through a 2 Ω resistor and a current of 0.3 A through a 7 Ω resistor. The internal resistance of the cell is :-

(1) 1.0 Ω (2) 0.5 Ω
(3) 2.0 Ω (4) 1.2 Ω

PCE076

77. A 10 V battery with internal resistance 0.5 Ω is connected across a variable resistance R. The value of R for which the power delivered to it is maximum, is equal to :-

(1) 0.5 Ω (2) 1 Ω
(3) 1.5 Ω (4) 2 Ω

PCE077

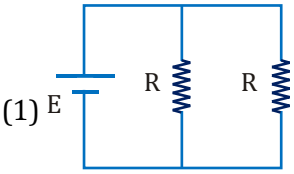
78. An electric bulb is designed to draw a power P_0 at voltage V_0. If the voltage is V, it draws a power P, then –

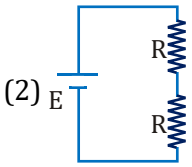
(1) $P = \left(\frac{V}{V_0}\right) P_0$ (2) $P = \left(\frac{V_0}{V}\right) P_0$

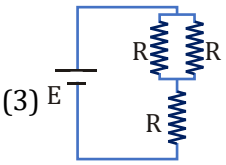
(3) $P = \left(\frac{V_0}{V}\right)^2 P_0$ (4) $P = \left(\frac{V}{V_0}\right)^2 P_0$

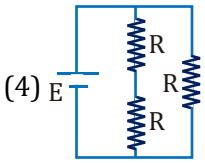
PCE078

79. Consider the four circuits shown in the figure below. In which circuit power dissipated maximum (Neglect the internal resistance of the power supply)

(1) 

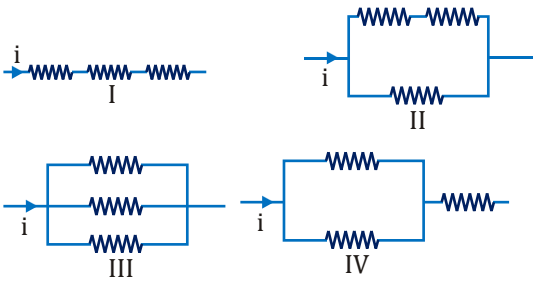
(2) 

(3) 

(4) 

PCE079

80. Three resistances of equal value are arranged in different combinations as shown below. Arrange them in the increasing order of power dissipation:



(1) III < II < IV < I (2) II < III < IV < I

(3) I < IV < III < II (4) I < III < II < IV

PCE080

81. 25 W, 200 V and 100 W, 200 V bulbs are connected in series to a source of 400 volts. Which bulb will fuse?

(1) 25 W

(2) 100 W

(3) Both will fuse at the same time

(4) None of the bulbs will fuse

PCE081

82. You are provided with 48 cells, each of emf 2 volts and internal resistance 4 ohms. What maximum current can flow in the circuit having an external resistance of 12 Ω ?

(1) 1 A (2) 1.2 A

(3) 0.96 A (4) 1.08 A

PCE082

83. Two electric bulbs of the same power, but with different marked voltages are connected in series across a power line. Their brightness will be :-

(1) directly proportional to their marked voltages

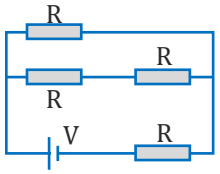
(2) inversely proportional to their marked voltages

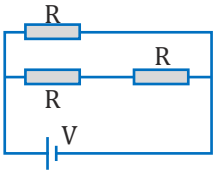
(3) directly proportional to the squares of their marked voltages

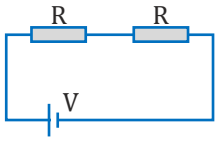
(4) inversely proportional to the squares of their marked voltages

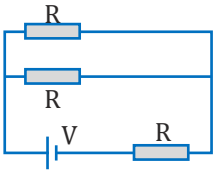
PCE083

84. Four circuits are shown below. All the batteries have the same voltage *V* and all resistors have the same resistance *R*. In which circuit does the battery delivers the most power?

(1) 

(2) 

(3) 

(4) 

PCE084

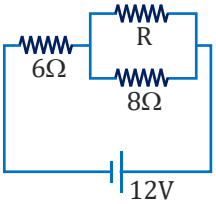
85. Two cells, each of e.m.f. E and internal resistance r, are connected in parallel across a resistor R. The power dissipated in the resistor is maximum if :

(1) R = r (2) R = 2r

(3) $R = \frac{3r}{2}$ (4) $R = \frac{r}{2}$

PCE085

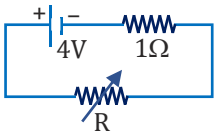
86. In the circuit shown in figure, the power which is dissipated as heat in the 6 Ω resistor is 6 W. What is the value of resistance R in the circuit?



(1) 6 Ω (2) 10 Ω (3) 13 Ω (4) 24 Ω

PCE086

87. For different values of resistance, R power consumptions in R are given. Then which of the following values are not possible?



(a) 2 W (b) 5 W (c) 8 W (d) 4 W

(1) Only c (2) b & c (3) a,b,c (4) All

PCE087

MEASURING DEVICES

88. An ammeter and a voltmeter are joined in series to a cell. Their readings are A and V respectively. If a resistance is now joined in parallel with the voltmeter

(1) both A and V will decrease

(2) both A and V will increase

(3) A will increase, V will decrease

(4) A will decrease, V will increase

PCE088

89. A galvanometer of 100 Ω resistance yields complete deflection when 10 mA current flows. What should be the value of shunt so that it can measure currents upto 100 mA?

(1) 11.11 Ω (2) 9.9 Ω

(3) 1.1 Ω (4) 4.4 Ω

PCE089

90. In order to change the range of a galvanometer of G Ω resistance from V volts to nV volts what will be the value of resistance in Ω connected in series with it :–

(1) (n–1)G (2) $\frac{G}{n}$

(3) nG (4) $\frac{G}{n-1}$

PCE090

91. Resistance in the two gaps of a meter bridge are 10 ohms and 30 ohms respectively. If the resistances are interchanged, the balance point shifts by :–

(1) 33.3 cm (2) 66.67 cm

(3) 25 cm (4) 50 cm

PCE091

92. A galvanometer acting as a voltmeter will have :

(1) a high resistance in series with its coil

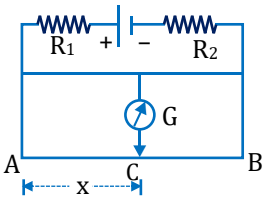
(2) a low resistance in parallel with its coil

(3) a low resistance in series with its coil

(4) a high resistance in parallel with its coil

PCE092

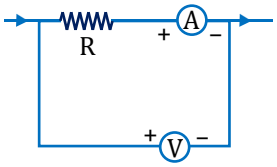
93. In the shown arrangement of the experiment of a meter bridge if AC, corresponding to null deflection of galvanometer, is x then what would be its value if the radius of the wire AB is doubled:-



(1) x (2) $\frac{x}{4}$ (3) 4x (4) 2x

PCE093

94. In the following circuit, the resistance of the voltmeter is 10,000 Ω and that of the ammeter is 20 Ω. If the reading of the ammeter is 0.1 A and that of the voltmeter is 12 V, then the value of R is :-



(1) 122 Ω (2) 100 Ω

(3) 118 Ω (4) 116 Ω

PCE094

95. The resistance of a galvanometer is G ohms and the range is 1 volt. The value of resistance (in Ω) used to convert it into a voltmeter of range 10 volts is :-

(1) 9 G (2) G (3) $\frac{1}{9}$ G (4) 10 G

PCE095

96. A galvanometer has 36 Ω resistance. If a 4 Ω shunt is added to this, the fraction of current that passes through the galvanometer is :-

(1) $\frac{1}{4}$ (2) $\frac{1}{9}$ (3) $\frac{1}{10}$ (4) $\frac{1}{40}$

PCE096

97. A galvanometer of resistance 100 Ω gives full deflection for a current of 10^{-5} A. The value of shunt required to convert it into an ammeter of range 1 ampere, is :-

(1) 1 Ω (2) 10^{-3} Ω

(3) 10^{-5} Ω (4) 100 Ω

PCE097

98. There are three voltmeters of the same range but of resistances 10000 Ω, 8000 Ω and 4000 Ω respectively. The best voltmeter among these is the one whose resistance is :–

(1) 10000 Ω

(2) 8000 Ω

(3) 4000 Ω

(4) all are equally good

PCE098

99. 20% of the main current passes through the galvanometer. If the resistance of the galvanometer is G, then the resistance of the shunt will be :-

(1) $\frac{G}{50}$ (2) $\frac{G}{4}$

(3) 50G (4) 9G

PCE099

100. An unknown resistance R_1 is connected in series with a resistance of 10 Ω. This combination is connected to one gap of a metre bridge while a resistance R_2 is connected in the other gap. The balance point is at 50 cm. Now, when the 10 Ω resistance is removed the balance point shifts to 40 cm. The value of R_1 is (in ohms) :-

(1) 20 (2) 10 (3) 60 (4) 40

PCE100

101. A 1 Ω voltmeter has a range of 1V. Find the additional resistance which has to be joined with the series in voltmeter to increase the range of voltmeter to 100 V :-

(1) 10 Ω (2) $\frac{1}{99}$ Ω

(3) 99 Ω (4) 100 Ω

PCE101

102. A galvanometer having a resistance G and current i_a flowing in it, produces full scale deflection. If S_1 is the value of shunt which converts it into an ammeter of range 0 – i and S_2 is the value of the shunt for the range 0 – 2i. Then the ratio $\frac{S_1}{S_2}$ will be :-

(1) 1 (2) 2

(3) $\frac{1}{2}\left(\frac{i-i_a}{2i-i_a}\right)$ (4) $\left(\frac{2i-i_a}{i-i_a}\right)$

PCE102

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	2	3	2	2	3	2	2	3	4	3	3	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	4	3	1	3	4	1	2	4	4	3	1	4	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	3	3	1	2	4	3	4	3	1	1	4	3	1	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	4	1	1	4	2	1	3	2	2	4	1	2	2	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	3	3	1	4	1	2	4	2	2	4	3	2	2	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	1	4	1	1	1	1	3	2	4	4	2	3	1	1
Question	91	92	93	94	95	96	97	98	99	100	101	102			
Answer	4	1	1	2	1	3	2	1	2	1	3	4			

Exercise - II (Previous Year Questions) AIPMT/NEET

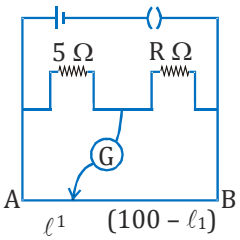
AIPMT 2014

1. Two cities are 150 km apart. Electric power is sent from one city to another city through copper wires. The fall of potential per km is 8 volts and the average resistance per km is 0.5 Ω. the power loss in the wires is :-

(1) 19.2 W (2) 19.2 kW
(3) 19.2 J (4) 12.2 kW

PCE146

2. The resistance in the two arms of a meter bridge are 5 Ω and R Ω, respectively. When the resistance R is shunted with an equal resistance, the new balance point is at 1.6 ℓ_1. The resistance 'R' is :-



(1) 10 Ω (2) 15 Ω (3) 20 Ω (4) 25 Ω

PCE147

3. In an ammeter 0.2% of main current passes through the galvanometer. If resistance of galvanometer is G, then resistance of ammeter will be :-

(1) $\frac{1}{499}$G (2) $\frac{499}{500}$G
(3) $\frac{1}{500}$G (4) $\frac{500}{499}$G

PCE149

Re-AIPMT 2015

4. Two metal wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of the metal wires respectively, the effective conductivity of the combination is :-

(1) $\frac{\sigma_1\sigma_2}{\sigma_1+\sigma_2}$ (2) $\frac{2\sigma_1\sigma_2}{\sigma_1+\sigma_2}$
(3) $\frac{\sigma_1+\sigma_2}{2\sigma_1\sigma_2}$ (4) $\frac{\sigma_1+\sigma_2}{\sigma_1\sigma_2}$

PCE151

5. A circuit contains an ammeter, a battery of 30 V and a resistance 40.8 ohm all connected in series. If the ammeter has a coil of resistance 480 ohm and a shunt of 20 ohm, the reading in the ammeter will be :-

(1) 1 A (2) 0.5 A (3) 0.25 A (4) 2 A

PCE152

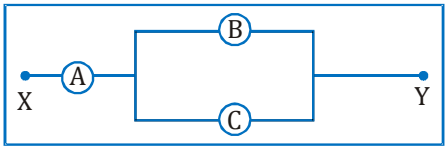
AIPMT - 2015

6. Across a metallic conductor of non-uniform cross section a constant potential difference is applied. The quantity which remains constant along the conductor is :

(1) current (2) drift velocity
(3) electric field (4) current density

PCE153

7. A, B and C are voltmeters of resistance R, 1.5 R and 3R respectively as shown in the figure. When some potential difference is applied between X and Y, the voltmeter readings are V_A, V_B and V_C respectively. Then :

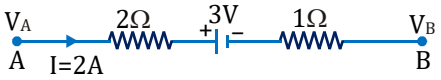


(1) $V_A \neq V_B = V_C$ (2) $V_A = V_B \neq V_C$
(3) $V_A \neq V_B \neq V_C$ (4) $V_A = V_B = V_C$

PCE155

NEET-II 2016

8. The potential difference ($V_A - V_B$) between the points A and B in the given figure is :-



(1) + 6 V (2) + 9 V (3) – 3 V (4) + 3 V

PCE157

9. A filament bulb (500 W, 100 V) is to be used in a 230 V main supply. When a resistance R is connected in series, it works perfectly and the bulb consumes 500 W. The value of R is :-

(1) 26 Ω (2) 13 Ω (3) 230 Ω (4) 46 Ω

PCE158

NEET(UG)-2017

10. The resistance of a wire is 'R' ohm. If it is melted and stretched to 'n' times its original length, its new resistance will be :-

(1) $\frac{R}{n}$ (2) n^2R (3) $\frac{R}{n^2}$ (4) nR

PCE159

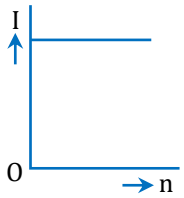
NEET(UG)-2018

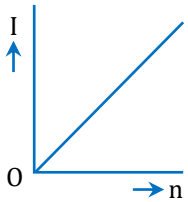
11. A set of 'n' equal resistors, of value 'R' each, are connected in series to a battery of emf 'E' and internal resistance 'R'. The current drawn is I. Now, the 'n' resistors are connected in parallel to the same battery. Then the current drawn from battery becomes 10 I. The value of 'n' is :-

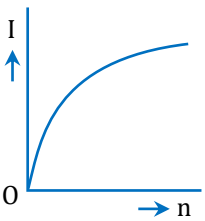
(1) 10 (2) 11 (3) 20 (4) 9

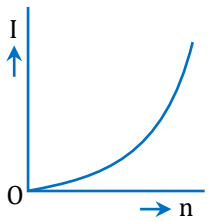
PCE161

12. A battery consists of a variable number 'n' of identical cells (having internal resistance 'r' each) which are connected in series. The terminals of the battery are short-circuited and the current I is measured. Which of the graphs shows the correct relationship between I and n?

(1) 

(2) 

(3) 

(4) 

PCE162

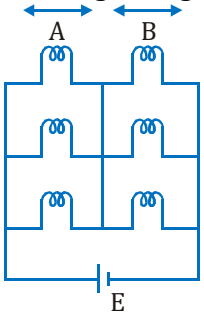
13. Current sensitivity of a moving coil galvanometer is 5 div/mA and its voltage sensitivity (angular deflection per unit voltage applied) is 20 div/V. The resistance of the galvanometer is

(1) 40 Ω (2) 25 Ω
(3) 250 Ω (4) 500 Ω

PCE164

NEET(UG)-2019

14. Six similar bulbs are connected as shown in the figure with a DC source of emf E, and zero internal resistance.
The ratio of power consumption by the bulbs when (i) all are glowing and (ii) in the situation when two from section A and one from section B are glowing, will be :



(1) 4 : 9 (2) 9 : 4 (3) 1 : 2 (4) 2 : 1

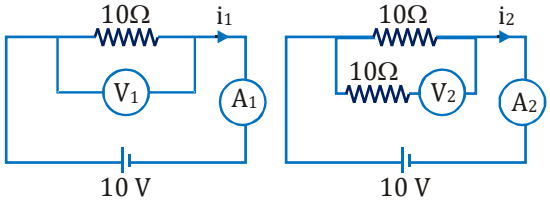
PCE165

15. Which of the following acts as a circuit protection device?

(1) conductor (2) inductor
(3) switch (4) fuse

PCE166

16. In the circuits shown below, the readings of the voltmeters and the ammeters will be :

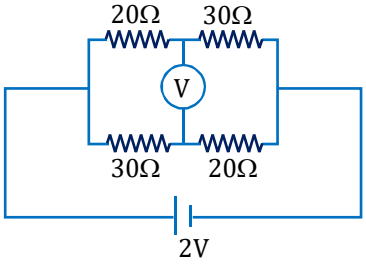


(1) $V_2 > V_1$ and $i_1 = i_2$
(2) $V_1 = V_2$ and $i_1 > i_2$
(3) $V_1 = V_2$ and $i_1 = i_2$
(4) $V_2 > V_1$ and $i_1 > i_2$

PCE167

NEET(UG)-2019 (Odisha)

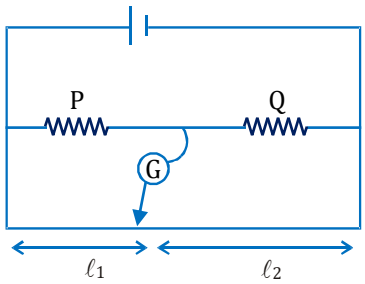
17. The reading of an ideal voltmeter in the circuit shown is :



(1) 0.6 V (2) 0 V (3) 0.5 V (4) 0.4 V

PCE168

18. The metre bridge shown is in balanced position with $\frac{P}{Q}=\frac{\ell_1}{\ell_2}$. If we now interchange the positions of galvanometer and cell, will the bridge work ? If yes, what will be balance condition ?



(1) yes, $\frac{P}{Q}=\frac{\ell_2-\ell_1}{\ell_2+\ell_1}$ (2) no, no null point

(3) yes, $\frac{P}{Q}=\frac{\ell_2}{\ell_1}$ (4) yes, $\frac{P}{Q}=\frac{\ell_1}{\ell_2}$

PCE169

NEET(UG)-2020

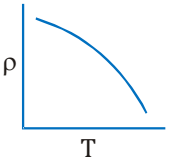
19. A charged particle having drift velocity of 7.5×10^{-4} ms^{-1} is an electric field of 3×10^{-10} Vm^{-1}, has a mobility in m^2 V^{-1} s^{-1} of :

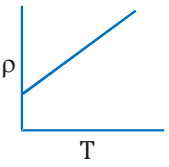
(1) 2.25×10^{-15} (2) 2.25×10^{15}

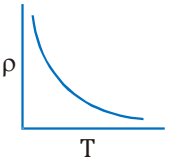
(3) 2.5×10^{6} (4) 2.5×10^{-6}

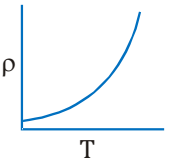
PCE171

20. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for copper?

(1) 

(2) 

(3) 

(4) 

PCE172

21. A resistance wire connected in the left gap of a metre bridge balances a 10 Ω resistance in the right gap at a point which divides the bridge wire in the ratio 3 : 2. If the length of the resistance wire is 1.5 m, then the length of 1 Ω of the resistance wire is :

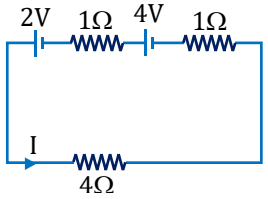
(1) 1.5×10^{-2} m (2) 1.0×10^{-2} m

(3) 1.0×10^{-1} m (4) 1.5×10^{-1} m

PCE173

NEET(UG)-2020 (Covid-19)

22. For the circuit shown in the figure, the current I will be

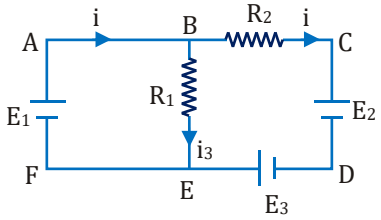


(1) 0.75 A (2) 1 A

(3) 1.5 A (4) 0.5 A

PCE174

23. For the circuit given below, the Kirchoff's loop rule for the loop BCDEB is given by the equation



(1) $-i_2R_2 + E_2 - E_3 + i_3R_1 = 0$

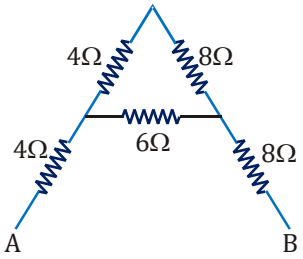
(2) $i_2R_2 + E_2 - E_3 - i_3R_1 = 0$

(3) $i_2R_2 + E_2 + E_3 + i_3R_1 = 0$

(4) $-i_2R_2 + E_2 + E_3 + i_3R_1 = 0$

PCE175

24. The equivalent resistance between A and B for the mesh shown in the figure is



(1) 7.2 Ω (2) 16 Ω

(3) 30 Ω (4) 4.8 Ω

PCE176

25. Two solid conductors are made up of same material, have same length and same resistance. One of them has a circular cross section of area A_1 and the other one has a square cross section of area A_2. The ratio A_1/A_2 is

(1) 1.5 (2) 1 (3) 0.8 (4) 2

PCE177

NEET(UG)-2021

26. **Column-I** gives certain physical terms associated with flow of current through a metallic conductor. **Column-II** gives some mathematical relations involving electrical quantities. Match **Column-I** and **Column-II** with appropriate relations.

Column-I		Column-II	
(A)	Drift Velocity	(P)	$\frac{m}{ne^2\rho}$
(B)	Electrical Resistivity	(Q)	nev_d
(C)	Relaxation Period	(R)	$\frac{eE}{m}\tau$
(D)	Current Density	(S)	$\frac{E}{J}$

(1) (A)-(R), (B)-(S), (C)-(P), (D)-(Q)
(2) (A)-(R), (B)-(S), (C)-(Q), (D)-(P)
(3) (A)-(R), (B)-(P), (C)-(S), (D)-(Q)
(4) (A)-(R), (B)-(Q), (C)-(S), (D)-(P)

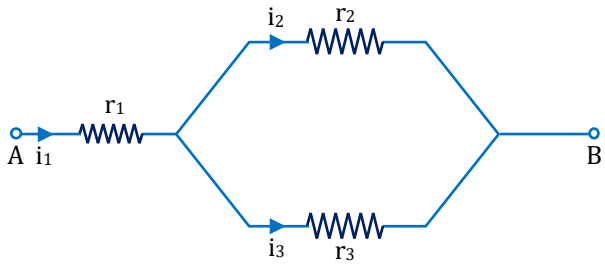
PCE178

27. The effective resistance of a parallel connection that consists of four wires of equal length, equal area of cross-section and same material is 0.25 Ω. What will be the effective resistance if they are connected in series?

(1) 0.25 Ω (2) 0.5 Ω (3) 1 Ω (4) 4Ω

PCE179

28. Three resistors having resistances r_1, r_2 and r_3 are connected as shown in the given circuit. The ratio $\frac{i_3}{i_1}$ of currents in terms of resistances used in the circuit is :



(1) $\frac{r_1}{r_2+r_3}$ (2) $\frac{r_2}{r_2+r_3}$ (3) $\frac{r_1}{r_1+r_2}$ (4) $\frac{r_2}{r_1+r_3}$

PCE181

NEET(UG)-2022

29. Two hollow conducting spheres of radii R_1 and R_2 ($R_1 >> R_2$) have equal charges. The potential would be :

(1) more on smaller sphere
(2) equal on both the spheres
(3) dependent on the material property of the sphere
(4) more on bigger sphere

PCE182

30. Two resistors of resistance, 100 Ω and 200 Ω are connected in parallel in an electrical circuit. The ratio of the thermal energy developed in 100 Ω to that in 200 Ω in a given time is :

(1) 2 : 1 (2) 1 : 4 (3) 4 : 1 (4) 1 : 2

PCE183

31. As the temperature increase, the electrical resistance :

(1) decreases for both conductors and semiconductors
(2) increases for conductors but decreases for semiconductors
(3) decreases for conductors but increase for semiconductors
(4) increases for both conductors and semiconductors.

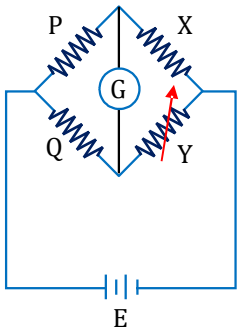
PCE184

32. A copper wire of length 10 m and radius $\left(10^{-2}/\sqrt{\pi}\right)$m has electrical resistance of 10 Ω. The current density in the wire for an electric field strength of 10 (V/m) is:

(1) 10^6 A/m^2 (2) 10^{-5} A/m^2
(3) 10^5 A/m^2 (4) 10^4 A/m^2

PCE185

33. A wheatstone bridge is used to determine the value of unknown resistance X by adjusting the variable resistance Y as shown in the figure. For the most precise measurement of X, the resistances P and Q :



(1) should be approximately equal and are small
(2) should be very large and unequal
(3) do not play any significant role
(4) should be approximately equal to 2X

PCE186

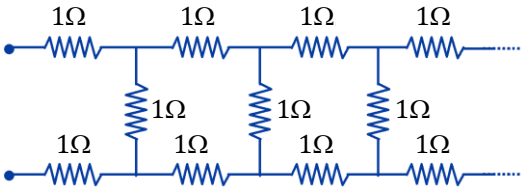
RE-NEET(UG)-2022

34. A cell of emf 4 V and internal resistance 0.5 Ω is connected to a 7.5 Ω external resistance. The terminal potential difference of the cell is :-

(1) 3.75 V (2) 4.25 V
(3) 4 V (4) 0.375 V

PCE187

35. The equivalent resistance of the infinite network given below is :



(1) 2 Ω (2) $(1+\sqrt{2})$ Ω
(3) $(1+\sqrt{3})\Omega$ (4) $(1+\sqrt{5})\Omega$

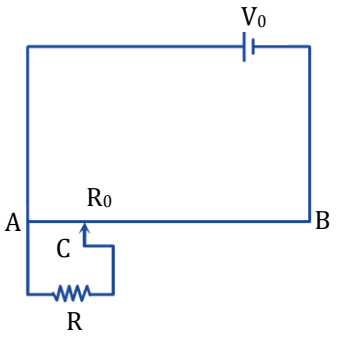
PCE188

36. The reciprocal of resistance is :

(1) reactance (2) mobility
(3) conductivity (4) conductance

PCE189

37. The sliding contact C is at one fourth of the length of the potentiometer wire (AB) from A as shown in the circuit diagram. If the resistance of the wire AB is R_0, then the potential drop (V) across the resistor R is:

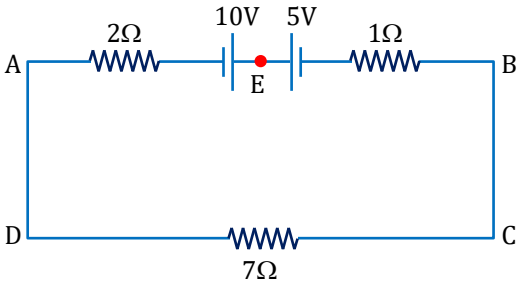


(1) $\frac{4V_0R}{3R_0+16R}$ (2) $\frac{4V_0R}{3R_0+R}$
(3) $\frac{2V_0R}{4R_0+R}$ (4) $\frac{2V_0R}{2R_0+3R}$

PCE190

NEET(UG)-2023

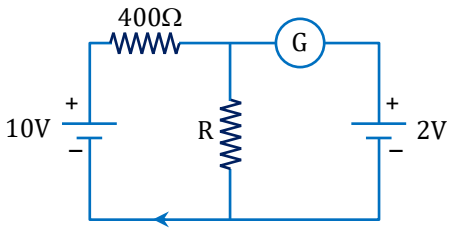
38. The magnitude and direction of the current in the following circuit is :-



(1) 0.5 A from A to B through E
(2) $\frac{5}{9}$ A from A to B through E
(3) 1.5 A from B to A through E
(4) 0.2 A from B to A through E

PCE223

39. If the galvanometer G does not show any deflection in the circuit shown, the value of R is given by :



(1) 50 Ω (2) 100Ω
(3) 400 Ω (4) 200 Ω

PCE224

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40. The resistance of platinum wire at 0°C is 2Ω and 6.8Ω at 80°C. The temperature coefficient of resistance of the wire is :

(1) $3 \times 10^{-3}\,°C^{-1}$ (2) $3 \times 10^{-2}\,°C^{-1}$

(3) $3 \times 10^{-1}\,°C^{-1}$ (4) $3 \times 10^{-4}\,°C^{-1}$

PCE225

41. 10 resistors, each of resistance R are connected in series to a battery of emf E and negligible internal resistance. Then those are connected in parallel to the same battery, the current is increased n times. The value of n is: -

(1) 100 (2) 1 (3) 1000 (4) 10

PCE226

RE-NEET(UG)-2023

42. A certain wire A has resistance 81 Ω. The resistance of another wire B of same material and equal length but of diameter thrice the diameter of A will be :

(1) 81 Ω (2) 9 Ω

(3) 729 Ω (4) 243 Ω

PCE227

43. A copper wire of radius 1 mm contains 10^{22} free electrons per cubic metre. The drift velocity for free electrons when 10 A current flows through the wire will be (Given, charge on electron = 1.6×10^{-19} C) :

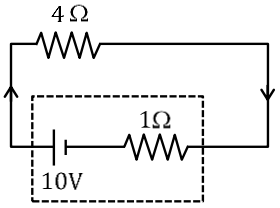
(1) $\frac{6.25\times10^{4}}{\pi} ms^{-1}$ (2) $\frac{6.25}{\pi}\times10^{3} ms^{-1}$

(3) $\frac{6.25}{\pi} ms^{-1}$ (4) $\frac{6.25\times10^{5}}{\pi} ms^{-1}$

PCE228

NEET(UG)-2024

44. The terminal voltage of the battery, whose emf is 10V and internal resistance 1Ω, when connected through an external resistance of 4Ω as shown in the figure.



(1) 4V (2) 6V (3) 8V (4) 10V

PCE229

45. A wire of length 'ℓ' and resistance 100Ω is divided into 10 equal parts. The first 5 parts are connected in series while the next 5 parts are connected in parallel. The two combinations are again connected in series. The resistance of this final combination is:

(1) 26Ω (2) 52Ω (3) 55Ω (4) 60Ω

PCE230

46. Two heaters A and B have power rating of 1 kW and 2 kW, respectively. Those two are first connected in series and then in parallel to a fixed power source. The ratio of power outputs for these two cases is :

(1) 1 :1 (2) 2 : 9 (3) 1 : 2 (4) 2 : 3

PCE231

RE-NEET(UG)-2024

47. A uniform wire of diameter d carries a current of 100 mA when the mean drift velocity of electrons in the wire is v. For a wire of diameter $\frac{d}{2}$ of the same material to carry a current of 200 mA, the mean drift velocity of electrons in the wire is:

(1) 4 v (2) 8 v (3) v (4) 2 v

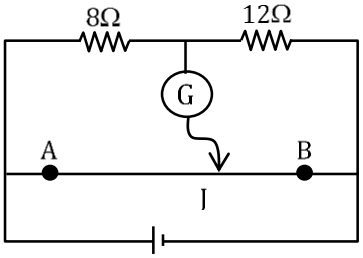
PCE232

48. A uniform metal wire of length ℓ has 10 Ω resistance. Now this wire is stretched to a length 2ℓ and then bent to form a perfect circle. The equivalent resistance across any arbitrary diameter of that circle is:

(1) 10Ω (2) 5Ω (3) 40Ω (4) 20Ω

PCE233

49. The given circuit shows a uniform straight wire AB of 40 cm length fixed at both ends. In order to get zero reading in the galvanometer G, the free end of J is to be placed from B at :



(1) 32 cm (2) 8 cm

(3) 16 cm (4) 24 cm

PCE234

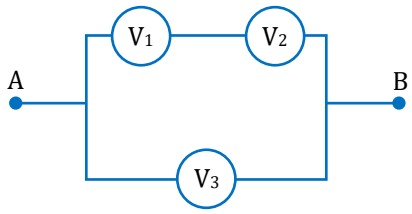
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EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	3	2	2	1	4	2	1	2	1	1	3	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	4	3	4	3	2	2	2	2	1	4	2	1	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	3	1	1	3	4	1	1	2	2	1	2	2	3	2
Question	46	47	48	49											
Answer	2	2	1	4											

Exercise - III (Analytical Questions) Master Your Understanding

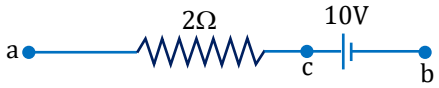
1. Three voltmeters all having different resistance, are joined as shown. When some potential difference is applied across A and B, their readings are V_1, V_2 and V_3. Then



(a) $V_1 = V_2$ (b) $V_1 \neq V_2$
(c) $V_1 + V_2 = V_3$ (d) $V_1 + V_2 > V_3$
(1) only a is correct
(2) only b is correct
(3) only c is correct
(4) only b & c are correct

PCE191

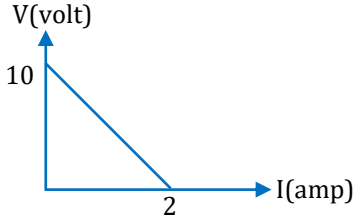
2. In the circuit shown in figure it is given that $V_b - V_a = 2$ volt. Choose the correct options.



(a) Current in the wire is 6 A
(b) Direction of current is from a to b
(c) $V_a - V_c = -12$ volt
(d) $V_c - V_a = -12$ volt
(1) only a & b (2) only b & c
(3) only a & c (4) all

PCE192

3. A battery of e.m.f E and internal resistance r is connected to a resistance R which can be adjusted to any value greater than or equal to zero. A graph is plotted between the current passing through the resistance (I) and potential difference (V) across it. Select the correct alternatives



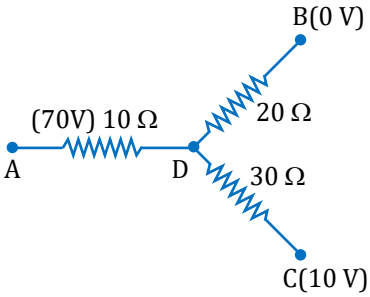
(a) Internal resistance of the battery is 5Ω
(b) emf of the battery is 10 V
(c) maximum current which can be taken from the battery is 2A
(d) V-I graph can never be a straight lines as shown in the figure
(1) only a & b (2) a, b & c
(3) only b & c (4) all

PCE193

4. Two heaters designed for the same voltage V have different power ratings. When connected individually across a source of voltage V, they produce H amount of heat each in times t_1 and t_2 respectively. When used together across the same source, they produce H amount of heat in time t. Then correct option is
(a) If they are in series, $t = t_1 + t_2$
(b) If they are in series, $t = 2(t_1 + t_2)$
(c) If they are in parallel, $t = \frac{t_1 t_2}{(t_1 + t_2)}$
(d) If they are in parallel, $t = \frac{t_1 t_2}{2(t_1 + t_2)}$
(1) only a & c (2) only a & b
(3) only c & d (4) all

PCE195

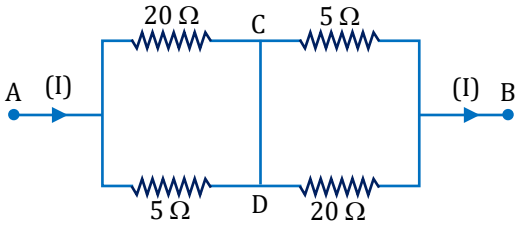
5. In the network shown in figure, points A, B and C are at potentials of 70 V, 0 and 10 V respectively.



(a) Point D is at a potential of 40V.
(b) The currents in the sections AD, DB, DC are in the ratio 3:2:1
(c) The currents in the sections AD, DB, DC are in the ratio 1:2:3.
(d) The network draws a total power of 200 W.
(1) only a is correct
(2) a and c are correct
(3) a, b, d are correct
(4) None of these

PCE196

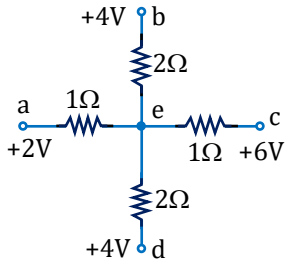
6. When some potential difference is maintained between A and B, current I enters the network at A and leaves at B see figure.



(a) The equivalent resistance between A and B is 8Ω
(b) C and D are at the same potential.
(c) No current flows between C and D
(d) Current 3I/5 flows from D to C.
(1) a, b, d are correct
(2) a, d are correct
(3) a, c, d are correct
(4) all are correct

PCE197

7. For the circuit shown in figure, match the two columns.

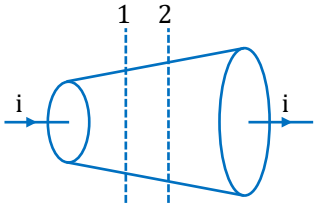


	Column-I		Column-II
(a)	Current in wire ae	(p)	1 A
(b)	Current in wire be	(q)	2 A
(c)	Current in wire ce	(r)	0.5 A
(d)	Current in wire de	(s)	Zero

(1) a-q, b-s, c-q, d-s (2) a-q, b-q, c-s, d-p
(3) a-p, b-r, c-s, d-q (4) a-q, b-r, c-p, d-s

PCE198

8. Current i is flowing through a wire of non-uniform cross section as shown. Match the following two columns.

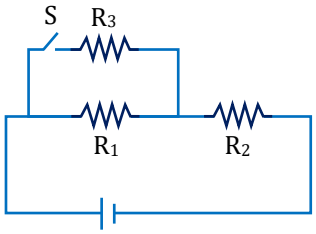


	Column-I		Column-II
(a)	Current density	(p)	is more at 1
(b)	Electric field	(q)	is more at 2
(c)	Resistance per unit length	(r)	is same at both sections 1 and 2
(d)	Potential difference per unit length	(s)	data insufficient

(1) a-p, b-p, c-p, d-p (2) a-q, b-r, c-s, d-p
(3) a-q, b-q, c-p, d-p (4) a-p, b-q, c-r, d-s

PCE199

9. In the circuit shown in figure, after closing the switch S, match the following two columns.



	Column-I		Column-II
(a)	Current through R_1	(p)	will increase
(b)	Current through R_2	(q)	will decrease
(c)	Potential difference across R_1	(r)	will remain same
(d)	Potential difference across R_2	(s)	data insufficient

(1) a-p, b-q, c-r, d-s (2) a-p, b-r, c-s, d-q
(3) a-p, b-r, c-s, d-p (4) a-q, b-p, c-q, d-p

PCE200

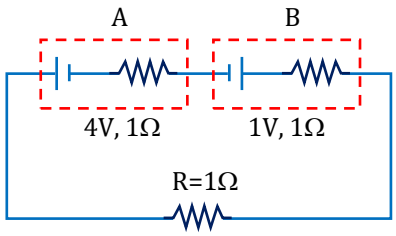
10. Match the following two columns.

	Column-I		Column-II
(a)	Electrical resistance	(p)	$[ML^3T^{-3}A^{-2}]$
(b)	Electrical potential	(q)	$[ML^2T^{-3}A^{-2}]$
(c)	Specific resistance	(r)	$[ML^2T^{-3}A^{-1}]$
(d)	Specific conductance	(s)	None of these

(1) a-q, b-s, c-r, d-p (2) a-q, b-r, c-p, d-s
(3) a-p, b-q, c-s, d-r (4) a-p, b-r, c-q, d-s

PCE201

11. In the circuit shown in the figure, match the following two columns:



	Column-I		Column-II (In SI units)
(a)	Potential difference across battery A	(p)	Zero
(b)	Potential difference across battery B	(q)	1
(c)	Net power consumed by R	(r)	2
(d)	Potential difference across R	(s)	3

(1) a-p, b-q, c-r, d-s (2) a-s, b-q, c-r, d-p

(3) a-s, b-r, c-q, d-q (4) a-q, b-r, c-s, d-s

PCE202

12. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Drift velocity of electron in a metallic wire will decrease if temperature of wire is increased

Reason (R) : On increasing temperature conductivity of metallic wire decreases.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PCE203

13. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Resistance of ammeter is less than resistance of milliammeter, (If made from same galvanometer)

Reason (R) : Value of shunt resistance in case of ammeter is more than a milliammeter.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PCE204

14. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Electric field is present inside the current carrying wire.

Reason (R) : The principle of conservation of charge is followed, when charges are in motion.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PCE205

15. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In balanced Wheatstone bridge, the current through cell depends on resistance of galvanometer.

Reason (R) : At balanced condition current through galvanometer is zero.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PCE206

16. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In series combination of electrical bulb, lower power rating bulb emits more light than that of higher power rating bulb.

Reason (R) : The lower power rating bulb in series gets more current than the higher power rating bulb.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PCE207

17. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When an external resistor of resistance R (connected across a cell of internal resistance r) is varied, power consumed by resistance R is maximum when R = r.

Reason (R) : Power consumed by a resistor of constant resistance R is maximum when current through it is maximum.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PCE208

18. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Ohm's law holds only for small currents in metallic wire not for high currents.

Reason (R) : For metallic wire resistance increases with increase in temperature.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PCE209

19. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In a Meter Bridge experiment, null point for an unknown resistance is measured. Now, the unknown resistance is put inside an enclosure maintained at a higher temperature. The null point can be obtained at the same point as before by decreasing the value of the standard resistance.

Reason (R) : Resistance of a metal increases with increase in temperature.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PCE210

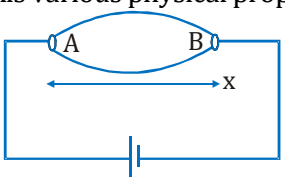
20. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Kirchoff's loop law represents conservation of energy
Reason (R) : If the sum of "Potential Differences" around a closed loop is zero, unlimited energy could be gained by repeatedly carrying a charge around a loop
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

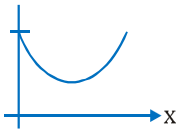
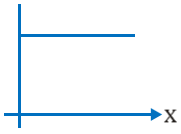
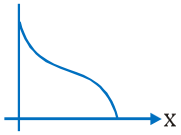
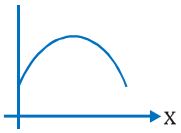
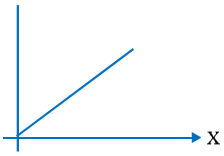
PCE211

21. If the current in resistance of uniform cross section is doubled then, (assume temperature to be approximately constant).
(A) the current density is doubled
(B) the conduction electron density is doubled
(C) the mean time between collision is constant
(D) the electron drift speed is doubled
(1) A, C, D (2) B, C, D
(3) A, B, C (4) A, B, C, D

PCE213

22. A resistor of varying cross–section is attached with a battery as shown. Variation of area from both ends are similar and areas are same at ends and maximum at centre. Match the approximate graphical representation of various physical properties in columns, where x axis represents position from end A of resistor and y–axis various physical properties.

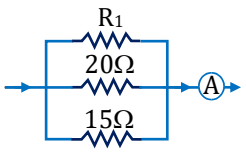


Column I	Column II
(A) i (current)	(P) 
(B) j (current density)	(Q) 
(C) E (electric field)	(R) 
(D) V_d (potential)	(S) 
	(T) 

Options :-
(1) A → S, B → P, C → P, D → R
(2) A → Q, B → P, C → P, D → R
(3) A → T, B → Q, C → P, D → S
(4) A → Q, B → P, C → P, D → T

PCE214

23. In the given circuit the current flowing through the resistance 20 ohms is 0.3 ampere while the ammeter reads 0.9 ampere. What is the value of R_1(in ohms).



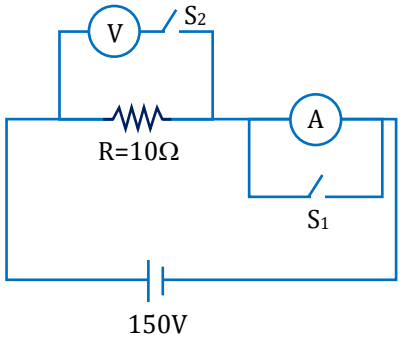
(1) 20Ω (2) 15Ω (3) 30Ω (4) 60Ω

PCE215

24. **Statement 1**: Voltage across a battery is always less than its emf.
and
Statement 2 : There is always some loss of energy due to internal resistance of the battery.
(1) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.
(2) Statement–1 is True, Statement–2 is True; Statement–2 is not a correct explanation for Statement–1.
(3) Statement–1 is True, Statement–2 is False.
(4) Statement–1 is False, Statement–2 is True.

PCE216

25. Voltmeter of resistance 10Ω and ammeter of resistance 10Ω are attached in the following circuit with switches S_1 and S_2



Column I (switch position)	Column II (reading of voltmeter & ammeter)
(A) S_1 open, S_2 closed	(P) 0, 7.5 A
(B) S_1 closed, S_2 open	(Q) 50 V, 10A
(C) S_1 open, S_2 open	(R) 0, 0
(D) S_1 closed, S_2 closed	(S) 150 V, 0

Options :-

(1) A → S, B → R, C → P, D → Q
(2) A → Q, B → R, C → P, D → S
(3) A → Q, B → P, C → R, D → S
(4) A → S, B → P, C → R, D → Q

PCE219

26.

	Column-I (Situration)		Column-II (Values in MKS units)
(A)	Magnitude of electric field at (1,1,1) if potential is given by $V=-\frac{1}{x}-\frac{1}{y}-\frac{1}{z}$, is	(P)	0.4
(B)	The charge flowing through a resistance 6Ω varies with t as Q = 4t – 2t². The total heat produced in resistance till the current reach to zero.	(Q)	$\sqrt{3}$
(C)	A milliammeter of range 10mA has a coil of resistance 0.99 Ω. To use it as an ammeter of range 1A, the required shunt must have a resistance of	(R)	0.01
(D)	Two unlike point charges of same magnitude 10 mC are placed on x-axis at a separation of 8 cm. If both charges placed symmetrically about origin in external electric field $\vec{E}=500\text{Vm}^{-1}\,\hat{i}$. Then potential energy of the system may be [Assume potential energy to be zero when dipole is perpendicular to electric field]	(S)	32

Options :-

(1) A → Q, B → S, C → P, D → R
(2) A → S, B → Q, C → R, D → P
(3) A → Q, B → S, C → R, D → P
(4) A → S, B → Q, C → P, D → R

PCE220

27. **Statement-1** : In real battery total chemical energy lost in the battery can not be obtained at load resistance.
Statement-2 : Two heaters with different resistance are connected in parallel. Higher resistance heater will glow more.

(1) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.
(2) Statement–1 is True, Statement–2 is True; Statement–2 is not a correct explanation for Statement–1.
(3) Statement–1 is True, Statement–2 is False.
(4) Statement–1 is False, Statement–2 is True.

PCE221

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	2	1	3	1	1	1	4	2	3	2	3	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27			
Answer	3	2	1	4	3	1	2	3	4	2	3	3			

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Exercise - I (Conceptual Questions) Build Up Your Understanding

CAPACITANCE, ENERGY STORED IN CAPACITOR & SPHERICAL CAPACITOR

1. The capacitance C of a capacitor is :-
(1) independent of the charge and potential of the capacitor.
(2) dependent on the charge and independent of potential.
(3) independent of the geometrical configuration of the capacitor.
(4) independent of the dielectric medium between the two conducting surfaces of the capacitor.

PCP001

2. To increase the charge on the plate of a capacitor implies to :-
(1) decrease the potential difference between the plates.
(2) decrease the capacitance of the capacitor.
(3) increase the capacitance of the capacitor.
(4) increase the potential difference between the plates.

PCP002

3. The net charge on a capacitor is :-
(1) 2q (2) q/2
(3) 0 (4) infinity

PCP003

4. The earth has Volume 'V' and Surface area 'A'; then its capacitance would be :
(1) $4\pi\epsilon_0\frac{A}{V}$ (2) $4\pi\epsilon_0\frac{V}{A}$
(3) $12\pi\epsilon_0\frac{V}{A}$ (4) $12\pi\epsilon_0\frac{A}{V}$

PCP004

5. Capacitors are used in electrical circuits where appliances need rapid :
(1) Current (2) Voltage
(3) Watt (4) Resistance

PCP005

6. Which of the following is called electrical energy tank?
(1) Resistor (2) Inductance
(3) Capacitor (4) Motor

PCP006

7. If the maximum circumference of a sphere is 2 m, then its capacitance in water would be :- (Dielectric constant of water = 81)
(1) 27.65 pF (2) 2385 pF
(3) 236.5 pF (4) 2865 pF

PCP007

8. The two parallel plates of a condenser have been connected to a battery of 300 V and the charge collected at each plate is 1 μC. The energy supplied by the battery is :
(1) 6×10^{-4}J (2) 3×10^{-4}J
(3) 1.5×10^{-4}J (4) 4.5×10^{-4}J

PCP008

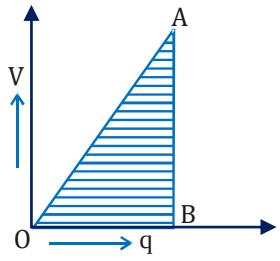
9. When a capacitor of value 200 μF charged to 200V is discharged separately through resistance of 2 ohms and 8 ohms, then heat produced in joule will respectively be:
(1) 4 and 16 (2) 16 and 4
(3) 4 and 8 (4) 4 and 4

PCP009

10. The potential to which a conductor is raised, depends on :-
(1) the amount of charge
(2) the geometry and size of the conductor
(3) both (1) and (2)
(4) None of these

PCP010

11. The charge q on a capacitor varies with voltage as shown in figure. The area of the triangle AOB represents :



(1) electric field between the plates
(2) electric flux between the plates
(3) energy density
(4) energy stored by the capacitor.

PCP011

12. An uncharged capacitor is connected to a battery. On charging the capacitor :-
(1) all the energy supplied is stored in the capacitor.
(2) half the energy supplied is stored in the capacitor.
(3) the energy stored depends upon the capacity of the capacitor only.
(4) the energy stored depends upon the time for which the capacitor is charged.

PCP012

CYLINDRICAL CAPACITOR PARALLEL PLATE CAPACITOR & EFFECT OF DIELECTRIC SLAB

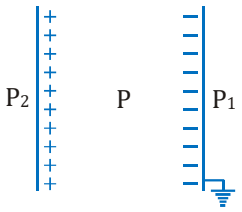
13. The capacity of parallel plate condenser depends on :
(1) the type of metal used
(2) the thickness of plates
(3) the potential difference applied across the plates
(4) the separation between the plates.

PCP013

14. A parallel plate capacitor has rectangular plates of 400 cm^2 area and are separated by a distance of 2 mm with air as the medium. What charge will appear on the plates if a 200 volt potential difference is applied across the capacitor?
(1) 3.54×10^{-6} C (2) 3.54×10^{-8} C
(3) 3.54×10^{-10} C (4) 1770.8×10^{-13} C

PCP014

15. There are two metallic plates of a parallel plate capacitor. One plate is given a charge +q while the other is earthed as shown. Points P, P_1 and P_2 are taken as shown in adjoining figure. Then the electric intensity is not zero at :



(1) P only (2) P_1 only
(3) P_2 only (4) P, P_1 and P_2

PCP015

16. The distance between the plates of a circular parallel plate capacitor of diameter 40 mm, whose capacity is equal to that of a metallic sphere of radius 1m will be :
(1) 0.01 mm (2) 0.1 mm
(3) 1.0 mm (4) 10 mm

PCP016

17. The energy density in a parallel plate capacitor is given as 2.1×10^{-9} J/m^3. The value of the electric field in the region between the plates is :
(1) 2.1 NC^{-1} (2) 21.6 NC^{-1}
(3) 72 NC^{-1} (4) 8.4 NC^{-1}

PCP017

18. A charged parallel plate capacitor of distance (d) has U_0 energy. A slab of dielectric constant (K) and thickness (d) is then introduced between the plates of the capacitor. The new energy of the system is given by :
(1) KU_0 (2) K^2U_0 (3) $\frac{U_0}{K}$ (4) $\frac{U_0}{K^2}$

PCP018

19. The energy and capacity of a charged parallel plate capacitor are U and C respectively. Now a dielectric slab of $\in_r = 6$ is inserted in it then energy and capacity becomes : (Assuming charge on plates remains constant)
(1) 6U, 6C (2) U, C
(3) $\frac{U}{6}$, 6C (4) U, 6C

PCP019

20. Distance between the plates of a parallel plate capacitor is 'd' and area of each plate is A. When a slab of dielectric constant K and thickness t is placed between the plates, its capacity becomes:
(1) $\frac{\varepsilon_0 A}{\left[d+t\left\{1-\frac{1}{K}\right\}\right]}$ (2) $\frac{\varepsilon_0 A}{\left[d+t\left\{1+\frac{1}{K}\right\}\right]}$
(3) $\frac{\varepsilon_0 A}{\left[d-t\left\{1+\frac{1}{K}\right\}\right]}$ (4) $\frac{\varepsilon_0 A}{\left[d-t\left\{1-\frac{1}{K}\right\}\right]}$

PCP020

21. When a slab of dielectric medium is placed between the plates of a parallel plate capacitor which is connected with a battery, then the charge on plates in comparision with earlier charge :
(1) is less
(2) is same
(3) is more
(4) depends on the nature of the material inserted

PCP021

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22. A glass slab is put within the plates of a charged parallel plate condenser. Which of the following quantities does not change?
(1) energy of the condenser
(2) capacity
(3) intensity of electric field
(4) charge

PCP022

23. A parallel plate capacitor is connected to a battery and a dielectric slab is inserted between the plates, then which quantity increase :
(1) potential difference
(2) electric field
(3) stored energy
(4) E.M.F. of battery

PCP023

24. A parallel plate capacitor is charged by a battery. After charging the capacitor, battery is disconnected and a dielectric plate is inserted between the plates. Then which of the following statements is not correct there is a/an?
(1) increase in the stored energy
(2) decrease in the potential difference
(3) decrease in the electric field
(4) increase in the capacitance

PCP024

25. A parallel plate capacitor is charged by a battery. After charging the capacitor, battery is disconnected and distance between the plates is decreased then which of the following statement is correct?
(1) electric field does not remain constant
(2) potential difference is increased
(3) the capacitance decreases
(4) the stored energy decreases

PCP025

26. A parallel plate capacitor is connected with a battery whose potential difference remains constant. If the plates of the capacitor are shifted apart then the intensity of electric field :
(1) decreases and charge on plates also decreases.
(2) remains constant but charge on plates decreases.
(3) remains constant but charge on the plates increases.
(4) increases but charge on the plates decreases.

PCP026

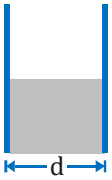
27. A parallel plate capacitor is charged with a battery and afterwards the battery is removed. if now, with the help of insulating handles, the distance between the plates is increased, then
(1) charge on capacitor increases and capacity decreases.
(2) potential difference between the plates increases.
(3) capacity of capacitor increases.
(4) value of energy stored in capacitor decreases.

PCP027

28. Can a metal be used as a medium for dielectric?
(1) Yes
(2) No
(3) Depends on its shape
(4) Depends on dielectric

PCP028

29. A parallel plate air capacitor has a capacitance C. When it is half filled with a dielectric of dielectric constant 5, the percentage increase in the capacitance will be :-

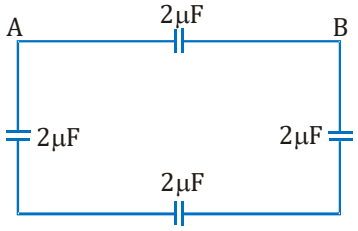


(1) 400% (2) 66.6% (3) 33.3% (4) 200%

PCP029

COMBINATION OF CAPACITORS & SHARING OF CHARGES

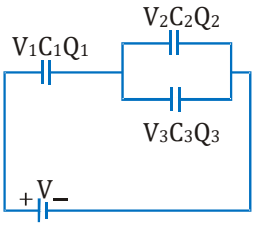
30. The equivalent capacitance between the points A and B in the given diagram is :



(1) 8 μF (2) 6 μF (3) $\frac{8}{3}$μF (4) $\frac{3}{8}$μF

PCP030

31. In an adjoining figure three capacitors C_1, C_2 and C_3 are joined to a battery. The correct condition will be :



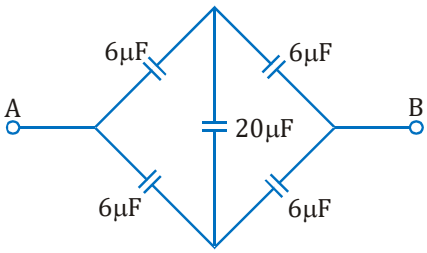
(1) $Q_1 = Q_2 = Q_3$ and $V_1 = V_2 = V_3 = V$
(2) $Q_1 = Q_2 + Q_3$ and $V = V_1 + V_2 + V_3$
(3) $Q_1 = Q_2 + Q_3$ and $V = V_1 + V_2$
(4) $Q_2 = Q_3$ and $V_2 = V_3$
(Symbols have their usual meanings)

PCP031

32. A number of capacitors, each of capacitance 1 μF and each one of which gets punctured if a potential difference just exceeding 500 volt is applied, are provided. Then an arrangement suitable for giving a capacitor of capacitance 3 μF across which 2000 volt may be applied requires at least :
(1) 4 component capacitors
(2) 12 component capacitors
(3) 48 component capacitors
(4) 16 component capacitors

PCP032

33. The effective capacity of the network between terminals A and B is :



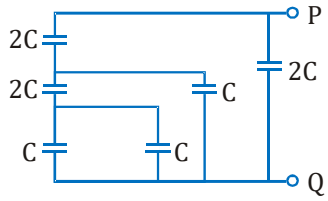
(1) 6 μF (2) 20 μF (3) 3 μF (4) 10 μF

PCP033

34. A series combination of two capacitances of value 0.1 μF and 1μF is connected with a source of voltage 500 volts. The potential difference in volts across the capacitor of value 0.1 μF will be:
(1) 50 (2) 500 (3) 45.5 (4) 454.5

PCP034

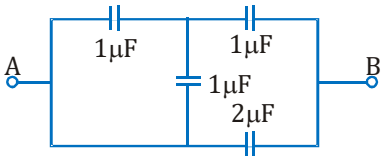
35. The value of equivalent capacitance of the combination shown in figure, between the points P and Q is :



(1) 3C (2) 2C (3) C (4) C/3

PCP035

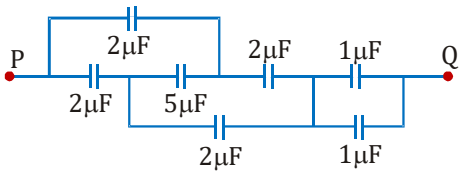
36. The equivalent capacitance between points A and B of the circuit shown will be :



(1) $\frac{2}{3}$ μF (2) $\frac{5}{3}$ μF
(3) $\frac{8}{3}$ μF (4) $\frac{7}{3}$ μF

PCP036

37. The effective capacitance between the points P and Q of the arrangement shown in the figure is :



(1) (1/2) μF (2)1 μF
(3) 2 μF (4) 1.33 μF

PCP037

38. Two spheres of radii R_1 and R_2 having equal charges are joined together with a copper wire. If V is the potential of each sphere after they are separated from each other, then the initial charge on both spheres was :

(1) $\frac{V}{k}(R_1 + R_2)$ (2) $\frac{V}{2k}(R_1 + R_2)$
(3) $\frac{V}{3k}(R_1 + R_2)$ (4) $\frac{V}{k}\frac{(R_1 R_2)}{(R_1 + R_2)}$

PCP038

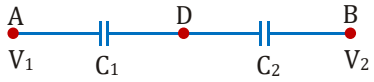
39. Two spheres of radii 1 cm and 2 cm have been charged with 1.5×10^{-8} and 0.3×10^{-7} coulombs of positive charge. When they are connected with a wire, charge :
(1) will flow from the first to the second
(2) will flow from the second to the first
(3) will not flow at all
(4) may flow either from first to second, or from the second to first, depending upon the length of the connecting wire

PCP039

40. Half of the space between a parallel plate capacitor is filled with a medium of dielectric constant K parallel to the plates. If initially the capacity was C, then the new capacity will be :
(1) 2KC/(1+K) (2) C (K+1)/2
(3) CK/(1+K) (4) KC

PCP040

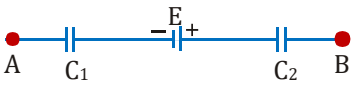
41. Two capacitances C_1 and C_2 in a circuit are joined as shown in figure The potential of point A is V_1 and that of B is V_2. The potential of point D will be :



(1) $\frac{1}{2}(V_1+V_2)$ (2) $\frac{C_2V_1+C_1V_2}{C_1+C_2}$
(3) $\frac{C_1V_1+C_2V_2}{C_1+C_2}$ (4) $\frac{C_2V_1-C_1V_2}{C_1+C_2}$

PCP041

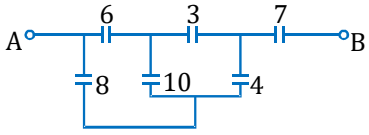
42. A circuit has a section AB as shown in the figure with E = 10 V, C_1 = 1.0 μF, C_2 = 2.0 μF and the potential difference $V_A - V_B$ = 5 V. The voltage across C_1 is :



(1) Zero (2) 5 V (3) 10 V (4) 15 V

PCP042

43. In the circuit diagram shown all the capacitors are in μF. The equivalent capacitance between points A & B is (in μF) :



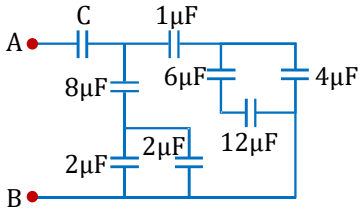
(1) 14/5 (2) 7.5
(3) 3/7 (4) None of these

PCP043

44. Two capacitances C_1 and C_2 are connected in series; assume that $C_1 < C_2$. The equivalent capacitance of this arrangement is C, where :
(1) $C < C_1/2$ (2) $C_1/2 < C < C_1$
(3) $C_1 < C < C_2$ (4) $C_2 < C < 2C_2$

PCP044

45. In the following circuit the resultant capacitance between A & B is 1μF. Find the value of C :

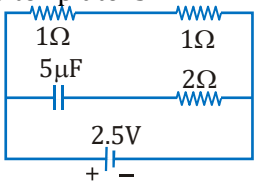


(1) $\frac{23}{32}$ μF (2) $\frac{32}{23}$ μF (3) $\frac{13}{23}$ μF (4) $\frac{23}{13}$ μF

PCP045

CHARGING AND DISCHARGING OF CAPACITORS & CAPACITOR'S CIRCUIT

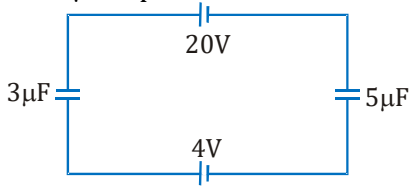
46. A capacitor of capacitance 5 μF is connected as shown in the figure. The internal resistance of the cell is 0.5 Ω. The amount of charge on the capacitor plate is :



(1) 0 (2) 5 μC (3) 10 μC (4) 25 μC

PCP046

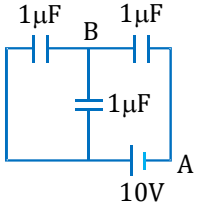
47. In the given circuit, the potential difference across 3 μF capacitor will be :



(1) 16 V (2) 10 V (3) 6 V (4) 4 V

PCP047

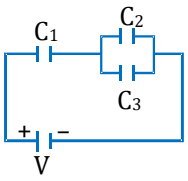
48. If potential of A is 10 V, then potential of B is :



(1) 25/3 V (2) 50/3 V
(3) 100/3 V (4) 50 V

PCP048

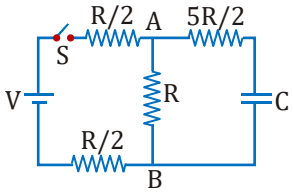
49. Capacitance $C_1 = 2C_2 = 2C_3$ and potential difference across C_1, C_2 and C_3 are V_1, V_2 and V_3 respectively then :



(1) $V_1 = V_2 = V_3$ (2) $V_1 = 2V_2 = 2V_3$
(3) $2V_1 = V_2 = V_3$ (4) $2V_1 = 2V_2 = V_3$

PCP049

50 In the circuit shown in figure, the battery is an ideal one with emf V. The capacitor is initially uncharged. Switch S is closed at time t = 0.



The final charge Q on the capacitor is :

(1) $\frac{CV}{2}$ (2) $\frac{CV}{3}$ (3) CV (4) $\frac{CV}{6}$

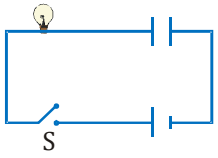
PCP050

51. In previous question, what is the current in the steady state is ?

(1) $\frac{V}{2R}$ (2) $\frac{V}{R}$ (3) $\frac{2V}{R}$ (4) $\frac{V}{3R}$

PCP051

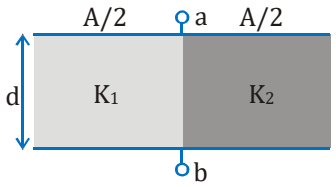
52. A bulb, a capacitor and a battery are connected together as shown here, with switch S initially open. When the switch S is closed, which one of the following is true?



(1) The bulb will light up for an instant when the capacitor starts charging.
(2) The bulb will light up when the capacitor is fully charged
(3) The bulb will not light up at all
(4) The bulb will light up and go off at regular intervals.

PCP052

53. The capacity of a parallel plate air capacitor is 10 μF. As shown in the figure this capacitor is divided into two equal parts; these parts are filled by media of dielectric constants $K_1 = 2$ and $K_2 = 4$. Capacity of this arrangement will be :



(1) 20 μF (2) 30 μF (3) 10 μF (4)40 μF

PCP053

54. Three capacitors, each of value 1 μF are so combined that the resultant capacity is 1.5 μF. Then :
(1) All three capacitors are connected in parallel.
(2) All three capacitors are connected in series.
(3)Third capacitor is in series with a parallel combination of the other two.
(4)Third capacitor is in parallel with a series combination of the other two.

PCP054

55. Two conducting spheres of radii R_1 and R_2 are charged with charges Q_1 and Q_2 respectively. On bringing them in contact there is :
(1) no change in the energy of the system
(2) an increase in the energy of the system if $Q_1R_2 \neq Q_2R_1$
(3) always a decrease in the energy of the system
(4) a decrease in the energy of the system if $Q_1R_2 \neq Q_2R_1$

PCP055

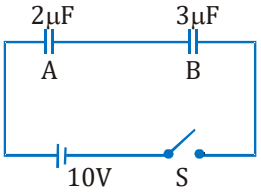
56. A capacitor is charged with a battery and energy stored is U. After disconnecting the battery another capacitor of same capacity is connected in parallel with it. The energy stored in each capacitor is:
(1) U/2 (2) U/4 (3) 4 U (4) 2 U

PCP056

57. Three capacitors of capacitances 3 μF, 10 μF and 15 μF are connected in series to a voltage source of 100 V. The charge on 15 μF is :
(1) 50 μC (2) 160 μC
(3) 200 μC (4) 280 μC

PCP057

58. Two capacitors A and B are connected in series with a battery as shown in the figure. When the switch S is closed and the two capacitors get charged fully, then :



(1) the potential difference across the plates of A is 4 V and across the plates of B is 6 V
(2) the p.d. across the plates of A is 6 V and across the plates of B is 4 V
(3) the ratio of electrical energies stored in A and B is 2 : 3
(4) the ratio of charges on A and B is 3 : 2

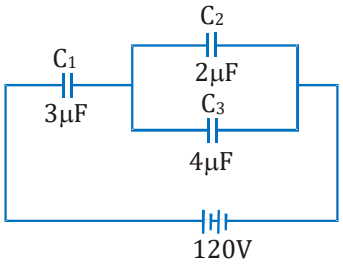
PCP058

59. An automobile spring extends 0.2 m for 5000 N load. The ratio of potential energy stored in this spring when it has been compressed by 0.2 m to the potential energy stored in a 10 μF capacitor at a potential difference of 10000 V will be :

(1) $\frac{1}{4}$ (2) 1 (3) $\frac{1}{2}$ (4) 2

PCP059

60. The charge on each capacitors shown in figure and the potential difference across them will be respectively :-



(1) 240μC, 80μC, 160μC and 80 V, 40 V, 40 V
(2) 300μC, 75μC, 150μC and 40 V, 80 V, 60 V
(3) 220μC, 70μC, 140μC and 60 V, 50 V, 40 V
(4) none of these

PCP060

61. Three capacitance 2 μF, 3 μF and 6 μF are connected in series with a 10 volt battery, then charge on 3 μF capacitor is :

(1) 5 μC (2) 10μC (3) 11μC (4) 15μC

PCP061

62. Two charged spheres having radii a and b are joined with a wire then the ratio of electric field E_a/E_b on their respective surfaces is :

(1) a/b (2) b/a (3) a^2/b^2 (4) b^2/a^2

PCP062

63. A solid conducting sphere of radius R_1 is surrounded by another concentric hollow conducting sphere of radius R_2. The capacitance of this assembly is proportional to:

(1) $\frac{R_2 - R_1}{R_1R_2}$ (2) $\frac{R_2 + R_1}{R_1R_2}$

(3) $\frac{R_1R_2}{R_1 + R_2}$ (4) $\frac{R_1R_2}{R_2 - R_1}$

PCP063

64. Two spherical conductors A and B of radius a and b (b > a) are placed in air concentrically. B is given a charge + Q coulombs and A is grounded. The equivalent capacitance of these is :

(1) $4\pi\in_0 \frac{ab}{(b-a)}$ (2) $4\pi\in_0 (a + b)$

(3) $4\pi\in_0 b$ (4) $4\pi\in_0 \frac{b^2}{(b-a)}$

PCP064

65. Time constant of a series R–C circuit is :-

(1) +RC (2) –RC (3) $\frac{R}{C}$ (4) $\frac{C}{R}$

PCP065

66. Energy per unit volume for a capacitor having area A and separation d kept at potential difference V is given by :

(1) $\frac{1}{2}\in_0 \frac{V^2}{d^2}$ (2) $\frac{1}{2\in_0} \frac{V^2}{d^2}$

(3) $\frac{\varepsilon_0 V^2 A^2}{2d^2}$ (4) $\frac{1}{2}\frac{V^2A^2}{\varepsilon_0 d^2}$

PCP066

67. A capacitor of capacity C_1 charged upto a voltage V and then connected to an uncharged capacitor of capacity C_2. Then final potential difference across each will be :

(1) $\frac{C_2V}{C_1 + C_2}$ (2) $\frac{C_1V}{C_1 + C_2}$

(3) $\left(1 + \frac{C_2}{C_1}\right)V$ (4) $\left(1 - \frac{C_2}{C_1}\right)V$

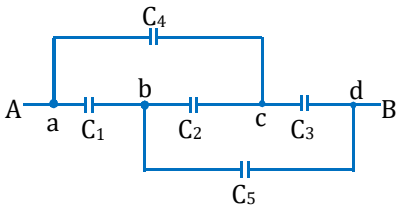
PCP067

68. A conducting sphere of radius 10 cm is charged with 10 μC. Another uncharged sphere of radius 20 cm is allowed to touch it for some time. After that, if the spheres are separated, then surface density of charge on the spheres will be in the ratio of :

(1) 1 : 4 (2) 1 : 3 (3) 2 : 1 (4) 1 : 1

PCP068

69. In the given figure, the capacitors C_1, C_3, C_4, C_5 have a capacitance 4 μF each. If the capacitor C_2 has a capacitance 10 μF, then effective capacitance between A and B will be:



(1) 2 μF (2) 4 μF (3) 6 μF (4) 8 μF

PCP069

70. Two capacitors of capacitances 3 μF and 6 μF are charged to a potential of 12 V each. They are now connected to each other with the positive plate of one joined to the negative plate of the other. The potential difference across each will be

(1) 3 V (2) Zero (3) 6 V (4) 4 V

PCP070

71. A capacitor of 0.2 μF capacitance is charged to 600 V. After removing the battery, it is connected with a 1.0 μF capacitor in parallel, then the potential difference across each capacitor will become :

(1) 300 V (2) 600 V

(3) 100 V (4) 120 V

PCP071

72. Mean electric energy density between the plates of a charged capacitor is :

Here q = Charge on capacitor A = Area of each plate of the capacitor

(1) $q^2/(2\in_0 A^2)$ (2) $q/(2\in_0 A^2)$

(3) $q^2/(2\in_0 A)$ (4) None of these

PCP072

73. If potential difference across a capacitor is changed from 15 V to 30 V, work done is W. The work done when potential difference is changed from 30 V to 60 V, will be :

(1) W (2) 4W (3) 3 W (4) 2 W

PCP073

74. Three capacitors each of capacity 4 μF are to be connected in such a way that the effective capacitance is 6 μF. This can be done by :-

(1) connecting all of them in series

(2) connecting all of them in parallel

(3) connecting two in series and one in parallel

(4) connecting two in parallel and one in series

PCP074

75. A capacitor is connected to a 10 V battery. The charge on plates is 40 μC when medium between plates is air. The charge on the plates become 100 μC when the space between the plates is filled with oil. The dielectric constant of oil is :

(1) 2.5 (2) 4 (3) 6.25 (4) 10

PCP075

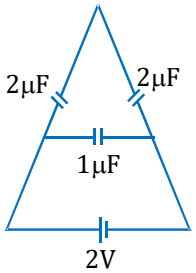
76. Two capacitor each having a capacitance C and breakdown voltage V are joined in series. The effective capacitance and maximum working voltage of the combination is :-

(1) 2C, 2V (2) $\frac{C}{2}, \frac{V}{2}$

(3) 2C, V (4) $\frac{C}{2}$,2V

PCP076

77. The charge (in μC) on any one of the 2μF capacitor and 1μF capacitor will be given respectively as :



(1) 1,2 (2) 2,1 (3) 1,1 (4) 2,2

PCP077

78. The electric field between the plates of a parallel plate capacitor when connected to a certain battery is E_0. If the space between the plates of the capacitor is filled by introducing a material of dielectric constant K without disturbing the battery connections; the field between the plates will be :

(1) KE_0

(2) E_0

(3) $\frac{E_0}{K}$

(4) None of the above

PCP078

79. A 40 μF capacitor in a defibrillator is charged to 3000 V. The energy stored in the capacitor is sent through the patient during a pulse of duration 2 ms. The power delivered to the patient is :

(1) 45 kW (2) 90 kW

(3) 180 kW (4) 360 kW

PCP079

80. Two capacitors with capacity C_1 and C_2, when connected in series, have a capacitance C_s and when connected in parallel have a capacitance C_p. Which of the following is true?

(1) $C_s = C_1+C_2$ (2) $C_p = \frac{C_1C_2}{C_1+C_2}$

(3) $\frac{C_s}{C_p} = \frac{C_1}{C_2}$ (4) $C_sC_p = C_1C_2$

PCP080

81. Two metallic spheres of radii R_1 and R_2 are connected by a thin wire. If + q_1 and + q_2 are the charges on the two spheres then :

(1) $\frac{q_1}{q_2} = \frac{R_1^2}{R_2^2}$ (2) $\frac{q_1}{q_2} = \frac{R_1}{R_2}$

(3) $\frac{q_1}{q_2} = \frac{R_1^3}{R_2^3}$ (4) $\frac{q_1}{q_2} = \frac{\left(R_1^2-R_2^2\right)}{\left(R_1^2+R_2^2\right)}$

PCP081

82. Two spheres have radii 10 cm & 20 cm. One of the sphere is given 150 μC charge and connected by a wire. Their common potential will be -

(1) 9×10^6 volts (2) $4{\cdot}5 \times 10^6$ volts

(3) $1{\cdot}8 \times 10^6$ volts (4) $1{\cdot}35 \times 10^9$ volts

PCP082

83. Total energy stored in a 900 μF capacitor at 100 volts is transferred into a 100 μF capacitor. The potential drop across the new capacitor is (in volts)

(1) 900 (2) 200

(3) 100 (4) 300

PCP083

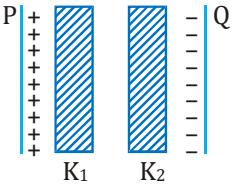
EXERCISE-I (Conceptual Questions) ANSWER KEY

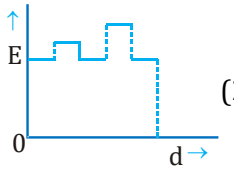
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	3	3	1	3	4	2	4	3	4	2	4	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	3	3	4	3	4	3	1	4	1	2	2	4	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	3	1	4	1	3	2	2	3	1	3	3	1	2	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	2	2	1	1	1	1	2	4	4	2	3	2	2	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	2	4	4	1	1	2	3	2	4	3	1	2	3	1
Question	76	77	78	79	80	81	82	83							
Answer	4	4	2	2	4	2	2	4							

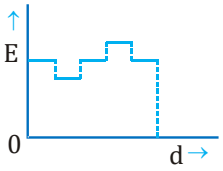
Exercise - II (Previous Year Questions) AIPMT/NEET

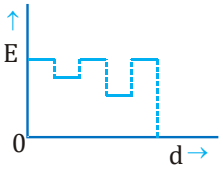
AIPMT 2014

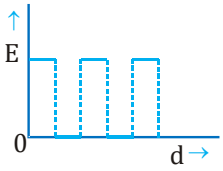
1. Two thin dielectric slabs of dielectric constants K_1 and K_2 ($K_1 < K_2$) are inserted between the plates of a parallel plate capacitor, as shown in the figure. The variation of electric field 'E' between the plates with distance 'd' as measured from plate P is correctly shown by :-



(1) 

(2) 

(3) 

(4) 

PCP091

AIPMT 2015

2. A parallel plate air capacitor of capacitance C is connected to a cell of emf V and then disconnected from it. A dielectric slab of dielectric constant K, which can just fill the air gap of the capacitor, is now inserted in it. Which of the following is **incorrect**?

(1) The energy stored in the capacitor decreases K times.

(2) The change in energy stored is $\frac{1}{2}CV^2\left(\frac{1}{K}-1\right)$.

(3) The charge on the capacitor is not conserved.

(4) The potential difference between the plates decreases K times.

PCP092

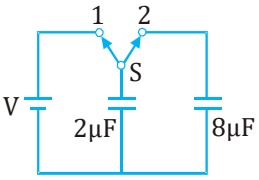
Re-AIPMT 2015

3. A parallel plate air capacitor has capacity 'C' and separation between the plates is 'd'. A potential difference 'V' is applied between the plates. Force of attraction between the plates of the parallel plate air capacitor is :-

(1) $\frac{C^2V^2}{2d^2}$ (2) $\frac{C^2V^2}{2d}$ (3) $\frac{CV^2}{2d}$ (4) $\frac{CV^2}{d}$

PCP093

NEET-I 2016

4. 

A capacitor of 2μF is charged as shown in the diagram. When the switch S is turned to position 2, the percentage of its stored energy dissipated is:

(1) 0% (2) 20% (3) 75% (4) 80%

PCP094

NEET(UG)-2017

5. A capacitor is charged by a battery. The battery is removed and another identical uncharged capacitor is connected in parallel. The total electrostatic energy of resulting system :-

(1) Decreases by a factor of 2

(2) Remains the same

(3) Increases by a factor of 2

(4) Increases by a factor of 4

PCP096

NEET(UG) 2018

6. The electrostatic force between the metal plates of an isolated parallel plate capacitor C having a charge Q and area A, is :-

(1) independent of the distance between the plates.

(2) linearly proportional to the distance between the plates

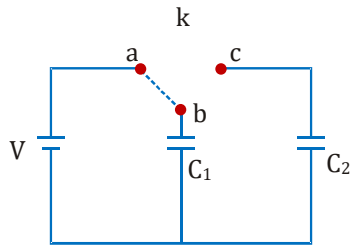
(3) proportional to the square root of the distance between the plates.

(4) inversely proportional to the distance between the plates.

PCP097

NEET(UG) 2019 (Odisha)

7. Two identical capacitors C_1 and C_2 of equal capacitance are connected as shown in the circuit. Terminals a and b of the key k are connected to charge capacitor C_1 using battery of emf V volt. Now disconnecting a and b the terminals b and c are connected. Due to this, what will be the percentage loss of energy?



(1) 75% (2) 0%

(3) 50% (4) 25%

PCP098

8. Two metal spheres, one of radius R and the other of radius 2R respectively have the same surface charge density σ. They are brought in contact and separated. What will be the new surface charge densities on them?

(1) $\sigma_1 = \frac{5}{6}\sigma,\ \sigma_2 = \frac{5}{2}\sigma$

(2) $\sigma_1 = \frac{5}{2}\sigma,\ \sigma_2 = \frac{5}{6}\sigma$

(3) $\sigma_1 = \frac{5}{2}\sigma,\ \sigma_2 = \frac{5}{3}\sigma$

(4) $\sigma_1 = \frac{5}{3}\sigma,\ \sigma_2 = \frac{5}{6}\sigma$

PCP099

NEET(UG) 2020

9. The capacitance of a parallel plate capacitor with air as medium is 6 μF. With the introduction of a dielectric medium, the capacitance becomes 30 μF. The permittivity of the medium is :

($\varepsilon_0 = 8.85 \times 10^{-12}\ C^2\ N^{-1}\ m^{-2}$)

(1) $5.00\ C^2\ N^{-1}\ m^{-2}$

(2) $0.44 \times 10^{-13}\ C^2\ N^{-1}\ m^{-2}$

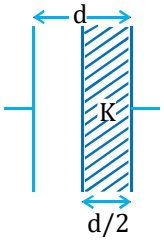
(3) $1.77 \times 10^{-12}\ C^2\ N^{-1}\ m^{-2}$

(4) $0.44 \times 10^{-10}\ C^2\ N^{-1}\ m^{-2}$

PCP100

NEET(UG) 2020 (Covid-19)

10. A parallel plate capacitor having cross-sectional area A and separation d has air in between the plates. Now an insulating slab of same area but thickness d/2 is inserted between the plates as shown in figure having dielectric constant K(= 4). The ratio of new capacitance to its original capacitance will be,



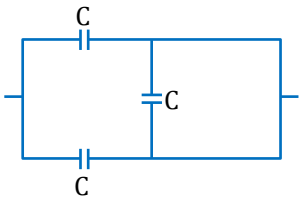
(1) 2 : 1 (2) 8 : 5

(3) 6 : 5 (4) 4 : 1

PCP101

NEET(UG) 2021

11. The equivalent capacitance of the combination shown in the figure is :



(1) 3C (2) 2C

(3) C/2 (4) 3C/2

PCP102

12. A parallel plate capacitor has a uniform electric field '$\vec{E}$' in the space between the plates. If the distance between the plates is 'd' and the area of each plate is 'A', the energy stored in the capacitor is :

(ε_0 = permittivity of free space)

(1) $\frac{1}{2}\varepsilon_0 E^2$ (2) $\varepsilon_0 EAd$

(3) $\frac{1}{2}\varepsilon_0 E^2 Ad$ (4) $\frac{E^2 Ad}{\varepsilon_0}$

PCP103

13. Polar molecules are the molecules :

(1) having zero dipole moment.

(2) acquire a dipole moment only in the presence of electric field due to displacement of charges.

(3) acquire a dipole moment only when magnetic field is absent.

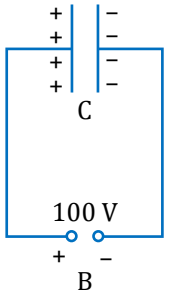
(4) having a permanent electric dipole moment.

PCP104

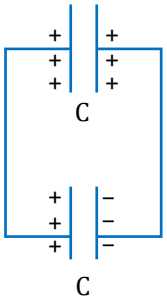
NEET(UG) 2022

14. A capacitor of capacitance C = 900 pF is charged fully by 100 V battery B as shown in figure (a). Then it is disconnected from the battery and connected to another uncharged capacitor of capacitance C = 900 pF as shown in figure (b). The electrostatic energy stored by the system (b) is:

(a)



(b)



(1) 3.25×10^{-6} J

(2) 2.25×10^{-6} J

(3) 1.5×10^{-6} J

(4) 4.5×10^{-6} J

PCP105

RE-NEET(UG) 2022

15. The distance between the two plates of a parallel plate capacitor is doubled and the area of each plate is halved. If C is its initial capacitance, its final capacitance is equal to:

(1) 2C (2) C/2

(3) 4C (4) C/4

PCP107

16. The effective capacitances of two capacitors are 3μF and 16 μF, when they are connected in series and parallel respectively. The capacitance of two capacitors are :

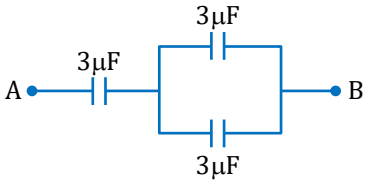
(1) 10 μF, 6 μF (2) 8 μF, 8 μF

(3) 12 μF, 4 μF (4) 1.2 μF, 1.8 μF

PCP108

NEET(UG) 2023

17. The equivalent capacitance of the system shown in the following circuit is :



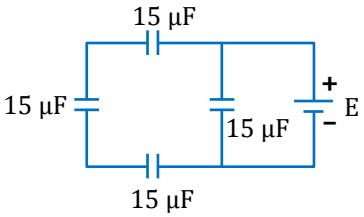
(1) 3μF (2) 6μF

(3) 9μF (4) 2μF

PCP128

RE-NEET(UG) 2023

18. The equivalent capacitance of the arrangement shown in figure is :



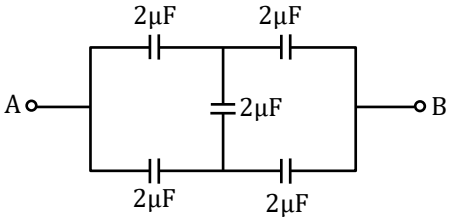
(1) 30 μF (2) 15 μF

(3) 25 μF (4) 20 μF

PCP129

NEET(UG) 2024

19. In the following circuit, the equivalent capacitance between terminal A and terminal B is :



(1) 2μF (2) 1μF (3) 0.5μF (4) 4μF

PCP130

20. If the plates of a parallel plate capacitor connected to a battery are moved close to each other, then

A. the charge stored in it, increases.
B. the energy stored in it, decreases.
C. its capacitance increases.
D. the ratio of charge to its potential remains the same.
E. the product of charge and voltage increases.

Choose the most appropriate answer from the options given below :

(1) A, B and E only (2) A, C and E only
(3) B, D and E only (4) A, B and C only

PCP131

RE-NEET(UG) 2024

21. A 12 pF capacitor is connected to a 50V battery, the electrostatic energy stored in the capacitor in nJ is :

(1) 15 (2) 7.5 (3) 0.3 (4) 150

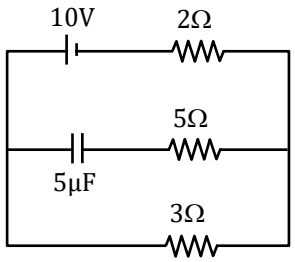
PCP132

22. The capacitance of a capacitor with charge q and a potential difference V depends on:

(1) both q and V
(2) the geometry of the capacitor
(3) q only
(4) V only

PCP133

23. The steady state current in the circuit shown below is:



(1) 0.67 A
(2) 1.5 A
(3) 2 A
(4) 1 A

PCP134

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	3	4	1	1	3	4	4	2	2	3	4	2	4
Question	16	17	18	19	20	21	22	23							
Answer	3	4	4	1	2	1	2	3							

Exercise – III (Analytical Questions) Master Your Understanding

1. A parallel plate air capacitor is connected to a battery. The quantities charge, voltage, electric field and energy associated with the capacitor are given by Q_0, V_0, E_0 and U_0 respectively. A dielectric slab is now inserted to fill the space between the plates with battery still in connection. The corresponding quantities now given by Q, V E and U are related to the previous one as :

(a) $Q > Q_0$ (b) $V > V_0$ (c) $E > E_0$ (d) $U > U_0$

(1) only (a) is correct

(2) only a & c are correct

(3) only a & d are correct

(4) all of above are correct

PCP109

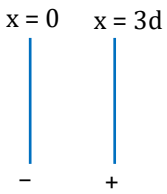
2. A dielectric slab of thickness d is inserted in a parallel plate capacitor whose negative plate is at x = 0 and positive plate is at x = 3d. The slab is equidistant from the plates. The capacitor is given some charge. As one goes from 0 to 3d:

(a) The magnitude of the electric field remains the same

(b) The direction of the electric field remains the same

(c) The electric potential increases continuously

(d) The electric potential increases at first, then decreases and again increases



(1) a and c are correct

(2) b and c are correct

(3) a, b and c are correct

(4) All are correct

PCP110

3. The plates of a parallel plate capacitor are, not exactly parallel. The surface charge density therefore:

(a) Is higher at the closer end

(b) The surface charge density will not be uniform

(c) Each plate will have the same potential at each point

(d) The electric field is smallest where the plates are closest

(1) only a is correct

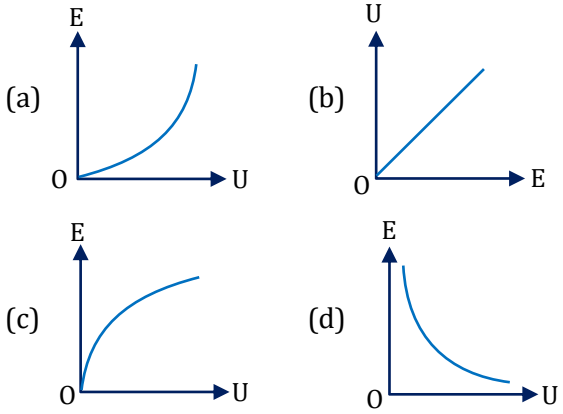
(2) only b & c are correct

(3) a & c are correct

(4) a, b & c are correct

PCP111

4. If at distance r from a positively charged particle, electric field strength and energy density are E and U, respectively, which of the following graphs is correct?



(1) only a

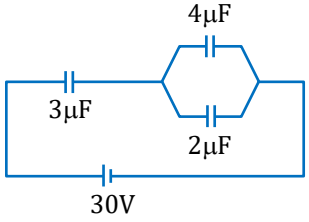
(2) only b

(3) only c

(4) a, b, d

PCP112

5. In the circuit shown in figure, match the following two columns.

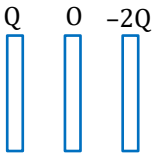


	Column-I		Column-II
(I)	Charge given by battery	(a)	20 μC
(II)	Charge of 4μF capacitor	(b)	30 μC
(III)	Charge of 2 μF capacitor	(c)	60 μC
(IV)	Charge of 3 μF capacitor	(d)	40 μC

(1) I-a, II-d, III-b, IV-c
(2) I-c, II-d, III-a, IV-c
(3) I-b, II-a, III-c, IV-d
(4) I-c, II-d, III-c, IV-a

PCP113

6. Three identical metal plates with large surface areas are kept parallel to each other as shown in fig. The leftmost plate is given charge Q, the rightmost given charge –2Q and the middle one remains neutral. Then



	Column-I		Column-II
i.	The charge appearing on outer surface of the rightmost plate	a.	$+\frac{Q}{2}$
ii.	The charge appearing on outer surface of the leftmost plate	b.	$-\frac{Q}{2}$
iii.	The charge appearing on left surface of the middle plate	c.	$-\frac{3Q}{2}$
iv.	The charge appearing on right surface of the middle plate	d.	$\frac{3Q}{2}$

(1) i-b, ii-b, iii-d, iv-c (2) i-c, ii-b, iii-d, iv-c
(3) i-b, ii-a, iii-c, iv-d (4) i-b, ii-b, iii-c, iv-d

PCP114

7. Match the following –

	List-I		List-II
(P)	Self-energy of a uniformly charged spherical shell	(1)	$\frac{1}{8\pi\varepsilon_0}\frac{Q^2}{R}$
(Q)	Energy stored per unit volume of a charged air filled capacitor	(2)	$\frac{1}{4\pi\varepsilon_0}\frac{2Q^2}{R^2}$
(R)	Force between two circular plates of a charged parallel plate capacitor, radius of the plates being R and separation R/32	(3)	$\frac{1}{4\pi\varepsilon_0}\frac{Q^2}{R}$
(S)	Work done by electric field in moving a point charge Q from the surface of a uniformly charged sphere of radius R and moving it to infinity, total charge on the sphere = Q	(4)	$\frac{1}{2}\varepsilon_0 E^2$

Codes :

	P	Q	R	S
(1)	3	1	4	2
(2)	2	1	3	4
(3)	1	4	2	3
(4)	4	1	2	3

PCP115

8. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Mutual force of attraction between two plates of isolated capacitor is independent of presence of dielectric slab between plates.

Reason (R) : Force of interaction between two charged bodies is independent of medium.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PCP116

9. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : It is not possible to make a spherical conductor of capacitance one farad.

Reason (R) : It is possible for earth as its radius is 6400 km.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PCP117

10. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When two charged capacitors are connected in parallel, then total potential energy of system always decreases.

Reason (R) : During recombination charge flow from one capacitor to another capacitor so energy is lost in the form of heat and electromagnetic radiation.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PCP118

11. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Circuits containing high capacity capacitors, charged to high voltage should be handled with caution, even when the current in the circuit is switched off.

Reason (R) : When an isolated capacitor is touched by hand or any other part of the human body, there is an easy path to the ground available for the discharge of the capacitor.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PCP119

12. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : If separation between plates of a parallel plate isolated charged capacitor is increased, its energy stored increases.

Reason (R) : Work done to more separate the plates get converted in electrostatic potential energy.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PCP120

13. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When a capacitor is charged by a battery, both the plates receive charge equal in magnitude, no matter size of plates are identical or not.

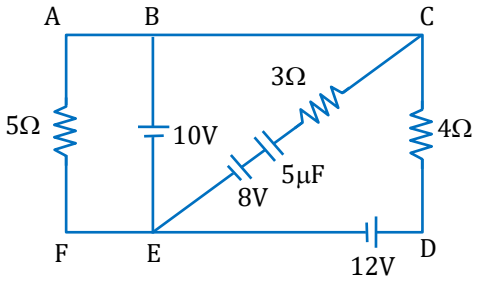
Reason (R) : The charge distribution on the plates of capacitor is in accordance with charge conservation principle.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PCP121

14. A network consisting of three resistors, three batteries, and a capacitor is shown in figure.



Column I	Column II
(A) Current in branch EB is	(P) 10 μC
(B) Current in branch CB is	(Q) 0.5 A
(C) Current in branch ED is	(R) 1.5 A
(D) Charge on capacitor is	(S) 5 μC
	(T) None

Options :-

(1) A → R, B → Q, C → Q, D → P
(2) A → R, B → Q, C → R, D → S
(3) A → Q, B → R, C → R, D → P
(4) A → R, B → Q, C → Q, D → S

PCP122

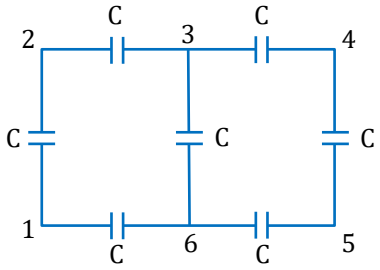
15. **Statement–1** : If capacitor is filled with, same thickness (t) of dielectric (t < d) and conducting sheet one after another, then capacitance are C_1 and C_2 respectively then $C_1 < C_2$

Statement–2 : Capacitance is more in presence of metal sheet in compare to dielectric sheet as $K_{Conduct} > K_{dielectric}$

(1) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1.
(2) Statement–1 is True, Statement–2 is True ; Statement–2 is not a correct explanation for Statement–1.
(3) Statement–1 is True, Statement–2 is False.
(4) Statement–1 is False, Statement–2 is True.

PCP123

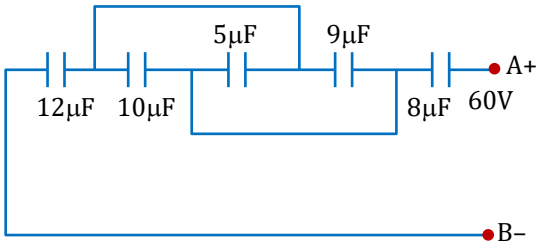
16. For the given circuit, select the incorrect alternative



(1) The equivalent capacitance between points 1 & 2 is $\frac{15C}{11}$
(2) The equivalent capacitance between points 3 & 6 is $\frac{5C}{3}$
(3) The equivalent capacitance between points 1 & 3 is $\frac{15C}{14}$
(4) The equivalent capacitance between points 3 & 5 is $\frac{14C}{15}$

PCP124

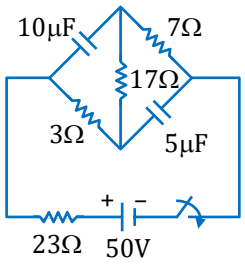
17. The equivalent capacitance between A and B is ..... and charge on 5μF capacitor is .....



(1) 4μF & 50 μC
(2) 5μF & 40 μC
(3) 8μF & 60 μC
(4) None

PCP125

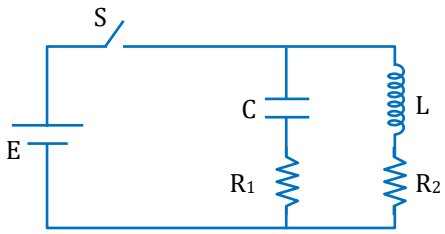
18. Write down the charge in μC on 10μF capacitor in steady state.



(1) 100 μC
(2) 150 μC
(3) 180 μC
(4) 200 μC

PCP126

19. In the circuit shown in figure E=25V, L=2H, C=60 μF, R_1 = 5Ω and R_2 = 10Ω. Switch S is closed at t = 0.



	Column-I		Column-II
(A)	Current through R_1 at t = 0	(P)	0
(B)	Current through R_2 at t = 0	(Q)	5A
(C)	Current through R_1 at t = ∞	(R)	2.5 A
(D)	Current through R_2 at t = ∞	(S)	None of these

Options :-

(1) A → P, B → Q, C → P, D → R

(2) A → Q, B → P, C → P, D → R

(3) A → Q, B → P, C → R, D → P

(4) A → P, B → R, C → P, D → Q

PCP127

EXERCISE-III (Analytical Questions) ANSWER KEY

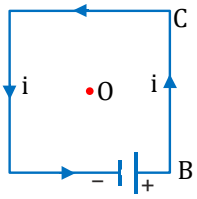
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	4	3	2	4	3	1	3	4	1	1	1	1	1
Question	16	17	18	19											
Answer	4	1	4	2											

Exercise - I (Conceptual Questions) Build Up Your Understanding

MAGNETIC FIELD & BIOT-SAVART LAW

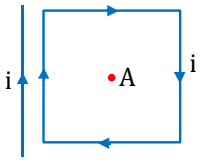
1. A stationary magnet does not interact with :-
(1) iron rod (2) moving charge
(3) moving magnet (4) stationary charge
PME001

2. Figure shows a square shape loop, whose one arm BC produces magnetic field B at the centre of coil. The resultant magnetic field due to all the arms will be :-



(1) 4B (2) B/2 (3) B (4) 2B
PME002

3. A wire is parallel to one arm of a square current carrying loop, which also carries current. Now at any point A within the coil the magnetic field will be :-



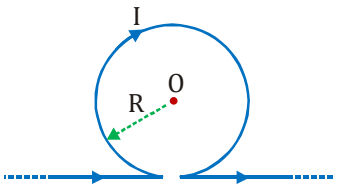
(1) less than the magnetic field produced due to loop only.
(2) more than the magnetic field produced due to loop only.
(3) equal to the earlier.
(4) zero
PME003

4. A current of 10 A is established in a long wire along positive z -direction. The magnetic field B at the point (1m, 0, 0) is :-
(1) 1μT along the –y direction
(2) 2μT along the +y direction
(3) 1μT along the –x direction
(4) 2μT along the +x direction
PME004

5. Radius of current carrying coil is 'R'. Then ratio of magnetic fields at the centre of the coil to the axial point, which is $R\sqrt{3}$ distance away from the centre of the coil :-
(1) 1 : 1 (2) 1 : 2 (3) 1 : 4 (4) 8 : 1
PME005

6. A coil of one loop is made by a wire of length L and there after a coil of two loops is made by same wire. The ratio of magnetic field at the centre of coils respectively :-
(1) 1 : 4 (2) 1 : 1 (3) 1 : 8 (4) 4 : 1
PME006

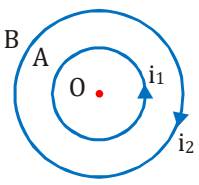
7. Magnetic field at point O will be :-



(1) $\frac{\mu_0 I}{2R}\otimes$ (2) $\frac{\mu_0 I}{2R}\odot$
(3) $\frac{\mu_0 I}{2R}\left(1-\frac{1}{\pi}\right)\otimes$ (4) $\frac{\mu_0 I}{2R}\left(1+\frac{1}{\pi}\right)\odot$
PME007

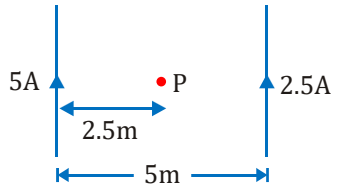
8. The vector form of Biot savart law for a current carrying element is :-
(1) $d\vec{B}=\frac{\mu_0}{4\pi}\frac{id\ell\sin\phi}{r^2}$ (2) $d\vec{B}=\frac{\mu_0}{4\pi}\frac{i\vec{d\ell}\times\hat{r}}{r^2}$
(3) $d\vec{B}=\frac{\mu_0}{4\pi}\frac{i\vec{d\ell}\times\hat{r}}{r^3}$ (4) $d\vec{B}=\frac{\mu_0}{4\pi}\frac{i\vec{d\ell}\times\vec{r}}{r^2}$
PME008

9. A and B are two concentric circular loop carrying current i_1 and i_2 as shown in figure. If ratio of their radii is 1:2 and ratio of the flux densities at the centre O due to A and B is 1:3 then the value of $\frac{i_1}{i_2}$ will be :-



(1) $\frac{1}{2}$ (2) $\frac{1}{3}$
(3) $\frac{1}{4}$ (4) $\frac{1}{6}$
PME009

10. For the given current distribution the magnetic field at point, 'P' is :-



(1) $\frac{\mu_0}{4\pi}\odot$ (2) $\frac{\mu_0}{\pi}\otimes$ (3) $\frac{\mu_0}{2\pi}\otimes$ (4) $\frac{\mu_0}{2\pi}\odot$

PME010

11. 1A current flows through an infinitely long straight wire. The magnetic field produced at a point 1m. away from it is :–

(1) 2×10^{-3} T (2) $\frac{2}{10}$ T

(3) 2×10^{-7} T (4) $2\pi \times 10^{-6}$ T

PME011

12. Two infinite long parallel wires carry equal currents in same direction. The magnetic field at mid point between the two wires is :–

(1) Twice the magnetic field produced due to each of the wires

(2) Half of the magnetic field produced due to each of the wires

(3) Square of the magnetic field produced due to each of the wires

(4) Zero

PME012

13. A closely wound flat circular coil of 25 turns of wire has diameter of 10 cm and carries a current of 4 ampere. Determine the magnetic flux density at the centre of the coil :-

(1) 1.679×10^{-5} T (2) 2.028×10^{-4} T

(3) 1.257×10^{-3} T (4) 1.512×10^{-6} T

PME013

14. π ampere current is flowing through a long straight wire. Due to this a field of 5×10^{-5} T produced, then distance of the point from the axis of the wire is :-

(1) $10^4\mu_0$ m (2) $10^5\mu_0$ m

(3) $10^6\mu_0$ m (4) $10^8\mu_0$ m

PME014

15. Radius of a current carrying coil is 'R'. The ratio of magnetic field at a axial point which is R distance away from the centre of the coil to the magnetic field at the centre of the coil :-

(1) $\left(\frac{1}{2}\right)^{1/2}$ (2) $\frac{1}{2}$ (3) $\left(\frac{1}{2}\right)^{3/2}$ (4) $\frac{1}{4}$

PME015

16. When the current flowing in a circular coil is doubled and the number of turns of the coil in it is halved, the magnetic field at its centre will become :-

(1) Same (2) Four times

(3) Half (4) Double

PME016

17. Which of the following statement is not true; magnetic field at the centre of current carrying loop :-

(1) proportional to current

(2) inversely proportional to radius

(3) proportional to number of turns

(4) none

PME017

18. A circular coil of radius R carries an electric current. The magnetic field due the coil at a point on the axis of the coil located at a distance r from the center of the coil, such that r >> R varies as :-

(1) $1/r$ (2) $1/r^{3/2}$ (3) $1/r^2$ (4) $1/r^3$

PME018

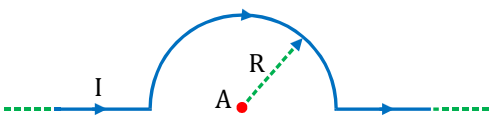
19. Two concentric circular loops of radii 0.08m and 0.1m carries current such that magnetic field at the centre is zero. If the current in the outer loop is 8A clockwise, current in the inner loop is :-

(1) 6.4 A anticlockwise (2) 6.4 A clockwise

(3) 8A anticlockwise (4) 3.2 A clockwise

PME019

20. In the shown figure magnetic field at point A will be :-



(1) $\frac{\mu_0 I}{4\pi}$ (2) $\frac{\mu_0 I}{4R}$ (3) $\frac{\mu_0 I}{4\pi R}$ (4) Zero

PME020

21. If radius of coil becomes two times and current becomes half then magnetic field at centre of the coil will be :-
(1) Two times (2) Four times
(3) Half (4) One fourth

PME021

22. 2A current is flowing in a circular loop of radius 1m. Magnitude of magnetic field at the centre of circular loop will be :-

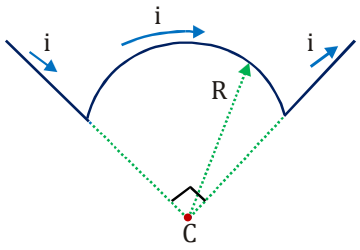
(1) $\frac{\mu_0}{2}$ (2) $2\mu_0$ (3) μ_0 (4) $\frac{\mu_0}{2\pi}$

PME022

23. A horizontal overhead powerline is at a height of 4m from the ground and carries a current of 100 A from east to west. The magnetic field directly below it on the ground is :-
(1) 2.5×10^{-7} T, southward
(2) 5×10^{-6} T, northward
(3) 5×10^{-6} T, southward
(4) 2.5×10^{-7} T, northward

PME023

24. The wire in the figure carries a current i and consists of a circular arc of radius R and central angle $\frac{\pi}{2}$ rad, and two straight sections whose extensions intersect the centre C of the arc. The magnetic field $\vec{B}$ that the current produces at C is :-



(1) $|\vec{B}| = \frac{\mu_0 i}{8R}$ into the plane of the figure

(2) $|\vec{B}| = \frac{\mu_0 i}{8R}$ out of the plane of the figure

(3) $|\vec{B}| = \frac{\mu_0 i}{8\pi R}$ into the plane of the figure

(4) $|\vec{B}| = \frac{\mu_0 i}{8\pi R}$ out of the plane of the figure

PME024

AMPERE'S LAW

25. For the hollow thin cylindrical current carrying pipe which statement is correct :-
(1) magnetic field inside the pipe is not zero
(2) magnetic field outside the pipe is zero
(3) electric field outside the pipe is zero
(4) electric field on the surface of pipe is zero

PME025

26. In a current carrying long solenoid the field produced does not depend on :-
(1) number of turns per unit length
(2) current in solenoid
(3) radius of cross section of the solenoid
(4) all of the above

PME026

27. If length and number of turns becomes half for a solenoid then value of magnetic field becomes:-
(1) twice (2) same
(3) half (4) one fourth

PME027

28. If number of turns and current become double for any solenoid, then value of magnetic field becomes :-
(1) twice (2) same
(3) half (4) four times

PME028

29. A current of $1/4\pi$ ampere is flowing through a toroid. It has 1000 number of turn per meter then value of magnetic field (in Wb/m^2) along its axis is :-
(1) 10^{-2} (2) 10^{-3} (3) 10^{-4} (4) 10^{-7}

PME029

30. Mean radius of a toroid is 10 cm and number of turns are 500. If current flowing through it is 0.1 ampere then value of magnetic field (in tesla) for toroid :-
(1) 10^{-2} (2) 10^{-5} (3) 10^{-3} (4) 10^{-4}

PME030

31. A long solenoid has length L, average diameter D and n layer of turns. Each layer contains N turns. If current flowing through the solenoid is i the value of magnetic field at the centre :-
(1) Proportional to D
(2) Inversely proportional to D
(3) Does not depend on D
(4) Proportional to L

PME031

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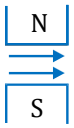
32. A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of the magnetic field is :-
(1) B/2 (2) B (3) 2B (4) 4B
PME032

33. A long solenoid having 200 turns/cm and carries current i. Magnetic field at its axis is $6{\cdot}28 \times 10^{-2}$ wb/m^2. An another solenoid having 100 turns/cm and carries $\frac{i}{3}$ current, then magnetic field at its axis will be :-
(1) 1.05×10^{-4} Wb/m^2
(2) 1.05×10^{-2} Wb/m^2
(3) 1.05×10^{-5} Wb/m^2
(4) 1.05×10^{-3} Wb/m^2
PME033

MAGNETIC FORCE & MOVING CHARGE IN MAGNETIC FIELD

34. A proton and an alpha particle are separately projected in a region where a uniform magnetic field exists. The initial velocities are perpendicular to the direction of magnetic field. If both the particles move along circles of equal radii, the ratio of momentum of proton to alpha particle $\left(\frac{P_p}{P_\alpha}\right)$ is :-
(1) 1 (2) 1/2 (3) 2 (4) 1/4
PME034

35. Cathode rays are moving between the poles of a magnet. Due to the effect of magnetic field of magnet :-



(1) velocity of rays increases
(2) velocity of rays decreases
(3) rays deflected towards south pole
(4) rays deflected in upward direction and perpendicular to the plane of the paper
PME035

36. Following charge has maximum frequency of rotation in uniform transverse magnetic field :-
(1) a proton (2) an alpha particle
(3) an electron (4) a neutron
PME036

37. Which of the following particle will experiences maximum magnetic force, when projected with the same velocity perpendicular to a magnetic field :-
(1) electron (2) proton
(3) He^+ (4) Li^{++}
PME037

38. When α and β rays are subjected to a magnetic field which is perpendicular to the direction of their motion, with their same speed. The curvature of path of both the particles are :-
(1) equal
(2) more for α particles
(3) more for β particles
(4) none
PME038

39. Two identically charged particles A and B initially at rest, are accelerated by a common potential difference V. They enters into a uniform transverse magnetic field B and describe a circular path of radii r_1 and r_2 respectively then their mass ratio is :-
(1) $\left(\frac{r_1}{r_2}\right)^2$ (2) $\left(\frac{r_2}{r_1}\right)^2$
(3) $\left(\frac{r_1}{r_2}\right)$ (4) $\left(\frac{r_2}{r_1}\right)$
PME039

40. An α-particle experiences a force of 3.84×10^{-14} N when it moves perpendicular to magnetic field of 0.2 Wb/m^2 then speed of the α - particle is :-
(1) 6.0×10^5 m/sec (2) 5.0×10^5 m/sec
(3) 1.2×10^6 m/sec (4) 3.8×10^6 m/sec
PME040

41. If an electron enters a magnetic field with its velocity pointing in the same direction as the magnetic field then :–
(1) the electron will turn towards right
(2) the electron will turn towards left
(3) the velocity of the electron will increase
(4) the velocity of the electron will remain unchanged
PME041

42. A charge having q/m equal to 10^8 C/kg and with velocity 3×10^5 m/s enters into a uniform magnetic field B = 0.3T at an angle 30° with direction of field. Then radius will be :-

(1) 0.01 cm (2) 0.5 cm
(3) 1 cm (4) 2 cm

PME042

43. When a charged particle enters in a uniform magnetic field its kinetic energy :-

(1) remains constant (2) increases
(3) decreases (4) becomes zero

PME043

44. Two particles x and y have equal charges and possessing equal kinetic energy enter in a uniform magnetic field and describe circular path of radius of curvature r_1 and r_2 respectively. The ratio of their masses is :-

(1) $\left(\frac{r_1}{r_2}\right)$ (2) $\left(\frac{r_1}{r_2}\right)^{1/2}$ (3) $\left(\frac{r_1}{r_2}\right)^2$ (4) $\left(\frac{r_2}{r_1}\right)$

PME044

45. An electron having mass 'm' and kinetic energy E enter in uniform magnetic field B perpendicularly, then its frequency of uniform circular motion will be :-

(1) $\frac{eE}{qVB}$ (2) $\frac{2\pi m}{eB}$ (3) $\frac{eB}{2\pi m}$ (4) $\frac{2m}{eBE}$

PME045

46. In a mass spectrograph an ion A of mass number 24 and charge +e and an ion B of mass number 22 and charge +2e are entered in transverse magnetic field with same velocity. The ratio of radii of their paths respectively :-

(1) $\frac{11}{24}$ (2) $\frac{12}{11}$ (3) $\frac{11}{22}$ (4) $\frac{24}{11}$

PME046

47. A charge particle is moving in the direction of a magnetic field. The magnetic force acting on the particle :-

(1) is in the direction of its velocity
(2) is in the direction opposite to its velocity
(3) is perpendicular to its velocity
(4) is zero

PME047

48. An electron of kinetic energy of 7.2×10^{-18} J is revolving in a circular path in magnetic field 9×10^{-5} Wb/m^2 then radius of its circular path is:–

(1) 1.25 cm. (2) 2.5 m.
(3) 2.5 cm. (4) 25.0 cm.

PME048

49. An electron moves with velocity v in uniform transverse magnetic field B on circular path of radius 'r', then e/m for it is :-

(1) $\frac{v}{Br}$ (2) $\frac{B}{rv}$ (3) Bvr (4) $\frac{vr}{B}$

PME049

50. A proton, deutron and an α–particle are accelerated by same potential, enter in uniform magnetic field perpendicularly. Ratio of radii of circular path respectively :-

(1) $1 : \sqrt{2} : \sqrt{2}$ (2) 2 : 2 : 1
(3) 1 : 2 : 1 (4) 1 : 1 : 1

PME050

51. A charged particle moves through a magnetic field in a direction perpendicular to it. Then the :-

(1) speed of the particle remains unchanged
(2) direction of motion of particle remains unchanged
(3) acceleration of particle remains unchanged
(4) velocity of particle remains unchanged

PME051

52. A magnetic field :-

(1) Always exerts a force on charged particle
(2) Never exerts a force on charged particle
(3) Exert a force, if the charged particle is moving across the magnetic field line
(4) Exerts a force, if the charged particle is moving along the magnetic field line

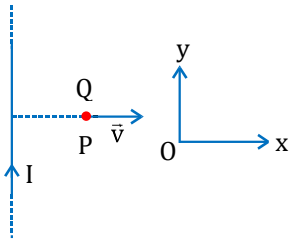
PME052

53. A proton and an α-particle moving with the same velocity and enter into a uniform magnetic field which is acting normal to the plane of their motion. The ratio of the radii of the circular paths described by the proton and α-particle respectively :-

(1) 1 : 2 (2) 1 : 4 (3) 1 : 16 (4) 4 : 1

PME053

54. A very long straight wire carries a current I. At the instant when a charge +Q at point P has velocity $\vec{v}$, as shown, the magnetic force on the charge is:-



(1) along ox (2) opposite to oy
(3) along oy (4) opposite to ox

PME054

LORENTZ FORCE & MOTION OF CHARGED PARTICALS IN B

55. In Thomson mass spectrograph $\vec{E} \perp \vec{B}$ then the velocity of undeflected electron beam will be :–

(1) $\frac{|\vec{E}|}{|\vec{B}|}$ (2) $\vec{E} \times \vec{B}$

(3) $\frac{|\vec{B}|}{|\vec{E}|}$ (4) $\frac{E^2}{B^2}$

PME055

56. A charge 'q' moves in a region where electric field and magnetic field both exist, then force on it :-

(1) $q(\vec{v}\times\vec{B})$ (2) $q\vec{E}+q(\vec{v}\times\vec{B})$

(3) $q\vec{E}+q(\vec{B}\times\vec{v})$ (4) $q\vec{B}+q(\vec{E}\times\vec{v})$

PME056

57. An electric field E and a magnetic field B applied on a proton which moves with velocity v, it goes undeflected through the region if :-

(1) $E \perp B$
(2) E is parallel to v and perpendicular to B
(3) E, B and v all three mutually perpendicular to each other and v = E/B
(4) E and B both are parallel but perpendicular to v

PME057

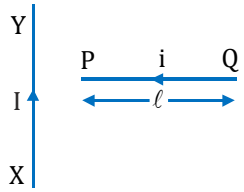
58. In a region constant uniform electric and magnetic field is present. Both field are parallel. In this region a charge released from rest, then path of particle is :-

(1) Circle (2) Helical
(3) Straight line (4) Ellipse

PME058

FORCE ON A CURRENT-CARRYING CONDUCTOR

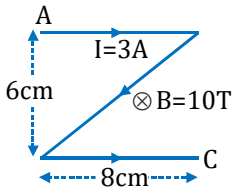
59. A wire PQ carries a current 'i' is placed perpendicular to a long wire XY carrying a current I. The direction of force on PQ will be:-



(1) towards right (2) towards left
(3) upwards (4) downwards

PME059

60. A current carrying wire AC is placed in uniform transverse magnetic field then the force on wire AC :-



(1) 3N (2) 4.2N (3) 6N (4) 4N

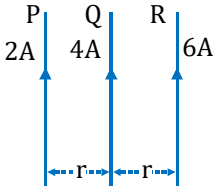
PME060

61. Force exist on a current carrying wire which is placed in external magnetic field, due to :–

(1) free electrons in wire
(2) free positive ions in wire
(3) (1) & (2) both
(4) none

PME061

62. P, Q and R long parallel straight wires in air, carrying currents as shown. The direction of resultant force on R is :-



(1) towards left
(2) towards right
(3) the same as that of current in Q
(4) perpendicular to plane of paper

PME062

63. Two parallel wires in free space are 10 cm apart and each carries a current of 10 A in the same direction. The magnetic force per unit length of each wire is :-
(1) 2×10^{-4} N, attractive
(2) 2×10^{-4} N, repulsive
(3) 2×10^{-7} N, attractive
(4) 2×10^{-7} N, repulsive

PME063

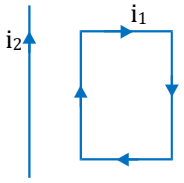
64. Wire in the form of a right angle ABC, with AB=3cm and BC = 4cm, carries a current of 10A. There is a uniform magnetic field of 5T perpendicular to the plane of the wire. The force on the wire will be :-
(1) 1.5N (2) 2.0N (3) 2.5N (4) 3.5N

PME064

65. When direct current passed through a spring then it :-
(1) Contracts (2) Expands
(3) Vibrate (4) Unchange

PME065

66. A rectangular loop carrying a current i_1, is situated near a long straight wire carrying a steady current i_2. The wire is parallel to one of the sides of the loop and is in the plane of the loop as shown in the figure. Then the current loop will :–



(1) move away from the wire
(2) move towards the wire.
(3) remain stationary.
(4) rotate about an axis parallel to the wire.

PME066

67. A current carrying wire is arranged at any angle in an uniform magnetic field, then
(1) only force acts on wire
(2) only torque acts on wire
(3) both
(4) none

PME067

68. Two parallel beams of positrons moving in the same direction will :-
(1) Repel to each other
(2) Will not interact with each other
(3) Attract to each other
(4) Be deflected normal to the plane containing the two beams

PME068

CURRENT LOOP & MOVING COIL GALVANOMETER

69. Due to the flow of current in a circular loop of radius R, the magnetic field produced at the centre of the loop is B. The magnetic moment of the loop is :-
(1) $BR^3/2\ \pi\mu_0$ (2) $2\pi\ BR^3/\mu_0$
(3) $BR^2/2\pi\mu_0$ (4) $2\pi\ BR^2/\mu_0$

PME069

70. A circular loop has a radius of 5 cm. and it is carrying a current of 0.1 A. its magnetic moment is :–
(1) 1.32×10^{-4} amp-m^2
(2) 2.62×10^{-4} amp-m^2
(3) 5.25×10^{-4} amp-m^2
(4) 7.85×10^{-4} amp-m^2

PME070

71. An electron is moving in a circle of radius 5.1×10^{-11} m at a frequency of 6.8×10^{15} rev./sec. The equivalent current is approximately :–
(1) 5.1×10^{-3} A (2) 6.8×10^{-3} A
(3) 1.1×10^{-3} A (4) 2.2×10^{-3} A

PME071

72. If number of turn, area and current through the coil is given by n, A and i respectively then its magnetic moment will be :–
(1) niA (2) n^2iA (3) niA^2 (4) $\frac{ni}{\sqrt{A}}$

PME072

73. An electron is moving around a proton in an orbit of radius 1Å and produces 16 Wb/m^2 of magnetic field at the centre, then find the angular velocity of electron :-
(1) $20\ \pi \times 10^{16}$ rad/sec
(2) 10×10^{16} rad/sec
(3) $\frac{5}{2\pi} \times 10^{16}$ rad/sec
(4) $\frac{5}{4\pi} \times 10^{16}$ rad/sec

PME073

74. Current I is flowing in a conducting circular loop of radius R. It is kept in a magnetic field B which is perpendicular to the plane of circular loop, the magnetic force acting on the loop is :-
(1) IRB (2) 2πIRB (3) Zero (4) πIRB
PME074

75. The unit of magnetic moment will be :-
(1) $\frac{A}{m}$ (2) A-m^2
(3) $\frac{T-m}{A}$ (4) $\frac{T-m}{A^2}$
PME075

76. A current carrying coil behave like tiny magnet. If area of coil is A and magnetic moment is 'M' then current through the coil is :-
(1) $\frac{M}{A}$ (2) $\frac{A}{M}$ (3) MA (4) $\frac{A^2}{M}$
PME076

77. An electron revolves with frequency 6.6×10^{15} r.p.s. around nucleus in circular orbit of radius 0.53 Å of hydrogen atom, then magnetic field produced at centre of orbit is :-
(1) 0.125 T (2) 1.25 T
(3) 12.5 T (4) 125 T
PME077

78. A current loop of area $0.01m^2$ and carrying a current of 10A is held perpendicular to a magnetic field of 0.1T, the torque in N–m acting on the loop is :-
(1) 0 (2) 0.001 (3) 0.01 (4) 1.1
PME078

79. A current carrying coil is placed in a constant uniform magnetic field B. Torque is maximum on this coil when plane of coil is :-
(1) perpendicular to B (2) parallel to B
(3) at 45° to B (4) at 60° to B
PME079

80. Magnetic field is parallel to the plane of coil then torque will be :-
(1) Maximum (2) Minimum
(3) Zero (4) None of these
PME080

BAR MAGNET

81. The work done in rotating a magnet of magnetic moment M by an angle of 90° from the external magnetic field direction is 'n' times the corresponding work done to turn it through an angle of 60°. Where 'n' gives by :-
(1) 1/2 (2) 2 (3) 1/4 (4) 1
PME081

82. Magnetic field lines produced by a bar magnet, cuts each other :–
(1) At neutral points
(2) Near the poles of the magnets
(3) At equatorial axis
(4) Never intersects to each other
PME082

MAGNETIC PROPERTIES OF MATERIALS

83. For protecting a magnetic needle it should be placed :-
(1) in an iron box (2) in wooden box
(3) in metallic box (4) none of these
PME083

84. Which of the following materials is repelled by an external magnetic field :-
(1) Iron (2) Cobalt
(3) Steel (4) Copper
PME084

85. If a diamagnetic material is placed in a magnetic field, the flux density inside the material compared to that outside will be :
(1) Slightly less (2) Slightly more
(3) Very much more (4) Same
PME085

86. To protect a sensitive instrument from external magnetic jerks, it should be placed in a container made of :-
(1) Nonmagnetic substance
(2) Diamagnetic substance
(3) Paramagnetic substance
(4) Ferromagnetic substance
PME086

87. Substances in which the magnetic moment of a single atom is not zero, are known as :-
(1) Diamagnetic (2) Ferromagnetic
(3) Paramagnetic (4) (2) and (3) both
PME087

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88. Susceptibility of a magnetic substance is found to depend on temperature and the strength of the magnetising field. The material is a :-
(1) Diamagnetic
(2) Ferromagnetic
(3) Paramagnetic
(4) Superconductor

PME088

89. Property possessed by only ferromagnetic substance is :-
(1) Attracting magnetic substance
(2) Hysteresis
(3) Susceptibility independent of temperature
(4) Directional property

PME089

90. The hard ferromagnetic material is characterized by :-
(1) Narrow hysteresis loop
(2) Broad hysteresis loop
(3) High mechanically hardness, all over
(4) Mechanically hard surface

PME090

91. The magnetic moment of paramagnetic materials is :-
(1) Infinity (2) Zero
(3) Constant but low (4) None of above

PME091

92. The cause of paramagnetism is :-
(1) Unparied electrons
(2) Electron excess and spin motion of electrons
(3) Paired electrons and orbital motion of electrons
(4) Electrons and orbital motion of electrons

PME092

93. The cause of diamagnetism is :-
(1) Orbital motion of electrons
(2) Spin motion of electrons
(3) Paired electrons
(4) None of the above

PME093

94. The magnetic moment of diamagnetic materials is :-
(1) Infinity
(2) Zero
(3) 100 amp–m^2
(4) None of the above

PME094

95. Which of the following statements is correct for diamagnetic materials :-
(1) $\mu_r < 1$
(2) χ is negative and low
(3) χ does not depend on temperature
(4) All of the above

PME095

96. The area of B–H loop for soft iron, as compared to that for steel is :-
(1) More
(2) Less
(3) Equal
(4) None of the above

PME096

97. The liquid in the watch glass in the following figure is -



(1) Ferromagnetic (2) Paramagnetic
(3) Diamagnetic (4) Nonmagnetic

PME097

98. Powerful permanent magnets are made of :-
(1) Cobalt (2) Aluminum
(3) Tin–coal (4) Cobalt–steel

PME098

99. Which of the following statements is correct for ferromagnetic material :-
(1) These become diamagnetic at Curie temperature
(2) These become paramagnetic at Curie temperature
(3) Their magnetic susceptibility becomes zero at Curie temperature
(4) Its magnetic properties are explained on the basis of electron principle

PME099

100. A material rod, when placed in a strong magnetic field, aligns itself at right angles to the magnetic field. The nature of material is :-
(1) Diamagnetic
(2) Paramagnetic
(3) Ferromagnetic
(4) Low ferromagnetic

PME100

101. The relative permeability of air is :-
(1) Zero (2) 1.04
(3) Infinity (4) 1

PME101

102. If the magnetic susceptibility of a magnetic material is –0.004 then its nature will be :-
(1) Diamagnetic (2) Paramagnetic
(3) Ferromagnetic (4) Non magnetic
PME102

103. The correct measure of magnetic hardness of a material is :-
(1) Remanent magnetism
(2) Hysteresis loss
(3) Coercivity
(4) Curie temperature
PME103

104. If the relative permeability of a material is 0.9999 then its nature will be :-
(1) Paramagnetic (2) Diamagnetic
(3) Ferromagnetic (4) Non-magnetic
PME104

105. The magnetic susceptibility of a paramagnetic material at –73°C is 0.0075 then its value at –173°C will be :-
(1) 0.0045 (2) 0.0030
(3) 0.015 (4) 0.0075
PME105

106. When a magnetic substance is heated, then it :
(1) Becomes a strong magnet
(2) Losses its magnetism
(3) Does not effect the magnetism
(4) Either (1) or (3)
PME106

107. Diamagnetic substance are :-
(1) Feebly attracted by magnets
(2) Strongly attracted by magnets
(3) Feebly repelled by magnets
(4) Strongly repelled by magnets
PME107

108. If a diamagnetic solution is poured into a U-tube and one arm of this U-tube placed between the poles of a strong magnet with the meniscus in a line with the field, then the level of the solution will :-
(1) Rise (2) Fall
(3) Oscillate slowly (4) Remain as such
PME108

109. Magnetic permeability is maximum for :
(1) Diamagnetic substance
(2) Paramagnetic substance
(3) Ferromagnetic substance
(4) All of these
PME109

110. Which one of the following is ferro-magnetic :-
(1) Co (2) Zn (3) Hg (4) Pt
PME110

111. For paramagnetic materials magnetic susceptibility is related with temperature as :-
(1) $\propto T^2$ (2) $\propto T^1$ (3) $\propto T^{-1}$ (4) $\propto T^{-2}$
PME111

112. According to Curie's law, the magnetic susceptibility of a substance at an absolute temperature T is proportional to :-
(1) $1/T$ (2) T (3) $1/T^2$ (4) T^2
PME112

113. A diamagnetic material in a magnetic field moves
(1) from stronger to the weaker parts of the field
(2) from weaker to the stronger parts of the field
(3) perpendicular to the field
(4) in none of the above directions
PME113

114. Diamagnetic substances characterise by :-
(1) low and negative magnetic susceptibility
(2) low and positive magnetic susceptibility
(3) high and negative magnetic susceptibility
(4) high and positive magnetic susceptibility
PME114

115. Magnetic susceptibility of a diamagnetic substance varies with absolute temperature as :-
(1) directly proportional to T
(2) inversely proportional to T
(3) remains unchanged with T
(4) exponential decrease with T
PME115

116. The material of permanent magnet has
(1) High retentivity, low coercivity
(2) Low retentivity, high coercivity
(3) Low retentivity, low coercivity
(4) High retentivity, high coercivity
PME116

117. Hysteresis property is shown by :-
(1) paramagnetic and diamagnetic
(2) diamagnetic
(3) paramagnetic
(4) ferromagnetic
PME117

118. Magnetic susceptibility of the following is :
(1) negative for diamagnetic
(2) positive for diamagnetic and paramagnetic
(3) negative for diamagnetic and zero for paramagnetic
(4) zero for paramagnetic and positive for ferromagnetic
PME118

119. Which statement is true :
(1) atomic magnetic dipole moment of diamagnet is zero
(2) atomic magnetic dipole moment of para-magnet is zero
(3) atomic magnetic dipole moment of ferro-magnet is zero
(4) ferromagnet is demagnetised rapidly after moving in magnetising field.

PME119

120. Curie-Weiss law is obeyed by iron at a temperature.....
(1) Below Curie temperature
(2) Above Curie temperature
(3) At Curie temperature only
(4) At all temperatures

PME120

121. Ferromagnetic substance contain :-
(1) empty subshell
(2) partially empty subshell
(3) full fill subshell
(4) none of these

PME121

122. Soft iron is used to make the core of transformer, because of its :
(1) low coercivity and low retentivity
(2) low coercivity and high retentivity
(3) high coercivity and high retentivity
(4) high coercivity and low retentivity

PME122

123. Above curie temperature ferromagnetic substance converts into :
(1) paramagnetic
(2) diamagnetic
(3) ferromagnetic
(4) non magnetic

PME123

124. Relation between μ_r and χ will be :
(1) $\mu_r = 1 + \chi$
(2) $\chi = \mu_r + 1$
(3) $\frac{\mu_0}{\mu}$
(4) $\mu_0\chi$

PME124

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	2	2	4	1	3	2	4	3	3	4	3	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	4	1	2	4	3	3	1	3	3	2	4	3	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	2	2	2	4	3	4	3	1	1	4	2	1	3	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	4	4	1	1	1	3	1	3	1	2	3	3	4	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	1	1	3	1	2	1	1	2	4	3	1	2	3	2
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	1	3	1	2	1	2	4	1	4	1	4	4	2	2	2
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	2	2	1	2	4	2	2	4	2	1	2	1	3	2	3
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	2	3	2	3	1	3	1	1	1	3	4	4	1	1	2
Question	121	122	123	124											
Answer	2	1	1	1											

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Two identical long conducting wires AOB and COD are placed at right angle to each other, with one above other such that 'O' is their common point for the two. The wires carry I_1 and I_2 currents respectively. Point 'P' is lying at distance 'd' from 'O' along a direction perpendicular to the plane containing the wires. The magnetic field at the point 'P' will be :-

(1) $\frac{\mu_0}{2\pi d}\left(I_1/I_2\right)$ (2) $\frac{\mu_0}{2\pi d}(I_1+I_2)$

(3) $\frac{\mu_0}{2\pi d}(I_1^2-I_2^2)$ (4) $\frac{\mu_0}{2\pi d}(I_1^2+I_2^2)^{1/2}$

PME125

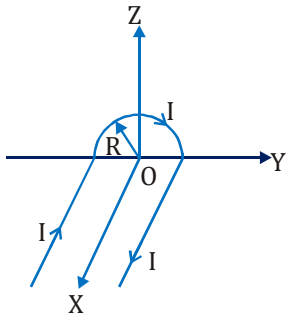
AIPMT 2015

2. An electron moving in a circular orbit of radius r makes n rotations per second. The magnetic field produced at the centre has magnitude :

(1) Zero (2) $\frac{\mu_0 n^2 e}{r}$ (3) $\frac{\mu_0 ne}{2r}$ (4) $\frac{\mu_0 ne}{2\pi r}$

PME126

3. A wire carrying current I has the shape as shown in adjoining figure. Linear parts of the wire are very long and parallel to X-axis while semi-circular portion of radius R is lying in Y-Z plane. Magnetic field at point O is :



(1) $\vec{B}=-\frac{\mu_0}{4\pi}\frac{I}{R}\left(\pi\hat{i}-2\hat{k}\right)$

(2) $\vec{B}=-\frac{\mu_0}{4\pi}\frac{I}{R}\left(\pi\hat{i}+2\hat{k}\right)$

(3) $\vec{B}=\frac{\mu_0}{4\pi}\frac{I}{R}\left(\pi\hat{i}-2\hat{k}\right)$

(4) $\vec{B}=\frac{\mu_0}{4\pi}\frac{I}{R}\left(\pi\hat{i}+2\hat{k}\right)$

PME127

Re-AIPMT 2015

4. A rectangular coil of length 0.12m and width 0.1m having 50 turns of wire is suspended vertically in a uniform magnetic field of strength 0.2 Weber/m^2. The coil carries a current of 2A. If the plane of the coil is inclined at an angle of 30° with the direction of the field, the torque required to keep the coil in stable equilibrium will be :

(1) 0.12 Nm (2) 0.15 Nm
(3) 0.20 Nm (4) 0.24 Nm

PME128

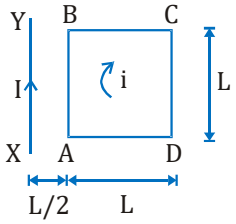
5. A proton and an alpha particle both enter a region of uniform magnetic field, B, moving at right angles to the field B. If the radius of circular orbits for both the particles is equal and the kinetic energy acquired by proton is 1 MeV, the energy acquired by the alpha particle will be :-

(1) 1 MeV (2) 4 MeV
(3) 0.5 MeV (4) 1.5 MeV

PME129

NEET-I 2016

6. A square loop ABCD carrying a current i, is placed near and coplanar with a long straight conductor XY carrying a current I, the net force on the loop will be :-



(1) $\frac{2\mu_0 Ii}{3\pi}$ (2) $\frac{\mu_0 Ii}{2\pi}$

(3) $\frac{2\mu_0 IiL}{3\pi}$ (4) $\frac{\mu_0 IiL}{2\pi}$

PME130

7. The magnetic susceptibility is negative for :

(1) diamagnetic material only
(2) paramagnetic material only
(3) ferromagnetic material only
(4) paramagnetic and ferromagnetic materials

PME131

8. A long straight wire of radius a carry a steady current I. The current is uniformly distributed over its cross-section. The ratio of the magnetic fields B and B', at radial distances $\frac{a}{2}$ and 2a respectively, from the axis of the wire is :

(1) $\frac{1}{4}$ (2) $\frac{1}{2}$ (3) 1 (4) 4

PME132

NEET-II 2016

9. A long wire carrying a steady current is bent into a circular loop of one turn. The magnetic field at the centre of the loop is B. It is then bent into a circular coil of n turns. The magnetic field at the centre of this coil of n turns will be :-

(1) 2nB (2) $2n^2B$
(3) nB (4) n^2B

PME133

10. A bar magnet is hung by a thin cotton thread in a uniform horizontal magnetic field and is in equilibrium state. The energy required to rotate it by 60° is W. Now the torque required to keep the magnet in this new position is :-

(1) $\frac{\sqrt{3}W}{2}$ (2) $\frac{2W}{\sqrt{3}}$
(3) $\frac{W}{\sqrt{3}}$ (4) $\sqrt{3}$ W

PME134

11. An electron is moving in a circular path under the influence of a transverse magnetic field of 3.57×10^{-2} T. If the value of e/m is 1.76×10^{11} C/kg, the frequency of revolution of the electron is :-

(1) 62.8 MHz (2) 6.28 MHz
(3) 1 GHz (4) 100 MHz

PME135

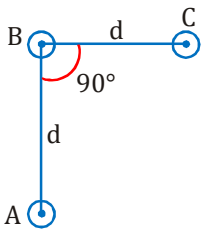
NEET(UG) 2017

12. A 250 turn rectangular coil of length 2.1 cm and width 1.25 cm carries a current of 85 μA and subjected to magnetic field of strength 0.85 T. Work done for rotating the coil by 180° against the torque is:-

(1) 4.55 μJ (2) 2.3 μJ
(3) 1.15 μ J (4) 9.1 μ J

PME136

13. An arrangement of three parallel straight wires placed perpendicular to plane of paper carrying same current 'I along the same direction is shown in fig. Magnitude of force per unit length on the middle wire 'B' is given by :-



(1) $\frac{2\mu_0 i^2}{\pi d}$ (2) $\frac{\sqrt{2}\mu_0 i^2}{\pi d}$
(3) $\frac{\mu_0 i^2}{\sqrt{2}\pi d}$ (4) $\frac{\mu_0 i^2}{2\pi d}$

PME137

NEET(UG) 2018

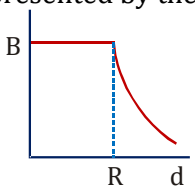
14. A thin diamagnetic rod is placed vertically between the poles of an electromagnet. When the current in the electromagnet is switched on, then the diamagnetic rod is pushed up, out of the horizontal magnetic field. Hence the rod gains gravitational potential energy. The work required to do this comes from

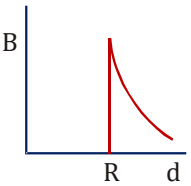
(1) the current source
(2) the magnetic field
(3) the lattice structure of the material of the rod
(4) the induced electric field due to the changing magnetic field

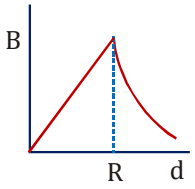
PME138

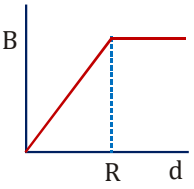
NEET(UG) 2019

15. A cylindrical conductor of radius R is carrying a constant current. The plot of the magnitude of the magnetic field, B with the distance d, from the centre of the conductor, is **correctly** represented by the figure :

(1) 

(2) 

(3) 

(4) 

PME139

16. Ionized hydrogen atoms and α-particles with same momenta enters perpendicular to a constant magnetic field B. The ratio of their radii of their paths $r_H : r_\alpha$ will be

(1) 2 :1 (2) 1 : 2

(3) 4 : 1 (4) 1 : 4

PME140

NEET(UG) 2019 (Odisha)

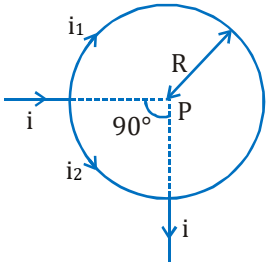
17. Two toroids 1 and 2 have total number of turns 200 and 100 respectively with average radii 40 cm and 20 cm respectively. If they carry same current i, then the ratio of the magnetic fields along the two is :

(1) 1 : 1 (2) 4 : 1

(3) 2 : 1 (4) 1 : 2

PME141

18. A straight conductor carrying current i splits into two parts as shown in the figure. The radius of the circular loop is R. The total magnetic field at the centre P of the loop is :



(1) Zero

(2) $3\mu_0 i/32$ R, outward

(3) $3\mu_0 i/32R$, inward

(4) $\mu_0 i/2R$, inward

PME142

NEET(UG) 2020

19. An iron rod of susceptibility 599 is subjected to a magnetising field of 1200 A m^{-1}. The permeability of the material of the rod is :

($\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1})

(1) $2.4\pi \times 10^{-7}$ T m A^{-1}

(2) $2.4\pi \times 10^{-4}$ T m A^{-1}

(3) 8.0×10^{-5} T m A^{-1}

(4) $2.4\pi \times 10^{-5}$ T m A^{-1}

PME143

20. A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is :

($\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1})

(1) 3.14×10^{-5} T (2) 6.28×10^{-4} T

(3) 3.14×10^{-4} T (4) 6.28×10^{-5} T

PME144

NEET(UG) 2020 (COVID-19)

21. A wire of length L metre carrying a current of I ampere is bent in the form of a circle. Its magnetic moment is,

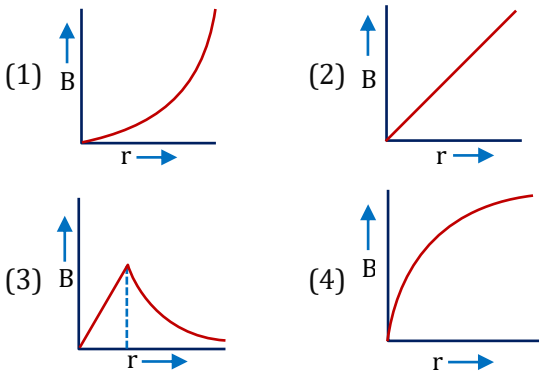
(1) $IL^2/4$ A m^2 (2) $I\pi L^2/4$ A m^2

(3) $2IL^2/\pi$ A m^2 (4) $IL^2/4\pi$ A m^2

PME145

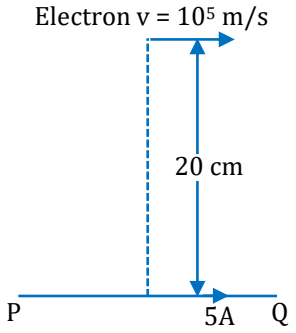
NEET(UG) 2021

22. A thick current carrying cable of radius 'R' carries current 'I' uniformly distributed across its cross-section. The variation of magnetic field B(r) due to the cable with the distance 'r' from the axis of the cable is represented by :



PME146

23. An infinitely long straight conductor carries a current of 5 A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant.



(1) 4×10^{-20} N (2) $8\pi \times 10^{-20}$ N

(3) $4\pi \times 10^{-20}$ N (4) 8×10^{-20} N

PME147

24. In the product

$\vec{F} = q\left(\vec{v} \times \vec{B}\right) = q\vec{v} \times \left(B\hat{i} + B\hat{j} + B_0\hat{k}\right)$

For q = 1 and $\vec{v} = 2\hat{i} + 4\hat{j} + 6\hat{k}$

and $\vec{F} = 4\hat{i} - 20\hat{j} + 12\hat{k}$

What will be the complete expression for $\vec{B}$?

(1) $-8\hat{i} - 8\hat{j} - 6\hat{k}$ (2) $-6\hat{i} - 6\hat{j} - 8\hat{k}$

(3) $8\hat{i} + 8\hat{j} - 6\hat{k}$ (4) $6\hat{i} + 6\hat{j} - 8\hat{k}$

PME148

25. A uniform conducting wire of length 12a and resistance 'R' is wound up as a current carrying coil in the shape of,

(i) an equilateral triangle of side 'a'.

(ii) a square of side 'a'.

The magnetic dipole moments of the coil in each case respectively are:

(1) $\sqrt{3}\,Ia^2$ and $3\,Ia^2$ (2) $3\,Ia^2$ and Ia^2

(3) $3\,Ia^2$ and $4\,Ia^2$ (4) $4\,Ia^2$ and $3\,Ia^2$

PME149

NEET(UG) 2022

26. Given below are two statements:

Statement I :

Biot-Savart's law gives us the expression for the magnetic field strength of an infinitesimal current element (Idl) of a current carrying conductor only.

Statement II :

Biot-Savart's law is analogous to Coulomb's inverse square law of charge q, with the former being related to the field produced by a scalar source, Idl while the latter being produced by a vector source, q.

In light of above statement choose the most appropriate answer from the options given below:

(1) Both statement I and Statement II are incorrect

(2) Statement I is correct and Statement II is incorrect

(3) Statement I is incorrect and Statement II is correct

(4) Both statement I and Statement II are correct

PME150

27. From Ampere's circuital law for a long straight wire of circular cross-section carrying a steady current, the variation of magnetic field in the inside and outside region of the wire is:

(1) a linearly increasing function of distance upto the boundary of the wire and then linearly decreasing for the outside region.

(2) a linearly increasing function of distance r upto the boundary of the wire and then decreasing one with 1/r dependence for the outside region.

(3) a linearly decreasing function of distance upto the boundary of the wire and then a linearly increasing one for the outside region.

(4) uniform and remains constant for both the regions.

PME151

RE-NEET (UG) 2022

28. A closely packed coil having 1000 turns has an average radius of 62.8 cm. If current carried by the wire of the coil is 1 A, the value of magnetic field produced at the centre of the coil will be (permeability of free space = $4\pi \times 10^{-7}$ H/m) nearly:

(1) 10^{-1} T (2) 10^{-2} T

(3) 10^{2} T (4) 10^{-3} T

PME152

29. The shape of the magnetic field lines due to an infinite long, straight current carrying conductor is :

(1) a straight line (2) circular

(3) elliptical (4) a plane

PME153

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30. Two very long, straight, parallel conductors A and B carry current of 5A and 10A respectively and are at a distance of 10 cm from each other. The direction of current in two conductors is same. The force acting per unit length between two conductors is:
($\mu_0 = 4\pi \times 10^{-7}$ SI unit)
(1) 2×10^{-4} Nm^{-1} and is attractive
(2) 2×10^{-4} Nm^{-1} and is repulsive
(3) 1×10^{-4} Nm^{-1} and is attractive
(4) 1×10^{-4} Nm^{-1} and is repulsive

PME154

31. The magnetic field on the axis of a circular loop of radius 100 cm carrying current $I = \sqrt{2}$ A, at point 1 m away from the centre of the loop is given by:
(1) 3.14×10^{-7} T (2) 6.28×10^{-7} T
(3) 3.14×10^{-4} T (4) 6.28×10^{-4} T

PME155

NEET (UG) 2023

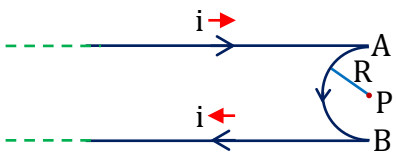
32. The net magnetic flux through any closed surface is :
(1) Positive (2) Infinity
(3) Negative (4) Zero

PME156

33. A wire carrying a current I along the positive x-axis has length L. It is kept in a magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} - 4\hat{k})$T. The magnitude of the magnetic force acting on the wire is :
(1) $\sqrt{5}$ IL (2) 5 IL
(3) $\sqrt{3}$ IL (4) 3 IL

PME157

34. A very long conducting wire is bent in a semi-circular shape from A to B as shown in figure. The magnetic field at point P for steady current configuration is given by :



(1) $\frac{\mu_0 i}{4R}$ pointed away from the page
(2) $\frac{\mu_0 i}{4R}\left[1-\frac{2}{\pi}\right]$ pointed away from page
(3) $\frac{\mu_0 i}{4R}\left[1-\frac{2}{\pi}\right]$ pointed into the page
(4) $\frac{\mu_0 i}{4R}$ pointed into the page

PME158

RE-NEET (UG) 2023

35. A long straight wire of length 2 m and mass 250 g is suspended horizontally in a uniform horizontal magnetic field of 0.7 T. The amount of current flowing through the wire will be ($g = 9.8$ ms^{-2}) :
(1) 2.45 A (2) 2.25 A
(3) 2.75 A (4) 1.75 A

PME159

36. A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected in the region such that its velocity is pointed along the direction of fields, then the electron:
(1) will turn towards right of direction of motion
(2) will turn towards left of direction of motion
(3) speed will decrease
(4) speed will increase

PME160

NEET (UG) 2024

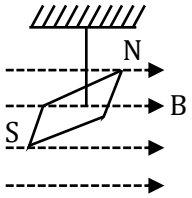
37. Match **List-I** with **List-II**

List-I (Material)	**List-II (Susceptibility(χ))**
A. Diamagnetic	I. $\chi = 0$
B. Ferromagnetic	II. $0 > \chi \geq -1$
C. Paramagnetic	III. $\chi >> 1$
D. Non-Magnetic	IV. $0 < \chi < \varepsilon$ (a small positive number)

Choose the correct answer from the options given below:
(1) A–II, B–III, C–IV, D–I
(2) A–II, B–I, C–III, D–IV
(3) A–III, B–II, C–I, D–IV
(4) A–IV, B–III, C–II, D–I

PME161

38. In a uniform magnetic field of 0.049 T, a magnetic needle performs 20 complete oscillations in 5 seconds as shown. The moment of inertia of the needle is 9.8×10^{-6} kg m^2. If the magnitude of magnetic moment of the needle is $x \times 10^{-5}$ Am^2; then the value of 'x' is :



(1) $5\ \pi^2$ (2) $128\ \pi^2$

(3) $50\ \pi^2$ (4) $1280\ \pi^2$

PME162

39. A tightly wound 100 turns coil of radius 10 cm carries a current of 7 A. The magnitude of the magnetic field at the centre of the coil is (Take permeability of free space as $4\pi \times 10^{-7}$ SI units) :

(1) 44 mT (2) 4.4 T

(3) 4.4 mT (4) 44 T

PME163

40. An iron bar of length L has magnetic moment M. It is bent at the middle of its length such that the two arms make an angle 60° with each other. The magnetic moment of this new magnet is :

(1) M (2) $\frac{M}{2}$

(3) 2M (4) $\frac{M}{\sqrt{3}}$

PME164

41. A sheet is placed on a horizontal surface in front of a strong magnetic pole. A force is needed to :

A. hold the sheet there if it is magnetic.

B. hold the sheet there if it is non-magnetic.

C. move the sheet away from the pole with uniform velocity if it is conducting.

D. move the sheet away from the pole with uniform velocity if it is both, non-conducting and non-polar.

Choose the correct statement(s) from the options given below:

(1) B and D only

(2) A and C only

(3) A, C and D only

(4) C only

PME165

RE-NEET (UG) 2024

42. The magnetic potential energy, when a magnetic bar of magnetic moment $\vec{m}$ is placed perpendicular to the magnetic field $\vec{B}$ is :

(1) $-\frac{mB}{2}$ (2) zero

(3) –m B (4) m B

PME166

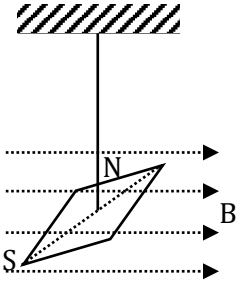
43. The incorrect relation for a diamagnetic material (all the symbols carry their usual meaning and ε is a small positive number) is :

(1) $\mu < \mu_0$ (2) $0 \leq \mu_r < 1$

(3) $-1 \leq \chi < 0$ (4) $1 < \mu_r < 1 + \varepsilon$

PME167

44. The magnetic moment and moment of inertia of a magnetic needle as shown are, respectively 1.0×10^{-2} A m^2 and $\frac{10^{-6}}{\pi^2}$ kg m^2. If it completes 10 oscillations in 10 s, the magnitude of the magnetic field is :



(1) 0.4 T (2) 4 T

(3) 0.4 mT (4) 4 mT

PME168

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45. The magnetic moment of an iron bar is M. It is now bent in such a way that it forms an arc section of a circle subtending an angle of 60° at the centre. The magnetic moment of this arc section is:

(1) $\frac{3M}{\pi}$ (2) $\frac{4M}{\pi}$

(3) $\frac{M}{\pi}$ (4) $\frac{2M}{\pi}$

PME169

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	2	3	1	1	1	3	4	4	3	4	3	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	1	1	2	2	4	3	4	2	1	2	2	4	2	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	4	2	2	4	3	1	4	3	2	2	2	4	3	1

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Exercise – III (Analytical Questions) Master Your Understanding

1. Two circular coils of radii 5 cm and 10 cm carry currents of 2 A. The coils have 50 and 100 turns respectively and are placed in such a way that their planes as well as their centres coincide. Magnitude of magnetic field at the common centre of coils is
(a) $8\pi \times 10^{-4}$ T if currents in the coils are in same sense
(b) $4\pi \times 10^{-4}$ T if currents in the coils are in opposite sense
(c) zero if currents in the coils are in opposite sense
(d) $8\pi \times 10^{-4}$ T if currents in the coils are in opposite sense
(1) b and d are correct (2) a and c are correct
(3) only a are correct (4) only d are correct

PME170

2. When a current carrying coil is placed in a uniform magnetic field with its magnetic moment anti-parallel to the field
(a) Torque on it is maximum
(b) Torque on it is zero
(c) Potential energy is maximum
(d) Dipole is in unstable equilibrium
(1) b, c, d are correct (2) only b is correct
(3) a and c are correct (4) none

PME171

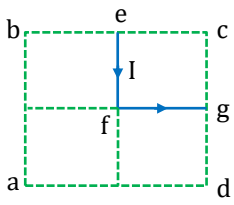
3. If a long cylindrical conductor carries a steady current parallel to its length
(a) the electric field along the axis is zero
(b) the magnetic field along the axis is zero
(c) the magnetic field outside the conductor is zero
(d) the electric field outside the conductor is zero
(1) b and d are correct (2) a and c are correct
(3) only d is correct (4) none

PME172

4. A charged particle revolves in circular path in uniform transverse magnetic field after being accelerated by a potential difference of V volt. If V is doubled, then choose the correct options.
(a) Kinetic energy of particle will become two times
(b) Radius of circular path will become two times
(c) Radius of circular path will become $\sqrt{2}$ times
(d) angular velocity will remain unchanged
(1) a and b are correct
(2) a and c are correct
(3) a, c, d are correct
(4) none

PME173

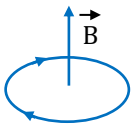
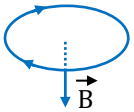
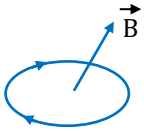
5. abcd is a square. There is a current I in wire efg as shown. Choose the correct options :



(a) Net magnetic field at a is inwards
(b) Net magnetic field at f is zero
(c) Net magnetic field at c is zero
(d) Net magnetic field at g is zero
(1) a and b are correct
(2) a, b, c are correct
(3) b and c are correct
(4) none

PME174

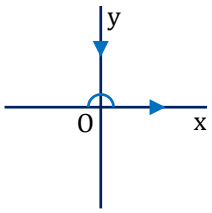
6. In column I, a current carrying loop and a uniform magnetic field are shown. Match this with column II.

	Column-I		Column-II
(a)		(p)	$F \neq 0$
(b)		(q)	Maximum torque
(c)		(r)	Minimum potential energy
(d)		(s)	Positive potential energy

(1) a-s, b-p, c-r, d-q (2) a-s, b-q, c-r, d-s
(3) a-s, b-r, c-p, d-q (4) a-q, b-p, c-r, d-s

PME175

7. Equal currents are flowing in two infinitely long separate wires lying along x and y-axes in the directions shown in figure. Match the following two columns.



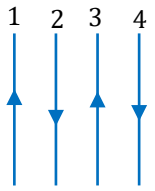
	Column-I		Column-II
(a)	Magnetic field at (a, a)	(p)	along positive y-axis
(b)	Magnetic field at (–a. – a)	(q)	along positive z-axis
(c)	Magnetic field at (a, –a)	(r)	along negative z-axis
(d)	Magnetic field at (–a, a)	(s)	zero

(1) a-p, b-q, c-s, d-r (2) a-q, b-r, c-s, d-s

(3) a-r, b-s, c-p, d-r (4) a-s, b-p, c-q, d-s

PME176

8. Equal currents are flowing in four infinitely long wires. Distance between two wires is same and directions of currents are shown in figure. Match the following two columns.



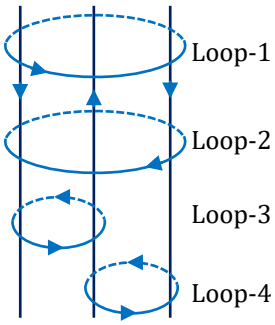
	Column-I		Column-II
(a)	Force on wire-1	(p)	Inwards
(b)	Force on wire-2	(q)	Leftwards
(c)	Force on wire-3	(r)	Rightwards
(d)	Force on wire-4	(s)	Zero

(1) a-q, b-r, c-q, d-r (2) a-r, b-q, c-p, d-r

(3) a-q, b-p, c-q, d-p (4) a-p, b-r, c-p, d-r

PME177

9. Three wires are carrying same constant current i in different directions. Four loops enclosing the wires in different manners are shown in figure. The direction of $d\vec{\ell}$ is shown in the figure.



	Column-I		Column-II
a	Along close Loop-1	p	$\oint \vec{B} d\vec{\ell} = \mu_0 i$
b	Along close Loop-2	q	$\oint \vec{B} \cdot d\vec{\ell} = -\mu_0 i$
c	Along close Loop-3	r	$\oint \vec{B} \cdot d\vec{\ell} = 0$
d	Along close Loop-4	s	$\oint \vec{B} \cdot d\vec{\ell} = 2\mu_0 i$

(1) a – q, b – p, c – r, d – r

(2) a – p, b – r, c – s, d – s

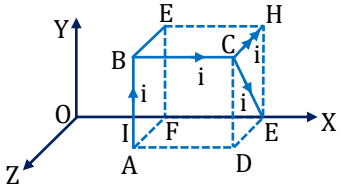
(3) a – r, b – p, c – s, d – p

(4) a – r, b – q, c – q, d – q

PME178

10. Side of cube is 'a' and wire AB, BC, CH and CE are insulated with each other. If each having current i, then match the following. External magnetic field applied along positive x-axis as $\vec{B} = B_0 \hat{i}$

given that OF = a



	List-I (Magnitude of force due to external magnetic field only)		List-II
(a)	On AB	(p)	Zero
(b)	On BC	(q)	$\sqrt{2}iB_0a$
(c)	On CH	(r)	iB_0a
(d)	On CE	(s)	$2iB_0a$

(1) a – r, b – p, c – r, d – q
(2) a – p, b – p, c – s, d – q
(3) a – q, b – s, c – r, d – s
(4) a – s, b – r, c – d, d – q

PME179

11. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Force on a current carrying wire of length $\overrightarrow{d\ell}$ placed in magnetic field $\vec{B}$ is given by $\overrightarrow{dF} = i\overrightarrow{d\ell} \times \vec{B}$

Reason (R) : Net force on a current carrying loop in a non-uniform magnetic field must be non-zero.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PME180

12. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The ferromagnetic substances obey Curie's law.

Reason (R) : At Curie point ferromagnetic substances start behaving as paramagnetic substances.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PME181

13. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The surface integral of magnetic field over any closed surface is always zero.

Reason (R) : Magnetic poles always exists in pair.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PME182

14. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A moving charge produces magnetic field.

Reason (B) : A current element produces magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PME183

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15. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Parallel wires carrying currents in same direction attracts each other due to magnetic force.

Reason (R) : Two electron beams moving parallel to each other repel each other due to electric force.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PME184

16. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Ampere's circuital law is not suitable for a finite straight wire carrying current I.

Reason (R) : One cannot calculate magnetic field due to a finite straight current carrying wire by using Biot Savart Law.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PME185

17. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The diamagnetic substances do not obey Curie's law.

Reason (R) : At Curie point ferromagnetic substances start behaving as paramagnetic substances.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PME186

18. **Statement–1** : In an isolated conductor, free electrons keep on moving but no net magnetic force acts on a conductor in a magnetic field.

Statement–2 : In a conductor, the average velocity of thermal motion of electrons is zero. Hence no current flows through the conductor.

(1) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1.

(2) Statement–1 is True, Statement–2 is True ; Statement–2 is not a correct explanation for Statement–1.

(3) Statement–1 is True, Statement–2 is False.

(4) Statement–1 is False, Statement–2 is True.

PME187

19. A particle with charge Q is moving in fields of combination given below. Match for the most correct possible path/s. Then match the following :-

	Column-I		Column-II
(A)	Uniform $\vec{E}$	(P)	Straight line
(B)	Uniform $\vec{B}$	(Q)	Parabola
		(R)	Circular

Options :-

(1) A $\rightarrow$ PQ, B $\rightarrow$ PR

(2) A $\rightarrow$ P, B $\rightarrow$ R

(3) A $\rightarrow$ Q, B $\rightarrow$ PR

(4) A $\rightarrow$ P, B $\rightarrow$ PQ

PME188

20. **Match the column I with column II -**

	Column-I		Column-II
(A)	Paramagnetic substance	(P)	$\chi_m = -1$
(B)	Diamagnetic substance	(Q)	$\chi_m < 0$
(C)	Super conductor	(R)	$\chi_m >> 1$
(D)	Ferromagnetic substance	(S)	$\mu_r > 1$

Options :-

(1) A → S, B → Q, C → P, D → R

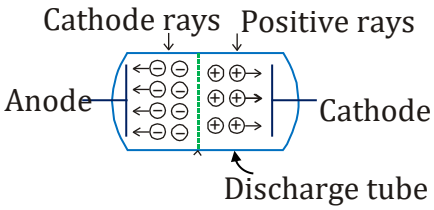
(2) A → R, B → P, C → Q, D → S

(3) A → S, B → P, C → Q, D → R

(4) A → R, B → Q, C → P, D → S

PME189

21. A magnetic field is applied at perpendicular to the axis of cylindrical discharge tube then cathode and positive rays deflected in –



(1) Same direction.

(2) Opposite direction.

(3) Mutually perpendicular direction.

(4) No deflection for both.

PME190

22. A square current carrying loop of side 'a' produced magnetic field at its centre is 'B'. Find magnetic moment of this loop in terms of 'B'.

(1) $\frac{\pi a^3 B}{2\sqrt{2}\mu_0}$ (2) $\frac{8\pi a^3 B}{\mu_0}$

(3) $\frac{\pi a^2 B}{2\sqrt{2}\mu_0}$ (4) $\frac{8\pi a^2 B}{\mu_0}$

PME191

23. Two long parallel wires carry i and 2i current in same direction. Magnetic field just between the wires is 'B'. If 2i current is switched off then magnetic field at the same point is :-

(1) 2B (2) B

(3) B/2 (4) $\sqrt{2}$ B

PME192

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	1	3	1	2	2	1	1	1	3	4	1	2	2
Question	16	17	18	19	20	21	22	23							
Answer	3	2	1	1	1	1	1	2							

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Exercise - I (Conceptual Questions) Build Up Your Understanding

MAGNETIC FLUX

1. 'SI' unit of magnetic flux is :-
(1) ampere/meter^2 (2) weber
(3) gauss (4) oersted

PEI001

2. A square coil of 0.01 m^2 area is placed perpendicular to the uniform magnetic field of 10^3 weber/m^2. The magnetic flux linked with the coil is :-
(1) 10 weber (2) 10^{-5} weber
(3) Zero (4) 100 weber

PEI002

FARADAY LAW & LENZ'S LAW

3. According to Faraday's Laws of electromagnetic induction:
(1) The direction of the induced current is such that it opposes itself
(2) The induced emf in the coil is proportional to the rate of change of magnetic flux associated with it
(3) The direction of induced emf is such that it opposes itself
(4) None of the above

PEI003

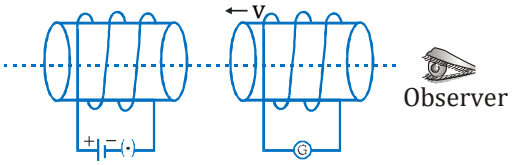
4. A coil having an area of 2m² is placed in a magnetic field which changes from 1 Weber/m² to 4 Weber/m² in 2 seconds. The e.m.f. induced in the coil will be :–
(1) 4 volt (2) 3 volt (3) 2 volt (4) 1 volt

PEI004

5. Magnetic field through a coil is changed with respect to time and an emf is induced in it. Select the incorrect statement regarding induced emf in coil :-
(1) Coil may be made up with wood
(2) Coil may be connected with an open circuit
(3) Coil must be of conducting nature
(4) Induced emf does not depends upon resistance of the coil

PEI005

6. The current flows in a circuit as shown below. If a second circuit is brought near the first circuit then the current in the second circuit will be :-



(1) Clock wise
(2) Anti clock wise
(3) Depending on the value of R_G
(4) None of the above

PEI006

7. A coil of resistance 10Ω and 1000 turns have the magnetic flux line of 5.5×10^{-4} Wb. If the magnetic flux changed to 5×10^{-4} Wb. in 0.1 sec, then the induced charge in coil is :–
(1) 50 μC (2) 5 μC
(3) 2 μC (4) 20 μC

PEI007

8. One coil of resistance 40Ω is connected to a galvanometer of 160Ω resistance. The coil has radius 6mm and turns 100. This coil is placed between the poles of a magnet such that magnetic field is perpendicular to coil. If coil is dragged out then the charge through the galvanometer is 32μC. The magnetic field is:-
(1) 6.55 T (2) 5.66 T
(3) 0.655 T (4) 0.566 T

PEI008

9. A square loop of side 22 cm is changed to a circle in time 0.4 s. The magnetic field present is 0.2 T. The emf induced is :-
(1) –6.6 mV (2) –13.2 mV
(3) +6.6 mV (4) +13.2 mV

PEI009

10. The magnetic flux in a coil of 100 turns increases by 12×10^3 Maxwell in 0.2 s due to the motion of a magnet. The emf induced in the coil will be:-
(1) 0.6 mV (2) 0.6 V
(3) 6 V (4) 60 V

PEI010

11. A closed coil consists of 500 turns on a rectangular frame of area 4.0 cm^2 and has a resistance of 50 ohms. The coil is kept with its plane perpendicular to a uniform magnetic field of 0.2 wb/m^2, the amount of charge flowing through the coil if it is turned over (rotated through 180°):-

(1) 1.6×10^{-3} C (2) 16×10^{-3} C
(3) 0.16×10^{-3} C (4) 160×10^{-3} C

PEI011

12. A coil of mean area 500 cm^2 and having 1000 turns is held perpendicular to a uniform field of 0.4 gauss. The coil is turned through 180° in $\frac{1}{10}$ second. The average induced e.m.f. :–

(1) 0.04 V (2) 0.4 V
(3) 4 V (4) 0.004 V

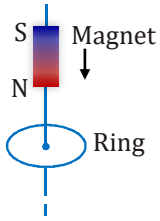
PEI012

13. An emf is induced in a coil. The linking magnetic flux

(1) Must be decreasing
(2) Must be increasing
(3) Must remain constant
(4) Can be either increasing or decreasing

PEI013

14. Consider a metal ring kept on a horizontal plane. A bar magnet is held above the ring with its length along the axis of the ring. If the magnet is dropped freely the acceleration of the falling magnet is (g is acceleration due to gravity) :-



(1) More than g
(2) Equal to g
(3) Less than g
(4) Depend on mass of magnet

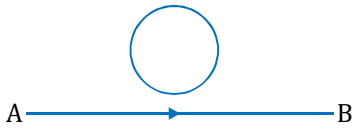
PEI014

15. An electron beam is moving near to a conducting loop then the induced current in the loop :-

(1) clockwise (2) anticlockwise
(3) both (4) no current

PEI015

16. The current flows from A to B as shown in the figure. The direction of the induced current in the loop is : -



(1) clockwise (2) anticlockwise
(3) straight line (4) none of these

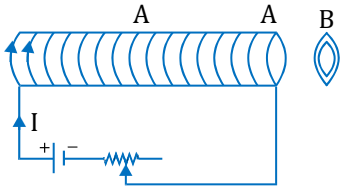
PEI016

17. Faraday law represents :-

(1) relation between I and B
(2) relation between magnetic force and magnetic field
(3) relation between e.m.f and rate of change of flux
(4) none of these

PEI017

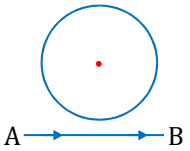
18. An aluminium ring B faces an electromagnet A. The current I through A can be altered. Then which of the following statement is correct :-



(1) If I decreases A will repel B
(2) Whether I increases or decreases, B will not experience any force
(3) If I increases, A will repel B
(4) If I increases, A will attract B

PEI018

19. A charge particle moves along the line AB, which lies in the same plane of a circular loop of conducting wire as shown in the fig. Then :-



(1) No current will be induced in the loop
(2) The current induced in the loop will change its direction as the charged particle passes by
(3) The current induced will be anticlockwise
(4) The current induced, will be clockwise

PEI019

20. The magnetic flux through a circuit of resistance R changes by an amount $\Delta\phi$ in a time Δt. The total quantity of electric charge Q that passes any point in the circuit during the time Δt is represented by:–

(1) $Q = \frac{\Delta\phi}{R}$ (2) $Q = \frac{\Delta\phi}{\Delta t}$

(3) $Q = R \cdot \frac{\Delta\phi}{\Delta t}$ (4) $Q = \frac{1}{R} \cdot \frac{\Delta\phi}{\Delta t}$

PEI020

21. If number of turns of $70cm^2$ coil is 200 and it is placed in a magnetic field of $0{\cdot}8\ Wb/m^2$ which is perpendicular to the plane of coil and it is rotated through an angle 180° in 0·1 sec, then induced emf in coil :-

(1) 11.2 V (2) 1.12 V

(3) 22.4 V (4) 2.24 V

PEI021

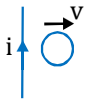
22. A circular loop of radius r is placed in a region where magnetic field increases with respect to time as B(t) = at then induced emf in coil :-

(1) $\pi r^2 a$ (2) $3\pi r^2 a$

(3) $2\pi r^2 a$ (4) $4\pi r^2 a$

PEI022

23. A circular loop of radius r is moved away from a current carrying wire then induced current in circular loop will be :-



(1) Clock wise (2) Anti clockwise

(3) Not induced (4) None of them

PEI023

SELF INDUCTION AND L–R DC CIRCUIT

24. When the current through a solenoid increases at a constant rate, the induced current

(1) is a constant and is in the direction of the inducing current

(2) is a constant and is opposite to the direction of the inducing current

(3) increase with time and is in the direction of the inducing current

(4) increase with time and opposite to the direction of the inducing current

PEI024

25. A solenoid of 10 henry inductance and 2 ohm resistance, is connected to a 10 volt battery. In how much time the magnetic energy will reach to 1/4th of the maximum value?

(1) 3.5 sec (2) 2.5 sec

(3) 5.5 sec (4) 7.5 sec

PEI025

26. An inductance coil have the time constant 4 sec, if it is cut into two equal parts and connected parallel then new time constant of the circuit :-

(1) 4 sec (2) 2 sec

(3) 1 sec (4) 0.5 sec

PEI026

27. Which statement is correct from following –

(a) Inductor store energy in the form of magnetic field

(b) Capacitor store energy in the form of electric field

(c) Inductor store energy in the form of electric and magnetic field both

(d) Capacitor store energy in the form of electric and magnetic field both

(1) a, b (2) a, c (3) b, d (4) b, c

PEI027

28. If a current of 2A give rise a magnetic flux of 5×10^{-5} weber/turn through a coil having 100 turns, then the magnetic energy stored in the medium surrounding by the coil is :-

(1) 5 joule (2) 5×10^{-7} joule

(3) 5×10^{-3} joule (4) 0.5 joule

PEI028

29. For a solenoid keeping the turn density constant its length makes halved and its cross section radius is doubled then the inductance of the solenoid increased by :–

(1) 200% (2) 100%

(3) 800% (4) 700%

PEI029

30. A constant current I is maintained in a solenoid. Which of the following quantities will increase if an iron rod is inserted in the solenoid along its axis:-

(a) Magnetic field at the centre

(b) Magnetic flux linked with the solenoid

(c) Self inductance of the solenoid

(d) Rate of Joule heating

(1) a, b, c (2) c, d

(3) a, b (4) Only b

PEI030

31. The inductance of a solenoid is 5H and its resistance is 5Ω. If it is connected to a 10 volt battery then time taken by the current to reach 9/10th of its maximum will be :-

(1) 4.0 s (2) 2.3 s (3) 1.4 s (4) 1.2 s

PEI031

32. An LR circuit with a battery is connected at t = 0. Which of the following quantities is not zero just after the connection :-

(a) Current in circuit

(b) Magnetic potential energy in the inductor

(c) Power delivered by the battery

(d) Emf induced in the inductor

(1) a, b (2) a, c

(3) c, d (4) Only d

PEI032

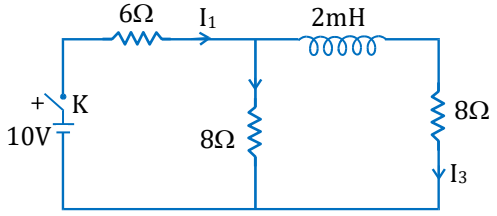
33. During 0.1s current in a coil increases from 1A to 1.5A. If inductance of this coil is 60 μH, induced current in external resistance of 600 μΩ is :-

(1) 1A (2) 4/3 A

(3) 2/3 A (4) 1/2 A

PEI033

34. In the circuit shown in figure what is the value of I_1 just after pressing the key K ?



(1) $\frac{5}{7}$ A

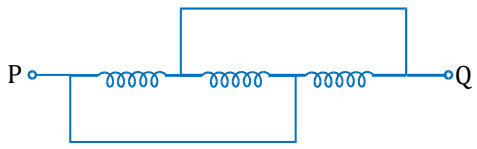
(2) $\frac{5}{11}$ A

(3) 1A

(4) None of the above

PEI034

35. Pure inductors each of inductance 3H are connected as shown. The equivalent induction of the circuit is :-



(1) 1H (2) 2H (3) 3H (4) 9H

PEI035

36. The time constant of an inductance coil is 2.0×10^{-3} s. When a 90 Ω resistance is joined in series, the time constant becomes 0.5×10^{-3} s. The inductance and resistance of the coil are :-

(1) 30mH ; 30Ω (2) 30mH ; 60Ω

(3) 60mH ; 30Ω (4) 60mH ; 60Ω

PEI036

37. A cylindrical iron core supports N turns. If a current I produces a magnetic flux ϕ across the core's cross section, then the magnetic energy is :-

(1) Iϕ (2) $\frac{1}{2}$ Iϕ

(3) $\frac{I^2\phi}{2}$ (4) $I^2\phi$

PEI037

38. The self-inductance of a toroid is

(N: no. of turns, r: radius of cross section, R_m: mean radius of toroid) :-

(1) $\frac{\mu_0 N^2 r^2}{2R_m}$ (2) $\frac{\mu_0 N^2 \pi r}{2R_m}$

(3) $\frac{\mu_0 N^2 r}{2R_m}$ (4) $\frac{\mu_0 N^2 r\pi}{R_m}$

PEI038

39. An inductance L and a resistance R are joined to a battery. After some time, battery is disconnected but L and R remains connected to the closed circuit. The current strength will be reduced to 37% of its initial value in :

(1) RL sec (2) R/L sec

(3) L/R sec (4) 1/LR sec

PEI039

40. Energy is stored in the choke coil in the form of:-

(1) Heat

(2) Electric field

(3) Magnetic field

(4) Electro-magnetic field

PEI040

41. When a current changes from 2A to 4A in 0.05 sec. in a coil, induced emf is 8 V. The self inductance of coil is :-
(1) 0.1 H (2) 0.2 H
(3) 0.4 H (4) 0.8 H
PEI041

42. An e.m.f. of 12 V is induced in a given coil when the current in it changes at the rate of 48 A/min. The inductance of the coil is :–
(1) 0.5 henry (2) 15 henry
(3) 1.5 henry (4) 9.6 henry
PEI042

43. Two conducting coils are placed co-axially now a cell is connected in one coil then they will :-
(1) attract to each other
(2) repel to each other
(3) both (1) & (2)
(4) they will not experience any force
PEI043

44. A coil of resistance 10Ω and an inductance 5H is connected to a 100V battery. Then energy stored in the coil is :-
(1) 125 erg (2) 125 J
(3) 250 erg (4) 250 J
PEI044

45. The energy density in magnetic field B is proportional to :-
(1) $\frac{1}{B}$ (2) $\frac{1}{B^2}$
(3) B (4) B^2
PEI045

46. Inductance of a solenoid is 3H and it consist of 500 turns. If number of turn make twice, then the value of self inductance becomes:-
(1) 1.5 H (2) 3 H
(3) 9 H (4) 12 H
PEI046

47. A coil of 40 henry inductance is connected in series with a resistance of 8 ohm and the combination is joined to the terminals of a 2 volt battery. The time constant of the circuit is :-
(1) $\frac{1}{5}$ sec (2) 40 sec
(3) 20 sec (4) 5 sec
PEI047

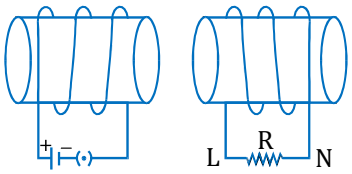
48. When current in a coil is reduced from 2A to 1A in 1ms, the induced emf is 5V. The inductance of coil is :
(1) 5 H (2) 5000 H
(3) 5 mH (4) 50 H
PEI048

49. A coil of inductance 300mH and resistance 2Ω is connected to a source of voltage 2V. The current reaches half of its steady state value in:-
(1) 0.3 s (2) 0.15 s
(3) 0.1 s (4) 0.05 s
PEI049

50. An ideal coil of 10H is connected in series with a resistance of 5Ω and a battery of 5V. 2 seconds after the connection is made, the current flowing in amperes in the circuit is :-
(1) e (2) e –1
(3) $(1-e^{-1})$ (4) (1–e)
PEI050

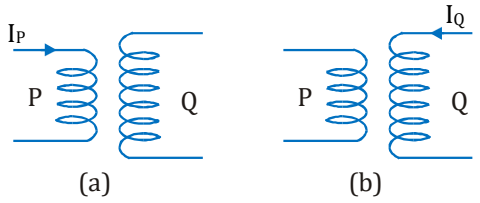
MUTUAL INDUCTION, TRANSFORMER AND EDDY CURRENTS

51. Two co–axial solenoids shown in figure. If key of primary suddenly opened then direction of instantaneous induced current in resistance 'R' which connected in secondary:



(1) L to N (2) N to L
(3) Alternating (4) Zero
PEI051

52. In figure (a) and figure (b) two air–cored solenoids P and Q have been shown. They are placed near each other. In figure (a), when I_P, the current in P, changes at the rate of $5As^{-1}$, an emf of 2 mV is induced in Q. The current in P is then switched off, and a current changing at $2A\ s^{-1}$ is fed through Q as shown in diagram. What emf will be induced in P:-



(1) 8×10^{-4} V (2) 2×10^{-8} V
(3) 5×10^{-3} V (4) 8×10^{-2} V

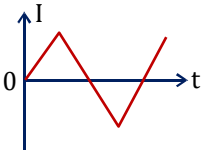
PEI052

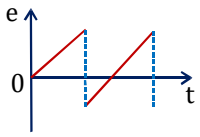
53. A small square loop of wire of side ℓ is placed inside a large square loop of wire of side L(L >> ℓ). The loops are coplanar and their centres coincide. The mutual inductance of the system is proportional to:-

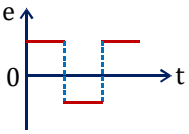
(1) $\frac{\ell}{L}$ (2) $\frac{\ell^2}{L}$
(3) $\frac{L}{\ell}$ (4) $\frac{L^2}{\ell}$

PEI053

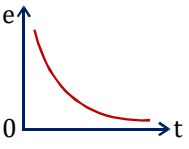
54. A current time curve is shown in the following diagram. This type of current is passed in the primary coil of transformer. The nature of induced emf in the secondary coil will be :-



(1) 

(2) 

(3) 

(4) 

PEI054

55. Two coil have a mutual inductance 0.005 H. The current changes in first coil according to equation $I = I_0 \sin\omega t$, where I_0 = 2A and $\omega = 100\pi$ rad/s. The maximum value of induced emf in second coil is

(1) 4π V (2) 3π V
(3) 2π V (4) π V

PEI055

56. The electric power is transferred to a far distance at high potential because:-
(1) It stops the wire theft
(2) To minimise power loss
(3) Generator gives only high potential
(4) Electric power is transferred early due to high potential

PEI056

57. If primary winding of a transformer were connected to a battery, the current in it will :-
(1) Increase (2) Remain constant
(3) Decrease (4) First (1) then (3)

PEI057

58. Which type of losses does not occur in transformer:-
(1) mechanical losses
(2) copper losses
(3) hysteresis losses
(4) eddy current losses

PEI058

59. The efficiency of a transformer is maximum, because :–
(1) No part of the transformer is in motion
(2) It creates maximum voltage
(3) It creates minimum voltage
(4) None of the above

PEI059

60. In order to avoid eddy currents in the core of a transformer :-
(1) The number of turns in the secondary coil is made considerably large
(2) A laminated core is used
(3) A step down transformer is used
(4) A high voltage alternating weak current is used

PEI060

61. The mutual inductance of two coils when magnetic flux changes by 2×10^{-2} Wb and current changes by 0.01 A is :-
(1) 2 H (2) 3 H
(3) 4 H (4) 8 H

PEI061

62. Primary winding and secondary winding of a transformer has 100 and 300 turns respectively. If its input power is 60 W then output power of the transformer will be:-
(1) 240 W (2) 180 W
(3) 60 W (4) 20 W

PEI062

63. The ratio of the secondary to the primary turns in a transformer is 3 : 2 and the output power is P. Neglecting all power losses, the input power must be :-

(1) $\frac{P}{2}$ (2) P (3) $\frac{2P}{3}$ (4) $\frac{3P}{2}$

PEI063

64. Mutual inductance of two coils depends on their self inductance L_1 and L_2 as :-

(1) $M_{12} = L_1/L_2$ (2) $M_{12} = L_2/L_1$

(3) $M_{12} = \sqrt{L_1 L_2}$ (4) $M_{12} = \sqrt{L_1/L_2}$

PEI064

65. In transformer, power of secondary coil is:-

(1) less than primary coil

(2) more than primary coil

(3) more in step up and less in step down than primary coil

(4) more in step down and less in step up than primary coil

PEI065

66. If the input voltage of a transformer is 2500 volts and output current is 80 ampere. The ratio of number of turns in the primary coil to that in secondary coil is 20 : 1. If efficiency of transformer is 100%, then the voltage in secondary coil is :

(1) $\frac{2500}{20}$ volt (2) 2500 × 20 volt

(3) $\frac{2500}{80\times20}$ volt (4) $\frac{2500\times20}{80}$ volt

PEI066

67. A step up transformer has turn ratio 10:1. A cell of e.m.f. 2 volts is fed to the primary. Secondary voltage developed is :-

(1) 20 V (2) 10 V

(3) 2 V (4) Zero

PEI067

68. The flux linked with a coil at any instant 't' is given by $\phi = 10t^2 - 50t + 250$. The induced emf at t = 3 s is :-

(1) 190 V (2) –190 V

(3) –10 V (4) 10 V

PEI068

69. Two coaxial solenoids are made by winding thin Cu wire over a pipe of cross-sectional area A = 10 cm^2 and length = 20 cm. If one of the solenoids has 300 turns and the other 400 turns, their mutual inductance is :-

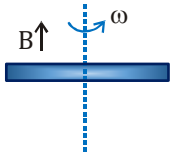
(1) $2.4\pi \times 10^{-5}$ H (2) $4.8\pi \times 10^{-4}$ H

(3) $4.8\pi \times 10^{-5}$ H (4) $2.4\pi \times 10^{-4}$ H

PEI069

DYNAMIC & ROTATIONAL E.M.F. GENERATOR

70. A conducting rod of length 2ℓ is rotating with constant angular speed ω about its perpendicular bisector. A uniform magnetic field $\vec{B}$ exists parallel to the axis of rotation. The emf induced between two ends of the rod is:-



(1) $B\omega\ell^2$ (2) $\frac{1}{2}B\omega\ell^2$

(3) $\frac{1}{8}B\omega\ell^2$ (4) Zero

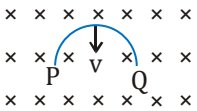
PEI070

71. A conducting rod of 1m length rotating with a frequency of 50 rev/sec. about its one of end inside the uniform magnetic field of 6.28 mT. The value of induced emf between end of rod is:

(1) 1 V (2) 2 V (3) 0.5 V (4) 0.25 V

PEI071

72. A semicircle loop PQ of radius 'R' is moved with velocity 'v' in transverse magnetic field as shown in figure. The value of induced emf. between the ends of loop is :-



(1) Bv (πr), end 'P' at high potential

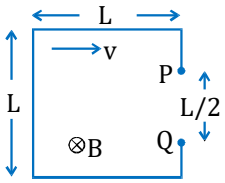
(2) 2 BRv, end P at high potential

(3) 2 BRv, end Q at high potential

(4) $B\frac{\pi R^2}{2}$ v, end P at high potential

PEI072

73. The loop shown moves with a constant velocity 'v' in a uniform magnetic field of magnitude 'B' directed into the paper. The potential difference between P and Q is 'e':-



(1) $e = \frac{BLv}{2}$, Q is positive with respect to P

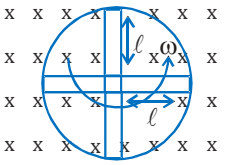
(2) $e = \frac{BLv}{2}$, P is positive with respect to Q

(3) e = 0

(4) e = BLv, Q is positive with respect to P

PEI073

74. A conducting wheel in which there are four rods of length ℓ as shown in figure is rotating with angular velocity ω in a uniform magnetic field B. The induced potential difference between its centre and rim will be :

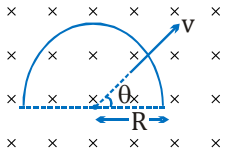


(1) $2B\omega\ell^2$ (2) $\sqrt{B\ell^2\omega}$

(3) $\frac{B\ell\omega}{2}$ (4) $\frac{B\ell^2\omega}{2}$

PEI074

75. A semi-circular loop of radius R is placed in a uniform magnetic field as shown. It is pulled with a constant velocity. The induced emf in the loop is :

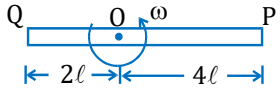


(1) Bv (πR) cosθ (2) Bv(pR) sinθ

(3) Bv(2R) cosθ (4) Bv(2R) sinθ

PEI075

76. A conducting rod rotates with a constant angular velocity 'ω' about the axis which passes through point 'O' and perpendicular to its length. A uniform magnetic field 'B' exists parallel to the axis of the rotation. Then potential difference between the two ends of the rod is :

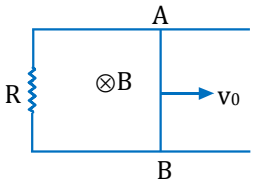


(1) $6B\omega\ell^2$ (2) $B\omega\ell^2$

(3) $10B\omega\ell^2$ (4) Zero

PEI076

77. Two long parallel metallic wires with a resistance 'R' form a horizontal plane. A conducting rod AB is on the wires shown in figure. The space has magnetic field pointing vertically downwards. The rod is given an initial velocity 'v_0'. There is no friction in the wires and the rod. After a time 't' the velocity v of the rod will be such that:–



(1) $v > v_0$ (2) $v < v_0$

(3) $v = v_0$ (4) $v = -v_0$

PEI077

78. The armature coil of dynamo is rotating. The generated induced emf varies and the number of magnetic lines of force also varies. Which of the following condition is correct:-

(1) lines of flux will be minimum, but induced emf will be zero.

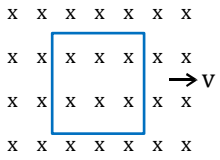
(2) lines of flux will be maximum, but the induced emf will be zero.

(3) lines of flux will be maximum, but induced emf will be not be zero.

(4) the lines of flux will be maximum, and the induced emf will be also maximum.

PEI078

79. A conducting square loop of side ℓ and resistance R moves in its plane with a uniform velocity perpendicular to one of its sides. A uniform and constant magnetic field B exists along the perpendicular to the plane of the loop as shown in the figure. The current induced in the loop is:-



(1) $B\ell v/R$, clockwise

(2) $B\ell v/R$, anticlockwise

(3) $2\ B\ell v/R$, anticlockwise

(4) zero

PEI079

80. If the rotational velocity of a dynamo armature is doubled, then induced e.m.f will become :-

(1) half (2) two times

(3) four times (4) unchanged

PEI080

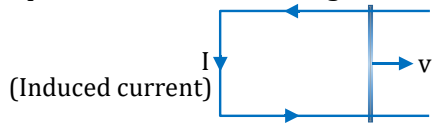
81. Dynamo which produces electricity, is a source of:

(1) gravity (2) magnetism

(3) e.m.f. (4) electrolysis

PEI081

82. For given arrangement (in horizontal plane) the possible direction of magnetic field:-



(1) towards right

(2) towards left

(3) vertically upward

(4) vertically downward

PEI082

83. A metallic disc of radius 'R' is rotating about its geometrical axis with constant angular speed 'ω' in external magnetic field B which is perpendicular to the plane of the disc then induced emf between the centre and any peripherical point of the disc is given by :-

(1) $\pi\omega BR^2$ (2) ωBR^2

(3) $\frac{\pi\omega BR^2}{2}$ (4) $\frac{\omega BR^2}{2}$

PEI083

84. Which of the following is correct for periodic electromagnetic induction :-

(1) maximum flux, zero emf

(2) zero flux, maximum emf

(3) zero flux, zero emf

(4) (1) & (2) both

PEI084

85. Dynamo is based on the principle of :-

(1) electro magnetic induction

(2) induced current

(3) induced magnetism

(4) Faraday effect

PEI085

86. Phase difference between induced emf and flux for a coil rotating in magnetic field :-

(1) 0 (2) $\pi/2$

(3) π (4) 2π

PEI086

87. In an AC generator, a coil with N turns, all of the same area A and total resistance R, rotates with frequency ω in a magnetic field B. The maximum value of emf generated in the coil is :-

(1) $NAB\omega$ (2) $NABR\omega$

(3) NAB (4) NABR

PEI087

88. The electric generator produce electric current based on which principle :-

(1) Ohm's law (2) Faraday's law of EMI

(3) Ampere's law (4) Biot - savart's law

PEI088

89. A rectangular loop of sides a & b is placed in magnetic field B. The emf induced in coil when normal of coil makes angle ωt with B:-

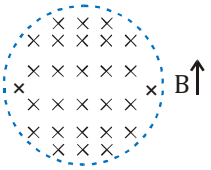
(1) $B(ab)\omega\cos\omega t$ (2) $B(ab)\omega\sin\omega t$

(3) $-B(ab)\omega\sin\omega t$ (4) $-B(ab)\omega\cos\omega t$

PEI089

INDUCED ELECTRIC FIELD

90. In this given figure if magnetic field increases with time then pattern of induced electric field lines will be :-



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(1) A.C.W. concentric circular field lines in the plane of the paper
(2) C.W. concentric circular field lines in the plane of the paper
(3) A.C.W. concentric circular field lines, perpendicular to the plane of the paper
(4) C.W. concentric circular field lines, perpendicular to the plane of the paper

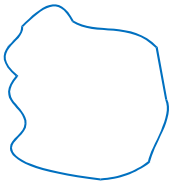
PEI090

91. A nonconducting circular ring of radius 4 cm is placed in a time varing magnetic field with rate of 0.2T/s. If 2C charge placed at its circumference then electric force on this charge will be :-

(1) 4×10^{-3} N (2) 8×10^{-3} N
(3) 6×10^{-2} N (4) 8×10^{-2} N

PEI091

92. As a result of change in the magnetic flux linked to the closed loop shown in the figure, an e.m.f. V volt is induced in the loop. The work done (joules) in taking a charge Q coulomb once along the loop is :-



(1) QV
(2) QV/2
(3) 2QV
(4) Zero

PEI092

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	2	2	3	2	2	4	1	1	1	1	4	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	3	2	1	3	1	1	2	1	1	1	3	2	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	4	4	1	1	3	2	1	3	3	2	2	2	4	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	4	3	3	3	1	1	2	2	4	2	2	1	1	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	3	2	3	1	1	4	3	4	4	1	3	2	4	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	1	2	2	4	2	3	4	4	4	1	2	1	2	2	1
Question	91	92													
Answer	2	1													

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. A transformer having efficiency of 90% is working on 200V and 3kW power supply. If the current in the secondary coil is 6A, the voltage across the secondary coil and the current in the primary coil respectively are :-

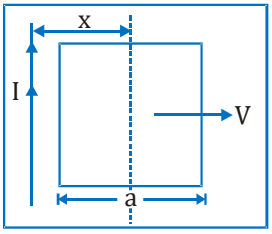
(1) 300 V, 15A (2) 450 V, 15A

(3) 450V, 13.5A (4) 600V, 15A

PEI105

AIPMT 2015

2. A conducting square frame of side 'a' and a long straight wire carrying current I are located in the same plane as shown in the figure. The frame moves to the right with a constant velocity 'V'. The emf induced in the frame will be proportional to:



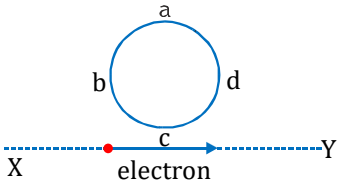
(1) $\frac{1}{(2x-a)^2}$ (2) $\frac{1}{(2x+a)^2}$

(3) $\frac{1}{(2x-a)(2x+a)}$ (4) $\frac{1}{x^2}$

PEI106

Re-AIPMT 2015

3. An electron moves on a straight line path XY as shown. The abcd is a coil adjacent to the path of electron. What will be the direction of current, if any, induced in the coil?



(1) No current induced

(2) abcd

(3) adcb

(4) The current will reverse its direction as the electron goes past the coil

PEI107

NEET-I 2016

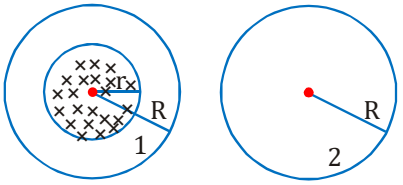
4. A long solenoid has 1000 turns. When a current of 4A flows through it, the magnetic flux linked with each turn of the solenoid is 4×10^{-3} Wb. The self inductance of the solenoid is :-

(1) 4H (2) 3H (3) 2H (4) 1H

PEI108

NEET-II 2016

5. A uniform magnetic field is restricted within a region of radius r. The magnetic field changes with time at a rate $\frac{d\vec{B}}{dt}$. Loop 1 of radius R > r encloses the region r and loop 2 of radius R is outside the region of magnetic field as shown in the figure below. Then the e.m.f. generated is :-



(1) $-\frac{d\vec{B}}{dt}\pi R^2$ in loop 1 and zero in loop 2

(2) $-\frac{d\vec{B}}{dt}\pi r^2$ in loop 1 and zero in loop 2

(3) Zero in loop 1 and zero in loop 2

(4) $-\frac{d\vec{B}}{dt}\pi r^2$ in loop 1 and $-\frac{d\vec{B}}{dt}\pi r^2$ in loop 2

PEI109

NEET(UG) 2017

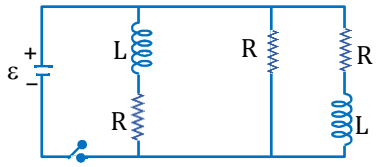
6. A long solenoid of diameter 0.1 m has 2×10^4 turns per meter. At the centre of the solenoid, a coil of 100 turns and radius 0.01 m is placed with its axis coinciding with the solenoid axis. The current in the solenoid reduces at a constant rate to 0A from 4 A in 0.05 s. If the resistance of the coil is $10\pi^2\,\Omega$ the total charge flowing through the coil during this time is :-

(1) 16 μC (2) 32 μC

(3) 16 π μC (4) 32 π μC

PEI110

7. Figure shows a circuit that contains three identical resistors with resistance R = 9.0 Ω each, two identical inductors with inductance L = 2.0 mH each, and an ideal battery with emf ε = 18 V. The current 'i' through the battery just after the switch closed is,..... :-



(1) 0.2 A (2) 2 A
(3) 0 ampere (4) 2 mA

PEI111

NEET(UG) 2018

8. The magnetic potential energy stored in a certain inductor is 25 mJ, when the current in the inductor is 60 mA. This inductor is of inductance :-

(1) 0.138 H (2) 138.88 H
(3) 1.389 H (4) 13.89 H

PEI112

NEET(UG) 2019

9. A 800 turn coil of effective area 0.05 m^2 is kept perpendicular to a magnetic field 5×10^{-5} T. When the plane of the coil is rotated by 90° around any of its coplanar axis in 0.1 s, the emf induced in the coil will be :

(1) 2 V (2) 0.2 V
(3) 2×10^{-3} V (4) 0.02 V

PEI113

10. In which of the following devices, the eddy current effect is **not** used?

(1) induction furnace
(2) magnetic braking in train
(3) electromagnet
(4) electric heater

PEI114

NEET(UG) 2019 (Odisha)

11. A cycle wheel of radius 0.5 m is rotated with constant angular velocity of 10 rad/s in a region of magnetic field of 0.1 T which is perpendicular to the plane of the wheel. The EMF generated between its centre and the rim is,

(1) 0.25 V (2) 0.125 V
(3) 0.5 V (4) zero

PEI115

NEET(UG) 2020 (Covid-19)

12. The magnetic flux linked with a coil (in Wb) is given by the equation

$\phi = 5t^2 + 3t + 16$

The magnitude of induced emf in the coil at the fourth second will be

(1) 33 V (2) 43 V
(3) 108 V (4) 10 V

PEI116

13. A wheel with 20 metallic spokes each 1 m long is rotated with a speed of 120 rpm in a plane perpendicular to a magnetic field of 0.4 G. The induced emf between the axle and rim of the wheel will be, (1 G = 10^{-4} T)

(1) 2.51×10^{-4} V
(2) 2.51×10^{-5} V
(3) 4.0×10^{-5} V
(4) 2.51 V

PEI117

NEET(UG) 2021

14. A step down transformer connected to an ac mains supply of 220 V is made to operate at 11V, 44 W lamp. Ignoring power losses in the transformer, what is the current in the primary circuit?

(1) 0.2 A (2) 0.4 A
(3) 2A (4) 4A

PEI118

15. Two conducting circular loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 >> R_2$, the mutual inductance M between them will be directly proportional to:

(1) $\frac{R_1}{R_2}$ (2) $\frac{R_2}{R_1}$

(3) $\frac{R_1^2}{R_2}$ (4) $\frac{R_2^2}{R_1}$

PEI119

NEET(UG) 2022

16. A big circular coil of 1000 turns and average radius 10 m is rotating about its horizontal diameter at 2 rad s^{-1}. If the vertical component of earth's magnetic field at that place is 2×10^{-5} T and electrical resistance of the coil is 12.56 Ω, then the maximum induced current in the coil will be :

(1) 1.5 A (2) 1 A

(3) 2 A (4) 0.25 A

PEI120

RE-NEET(UG) 2022

17. The magnetic flux linked to a circular coil of radius R is :

$$\phi = 2t^3 + 4t^2 + 2t + 5 \text{ Wb}$$

The magnitude of induced emf in the coil at t = 5s is:

(1) 108 V (2) 197 V

(3) 150 V (4) 192 V

PEI121

NEET(UG) 2023

18. A 12 V, 60 W lamp is connected to the secondary of a step down transformer, whose primary is connected to ac mains of 220 V. Assuming the transformer to be ideal, what is the current in the primary winding?

(1) 2.7 A (2) 3.7 A

(3) 0.37 A (4) 0.27 A

PEI136

19. The magnetic energy stored in an inductor of inductance 4 μH carrying a current of 2 A is :

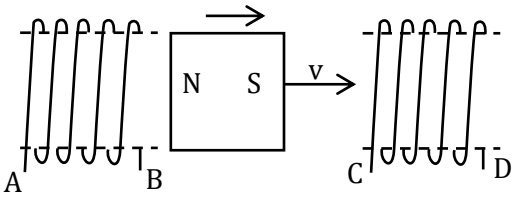
(1) 4 mJ (2) 8 mJ

(3) 8 μJ (4) 4 μJ

PEI137

NEET(UG) 2024

20.



Solenoid-1 Solenoid-2

In the above diagrams, a strong bar magnet is moving towards solenoid-2 from solenoid-1. The direction of induced current in solenoid-1 and that in solenoid-2, respectively, are through the directions :

(1) AB and DC (2) BA and CD

(3) AB and CD (4) BA and DC

PEI138

21. In a ideal transformer, the turns ratio $\frac{N_p}{N_s} = \frac{1}{2}$.

The ratio $V_s : V_p$ is equal to (the symbols carry their usual meaning) :

(1) 1 : 2 (2) 2 : 1

(3) 1 : 1 (4) 1 : 4

PEI139

RE-NEET(UG) 2024

22. Let us consider two solenoids A and B, made from same magnetic material of relative permeability μ_r and equal area of cross-section. Length of A is twice that of B and the number of turns per unit length in A is half that of B. The ratio of self inductance of the two solenoids, $L_A : L_B$ is :

(1) 1 : 2 (2) 2 : 1 (3) 8 : 1 (4) 1 : 8

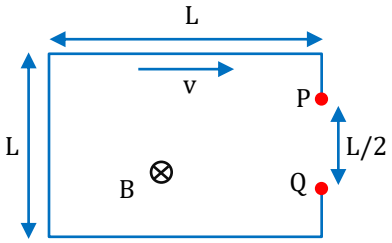
PEI140

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	4	4	2	2	2	4	4	4	2	2	1	1	4
Question	16	17	18	19	20	21	22								
Answer	2	4	4	3	1	2	1								

Exercise – III (Analytical Questions) Master Your Understanding

1. The loop shown moves with a velocity v in a uniform magnetic field of magnitude B, directed into the paper as shown in the figure. The potential difference between points P and Q is e. Then



(a) $e = \frac{1}{2}BL\nu$

(b) $e = BL\nu$

(c) P is positive with respect to Q

(d) Q is positive with respect to P

(1) only a is correct

(2) only a & c are correct

(3) only b is correct

(4) only b & d are correct

PEI122

2. Choose the correct options

(a) SI unit of magnetic flux is henry-ampere

(b) SI unit of coefficient of self-induction is J/A

(c) SI unit of coefficient of self-induction is $\frac{\text{volt}-\text{second}}{\text{ampere}}$

(d) SI unit of magnetic induction is weber

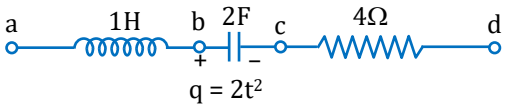
(1) only (a) (2) only (b)

(3) (a) and (c) (4) (b) and (d)

PEI123

3. In the figure shown, q is in coulomb and t in second.

At time t = 1s



(a) $V_a - V_b = 4V$

(b) $V_b - V_c = 1V$

(c) $V_c - V_d = 16V$

(d) $V_a - V_d = 20V$

(1) only (a) is correct

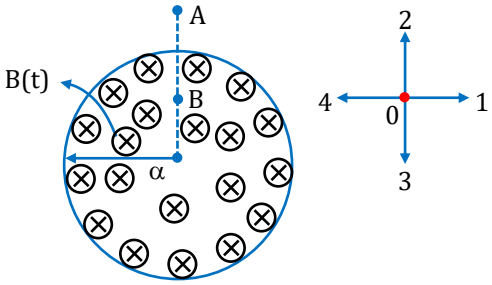
(2) only (b) is correct

(3) (a), (b) and (c) are correct

(4) only (d) is correct

PEI124

4. A uniform but time varying magnetic field B(t) exists in a cylindrical region of radius α and is directed into the plane of the paper, as shown in fig. The magnetic field decreases at a constant rate inside the region. If r is the distance from the axis of the cylindrical region then match column I to column II.



	Column-I		Column-II
I.	Induced electric field at point A	a.	0 to 1
II.	Induced electric field at point B	b.	Zero
III.	Force on an electron placed at point A	c.	0 to 4
IV.	Force on an electron placed at point B	d.	0 to 2

(1) I – a, II – d, III – d, IV – a

(2) I – d, II – a, III – c, IV – a

(3) I – a, II – b, III – c, IV – d

(4) I – a, II – a, III – c, IV – c

PEI125

5. Match the column.

	Column-I		Column-II
(a)	Magnetic field	(p)	$[M^1T^{-2}A^{-1}]$
(b)	Coefficient of self induction	(q)	$[T^2]$
(c)	LC	(r)	$[ML^2T^{-2}A^{-2}]$
(d)	Magnetic flux	(s)	None of these

(1) a – p, b – r, c – q, d – s

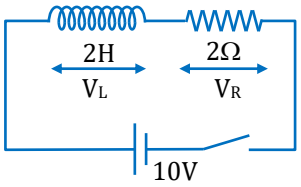
(2) a – p, b – q, c – r, d – s

(3) a – p, b – s, c – q, d – r

(4) a – s, b – s, c – s, d – r

PEI126

6. In the circuit shown in figure, switch is closed at time t = 0. Match the following two columns.

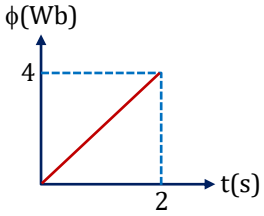


	Column-I		Column-II
(a)	V_L at t = 0	(p)	zero
(b)	V_R at t = 0	(q)	10V
(c)	V_L at t = 1s	(r)	$\frac{10}{e}$V
(d)	V_R at t = 1s	(s)	$\left(1-\frac{1}{e}\right)$10V

(1) a – q, b – p, c – r, d – s
(2) a – q, b – r, c – p, d – s
(3) a – r, b – q, c – r, d – s
(4) a – s, b – r, c – p, d – q

PEI127

7. Magnetic flux passing through a coil of resistance 2Ω is as shown in figure. Match the following two columns. In column II all physical quantities are in SI units.



	Column-I		Column-II
(a)	Induced emf produced	(p)	4
(b)	Induced current	(q)	1
(c)	Charge flow in 2s	(r)	8
(d)	Heat generation in 2s	(s)	2

(1) a-s, b-q, c-s, d-p (2) a-q, b-r, c-p, d-s
(3) a-p, b-q, c-r, d-s (4) a-r, b-q, c-s, d-p

PEI128

8. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Transformer can't be used to step up or step down the steady dc voltage
Reason (R) : Transformers are based upon the principle of mutual induction for which a time varying voltage is necessary.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PEI129

9. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : The direction of induced current is always such as to oppose the change that causes it.
Reason (R) : The direction of induced current is given by Lenz's law.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PEI130

10. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : A time varying magnetic field induces an electric field.
Reason (R) :- An inductor always tends to keep the flux constant.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PEI131

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11. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The mutual inductance of coil 1 with respect to coil 2 is the same as mutual inductance of coil 2 with respect to coil 1.

Reason (R) : Mutual inductance between coils is independent of current passing through the coil

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PEI132

12. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The electric field created by time-varying magnetic field is non-conservative.

Reason (R) : The line integral of induced electric field in a closed loop is always equal to zero.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PEI133

13. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Magnetic flux is a scalar quantity.

Reason (R) : Value of magnetic flux cannot be negative.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PEI134

14. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When a coil is rotated in a uniform magnetic field about an diametrical axis perpendicular to the field, emf is induced in it which is maximum for the orientation of coil in which magnetic flux through the coil is zero.

Reason (R) : In an electric generator, electrical energy is generated by rotating a coil in a magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PEI135

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Answer	2	3	3	4	1	1	1	1	1	2	2	3	3	2	

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Exercise - I (Conceptual Questions) Build Up Your Understanding

PEAK, AVERAGE AND RMS VALUE

1. What is the r.m.s. value of an alternating current which when passed through a resistor produces heat which is thrice of that produced by a direct current of 2 amperes in the same resistor :-
(1) 6 amp (2) 2 amp
(3) 3.46 amp (4) 0.66 amp
PAC001

2. The peak value of an alternating e.m.f. which is given by $E = E_0 \cos\omega t$ is 10 volts and its frequency is 50 Hz. At time $t = \frac{1}{600}$ s, the instantaneous e.m.f. is
(1) 10 V (2) $5\sqrt{3}$ V (3) 5 V (4) 1V
PAC002

3. The phase difference between current and voltage in an AC circuit is $\frac{\pi}{4}$ radian. If the frequency of AC is 50 Hz, then the phase difference is equivalent to the time difference:-
(1) 0.78 s (2) 15.7 ms
(3) 2.5 s (4) 2.5 ms
PAC003

4. A current in circuit is given by $i = 3 + 4 \sin \omega t$. Then the effective value of current is :
(1) 5 (2) $\sqrt{7}$ (3) $\sqrt{17}$ (4) $\sqrt{10}$
PAC004

5. Incorrect statement are :
(a) A.C. meters can measure D.C also
(b) If A.C. meter measures D.C. there scale must be linear and uniform
(c) A.C. and D.C. meters are based on heating effect of current
(d) A.C. meter reads rms value of current
(1) a, b (2) b, c (3) c, d (4) d, a
PAC005

6. The r.m.s. value of current for a variable current $i = i_1 \cos\omega t + i_2 \sin\omega t$:–
(1) $\frac{1}{\sqrt{2}}(i_1 + i_2)$ (2) $\frac{1}{\sqrt{2}}(i_1 + i_2)^2$
(3) $\frac{1}{\sqrt{2}}(i_1^2 + i_2^2)^{1/2}$ (4) $\frac{1}{2}(i_1^2 + i_2^2)^{1/2}$
PAC006

7. The relation between an A.C. voltage source and time in SI units is :
$V = 120 \sin(100\pi t) \cos(100\pi t)$ volt
value of peak voltage and frequency will be respectively :–
(1) 120 volt and 100 Hz
(2) $\frac{120}{\sqrt{2}}$ volt and 100 Hz
(3) 60 volt and 200 Hz
(4) 60 volt and 100 Hz
PAC007

8. If an A.C. main supply is given to be 220 V. What would be the average e.m.f. during a positive half cycle :-
(1) 198 V (2) 386 V
(3) 256 V (4) None of these
PAC008

9. The hot wire ammeter measures :-
(1) D.C. current (2) A.C. current
(3) None of above (4) both (1) & (2)
PAC009

10. Frequency of A.C. in India is –
(1) 45 Hz
(2) 60 Hz
(3) 50 Hz
(4) None of the above
PAC010

11. For an alternating current $I = I_0 \cos\omega t$, the rms value and peak value of current is :-
(1) $I_0, \frac{I_0}{\sqrt{2}}$ (2) $\frac{I_0}{\sqrt{2}}, I_0$
(3) $I_0, \frac{I_0}{2}$ (4) $2I_0, \frac{I_0}{\sqrt{2}}$
PAC011

12. If a step-up transformer have turn ratio 5, frequency 50 Hz root mean square value of potential difference on primary 100 volts and the resistance of the secondary winding is 500 Ω then the peak value of voltage in secondary winding will be (the efficiency of the transformer is hundred percent)
(1) $500\sqrt{2}$ (2) $10\sqrt{2}$
(3) $50\sqrt{2}$ (4) $20\sqrt{2}$
PAC012

SIMPLE AC CIRCUIT

13. A resonant A.C. circuit contains a capacitor of capacitance 10^{-6} F and an inductor of 10^{-4} H. The frequency of electrical oscillations will be :-

(1) 10^5 Hz (2) 10 Hz

(3) $\frac{10^5}{2\pi}$ Hz (4) $\frac{10}{2\pi}$ Hz

PAC013

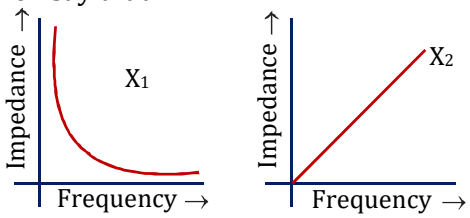
14. A resistance of 300Ω and an inductance of $\frac{1}{\pi}$ henry are connected in series to a A.C. voltage of 20 volts and 200 Hz frequency. The phase angle between the voltage and current is :-

(1) $\tan^{-1}\left(\frac{4}{3}\right)$ (2) $\tan^{-1}\left(\frac{3}{4}\right)$

(3) $\tan^{-1}\left(\frac{3}{2}\right)$ (4) $\tan^{-1}\left(\frac{2}{3}\right)$

PAC014

15. The graphs given below depict the dependence of two reactive impedances X_1 and X_2 on the frequency of the alternating e.m.f. applied individually to them. We can then say that :



(1) X_1 is an inductor and X_2 is a capacitor
(2) X_1 is a resistor and X_2 is a capacitor
(3) X_1 is a capacitor and X_2 is an inductor
(4) X_1 is an inductor and X_2 is a resistor

PAC015

16. A 12 ohm resistor and a 0.21 henry inductor are connected in series to an AC source operating at 20 volts, 50 cycle/second. The phase angle between the current and the source voltage is:

(1) 30° (2) 40° (3) 80° (4) 90°

PAC016

17. A 110 V, 60 W lamp is run from a 220 V AC mains using a capacitor in series with the lamp, instead of a resistor then the voltage across the capacitor is about:-

(1) 110 V (2) 190 V
(3) 220 V (4) 311 V

PAC017

18. The resistance that must be connected in series with inductance of 0.2 H in order that the phase difference between current and e.m.f. may be 45° when the frequency is 50 Hz, is:-

(1) 6.28 ohm. (2) 62.8 ohm.
(3) 628 ohm. (4) 31.4 ohm.

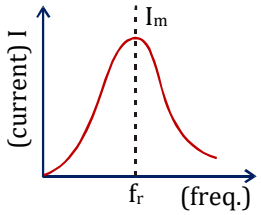
PAC018

19. A student connects a long air cored - coil of manganin wire to a 100 V D.C. supply and records a current of 25 amp. When the same coil is connected across 100 V. 50 Hz a.c. the current reduces to 20 A , the reactance of the coil is :-

(1) 4 Ω (2) 3 Ω (3) 5 Ω (4) None

PAC019

20. The graph shows variation of I with f for a series R-L-C network. Keeping L and C constant, if R decreases :



(a) Maximum current (I_m) increases
(b) Sharpness of the graph increases
(c) Quality factor increases
(d) Band width increases

(1) a, b, c (2) b, c, d (3) c, d, a (4) All

PAC020

21. Alternating current is flowing in inductance L and resistance R. The frequency of source is $\omega/2\pi$. Which of the following statement is correct :

(1) For low frequency the limiting value of impedance is L.
(2) For high frequency the limiting value of impedance is ωL.
(3) For high frequency the limiting value of impedance is R.
(4) For low frequency the limiting value of impedance is ωL.

PAC021

22. A bulb and a capacitor are connected in series to a source of alternating current. If its frequency is increased, while keeping the voltage of the source constant, then
(1) Bulb will give more intense light.
(2) Bulb will give less intense light.
(3) Bulb will give light of same intensity as before
(4) Bulb will stop radiating light.

PAC022

23. In an A.C. circuit resistance and inductance are connected in series. The potential and current in inductance is:
(1) $V_0 \sin \omega t, \frac{V_0}{\omega L} \sin\omega t$
(2) $V_0 \sin \omega t, \frac{V_0}{\omega L} \sin(\omega t + \pi/2)$
(3) $V_0 \sin (\omega t + \pi/2), \frac{V_0}{\omega L} \sin\omega t$
(4) $V_0 \sin(\omega t + \pi/2), \frac{V_0}{\omega L} \sin(\omega t - \pi/2)$

PAC023

24. An a.c. source of voltage V and of frequency 50 Hz is connected to an inductor of 2H and negligible resistance. A current of r.m.s value I flows in the coil. When the frequency of the voltage is changed to 400 Hz keeping the magnitude of V the same, the current is now :-
(1) 8I in phase with V
(2) 4I and leading by 90° from V
(3) $\frac{I}{4}$ and lagging by 90° from V
(4) $\frac{I}{8}$ and lagging by 90° from V

PAC024

25. A capacitor of capacity C is connected in A.C. circuit. The applied emf is $V=V_0 \sin\omega t$, then the current is :
(1) $I = \frac{V_0}{\omega L} \sin\omega t$
(2) $I = \frac{V_0}{\omega L} \sin(\omega t + \pi/2)$
(3) $I = V_0 \omega C \sin\omega t$
(4) $I = V_0 \omega C \sin (\omega t + \pi/2)$

PAC025

26. The impedence of a circuit, when a resistance R and an inductor of inductance L are connected in series in an A.C. circuit of frequency (f) is :-
(1) $\sqrt{R + 4\pi f L^2}$ (2) $\sqrt{R + 4\pi^2 f^2 L^2}$
(3) $\sqrt{R^2 + 4\pi^2 f^2 L^2}$ (4) $\sqrt{R^2 + 2\pi^2 f^2 L^2}$

PAC026

27. A capacitor of capacity C and reactance X if capacitance and frequency become double then reactance will be :–
(1) 4X (2) $\frac{X}{2}$ (3) $\frac{X}{4}$ (4) 2X

PAC027

28. The coil of choke in a circuit :
(1) increases the current
(2) controls the current
(3) has high resistance to d.c. circuit
(4) does not change the current

PAC028

29. The inductive reactance of an inductive coil with $\frac{1}{\pi}$ henry and 50 Hz :–
(1) $\frac{50}{\pi}$ ohm (2) $\frac{\pi}{50}$ ohm
(3) 100 ohm (4) 50 ohm

PAC029

30. In the L–R circuit R = 10Ω and L = 2H. If 120V, 60Hz alternating voltage is applied then the flowing current in this circuit will be :-
(1) 0.32 A (2) 0.16 A
(3) 0.48 A (4) 0.80 A

PAC030

31. An inductance of 0.4 Henry and a resistance of 100 ohm are connected to a A.C. voltage source of 220 V and 50 Hz. Then find out the phase difference between the voltage and current flowing in the circuit :
(1) $\tan^{-1}(2.25\,\pi)$ (2) $\tan^{-1}(0.4\,\pi)$
(3) $\tan^{-1}(1.5\,\pi)$ (4) $\tan^{-1}(0.5\,\pi)$

PAC031

32. A capacitor of capacitance 100 μF & a resistance of 100Ω is connected in series with AC supply of 220V, 50Hz. The current leads the voltage by

(1) $\tan^{-1}\left(\frac{1}{2\pi}\right)$ (2) $\tan^{-1}\left(\frac{1}{\pi}\right)$

(3) $\tan^{-1}\left(\frac{2}{\pi}\right)$ (4) $\tan^{-1}\left(\frac{4}{\pi}\right)$

PAC032

33. If the current through an inductor of inductance L is given by $I = I_0 \sin\omega t$, then the voltage across inductor will be :-

(1) $I_0 \omega L \sin(\omega t - \pi/2)$

(2) $I_0 \omega L \sin(\omega t + \pi/2)$

(3) $I_0 \omega L \sin(\omega t - \pi)$

(4) None of these

PAC033

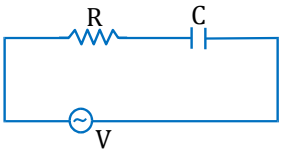
34. There is a 5 Ω resistance in an A.C. circuit. Inductance of 0.1 H is connected with it in series. If equation of A.C. e.m.f. is V = 5 sin50t then the phase difference between current and e.m.f. is :-

(1) $\frac{\pi}{2}$ (2) $\frac{\pi}{6}$

(3) $\frac{\pi}{4}$ (4) 0

PAC034

35. A 50 Hz a.c. source of 20V is connected across R and C as shown in figure below. The voltage across R is 12V. The voltage across C is



(1) 8 V

(2) 16V

(3) 10 V

(4) Not possible to determine unless values of R and C are given

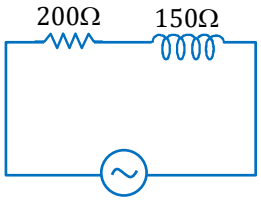
PAC035

36. 200Ω resistance and 1H inductance is connected in series with an A.C. circuit. The frequency of the source is $\frac{200}{2\pi}$ Hz. Then phase difference in between V and I will be :-

(1) 30° (2) 60° (3) 45° (4) 90°

PAC036

37. Impedance of the following circuit will be :

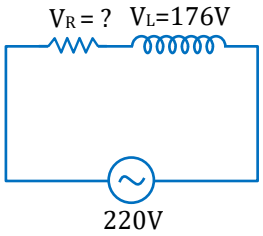


(1) 150Ω (2) 200Ω

(3) 250Ω (4) 340Ω

PAC037

38. In the figure shown, find V_R :



(1) 132V (2) 396V

(3) 185 V (4) $\sqrt{220\times176}$V

PAC038

39. If alternating current of 60 Hz frequency is flowing through inductance of L = 1 mH and drop in ΔV_L is 0.6 V then alternating current :-

(1) $\frac{1}{\pi}$ A (2) $\frac{5}{\pi}$ A (3) $\frac{50}{\pi}$ A (4) $\frac{20}{\pi}$ A

PAC039

LCR SERIES CIRCUIT, RESONANCE

40. An inductance of 1mH, a condenser of 10μF and a resistance of 50Ω are connected in series. The reactance of inductor and condensers are same. The reactance of either of them will be :-

(1) 100 Ω (2) 30 Ω

(3) 3.2 Ω (4) 10 Ω

PAC040

41. L, C and R represent physical quantities inductance, capacitance and resistance respectively. The combination representing dimension of frequency is

(1) LC (2) $(LC)^{-1/2}$

(3) $\left(\frac{L}{C}\right)^{-1/2}$ (4) $\frac{C}{L}$

PAC041

42. A circuit contains R, L and C connected in series with an A. C. source. The values of the reactance for inductor and capacitor are 200Ω and 600Ω respectively and the impedance of the circuit is Z_1. What happens to the impedance of the same circuit if the values of the reactance are interchanged:-

(1) The impedance will remain unchanged

(2) The impedance will increase

(3) The impedance will decrease

(4) Information insufficient

PAC042

43. When V = 100 sinωt is applied across a series (R-L-C) circuit, at resonance the current in resistance (R = 100 Ω) is i = i_0 sinωt, then power dissipation in circuit is:-

(1) 50 W

(2) 100 W

(3) 25 W

(4) Can't be calculated

PAC043

44. At resonance in a series LCR circuit, which of the following statements is true: -

(1) Current in the circuit is maximum and phase difference between E and I is π/2

(2) Current in the circuit is maximum and phase difference between E and I is zero

(3) Voltage is maximum and phase difference between E and I is π/2

(4) Current is minimum and phase difference between E and I is zero

PAC044

45. An alternating voltage is connected in series with a resistance R and an inductance L. If the potential drop across the resistance is 200 volt and across the inductance is 150 volt, the applied voltage:

(1) 350 volt (2) 250 volt

(3) 500 volt (4) 300 volt

PAC045

46. For a series R-L-C circuit :-

(a) Voltage across L and C are differ by π

(b) Current through L and R are in same phase

(c) Voltage across R and L differ by π/2

(d) Voltage across L and current through C are differ by π/2

(1) a, b, c (2) b, c, d

(3) c, d, a (4) All

PAC046

47. A series R–L–C (R = 10Ω, X_L = 20Ω, X_C = 20 Ω) circuit is supplied by V = 10 sin ωt volt then power dissipation in circuit is :-

(1) Zero (2) 10 W

(3) 5 W (4) 2.5 W

PAC047

48. The self inductance of the motor of an electric fan is 10 H. In order to impart maximum power at 50Hz. It should be connected to a capacitance of :

(1) 2×10^{-6} F (2) 3×10^{-6} F

(3) 10^{-4} F (4) 10^{-6} F

PAC048

49. In a series resonant R-L-C circuit, if L is increased by 25% and C is decreased by 20%, then the resonant frequency will :

(1) Increases by 10%

(2) Decreases by 10%

(3) Remain unchanged

(4) Increases by 2.5%

PAC049

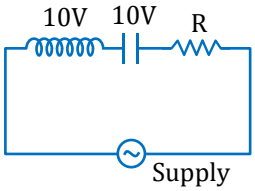
50. The value of quality factor is :-

(ω_0 : resonating frequency)

(1) $\frac{\omega_0 L}{R}$ (2) $\frac{\omega_0}{RC}$ (3) $\sqrt{LC}$ (4) L/R

PAC050

51. If value of R is changed, then :-



(1) Voltage across L remains same
(2) Voltage across C remains same
(3) Voltage across LC combination remains same
(4) Voltage across LC combination changes

PAC051

52. In a series LCR circuit voltage across resistor, inductor and capacitor are 1V, 3V and 2V respectively. At the instant t when the source voltage is given by :
$V = V_0 \cos \omega t$, the current in the circuit will be :

(1) $I = I_0 \cos\left(\omega t + \frac{\pi}{4}\right)$

(2) $I = I_0 \cos\left(\omega t - \frac{\pi}{4}\right)$

(3) $I = I_0 \cos\left(\omega t + \frac{\pi}{3}\right)$

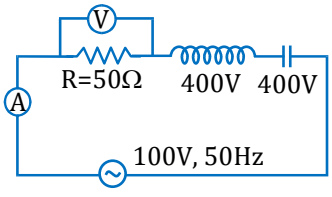
(4) $I = I_0 \cos\left(\omega t - \frac{\pi}{3}\right)$

PAC052

53. In an AC Circuit decrease in impedance with increase in frequency indicates that circuit has/have :-
(1) Only resistance
(2) Resistance & inductance.
(3) Resistance & capacitance
(4) Resistance, capacitance & inductance.

PAC053

54. In given LCR circuit, the voltage across the terminals of a resistance & current will be–



(1) 400V, 2A (2) 800V, 2A
(3) 100V, 2A (4) 100V, 4A

PAC054

55. Phase of current in LCR circuit –
(1) Is in the phase of potential
(2) Leading from the phase of potential
(3) Lagging from the phase of potential
(4) Before resonance frequency, leading from the phase of potential and after resonance frequency, lagging from the phase of potential

PAC055

56. In LCR circuit, the voltage across the terminals of a resistance, inductance & capacitance are 40V, 30V & 60V, then the voltage across the main source will be –
(1) 130 volt (2) 100 volt
(3) 70 volt (4) 50 volt

PAC056

57. For an alternating current of frequency $\frac{500}{\pi}$ Hz in L-C-R series circuit with L = 1H, C = 1 μF, R = 100Ω, impedance is :-
(1) 100 Ω (2) $100\sqrt{\pi}\,\Omega$
(3) $100\sqrt{2\pi}\,\Omega$ (4) 100 π Ω

PAC057

POWER IN AC CIRCUIT

58. A sinusoidal A.C. current flows through a resistor of resistance R. If the peak current is I_P, then the power dissipated is :-
(1) $I_p^2 R\cos\theta$ (2) $\frac{1}{2} I_p^2 R$
(3) $\frac{4}{\pi} I_p^2 R$ (4) $\frac{1}{\pi^2} I_p^2 R$

PAC058

59. An AC circuit draws 5A at 160 V and the power consumption is 600 W. Then the power factor is:-
(1) 1 (2) 0.75
(3) 0.50 (4) Zero

PAC059

60. Which is not correct for average power P at resonance :
(1) $P = I_{rms} V_{rms}$ (2) $P = \frac{V}{\sqrt{2}}\frac{I}{\sqrt{2}}$
(3) P = VI (4) $P = I^2_{rms} R$

PAC060

61. In an A.C. circuit inductance, capacitance and resistance are connected. If the effective voltage across inductance is V_L, across capacitance is V_c and across resistance is V_R, then the total effective value of voltage is :

(1) $V_R+V_L+V_c$ (2) $V_R+V_L-V_c$

(3) $\sqrt{V_R^2+(V_L-V_C)^2}$ (4) $\sqrt{V_R^2-(V_L-V_C)^2}$

PAC061

62. In an a.c. circuit V and I are given by

V = 100 sin (100 t) volts

I = 100 sin (100t + π/3) mA

The power dissipated in the circuit is

(1) 10^4 watt (2) 10 watt

(3) 2.5 watt (4) 5.0 watt

PAC062

63. For a series LCR circuit the power loss at resonance is :-

(1) $\frac{V^2}{\left[\omega L-\frac{1}{\omega C}\right]}$ (2) $I^2L\omega$

(3) I^2R (4) $\frac{V^2}{C\omega}$

PAC063

64. In an alternating circuit applied voltage and current are $E = E_0 \sin\omega t$ and $I = I_0 \sin(\omega t+\pi/2)$ respectively. Then the power consumed in the circuit will be :

(1) Zero (2) $E_0I_0/2$

(3) $E_0I_0/\sqrt{2}$ (4) $E_0I_0/4$

PAC064

65. In which of the following case power factor will be negligible :–

(1) Inductance and resistance both high

(2) Inductance and resistance both low.

(3) Low resistance and high inductance

(4) High resistance and low inductance

PAC065

66. If V = 100 sin100t volt, and

I = 100 $\sin(100t + \frac{\pi}{6})$ A. then find the watt less power in watt :-

(1) 10^4 (2) 10^3

(3) 10^2 (4) 2.5×10^3

PAC066

67. An A.C. supply gives 30V r.m.s. voltage which passes through a 10Ω resistance. The power dissipated in it is :-

(1) $90\sqrt{2}$ W (2) 90W

(3) $45\sqrt{2}$ W (4) 45 W

PAC067

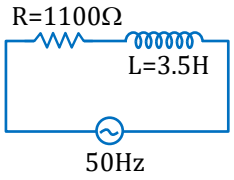
68. An inductor of inductance L and resistor of resistance R are joined in series and connected by a source of frequency ω. Power dissipated in the circuit is :-

(1) $\frac{\left(R^2+\omega^2L^2\right)}{V}$ (2) $\frac{V^2R}{\left(R^2+\omega^2L^2\right)}$

(3) $\frac{V}{\left(R^2+\omega^2L^2\right)}$ (4) $\frac{\sqrt{R^2+\omega^2L^2}}{V^2}$

PAC068

69. For given circuit the power factor is :



(1) 0 (2) 1/2

(3) $1/\sqrt{2}$ (4) None of these

PAC069

70. In a purely capacitive circuit average power dissipated in the circuit is -

(1) $V_{rms}\, I_{rms}$

(2) Depends on capacitance

(3) Infinite

(4) Zero

PAC070

71. Energy loss in pure capacitance in A.C. circuit is

(1) $\frac{1}{2}CV^2$ (2) CV

(3) $\frac{1}{4}CV^2$ (4) Zero

PAC071

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72. Power dissipated in pure inductance will be :

(1) $\frac{LI^2}{2}$ (2) $2LI^2$

(3) $\frac{LI^2}{4}$ (4) Zero

PAC072

73. The power factor of L-R circuit is :

(1) $\frac{\omega L}{R}$ (2) $\frac{R}{\sqrt{(\omega L)^2 + R^2}}$

(3) ωLR (4) $\sqrt{\omega LR}$

PAC073

74. If alternating current of rms value 'a' flows through resistance R then power loss in resistance is :

(1) Zero (2) a^2R

(3) $\frac{a^2R}{2}$ (4) $2a^2R$

PAC074

75. Which of the following device in alternating circuit provides maximum power :-

(1) Only capacitor

(2) Capacitor and resistor

(3) Only inductor

(4) Only resistor

PAC075

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	4	3	2	3	4	1	4	3	2	1	3	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	2	2	2	1	2	1	3	4	4	3	3	2	3	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	2	2	3	2	3	3	1	2	4	2	1	1	2	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	3	4	3	1	3	2	3	3	4	4	1	2	2	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	3	3	3	1	3	4	2	2	3	4	4	4	2	2	4

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Exercise - II (Previous Year Questions) AIPMT/NEET

Re-AIPMT 2015

1. A series R-C circuit is connected to an alternating voltage source. Consider two situations:
(a) When capacitor is air filled.
(b) When capacitor is mica filled.
Current through resistor is i and voltage across capacitor is V then :-
(1) $V_a = V_b$ (2) $V_a < V_b$
(3) $V_a > V_b$ (4) $i_a > i_b$

PAC095

AIPMT 2015

2. A resistance 'R' draws power 'P' when connected to an AC source. If an inductance is now placed in series with the resistance, such that the impedance of the circuit becomes 'Z', the power drawn will be:
(1) $P\sqrt{\frac{R}{Z}}$ (2) $P\left(\frac{R}{Z}\right)$
(3) P (4) $P\left(\frac{R}{Z}\right)^2$

PAC096

NEET-I 2016

3. An inductor 20 mH, a capacitor 50 μF and a resistor 40Ω are connected in series across a source of emf V = 10 sin 340 t. The power loss in A.C. circuit is:-
(1) 0.51 W (2) 0.67 W
(3) 0.76 W (4) 0.89 W

PAC097

4. A small signal voltage $V(t) = V_0 \sin\omega t$ is applied across an ideal capacitor C :-
(1) Current I(t), lags voltage V(t) by 90°.
(2) Over a full cycle the capacitor C does not consume any energy from the voltage source.
(3) Current I(t) is in phase with voltage V(t).
(4) Current I(t) leads voltage V(t) by 180°.

PAC098

NEET-II 2016

5. Which of the following combinations should be selected for better tuning of an L-C-R circuit used for communication?
(1) R = 15 Ω, L = 3.5 H, C = 30 μF
(2) R = 25 Ω, L = 1.5 H, C = 45 μF
(3) R = 20 Ω, L = 1.5 H, C = 35 μF
(4) R = 25 Ω, L = 2.5 H, C = 45 μF

PAC099

6. The potential differences across the resistance, capacitance and inductance are 80 V, 40 V and 100 V respectively in an L-C-R circuit. The power factor of this circuit is :-
(1) 0.8 (2) 1.0 (3) 0.4 (4) 0.5

PAC100

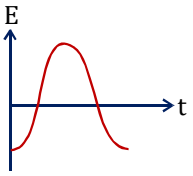
NEET (UG) 2018

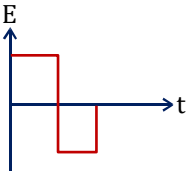
7. An inductor 20 mH, a capacitor 100 μF and a resistor 50 Ω are connected in series across a source of emf, V = 10 sin 314 t. The power loss in the circuit is
(1) 0.79 W (2) 0.43 W
(3) 2.74 W (4) 1.13 W

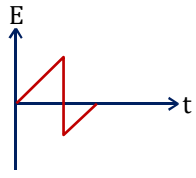
PAC101

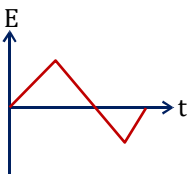
NEET (UG) 2019 (Odisha)

8. The variation of EMF with time for four types of generators are shown in the figures. Which amongst them can be called AC?

(a) 

(b) 

(c) 

(d) 

(1) (a) & (d) (2) (a), (b), (c) & (d)
(3) (a) & (b) (4) only (a)

PAC102

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9. A circuit when connected to an AC source of 12 V gives a current of 0.2 A. The same circuit when connected to a DC source of 12 V, gives a current of 0.4 A. The circuit is

(1) series LR (2) series RC
(3) series LC (4) series LCR

PAC103

NEET (UG) 2020

10. A 40 μF capacitor is connected to a 200 V, 50 Hz ac supply. The rms value of the current in the circuit is, nearly :

(1) 25.1 A (2) 1.7 A
(3) 2.05 A (4) 2.5 A

PAC104

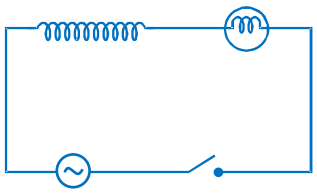
11. A series LCR circuit is connected to an ac voltage source. When L is removed from the circuit, the phase difference between current and voltage is $\frac{\pi}{3}$. If instead C is removed from the circuit, the phase difference is again $\frac{\pi}{3}$ between current and voltage. The power factor of the circuit is :

(1) –1.0 (2) zero
(3) 0.5 (4) 1.0

PAC105

NEET (UG) 2020(Covid-19)

12. A light bulb and an inductor coil are connected to an ac source through a key as shown in the figure below. The key is closed and after sometime an iron rod is inserted into the interior of the inductor. The glow of the light bulb



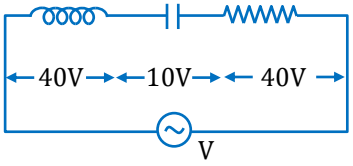
(1) decreases
(2) remains unchanged
(3) will fluctuate
(4) increases

PAC106

NEET (UG) 2021

13. An inductor of inductance L, a capacitor of capacitance C and a resistor of resistance 'R' are connected in series to an ac source of potential difference 'V' volts as shown in figure.

Potential difference across L, C and R is 40 V, 10 V and 40 V, respectively. The amplitude of current flowing through LCR series circuit is $10\sqrt{2}$ A. The impedance of the circuit is :-



(1) $4\sqrt{2}\,\Omega$ (2) $5/\sqrt{2}\,\Omega$
(3) 4 Ω (4) 5 Ω

PAC107

14. A series LCR circuit containing 5.0 H inductor, 80 μF capacitor and 40 Ω resistor is connected to 230 V variable frequency ac source. The angular frequencies of the source at which power transferred to the circuit is half the power at the resonant angular frequency are likely to be :

(1) 25 rad/s and 75 rad/s
(2) 50 rad/s and 25 rad/s
(3) 46 rad/s and 54 rad/s
(4) 42 rad/s and 58 rad/s

PAC108

NEET (UG) 2022

15. The peak voltage of the ac source is equal to:

(1) the rms value of the ac source
(2) $\sqrt{2}$ times the rms value of the ac source
(3) $1/\sqrt{2}$ time the rms value of the ac source
(4) the value of voltage supplied to the circuit.

PAC109

16. A series LCR circuit with inductance 10H, capacitance 10 μF, resistance 50 Ω is connected to an ac source of voltage, V = 200 sin (100 t) volt. If the resonant frequency of the LCR circuit is ν_o and the frequency of the ac source is ν, then

(1) $\nu_o = \nu = \frac{50}{\pi}$ Hz

(2) $\nu_o = \frac{50}{\pi}$ Hz, $\nu = 50$ Hz

(3) $\nu = 100$ Hz ; $\nu_o = \frac{100}{\pi}$ Hz

(4) $\nu_o = \nu = 50$ Hz

PAC110

RE-NEET (UG) 2022

17. An inductor of inductance 2 mH is connected to a 220 V, 50 Hz a.c. source. Let the inductive reactance in the circuit is X_1. If a 220 V dc source replaces the ac source in the circuit, then the inductive reactance in the circuit is X_2. X_1 and X_2 respectively are :

(1) 6.28 Ω, zero (2) 6.28 Ω, infinity

(3) 0.628 Ω, zero (4) 0.628 Ω, infinity

PAC111

18. A standard filament lamp consumes 100 W when connected to 200 V ac mains supply. The peak current through the bulb will be :

(1) 0.707 A (2) 1 A

(3) 1.414 A (4) 2 A

PAC112

NEET (UG) 2023

19. In a series LCR circuit, the inductance L is 10 mH, capacitance C is 1μF and resistance R is 100 Ω. The frequency at which resonance occurs is :-

(1) 15.9 kHz (2) 1.59 rad/s

(3) 1. 59 kHz (4) 15.9 rad/s

PAC125

20. An ac source is connected to a capacitor C. Due to decrease in its operating frequency :

(1) displacement current increases.

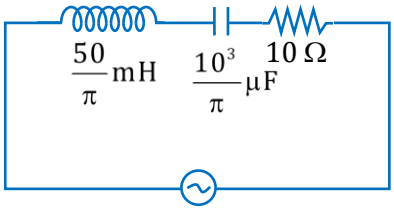
(2) displacement current decreases.

(3) capacitive reactance remains constant.

(4) capacitive reactance decreases.

PAC126

21. The net impedance of circuit (as shown in figure) will be :



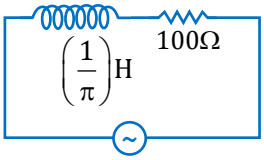
(1) 15 Ω (2) $5\sqrt{5}\,\Omega$

(3) 25 Ω (4) $10\sqrt{2}\,\Omega$

PAC127

RE-NEET (UG) 2023

22. An ac source is connected in the given circuit. The value of ϕ will be :



(1) 60°

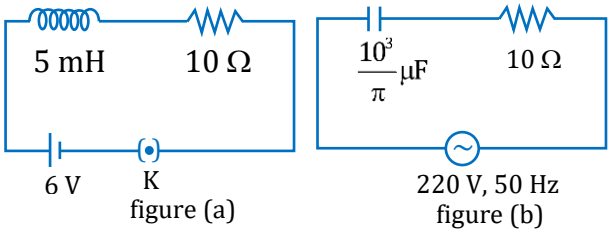
(2) 90°

(3) 30°

(4) 45°

PAC128

23. If Z_1 and Z_2 are the impedances of the given circuits (a) and (b) as shown in figures, then choose the **correct** option



(1) $Z_1 < Z_2$ (2) $Z_1 + Z_2 = 20\ \Omega$

(3) $Z_1 = Z_2$ (4) $Z_1 > Z_2$

PAC129

24. The maximum power is dissipated for an ac in a/an:

(1) resistive circuit (2) LC circuit

(3) inductive circuit (4) capacitive circuit

PAC130

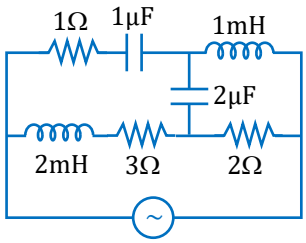
25. An emf is generated by an ac generator having 100 turn coil, of loop area 1 m^2. The coil rotates at a speed of one revolution per second and placed in a uniform magnetic field of 0.05 T perpendicular to the axis of rotation of the coil. The maximum value of emf is :-

(1) 3.14 V (2) 31.4 V

(3) 62.8 V (4) 6.28 V

PAC131

26. For very high frequencies, the effective impedance of the circuit (shown in the figure) will be :-

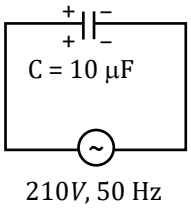


(1) 4 Ω (2) 6 Ω (3) 1 Ω (4) 3 Ω

PAC132

NEET (UG) 2024

27. A 10 μF capacitor is connected to a 210 V, 50 Hz source as shown in figure. The peak current in the circuit is nearly (π = 3.14) :



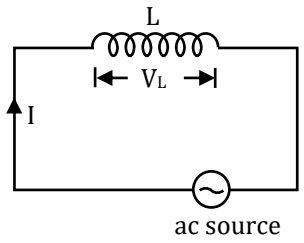
(1) 0.58 A (2) 0.93 A

(3) 1.20 A (4) 0.35 A

PAC133

RE-NEET (UG) 2024

28. In the circuit shown below, the inductance L is connected to an ac source. The current flowing in the circuit is $I = I_0 \sin\omega t$. The voltage drop (V_L) across L is :



(1) $\omega L\, I_0 \sin\omega t$ (2) $\frac{I_0}{\omega L}\sin\omega t$

(3) $\frac{I_0}{\omega L}\cos\omega t$ (4) $\omega L\, I_0 \cos\omega t$

PAC134

29. A step up transformer is connected to an ac mains supply of 220 V to operate at 11000 V, 88 watt. The current in the secondary circuit, ignoring the power loss in the transformer, is:

(1) 8 mA (2) 4 mA

(3) 0.4 A (4) 4 A

PAC135

30. The amplitude of the charge oscillating in a circuit decreases exponentially as $Q = Q_0 e^{-Rt/2L}$, where Q_0 is the charge at t = 0 s. The time at which charge amplitude decreases to 0.50 Q_0 is nearly :

[Given that R = 1.5Ω, L = 12 mH, *ln* (2) = 0.693]

(1) 19.01 ms (2) 11.09 ms

(3) 19.01 s (4) 11.09 s

R5PAC136

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	1	2	1	1	1	2	1	4	4	1	4	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	3	1	3	2	2	4	1	1	2	4	2	4	1	2

Exercise – III (Analytical Questions) Master Your Understanding

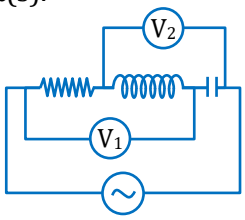
1. Resonance occurs in a series LCR circuit when the frequency of the applied emf is 1000 Hz. Then
(a) when frequency = 900 Hz, then the current through the voltage source will be ahead of e.m.f. of the source
(b) the impedance of the circuit is minimum at f = 1000 Hz
(c) at resonance the voltage across L and current in C differ in phase by 180°
(d) If the value of C is doubled resonance occurs at f = 2000 Hz
(1) only a & b are correct
(2) a, b, c are correct
(3) c & d are correct
(4) all are correct

PAC113

2. If R, C, and L be the resistance, capacitance, and inductance in the circuit in which ac of frequency f is set up, then which of the following have the dimensions of R?
(a) fC (b) fL (c) 1/fC (d) L/f
(1) only a & b (2) only b & c
(3) only c & d (4) all

PAC114

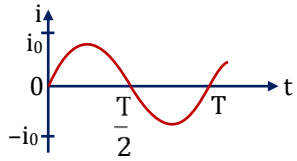
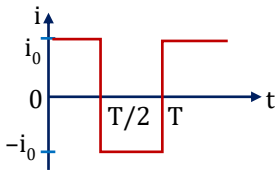
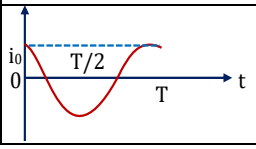
3. In an RLC series circuit shown in figure, the readings of voltmeters V_1 and V_2 are 100 V and 120 V respectively. The source voltage is 130 V. For this situation mark out the correct statement(s).



(a) Voltage across resistor, inductor and capacitor are 50 V, 86.6 V and 206.6 V, respectively
(b) Voltage across resistor, inductor and capacitor are 10V,90 V and 30 V, respectively
(c) Power factor of the circuit is $\frac{5}{13}$
(d) The circuit is capacitive in nature
(1) a, c, d (2) b, c, d
(3) c, d (4) None

PAC115

4. Match the column :
In column I, variation of current I with t is given in figures. In column II root mean square current i_{rms}, and average current are given. Match the column I with corresponding quantities given in Column II.

	Column-I		Column-II
(A)		(P)	$i_{rms} = \frac{i_0}{\sqrt{2}}$
(B)		(Q)	Average current is i_0
(C)		(R)	$I_{rms} = \frac{i_0}{\sqrt{3}}$
(D)		(S)	None of above

(1) a – S, b – Q, c – Q, d – P
(2) a – P, b – Q, c – S, d – P
(3) a – Q, b – P, c – R, d – S
(4) a – P, b – P,Q, c – P, d – P

PAC116

5. Match the following :

	Column-I		Column-II
(A)	LCR circuits	(P)	Resonant curve will be flattened
(B)	Inductor	(Q)	Sharpness indicates sensitivity
(C)	More resistance	(R)	Have resonant frequency $\omega = \frac{1}{\sqrt{LC}}$ with maximum current
(D)	Radio Tuner's characteristic curve (I – f)	(S)	Analogous to mass

(1) A-R, B-S, C-Q, D-P (2) A-Q, B-P, C-Q, D-R
(3) A-S, B-Q, C-R, D-P (4) A-R, B-S, C-P, D-Q

PAC117

6. Match the column

	Column-I		Column-II
(a)	In L-R series AC circuit	(p)	Current lags voltage by $\pi/2$
(b)	In R-C series AC circuit	(q)	Current lags voltage by an angle less than $\pi/2$
(c)	In L-C series AC circuit ($X_L > X_C$)	(r)	Current leads voltage by an angle less than $\pi/2$
(d)	In purely resistive AC circuit	(s)	Current and voltage are in phase

(1) a-q, b-r, c-p, d-s (2) a-s, b-q, c-p, d-r
(3) a-s, b-r, c-q, d-p (4) a-p, b-s, c-r, d-q

PAC118

7. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Smaller the band width, sharper the resonance and easier it is to tune an LCR circuit.

Reason (R) : Resonant frequency is arithmetic mean of half power frequencies.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PAC119

8. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The impedance of series LCR circuit can be greater, equal or less than the resistance.

Reason (R) : The impedance of series LCR circuit depends over angular frequency of applied emf.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PAC120

9. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In a series LCR circuit at resonance, the voltage across the capacitor or inductor may be more than the applied voltage.

Reason (R) : At resonance in a series LCR circuit, the voltages across inductor and capacitor are equal & 180° out of phase.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PAC121

10. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Average power consumed in an ac circuit is equal to average power consumed by resistors in the circuit.

Reason (R) : Average power consumed by capacitor and inductor is zero

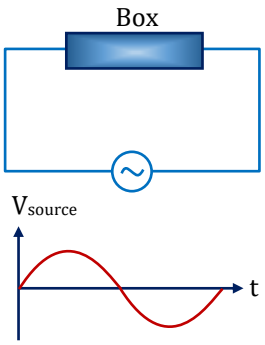
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

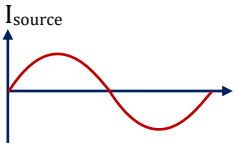
PAC122

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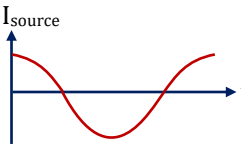
11. Box may have any series combination of L, C and R. Column-I represents source current and column-II represents possible statements.



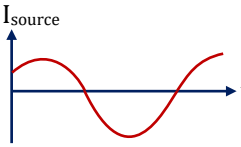
Column-I **Column-II**

(A) 

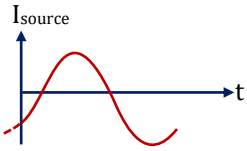
(P) Box may contain LCR.

(B) 

(Q) Box may contain only LR.

(C) 

(R) Power factor of box is zero.

(D) 

(S) Box may contain only CR.

(T) State of resonance.

Options :-

(1) A → P, B → R, C → PS, D → P

(2) A → PT, B → R, C → PS, D → PQ

(3) A → PT, B → PR, C → P, D → PQ

(4) A → P, B → PR, C → P, D → T

PAC124

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	1	2	1	2	4	1	2	4	2	1	2

Important Notes

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Exercise - I (Conceptual Questions) Build Up Your Understanding

1. If $\vec{E}$ and $\vec{B}$ are the electric and magnetic field vectors of electromagnetic waves then the direction of propagation of electromagnetic wave is along the direction of –
(1) $\vec{E}$ (2) $\vec{B}$
(3) $\vec{E} \times \vec{B}$ (4) none of these
PEW001

2. The electromagnetic waves do not transport -
(1) energy (2) charge
(3) momentum (4) information
PEW002

3. The wave function (in S.I. units) for an electromagnetic wave is given as –
$\psi(x,t)=10^3 \sin\pi(3\times10^6 x-9\times10^{14}t)$
The speed of the wave is –
(1) 9×10^{14} m/s (2) 3×10^8 m/s
(3) 3×10^6 m/s (4) 3×10^7 m/s
PEW003

4. In the above problem, wavelength of the wave is –
(1) 666 nm (2) 666 Å
(3) 666 μm (4) 6.66 nm
PEW004

5. In an electromagnetic wave the average energy density is associated with –
(1) electric field only
(2) magnetic field only
(3) equally with electric and magnetic fields
(4) average energy density is zero
PEW005

6. In an electromagnetic wave the energy density associated with magnetic field will be
(1) $\frac{1}{2}LI^2$ (2) $\frac{B^2}{2\mu_0}$
(3) $\frac{1}{2}\mu_0 B^2$ (4) $\frac{1}{2}\frac{\mu_0}{B^2}$
PEW006

7. In the above problem, the energy density associated with the electric field will be –
(1) $\frac{1}{2}CV^2$ (2) $\frac{1}{2}\frac{q^2}{C}$
(3) $\frac{1}{2}\frac{\varepsilon^2}{E}$ (4) $\frac{1}{2}\varepsilon_0 E^2$
PEW007

8. If there were no atmosphere, the average temperature on earth surface would be –
(1) lower (2) higher
(3) same (4) 0° C
PEW008

9. In which part of earth's atmosphere is the ozone layer present?
(1) troposphere (2) stratosphere
(3) ionosphere (4) mesosphere
PEW009

10. The ozone layer is earth's atmosphere is crucial for human survival because it –
(1) contains ions
(2) reflects radio signals
(3) reflects ultraviolet rays
(4) reflects infrared rays
PEW010

11. The frequency from 3×10^9 Hz to 3×10^{10} Hz is –
(1) high frequency band
(2) super high frequency band
(3) ultra high frequency band
(4) very high frequency band
PEW011

12. The frequency from 3MHz to 30 MHz is known as –
(1) audio band
(2) medium frequency band
(3) very high frequency band
(4) high frequency band
PEW012

13. The AM range of radiowaves have frequency –
(1) less than 30 MHz
(2) more than 30 MHz
(3) less than 20000Hz
(4) more than 20000Hz
PEW013

14. Select wrong statement from the following for EMW-
(1) are transverse
(2) travel with same speed in all medium
(3) travel with the speed of light
(4) are produced by accelerating charge
PEW014

15. The waves related to tele-communication are -
(1) infrared (2) visible light
(3) microwaves (4) ultraviolet rays
PEW015

16. The nature of electromagnetic wave is –
(1) longitudinal
(2) longitudinal stationary
(3) transverse
(4) transverse stationary
PEW016

17. Greenhouse effect keeps the earth surface –
(1) cold at night (2) dusty and cold
(3) warm at night (4) moist
PEW017

18. The speed of electromagnetic radiation in vacuum is :-
(1) $\mu_0\varepsilon_0$ (2) $\sqrt{\mu_0\varepsilon_0}$
(3) $\frac{1}{\mu_0\varepsilon_0}$ (4) $\frac{1}{\sqrt{\mu_0\varepsilon_0}}$
PEW018

19. What is the cause of greenhouse effect?
(1) Infrared rays (2) Ultraviolet rays
(3) X-rays (4) Radio waves
PEW019

20. The conduction current is the same as displacement current when source is :-
(1) ac only (2) dc only
(3) both ac and dc (4) neither dc nor ac
PEW020

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	2	1	3	2	4	1	2	3	2	4	1	2	3
Question	16	17	18	19	20										
Answer	3	3	4	1	3										

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Exercise - II (Previous Year Questions) AIPMT/NEET

Re-AIPMT 2015

1. The energy of the em waves is of the order of 15 keV. To which part of the spectrum does it belong?

(1) γ-rays (2) X-rays

(3) Infra-red rays (4) Ultraviolet rays

PEW028

AIPMT 2015

2. A radiation of energy 'E' falls normally on a perfectly reflecting surface. The momentum transferred to the surface is

(c = Speed of light) :-

(1) $\frac{2E}{c}$ (2) $\frac{2E}{c^2}$ (3) $\frac{E}{c^2}$ (4) $\frac{E}{c}$

PEW029

NEET-I 2016

3. Out of the following options which one can be used to produce a propagating electromagnetic wave?

(1) A charge moving at constant velocity

(2) A stationary charge

(3) A chargeless particle

(4) An accelerating charge

PEW030

4. A 100 Ω resistance and a capacitor of 100 Ω reactance are connected in series across a 220 V source. When the capacitor is 50% charged, the peak value of the displacement current is :-

(1) 4.4 A (2) $11\sqrt{2}$ A

(3) 2.2 A (4) 11 A

PEW031

NEET (UG) 2017

5. In an electromagnetic wave in free space the root mean square value of the electric field is E_{rms} = 6V/m. The peak value of the magnetic field is :-

(1) 2.83×10^{-8} T (2) 0.70×10^{-8} T

(3) 4.23×10^{-8} T (4) 1.41×10^{-8} T

PEW032

NEET (UG) 2018

6. An em wave is propagating in a medium with a velocity $\vec{V} = V\hat{i}$. The instantaneous oscillating electric field of this em wave is along +y axis. Then the direction of oscillating magnetic field of the em wave will be along :-

(1) –z direction (2) +z direction

(3) –y direction (4) –x direction

PEW033

NEET (UG) 2019 (Odisha)

7. For a transparent medium relative permeability and permittivity, μ_r and $\in_r$ are 1.0 and 1.44 respectively. The velocity of light in this medium would be,

(1) 2.5×10^8 m/s (2) 3×10^8 m/s

(3) 2.08×10^8 m/s (4) 4.32×10^8 m/s

PEW034

NEET (UG) 2020

8. Light with an average flux of 20 W/cm² falls on a non-reflecting surface at normal incidence having surface area 20 cm². The energy received by the surface during time span of 1 minute is :

(1) 48×10^3 J (2) 10×10^3 J

(3) 12×10^3 J (4) 24×10^3 J

PEW035

9. The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is :

(c = speed of electromagnetic waves)

(1) $1 : c^2$ (2) $c : 1$

(3) 1 : 1 (4) 1 : c

PEW036

NEET (UG) 2020 (COVID-19)

10. The E.M. wave with shortest wavelength among the following is

(1) Ultraviolet rays (2) X-rays

(3) Gamma-rays (4) Microwaves

PEW037

11. The magnetic field in a plane electromagnetic wave is given by :

$B_y = 2 \times 10^{-7} \sin(\pi \times 10^3 x + 3\pi \times 10^{11} t)T$

Calculate the wavelength.

(1) $\pi \times 10^3$ m (2) 2×10^{-3} m

(3) 2×10^3 m (4) $\pi \times 10^{-3}$ m

PEW038

NEET (UG) 2021

12. A capacitor of capacitance 'C', is connected across an ac source of voltage V, given by $V = V_0 \sin \omega t$. The displacement current between the plates of the capacitor, would then be given by :

(1) $I_d = V_0 \omega C \cos \omega t$ (2) $I_d = \frac{V_0}{\omega C} \cos \omega t$

(3) $I_d = \frac{V_0}{\omega C} \sin \omega t$ (4) $I_d = V_0 \omega C \sin \omega t$

PEW039

13. For a plane electromagnetic wave propagating in x-direction, which one of the following combinations gives the correct possible directions for electric field (E) and magnetic field (B) respectively?

(1) $\hat{j}+\hat{k}, \hat{j}+\hat{k}$ (2) $-\hat{j}+\hat{k}, -\hat{j}-\hat{k}$

(3) $\hat{j}+\hat{k}, -\hat{j}-\hat{k}$ (4) $-\hat{j}+\hat{k}, -\hat{j}+\hat{k}$

PEW040

NEET (UG) 2022

14. When light propagates through a material medium of relative permittivity ε_r and relative permeability μ_r, the velocity of light, v is given by : (c = velocity of light in vacuum)

(1) $v = \sqrt{\frac{\mu_r}{\varepsilon_r}}$ (2) $v = \sqrt{\frac{\varepsilon_r}{\mu_r}}$

(3) $v = \frac{c}{\sqrt{\varepsilon_r \mu_r}}$ (4) $v = c$

PEW041

15. Match List - I with List -II

List -I (Electromagnetic waves)		List-II (Wavelength)	
(a)	AM radio waves	(i)	10^{-10} m
(b)	Microwaves	(ii)	10^2 m
(c)	Infrared radiations	(iii)	10^{-2} m
(d)	X-rays	(iv)	10^{-4} m

Choose the correct answer from the options given below :

(1) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

(2) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

(3) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

(4) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

PEW042

RE-NEET (UG) 2022

16. The magnetic field of a plane electromagnetic wave is given by

$\vec{B} = 3\times10^{-8} \cos(1.6\times10^3 x + 48\times10^{10} t)\hat{j}$, then the associated electric field will be :

(1) $3\times10^{-8} \cos(1.6\times10^3 x + 48\times10^{10} t)\hat{i}\, V/m$

(2) $3\times10^{-8} \sin(1.6\times10^3 x + 48\times10^{10} t)\hat{i}\, V/m$

(3) $9\sin(1.6\times10^3 x - 48\times10^{10} t)\hat{k}\, V/m$

(4) $9\cos(1.6\times10^3 x + 48\times10^{10} t)\hat{k}\, V/m$

PEW043

17. The ratio of the magnitude of the magnetic field and electric field intensity of a plane electromagnetic wave in free space of permeability μ_0 and permittivity ε_0 is

(c - velocity of light in free space)

(1) c (2) $\frac{1}{c}$

(3) $\frac{c}{\sqrt{\mu_0 \varepsilon_0}}$ (4) $\frac{\sqrt{\mu_0 \varepsilon_0}}{c}$

PEW044

NEET (UG) 2023

18. In a plane electromagnetic wave travelling in free space, the electric field component oscillates sinusoidally at a frequency of 2.0×10^{10} Hz and amplitude 48 Vm^{-1}. Then the amplitude of oscillating magnetic field is : (Speed of light in free space = 3×10^8 m s^{-1})

(1) 1.6×10^{-8} T (2) 1.6×10^{-7} T

(3) 1.6×10^{-6} T (4) 1.6×10^{-9} T

PEW049

RE-NEET (UG) 2023

19. $\in_0$ and μ_0 are the electric permittivity and magnetic permeability of free space respectively. If the corresponding quantities of a medium are $2\in_0$ and $1.5\mu_0$ respectively, the refractive index of the medium will nearly be :

(1) $\sqrt{2}$ (2) $\sqrt{3}$ (3) 3 (4) 2

PEW050

20. To produce an instantaneous displacement current of 2 mA in the space between the parallel plates of a capacitor of capacitance 4 μF, the rate of change of applied variable potential difference $\left(\frac{dV}{dt}\right)$ must be :

(1) 800 V/s (2) 500 V/s
(3) 200 V/s (4) 400 V/s

PEW051

NEET (UG) 2024

21. A parallel plate capacitor is charged by connecting it to a battery through a resistor. If I is the current in the circuit, then in the gap between the plates :

(1) there is no current.
(2) displacement current of magnitude equal to I flows in the same direction as I.
(3) displacement current of magnitude equal to I flows in a direction opposite to that of I.
(4) displacement current of magnitude greater than I flows but can be in any direction.

PEW052

22. The property which is not of an electromagnetic wave travelling in free space is that :

(1) they are transverse in nature.
(2) the energy density in electric field is equal to energy density in magnetic field.
(3) they travel with a speed equal to $\frac{1}{\sqrt{\mu_0 \in_0}}$
(4) they originate from charges moving with uniform speed.

PEW053

RE-NEET (UG) 2024

23. The electromagnetic radiation which has the smallest wavelength are :

(1) X-rays
(2) Gamma rays
(3) Ultraviolet rays
(4) Microwaves

PEW054

24. If the ratio of relative permeability and relative permittivity of a uniform medium is 1 : 4. The ratio of the magnitudes of electric field intensity (E) to the magnetic field intensity (H) of an EM wave propagating in that medium is

$\left(\text{Given that } \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi\right)$:

(1) $30\pi : 1$ (2) $1 : 120\pi$
(3) $60\pi : 1$ (4) $120\pi : 1$

PEW055

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	4	3	1	2	1	4	3	3	2	1	2	3	3
Question	16	17	18	19	20	21	22	23	24						
Answer	4	2	2	2	2	2	4	2	3						

Exercise – III (Analytical Questions) Master Your Understanding

1. For EM waves

(a) γ rays have higher penetrating power than x rays

(b) x rays have higher penetrating power than γ rays

(c) Infrared has higher wavelength than x-rays

(d) Range of visible light is 4000 Å to 7000 Å

then correct options are

(1) a, b and d (2) a, c and d

(3) a and c (4) a, b and d

PEW045

2. Electric field associated with EM wave in free space is given by $\vec{E}=30\sin\left(6\times10^8 t+kz\right)\hat{i}$ then match the following columns.

	Column-I		Column-II
(a)	Direction of propagation of EM wave	(p)	–y axis
(b)	Wave length of EM wave	(q)	2 m^{-1}
(c)	Direction of magnetic field of EM wave	(r)	3.14 m
(d)	Propagation constant 'k' of EM wave	(s)	'–z' axis

(1) a-p, b-r, c-s, d-r (2) a-s, b-r, c-p, d-q

(3) a-p, b-q, c-s, d-s (4) a-s, b-r, c-p, d-p

PEW046

3. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : MRI is less harmful medical diagnosis technique than CT-scan

Reason (R) : X-rays are used in CT-scan, while MRI uses radio waves.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PEW047

4. **Match the column I with column II -**

	Column-I (Law)		Column-II (Mathematical formula)
(A)	Faraday law	(P)	$\oint\vec{E}.d\vec{A}=\frac{q}{\epsilon_0}$
(B)	Ampere Maxwell law	(Q)	$\oint\vec{B}.d\vec{A}=0$
(C)	Gauss law for magnetism	(R)	$\oint\vec{E}.d\vec{\ell}=-\frac{d\phi_B}{dt}$
(D)	Gauss law for electrostatics	(S)	$\oint\vec{B}.d\vec{\ell}=\mu_0 i+\epsilon_0\mu_0\frac{d\phi_E}{dt}$

Options :-

(1) A → R, B → Q, C → S, D → P

(2) A → P, B → Q, C → S, D → R

(3) A → P, B → S, C → Q, D → R

(4) A → R, B → S, C → Q, D → P

PEW048

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4
Answer	2	2	1	4

Exercise - I (Conceptual Questions) Build Up Your Understanding

REFLECTION OF LIGHT AT PLANE SURFACE

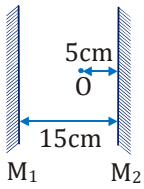
1. When a ray of light is incident normally on a plane mirror then the angle of reflection will be
(1) 0° (2) 90° (3) 180° (4) – 90°
PRI001

2. A ray is incident at 30° angle on plane mirror. What will be deviation after reflection from mirror.
(1) 120° (2) 60° (3) 30° (4) 45°
PRI002

3. Two plane mirrors are lying perpendicular to each other, there is lamp in between mirrors. Then number of images of lamp will be
(1) 3 (2) 4 (3) 5 (4) 6
PRI003

4. A man runs towards stationary plane mirror at a speed of 15 m/s. What is the speed of his image with respect to mirror :–
(1) 7.5 m/s (2) 15 m/s
(3) 30 m/s (4) 45 m/s
PRI004

5. Figure shows two plane mirrors parallel to each other and an object O placed between them. Then the distance of the first three images from the mirror M_2 will be :(in cm)



(1) 5, 10, 15 (2) 5, 15, 30
(3) 5, 25, 35 (4) 5, 15, 25
PRI005

6. A ray gets successively reflection from two mirrors inclined at an angle of 40°. If the angle of incidence on the first mirror is 30° then the net deviation of this ray after two reflections.
(1) 40° (2) 280° (3) 90° (4) 240°
PRI006

7. A tall man of height 6 feet, want to see his full image. Then required minimum length of the mirror will be :-
(1) 12 feet (2) 3 feet
(3) 6 feet (4) Any length
PRI007

8. A man is 180 cm tall and his eyes are 10 cm below the top of his head. In order to see his entire height right from toe to head, he uses a plane mirror kept at a distance of 1 m from him. The minimum length of the plane mirror required is
(1) 180 cm (2) 90 cm
(3) 85 cm (4)170 cm
PRI008

9. Two plane mirrors are at 45° to each other. If an object is placed between them then the number of images will be
(1) 5 (2) 9 (3) 7 (4) 8
PRI009

REFLECTION AT SPHERICAL SURFACE

10. The focal length of a concave mirror is 50 cm. where an object be placed so that its image is two times magnified, real and inverted -
(1) 75 cm (2) 72 cm
(3) 63 cm (4) 50 cm
PRI010

11. An object of height 7.5 cm is placed in front of a convex mirror of radius of curvature 25 cm at a distance of 40 cm. The height of the image should be -
(1) 2.3 cm (2) 1.78 cm
(3) 1 cm (4) 0.8 cm
PRI011

12. A square of side 3 cm is placed at a distance of 25 cm from a concave mirror of focal length 10 cm. The centre of the square is at the axis of the mirror and the plane is normal to the axis. The area enclosed by the image of the wire is -
(1) 4 cm^2 (2) 6 cm^2
(3) 16 cm^2 (4)36 cm^2
PRI012

13. The focal length of a concave mirror is 12 cm. Where should an object of height 4 cm be placed, so that a real image of 1 cm height is formed ?
(1) 48 cm (2) 3 cm
(3) 60 cm (4) 15 cm
PRI013

14. An object is lying at a distance of 90 cm from a concave mirror of focal length 30 cm. The position and nature of image formed by it will be
(1) 45 cm, of the size of object
(2) 90 cm, smaller than object
(3) 30 cm, bigger than object
(4) – 45 cm smaller than object

PRI014

15. An object of height 1.5 cm is situated at a distance of 15 cm from a concave mirror. The concave mirror forms its real image of height 3.0 cm. The focal length of concave mirror will be
(1) – 10 cm (2) – 20 cm
(3) 20 cm (4) 30 cm

PRI015

16. A boy stands straight infront of a mirror at a distance of 30 cm away from it. He sees his erect image whose height is $1/5^{th}$ of his real height. The mirror he is using is -
(1) Plane mirror (2) Convex mirror
(3) Concave mirror (4) None

PRI016

17. A point object is moving on the principal axis of a concave mirror of focal length 24 cm towards the mirror. When it is at a distance of 60 cm from the mirror, its velocity is 9 cm/sec. What is the velocity of the image at that instant -
(1) 5 cm/sec. (2) 12 cm/sec
(3) 4 cm/sec (4) 9 cm/sec

PRI017

18. A concave mirror gives an image three times as large as the object placed at a distance of 20 cm from it. For the image to be real, the focal length should be -
(1) –10 cm (2) –15 cm
(3) –20 cm (4) –30 cm

PRI018

19. The minimum distance between the object and its real image for concave mirror is
(1) f (2) 2f (3) 4f (4) Zero

PRI019

20. A convex mirror has a radius of curvature of 22 cm. If an object is placed 14 cm away from the mirror then its image is formed at-
(1) 6.2 cm on the front side of the mirror
(2) 6.2 cm on the back side of the mirror.
(3) 51.3 cm on the front of the mirror.
(4) 51.3 cm on the back side of the mirror.

PRI020

21. The focal length of a spherical mirror is :-
(1) Maximum for red light
(2) Maximum for blue light
(3) Maximum for white light
(4) Same for all lights

PRI021

22. A point object is placed at a distance of 10 cm and its real image is formed at a distance of 20 cm from a concave mirror. If the object is moved by 0.1 cm towards the mirror, the image will shift by about.
(1) 0.4 cm away from the mirror
(2) 0.4 cm towards the mirror
(3) 0.8 cm away from the mirror
(4) 0.8 cm towards the mirror

PRI022

23. The focal length of a convex mirror is 20 cm its radius of curvature will be
(1) 10 cm (2) 20 cm
(3) 30 cm (4) 40 cm

PRI023

24. A diminished virtual image can be formed only in
(1) Plane mirror (2) A concave mirror
(3) A convex mirror (4) None

PRI024

REFRACTION AT PLANE SURFACE : SNELL'S LAWS TIR

25. The colour are characterised by which of following character of light-
(1) Frequency (2) Amplitude
(3) Wavelength (4) Velocity

PRI025

26. The wavelength of light in two liquids 'x' and 'y' is 3500 Å and 7000Å, then the critical angle will be
(1) 60° (2) 45° (3) 30° (4) 15°

PRI026

27. A microscope is focused on a mark, then a glass slab of refractive index 1.5 and thickness of 6 cm is placed on the mark to get the mark again in focus, the microscope should be moved

(1) 4 cm (2) 2 cm (3) 6 cm (4) 8 cm

PRI027

28. Velocity of light in glass, whose refractive index w.r.t. air is 1.5, is 2×10^8 m/Sec. In a certain liquid the velocity of light is found to be 2.5×10^8 m/Sec. The refractive index of liquid w.r.t. air is

(1) 0.64 (2) 0.80 (3) 1.20 (4) 1.44

PRI028

29. A point source of light is place 4 m below the surface of water of refractive index 5/3. The minimum diameter of a disc which should be placed over the source on the surface of water to cut-off all light coming out of water is.

(1) 2 m (2) 6 m (3) 4 m (4) 3 m

PRI029

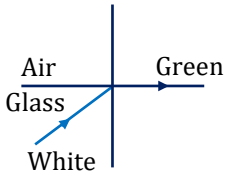
30. When a ray of light enters a medium of refractive index μ, it is observed that the angle of refraction is half of the angle of incidence is than angle of incidence is

(1) $2\cos^{-1}(\mu/2)$ (2) $\cos^{-1}(\mu/2)$

(3) $2\cos^{-1}(\mu)$ (4) $2\sin^{-1}(\mu/2)$

PRI030

31. White light is incident on the interface of glass and air as shown in the figure. If green light is just totally internally reflected then the emerging ray in air contains



(1) Yellow, orange, red

(2) Violet, indigo, blue

(3) All colours

(4) All colours except green

PRI031

32. A plane glass slab is kept over various coloured letters; the letter which appears least raised is

(1) blue (2) voilet (3) green (4) red

PRI032

33. A bubble in glass slab (μ = 1.5) when viewed from one side appears at 5 cm and 2cm from other side, then thickness of slab is:-

(1) 3.75 cm (2) 3 cm

(3) 10.5 cm. (4) 2.5 cm

PRI033

34. A ray of light travelling in air have wavelength λ , frequency n, velocity V and intensity I. If this ray enters into water than these parameters are λ' , n', v' and I' respectively. Which relation is correct from following-

(1) $\lambda = \lambda'$ (2) $n = n'$ (3) $v = v'$ (4) $I = I'$

PRI034

35. Light travels through a glass plate of thickness t and having refractive index n. If c is the velocity of light in vacuum. the time taken by the light to travel this thickness of glass is

(1) $\frac{t}{nc}$ (2) tnc (3) $\frac{nt}{c}$ (4) $\frac{tc}{n}$

PRI035

36. A ray of light propagates from glass (refractive index = 3/2) to water (refractive index = 4/3). The value of the critical angle

(1) $\sin^{-1}(1/2)$ (2) $\sin^{-1}\left(\frac{\sqrt{8}}{9}\right)$

(3) $\sin^{-1}(8/9)$ (4) $\sin^{-1}(5/7)$

PRI036

37. Relation between critical angles of water and glass is

(1) $C_w > C_g$ (2) $C_w < C_g$

(3) $C_w = C_g$ (4) $C_w = C_g = 0$

PRI037

38. Critical angle of light passing from glass to air is minimum for

(1) Red (2) Green

(3) Yellow (4) Violet

PRI038

39. Which of the following is used in optical fibres
(1) T.I.R.
(2) Scattering
(3) Diffraction
(4) Refraction

PRI039

40. Brilliance of diamond is due to
(1) Shape (2) Cutting
(3) Reflection (4) T.I.R.

PRI040

41. 'Mirage' is a phenomenon due to
(1) Reflection of light
(2) Refraction of light
(3) T.I.R. of light
(4) Diffraction of light

PRI041

42. An object is immersed in a fluid. In order that the object becomes invisible, it should.
(1) Bahave as a perfect reflector
(2) Absorb all light falling on it
(3) Have refractive index one
(4) Have refractive index exactly matching with that of the surrounding fluid

PRI042

PRISM

43. The angle of a glass prism is 4.5° and its refractive index is 1.52. The angle of minimum deviation will be –
(1) 1.5° (2) 2.3°
(3) 4.5° (4) 2°

PRI043

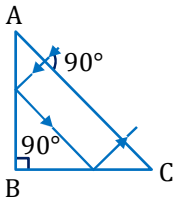
44 A ray of light passes through equilateral Prism (μ =1.5) such that angle of incidence is equal to angle of emergence and the later is equal to 3/4th of Prism angle. The angle of deviation is
(1) 60° (2) 30°
(3) 45° (4) 120°

PRI044

45. The angle of minimum deviation measured with a prism is 30° and the angle of prism is 60°. The refractive index of prism material is -
(1) $\sqrt{2}$ (2) 2
(3) $\frac{3}{2}$ (4) $\frac{4}{3}$

PRI048

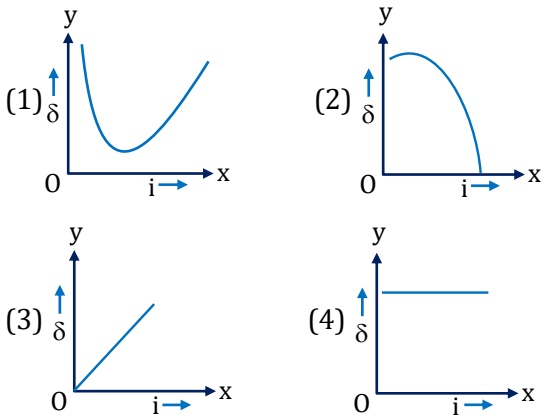
46. A ray falls on a prism ABC (AB = BC) and travels as shown in figure. The minimum refractive index of the prism material should be



(1) $\frac{4}{3}$ (2) $\sqrt{2}$
(3) 1.5 (4) $\sqrt{3}$

PRI050

47. For a prism, graph between angle of deviation (δ) and angle of incidence will be



PRI051

48. Three prisms 1,2 and 3 have the prism angle A = 60°, but their refractive indices are respectively 1.4, 1.5 and 1.6. If δ_1, δ_2,δ_3, be their respective angles of deviation then :–
(1) $\delta_3 > \delta_2 > \delta_1$ (2) $\delta_1 > \delta_2 > \delta_3$
(3) $\delta_1 = \delta_2 = \delta_3$ (4) $\delta_2 > \delta_1 > \delta_3$

PRI052

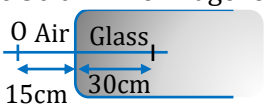
49. The refractive index of the material of a prism is $\sqrt{2}$ and its refracting angle is 30°. One of the refracting surfaces of the prism is silvered. A beam of monochromatic light entering the prism from the other face will retrace its path after reflection from the silvered surface if its angle of incidence on the prism is :–

(1) 60° (2) 0° (3) 30° (4) 45°

PRI054

REFRACTION AT SPHERICAL SURFACES

50. A point object O is placed in front of a glass rod having spherical end of radius of curvature 30 cm. The image formed would be



(1) 30 cm left (2) infinity

(3) 1 cm to the right (4) 18 cm to the left

PRI055

51. A spherical surface of radius of curvature R separates air (refractive index 1.0) from glass (refractive index 1.5). The centre of curvature is in glass. A point object P placed in air is found to have a real image Q in the glass. The line PQ cuts the surface at a point O and PO = OQ. The distance PO is equal to

(1) R (2) 3 R (3) 2 R (4) 5 R

PRI056

LENSES

52. Two convex lens of focal length 20 cm and 25 cm are placed in contact with each other, then power of this combination is –

(1) + 1 D (2) + 9 D

(3) – 1 D (4) – 9 D

PRI057

53. Two lenses of power +2.50 D and –3.75 D are combined to form a compound lens. Its focal length in cm will be -

(1) 40 (2) –40 (3) –80 (4) 160

PRI058

54. Lenses of powers 3D and –5D are combined to form a compound lens. An object is placed at a distance of 50 cm from this lens. Calculate the position of its image.

(1) –10 cm (2) +10 cm

(3) –25 cm (4) + 25 cm.

PRI059

55. A convex lens of Focal length of 40cm is in contact with a concave lens of focal length 25cm. The power of the combination is.

(1) –1.5 D (2) -6.5 D

(3) +6.5 D (4) +6.67 D

PRI060

56. An object is put at a distance of 5 cm from the first focus of a convex lens of focal length 10cm. If a real image is formed it's distance from the lens will be :-

(1) 15 cm (2) 20 cm

(3) 25 cm (4) 30 cm

PRI061

57. An equiconvex lens has a power of 5 dioptre. If it is made of glass of refractive index 1.5. then radius of curvature of its each surface will be ?

(1) 20 cm (2) 10 cm (3) 40 cm (4) ∞

PRI062

58. An object placed at a distance of 9cm from first principal focus of convex lens, produces a real image at a distance of 25cm from its second principal focus. Then focal length of lens is

(1) 9 cm (2) 25 cm

(3) 15 cm (4) 17 cm

PRI063

59. A glass convex lens (μ_g = 1.5) has a focal length of 8 cm when placed in air. What would be the focal length of the lens when it is immersed in water (μ_w = 1.33)

(1) 2 m (2) 4 cm

(3) 16 cm (4) 32 cm

PRI064

60. Two thin convex lenses of focal length 10 cm and 15 cm are separated by a distance of 10 cm. The focal length of the combination is :-

(1) 4.2 cm (2) 6 cm

(3) 10 cm (4) 15 cm

PRI065

61. A convex lens of power P is immersed in water. how will its power change ?

(1) Increases

(2) Decreases

(3) Remains unchanged

(4) Increases for red colour and decreases for blue colour

PRI066

62. A convex lens is made up of three different materials as shown in the figure. For a point object placed on its axis, the number of images formed are



(1) 1 (2) 3 (3) 4 (4) 5

PRI067

63. The focal length of a convex lens is 10 cm and its refractive index is 1.5. If the radius of curvature of one surface is 7.5 cm, the radius of curvature of the second surface will be

(1) 7.5 cm (2) 15.0 cm
(3) 75 cm (4) 5.0 cm

PRI068

64. A plano convex lens is made of refractive index 1.6. The radius of curvature of the curved surface is 60 cm. The focal length of the lens is

(1) 50 cm (2) 100 cm
(3) 200 cm (4) 400 cm

PRI070

65. If in a plano-convex lens. the radius of curvature of the convex surface is 10 cm and the focal length of the lens is 30 cm, then the refractive index of the material of lens will be

(1) 1.5 (2) 1.66 (3) 1.33 (4) 3

PRI071

66. If a convex lens of focal length 80 cm and a concave lens of focal length 50 cm are combined together, what will be their resulting power

(1) + 6.5 D (2) – 6.5 D
(3) + 7.5 D (4) –0.75 D

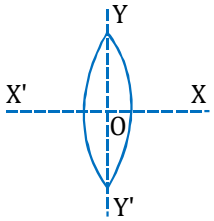
PRI072

67. A convex lens is dipped in a liquid whose refractive index is equal to the refractive index of the lens. Then its focal length will :-

(1) Become zero
(2) Become infinite
(3) Become small, but non–zero
(4) Remain unchanged

PRI073

68. An equiconvex lens is cut into two halves along (i) XOX′ and (ii) YOY′ as shown in the figure. Let f, f′, f″ be the focal lengths of the complete lens, of each half in case (i), and of each half in case (ii), respectively Choose the correct statement from the following :–



(1) f′ = f, f″ = 2f (2) f′ = 2f, f″ = f
(3) f′ = f, f″ = f (4) f′ = 2f, f″ = 2f

PRI074

69. An object and its real image are located at distances 25 cm and 40 cm respectively from the two principal focii of a convex lens. The linear magnification of the image is near to

(1) + 1.3 (2) –1.3
(3) + 1.8 (4) –1.8

PRI075

MICROSCOPES AND TELESCOPES

70. The focal length of convex lens is 2.5 cm. Its magnifying power for minimum distance of distinct vision will be -

(1) 25 (2) 52 (3) 11 (4) 1.1

PRI082

71. An astronomical telescope of magnifying power 8 is made using two lenses spaced 45 cm apart. The focal length of the lenses used are

(1) f = 40 cm, f = 5 cm (2) f = 8 cm, f = 5 cm
(3) f = 5 cm, f = 47 cm (4) f = 20 cm, f = 5 cm

PRI083

72. The magnifying power of the objective of a compound microscope is 7 if the magnifying power of the microscope is 35, then the magnifying power of eyepiece will be -

(1) 245 (2) 5 (3) 28 (4) 42

PRI084

73. An astronomical telescope has focal length 100 cm and 10 cm of objective and eyepiece lens respectively when final image is formed at least distance of distinct vision, magnification power of telescope will be :

(1) –10 (2) –11 (3) –14 (4) –15

PRI085

74. If tube length of astronomical telescope is 105 cm and magnifying power is 20 for normal setting. Calculate the focal length of objective.
(1) 100 cm (2) 10 cm
(3) 20 cm (4) 25 cm

PRI086

75. Least distance of distinct vision is 25 cm, what will be Magnifying power of simple microscope of focal length 5 cm, if final image is formed at minimum distance of distinct vision –
(1) $\frac{1}{5}$ (2) 5 (3) $\frac{1}{6}$ (4) 6

PRI087

76. In a compound microscope, the intermediate image, in normal use is
(1) Virtual, erect and magnified
(2) Real, erect and magnified
(3) Real, inverted and magnified
(4) Virtual, inverted and magnified

PRI088

77. The focal lengths of the objective and eye-piece of a telescope are respectively 100 cm and 2 cm. The moon subtends an angle of 0.5° at the eye. If it is looked through the telescope, the angle subtended by the moon's image will be
(1) 100° (2) 50° (3) 25° (4) 10°

PRI089

78. In a laboratory four convex lenses L_1, L_2, L_3 and L_4 of focal lengths 2,4,6 and 8 cm respectively are available. Two of these lenses form a telescope of length 10 cm and magnifying power 4. The objective and eye lenses are
(1) L_2 , L_3 (2) L_1 , L_4
(3) L_3 , L_2 (4) L_4 , L_1

PRI090

79. The magnifying power of a simple microscope is 6. The focal length of its lens in metres will be, if least distance of distinct vision is 25 cm
(1) 0.05 (2) 0.06 (3) 0.25 (4) 0.12

PRI091

80. An astronomical telescope has a magnifying power 10. The focal length of eyepiece is 20 cm. The focal length of objective is
(1) 2 cm (2) 200cm
(3) $\frac{1}{2}$ cm (4) $\frac{1}{200}$ cm

PRI092

81. A telescope has an objective lens of focal length 200 cm and an eye piece with focal length 2cm. If this telescope is used to see a 50 meter tall building at a distance of 2km, what is the height of the image of the building formed by the objective lens
(1) 5 cm (2) 10 cm
(3) 1 cm (4) 2 cm

PRI093

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	1	2	3	2	2	2	3	1	2	1	3	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	2	4	2	4	1	4	3	1	3	2	3	2	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	4	3	2	3	3	1	4	1	4	3	4	2	2	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	1	1	4	1	4	2	3	3	1	4	1	3	4	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	2	2	2	3	4	2	1	2	3	1	2	3	1	4
Question	76	77	78	79	80	81									
Answer	3	3	4	1	2	1									

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. If the focal length of objective lens is increased then magnifying power of :-
(1) microscope will increase but that of telescope decrease.
(2) microscope and telescope both will increase.
(3) microscope and telescope both will decrease
(4) microscope will decrease but that of telescope increase.

PRI121

2. The angle of a prism is 'A'. One of its refracting surfaces is silvered. Light rays falling at an angle of incidence 2A on the first surface returns back through the same path after suffering reflection at the silvered surface. The refractive index μ, of the prism is :-
(1) 2sinA (2) 2cos A
(3) $\frac{1}{2}$ cosA (4) tanA

PRI122

AIPMT 2015

3. The refracting angle of a prism is A, and refractive index of the material of the prism is cot(A/2). The angle of minimum deviation is :-
(1) 180° – 2A (2) 90° – A
(3) 180° + 2A (4) 180° – 3A

PRI123

4. Two identical thin plano-convex glass lenses (refractive index 1.5) each having radius of curvature of 20 cm are placed with their convex surfaces in contact at the centre. The intervening space is filled with oil of refractive index 1.7. The focal length of the combination is :-
(1) –25 cm (2) –50 cm
(3) 50 cm (4) –20 cm

PRI124

NEET-I 2016

5. The angle of incidence for a ray of light at a refracting surface of a prism is 45°. The angle of prism is 60°. If the ray suffers minimum deviation through the prism, the angle of minimum deviation and refractive index of the material of the prism respectively, are :
(1) 45°, $\frac{1}{\sqrt{2}}$ (2) 30°, $\sqrt{2}$
(3) 45°, $\sqrt{2}$ (4) 30°, $\frac{1}{\sqrt{2}}$

PRI126

6. An astronomical telescope has objective and eyepiece of focal lengths 40 cm and 4 cm respectively. To view an object 200 cm away from the objective, the lenses must be separated by a distance :-
(1) 37.3 cm (2) 46.0 cm
(3) 50.0 cm (4) 54.0 cm

PRI127

7. Match the corresponding entries of **column-1** with **coloumn-2** (Where m is the magnification produced by the mirror) :-

Column-1	**Column-2**
(A) m = –2	(a) Convex mirror
(B) m = $-\frac{1}{2}$	(b) Concave mirror
(C) m = +2	(c) Real image
(D) m = $+\frac{1}{2}$	(d) Virtual image

(1) A → b and c, B → b and c, C → b and d, D → a and d.
(2) A → a and c, B → a and d, C → a and b, D → c and d
(3) A → a and d, B → b and c, C → b and d, D → b and c
(4) A → c and d, B → b and d, C → b and c, D → a and d

PRI128

NEET-II 2016

8. Two identical glass (μ_g = 3/2) equiconvex lenses of focal length f each are kept in contact. The space between the two lenses is filled with water (μ_w = 4 / 3). The focal length of the combination is :-
(1) 4f/3 (2) 3f/4 (3) f/3 (4) f

PRI129

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9. An air bubble in a glass slab with refractive index 1.5 (near normal incidence) is 5 cm deep when viewed from one surface and 3 cm deep when viewed from the opposite face. The thickness (in cm) of the slab is :-

(1) 12 (2) 16 (3) 8 (4) 10

PRI130

NEET(UG) 2017

10. A beam of light from a source L is incident normally on a plane mirror fixed at a certain distance x from the source. The beam is reflected back as a spot on a scale placed just above the source L. When the mirror is rotated through a small angle θ, the spot of the light is found to move through a distance y on the scale. The angle θ is given by :-

(1) $\frac{y}{x}$ (2) $\frac{x}{2y}$ (3) $\frac{x}{y}$ (4) $\frac{y}{2x}$

PRI132

NEET(UG) 2018

11. The refractive index of the material of a prism is $\sqrt{2}$ and the angle of the prism is 30°. One of the two refracting surfaces of the prism is made a mirror inwards, by silver coating. A beam of monochromatic light entering the prism from the other face will retrace its path (after reflection from the silvered surface) if its angle of incidence on the prism is :-

(1) 60° (2) 45° (3) 30° (4) zero

PRI134

12. An object is placed at a distance of 40 cm from a concave mirror of focal length 15 cm. If the object is displaced through a distance of 20 cm towards the mirror, the displacement of the image will be:-

(1) 30 cm away from the mirror
(2) 36 cm away from the mirror
(3) 30 cm towards the mirror
(4) 36 cm towards the mirror

PRI135

NEET(UG) 2019

13. Which colour of the light has the longest wavelength ?

(1) red (2) blue
(3) green (4) violet

PRI136

14. Two similar thin equiconvex lenses, of focal length f each, are kept coaxially in contact with each other such that the focal length of the combination is F_1. When the space between the two lenses is filled with glycerine (which has the same refractive index (μ = 1.5) as that of glass) then the equivalent focal length is F_2. The ratio $F_1 : F_2$ will be :

(1) 2 : 1 (2) 1 : 2 (3) 2 : 3 (4) 3 : 4

PRI138

15. In total internal reflection when the angle of incidence is equal to the critical angle for the pair of media in contact, what will be angle of refraction?

(1) 180°
(2) 0°
(3) equal to angle of incidence
(4) 90°

PRI139

NEET(UG) 2019 (Odisha)

16. An equiconvex lens has power P. It is cut into two symmetrical halves by a plane containing the principal axis. The power of one part will be:

(1) 0 (2) $\frac{P}{2}$ (3) $\frac{P}{4}$ (4) P

PRI140

17. A double convex lens has focal length 25 cm. The radius of curvature of one of the surfaces is double of the other. Find the radii if the refractive index of the material of the lens is 1.5:

(1) 100 cm, 50 cm (2) 25 cm, 50 cm
(3) 18.75 cm, 37.5 cm (4) 50 cm, 100 cm

PRI141

NEET(UG) 2020

18. A ray is incident at an angle of incidence i on one surface of a small angle prism (with angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ, then the angle of incidence is nearly equal to:

(1) $\frac{\mu A}{2}$ (2) $\frac{A}{2\mu}$ (3) $\frac{2A}{\mu}$ (4) μA

PRI142

NEET(UG) 2020 (COVID-19)

19. A plano-convex lens of unknown material and unknown focal length is given. With the help of a spherometer we can measure the
(1) focal length of the lens
(2) radius of curvature of the curved surface
(3) aperture of the lens
(4) refractive index of the material

PRI143

20. The power of a biconvex lens is 10 dioptre and the radius of curvature of each surface is 10 cm. Then the refractive index of the material of the lens is,

(1) $\frac{4}{3}$ (2) $\frac{9}{8}$ (3) $\frac{5}{3}$ (4) $\frac{3}{2}$

PRI144

21. An object is placed on the principal axis of a concave mirror at a distance of 1.5 f (f is the focal length). The image will be at,

(1) –3 f (2) 1.5 f (3) –1.5 f (4) 3 f

PRI145

22. If the critical angle for total internal reflection from a medium to vacuum is 45°, then velocity of light in the medium is,

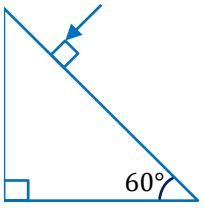
(1) 1.5×10^8 m/s (2) $\frac{3}{\sqrt{2}} \times 10^8$ m/s

(3) $\sqrt{2} \times 10^8$ m/s (4) 3×10^8 m/s

PRI146

NEET(UG) 2021

23. Find the value of the angle of emergence from the prism. Refractive index of the glass is $\sqrt{3}$.



(1) 60° (2) 30° (3) 45° (4) 90°

PRI147

24. A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance 'd' between them. If a parallel beam of light falling on 'A' leaves 'B' as a parallel beam, then the distance 'd' in cm will be :-

(1) 25 (2) 15 (3) 50 (4) 30

PRI148

NEET(UG) 2022

25. A biconvex lens has radii of curvature, 20 cm each. if the refractive index of the material of the lens is 1.5, the power of the lens is :-

(1) +20 D (2) +5D
(3) infinity (4) +2D

PRI150

26. Two transparent media A and B are separated by a plane boundary. The speed of light in those media are 1.5×10^8 m/s and 2.0×10^8 m/s, respectively. The critical angle for a ray of light for these two media is:

(1) $\sin^{-1}(0.750)$ (2) $\tan^{-1}(0.500)$
(3) $\tan^{-1}(0.750)$ (4) $\sin^{-1}(0.500)$

PRI152

RE-NEET(UG) 2022

27. An astronomical refracting telescope is being used by an observer to observe planets in normal adjustment. The focal lengths of the objective and eye piece used in the construction of the telescope are 20 m and 2cm respectively. Consider the following statements about the telescope :

(a) The distance between the objective and eye piece is 20.02 m
(b) The magnification of the telescope is (–) 1000
(c) The image of the planet is erect and diminished
(d) The aperture of eye piece is smaller than that of objective

The correct statements are :

(1) (a), (b) and (c) (2) (b), (c) and (d)
(3) (c), (d) and (a) (4) (a), (b) and (d)

PRI154

NEET(UG) 2023

28. Light travels a distance x in time t_1 in air and 10x in time t_2 in another denser medium. What is the critical angle for this medium ?

(1) $\sin^{-1}\left(\frac{10t_2}{t_1}\right)$ (2) $\sin^{-1}\left(\frac{t_1}{10t_2}\right)$

(3) $\sin^{-1}\left(\frac{10t_1}{t_2}\right)$ (4) $\sin^{-1}\left(\frac{t_2}{t_1}\right)$

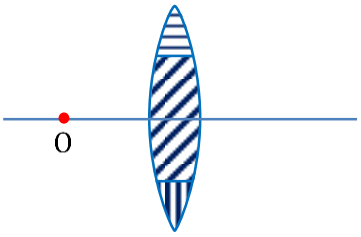
PRI171

29. Two thin lenses are of same focal lengths (f), but one is convex and the other one is concave. When they are placed in contact with each other, the equivalent focal length of the combination will be:
(1) $f/4$ (2) $f/2$
(3) Infinite (4) Zero

PRI172

RE-NEET(UG) 2023

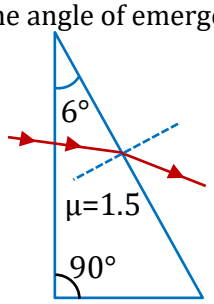
30. A lens is made up of 3 different transparent media as shown in figure. A point object O is placed on its axis beyond 2f. How many real images will be obtained on the other side?



(1) 2
(2) 1
(3) No image will be formed
(4) 3

PRI173

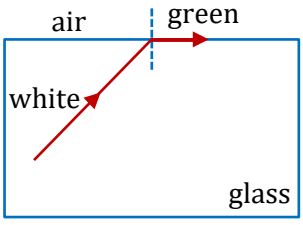
31. A horizontal ray of light is incident on the right angled prism with prism angle 6°. If the refractive index of the material of the prism is 1.5, then the angle of emergence will be:



(1) 9° (2) 10° (3) 4° (4) 6°

PRI174

32. Which set of colours will come out in air for a situation shown in figure?



(1) Yellow, Orange and Red
(2) All
(3) Orange, Red and Violet
(4) Blue, Green and Yellow

PRI175

33. An object is mounted on a wall. Its image of equal size is to be obtained on a parallel wall with the help of a convex lens placed between these walls. The lens is kept at distance x in front of the second wall. The required focal length of the lens will be :

(1) less than $\frac{x}{4}$

(2) more than $\frac{x}{4}$ but less than $\frac{x}{2}$

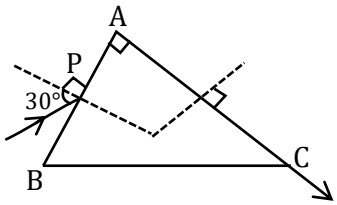
(3) $\frac{x}{2}$

(4) $\frac{x}{4}$

PRI176

NEET(UG) 2024

34. A light ray enters through a right angled prism at point P with the angle of incidence 30° as shown in figure. It travels through the prism parallel to its base BC and emerges along the face AC. The refractive index of the prism is :



(1) $\frac{\sqrt{5}}{4}$ (2) $\frac{\sqrt{5}}{2}$
(3) $\frac{\sqrt{3}}{4}$ (4) $\frac{\sqrt{3}}{2}$

PRI177

35. A small telescope has an objective of focal length 140 cm and an eye piece of focal length 5.0 cm. The magnifying power of telescope for viewing a distant object is :
(1) 34 (2) 28
(3) 17 (4) 32

PRI178

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RE-NEET(UG) 2024

36. Given below are two statements:

Statement I : Image formation needs regular reflection and/or refraction.

Statement II : The variety in colour of objects we see around us is due to the constituent colours of the light incident on them.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) **Statement I** is correct but **Statement II** is incorrect

(2) **Statement I** is incorrect but **Statement II** is correct.

(3) Both **Statement I** and **Statement II** are correct

(4) Both **Statement I** and **Statement II** are incorrect

PRI179

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	1	2	2	4	1	2	1	4	2	2	1	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	4	2	4	1	2	1	2	2	1	4	3	3	4
Question	31	32	33	34	35	36									
Answer	1	1	3	2	2	3									

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Exercise - III (Analytical Questions) Master Your Understanding

1. Choose correct answer :

	Column-I		Column-II
(A)	Human Eye	(P)	Enlarged and inverted image
(B)	Simple microscope	(Q)	Enlarged and erect image
(C)	Compound microscope	(R)	Diminished and inverted image
(D)	Astronomical telescope	(S)	Diminished and erect image

Optics

(1) A → P, B → Q, C → P, D → P
(2) A → S, B → P, C → Q, D → R
(3) A → R, B → P, C → R, D → Q
(4) A → R, B → Q, C → P, D → R

PRI155

2. Given blow are two statements:

Statement-1 : Magnifying power of a simple microscope is positive.

Statement-2 : Magnifying power of a compound microscope is negative.

In light of above statement choose the most appropriate answer from the options given below:

(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

PRI156

3. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Magnifying power of a simple microscope is inversely proportional to focal length of lens in normal adjustment.

Reason (R) : Magnifying power of simple microscope can be infinite.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PRI157

4. A bird is flying at the height of 12 cm from the surface of a lake and a fish is swimming at a depth of 24 cm from the surface.

(Take is $\mu = 4/3$)

	Column-A		Column-B
(A)	Distance of fish from the surface as seen by bird	(P)	16 cm
(B)	Distance of bird from the surface as seen by fish	(Q)	40 cm
(C)	Distance between fish and bird as seen by bird	(R)	18 cm
(D)	Distance between fish and bird as seen by fish	(S)	30 cm

(1) A → P, B → R, C → S, D → Q
(2) A → R, B → P, C → Q, D → S
(3) A → R, B → P, C → S, D → Q
(4) A → P, B → R, C → Q, D → S

PRI159

5. (1) When a light ray goes from denser to rarer medium it may suffer TIR
(2) For TIR to take place angle of incidence should be greater than critical angle.
(3) Laws of reflection are not valid for TIR
(4) Intensity gets decreased during TIR

Incorrect statements are -

(1) 1 and 2
(2) 3 and 4
(3) 1, 3 and 4
(4) 1, 2, 3 and 4

PRI160

6. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Keeping a point object fixed,

if a plane mirror is moved, the image will also move always.

Reason (R) : In case of a plane mirror, distance of object and its image is equal from any point on the mirror.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PRI161

7. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : A thin lens has two principal focal lengths which may be different in magnitude.

Reason (R) : The distance of both principal focus from optical centre of thin lens depend on the two radii of curvature of the lens. Distance of both principal focus from optical centre of a lens are same only if radii of curvature of both sides of lens are same.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) Both **(A)** and **(R)** are false.

PRI162

8. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : When the upper half of a converging lens is missing, a real image formed by the lens for a real object will lack its lower half.

Reason (R) : The real image formed by a thin lens for a real object will be always erected.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) Both **(A)** and **(R)** are false.

PRI163

9. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A simple microscope may have different magnification for different persons.

Reason (R) : All persons must have the same near point distance of 25 cm.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PRI164

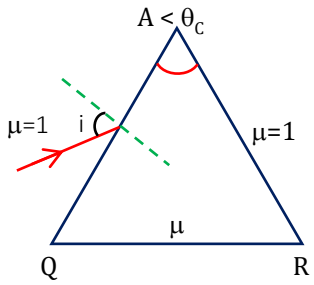
10. Column Matching :-

	Column-I		Column-II
(A)	An object is placed at a distance equal to focal length from pole before convex mirror	(P)	Magnification is ∞
(B)	An object is placed at focus before a concave mirror	(Q)	Magnification is 0.5
(C)	An object is placed at the center of curvature before a concave mirror	(R)	Magnification is (1/3)
(F)	As object is placed at a distance equal to radius of curvature before a convex mirror	(S)	Magnification is –1

(1) A → P, B → Q, C → R, D → S
(2) A → Q, B → P, C → S, D → R
(3) A → Q, B → P, C → R, D → S
(4) A → P, B → R, C → Q, D → S

PRI165

11. Match the column (θ_C is the critical angle b/w prism medium and surrounding medium)

(A)		(1)	Light ray always crosses the prism for any value of i
(B)	A > 2θC	(2)	Light ray never crosses the prism for any value of i
(C)	θC < A < 2θC	(3)	Light ray may crosses the prism depends on i

Option :
(1) A→1, B→1, C→3 (2) A→3, B→2, C→1
(3) A→2, B→1, C→2 (4) A→1, B→2, C→3

PRI166

12. All the following statements are correct except : (For real object)
(1) The magnification produced by a convex mirror is always less than 1.
(2) A virtual, erect, same sized image can be obtained by using only a plane mirror.
(3) A virtual, erect, magnified image can be formed by using concave mirror.
(4) A real inverted, same sized image can be formed by using a convex mirror.

PRI167

13. Match the following :-

(A)	$\mu = \tan i$	(P)	Snell's law
(B)	$\mu = \frac{1}{\sin\theta_C}$	(Q)	Brewsters angle
(C)	$\mu = \frac{\sin i}{\sin r}$	(R)	Prism
(D)	$\mu = \frac{\sin\left(\frac{\delta_{min}+A}{2}\right)}{\sin\frac{A}{2}}$	(S)	Total internal reflection

(1) A → P, B → Q, C → R, D → S
(2) A → Q, B → P, C → S, D → R
(3) A → Q, B → S, C → P, D → R
(4) A → P, B → S, C → S, D → R

PRI169

14. The angle of a prism is 30°. The rays incident at 60° on one refracting face suffer a deviation of 30°. Then the angle of emergence is 3K°. Find the value of K.
(1) 40° (2) 20°
(3) 10° (4) 0°

PRI170

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Answer	4	1	3	3	2	4	4	4	3	2	4	4	3	4

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Important Notes

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Exercise - I (Conceptual Questions) Build Up Your Understanding

HUYGEN'S WAVE THEORY OF LIGHT

1. Which of the following phenomenon can not be explained by the Huygen's theory -
(1) Refraction (2) Reflection
(3) Diffraction (4) Polarization
PWO001

2. Huygen's principle is applicable to -
(1) Only light waves
(2) Only sound waves
(3) Only mechanical waves
(4) For all the above waves
PWO002

3. According to huygen's theory of secondary wavelets, following can be explained -
(1) Propagation of light in medium
(2) Reflection of light
(3) Refraction of light
(4) All of the above
PWO003

4. Huygen's theory of secondary waves can be used to find-
(1) Velocity of light
(2) The wavelength of light
(3) Wave front geometrically
(4) Magnifying power of microscope
PWO004

5. The main drawback of huygen's theory was-
(1) Failure in explanation of rectilinear propagation of light
(2) Failure of explain the spectrum of white light
(3) Failure to explain the formation of newton's rings
(4) A failure of experimental verification of ether medium
PWO005

6. Light has a wave nature, because-
(1) The light travel in a straight line
(2) Light exhibits phenomenon of reflection and refraction
(3) Light exhibits phenomenon of interference
(4) Light exhibits phenomenon of photo electric effect
PWO006

7. Wave nature of light is verified by-
(1) Interference
(2) Photo electric effect
(3) Reflection
(4) Refraction
PWO007

INTERFERENCE

8. The energy in the phenomenon of interference:
(1) is conserved, gets redistributed
(2) is equal at every point
(3) is destroyed in regions of dark fringes
(4) is created at the place of bright fringes
PWO008

9. The resultant amplitude in interference with two coherent sources depends upon :
(1) only amplitude
(2) only phase difference
(3) on both the above
(4) none of the above
PWO009

10. Which of following nature of light waves is supported by the phenomenon of interference:
(1) longitudinal
(2) transverse
(3) both transverse and longitudinal
(4) None of the above
PWO010

11. For distinct interference pattern to be observed, necessary condition is that ratio of intensity of light emission by both the sources should be :
(1) 2 : 1 (2) 1 : 2 (3) 1 : 1 (4) 1 : 4
PWO011

12. The phase difference corresponding to path difference of x is :
(1) $\frac{2\pi x}{\lambda}$ (2) $\frac{2\pi\lambda}{x}$ (3) $\frac{\pi x}{\lambda}$ (4) $\frac{\pi\lambda}{x}$
PWO012

13. The coherent source of light produces constructive interference when phase difference between them is :
(1) π (2) $\frac{1}{2}\pi$ (3) $\frac{3}{2}\pi$ (4) 2π
PWO013

14. Phenomenon of interference is not observed by two sodium lamps of same power. It is because both waves have :
(1) not constant phase difference
(2) zero phase difference
(3) different intensity
(4) different frequencies

PWO014

15. Coherent sources can be obtained :
(1) only by division of wave front
(2) only by division of amplitude
(3) both by division of amplitude and wave front
(4) none of the above

PWO015

16. In an interference of light derived from two slit apertures, if at some point on the screen, yellow light has a path difference of $\frac{3\lambda}{2}$, then the fringe at that point will be :
(1) yellow in colour (2) white in colour
(3) dark (4) bright

PWO016

17. Two beams of light having intensities I and 4I interfer to produce a fringe pattern on a screen. The phase difference between the beam is $\frac{\pi}{2}$ at point A and 2π at point B. Then find out the difference between the resultant intensities at A and B.
(1) 2I (2) 5I (3) I (4) 4I

PWO017

18. Amplitude of waves observed by two light sources of same wave length are a and 2a and have a phase difference of π between them. Then minimum intensity of light will be proportional to :
(1) 0 (2) $5a^2$ (3) a^2 (4) $9a^2$

PWO018

19. If the intensity of the waves observed by two coherent sources is I. Then the intensity of resultant wave in constructive interference will be :-
(1) 2I (2) 4I
(3) I (4) None of the above

PWO019

20. If intensity of each of the two waves is I and they are having phase difference of 120^0, when the waves are superimposed, then the resultant intensity will be :
(1) I (2) 2I (3) I/2 (4) 4I

PWO020

21. Ratio of intensity of two waves is 25 : 1. If interference occurs, then ratio of maximum and minimum intensity should be :
(1) 25 : 1 (2) 5 : 1 (3) 9 : 4 (4) 4 : 9

PWO021

22. The intensity of two waves is 2 and 3 unit, then average intensity of light in the overlapping region will have the value :
(1) 2.5 (2) 6 (3) 5 (4) 13

PWO022

23 The light waves from two independent monochromatic light sources are given by - $y_1 = 2\sin(\omega t - kx)$ and $y_2 = 3\cos(\omega t - kx)$, then the following statement is correct
(1) Both the waves are coherent
(2) Both the waves are incoherent
(3) Both the waves have different time periods
(4) None of the above

PWO023

24. The phenomenon of interference is shown by :
(1) Longitudinal mechanical waves only
(2) Transverse mechanical waves only
(3) Electromagnetic waves only
(4) All the above type of waves

PWO024

25. For the sustained interference of light, the necessary condition is that the two sources should :
(1) Have constant phase difference
(2) Be narrow
(3) Be close to each other
(4) Of same amplitude

PWO025

26. If ratio of amplitude of two interfering source is 3 : 5. Then ratio of intensity of maxima and minima in interference pattern will be :
(1) 25 : 16 (2) 5 : 3 (3) 16 : 1 (4) 25 : 9

PWO026

27. Two coherent light beams of intensity I and 4I are superposed. The maximum and minimum possible intensities in the resulting beam are :
(1) 5I and 3I (2) 5I and I
(3) 9I and 3I (4) 9I and I

PWO027

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28. Two coherent sources of intensities I_1 and I_2 produce an interference pattern. The maximum intensity in the interference pattern will be :
(1) $I_1 + I_2$ (2) $I_1^2 + I_2^2$
(3) $(I_1 + I_2)^2$ (4) $\left(\sqrt{I_1} + \sqrt{I_2}\right)^2$
PWO028

29. Two wave are represented by the equations $y_1 = a \sin \omega t$ and $y_2 = a \cos \omega t$. The first wave :
(1) Leads the second by π
(2) Lags the second by π
(3) Leads the second by $\frac{\pi}{2}$
(4) Lags the second by $\frac{\pi}{2}$
PWO029

30. The resultant amplitude of a vibrating particle by the superposition of the two waves $y_1 = a\sin\left(\omega t + \frac{\pi}{3}\right)$ and $y_2 = a \sin \omega t$ is :-
(1) a (2) $\sqrt{2}$ a (3) 2a (4) $\sqrt{3}$ a
PWO030

31. Two coherent sources of different intensities send waves which interfere. If the ratio of maximum and minimum intensity in the interference pattern is 25 then find ratio of intensities of sources :
(1) 25 : 1 (2) 5 : 1
(3) 9 : 4 (4) 25 : 16
PWO031

32. What is the path difference of destructive interference :
(1) $n\lambda$ (2) $n(\lambda + 1)$
(3) $\frac{(n+1)\lambda}{2}$ (4) $\frac{(2n+1)\lambda}{2}$
PWO032

33. If an interference pattern have maximum and minimum intensities in 36 :1 ratio then what will be the ratio of amplitudes :
(1) 5 : 7 (2) 7 : 4 (3) 4 : 7 (4) 7 : 5
PWO033

34. When a thin transparent plate of thickness t and refractive index μ is placed in the path of one of the two interfering waves of light, then the path difference changes by :
(1) $(\mu + 1)t$ (2) $(\mu - 1)t$
(3) $\frac{(\mu+1)}{t}$ (4) $\frac{(\mu-1)}{t}$
PWO034

35. Due to effect of interference, floating oil layer in water is visible coloured, due to observation of this event the thickness of oil layer should be :
(1) 10 nm (2) 0.1 μm
(3) 1 mm (4) 10 mm
PWO035

36. If intensity ratio of two interfering waves is 9 : 1 then ratio of maximum to minimum amplitude of resultant wave is :-
(1) 2 : 1 (2) 3 : 2 (3) 1 : 3 (4) 5 : 2
PWO036

37. For coherent sources which is essential :-
(1) colour same (2) ϕ constant
(3) ν different (4) Amplitude same
PWO037

38. When exposed to sunlight, thin films of oil on water often exhibit brilliant colors due to the phenomenon of –
(1) interference (2) diffraction
(3) dispersion (4) polarisation
PWO038

39. If $\frac{I_1}{I_2} = \frac{9}{1}$ then $\frac{I_{max}}{I_{min}} = ?$
(1) 100 : 64 (2) 64 : 100
(3) 4 : 1 (4) 1 : 4
PWO039

40. Soap bubble appears coloured due to the phenomenon of :-
(1) Total internal reflection
(2) Interference by division of amplitude
(3) Interference by division of wavefront
(4) Diffraction of light
PWO040

41. Two coherent light sources emit light of the
(1) same intensity
(2) different frequency
(3) constant phase difference but different wavelengths
(4) same frequency having constant phase difference
PWO041

YDSE

42. In Young's experiment, if the amplitude of interferring waves are unequal then the :
(1) contrast in the fringes decreases
(2) contrast in the fringes increase
(3) number of fringes will increase
(4) number of fringes will decrease

PWO042

43. Young's experiment proves that which of the following fact :
(1) light is made up of particles
(2) light is made up of waves
(3) light is made up of neither waves nor particles
(4) fringe width doesn't depend upon the spacing between slits.

PWO043

44. Which of the following statement is true, in Young's experiment, separation between the slits is gradually increased :
(1) fringe width increases and fringes disappear
(2) fringe width decreases and fringes disappear
(3) fringes become blurred
(4) fringe width remains constant and fringes are more bright

PWO044

45. In Young's double slit experiment :
(1) only interference occurs
(2) only diffraction occurs
(3) both interference and diffraction occurs
(4) none of the above

PWO045

46. In Young's double slit experiment, one of the slits is so painted that intensity of light emitted from it is half of that of the light emitted from other slit. Then:
(1) fringe system will disappear
(2) bright fringes will become brighter and dark fringes will be darker
(3) both bright and dark fringes will become darker
(4) dark fringes will become less dark and bright fringes will become less bright.

PWO046

47. In white light interference, nearest to the central (bright) fringe, will have which of the following colour
(1) violet (2) yellow
(3) red (4) green

PWO047

48. In Young's double slit experiment, wavelength of light is 6000Å. Then the phase difference between the light waves reaching the third bright fringe from the central fringe will be :
(1) zero (2) 2π (3) 4π (4) 6π

PWO048

49. If intensity of each wave in the observed interference pattern in Young's double slit experiment is I_0. then for some point P where the phase difference is ϕ , intensity I will be :
(1) $I = I_0 \cos\phi$ (2) $I = I_0 \cos^2 \phi$
(3) $I = I_0 (1+\cos\phi)$ (4) $I = 2I_0(1+\cos\phi)$

PWO049

50. In Young's double slit experiment, bright fringes are of :
(1) equal widths and unequal intensities
(2) unequal widths and equal intensities
(3) equal widths and equal intensities
(4) unequal widths and unequal intensities

PWO050

51. In Young's experiment, monochromatic light through a single slit S is used to illuminate the two slits S_1 and S_2. Interference fringes are obtained on a screen. The fringe width is found to be w. Now a thin sheet of mica (thickness t and refractive index μ) is placed near and in front of one of the two slits. Now the fringe width is found to be w′, then :
(1) $w' = w/\mu$ (2) $w' = w\mu$
(3) $w' = (\mu - 1)\, tw$ (4) $w' = w$

PWO051

52. In Young's double slit experiment, the two slits act as coherent sources of equal amplitude A and wavelength λ. In another experiment with the same set up the two slits are sources of equal amplitude A and wavelength λ but are incoherent. The ratio of the intensity of light at the mid-point of the screen in the first case to that in the second case is :
(1) 4 : 1
(2) 2 : 1
(3) 1 : 1
(4) None of the above

PWO052

53. In Young's double slit experiment, if the width of the slits are in the ratio 4 : 9 the ratio of the intensity of maxima to the intensity at minima will be :
(1) 169 : 25 (2) 81 : 16
(3) 25 : 1 (4) 9 : 4
PWO0053

54. In an interference experiment, the spacing between successive maxima or minima is :
(1) $\frac{\lambda d}{D}$ (2) $\frac{\lambda D}{d}$ (3) $\frac{dD}{\lambda}$ (4) $\frac{\lambda d}{4D}$
PWO0054

55. Young's experiment is performed in air and then performed in water, the fringe width :
(1) Will remain same (2) Will decrease
(3) Will increase (4) Will be infinite
PWO0055

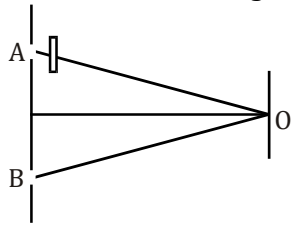
56. In Young's experiment, one slit is covered with a blue filter and the other with a yellow filter. Then the interference pattern :
(1) Will be blue
(2) Will be yellow
(3) Will be green
(4) Will not be formed
PWO0056

57. In Young's double slit experiment, a mica sheet of thickness t and refractive index μ is introduced in the path of ray from the first source S_1. By how much distance the fringe pattern will be displaced:-
(1) $\frac{d}{D}(\mu-1)t$ (2) $\frac{D}{d}(\mu-1)t$
(3) $\frac{d}{(\mu-1)D}$ (4) $\frac{D}{d}(\mu-1)$
PWO0057

58. In Young's experiment, light of wavelength 6000Å is used to produce fringes of width 0.8 mm at a distance of 2.5 m. If the whole experiment is deep in a liquid of refractive index 1.6, then fringe width will be :
(1) 0.5 mm (2) 0.6 mm
(3) 0.4 mm (4) 0.2 mm
PWO0058

59. If a transparent medium of refractive index μ = 1.5 and thickness t = 2.5 × 10^{-5} m is inserted in front of the slits of Young's Double slit experiment, how much will be the shift in the interference pattern? The distance between the slits is 0.5 mm and that between slits and screen is 100 cm :
(1) 5 cm (2) 2.5 cm
(3) 0.25 cm (4) 0.1 cm
PWO0059

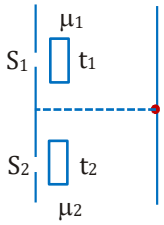
60. In Young's experiment, monochromatic light is used to illuminate the two slits A and B. Interference fringes are observed on a screen placed in front of the slits.
Now if a thin glass plate is placed normally in the path of the beam coming from the slit then :



(1) The fringes will disappear
(2) The fringe width will decrease
(3) The fringe width will increase
(4) There will be no change in the fringe width
PWO0060

61. The central fringe of interference pattern produced by light of wavelength 6000Å is found to shift to the position of 4th bright fringe, after a glass plate of μ = 1.5 is introduced. The thickness of the glass plate is :
(1) 4.8 μm (2) 8.23 μm
(3) 14.98 μm (4) 3.78 μm
PWO0061

62. In a Young's double slit experiment, a slab of thickness 1.2 μm and refractive index 1.5 is placed in front of one slit and another slab of thickness t and refractive index 2.5 is placed in front of the second slit. If the position of the central fringe remains unaltered, then the thickness t is–



(1) 0.4 μm (2) 0.8 μm
(3) 1.2 μm (4) 7 μm
PWO0062

63. In Y.D.S.E. the fringe width is 0.2 mm. If wavelength of light is increase by 10% and separation between the slits is increased by 10% then fringe width will be :
(1) 0.20 mm (2) 0.165 mm
(3) 0.401 mm (4) 0.242 mm
PWO063

64. A very thin transparent film of soap solution (thickness $\rightarrow$ 0) is seen under reflection of white light. Then the colour of the film appear to be :
(1) blue (2) black
(3) red (4) yellow
PWO064

65. In Young's double slit experiment, if monochromatic light is replaced by white light :
(1) All bright fringes become white
(2) All bright fringes have coloures between violet and red
(3) Only the central fringe is white, all other fringes are coloured
(4) No fringes are observed
PWO065

66. The fringe width in Young's double slit experiment increases when :
(1) Wavelength increases
(2) Distance between the slits increases
(3) Distance between the source and screen decreases
(4) Frequency of incident light increases
PWO066

67. Young's double slit experiment is performed with light of wavelength 550 nm. The separation between the slits is 1.10 mm and screen is placed at distance of 1m. What is the distance between the consecutive bright or dark fringes.
(1) 1.5 mm (2) 1.0 mm
(3) 0.5 mm (4) None of these
PWO067

68. In the Young's double slit experiment, for which colour the fringe width is least ?
(1) Red (2) Green
(3) Blue (4) Yellow
PWO068

69. If the sodium light in Young's double slit experiment is replaced by red light, the fringe width will :
(1) Decrease
(2) Increase
(3) Remain unaffected
(4) First increase, then decrease
PWO069

70. A double slit experiment is performed with light of wavelength 500 nm. A thin film of thickness 2μm and refractive index 1.5 is introduced in the path of the upper beam. The location of the central maximum will :
(1) Remain unshifted
(2) Shift downward by nearly two fringes
(3) Shift upward by nearly two fringes
(4) Shift downward by 10 fringes
PWO070

71. In YDSE experiment, when two light waves make third minima, then they have :-
(1) Phase difference of 3π
(2) Phase difference of $\frac{5\pi}{2}$
(3) Path difference of 3λ
(4) Path difference of $\frac{5\lambda}{2}$
PWO071

72. A monochromatic beam of light is used for the formation of fringes on the screen by illuminating the two slits in the Young's double slit interference experiment. When a thin film of mica is interposed in the path of one of the interfering beams then :
(1) The fringe width increases
(2) The fringe width decreases
(3) The fringe width remains the same but the pattern shifts
(4) The fringe pattern disappears
PWO072

73. In an interference experiment, third bright fringe is obtained at a point on the screen with a light of 700 nm. What should be the wavelength of the light in order to obtain 5th bright fringe at the same point?
(1) 500 nm (2) 630 nm
(3) 750 nm (4) 420 nm
PWO073

74. In a double slit experiment if light of wavelength 5000 Å is used then fringe width of 1 mm is obtained. If now light of wavelength 6000 Å is used without altering the system then new fringe width will be :
(1) 1 mm (2) 0·5 mm
(3) 1·2 mm (4) 1·5 mm
PWO074

75. Monochromatic green light has wavelength 5×10^{-7} m. The separation between slits is 1 mm. The fringe width of interference pattern obtained on screen at a distance of 2 meter is :
(1) 1 mm (2) 0.5 mm
(3) 2 mm (4) 0.1 mm
PWO075

76. In Young's double slit experiment when wavelength of 700 nm is used then fringe width of 0.7 mm is obtained. If wavelength of 500nm is used then what is the fringe width?
(1) 0.35 mm (2) 0.5 mm
(3) 3.5 mm (4) 5 mm
PWO076

77. What will be the effect on fringe width, when distance between slits become doubled-
(1) 1/2 times (2) 2 times
(3) 1/4 times (4) Unchanged
PWO077

DIFFRACTION

78. The conversation going on, in some room, can be heared by the person outside the room. The reason for it is :
(1) Interference of sound
(2) Reflection of sound
(3) Diffraction of sound
(4) Refraction of sound
PWO078

79. Phenomenon of diffraction occurs :
(1) only in case of light and sound waves
(2) for all kinds of waves
(3) for electro-magnetic waves and not for matter waves
(4) for light waves but not is case of X rays
PWO079

80. Which of the following ray gives more distinct diffraction :
(1) X-ray (2) light ray
(3) γ –ray (4) Radio wave
PWO080

81. All fringes of diffraction are of :
(1) the same intensity (2) unequal width
(3) the same width (4) full darkness
PWO081

82. What happens, when the width of the slit aperture is increased in an experiment of single slit diffraction experiment :
(1) spread of diffraction region is increased
(2) spread of diffraction region is decreased
(3) spread of diffraction region will be decreased and mid-band becomes narrow
(4) none of the above
PWO082

83. Light waves do not travels strictly in straight line, can be best explained by :
(1) Particle nature of light
(2) Diffraction
(3) Interference
(4) Polarisation
PWO083

84. In the diffraction pattern of a single slit aperture, the width of the central fringe compared to widths of the other fringes, is :
(1) equal (2) less
(3) little more (4) double
PWO084

85. Diffracted fringes obtained from the slit aperture are of :-
(1) same width
(2) different width
(3) uniform intensity
(4) non-uniform width & non uniform intensity
PWO085

86. Central fringe obtained in diffraction pattern due to a single slit :
(1) is of minimum intensity
(2) is of maximum intensity
(3) intensity does not depend upon slit width
(4) none of the above
PWO086

87. In a single slit diffraction pattern, if the light source is used of less wavelength than previous one. Then width of the central fringe will be :
(1) less (2) increase
(3) unchanged (4) none of the above
PWO087

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88. In the laboratory, diffraction of light by a single slit is being observed. If slit is made slightly narrow, then diffraction pattern will :
(1) be more spreaded than before
(2) be less spreaded than before
(3) be spreaded as before
(4) be disappeared

PWO088

89. For Fraunhofer single slit diffraction :
(1) width of central maxima is proportional to λ
(2) on increasing the slit width, the width of central maxima decreases
(3) on making the slit width a = λ, central fringe spreads in the range $\pm 90°$
(4) all of the above are correct

PWO089

90. In a Fraunhofer's diffraction by a slit, if slit width is a, wavelength λ, focal length of lens is f, linear width of central maxima is :
(1) $\frac{f\lambda}{a}$ (2) $\frac{fa}{\lambda}$ (3) $\frac{2f\lambda}{a}$ (4) $\frac{f\lambda}{2a}$

PWO090

91. In a Fraunhofer's diffraction obtained by a single slit aperture, the value of path difference for n^{th} order of minima is :
(1) $n\lambda$ (2) $2n\lambda$
(3) $(2n-1)\lambda/2$ (4) $(2n-1)\lambda$

PWO091

92. A light source of 5000Å wavelength produces a single slit diffraction. The first minima in diffraction pattern is seen, at a distance of 5mm from central maxima. The distance between screen and slit is 2 metre. The width of slit in mm will be :
(1) 0.1 (2) 0.4 (3) 0.2 (4) 2

PWO092

93. A plane wave front of wavelength 6000 Å is incident upon a slit of 0.2mm width, which enables Fraunhofer's diffraction pattern to be obtained on a screen 2 metre away. Width of the central maxima in mm will be :
(1) 10 (2) 12 (3) 8 (4) 2

PWO093

94. The waves of 600μm wave length are incident normally on a slit of 1.2mm width. The value of diffraction angle corresponding to the first minima will be (in radian) :
(1) $\frac{\pi}{2}$ (2) $\frac{\pi}{6}$ (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{4}$

PWO094

95. In Fraunhoffer diffraction the centre of diffraction image is :
(1) always bright
(2) always dark
(3) sometimes bright and sometimes dark
(4) bright for large wavelength and dark for low wavelength

PWO095

96. A single slit of width d is placed in the path of beam of wavelength λ. The angular width of the principal maximum obtained is :
(1) $\frac{d}{\lambda}$ (2) $\frac{\lambda}{d}$ (3) $\frac{2\lambda}{d}$ (4) $\frac{2d}{\lambda}$

PWO096

97. Bending of light waves at the sharp edges of an opaque obstacle is known as
(1) refraction (2) reflection
(3) diffraction (4) interference

PWO097

98. Diffraction and interference of light refers to:
(1) quantum nature of light
(2) wave nature of light
(3) transverse nature of light
(4) electromagnetic nature of light

PWO098

99. The phenomenon of diffraction of light was discovered by :
(1) Huygens (2) Newton
(3) Fresnel (4) Grimaldi

PWO099

100. Angular width (θ) of central maximum of a diffraction pattern of a single slit does not depend upon :
(1) Distance between slit and source
(2) Wavelength of light used
(3) Width of the slit
(4) Frequency of light used

PWO100

101. Red light is generally used to observe diffraction pattern from single slit. If green light is used instead of red light, then diffraction pattern :
(1) Will be more clear
(2) Will be contract
(3) Will be expanded
(4) Will not visualize

PWO101

102. Diffraction of sound waves is more evident than light waves in daily life because :-
(1) $\lambda_{Sound} > \lambda_{light}$
(2) $\lambda_{sound} = \lambda_{light}$
(3) $\lambda_{sound} < \lambda_{light}$
(4) Sound waves are longitudinal but light waves are transverse.

PWO102

103. In single slit Fraunhoffer diffraction which type of wavefront is required :
(1) cylindrical (2) spherical
(3) elliptical (4) plane

PWO103

104. If in Fraunhofer diffraction due to a single slit, the slit width is increased, then the width of the central maximum will
(1) increase
(2) decrease
(3) not change
(4) change depends on the wavelength of light used

PWO104

POLARISATION, BREWSTER LAW AND MALUS LAW

105. A polariser is used to :
(1) Reduce intensity of light
(2) Produce polarised light
(3) Increase intensity of light
(4) Produce unpolarised light

PWO105

106. Light waves can be polarised as they are :
(1) Transverse
(2) Of high frequency
(3) Longitudinal
(4) Reflected

PWO106

107. Through which character we can distinguish the light waves from sound waves :
(1) Interference (2) Refraction
(3) Polarisation (4) Reflection

PWO107

108. Which of following can not be polarised :
(1) Radio waves (2) Ultraviolet rays
(3) Infrared rays (4) Ultrasonic waves

PWO108

109. The transverse nature of light is shown by
(1) Interference of light
(2) Refraction of light
(3) Polarisation of light
(4) Dispersion of light

PWO109

110. The angle of polarisation for any medium is 60°, what will be critical angle for this :
(1) $\sin^{-1}\sqrt{3}$ (2) $\tan^{-1}\sqrt{3}$
(3) $\cos^{-1}\sqrt{3}$ (4) $\sin^{-1}\frac{1}{\sqrt{3}}$

PWO110

111. The angle of incidence at which reflected light is totally polarized for reflection from air to glass (refractive index n)
(1) $\sin^{-1}(n)$ (2) $\sin^{-1}\left(\frac{1}{n}\right)$
(3) $\tan^{-1}\left(\frac{1}{n}\right)$ (4) $\tan^{-1}(n)$

PWO111

112. A ray of light is incident on the surface of a glass plate at an angle of incidence equal to Brewster's angle ϕ. If μ represents the refractive index of glass with respect to air, then the angle between reflected and refracted rays is :
(1) $90 + \phi$
(2) $\sin^{-1}(\mu\cos\phi)$
(3) 90°
(4) $90° - \sin^{-1}(\sin\phi/\mu)$

PWO112

113. Refractive index of material is equal to tangent of polarizing angle. It is called.
(1) Brewster's law (2) Lambert's law
(3) Malus's law (4) Bragg's law

PWO113

114. When unpolarized light beam is incident from air onto glass (n = 1.5) at the polarizing angle:
(1) Reflected beam is 100 percent polarized
(2) Reflected and refracted beams are partially polarized
(3) The reflected and refracted ray will not perpendicular to each other
(4) All of the above

PWO114

115. When the angle of incidence on a material is 60°, the reflected light is completely polarized. The velocity of the refracted ray inside the material is (in ms^{-1}) :

(1) 3×10^8 (2) $\left(\frac{3}{\sqrt{2}}\right) \times 10^8$

(3) $\sqrt{3} \times 10^8$ (4) 0.5×10^8

PWO115

116. A polaroid is placed at 45° to an incoming light of intensity I_0. Now the intensity of light passing through polaroid after polarisation would be:

(1) I_0 (2) $I_0/2$ (3) $I_0/4$ (4) Zero

PWO116

117. Plane polarised light is passed through a polaroid. On viewing through the polaroid we find that when the polaroid is given one complete rotation about the direction of the light, one of the following is observed.

(1) The intensity of light gradually decreases to zero and remains at zero

(2) The intensity of light gradually increases to a maximum and remains at maximum

(3) There is no change in intensity

(4) The intensity of light is twice maximum and twice zero

PWO117

118. Polarised glass is used in sun glasses because :

(1) It reduces the light intensity to half on account of polarisation

(2) It is fashionable

(3) It has good colour

(4) It is cheaper

PWO118

119. When a plane polarised light is passed through an analyser and analyser is rotated from 0 to 90°, the intensity of the emerging light :

(1) Varies between a maximum and minimum

(2) Becomes zero

(3) Does not vary

(4) Varies between a maximum and zero

PWO119

120. When an unpolarized light of intensity I_0 is incident on a polarizing sheet, the intensity of the light which does not get transmitted is :

(1) Zero (2) I_0

(3) $\frac{1}{2} I_0$ (4) $\frac{1}{4} I_0$

PWO120

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	4	3	4	3	1	1	3	4	3	1	4	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	3	2	1	3	3	2	4	1	3	4	4	4	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	4	4	2	2	1	2	1	3	2	4	1	2	2	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	3	4	4	3	4	2	3	2	2	4	2	1	2	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	1	1	2	3	1	3	3	2	3	4	3	4	3	1
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	1	3	2	4	2	3	2	4	4	2	1	1	4	3
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	1	3	2	2	1	3	3	2	4	1	2	1	4	2	2
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	1	3	4	3	4	4	3	1	1	3	2	4	1	4	3

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. In the Young's double-slit experiment, the intensity of light at a point on the screen where the path difference is λ is K, (λ being the wave length of light used). The intensity at a point where the path difference is λ/4, will be :-

(1) K (2) K/4 (3) K/2 (4) Zero

PWO123

2. A beam of light of λ = 600 nm from a distant source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. The distance between first dark fringes on either side of the central bright fringe is :-

(1) 1.2 cm (2) 1.2 mm

(3) 2.4 cm (4) 2.4 mm

PWO124

AIPMT 2015

3. In a double slit experiment, the two slits are 1 mm apart and the screen is placed 1 m away. A monochromatic light of wavelength 500 nm is used. What will be the width of each slit for obtaining ten maxima of double slit within the central maxima of single slit pattern?

(1) 0.1 mm (2) 0.5 mm

(3) 0.02 mm (4) 0.2 mm

PWO125

4. For a parallel beam of monochromatic light of wavelength 'λ', diffraction is produced by a single slit whose width 'a' is of the order of the wavelength of the light. If 'D' is the distance of the screen from the slit, the width of the central maxima will be :

(1) $\frac{D\lambda}{a}$ (2) $\frac{Da}{\lambda}$ (3) $\frac{2Da}{\lambda}$ (4) $\frac{2D\lambda}{a}$

PWO126

Re-AIPMT 2015

5. At the first minimum adjacent to the central maximum of a single-slit diffraction pattern the phase difference between the Huygen's wavelet from the edge of the slit and the wavelet from the mid point of the slit is :-

(1) $\frac{\pi}{8}$ radian (2) $\frac{\pi}{4}$ radian

(3) $\frac{\pi}{2}$ radian (4) π radian

PWO127

6. Two slits in Young's experiment have widths in the ratio 1 : 25. The ratio of intensity at the maxima and minima in the interference pattern, $\frac{I_{max}}{I_{min}}$ is :

(1) $\frac{4}{9}$ (2) $\frac{9}{4}$ (3) $\frac{121}{49}$ (4) $\frac{49}{121}$

PWO128

NEET-I 2016

7. In a diffraction pattern due to a single slit of width 'a', the first minimum is observed at an angle 30° when light of wavelength 5000 Å is incident on the slit. The first secondary maximum is observed at an angle of :

(1) $\sin^{-1}\left(\frac{1}{4}\right)$ (2) $\sin^{-1}\left(\frac{2}{3}\right)$

(3) $\sin^{-1}\left(\frac{1}{2}\right)$ (4) $\sin^{-1}\left(\frac{3}{4}\right)$

PWO129

8. The intensity at the maximum in a Young's double slit experiment is I_0. Distance between two slits is d = 5λ, where λ is the wavelength of light used in the experiment. What will be the intensity in front of one of the slits on the screen placed at a distance D = 10 d?

(1) I_0 (2) $\frac{I_0}{4}$ (3) $\frac{3}{4}I_0$ (4) $\frac{I_0}{2}$

PWO130

NEET-II 2016

9. The interference pattern is obtained with two coherent light sources of intensity ratio n. In the interference pattern, the ratio $\frac{I_{max}-I_{min}}{I_{max}+I_{min}}$ will be:-

(1) $\frac{\sqrt{n}}{(n+1)^2}$ (2) $\frac{2\sqrt{n}}{(n+1)^2}$

(3) $\frac{\sqrt{n}}{n+1}$ (4) $\frac{2\sqrt{n}}{n+1}$

PWO131

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10. A linear aperture whose width is 0.02 cm is placed immediately in front of a lens of focal length 60 cm. The aperture is illuminated normally by a parallel beam of wavelength 5×10^{-5} cm. The distance of the first dark band of the diffraction pattern from the centre of the screen is :-

(1) 0.20 cm (2) 0.15 cm
(3) 0.10 cm (4)0.25 cm

PWO132

NEET(UG) 2017

11. Young's double slit experiment is first performed in air and then in a medium other than air. It is found that 8th bright fringe in the medium lies where 5th dark fringe lies in air. The refractive index of the medium is nearly :-

(1) 1.59 (2) 1.69 (3) 1.78 (4) 1.25

PWO134

12. Two Polaroids P_1 and P_2 are placed with their axis perpendicular to each other. Unpolarised light I_0 is incident on P_1 . A third polaroid P_3 is kept in between P_1 and P_2 such that its axis makes an angle 45° with that of P_1. The intensity of transmitted light through P_2 is :-

(1) $\frac{I_0}{4}$ (2) $\frac{I_0}{8}$ (3) $\frac{I_0}{16}$ (4) $\frac{I_0}{2}$

PWO135

NEET(UG) 2018

13. Unpolarised light is incident from air on a plane surface of a material of refractive index 'μ'. At a particular angle of incidence 'i', it is found that the reflected and refracted rays are perpendicular to each other. Which of the following options is correct for this situation?

(1) Reflected light is polarised with its electric vector parallel to the plane of incidence

(2) Reflected light is polarised with its electric vector perpendicular to the plane of incidence

(3) $i = \sin^{-1}\left(\frac{1}{\mu}\right)$

(4) $i = \tan^{-1}\left(\frac{1}{\mu}\right)$

PWO136

14. In Young's double slit experiment the separation d between the slits is 2 mm, the wavelength λ of the light used is 5896 Å and distance D between the screen and slits is 100 cm. It is found that the angular width of the fringes is 0.20°. To increase the fringe angular width to 0.21° (with same λ and D) the separation between the slits needs to be changed to :-

(1) 1.8 mm (2) 1.9 mm
(3) 2.1 mm (4) 1.7 mm

PWO137

15. An astronomical refracting telescope will have large angular magnification and high angular resolution, when it has an objective lens of :-

(1) small focal length and large diameter
(2) large focal length and small diameter
(3) large focal length and large diameter
(4) small focal length and small diameter

PWO138

NEET(UG) 2019

16. In a double slit experiment, when light of wavelength 400 nm was used, the angular width of the first minima formed on a screen placed 1m away, was found to be 0.2°. What will be the angular width of the first minima, if the entire experimental apparatus is immersed in water ($\mu_{water} = 4/3$)

(1) 0.266° (2) 0.15° (3) 0.05° (4) 0.1°

PWO139

NEET(UG) 2019 (Odisha)

17. In a Young's double slit experiment if there is no initial phase difference between the light from the two slits, a point on the screen corresponding to the fifth minimum has path difference.

(1) $5\frac{\lambda}{2}$ (2) $10\frac{\lambda}{2}$ (3) $9\frac{\lambda}{2}$ (4) $11\frac{\lambda}{2}$

PWO140

18. Angular width of the central maxima in the Fraunhofer diffraction for λ = 6000 Å is θ_0. When the same slit is illuminated by another monochromatic light, the angular width decreases by 30%. The wavelength of this light is,

(1) 1800 Å (2) 4200 Å
(3) 6000 Å (4) 420 Å

PWO141

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NEET(UG) 2020

19. The Brewsters angle i_b for an interface should be:

(1) $i_b = 90°$ (2) $0° < i_b < 30°$

(3) $30° < i_b < 45°$ (4) $45° < i_b < 90°$

PWO143

20. In Young's double slit experiment, if the separation between coherent sources is halved and the distance of the screen from the coherent sources is doubled, then the fringe width becomes :

(1) one-fourth (2) double

(3) half (4) four times

PWO144

NEET(UG) 2020 (COVID-19)

21. Two coherent sources of light interfere and produce fringe pattern on a screen. For central maximum, the phase difference between the two waves will be

(1) zero (2) π

(3) $3\pi/2$ (4) $\pi/2$

PWO145

NEET(UG) 2021

22. A lens of large focal length and large aperture is best suited as an objective of an astronomical telescope since :

(1) a large aperture contributes to the quality and visibility of the images.

(2) a large area of the objective ensures better light gathering power.

(3) a large aperture provides a better resolution.

(4) all of the above.

PWO146

NEET(UG) 2022

23. In a Young's double slit experiment, a student observes 8 fringes in a certain segment of screen when a monochromatic light of 600 nm wavelength is used. If the wavelength of light is changed to 400 nm, then the number of fringes he would observe in the same region of the screen is :

(1) 8 (2) 9 (3) 12 (4) 6

PWO147

RE-NEET(UG) 2022

24. After passing through a polariser a linearly polarised light of intensity I is incident on an analyser making an angle of 30° with that of the polariser. The intensity of light emitted from the analyser will be :

(1) $\frac{I}{2}$ (2) $\frac{I}{3}$ (3) $\frac{3I}{4}$ (4) $\frac{2I}{3}$

PWO148

25. If the screen is moved away from the plane of the slits in a Young's double slit experiment, then the :

(1) angular separation of the fringes increases

(2) angular separation of the fringes decreases

(3) linear separation of the fringes increases

(4) linear separation of the fringes decreases

PWO149

NEET(UG) 2023

26. For Young's double slit experiment, two statements are given below :

Statement I : If screen is moved away from the plane of slits, angular separation of the fringes remains constant.

Statement II : If the monochromatic source is replaced by another monochromatic source of larger wavelength, the angular separation of fringes decreases.

In the light of the above statements, choose the *correct* answer from the options given below :

(1) Both **Statement I** and **Statement II** are false

(2) **Statement I** is true but **Statement II** is false

(3) **Statement I** is false but **Statement II** is true

(4) Both **Statement I** and **Statement II** are true

PWO166

NEET(UG) 2024

27. An unpolarised light beam strikes a glass surface at Brewster's angle. Then :-

(1) the reflected light will be partially polarised.

(2) the refracted light will be completely polarised.

(3) both the reflected and refracted light will be completely polarised.

(4) the reflected light will be completely polarised but the refracted light will be partially polarised.

PWO167

28. If the monochromatic source in Young's double slit experiment is replaced by white light, then

(1) interference pattern will disappear.

(2) there will be a central dark fringe surrounded by a few coloured fringes.

(3) there will be a central bright white fringe surrounded by a few coloured fringes.

(4) all bright fringes will be of equal width.

PWO168

RE-NEET(UG) 2024

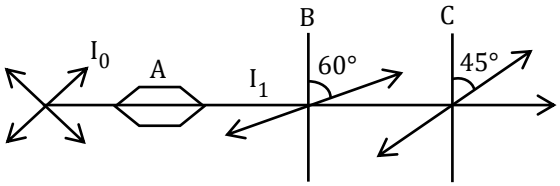
29. Two slits in Young's double slit experiment are 1.5 mm apart and the screen is placed at a distance of 1m from the slits. If the wavelength of light used is 600×10^{-9}m then the fringe separation is :

(1) 4×10^{-5} m (2) 9×10^{-8} m

(3) 4×10^{-7} m (4) 4×10^{-4} m

PWO169

30. A beam of unpolarized light of intensity I_0 is passed through a polaroid A, then through another polaroid B, oriented at 60° and finally through another polaroid C, oriented at 45° relative to B as shown. The intensity of emergent light is :



(1) $\frac{I_0}{16}$ (2) $\frac{I_0}{4}$

(3) $\frac{I_0}{2}$ (4) $\frac{I_0}{32}$

PWO170

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	4	4	4	2	4	4	4	2	3	2	2	2	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	2	4	4	1	4	3	3	3	2	4	3	4	1

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Exercise - III (Analytical Questions) Master Your Understanding

INTERFERENCE

1. (A) Phase difference between various particles on a wave front is zero
(B) Normal to the wave front represents a ray of light.
(C) A point source of light gives rise to spherical wavefronts.
(D) A wave front always travels in forward direction in a medium.
Correct statements are –
(1) A and B only (2) A, B and C only
(3) C and D only (4) A, B, C and D

PWO150

2. In YDSE, match the column-I & column-II

	Column-I		Column-II
(i)	Fringe width	(a)	$\frac{3\lambda D}{d}$
(ii)	Angular fringe width	(b)	$\frac{7\lambda D}{2d}$
(iii)	Position of 3rd bright w.r.t. centre	(c)	$\frac{\lambda}{d}$
(iv)	Position of 4th dark w.r.t. centre	(d)	$\frac{\lambda D}{d}$

(1) (i – a), (ii – b), (iii – c), (iv – d)
(2) (i – d), (ii – c), (iii – a), (iv – b)
(3) (i – d), (ii – c), (iii – b), (iv – a)
(4) (i – c), (ii – d), (iii – a), (iv – b)

PWO151

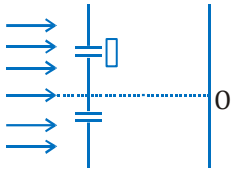
3. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : Wave nature can be proved by phenomena of interference and diffraction.
Reason (R) : Only transverse wave can show the phenomena of polarization.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PWO152

4. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : When width of one of the slits in Young's double slit experiment is four times that of the other, then brighter frings are nine times brighter than the dark fringes.
Reason (R) : The amplitude of the wave is proportional to the width of the slit.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PWO153

5. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : In the figure shown zero order maxima will be above point "O".



Reason (R) : Zero order maxima means a point where the phase difference is zero.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PWO154

6. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : If white light is used in place of monochromatic light in YDSE then central point is white. Although at other places coloured fringes will be obtained.

Reason (R) : At centre, path difference is zero for each wave length. Hence all wave will interfere constructively.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PWO155

DIFFRACTION

7. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : In case of single slit diffraction intensity of higher order maxima decreases.

Reason (R) : Higher order maxima are at larger distance.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PWO156

8. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Out of radio waves and microwaves, the radio waves are more diffracted in daily life.

Reason (R) : Practically diffraction is more for waves which have longer wavelength.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PWO157

9. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Diffraction takes place for all types of waves mechanical or non-mechanical, transverse or longitudinal.

Reason (R) : Diffraction's effects are perceptible only if wavelength of wave is comparable to dimensions of diffracting device.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PWO158

10. Match the column :

Column-A		Column-B	
(A)	Fresnel distance equal to	(P)	$\theta = 90°$
(B)	In a single slit experiment, when $\Delta = a$, then angular width	(Q)	$\frac{a^2}{\lambda}$
(C)	In single slit experiment, angular width for first dark fringe.	(R)	$n\lambda$
(D)	Path difference for dark fringe in single slit diffraction	(S)	$\frac{\lambda}{a}$

(1) $A \rightarrow Q, B \rightarrow P, C \rightarrow S, D \rightarrow R$

(2) $A \rightarrow R, B \rightarrow Q, C \rightarrow P, D \rightarrow S$

(3) $A \rightarrow P, B \rightarrow R, C \rightarrow Q, D \rightarrow P$

(4) $A \rightarrow S, B \rightarrow S, C \rightarrow R, D \rightarrow Q$

PWO159

11. In a single slit diffraction, for first maxima:

(i) $\Delta x = \lambda$ (ii) $\Delta\varphi = 2\pi$

(iii) $\Delta x = \frac{3\lambda}{2}$ (iv) $\Delta\varphi = 3\pi$

(1) (i) and (ii) (2) (i) and (iv)
(3) (ii) and (iii) (4) (iii) and (iv)

PWO160

POLARISATION

12. Match the column :

($R \rightarrow$ Rarer medium, $D \rightarrow$ Denser medium)

	Column-I		Column-II
(A)	When light is going from R to D and angle of polarisation θ_p then	(P)	$\theta_p = \tan^{-1}(\text{cosec } i_c)$
(B)	When light is going from D to R and angle of polarisation θ_p then	(Q)	$i_c = \text{cosec}^{-1}(\cot \theta_p)$
(C)	When light is going from R to D and i_c is critical angle then	(R)	$\theta_p = \tan^{-1}(\sin i_c)$
(D)	When light is going from for D to R and i_c is critical angle then	(S)	$i_c = \sin^{-1}(\cot \theta_p)$

Which one of the following can be taken correct?

(1) (A, S), (B, Q), (C, P), (D, R)
(2) (A, Q), (B, S), (C, P), (D, C)
(3) (A, S), (B, Q), (C, R), (D, P)
(4) (A, Q), (B, S), (C, R), (D, A)

PWO161

13. When incident light is polarized in a plane perpendicular to the plane of incidence and it is incident at the angle of polarization

(A) Reflected light is perfectly polarised
(B) Refracted light will be partially polarised
(C) Reflected light is partially polarised
(D) There will be no reflection

(1) (A, B) (2) (B, C) (3) (B, D) (4) (D)

PWO162

14. In Nicol prism we get polarized light using the phenomenon of :

(A) Total internal reflection
(B) Selective absorption
(C) Scattering
(D) Double refraction

(1) A and B (2) B and D
(3) A and D (4) C and D

PWO163

15. When sun light is incident on water surface at an angle of incidence 53°, the reflected ray is found to be completely polarized then

(A) The angle of refraction is 37°
(B) The refractive index of water is $\frac{3}{2}$
(C) The angle of refraction is 53°
(D) The refractive index of water is $\frac{4}{3}$

(1) A, B (2) B, C (3) C, D (4) A, D

PWO164

16. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : If three polarisers are arranged such that the axis of any two successive polarisers make equal angle of 45° with each other. If unpolarised light of intensity I_0 incident on first polariser then intensity of emergent light after 3rd polariser is $\frac{I_0}{8}$.

Reason (R) : In given condition each time intensity becomes 50% by Malus law.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PWO165

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	2	3	1	1	2	1	2	1	4	1	4	3	4
Question	16														
Answer	1														

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Important Notes

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Exercise - I (Conceptual Questions) Build Up Your Understanding

PHOTO ELECTRIC EFFECT

1. The energy of photon of visible light with maximum wavelength in eV is :
(1) 1 (2) 1.6 (3) 3.2 (4) 7
PMD001

2. What is the momentum of a photon having frequency 1.5×10^{13} Hz :
(1) 3.3×10^{-29} kg m/s (2) 3.3×10^{-34} kg m/s
(3) 6.6×10^{-34} kg m/s (4) 6.6×10^{-30} kg m/s
PMD002

3. The strength of photoelectric current is directly proportional to :
(1) Frequency of incident radiation
(2) Intensity of incident radiation
(3) Angle of incidence of radiation
(4) Distance between anode and cathode
PMD003

4. When light is incident on surface, photo electrons are emitted. For photoelectrons :
(1) The value of kinetic energy is same for all
(2) Maximum kinetic energy do not depend on the wave length of incident light
(3) The value of kinetic energy is equal to or less than a maximum kinetic energy
(4) None of the above.
PMD004

5. When light falls on a photosensitive surface, electrons are emitted from the surface. The kinetic energy of these electrons does not depend on the:
(1) Wavelength of light
(2) Frequency of light
(3) Type of material used for the surface
(4) Intensity of light
PMD005

6. The work- function of a substance is 4.0 eV. The longest wavelength of light that can cause photoelectron emission from this substance is approximately :
(1) 540 nm (2) 400 nm
(3) 310 nm (4) 220 nm
PMD006

7. Photoelectric effect takes place in element A. Its work function is 2.5 eV and threshold wavelength is λ. An other element B is having work function of 5 eV. Then find out the maximum wavelength that can produce photoelectric effect in B :
(1) $\lambda/2$ (2) 2λ (3) λ (4) 3λ
PMD007

8. When light of wavelength lesser than 6000 Å is incident on a metal, electrons are emitted. The approximate work-function of the metal is:
(1) 1 eV (2) 2 eV (3) 4 eV (4) 6 eV
PMD008

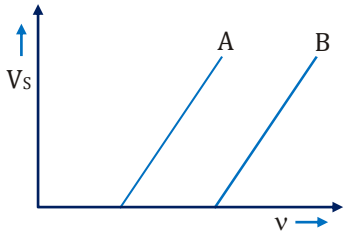
9. Surface of sodium is illuminated by a light of 6000 Å wavelength. Work function of sodium is 1.6 eV. Then minimum K.E. of emitted electrons is :
(1) 0 eV (2) 1.53 eV
(3) 2.46 eV (4) 4.14 eV
PMD009

10. The maximum kinetic energy of photoelectrons emitted from a surface when photons of energy 6 eV fall on it is 4 eV. The stopping potential in volt is :
(1) 4 (2) 6 (3) 8 (4) 10
PMD010

11. When a point source of monochromatic light is at a distance of 0.2 m from a photoelectric cell, the cut - off voltage and the saturation current are 0.6 volt and 18 mA respectively. If the same source is placed 0.6 m away from the photoelectric cell, then :-
(1) The stopping potential will be 0.2 V
(2) The stopping potential will be 0.6 V
(3) The saturation current will be 6 mA
(4) The saturation current will be 18 mA
PMD011

12. The maximum wavelength of light for photoelectric effect from a metal is 200 nm. The maximum kinetic energy of electron which is emitted by the radiation of wavelength 100 nm will be :
(1) 12.4 eV (2) 6.2 eV
(3) 100 eV (4) 200 eV
PMD012

13. The stopping potential as a function of frequency of incident radiation is plotted for two different surfaces A and B. The graphs show that the work function of A is



(1) Greater than that of B
(2) Smaller than that of B
(3) Same as that of B
(4) No comparison can be done from given graphs

PMD013

14. The slope of graph drawn between stopping potential and frequency of incident light for a given surface will be :-

(1) h (2) h/e (3) eh (4) e

PMD014

15. By photo electric effect, Einstein proved :-

(1) $E = h\nu$ (2) $KE = \frac{1}{2}mv^2$

(3) $E = mc^2$ (4) $E = \frac{-Rhc^2}{n^2}$

PMD015

16. Which one among shows particle nature of light?

(1) P.E.E. (2) Interference
(3) Refraction (4) Polarization

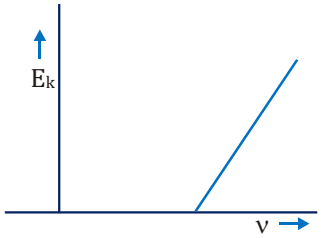
PMD016

17. A photo-cell is illuminated by a source of light, which is placed at a distance d from the cell, If the distance become d/2, then number of electrons emitted per second will be :-

(1) Remain same (2) Four times
(3) Two times (4) One-fourth

PMD017

18. Graph is plotted between maximum kinetic energy of electron with frequency of incident photon in Photo electric effect. The slope of curve will be :



(1) Charge of electron
(2) Work function of metal
(3) Planck's constant
(4) Ratio of Planck constant and charge of electron

PMD018

19. Photon of energy 6 eV is incident on a metal surface of work function 4 eV. Maximum KE of emitted photo electrons will be :-

(1) 0 eV (2) 1 eV
(3) 2 eV (4) 10 eV

PMD019

20. Light of frequency ν is incident on a metal of threshold frequency ν_0. Then work function of metal will be:-

(1) $h\nu$ (2) $h\nu_0$
(3) $h(\nu - \nu_0)$ (4) $h(\nu + \nu_0)$

PMD020

21. The work function of a photo electric material is 3.3 eV. Its threshold frequency will be :

(1) 4×10^{23} Hz (2) 8×10^{12} Hz
(3) 4×10^{11} Hz (4) 8×10^{14} Hz

PMD021

22. The value of planck's constant is :–

(1) 6.63×10^{-34} J/s
(2) 6.63×10^{-34} kg–m^2/s
(3) 6.63×10^{-34} kg–m^2
(4) 6.63×10^{-34} J–s^{-1}

PMD022

23. When ultraviolet rays incident on metal plate then photoelectric effect does not occur, it occurs by incidence of :–

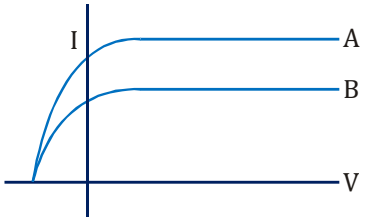
(1) Infrared rays (2) X–rays
(3) Radio wave (4) Light wave

PMD023

24. A photoelectric cell is illuminated by a point source of light 1 m away. When the source is shifted to 2m then
(1) each emitted electron carries one quarter of the initial energy
(2) number of electrons emitted is half the initial number
(3) each emitted electron carries half the initial energy
(4) number of electrons emitted is a quarter of the initial number

PMD024

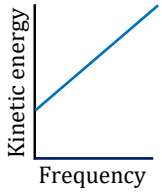
25. The graphs show the variation of current I (y-axis) in two photocell A & B as a function of the applied voltage V(x-axis) when light of same frequency is incident on the cell. Which of the following is the correct conclusion drawn from the data ?

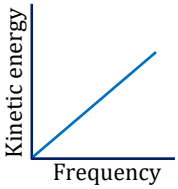


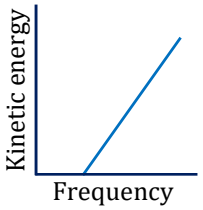
(1) Cathodes of the two cells are made from the same substance, the intensity of light used are different
(2) Cathodes are made from different substances and the intensity of light is the same
(3) Cathode substances as well as intensity of light are different
(4) no conclusion can be drawn

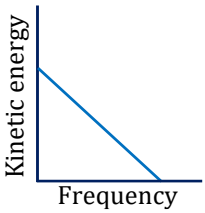
PMD025

26. According to Einstein's photoelectric equation, the graph between the kinetic energy of photoelectrons ejected and the frequency of incident radiation is

(1) 

(2) 

(3) 

(4) 

PMD026

27. According to Einstein's photoelectric equation, the plot of the kinetic energy of the emitted photoelectrons from a metal v/s the frequency of the incident radiation gives a straight line whose slope :
(1) depends on the intensity of the radiation
(2) depends of the nature of the metal used
(3) depends both on the intensity of the radiation and the metal used.
(4) is the same for all metals and independent of the intensity of the radiation.

PMD027

28. A photon of energy 4 eV is incident on a metal surface whose work function is 2eV. The minimum reverse potential to be applied for stopping the current is :-
(1) 2V (2) 4V (3) 6V (4) 8V

PMD028

MATTER WAVES

29. If E and P are the energy and the momentum of a photon respectively then on reducing the wavelength of photon -
(1) P and E both will decrease
(2) P and E both will increase
(3) P will increase and E will decrease
(4) P will decrease and E will increase

PMD029

30. If the kinetic energy of a moving particle is E, then the De Broglie wavelength is :

(1) $h\sqrt{2mE}$ (2) $\sqrt{\frac{2mE}{h}}$

(3) $\frac{h}{\sqrt{2mE}}$ (4) $\frac{hE}{\sqrt{2mE}}$

PMD030

31. Electron has energy of 100 eV what will be its wavelength :
(1) 1.2 Å (2) 10 Å
(3) 100 Å (4) 1 Å

PMD031

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32. The ratio of wavelength of deutron and proton accelerated through the same potential difference will be -

(1) $\frac{1}{\sqrt{2}}$ (2) $\sqrt{\frac{2}{1}}$ (3) $\frac{1}{2}$ (4) $\frac{2}{1}$

PMD032

33. An electron is accelerated from rest, between two points A and B at which the potentials are 20V and 40 V respectively. The De Broglie wavelength associated with the electron at B will be -

(1) 0.75 Å (2) 7.5 Å
(3) 2.75 Å (4) 2.75 m

PMD033

34. An electron is moving with velocity 6.6×10^3 m/s. The De-Broglie wavelength associated with electron is (mass of electron = 9×10^{-31} Kg, Plank's Constant = 6.62×10^{-34} J-S)

(1) 1×10^{-19} m (2) 1×10^{-5} m
(3) 1×10^{-7} m (4) 1×10^{-10} m

PMD034

35. The energy that should be added to an electron to reduce its De-Broglie wavelength from 10^{-10}m to 0.5×10^{-10} m will be :

(1) Four times the initial energy
(2) Equal to initial energy
(3) Twice the initial energy
(4) Thrice the initial energy

PMD035

36. The magnitude of De broglie wavelength (λ) of electron (e), proton (p), neutron (n) and α - particle (α) all having the same kinetic energy of 1MeV, in the increasing order will follow the sequence :

(1) $\lambda_e, \lambda_p, \lambda_n, \lambda_\alpha$ (2) $\lambda_e, \lambda_n, \lambda_p, \lambda_\alpha$
(3) $\lambda_\alpha, \lambda_n, \lambda_p, \lambda_e$ (4) $\lambda_p, \lambda_e, \lambda_\alpha, \lambda_n$

PMD036

37. The accelerating voltage of an electron gun is 50,000 volt. De-Broglie wavelength of the electron will be :

(1) 0.55 Å (2) 0.055 Å
(3) 0.077Å (4) 0.095 Å

PMD037

38. If the mass of neutron = 1.7×10^{-27} kg. then the De broglie wavelength of neutron of energy 3eV is :

(1) 1.6×10^{-10} m (2) 1.6×10^{-11} m
(3) 1.4×10^{-10} m (4) 1.4×10^{-11} m

PMD038

39. A proton and an α-particle accelerated through same voltage. The ratio of their De-broglie wavelength will be :

(1) 1 : 2 (2) $2\sqrt{2}:1$
(3) $\sqrt{2}:1$ (4) 2 : 1

PMD039

40. The De Broglie wavelength of an atom at absolute temperature T K will be -

(1) $\frac{h}{mKT}$ (2) $\frac{h}{\sqrt{3mKT}}$
(3) $\frac{\sqrt{3mKT}}{h}$ (4) $\sqrt{3mKT}$

PMD040

41. The De-Broglie wavelength associated with electrons revolving round the nucleus in a hydrogen atom in ground state, will be-

(1) 0.3 Å (2) 3.3 Å
(3) 6.62 Å (4) 10 Å

PMD041

42. The wavelength of very fast moving electron ($v \approx c$) is :

(1) $\lambda = \frac{h}{m_0 v}$ (2) $\lambda = \frac{h}{\sqrt{2mE}}$
(3) $\lambda^2 = \frac{h^2}{\sqrt{2mE}}$ (4) $\lambda = \frac{h\sqrt{1-\frac{v^2}{c^2}}}{m_0 v}$

PMD042

43. Which experiment explains the wave nature of electron :-

(1) Michelson's experiment
(2) Davisson Germer experiment
(3) Roentgen experiment
(4) Rutherford experiment

PMD043

44. An electron and a proton have the same De-Broglie wavelength. Then the kinetic energy of the electron is :
(1) zero
(2) Infinity
(3) Equal to kinetic energy of proton
(4) Greater than the kinetic energy of proton

PMD044

45. De Broglie equation for an electron shows is :
(1) Particle nature (2) Wave nature
(3) Dual nature (4) None of these

PMD045

46. What will happen to De Broglie's wavelength if the velocity of electron is increased :
(1) It will increase
(2) It will decrease
(3) It will remain same
(4) It will become twice

PMD046

47. A photon of wavelength 4400Å is passing through vaccum. The effective mass and momentum of the photon are respectively
(1) 5×10^{-36} kg, 1.5×10^{-27} kg – m/s
(2) 5×10^{-35} kg, 1.5×10^{-26} kg – m/s
(3) zero, 1.5×10^{-26} kg – m/s
(4) 5×10^{-36} kg, 1.67×10^{-43} kg – m/s

PMD047

48. Which of the following is true for photon :-
(1) $E = \frac{hc}{\lambda}$ (2) $E = \frac{1}{2}mv^2$
(3) $P = \frac{E}{2V}$ (4) $E = \frac{1}{2}mc^2$

PMD048

49. For a moving particle having kinetic energy E, the correct de Broglie wavelength is :
(1) It is not applicable for a particle
(2) $\frac{h}{\sqrt{2mE}}$
(3) $E\sqrt{\frac{h}{2m}}$
(4) $\frac{h}{2mE}$

PMD049

50. The De Broglie wavelength of an electron in the first bohr orbit is :
(1) Equal to the circumference of the first orbit
(2) Equal to twice the circumference of the first orbit
(3) Equal to half the circumference of the first orbit
(4) Equal to one fourth the circumference of first orbit

PMD050

51. If given particles are moving with same velocity, then maximum de-Broglie wavelength for :
(1) Proton (2) α - particle
(3) Neutron (4) β - particle

PMD051

52. A proton is about 1840 times heavier than an electron. When it is accelerated by a potential difference of 1 kV, its kinetic energy will be :–
(1) 1840 keV. (2) 1/1840 keV.
(3) 1 keV. (4) 920 keV.

PMD052

53. If an electron and a photon propagate with same wavelength, it implies that they can have the same:-
(1) Energy
(2) Momentum
(3) Velocity
(4) Angular momentum

PMD053

54. According to De Broglie, wavelength of electron in second orbit is 10^{-9} metre. Then the circumference of orbit is :–
(1) 10^{-9} m (2) 2×10^{-9} m
(3) 3×10^{-9} m (4) 4×10^{-9}m

PMD054

55. If the mass of a microscopic particle as well as its speed are halved, the de Broglie wavelength associated with the particle will
(1) increased by a factor more than 2
(2) increase by a factor of 2
(3) decrease by a factor of 2
(4) decrease by a factor more than 2

PMD055

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56. An electron and proton are accelerated through same potential, then λ_e/λ_p will be

(1) 1 (2) m_e/m_p

(3) m_p/m_e (4) $\sqrt{m_p/m_e}$

PMD056

57. An electron, proton and alpha particle have same kinetic energy. The corresponding de-Broglie wavelength would have the following relationship

(1) $\lambda_e > \lambda_p > \lambda_\alpha$ (2) $\lambda_p > \lambda_e > \lambda_\alpha$

(3) $\lambda_\alpha > \lambda_e > \lambda_p$ (4) $\lambda_\alpha > \lambda_p > \lambda_e$

PMD057

58. What is the de Broglie wavelength of an electron with a kinetic energy of 120 eV ?
(Given : $h = 6.63 \times 10^{-34}$ J s, $m_e = 9.11 \times 10^{-31}$kg and $e = 1.6 \times 10^{-19}$ coulomb)

(1) 725 pm (2) 500 pm

(3) 322 pm (4) 112 pm

PMD058

59. If λ_p and λ_∞ be the wavelengths of protons and α-particles of equal kinetic energies, then

(1) $\lambda_p = \frac{\lambda_\alpha}{4}$ (2) $\lambda_p = \frac{\lambda_\alpha}{2}$

(3) $\lambda_p = \lambda_\alpha$ (4) $\lambda_p = 2\lambda_\alpha$

PMD059

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	2	3	4	3	1	2	1	1	2	2	2	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	3	3	2	4	2	2	4	1	3	4	1	2	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	1	3	3	4	3	2	2	2	2	2	4	2	4	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	
Answer	2	1	1	2	1	4	3	2	2	1	4	1	4	4	

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Light with an energy flux of 25×10^4 Wm^{-2} falls on a perfectly reflecting surface at normal incidence. If the surface area is 15 cm^2, the average force exerted on the surface is :-

(1) 1.25×10^{-6} N (2) 2.50×10^{-6} N

(3) 1.20×10^{-6} N (4) 3.0×10^{-6} N

PMD060

2. When the energy of the incident radiation is increased by 20%, the kinetic energy of the photoelectrons emitted from a metal surface increased from 0.5 eV to 0.8 eV. The work function of the metal is :-

(1) 0.65 eV (2) 1.0 eV

(3) 1.3 eV (4) 1.5 eV

PMD061

3. If the kinetic energy of the particle is increased to 16 times its previous value, the percentage change in the de-Broglie wavelength of the particle is :-

(1) 25 (2) 75 (3) 60 (4) 50

PMD062

AIPMT 2015

4. A certain metallic surface is illuminated with monochromatic light of wavelength, λ. The stopping potential for photo-electric current for this light is $3V_0$. if the same surface is illuminated with light of wavelength 2λ, the stopping potential is V_0. The threshold wavelength for this surface for photoelectric effect is :-

(1) 4λ (2) $\frac{\lambda}{4}$ (3) $\frac{\lambda}{6}$ (4) 6λ

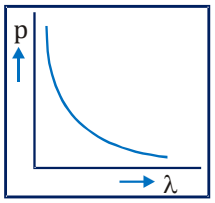
PMD063

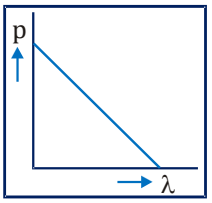
5. A radiation of energy 'E' falls normally on a perfectly reflecting surface. The momentum transferred to the surface is (C=Velocity of light):-

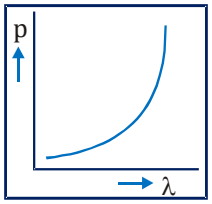
(1) $\frac{2E}{C}$ (2) $\frac{2E}{C^2}$ (3) $\frac{E}{C^2}$ (4) $\frac{E}{C}$

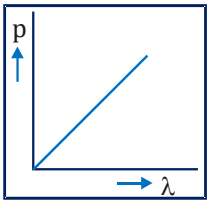
PMD064

6. Which of the following figures represent the variation of particle momentum and the associated de-Broglie wavelength ?

(1) 

(2) 

(3) 

(4) 

PMD065

RE-AIPMT 2015

7. Light of wavelength 500 nm is incident on a metal with work function 2.28 eV. The de Broglie wavelength of the emitted electron is :-

(1) $\leq 2.8 \times 10^{-12}$m (2) $< 2.8 \times 10^{-10}$m

(3) $< 2.8 \times 10^{-9}$ m (4) $\geq 2.8 \times 10^{-9}$ m

PMD066

8. A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and $\lambda/2$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface of the material is: (h = Plank's constant, c = speed of light)

(1) $\frac{hc}{3\lambda}$ (2) $\frac{hc}{2\lambda}$ (3) $\frac{hc}{\lambda}$ (4) $\frac{2hc}{\lambda}$

PMD067

NEET-I 2016

9. An electron of mass m and a photon have same energy E. The ratio of de-Broglie wavelengths associated with them is :

(1) $\frac{1}{c}\left(\frac{E}{2m}\right)^{\frac{1}{2}}$ (2) $\left(\frac{E}{2m}\right)^{\frac{1}{2}}$

(3) $c(2mE)^{\frac{1}{2}}$ (4) $\frac{1}{xc}\left(\frac{2m}{E}\right)^{\frac{1}{2}}$

PMD068

10. When a metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V. If the same surface is illuminated with radiation of wavelength 2λ, the stopping potential is $\frac{V}{4}$. The threshold wavelength for the metallic surface is :-

(1) 4 λ (2) 5 λ (3) $\frac{5}{2}\lambda$ (4) 3 λ

PMD069

NEET-II 2016

11. Photons with energy 5 eV are incident on a cathode C in a photoelectric cell. The maximum energy of emitted photoelectrons is 2 eV. When photons of energy 6 eV are incident on C, no photoelectrons will reach the anode A, if the stopping potential of A relative to C is :-

(1) – 1 V (2) – 3 V (3) + 3 V (4) + 4 V

PMD070

NEET(UG) 2017

12. The de-Broglie wavelength of a neutron in thermal equilibrium with heavy water at a temperature T (Kelvin) and mass m, is :-

(1) $\frac{h}{\sqrt{3mkT}}$ (2) $\frac{2h}{\sqrt{3mkT}}$

(3) $\frac{2h}{\sqrt{mkT}}$ (4) $\frac{2h}{\sqrt{mkT}}$

PMD071

13. The photoelectric threshold wavelength of silver is 3250×10^{-10}m. The velocity of the electron ejected from a silver surface by ultraviolet light of wavelength 2536×10^{-10} m is :-

(Given h = 4.14×10^{-15} eVs and c = 3×10^{8} ms^{-1})

(1) $\approx 0.6 \times 10^{4}$ ms^{-1} (2) $\approx 61 \times 10^{3}$ ms^{-1}

(3) $\approx 0.3 \times 10^{6}$ ms^{-1} (4) $\approx 6 \times 10^{5}$ ms^{-1}

PMD072

NEET(UG) 2018

14. An electron of mass m with an initial velocity $\vec{V} = V_0\hat{i}(V_0 > 0)$ enters an electric field $\vec{E} = -E_0\hat{i}$ (E_0 = constant >0) at t = 0. If λ_0 is its de-Broglie wavelength initially, then its de-Broglie wavelength at time t is :-

(1) $\frac{\lambda_0}{\left(1+\frac{eE_0}{mV_0}t\right)}$ (2) $\lambda_0\left(1+\frac{eE_0}{mV_0}y\right)$

(3) $\lambda_0 t$ (4) λ_0

PMD073

15. When the light of frequency $2\nu_0$ (where ν_0 is threshold frequency), is incident on a metal plate, the maximum velocity of electrons emitted is v_1. When the frequency of the incident radiation is increased to $5\nu_0$, the maximum velocity of electrons emitted from the same plate is v_2. The ratio of v_1 to v_2 is :-

(1) 1 : 2 (2) 1 : 4 (3) 4 : 1 (4) 2 : 1

PMD074

NEET(UG) 2019

16. An electron is accelerated through a potential difference of 10,000 V. Its de Broglie wavelength is, (nearly): ($m_e = 9 \times 10^{-31}$ kg)

(1) 12.2×10^{-13} m (2) 12.2×10^{-12} m

(3) 12.2×10^{-14} m (4) 12.2 nm

PMD075

NEET(UG) 2019 (Odisha)

17. The work function of a photosensitive material is 4.0 eV. The longest wavelength of light that can cause photon emission from the substance is (approximately)

(1) 3100 nm (2) 966 nm

(3) 31 nm (4) 310 nm

PMD076

18. A proton and an α-particle are accelerated from rest to the same energy. The de Broglie wavelengths λ_p and λ_α are in the ratio,

(1) 2 : 1 (2) 1 : 1 (3) $\sqrt{2}$:1 (4) 4 : 1

PMD077

NEET(UG) 2020

19. Light with an average flux of 20 W/cm^2 falls on a non-reflecting surface at normal incidence having surface area 20 cm^2. The energy received by the surface during time span of 1 minute is :

(1) 48×10^3 J (2) 10×10^3 J

(3) 12×10^3 J (4) 24×10^3 J

PMD078

20. The energy required to break one bond in DNA is 10^{-20} J. This value in eV is nearly :

(1) 0.006 (2) 6 (3) 0.6 (4) 0.06

PMD079

21. An electron is accelerated from rest through a potential difference of V volt. If the de Broglie wavelength of the electron is 1.227×10^{-2} nm, the potential difference is :

(1) 10^4 V (2) 10 V (3) 10^2 V (4) 10^3 V

PMD080

22. Light of frequency 1.5 times the threshold frequency is incident on a photosensitive material. What will be the photoelectric current if the frequency is halved and intensity is doubled ?

(1) zero (2) doubled

(3) four times (4) one-fourth

PMD081

NEET(UG) 2020 (COVID-19)

23. The de Broglie wavelength of an electron moving with kinetic energy of 144 eV is nearly

(1) $10^2 \times 10^{-3}$ nm (2) $10^2 \times 10^{-4}$ nm

(3) $10^2 \times 10^{-5}$ nm (4) $10^2 \times 10^{-2}$ nm

PMD082

24. The wave nature of electrons was experimentally verified by,

(1) de Broglie

(2) Hertz

(3) Einstein

(4) Davisson & Germer

PMD083

NEET(UG) 2021

25. The number of photons per second on an average emitted by the source of monochromatic light of wavelength 600 nm, when it delivers the power of 3.3×10^{-3} watt will be : ($h = 6.6 \times 10^{-34}$ Js)

(1) 10^{18} (2) 10^{17} (3) 10^{16} (4) 10^{15}

PMD084

26. An electromagnetic wave of wavelength 'λ' is incident on a photosensitive surface of negligible work function. If 'm' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d, then :

(1) $\lambda = \left(\frac{2m}{hc}\right)\lambda_d^2$ (2) $\lambda_d = \left(\frac{2mc}{h}\right)\lambda^2$

(3) $\lambda = \left(\frac{2mc}{h}\right)\lambda_d^2$ (4) $\lambda = \left(\frac{2h}{mc}\right)\lambda_d^2$

PMD085

NEET(UG) 2022

27. When two monochromatic lights of frequency, ν and $\frac{\nu}{2}$ are incident on a photoelectric metal, their stopping potential becomes $\frac{V_s}{2}$ and V_s respectively. The threshold frequency for this metal is:

(1) 3ν (2) $\frac{2}{3}\nu$ (3) $\frac{3}{2}\nu$ (4) 2ν

PMD086

RE-NEET(UG) 2022

28. The light rays having photons of energy 4.2 eV are falling on a metal surface having a work function of 2.2 eV. The stopping potential of the surface is :

(1) 2 eV (2) 2 V (3) 1.1 V (4) 6.4 V

PMD087

29. The threshold frequency of a photoelectric metal is ν_0. If light of frequency $4\nu_0$ is incident on this metal, then the maximum kinetic energy of emitted electrons will be :

(1) $h\nu_0$ (2) $2\ h\nu_0$ (3) $3\ h\nu_0$ (4) $4\ h\nu_0$

PMD088

NEET(UG) 2023

30. The work functions of Caesium (Cs), potassium (K) and Sodium (Na) are 2.14 eV, 2.30 eV and 2.75 eV respectively. If incident electromagnetic radiation has an incident energy of 2.20 eV, which of these photosensitive surfaces may emit photoelectrons ?

(1) Both Na and K (2) K only
(3) Na only (4) Cs only

PMD089

RE-NEET(UG) 2023

31. The de Broglie wavelength associated with an electron, accelerated by a potential difference of 81 V is given by:

(1) 13.6 nm (2) 136 nm
(3) 1.36 nm (4) 0.136 nm

PMD090

32. The maximum kinetic energy of the emitted photoelectrons in photoelectric effect is independent of :

(1) work function of material
(2) intensity of incident radiation
(3) frequency of incident radiation
(4) wavelength of incident radiation

PMD091

NEET(UG) 2024

33. If c is the velocity of light in free space, the correct statements about photon among the following are :

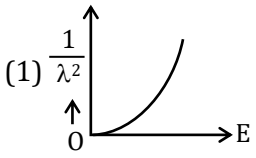
A. The energy of a photon is $E = h\nu$
B. The velocity of a photon is c.
C. The momentum of a photon, $p = \frac{h\nu}{c}$
D. In a photon-electron collision, both total energy and total momentum are conserved.
E. Photon possesses positive charge.

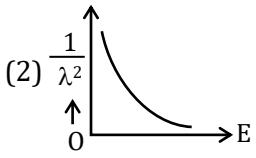
Choose the correct answer from the options given below :

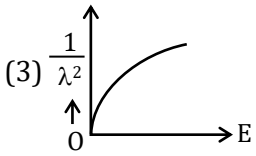
(1) A and B only (2) A, B, C and D only
(3) A, C and D only (4) A, B, D and E only

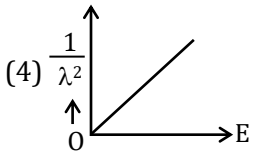
PMD092

34. The graph which shows the variation of $\left(\frac{1}{\lambda^2}\right)$ and its kinetic energy, E is (where λ is de Broglie wavelength of a free particle) :

(1) 

(2) 

(3) 

(4) 

PMD093

RE-NEET(UG) 2024

35. An electron and an alpha particle are accelerated by the same potential difference. Let λ_e and λ_α denote the de Broglie wavelengths of the electron and the alpha particle, respectively then :

(1) $\lambda_e > \lambda_\alpha$ (2) $\lambda_e = 4\,\lambda_\alpha$
(3) $\lambda_e = \lambda_\alpha$ (4) $\lambda_e < \lambda_\alpha$

PMD094

36. If ϕ is the work function of photosensitive material in eV and light of wavelength of numerical value $\lambda = \frac{hc}{e}$ metre, is incident on it with energy above its threshold value at an instant then the maximum kinetic energy of the photo-electron ejected by it at that instant (Take h - Planck's constant, c - velocity of light in free space) is (in SI units) :

(1) $e + 2\phi$ (2) $2e - \phi$ (3) $e - \phi$ (4) $e + \phi$

PMD095

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	2	1	1	1	4	2	1	4	2	1	4	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	1	4	4	1	1	1	4	3	3	Bonus	2	3	4
Question	31	32	33	34	35	36									
Answer	4	2	2	4	1	3									

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Exercise - III (Analytical Questions) Master Your Understanding

1. The work function of caesium (Cs) metal is 2 eV. When monochromatic radiation beam of intensity 60 kWm^{-2} and frequency 6×10^{14}Hz, is falling normally on Cs plate of 6.6 cm^2 in a photocell then
(a) Energy of a photoelectron may be 8×10^{-20} J
(b) Stopping potential is 0.48 V
(c) Saturation current is 16 A
Correct choices are
(1) Only b (2) Only b & c
(3) Only a & b (4) a, b, & c

PMD096

2. Two statements are given below :
Statement I : Photosensitivity of a metal is high, if its work function is low.
Statement II : Work function for a metal is given by hv_0, where v_o is threshold frequency.
Considering the above, choose the appropriate answer from the options given below :
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

PMD097

3. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : Two photons each of energy 3 eV, can't eject electrons from a surface of work function 5 eV.
Reason (R) : Only a photon of energy greater than work function may emit photoelectron.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PMD098

4. On a photosensitive surface of threshold wavelength 5λ, two different monochromatic beams of wavelengths 3λ and λ are incident, one after another, then match the ratio of physical quantities specified in column – I to the ratio values given in column – II

	Column I		Column II
(a)	Maximum KE of photo electrons	(i)	1 : 3
(b)	Speed of incident photons	(ii)	1 : 6
(c)	Momentum of incident photons	(iii)	$1:\sqrt{6}$
(d)	Fastest speeds of photo electrons	(iv)	1 : 1

Choose the correct answer from the option given below :
(1) (a) – iii, (b) – ii, (c) – i, (d) – iv
(2) (a) – ii, (b) – iv, (c) – iii, (d) – i
(3) (a) – ii, (b) – iv, (c) – i, (d) – iii
(4) (a) – i, (b) – ii, (c) – iv, (d) – iii

PMD099

5. Match the column I with column II –

	Column-I		Column-II
(A)	Force between a proton and a proton inside nucleus	(P)	Gravitational force
(B)	Force between a proton and a neutron inside nucleus	(Q)	Electrostatic force
(C)	Force on an electron in an atom	(R)	Strong force
(D)	Conservative force	(S)	Weak force

Options :-
(1) A → PQR, B → PR, C → PQ, D → PQ
(2) A → R, B → R, C → Q, D → RS
(3) A → PQR, B → PQR, C → PQ, D → RS
(4) A → PQR, B → P, C → PQ, D → RS

PMD100

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6. A proton and an electron both have energy 50 eV.

Statement I : Both have different wavelengths.

Statement II : Wavelength depends on energy, not on mass.

In light of above statement choose the most **appropriate** answer from the options given below:

(1) Both statement I and statement II are correct.

(2) Statement I is correct and statement II is incorrect.

(3) Statement I is incorrect and statement II is correct.

(4) Both statements I and statements II are incorrect.

PMD101

7. Match the following ratio of De–Broglie wavelengths of a proton and α particle, when both

	List-I		List-II
A	Have same speed	(p)	2 : 1
B	Have same potential difference	(q)	1 : 1
C	Have same kinetic energy.	(r)	4 : 1
D	Have same linear momentum	(s)	$2\sqrt{2}:1$

	A	B	C	D
1	p	q	r	s
2	r	s	p	q
3	p	r	q	s
4	r	q	s	p

PMD102

8. The wavelength of a photon is twice the de Broglie wavelength of an electron. The speed of the electron is $v_e = \frac{c}{100}$ then,

(a) $\frac{E_e}{E_p} = 10^{-4}$

(b) $\frac{E_e}{E_p} = 10^{-2}$

(c) $\frac{p_e}{m_e c} = 10^{-2}$

(d) $\frac{p_e}{m_e c} = 10^{-4}$

(1) a, c, d (2) b, c

(3) c, d (4) a, b, c

PMD103

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	4	1	1	3	1	2	2	2

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Exercise - I (Conceptual Questions) Build Up Your Understanding

NUCLEAR PHYSICS

1. Let F_{pp}, F_{pn} and F_{nn} denote the nuclear force between proton–proton, proton–neutron and neutron–neutron pair respectively. When separation is 1 fm :-

(1) $F_{pp} < F_{pn} = F_{nn}$ (2) $F_{pp} > F_{pn} = F_{nn}$

(3) $F_{pp} = F_{pn} = F_{nn}$ (4) $F_{pp} < F_{pn} < F_{nn}$

PMD127

2. A nuclear fission is given below

$A^{240} \rightarrow B^{100} + C^{140} + Q(\text{energy})$

Let binding energy per nucleon of nucleus A, B and C is 7.6 MeV, 8.1 MeV and 8.1 MeV respectively. Value of Q is :-(Approximately)

(1) 20 MeV (2) 220 MeV

(3) 120 MeV (4) 240 MeV

PMD128

3. How much energy will be released when 10 kg of U^{235} is completely converts into energy :-

(1) 5×10^{27} MeV (2) 5×10^{24} MeV

(3) 9×10^{17} J (4) All of these

PMD129

4. How much energy is released when 2 mole of U^{235} is fissioned :-

(1) 10^{24} MeV (2) 24×10^{25} MeV

(3) 10^{24} J (4) 10^{24} kWh

PMD130

5. As the mass number increases, binding energy per nucleon :-

(1) increases

(2) decreases

(3) remains same

(4) may increase or may decrease

PMD131

6. Possible forces on a proton by a proton in a nucleus is/are :-

(1) Coulomb force (2) Nuclear force

(3) Gravitational force (4) All of these

PMD132

7. The energy radiated by a red giant star produces by :-

(1) Fission process

(2) Fusion process

(3) Chemical burning of hydrogen

(4) Gravitational contraction

PMD133

8. In the process of atomic explosion, the most of fission energy is released in the form of :–

(1) γ - rays

(2) Kinetic energy of products

(3) Infra red rays

(4) Visible light

PMD134

9. Which of the following nucleus is fissionable by slow neutrons :-

(1) $_{92}U^{238}$ (2) $_{93}Np^{239}$

(3) $_{92}U^{235}$ (4) $_{2}He^{4}$

PMD135

10. The example of nuclear fusion is .

(1) formation of barium and krypton from uranium

(2) formation of helium from hydrogen

(3) formation of plutonium –235 from uranium –235

(4) formation of water from hydrogen and oxygen

PMD136

11. Electron - positron pair can be created by γ-rays. In this process the minimum energy of γ-rays should be :-

(1) 5.0 MeV (2) 4.02 MeV

(3) 15.0 MeV (4) 1.02 MeV

PMD137

12. For nuclear reaction :

$_{92}U^{235} + {}_{0}n^{1} \rightarrow {}_{56}Ba^{144} + ...+3\ {}_{0}n^{1}$

(1) $_{26}Kr^{89}$ (2) $_{36}Kr^{89}$

(3) $_{26}Sr^{90}$ (4) $_{38}Sr^{89}$

PMD138

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13. For the given reaction, the particle X is -

$_6C^{11} \rightarrow {}_5B^{11} + \beta^+ + X$

(1) Neutron (2) Anti neutrino
(3) Neutrino (4) Proton

PMD139

14. In a breeder reactor, useful fuel obtained from U^{238} is :

(1) Ac^{233} (2) Th^{238}
(3) U^{235} (4) Pu^{239}

PMD140

15. Boron used in Atomic Reactor for:-

(1) absorbption of neutrons
(2) absorbption of α - particles
(3) speed up the reaction
(4) change the reaction

PMD141

16. Who discovered the nuclear fission :-

(1) Otto Hahn and strassman
(2) Fermi
(3) Baithe
(4) Rutherford

PMD142

17. Which one is best neutron moderator in all respects:

(1) Barium oxide (2) Water
(3) Graphite (4) Heavy water

PMD143

18. X(n, α) ^7_3Li , then the element X will be :–

(1) $^{10}_5B$ (2) ^9_5B (3) $^{11}_4Be$ (4) ^4_2He

PMD144

19. M_n and M_p represent the mass of neutron and proton respectively. An element having nuclear mass M has N neutrons and Z-protons, then the correct relation will be :-

(1) $M<\{N.M_n+Z.M_p\}$ (2) $M>\{N.M_n+Z.M_p\}$
(3) $M=\{N.M_n+Z.M_p\}$ (4) $M=N\{M_n + M_p\}$

PMD145

20. Energy is released in nuclear fission is due to

(a) Few mass is converted into energy
(b) Total binding energy of fragments is more than the B.E. of parental element
(c) Total B.E. of fragments is less than the B.E. of parental element
(d) Total B.E. of fragments is equals to the B.E. of parental element is

(1) a,c (2) a,b (3) a,d (4) All

PMD146

21. Energy in an atom bomb is produced by the process of :

(1) nuclear fusion
(2) nuclear fission
(3) combination of hydrogen atoms
(4) combination of electrons and protons

PMD147

22. Assuming that 200 MeV of energy is released per fission of $_{92}U^{235}$ atom. Find the number of fission per second required to release 1 kW power :-

(1) 3.125×10^{13} (2) 3.125×10^{14}
(3) 3.125×10^{15} (4) 3.125×10^{16}

PMD148

23. 1 a.m.u. (1.66×10^{-27} kg) is equal to

(1) 139 MeV/c^2 (2) 39 MeV/c^2
(3) 93 MeV/c^2 (4) 931 MeV/c^2

PMD149

24. In nuclear fission the percentage of mass converted into energy is about :-

(1) 0.1% (2) 1%
(3) 10% (4) 0.01%

PMD150

25. Which of the following are suitable for the fusion process :-

(1) Light nuclei
(2) heavy nuclei
(3) Element must be lying in the middle of the periodic table
(4) Middle elements, which are lying on binding energy curve

PMD151

26. Which one of the following particle is unstable?

(1) α-particle (2) electron
(3) proton (4) neutron

PMD152

27. Which of the following is weakest force :-
(1) Gravitational force
(2) Electric force
(3) Magnetic force
(4) Nuclear force

PMD153

28. The volume occupied by an atom is greater than the volume of the nucleus by a factor of about :-
(1) 10^1 (2) 10^5 (3) 10^{10} (4) 10^{15}

PMD154

29. The mass of proton is 1.0073 u and that of neutron is 1.0087 u (u = atomic mass unit). The binding energy of $_2{}^4He$ is :-
(1) 0.0305 J (2) 0.0305 erg
(3) 28.4 MeV (4) 0.061 u
(Given:- mass of helium nucleus 4.0015 u)

PMD155

30. The mass number of a nucleus is
(1) always less than its atomic number
(2) always more than its atomic number
(3) may equal to its atomic number
(4) sometimes less than and sometimes more than its atomic number

PMD156

31. In the following reaction X is :–
$_{20}Ca^{40} + X \rightarrow {}_{21}Sc^{43} + {}_1H^1$
(1) Electron (2) Positron
(3) alpha particle (4) Proton

PMD157

32. Nuclear fusion is possible :–
(1) only between light nuclei.
(2) only between heavy nuclei.
(3) between both light and heavy nuclei.
(4) only between nuclei which are stable against β–decay.

PMD158

33. The order of nuclear density is
(1) 10^{13} kg/m^3 (2) 10^{15} kg/m^3
(3) 10^{17} kg/m^3 (4) 10^{19} kg/m^3

PMD159

34. Two light nuclei of masses m_1 and m_2 are fused to form a more stable nucleus of mass m_3 then:-
(1) $m_3 = |m_1 - m_2|$ (2) $m_3 < (m_1 + m_2)$
(3) $m_3 > (m_1 + m_2)$ (4) $m_3 = m_1 + m_2$

PMD160

35. A nucleus represented by the symbol ^A_ZX has :-
(1) Z protons and A – Z neutrons
(2) Z protons and A neutrons
(3) A protons and Z – A neutrons
(4) Z neutrons and A – Z protons

PMD161

36. M_p denotes the mass of a proton and M_n that of a neutron. A given nucleus, of binding energy B, contains Z protons and N neutrons. The mass M (N, Z) of the nucleus is given by (c is velocity of light) :-
(1) $M(N, Z) = NM_n + ZM_p + Bc^2$
(2) $M(N, Z) = NM_n + ZM_p - B/c^2$
(3) $M(N, Z) = NM_n + ZM_p + B/c^2$
(4) $M(N, Z) = NM_n + ZM_p - Bc^2$

PMD162

37. Mass equivalent to energy 931 MeV is
(1) 6.02×10^{-27} kg (2) 1.66×10^{-27} kg
(3) 16.66×10^{-26} kg (4) 6.02×10^{-26} kg

PMD163

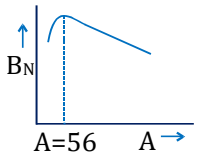
38. One milligram of matter converted into energy will give
(1) 9 J (2) 9×10^{13} J
(3) 9×10^5 J (4) 9×10^{10} J

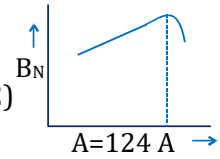
PMD164

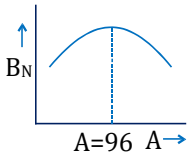
39. Force acting on proton-proton inside a nucleus is–
(1) Nuclear force > electric force
(2) Electric force > Nuclear force
(3) Gravitational force > Nuclear force
(4) None of the above

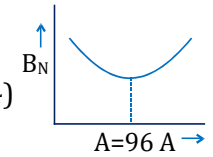
PMD165

40. The dependence of binding energy per nucleon (B_N) on the mass number (A), is represented by

(1) 

(2) 

(3) 

(4) 

PMD166

41. Which one of the following pairs of nuclei are isotones :-

(1) $_{34}Se^{74}$, $_{31}Ga^{71}$ (2) $_{38}Sr^{84}$, $_{38}Sr^{86}$

(3) $_{42}Mo^{92}$, $_{40}Zr^{92}$ (4) $_{20}Ca^{40}$, $_{16}S^{32}$

PMD167

42. Which one of the following is a possible nuclear reaction :-

(1) $^{10}_{5}B + {}^{4}_{2}He \longrightarrow {}^{13}_{7}N + {}^{1}_{1}H$

(2) $^{23}_{11}Na + {}^{1}_{1}H \longrightarrow {}^{22}_{10}Ne + {}^{4}_{2}He$

(3) $^{239}_{93}Np \longrightarrow {}^{239}_{94}Pu + \beta^- + \bar{\nu}$

(4) $^{11}_{7}N + {}^{1}_{1}H \longrightarrow {}^{12}_{6}C + \beta^- + \nu$

PMD168

43. In the reaction $^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + {}^{1}_{0}n$. If the binding energies of $^{2}_{1}H$, $^{3}_{1}H$ and $^{4}_{2}He$ are respectively a, b and c (in MeV), then the energy (in MeV) released in this reaction is

(1) a + b + c (2) c + a – b

(3) c – a – b (4) a + b + c

PMD169

44. In any fission process the ratio $\frac{\text{mass of fission products}}{\text{mass of parent nucleus}}$ is –

(1) Greater than 1

(2) Depends on the mass of the parent nucleus

(3) Equal to 1

(4) Less than 1

PMD170

45. Fission of nuclei is possible because the binding energy per nucleon in them –

(1) Decreases with mass number at low mass numbers.

(2) Increases with mass number at low mass numbers.

(3) Decreases with mass number at high mass numbers.

(4) Increases with mass number at high mass numbers.

PMD171

46. The main function of moderators in nuclear reactors is to :-

(1) decrease the energy of neutrons

(2) absorb the extra neutrons

(3) provide shield from nuclear radiations

(4) provide cooling

PMD172

RADIOACTIVITY

47. Which spectrum is continuous

(1) α–rays (2) β–rays

(3) γ–rays (4) All of these

PMD173

48. Which statement about radioactive radiations is true

(1) Speed of α-particles is a characteristic property.

(2) Speed of β-particles is a characteristic property.

(3) Speed of γ-photon is a characteristic property.

(4) All of these.

PMD174

49. Which one moves with greatest speed :-

(1) α-rays (2) β–rays

(3) γ–rays (4) cathode rays

PMD175

50. If a nucleus is emitting e^- particle, its neutron to proton ratio (n/p) will :–

(1) Increase.

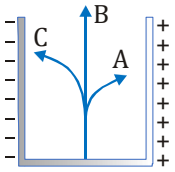
(2) Decrease

(3) Remain unchanged

(4) Can't be determined.

PMD176

51. A radioactive source is kept in an uniform electric field α, β and γ- particle are emitting. α,β,γ are respectively :-



(1) A,B,C (2) A,C,B (3) C,A,B (4) C,B,A

PMD177

52. A radioactive element ${}_{90}X^{238}$ decays in to ${}_{83}Y^{222}$. The number of β - particles emitted is:

(1) 2 (2) 4 (3) 6 (4) 1

PMD178

53. A radioactive reaction is ${}_{92}U^{238} \rightarrow {}_{82}Pb^{206}$. How many α and β particles are emitted ?

(1) 10α, 6β

(2) 4 proton, 8 neutron

(3) 6 electron, 8 proton

(4) 6β, 8α

PMD179

54. Which rays contain (+ Ve) charged particle:–

(1) α–rays (2) β–rays

(3) γ–rays (4) X–rays

PMD180

55. When ${}_{90}Th^{238}$ changes into ${}_{83}Bi^{222}$, then the number of emitted α and β-particles are :-

(1) 8α, 7β (2) 4α, 7β

(3) 4α, 4β (4) 4α, 1β

PMD181

56. A radioactive nucleus decay as follows :-

$X \xrightarrow{\alpha} X_1 \xrightarrow{\beta} X_2 \xrightarrow{\alpha} X_3 \xrightarrow{\gamma} X_4$,

if the atomic number and the mass number of X are 72 and 180 then the mass number and atomic number of X_4 are :-

(1) 172, 70 (2) 171, 69

(3) 172, 69 (4) 172, 68

PMD182

57. Which of the following ray are **not** electromagnetic waves :-

(1) X-rays (2) γ-rays

(3) β-rays (4) Heat rays

PMD183

58. A nuclear reaction given by

${}_{Z}X^{A} \rightarrow {}_{Z+1}Y^{A} + {}_{-1}e^{0} + \bar{\nu}$ represents.

(1) β-decay (2) γ-decay

(3) fusion (4) fission

PMD184

59. Which of the following radiations gets deflected by a magnetic field ?

(1) X-rays (2) γ-rays

(3) β-rays (4) radio waves

PMD185

60. α–Particles can be detected using :–

(1) Thin aluminium sheet

(2) Barium sulphate

(3) Zinc sulphide screen

(4) Gold foil

PMD186

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	3	2	4	4	2	2	3	2	4	2	3	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	1	1	2	2	1	4	1	1	4	1	4	3	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	1	3	2	1	2	2	4	1	1	1	3	3	4	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	2	1	3	2	3	4	4	1	4	3	3	1	3	3

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. The Binding energy per nucleon of $^{7}_{3}Li$ and $^{4}_{2}He$ nuclei are 5.60 MeV and 7.06 MeV, respectively. In the nuclear reaction $^{7}_{3}Li + {}^{1}_{1}H \rightarrow 2\,{}^{4}_{2}He + Q$, the value of energy Q released is:-

(1) 19.6 MeV (2) –2.4 MeV

(3) 8.4 MeV (4) 17.3 MeV

PMD187

AIPMT 2015

2. If radius of the $^{27}_{13}Al$ nucleus is taken to be R_{Al} then the radius of $^{125}_{53}Te$ nucleus is nearly :

(1) $\frac{5}{3}R_{Al}$ (2) $\frac{3}{5}R_{Al}$

(3) $\left(\frac{13}{53}\right)^{1/3}R_{Al}$ (4) $\left(\frac{53}{13}\right)^{1/3}R_{Al}$

PMD188

RE-AIPMT 2015

3. A nucleus of uranium decays at rest into nuclei of Thorium and Helium. Then :-

(1) The Helium nucleus has less kinetic energy than the Thorium nucleus

(2) The Helium has more kinetic energy than the Thorium nucleus.

(3) The Helium nucleus has less momentum than the Thorium nucleus.

(4) The Helium nucleus has more momentum than the Thorium nucleus.

PMD189

NEET(UG) 2019

4. α-particle consists of :

(1) 2 protons and 2 neutrons only

(2) 2 electrons, 2 protons and 2 neutrons

(3) 2 electrons and 4 protons only

(4) 2 protons only

PMD190

NEET(UG) 2020

5. When a uranium isotope $^{235}_{92}U$ is bombarded with a neutron, it generates $^{89}_{36}Kr$, three neutrons and:

(1) $^{103}_{36}Kr$ (2) $^{144}_{56}Ba$

(3) $^{91}_{40}Zr$ (4) $^{101}_{36}Kr$

PMD191

6. The energy equivalent of 0.5 g of a substance is:

(1) 0.5×10^{13} J (2) 4.5×10^{16} J

(3) 4.5×10^{13} J (4) 1.5×10^{13} J

PMD192

NEET(UG) 2020 (COVID-19)

7. What happens to the mass number and atomic number of an element when it emits γ-radiation?

(1) Mass number decreases by four and atomic number decreases by two.

(2) Mass number and atomic number remain unchanged.

(3) Mass number remains unchanged while atomic number decreases by one.

(4) Mass number increases by four and atomic number increases by two.

PMD193

NEET(UG) 2021

8. A nucleus with mass number 240 breaks into two fragments each of mass number 120, the binding energy per nucleon of unfragmented nuclei is 7.6 MeV while that of fragments is 8.5 MeV. The total gain in the Binding Energy in the process is :

(1) 0.9 MeV (2) 9.4 MeV

(3) 804 MeV (4) 216 MeV

PMD194

9. A radioactive nucleus $^{A}_{Z}X$ undergoes spontaneous decay in the sequence $^{A}_{Z}X \rightarrow {}_{Z-1}B \rightarrow {}_{Z-3}C \rightarrow {}_{Z-2}D$, where Z is the atomic number of element X. The possible decay particles in the sequence are :

(1) α, β^-, β^+ (2) α, β^+, β^-

(3) β^+, α, β^- (4) β^-, α, β^+

PMD195

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NEET(UG) 2022

10. In the given nuclear reaction, the element X is:

$^{22}_{11}Na \rightarrow X + e^{+} + \nu$

(1) $^{23}_{10}Ne$ (2) $^{22}_{10}Ne$ (3) $^{22}_{12}Mg$ (4) $^{23}_{11}Na$

PMD196

11. A nucleus of mass number 189 splits into two nuclei having mass number 125 and 64. The ratio of radius of two daughter nuclei respectively is:

(1) 4 : 5 (2) 5 : 4

(3) 25 : 16 (4) 1 : 1

PMD197

NEET(UG) 2024

12. $^{290}_{82}X \xrightarrow{\alpha} Y \xrightarrow{e^{+}} Z \xrightarrow{\beta^{-}} P \xrightarrow{e^{-}} Q$

In the nuclear emission stated above, the mass number and atomic number of the product Q respectively, are :

(1) 280, 81 (2) 286, 80

(3) 288, 82 (4) 286, 81

PMD198

RE-NEET(UG) 2024

13. Select the correct statements among the following :

A. Slow neutrons can cause fission in $^{235}_{92}U$ than fast neutrons.

B. α-rays are helium nuclei

C. β-rays are fast moving electrons or positrons.

D. γ-rays are electromagnetic radiations of wavelengths larger than X-rays.

Choose the **most appropriate** answer from the options given below :

(1) A, B and C only

(2) A, B and D only

(3) A and B only

(4) C and D only

PMD199

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer	4	1	2	1	2	3	2	4	3	2	2	4	1

Exercise - III (Analytical Questions) Master Your Understanding

1. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Radioactive nuclei emits β^-particle.

Reason (R) : Electrons exist inside the nucleus

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PMD200

2. Match the following

	List-I		List-II
a	$n \rightarrow p +$	(p)	α–decay
b	$p \rightarrow n +$	(q)	Positron emission
c	x–ray emission	(r)	β^-emission
d	$^{226}_{88}Ra \rightarrow ^{222}_{86}Rn +$ ___	(s)	K electron capture

(1) $a \rightarrow r, b \rightarrow q, c \rightarrow q, d \rightarrow s$

(2) $a \rightarrow q, b \rightarrow r, c \rightarrow s, d \rightarrow p$

(3) $a \rightarrow r, b \rightarrow q, c \rightarrow s, d \rightarrow p$

(4) $a \rightarrow q, b \rightarrow s, c \rightarrow r, d \rightarrow p$

PMD201

3. When a nucleus with atomic number Z and mass number A undergoes a radioactive decay process:

(A) both Z and A will decrease if the process is α– decay

(B) Z will decrease but A will not change if process is β^+ decay

(C) Z will decrease but A will not change if process is β^- decay.

(D) Z and A will remain unchanged if process is γ–decay.

Choose the correct statement :

(1) Only A (2) B, C, D

(3) A, D (4) A, B, D

PMD202

4. Choose correct statement/statements :

(a) Nuclear force is independent of charge of nucleons.

(b) Nuclear force has saturation property.

(c) Nuclear force is always Attractive is nature.

(d) Nuclear force exists in range of femtometre.

(1) a,b,c (2) a,b,d

(3) b,c,d (4) a, c, d

PMD203

5. Match the following column:

For two nuclei having mass number 2 and 16

(i) Ratio of nuclear radii	(p) 1 : 8
(ii) Ratio of densities	(q) 1 : 2
(iii) Ratio of volumes	(r) 1 : 1
(iv) Ratio of masses	(s) 1 : 8

(1) (i) $\rightarrow$ q, (ii) $\rightarrow$ r, (iii) $\rightarrow$ s, (iv) $\rightarrow$ p

(2) (i) $\rightarrow$ r, (ii) $\rightarrow$ s, (iii) $\rightarrow$ p, (iv) $\rightarrow$ q

(3) (i) $\rightarrow$ p, (ii) $\rightarrow$ q, (iii) $\rightarrow$ r, (iv) $\rightarrow$ s

(4) (i) $\rightarrow$ r, (ii) $\rightarrow$ p, (iii) $\rightarrow$ s, (iv) $\rightarrow$ q

PMD204

6. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Fission reactions are exothermic

Reason (R) : Total mass of fragments is lesser than the total mass of reactants.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PMD205

7. Given below are two statements

Statement I : Binding energy per nucleon is more important than binding energy.

Statement II : Binding energy per nucleon is measurement of nuclear stability.

In light of above statement choose the most **appropriate** answer from the options given below:

(1) Both statement I and statement II are correct.

(2) Statement I is correct and statement II is incorrect.

(3) Statement I is incorrect and statement II is correct.

(4) Both statements I and statements II are incorrect.

PMD206

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	3	3	4	2	1	1	1

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Important Notes

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Exercise - I (Conceptual Questions) Build Up Your Understanding

ATOMIC STRUCTURE

1. What is the wavelength of the least energetic photon emitted in the Lyman series of the hydrogen atom spectrum ?
(1) 150 nm (2) 122 nm
(3) 102 nm (4) 82 nm
PMD220

2. What is the ratio of the shortest wavelength of the Balmer series to the shortest wavelength of the Lyman series ?
(1) 4 : 1 (2) 4 : 3 (3) 4 : 9 (4) 5 : 9
PMD221

3. Kinetic energy for Hydrogen atom in first Bohr's orbit is–
(1) – 13.6 eV (2) 13.6 eV
(3) –27.2 eV (4) –6.5 eV
PMD222

4. According to Bohr Model for Hydrogen, energy is proportional to :
(1) $-Z^2 / n$ (2) $- n/Z^2$
(3) $-Z^2/n^2$ (4) $-n^2/Z$
PMD223

5. In above question radius is related as :-
(1) n^2 / Z (2) $\frac{n}{Z}$
(3) $\frac{n}{Z^2}$ (4) $\frac{n^2}{Z^2}$
PMD224

6. If ionization potential of Hydrogen atom is 13.6 V then what is ionization potential of He atom?
(1) 27.6 V (2) 13.6 V
(3) 54.2 V (4) None of these
PMD225

7. Which of the following statements is correct?
(1) Lyman series is continuous
(2) Balmer series lies in ultraviolet region
(3) Paschen series lies in infrared region
(4) Brackett series lies in visible region
PMD226

8. According to the Bohr theory of Hydrogen atom, the speed of the electron, its energy and the radius of its orbit varies with the principal quantum number n, respectively, as
(1) $\frac{1}{n}, \frac{1}{n^2}, n^2$ (2) $\frac{1}{n}, n^2, \frac{1}{n^2}$
(3) $n^2, \frac{1}{n^2}, n^2$ (4) $n, \frac{1}{n^2}, \frac{1}{n^2}$
PMD227

9. If the ionization potential of hydrogen atom is 13.6 eV, its energy in the n = 3 is approximately
(1) – 1.14 eV (2) – 1.51 eV
(3) –3.4 eV (4) – 4.53 eV
PMD228

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	2	1	2	3	1	3	3	1	2

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Hydrogen atom is ground state is excited by a monochromatic radiation of λ = 975 Å. Number of spectral lines in the resulting spectrum emitted will be :-

(1) 3 (2) 2 (3) 6 (4) 10

PMD229

AIPMT 2015

2. Consider 3rd orbit of He^+ (Helium), using non-relativistic approach, the speed of electron in this orbit will be [given K = 9×10^9 constant, Z = 2 and h (Planck's Constant) = 6.6×10^{-34} J s]

(1) 1.46×10^6 m/s

(2) 0.73×10^6 m/s

(3) 3.0×10^8 m/s

(4) 2.92×10^6 m/s

PMD230

RE-AIPMT 2015

3. In the spectrum of hydrogen, the ratio of the longest wavelength in the Lyman series to the longest wavelength in the Balmer series is :

(1) $\frac{5}{27}$ (2) $\frac{4}{9}$

(3) $\frac{9}{4}$ (4) $\frac{27}{5}$

PMD231

NEET-I 2016

4. When an α-particle of mass 'm' moving with velocity 'v' bombards on a heavy nucleus of charge 'Ze', its distance of closest approach from the nucleus depends on m as :

(1) $\frac{1}{m}$ (2) $\frac{1}{\sqrt{m}}$

(3) $\frac{1}{m^2}$ (4) m

PMD232

5. Given the value of Rydberg constant is $10^7 m^{-1}$, the wave number of the last line of the Balmer series in hydrogen spectrum will be :-

(1) 0.025×10^4 m^{-1} (2) 0.5×10^7 m^{-1}

(3) 0.25×10^7 m^{-1} (4) 2.5×10^7 m^{-1}

PMD233

NEET-II 2016

6. If an electron in a hydrogen atom jumps from the 3rd orbit to the 2nd orbit, it emits a photon of wavelength λ. When it jumps from the 4th orbit to the 3rd orbit, the corresponding wavelength of the photon will be :-

(1) $\frac{20}{7}\lambda$ (2) $\frac{20}{13}\lambda$

(3) $\frac{16}{25}\lambda$ (4) $\frac{9}{16}\lambda$

PMD234

NEET(UG) 2017

7. The ratio of wavelengths of the last line of Balmer series and the last line of Lyman series is:-

(1) 1 (2) 4 (3) 0.5 (4) 2

PMD235

NEET(UG) 2018

8. The ratio of kinetic energy to the total energy of an electron in a Bohr orbit of the hydrogen atom, is :-

(1) 1 : 1 (2) 1 : –1 (3) 2 : –1 (4) 1 : –2

PMD236

NEET(UG) 2019

9. The total energy of an electron in an atom in an orbit is –3.4 eV. Its kinetic and potential energies are, respectively:

(1) –3.4 eV, –3.4 eV

(2) –3.4 eV, –6.8 eV

(3) 3.4 eV, –6.8 eV

(4) 3.4 eV, 3.4 eV

PMD237

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NEET(UG) 2019 (Odisha)

10. The radius of the first permitted Bohr orbit for the electron, in a hydrogen atom equals 0.51 Å and its ground state energy equals –13.6 eV. If the electron in the hydrogen atom is replaced by muon(μ^-) [charge same as electron and mass 207 m_e], the first Bohr radius and ground state energy will be :

(1) 0.53×10^{-13} m, –3.6 eV

(2) 25.6×10^{-13} m, –2.8 eV

(3) 2.56×10^{-13} m, –2.8 keV

(4) 2.56×10^{-13} m, –13.6 eV

PMD238

NEET(UG) 2020

11. For which one of the following, Bohr model in **not** valid ?

(1) Singly ionised neon atom (Ne^+)

(2) Hydrogen atom

(3) Singly ionised helium atom (He^+)

(4) Deuteron atom

PMD239

NEET(UG) 2020 (COVID-19)

12. The total energy of an electron in the nth stationary orbit of the hydrogen atom can be obtained by

(1) $E_n = \frac{13.6}{n^2}$ eV (2) $E_n = -\frac{13.6}{n^2}$ eV

(3) $E_n = -\frac{1.36}{n^2}$ eV (4) $E_n = -13.6 \times n^2$ eV

PMD240

NEET(UG) 2021

13. An electromagnetic wave of wavelength 'λ' is incident on a photosensitive surface of negligible work function. If 'm' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d, then :

(1) $\lambda = \left(\frac{2m}{hc}\right)\lambda_d^2$ (2) $\lambda_d = \left(\frac{2mc}{h}\right)\lambda^2$

(3) $\lambda = \left(\frac{2mc}{h}\right)\lambda_d^2$ (4) $\lambda = \left(\frac{2h}{mc}\right)\lambda_d^2$

PMD241

RE-NEET(UG) 2022

14. Let R_1 be the radius of the second stationary orbit and R_2 be the radius of the fourth stationary orbit of an electron in Bohr's model. The ratio $\frac{R_1}{R_2}$ is :

(1) 0.25 (2) 0.5

(3) 2 (4) 4

PMD242

NEET(UG) 2023

15. The radius of inner most orbit of hydrogen atom is 5.3×10^{-11} m. What is the radius of third allowed orbit of hydrogen atom ?

(1) 1.06 Å (2) 1.59 Å

(3) 4.77 Å (4) 0.53 Å

PMD243

RE-NEET(UG) 2023

16. The ground state energy of hydrogen atom is –13.6 eV. The energy needed to ionize hydrogen atom from its second excited state will be :

(1) 13.6 eV (2) 6.8 eV

(3) 1.51 eV (4) 3.4 eV

PMD244

17. The wavelength of Lyman series of hydrogen atom appears in:

(1) visible region

(2) far infrared region

(3) ultraviolet region

(4) infrared region

PMD245

NEET(UG) 2024

18. Given below are two statements :

Statement I : Atoms are electrically neutral as they contain equal number of positive and negative charges.

Statement II : Atoms of each element are stable and emit their characteristic spectrum.

In the light of the above statements, choose the *most appropriate* answer from the options given below :

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(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

PMD246

19. Match List I with List II.

List I (Spectral Lines of Hydrogen for transitions from)	**List II (Wavelengths (nm))**
A. $n_2 = 3$ to $n_1 = 2$	I. 410.2
B. $n_2 = 4$ to $n_1 = 2$	II. 434.1
C. $n_2 = 5$ to $n_1 = 2$	III. 656.3
D. $n_2 = 6$ to $n_1 = 2$	IV. 486.1

Choose the correct answer from the options given below :
(1) A–II, B–I, C–IV, D–III
(2) A–III, B–IV, C–II, D–I
(3) A–IV, B–III, C–I, D–II
(4) A–I, B–II, C–III, D–IV

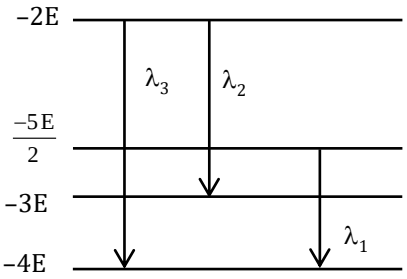
PMD247

RE-NEET(UG) 2024

20. The spectral series which corresponds to the electronic transition from the levels $n_2 = 5,6,...$ to the level $n_1 = 4$ is:
(1) Pfund series
(2) Brackett series
(3) Lyman series
(4) Balmer series

PMD248

21. Some energy levels of a molecule are shown in the figure with their wavelengths of transitions. Then :



(1) $\lambda_3 > \lambda_2, \lambda_1 = 2\lambda_2$
(2) $\lambda_3 > \lambda_2, \lambda_1 = 4\lambda_2$
(3) $\lambda_1 > \lambda_2, \lambda_2 = 2\lambda_3$
(4) $\lambda_2 > \lambda_1, \lambda_2 = 2\lambda_3$

PMD249

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	1	1	3	1	2	2	3	3	1	2	3	1	3
Question	16	17	18	19	20	21									
Answer	3	3	3	2	2	4									

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Exercise - I (Conceptual Questions) Build Up Your Understanding

SEMICONDUCTORS

1. On increasing the temperature the specific resistance of a semiconductor :–
(1) increases
(2) decreases
(3) does not change
(4) first decreases and then increases

PSE001

2. Platinum and silicon are cooled after heating up to 25° C then :–
(1) resistance of platinum will increase and that of silicon decreases.
(2) resistance of silicon will increase and that of platinum decreases.
(3) resistance of both will decrease.
(4) resistance of both increases.

PSE002

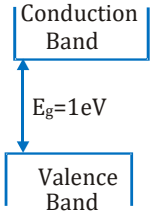
3. Electric conduction in a semiconductor takes place due to :–
(1) electrons only
(2) holes only
(3) both electrons and holes
(4) neither electrons nor holes

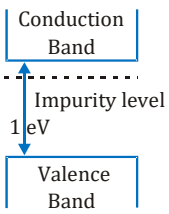
PSE003

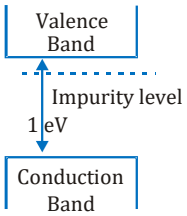
4. The atomic bonding is same for which of the following pairs :–
(1) Ag and Si (2) Ge and Si
(2) Ne and Ge (4) NaCl and Ge

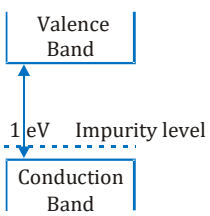
PSE004

5. Which of the following energy band diagram shows the n-type semiconductor :–

(1)


(2)


(3)


(4)


PSE005

6. Let n_p and n_e be the numbers of holes and conduction electrons in an extrinsic semiconductor.
(1) $n_p > n_e$ (2) $n_p = n_e$
(3) $n_p < n_e$ (4) $n_p \neq n_e$

PSE006

7. A p-type semiconductor is :–
(1) positively charged
(2) negatively charged
(3) uncharged
(4) uncharged at 0K but charged at higher temperatures

PSE007

8. Which statement is correct for p-type semiconductor
(1) the number of electrons in conduction band is more than the number of holes in valence band at room temperature
(2) the number of holes in valence band is more than the number of electrons in conduction band at room temperature
(3) there are no holes and electrons at room temperature
(4) number of holes and electrons is equal in valence and conduction band

PSE008

9. When an impurity is doped into an intrinsic semiconductor, the conductivity of the semiconductor :
(1) increases (2) decreases
(3) remains the same (4) become zero

PSE009

10. When we convert pure semiconductor into N type the number of hole :–
(1) Increases (2) Decreases
(3) Remains constant (4) None

PSE010

11. A semiconductor is damaged by a strong current, because :–
(1) lack of free electrons
(2) decrease in electrons
(3) excess of electrons
(4) none of these

PSE011

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12. If n_e and n_h are the number of electrons and holes in a semiconductor heavily doped with phosphorus, then :–
(1) $n_e > > n_h$ (2) $n_e < < n_h$
(3) $n_e \leq n_h$ (4) $n_e = n_h$
PSE012

13. Two wires P and Q made up of different materials have same resistance at room temperature. When heated, resistance of P increases and that of Q decreases. We conclude that:–
(1) P and Q both are conductors but because of being made of different materials it happens so.
(2) P is n-type semiconductor and Q is p-type semiconductor.
(3) P is semiconductor and Q is conductor.
(4) P is conductor and Q is semiconductor.
PSE013

14. When the conductivity of a semiconductor is only due to breaking of covalent bonds, the semiconductor is called :–
(1) intrinsic (2) extrinsic
(3) p-type (4) n-types
PSE014

15. In an intrinsic semiconductor, number of electrons and holes at room temperature are :-
(1) equal (2) zero
(3) unequal (4) infinity
PSE015

16. A semiconductor wire is connected in an electric circuit in series and temperature of semiconductor increased then the current in the circuit :–
(1) decreases (2) constant
(3) increases (4) will not flow
PSE016

17. In germanium crystal, the forbidden energy gap in joule is :–
(1) 1.6×10^{-19} (2) zero
(3) 1.12×10^{-19} (4) 1.76×10^{-19}
PSE017

18. In semiconductor, at room temperature :–
(1) valence band are partially empty and conduction band are partially filled
(2) valence band are fully filled and conduction band are partially empty
(3) valence band are fully filled
(4) conduction band are fully empty
PSE018

19. The probability of electrons to be found in the conduction band of an intrinsic semiconductor at a finite temperature :–
(1) decreases exponentially with increasing band gap.
(2) increases exponentially with increasing band gap.
(3) decrease with increasing temperature.
(4) is independent of the temperature and the band gap.
PSE019

20. In a p-type semiconductor, there are mainly :-
(1) free electrons (2) holes
(3) both (1) and (2) (4) none of these
PSE020

21. A conducting wire of Copper and Germanium are cooled from room temperature to temperature 80K, then their resistance will :–
(1) increase
(2) decrease
(3) copper's increase and Germanium's decrease
(4) copper's decrease and Germanium's increase
PSE021

22. Choose the false statement from the following :-
(1) the resistivity of a semiconductor increases with increase in temperature.
(2) substances with energy gap of the order of 10 eV are insulators.
(3) in conductors the valence and conduction bands may overlap.
(4) the conductivity of a semiconductor increases with increases in temperature.
PSE022

23. Carbon, Silicon and Germanium atoms have four valence electrons each. Their valence and conduction bonds are separated by energy band gaps represented by $(E_g)_C$, $(E_g)_{Si}$ and $(E_g)_{Ge}$ respectively. Which one of the following relationships is true in their case :–
(1) $(E_g)_C < (E_g)_{Ge}$ (2) $(E_g)_C > (E_g)_{Si}$
(3) $(E_g)_C = (E_g)_{Si}$ (4) $(E_g)_C < (E_g)_{Si}$
PSE023

24. In semiconducting material the mobilities of electrons and holes are μ_e and μ_h respectively. Which of the following is true :–
(1) $\mu_e > \mu_h$ (2) $\mu_e < \mu_h$
(3) $\mu_e = \mu_h$ (4) $\mu_e < 0; \mu_h > 0$
PSE024

25. Impurity energy level of n-type semiconductor lies in:-
(1) just above valence band
(2) just below conduction band
(3) between valence and conduction band
(4) none of these
PSE025

26. What is the energy gap in Si semiconductor?
(1) 4.4 eV (2) 0.3 eV
(3) 0.7 eV (4) 1.1 eV
PSE026

PN JUNCTION & BIASING OF DIODE

27. Region which have no free electrons and holes in a p-n junction is :–
(1) p - region (2) n - region
(3) junction (4) depletion region
PSE027

28. In p-n junction at the near at junction there are :-
(1) positive Ions
(2) negative Ions
(3) positive and negative Ions
(4) electrons and holes
PSE028

29. Depletion layer in p-n junction region is caused by
(1) drift holes
(2) diffusion of free carriers
(3) migration of impurity ions
(4) drift of electrons
PSE029

30. In a P–N Junction diode not connected to any circuit
(1) potential is the same everywhere.
(2) the P - type side is at a higher potential than the N - type side.
(3) there is an electric field at the junction directed from the N-type side to the P-type side.
(4) there is an electric field at the junction directed from the P-type side to the N-type side.
PSE030

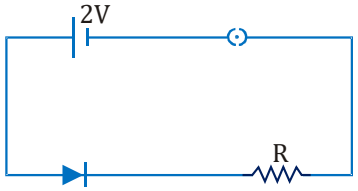
31. The minority current in a p-n junction is :–
(1) from the n-side to the p-side
(2) from the p-side to the n-side
(3) from the n-side to the p-side if the junction is forward-biased and in the opposite direction if it is reverse biased.
(4) from the p-side to the n-side if the junction is forward-biased and in the opposite direction if it is reverse biased
PSE031

32. The majority current in a p-n junction is :–
(1) from the n-side to the p-side
(2) from the p-side to the n-side
(3) from the n-side to the p-side if the junction is forward-biased and in the opposite direction if it is reverse biased
(4) from the p-side to the n-side if the junction is forward-biased and in the opposite direction if it is reverse biased
PSE032

33. Diffusion current in a p-n junction is greater than the drift current in magnitude :–
(1) if the junction is forward-biased
(2) if the junction is reverse-biased
(3) if the junction is unbiased
(4) in no case
PSE033

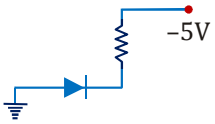
34. In a ..... biased P-N junction, the net flow holes is from N-region to the P-region :–
(1) F.B. (2) R. B.
(3) Unbiased (4) both 1 & 2
PSE034

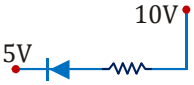
35. A 2 V battery forward biases a diode however there is a drop of 0.5 V across the diode which is independent of current. Also a current greater then 10 mA produces large joule loss and damages diode. If diode is to be operated at 5 mA, the series resistance to be put is :–

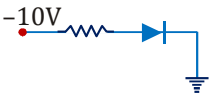


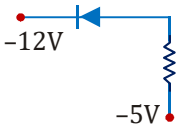
(1) 3 kΩ (2) 300 kΩ
(3) 300 Ω (4) 200 kΩ
PSE035

36. Which of the following diode is reverse biased:-

(1) 

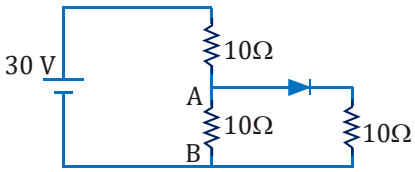
(2) 

(3) 

(4) 

PSE036

37. Find V_{AB} :–



(1) 10 V (2) 20 V (3) 30 V (4) none

PSE037

38. Assuming that the junction diode is ideal then the current through the diode is:-



(1) 200 mA (2) 20 mA (3) 2 mA (4) zero

PSE038

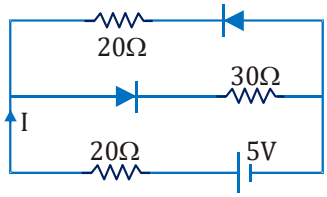
39. The resistance of a reverse biased pn junction diode is about :–

(1) 1 ohm (2) 10^2 ohm

(3) 10^3 ohm (4) 10^6 ohm

PSE039

40. Current I in the circuit will be :–



(1) $\frac{5}{40}$ A (2) $\frac{5}{50}$ A

(3) $\frac{5}{10}$ A (4) $\frac{5}{20}$ A

PSE040

41. In a p-n junction the depletion layer of thickness 10^{-6} m has potential across it is 0.1 V. The average electric field is (V/m) :–

(1) 10^7 (2) 10^{-6} (3) 10^5 (4) 10^{-5}

PSE041

42. In a unbias p-n junction :-

(1) high potential is at n side and low potential is at p side.

(2) high potential is at p side and low potential is at n side.

(3) p and n both are at same potential.

(4) undetermined.

PSE042

43. On increasing the reverse bias to a large value in p-n junction diode then value of current

(1) remains fixed

(2) increases slowly

(3) decrease slowly

(4) suddenly increase

PSE043

44. Reverse bias applied to a junction diode :–

(1) lowers the potential barrier.

(2) raises the potential barrier.

(3) increases the majority carrier current.

(4) increases the minority carrier current.

PSE044

45. Correct statement for diode is :–

(1) in full wave rectifier both diodes work alternatively.

(2) in full wave rectifier both diodes work simultaneously.

(3) efficiency of full wave rectifier and half wave rectifier is same.

(4) full wave rectifier in bidirectional.

PSE045

46. When a junction diode is reverse biased, the flow of current across the junction is mainly due to :–

(1) diffusion of charges

(2) depends upon the nature of material

(3) drift of charges

(4) both drift and diffusion of charges

PSE046

47. The width of depletion region in a p-n junction diode

(1) increases when reverse bias is applied.

(2) increases when a forward bias is applied.

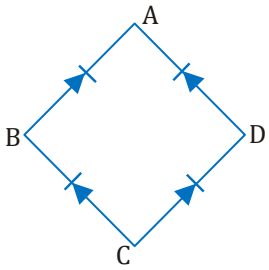
(3) decreases when a reverse bias is applied.

(4) remains the same irrespective of the bias voltage.

PSE047

SPECIAL TYPES OF DIODES

48. For the given circuit shown in fig, to act as full wave rectifier :– a.c. input should be connected across ..... and ..... the d.c. output would appear across ..... and .....



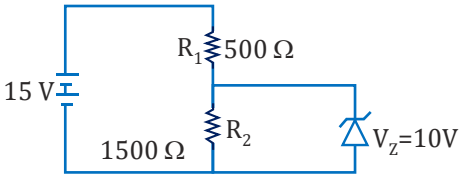
(1) A, C, B, D (2) B, D, A, C
(3) A, B, C, D (4) C, A, D, B

PSE048

49. Forbidden energy gap of Ge is 0.75 eV, maximum wave length of incident radiation of photon for producing electron - hole pair in germanium semiconductor is :–
(1) 4200 Å (2) 16500 Å
(3) 4700 Å (4) 4000 Å

PSE049

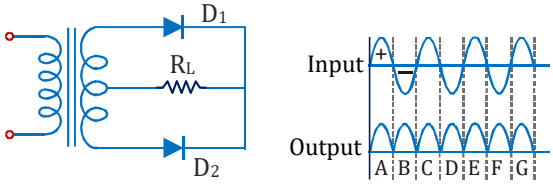
50. In the circuit given the current through the zener diode is :-



(1) 10 mA (2) 6.67 mA
(3) 5 mA (4) 3.33 mA

PSE050

51. A full wave rectifier circuit along with the input and output voltage is shown in the figure then output due to diode D_2 is :–



(1) A, C (2) B, D
(3) B, C (4) A, D

PSE051

52. If a full wave rectifier circuit is operating from 50 Hz mains, the fundamental frequency in the ripple will be :–
(1) 25 Hz (2) 50 Hz
(3) 70.7 Hz (4) 100 Hz

PSE052

53. The electrical circuit used to get smooth DC output from a rectifier circuit is called :–
(1) filter (2) oscillator
(3) logic gate (4) amplifier

PSE053

54. In p-n junction photocell electromotive force due to monochromatic light is proportional to
(1) p-n potential barrier
(2) intensity of light
(3) frequency of light
(4) p-n applied voltage

PSE054

55. Efficiency of a half wave rectifier is nearly :–
(1) 80 % (2) 60 % (3) 40 % (4) 20 %

PSE055

56. Zener dode is used for :–
(1) rectification
(2) stabilization
(3) amplification
(4) producing oscillations in an oscillator

PSE056

LOGIC GATE

57. The output of the given logic gate is 1 when inputs A, B and C are such that :–



(1) A = 1, B = 0, C = 1 (2) A = 1, B = 1, C = 0
(3) A = B = C = 0 (4) A = B = C = 1

PSE081

58. A two inputed XOR gate produces an high output only when its both inputs are :–
(1) same (2) different
(3) low (4) high

PSE082

59. Which of the following Boolean expression is not correct :–
(1) $\overline{\bar{A}.\bar{B}} = A + B$ (2) $\overline{\bar{A}+\bar{B}} = A \,.\, B$
(3) $\overline{\overline{A.B}} = A.B$ (4) $\bar{1}+\bar{1}=1$

PSE083

60. In Boolean algebra, which of the following is not equal to zero :–
(1) $A.\bar{A}$ (2) A.0 (3) $\overline{A+\bar{A}}$ (4) $\overline{\bar{A}.0}$

PSE084

61. Digital circuits can be made by respectative use of :–
(1) OR gate (2) AND gate
(3) NOT gate (4) NAND gate

PSE085

62. The truth table shown below is for which of the following gates :–

A	B	Y
1	1	1
0	1	0
1	0	0
0	0	1

(1) XNOR (2) AND
(3) XOR (4) NOR

PSE086

63. When all the inputs of a NAND gate are connected together, the resulting circuit is :–
(1) a NOT gate (2) an AND gate
(3) an OR gate (4) a NOR gate

PSE087

64. A NAND gate followed by a NOT gate is :–
(1) an OR gate (2) an AND gate
(3) a NOR gate (4) a XOR gate

PSE088

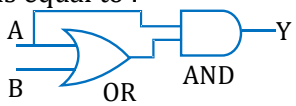
65. The NOR gate is logically equivalent to an OR gate followed by :–
(1) an inverter (2) a NOR gate
(3) a NAND gate (4) All of above

PSE089

66. The output of a two input NOR gate is in state 1 when :–
(1) either input terminals is at 0 state
(2) either input terminals is at 1 state
(3) both input terminals are at 0 state
(4) both input terminals are at 1 state

PSE090

67. The output Y of the combination of gates shown is equal to :–



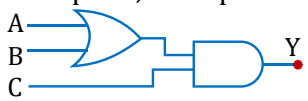
(1) A (2) $\bar{A}$ (3) A + B (4) AB

PSE091

68. What would be the output of the circuit whose Boolean expression $Y = A\bar{B} + AB$ when A = 1, B = 0 :–
(1) 1 (2) 0
(3) both (1) & (2) (4) none of these

PSE092

69. To get an output 1, the input ABC should be :–



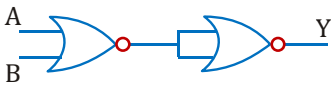
(1) 101 (2) 100 (3) 110 (4) 010

PSE093

70. The output of 2 input gate is 1 only if its inputs are equal. It is true for :–
(1) NAND (2) AND
(3) EX-NOR (4) EX-OR

PSE094

71. The circuit shown here is logically equivalent to:–



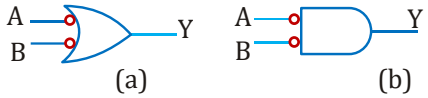
(1) OR gate (2) AND gate
(3) NOT gate (4) NAND gate

PSE095

72. A two-input NAND gate is followed by a single-input NOR gate. This logic circuit will function as :–
(1) an AND gate (2) an OR gate
(3) a NOT gate (4) a NOR gate

PSE096

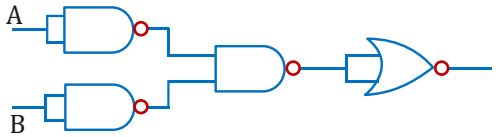
73. The logic symbols shown here are logically equivalent to :–



(1) (a) AND and (b) OR gate
(2) (a) NOR and (b) NAND gate
(3) (a) OR and (b) AND gate
(4) (a) NAND and (b) NOR gate

PSE097

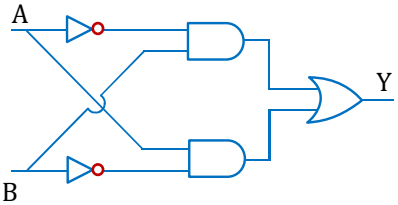
74. The combination of the gates shown will produce



(1) OR gate (2) AND gate
(3) NOR gate (4) NAND gate

PSE098

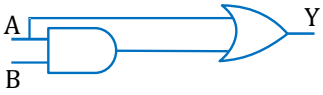
75. The combination of the gates shown will produce



(1) AND gate (2) NAND gate
(3) NOR gate (4) XOR gate

PSE099

76. The truth table for the following combination of gates is :-



(1)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

(2)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(3)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(4)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

PSE100

77. In the Boolean algebra $\bar{A}.\bar{B}$ equals :–

(1) A + B (2) $\overline{A+B}$ (3) A . B (4) $\bar{A}.B$

PSE101

78. How many NOR gates are required to form NAND gate :–

(1) 1 (2) 3 (3) 2 (4) 4

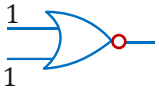
PSE102

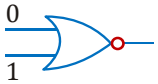
79. How many NAND gates are used to form AND gate :–

(1) 3 (2) 2 (3) 1 (4) 4

PSE103

80. Which of the following gates will have an output of 1 :–

(a) 

(b) 

(c) 

(d) 

(1) (a) and (b) (2) (b) and (c)
(3) (c) and (d) (4) (a) and (d)

PSE104

81. The logic behind 'NOR' gate is that it gives :–

(1) high output when both inputs are high
(2) high output when both inputs are low
(3) low output when both inputs are low
(4) none of these

PSE105

82. Logic gates are the building blocks of a :–

(1) abacus system (2) analog system
(3) digital system (4) none of these

PSE106

83. Boolean algebra is essentially based on :–

(1) logic (2) truth
(3) numbers (4) symbol

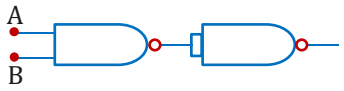
PSE107

84. Out of the following, universal gate is :–

(1) NOT (2) OR
(3) AND (4) NAND

PSE108

85. Identify the logic operation of the following logic circuit



(1) NAND (2) AND (3) NOR (4) OR

PSE109

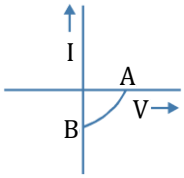
EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	3	2	2	4	3	2	1	2	3	1	4	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	1	1	2	4	1	2	1	2	4	4	3	2	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	2	1	2	3	3	1	2	4	2	3	1	4	2	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	1	2	2	4	2	4	1	2	3	2	4	2	4	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	1	1	2	4	3	1	1	1	3	1	1	4	3	4
Question	76	77	78	79	80	81	82	83	84	85					
Answer	1	2	4	2	3	2	3	1	4	2					

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. The given graph represents V-I characteristic for a semiconductor device.



Which of the following statement is **correct** ?

(1) It is V - I characteristic for solar cell where, point A represents open circuit voltage and point B short circuit current.

(2) It is a for a solar cell and point A and B represent open circuit voltage and current, respectively.

(3) It is for a photodiode and points A and B represent open circuit voltage and current, respectively.

(4) It is for a LED and points A and B represent open circuit voltage and short circuit current, respectively.

PSE138

2. The barrier potential of a p-n junction depends on:

(a) type of semiconductor material

(b) amount of doping

(c) temperature

Which one of the following is correct?

(1) (a) and (b) only

(2) (b) only

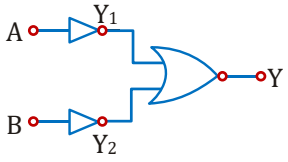
(3) (b) and (c) only

(4) (a), (b) and (c)

PSE139

AIPMT-2015

3. Which logic gate is represented by the following combination of logic gates ?

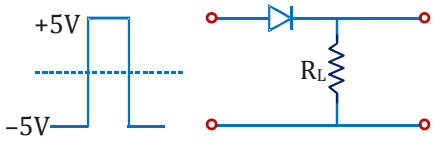


(1) NAND (2) AND

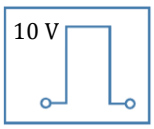
(3) NOR (4) OR

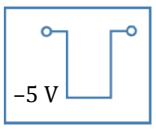
PSE140

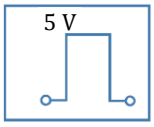
4. If in a p-n junction, a square input signal of 10 V is applied as shown,

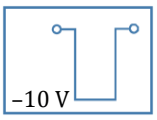


then the output across R_L will be :

(1) 

(2) 

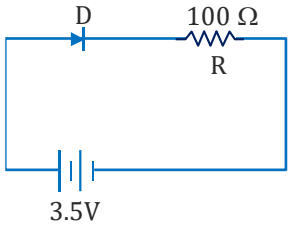
(3) 

(4) 

PSE141

RE–AIPMT-2015

5. In the given figure, a diode D is connected to an external resistance R =100 Ω and an e.m.f of 3.5 V. If the barrier potential developed across the diode is 0.5 V, the current in the circuit will be :



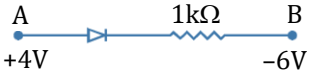
(1) 35 mA (2) 30 mA

(3) 40 mA (4) 20 mA

PSE142

NEET-I 2016

6. Consider the junction diode as ideal. The value of current flowing through AB is :

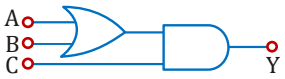


(1) 0 A (2) 10^{-2} A

(3) 10^{-1} A (4) 10^{-3} A

PSE144

7. To get output 1 for the following circuit, the correct choice for the input is

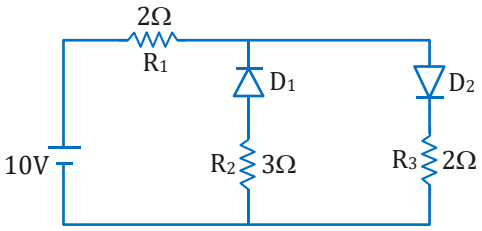


(1) A = 0, B = 1, C = 0
(2) A = 1, B = 0, C = 0
(3) A = 1, B = 1, C = 0
(4) A = 1, B = 0, C = 1

PSE146

NEET-II 2016

8. The given circuit has two ideal diodes connected as shown in the figure below. The current flowing through the resistance R_1 will be :-



(1) 1.43 A (2) 3.13 A
(3) 2.5 A (4) 10.0 A

PSE148

9. What is the output Y in the following circuit, when all the three inputs A, B, C are first 0 and then 1 ?



(1) 1,0 (2) 1,1 (3) 0,1 (4) 0.0

PSE149

NEET(UG) 2017

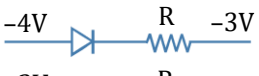
10. The given electrical network is equivalent to :

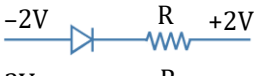


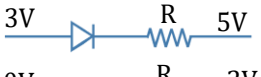
(1) OR gate (2) NOR gate
(3) NOT gate (4) AND gate

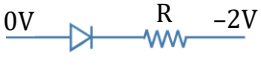
PSE151

11. Which one of the following represents forward bias diode ?

(1) 

(2) 

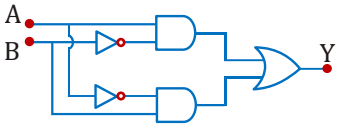
(3) 

(4) 

PSE152

NEET(UG) 2018

12. In the combination of the following gates the output Y can be written in terms of inputs A and B as :-



(1) $\overline{A \cdot B}$ (2) $A \cdot \overline{B} + \overline{A} \cdot B$
(3) $\overline{A \cdot B} + A.B$ (4) $\overline{A + B}$

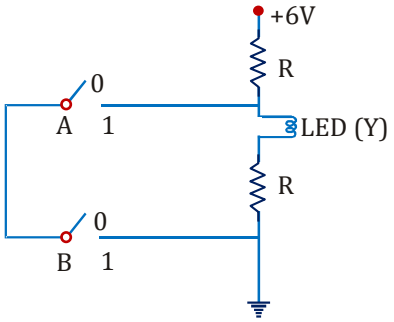
PSE153

13. In a p-n junction diode, change in temperature due to heating :-
(1) affects only reverse resistance
(2) affects only forward resistance
(3) does not affect resistance of p-n junction
(4) affects the overall V–I characteristics of p-n junction

PSE155

NEET(UG) 2019

14.



The correct Boolean operation represented by the circuit diagram drawn is :
(1) AND (2) OR (3) NAND (4) NOR

PSE156

15. For a p-type semiconductor which of the following statements is **true** ?
(1) Electrons are the majority carriers and trivalent atoms are the dopants.
(2) Holes are the majority carriers and trivalent atoms are the dopants.
(3) Holes are the majority carriers and pentavalent atoms are the dopants.
(4) Electrons are the majority carriers and pentavalent atoms are the dopants.

PSE157

NEET(UG) 2019 (Odisha)

16. An LED is constructed from a p-n junction diode using GaAsP. The energy gap is 1.9 eV. The wavelength of the light emitted will be equal to :-

(1) 10.4×10^{-26} m

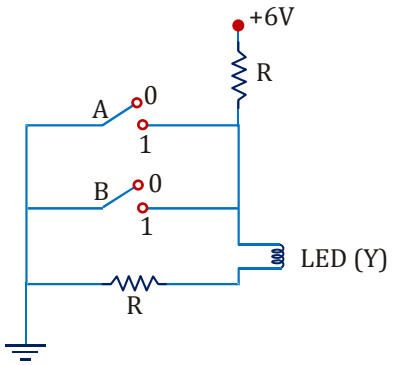
(2) 654 nm

(3) 654 Å

(4) 654×10^{-11} m

PSE158

17. The circuit diagram shown here corresponds to. the logic gate,



(1) NOR (2) AND

(3) OR (4) NAND

PSE159

NEET(UG) 2020

18. The increase in the width of depletion region in a p-n junction diode is due to :

(1) increase in forward current

(2) forward bias only

(3) reverse bias only

(4) both forward bias and reverse bias

PSE161

19. The solids which have the negative temperature coefficient of resistance are :

(1) insulators and semiconductors

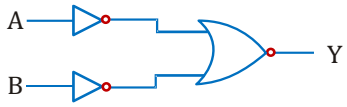
(2) metals

(3) insulators only

(4) semiconductors only

PSE162

20. For the logic circuit shown, the truth table is :



(1)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(2)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(3)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(4)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

PSE163

NEET(UG) 2020 (COVID-19)

21. Out of the following which one is a forward biased diode ?

(1) 

(2) 

(3) 

(4) 

PSE164

22. Which of the following gate is called universal gate?

(1) OR gate (2) AND gate

(3) NAND gate (4) NOT gate

PSE166

23. An intrinsic semiconductor is converted into n-type extrinsic semiconductor by doping it with :-

(1) Phosphorous (2) Aluminium

(3) Silver (4) Germanium

PSE167

NEET(UG) 2021

24. The electron concentration in an n-type semiconductor is the same as hole concentration in a p-type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them.

(1) current in n-type = current in p-type

(2) current in p-type > current in n-type

(3) current in n-type > current in p-type

(4) No current will flow in p-type, current will only flow in n-type

PSE168

25. Consider the following **statements (A)** and **(B)** and identify the **correct** answer.

(A) A zener diode is connected in reverse bias, when used as a voltage regulator.

(B) The potential barrier of p-n junction lies between 0.1 V to 0.3 V.

(1) **(A)** and **(B)** both are correct.

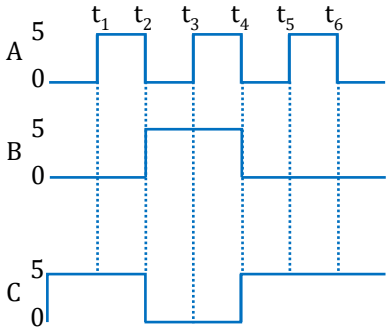
(2) **(A)** and **(B)** both are incorrect.

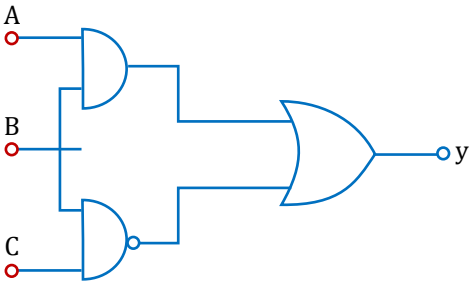
(3) **(A)** is correct and **(B)** is incorrect.

(4) **(A)** is incorrect but **(B)** is correct.

PSE169

26. For the given circuit, the input digital signals are applied at the terminals A, B and C. What would be the output at the terminal y ?





(1) 

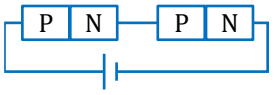
(2) 

(3) 

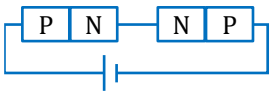
(4) 

PSE170

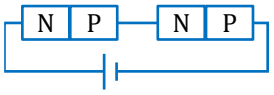
NEET(UG) 2022



(a)

27. 

(b)



(c)

In the given circuits (a), (b) and (c), the potential drop across the two p-n junctions are equal in :

(1) (b) only

(2) (c) only

(3) Both (a) and (c)

(4) (a) only

PSE171

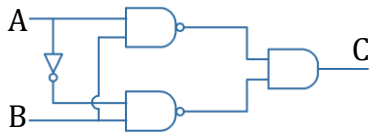
28. In half wave rectification, if the input frequency is 60 Hz, then the output frequency would be :

(1) 30 Hz (2) 60 Hz

(3) 120 Hz (4) Zero

PSE172

29.



The truth table for the given logic circuit is :

(1)

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

(2)

A	B	C
0	0	1
0	1	0
1	0	1
1	1	0

(3)

A	B	C
0	0	0
0	1	1
1	0	0
1	1	1

(4)

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

PSE173

RE-NEET(UG) 2022

30. The incorrect statement about the property of a Zener diode is :-

(1) Zener voltage remains constant at breakdown

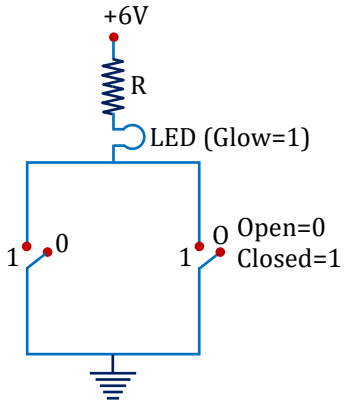
(2) It is designed to operate under reverse bias

(3) Depletion region formed is very wide

(4) p and n regions of zener diode are heavily doped

PSE174

31.



Identify the equivalent logic gate represented by the given circuit :

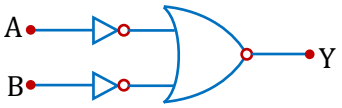
(1) OR (2) NOR

(3) AND (4) NOT

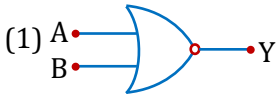
PSE175

RE-NEET(UG) 2023

32. The given circuit is equivalent to :



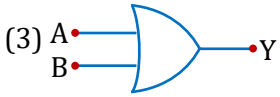
(1)



(2)



(3)



(4)



PSE185

33. A p-type extrinsic semiconductor is obtained when Germanium is doped with:

(1) Antimony

(2) Phosphorous

(3) Arsenic

(4) Boron

PSE186

NEET(UG) 2024

34. A logic circuit provides the output Y as per the following truth table :

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

The expression for the output Y is

(1) $A.B+\bar{A}$

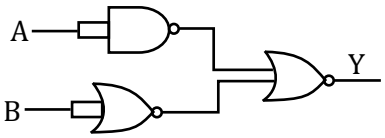
(2) $A.\bar{B}+\bar{A}$

(3) $\bar{B}$

(4) B

PSE187

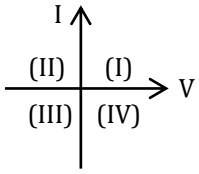
35. The output (Y) of the given logic gate is similar to the output of an/a :



(1) NAND gate
(2) NOR gate
(3) OR gate
(4) AND gate

PSE188

36. Consider the following statements A and B and identify the correct answer :



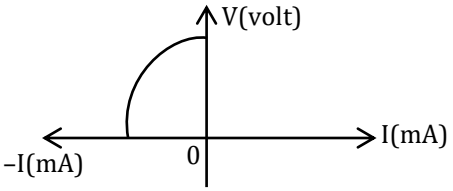
A. For a solar-cell, the I-V characteristics lies in the IV quadrant of the given graph.
B. In a reverse biased pn junction diode, the current measured in (μA), is due to majority charge carriers.

(1) A is correct but B is incorrect.
(2) A is incorrect but B is correct.
(3) Both A and B are correct.
(4) Both A and B are incorrect.

PSE189

RE-NEET(UG) 2024

37.



The I - V characteristics shown above are exhibited by a:
(1) light emitting diode
(2) zener diode
(3) photodiode
(4) solar cell

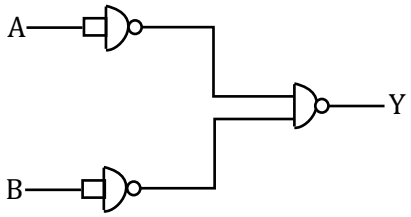
PSE190

38. When the output of an OR gate is applied as input to a NOT gate, then the combination acts as a :

(1) NAND gate (2) NOR gate
(3) AND gate (4) OR gate

PSE191

39. The output Y for the inputs A and B of the given logic circuit is :



(1) $A \cdot B$ (2) $\bar{A} \cdot \bar{B}$
(3) $A + B$ (4) $\bar{A} + \bar{B}$

PSE192

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	2	3	2	2	4	3	1	2	4	2	4	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	3	1	2	4	3	1	3	3	3	3	2	2	3
Question	31	32	33	34	35	36	37	38	39						
Answer	1	4	4	3	4	1	4	2	3						

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Exercise - III (Analytical Questions) Master Your Understanding

1. Given blow are two statements:

Statement-I : Doping of indium in Si crystal leads to p-type semiconductor.

Statement-II : Indium is a pentavalent atom.

In light of above statement choose the most **appropriate** answer from the options given below:

(1) Both statement I and statement II are correct.

(2) Statement I is correct and statement II is incorrect.

(3) Statement I is incorrect and statement II is correct.

(4) Both statements I and statements II are incorrect.

PSE177

2. Match the column-I of special diodes with their utilization given in column-II:

Column I	**Column II**
(a) Zener diode	(1) As a light sensor
(b) LED	(2) Voltage regulator circuit
(c) Photodiode	(3) As a bulb of light

(1) $a \rightarrow 3, b \rightarrow 2, c \rightarrow 1$

(2) $a \rightarrow 3, b \rightarrow 1, c \rightarrow 2$

(3) $a \rightarrow 1, b \rightarrow 3, c \rightarrow 2$

(4) $a \rightarrow 2, b \rightarrow 3, c \rightarrow 1$

PSE178

3. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : The number of electrons in n-type silicon semiconductor is higher than the number of electrons in a pure silicon semiconductor.

Reason (R) : The law of mass action is applicable only to n-type semiconductors.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PSE180

4. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Temperature coefficient of resistance for semiconductor is negative

Reason (R) : Conductivity of semiconductor increases with increment in temperature.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PSE182

5. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : The OR gate function corresponds to switches in parallel.

Reason (R) : The AND gate function corresponds to switches in series.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PSE183

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6. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : In an unbiased p–n junction, holes diffuse from the p–region to n–region.

Reason (R) : Hole concentration in p–region is more as compared to n–region.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PSE184

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6
Answer	2	4	3	1	2	1

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Electrostatics

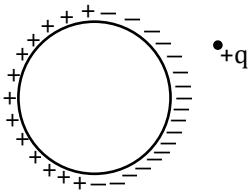
SOLUTIONS

Exercise-I (Conceptual Questions)

1. $N = \frac{Q}{e}$, if N is not an integer then Q is not possible.

2. Charge is quantized it always present in integral multiple of elementary charge.

3.



Net charge = 0

4. $F \propto q_1q_2$

$F \propto q_2$ ...(i) $[q_1 = q_2]$

50% charge, transferred from one to another sphere.

$$q_1' = q + \frac{q}{2} = \frac{3q}{2} \Rightarrow q_2' = q - \frac{q}{2} = \frac{q}{2}$$

$$F' \propto \left(\frac{3q}{2}\right)\left(\frac{q}{2}\right) \propto \frac{3}{4}q^2 \quad ...(ii)$$

From (i) & (ii)

$$F' = \frac{3}{4}F$$

5. $$F = \frac{1}{4\pi\epsilon_0}\frac{q_1q_2}{r^2} \quad .....(1)$$

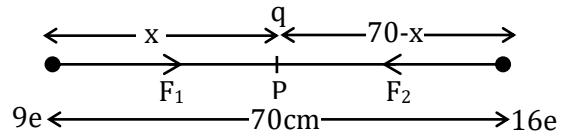
$$F' = \frac{1}{4\pi\epsilon_0 K}\frac{q_1q_2}{r'^2} \quad .....(2)$$

$F' = F$

$$\frac{1}{4\pi\epsilon_0 K}\frac{q_1q_2}{r'^2} = \frac{1}{4\pi\epsilon_0}\frac{q_1q_2}{r^2}$$

$$r' = \frac{r}{\sqrt{k}}$$

6.



For charge, which placed at point P :-

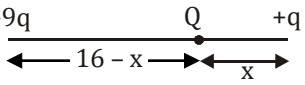
$F_{net} = 0$

$F_1 = F_2$

$$\frac{k(9e)(q)}{x^2} = \frac{k(16e)(q)}{(70-x)^2}$$

$\Rightarrow$ x = 30 cm from 9e or 40 cm from 16e.

7.



$$\frac{kqQ}{x^2} = \frac{k(9q)Q}{(16-x)^2} ; \frac{1}{x} = \frac{3}{16-x}$$

x = 4 cm from (+q) and 12 cm from (+9q)

8.

$$F = \frac{k(7\mu C)(-5\mu C)}{x^2}$$

After adding (–2μC) charge

$$F' = \frac{k(5\mu C)(-7\mu C)}{x^2} = F$$

9. $$F = \frac{kQ^2}{r^2} \Rightarrow 0.144 = \frac{9\times10^9 Q^2}{\left(5\times10^{-2}\right)^2}$$

$$Q^2 = \frac{0.144\times25\times10^{-4}}{9\times10^9} = \frac{144\times25}{9}\times10^{-16}$$

$$Q = \frac{12\times5}{3}\times10^{-8} = 0.2 \times 10^{-6}\ C.$$

10. $F_1 = \frac{k(2\mu C)(6\mu C)}{x^2}$ = 12N repulsion

After adding charge (–4μC)

$$F_2 = \frac{k(-2\mu C)(2\mu C)}{x^2} = -4N \text{ attractive}$$

11. $$\frac{kq^2}{R^2} = \frac{GM_eM_m}{R^2}$$

$$q = \sqrt{\frac{6.67\times10^{-11}\times10^{25}\times10^{23}}{9\times10^9}} = 8.6 \times 10^{13}\ C$$

12. Electrostatic force b/w two charge particles does not affected due to presence or absence of another charges.

13. $Q = Q_1 + Q_2$

$$F = \frac{kQ_1Q_2}{R^2} = \frac{kQ_1(Q-Q_1)}{R^2}$$

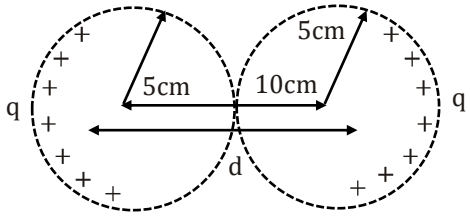
For maximum repulsion $\frac{dF}{dQ_1} = 0$

$$\therefore \frac{k}{R^2}\frac{d}{dQ_1}\left(Q_1Q - Q_1^2\right) = \frac{k}{R^2}(Q - 2Q_1) = 0$$

$$Q - 2Q_1 = 0 ; Q_1 = \frac{Q}{2}$$

$$Q_2 = Q - Q_1 = \frac{Q}{2}$$

14. q — 10cm — q, $f = \frac{Kq^2}{(0.1)^2}$



$f' = \frac{Kq^2}{\left(d \times 10^{-2}\right)^2} < f$

charges moves away from each other due to repulsion force thats why separation b/w them increases and force decreases

15. +4q, −Q, F_Q, +q, F_{4q}; $\ell/2$, $\ell/2$

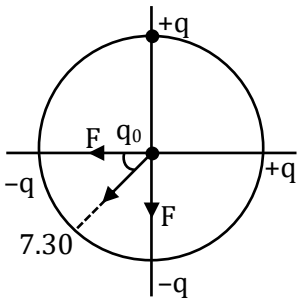
$\Rightarrow \frac{k.4q.q}{\ell^2} = \frac{kqQ}{(\ell/2)^2} \Rightarrow Q = q$ (but unlike)

16. $F = \frac{kq^2}{r^2}$. −q ——— +q

−q, +q, +q; r/2, r/2

$F' = \frac{kq.q}{(r/2)^2} + \frac{kq.q}{(r/2)^2} = 8\frac{kq^2}{r^2} = 8F$ (towards −q)

17.



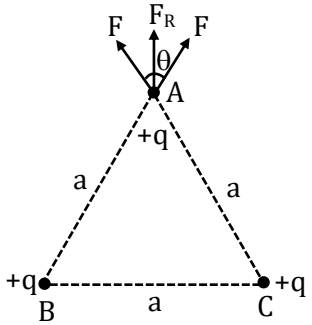
18. L, L; +Q, +Q; 180°

$F = \frac{kQ^2}{4L^2}$ (No. gravitational effect)

19. Wrong statement → 2

Proton and Neutron bound by neuclear force which is most powerful force in short range.

20.



$F_R = 2F \cos \theta/2$,

where $\theta = 60°$

$= 2\left(\frac{kq^2}{a^2}\right)\cos 30°$

$= \frac{\sqrt{3}kq^2}{a^2}$

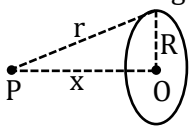
21. 9e — x — P — r−x — 3e; r

$x = \frac{\sqrt{9}}{\sqrt{9}+\sqrt{3}}$ $r = \frac{3r}{3+\sqrt{3}}$

$x = \frac{r}{1+\sqrt{3}/3} = \frac{r}{1+1/\sqrt{3}}$

From 9e charge

22.



$E = \frac{kQx}{\left(R^2+x^2\right)^{3/2}}$

$r^2 = R^2 + x^2$

$x^2 = r^2 - R^2$

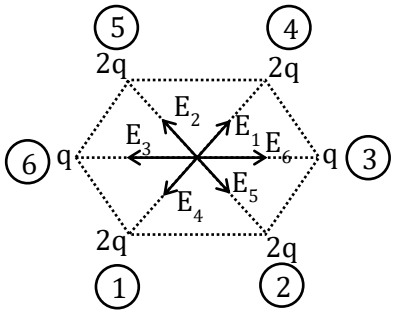
23. $E = \frac{k(2\mu C)}{(1/2)^2} - \frac{k(1\mu C)}{(1/2)^2}$

$= 8 \times 9 \times 10^9 \times 10^{-6} - 4 \times 9 \times 10^9 \times 10^{-6}$

$= 3.6 \times 10^4$ N/C

24. $E = \frac{K}{x^3}, K = Ex^3 = \frac{N}{C} \times m^3$

25.



In case III

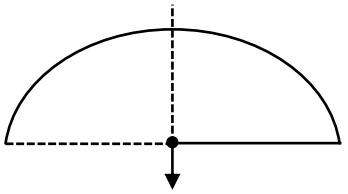
$E_1 = E_2 = E_4 = E_5 = \frac{2kq}{r^2}$

$E_3 = E_6 = \frac{kq}{r^2}$

$E_{net} = 0$

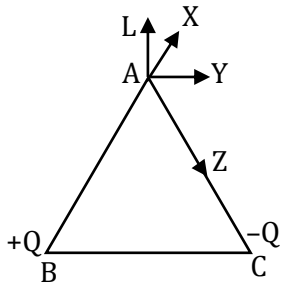
26. $E = \frac{2k\lambda}{a}\sin\alpha/2$, where $\alpha = 180°$

$= \frac{2k\lambda}{a}\sin 90° = \frac{\lambda}{2\pi\in_0 a}$

27.  $\leftarrow \lambda = \frac{Q}{\pi R}$

$= \frac{2k\lambda}{R} = \frac{2\times 9\times 10^9 \times 1.4\times 10^{-9}}{3.14\times 0.5\times 0.5}$

$= 32$ v/m

28. 

Direction of electric Field towards AY

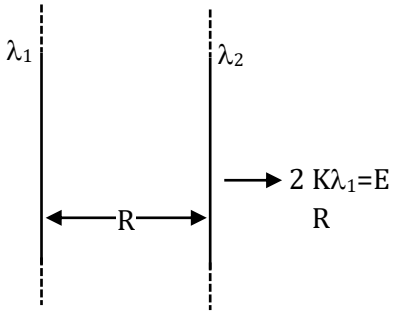
29. For balancing condⁿ

$qE = mg$

$E = \frac{mg}{q} = \frac{10^{-6}\times 10}{1\times 10^{-6}}$

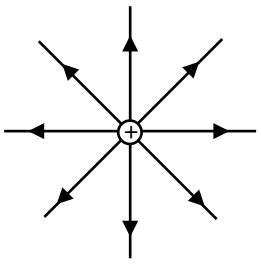
$E = 10\frac{V}{m}$ downward

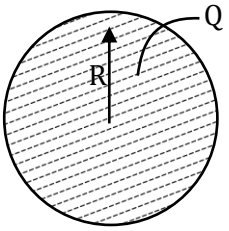
30. $\tan\theta = \frac{F_e}{mg} = \frac{kQ^2}{x^2 mg}$

31. 

Force per unit length

$\frac{F}{\ell} = E\lambda_2 = \frac{2k\lambda_1\lambda_2}{R}$

32. 

33. 

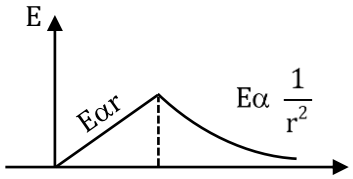
inside $E = \frac{kQr}{R^3}$; outside $E = \frac{kQ}{r^2}$

34. Electrostatic field lines never make closed loop.

35. $Q = \frac{q_{in}}{\varepsilon_0} = \frac{Q}{\varepsilon_0}$

36. $E_{inside} = \frac{kQr}{R^3}$ $E \propto r$ $0 < r < R$

$E_{outside} = \frac{kQ}{r^2}$ $E \propto \frac{1}{r^2}$ $R < r < \infty$



37. $\phi \propto q$ ...(i)

After addition flux becomes ϕ'.

$\phi' \propto \theta'$ ...(ii)

$\frac{\phi'}{\phi} = \frac{q^1}{q} = \frac{100}{20} = 5$

$\phi' = 5\phi$

change influx $\Delta\phi = \phi' - \phi = 4\phi$

38. $\vec{E} = 8\hat{i} + 4\hat{j} + 3\hat{k}$

$\vec{A} = 100\,\hat{k}$

$\phi = \vec{E}\cdot\vec{A} = (8\hat{i} + 4\hat{j} + 3\hat{k})\cdot(100\hat{k})$

$\phi = 300$ unit

i & $\hat{j}$ component of E.F. do not give flux in the given area.

39. $\phi = \frac{\text{total enclosed charge}}{\in_0} = \frac{100Q}{\in_0}$

40. In case of net charge is zero then incoming lines = outgoing lines

41. $E_1 = \frac{kQ}{r_1^2}$

$E_2 = \frac{kQr_2}{R^3}$

$\frac{E_2}{E_1} = \frac{r_2 . r_1^2}{R^3} = 120\ V/m$

42. $\phi = \frac{q_{in}}{\varepsilon_0}$

$q_{in} = (\phi_2 - \phi_1) \times \varepsilon_0$

43. $\vec{E} = (5\hat{i} + 2\hat{j})\ N/C$

$\vec{A}$ = $2m^2$ in y-z plane = $2m^2\,\hat{i}$

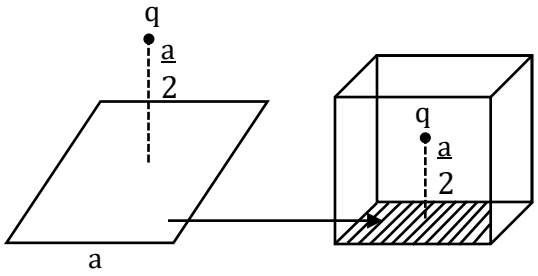
$\phi = \vec{E} \cdot \vec{A}$ = 10 units

44. $\phi = \frac{1}{8\in_0}(\Sigma q) = \frac{1}{8\in_0}(36\mu C)$

$= \frac{4\pi \times 9 \times 10^9}{8} \times 36 \times 10^{-6}$

$= 162\pi \times 10^3 Nm^2/C$

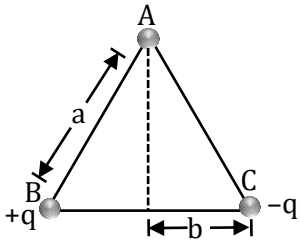
45.



net flux through cube = $\frac{q}{\varepsilon_0}$

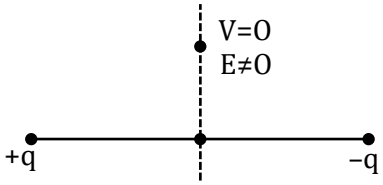
than flux through base = $\frac{q}{6\varepsilon_0}$

46.



$V_A = \frac{k(+q)}{a} + \frac{k(-q)}{a} = 0$

47.



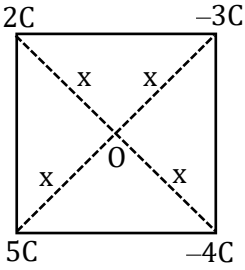
48. Inside potential = potential at surface

49. remain same (equipotential surface)

50. uniform

51. Potential difference between outer and inner spheres doesn't depend upon the charge given to outer sphere hence remain same.

52.



$V_O = \frac{k(2C)}{x} + \frac{k(-3C)}{x} + \frac{k(-4C)}{x} + \frac{k(5C)}{x}$

$V_O = 0,\ E_O \neq 0$

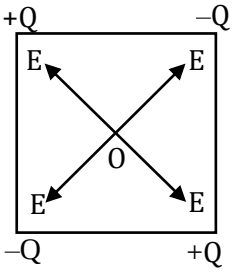
53. $\int_{\ell=-\infty}^{0} -\vec{E}.d\vec{\ell} = \frac{kq}{R} = V_0$

$V_0 = \frac{kq}{R} = \frac{9 \times 10^9 \times 1.11 \times 10^{-10}}{0.5} = 2\ volt$

54. E inside = 0

55. $V = \frac{kQ}{R}$ $(V_{in} = V_S)$

56.



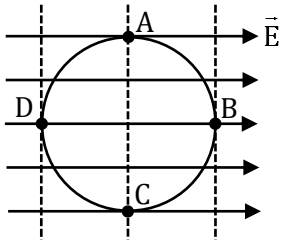
E = 0 ; V = 0

57. $V_{10} - V_{20} = -\int_{20}^{10} \vec{E}.d\vec{x}$

$V_{10} - V_{20} = -\int_{20}^{10} \left(\frac{100}{x^2}\right) dx = 100\left[\frac{1}{x}\right]_{20}^{10}$

$= 100\left[\frac{1}{10} - \frac{1}{20}\right] = 5\ volt$

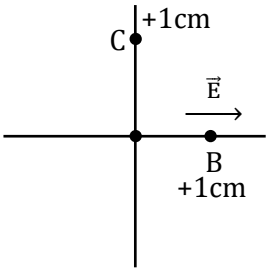
58.



$V_D > V_A = V_C > V_B$

Along the direction of E.F., potential decreases.

59.



Along direction of E.F. potential decrease then
$(V_A = V_C > V_B)$

60. $E = -\frac{dV}{dx} = -10x + 10 = 0$

61. $\frac{kQ}{r} = 600$ ...(i) $\frac{KQ}{r^2} = 200$ ...(ii)

From (i) & (ii)

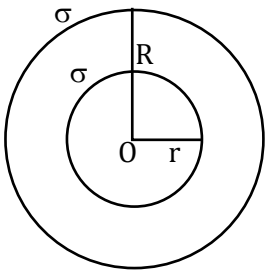
$200r^2 = 600r$

$r = 3$

from (i)

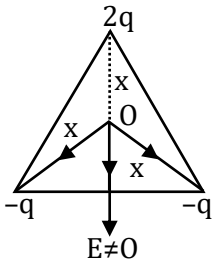
$Q = \frac{600\times 3}{9\times 10^9} = 2 \times 10^{-7}\ C = 0.2\mu C$

62.



$$V_0 = \frac{k\left(\sigma 4\pi r^2\right)}{r} + \frac{k\left(\sigma 4\pi R^2\right)}{R} = \frac{\sigma}{\varepsilon_0}(r+R)$$

63.



$$V = \frac{k(2q)}{x} + \frac{k(-q)}{x} + \frac{k(-q)}{x}$$

$V = 0$

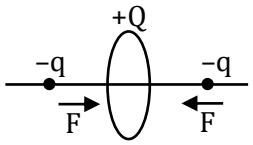
But $E \neq 0$

64. $\Delta V = -\vec{E}\cdot\overrightarrow{dx} = Ed\cos 60$

65. $\vec{E} = -\frac{dV}{dx}\hat{i} - \frac{dV}{dy}\hat{j} - \frac{dV}{dz}\hat{k} = -6\hat{i} + 8\hat{j}$

$\Rightarrow |\vec{E}| = 10 \Rightarrow |\vec{F}| = q|\vec{E}| = 20N$

66.



(i) Displacement is not small so no SHM execution

(ii) e⁻s move from lower to higher potential.

67. $E_1 = \frac{KQ_1}{r_1^2}$ $E_2 = \frac{KQ_2}{r_2^2}$

$\because$ $E_1 = E_2$

$$\frac{Q_1}{r_1^2} = \frac{Q_2}{r_2^2}$$

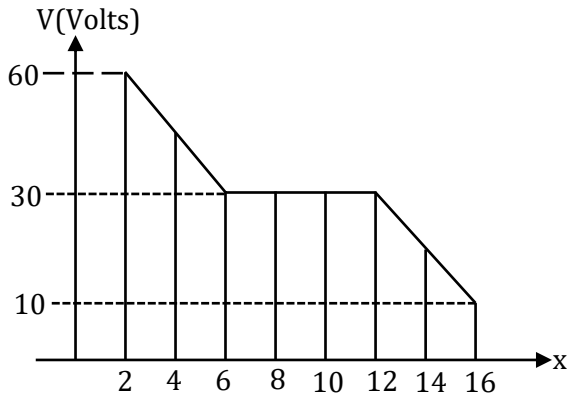
$V_1 = \frac{kQ_1}{r_1}$ $V_2 = \frac{kQ_2}{r_2}$

$$\frac{V_1}{V_2} = \frac{Q_1}{r_1}\left(\frac{r_2}{Q_2}\right) = \frac{r_2}{r_1}\left(\frac{r_1}{r_2}\right)^2 = \frac{r_1}{r_2}$$

68. $\Delta V = \left|\frac{KQ}{3r} - \frac{KQ}{r}\right| = \frac{2KQ}{3r} = V \Rightarrow KQ = \frac{3Vr}{2}$

$$E = \frac{KQ}{9r^2} = \frac{3Vr}{2\left(9r^2\right)} = \frac{V}{6r}$$

69.



$E = -\frac{dV}{dx}$ (−ve slope of V-x curve)

electric field at 13m is

$$= \frac{30-10}{16-12} = \frac{20}{4} = 5(V/m)$$

70. Electric field is directed from high potential to low potential.

71. Points A,B, C,D, E are equal distance from charge Q (at centre) so :- (Equipotential points)

$V_A = V_B = V_C = V_D = V_E = V$

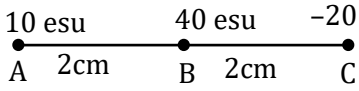
$W = q(V_f - V_i)$

$= q(V - V)$

$= 0$

72. Option (4) is incorrect

73.



$$U_A = q_A V_A = 10\left[\frac{kq_B}{r_1}+\frac{kq_C}{r_2}\right]$$

$$=10\left[\frac{1\times(40)}{2}+\frac{1\times(-20)}{4}\right]=150 \text{ ergs}$$

74. For closed path : -

$W_{AB} + W_{BC} + W_{CA} = 0$

$2 + (-3) + W_{CA} = 0$

$W_{CA} = +1$ J

75. $W = Q(V_B - V_A)$

$15 = 0.01(V_B - V_A)$

$V_B - V_A = 1500$ V

76. $F = qE = q\frac{\Delta V}{d}$

$2000 = 5 \times \frac{\Delta V}{2\times10^{-2}} \Rightarrow \Delta V = 8$ volt

77. Depending upon nature of charges, may increase or may decrease.

78. Velocity will be decreased.

79. $qE = mg$

$a = \frac{qE}{m}$ $(q,E \rightarrow \text{constant})$

$\boxed{a \propto \frac{1}{m}} \Rightarrow \frac{a_e}{a_P} = \frac{m_p}{m_e}$

80. speed of charge after time 't'

$v = u + at$, where $u = 0$ and $a = \frac{qE}{m}$

$v = \left(\frac{qE}{m}\right)t$

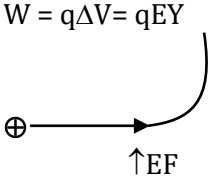
so $KE = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{qEt}{m}\right)^2$

$KE = \frac{1}{2}\frac{q^2E^2t^2}{m}$

81. $\Delta V = EY$

$W = q\Delta V = qEY$

82.



direction of E.F. will be down to top

$3\oplus \Rightarrow \uparrow$

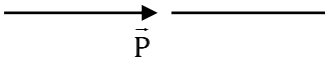
$2\ominus \Rightarrow \downarrow$ $4\ominus \Rightarrow \downarrow$

83. $P = q\,d$

$= 10^{-10} \times 1 \times 10^{-8}$

$= 10^{-18} = 1$ debye

84.



$V = \frac{kP}{r^2}$ $E = \frac{2kP}{r^3}$

$V \propto \frac{1}{r^2}$ $E \propto \frac{1}{r^3}$

85. $\vec{\tau} = \vec{P} \times \vec{E}$

$\tau = PE\sin\theta$

for maximum value $\theta = 90°$

86. $E_{axial} = \frac{2kp}{r^3}$

$E_{equitorial} = \frac{kp}{r^3}$

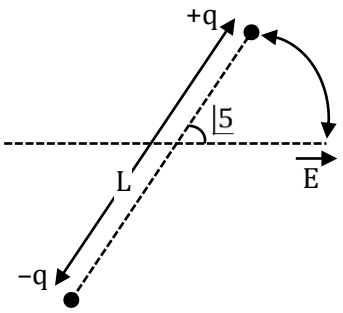
$\frac{E_{axial}}{E_{equtorial}} = \frac{2}{1}$

87. $P = qd = 2 \times 10^{-8}$ (C–m)

for maximum $\theta = 90° \Rightarrow \sin\theta = 1$

$\tau = PE = 10 \times 10^{-3}$ Nm

88.



$I\alpha = -pE\sin\theta$

for small angle $\sin\theta \simeq \theta$

$I\alpha = -pE\theta$

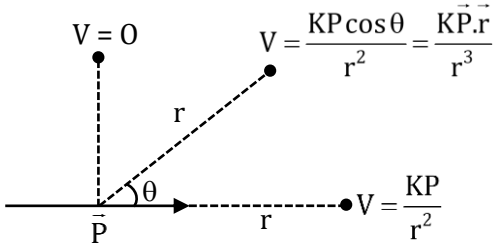
$\alpha = -\left(\frac{pE}{I}\right)\theta = -\omega^2\theta$

Hence $\omega = \sqrt{\frac{pE}{I}}$

$T = 2\pi\sqrt{\frac{I}{pE}}$

$= 2\pi\sqrt{\frac{ML^2/2}{qLE}} = 2\pi\sqrt{\frac{ML}{2qE}}$

89.



90. $F \propto \frac{1}{d^3}$

$F' = F/8$

91. $V=\frac{kp\cos\theta}{r^2}$

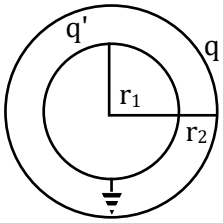
92. τ may be zero because it depends upon angle.

93. Same charge induced in both spheres.

($\varepsilon_r \rightarrow$ same)

94. Direction of charge flow from inside to outside in conducting concentric hollow metallic spheres.

95.



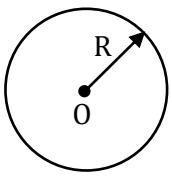
Potential of inner sphere in zero due to earthing V = 0

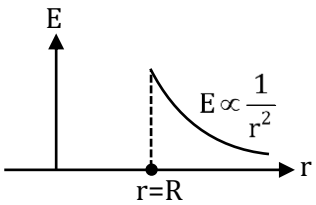
$$\frac{Kq}{r_2}+\frac{Kq'}{r_1}=0 \Rightarrow q'=-\frac{qr_1}{r_2}$$

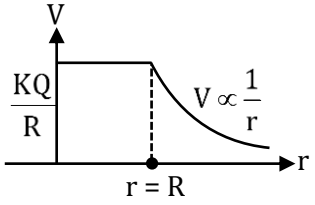
96. Minimum PE $\rightarrow$ more stability

97. electric field lines perpendicular to the conducting material.

98.







Exercise-II (Previous Year Questions)

1. For conducting sphere

At centre, E = 0 & $V=\frac{Q}{4\pi\in_0 R}$

2. $\vec{E}=-\frac{\partial V}{\partial x}\hat{i}-\frac{\partial V}{\partial y}\hat{j}-\frac{\partial V}{\partial z}\hat{k}$

$= -[(6 - 8y)\,\hat{i} + (-8x - 8 + 6z)\,\hat{j} + (6y)\,\hat{k}]$

At (1, 1, 1), $\vec{E}=2\hat{i}+10\hat{j}-6\hat{k}$

$\Rightarrow \quad (\vec{E})=\sqrt{2^2+10^2+6^2}=\sqrt{140}=2\sqrt{35}$

Force = qE = $2 \times 2\sqrt{35} = 4\sqrt{35}$ N

3. Flux linked with sphere = $\vec{E}.d\vec{s}$

since electric field is radial. It is always perpendicular to the surface.

so $\phi = Ar.(4\pi r^2)$

$\phi = A(a)(4\pi r^2)$ (as r = a)

$\phi = A4\pi a^3$

Now according to gauss law

$$\phi=\frac{q_{in}}{\in_0} \Rightarrow q_{in}=\phi.\in_0$$

So $q_{in}=A4\pi a^3\in_0$

4. $\vec{E}=-\frac{\partial V}{\partial x}\hat{i}-\frac{\partial V}{\partial y}\hat{j}-\frac{\partial V}{\partial z}\hat{k}$

$\vec{E}=-(6y)\hat{i}-(6x-1+2z)\hat{j}-(2y)\hat{k}$

at point (1,1,0)

$\vec{E}=-6\hat{i}-5\hat{j}-2\hat{k}=-(6\hat{i}+5\hat{j}+2\hat{k})$

5. $\tau = pE\sin\theta$

$\tau = q\ell E \sin\theta$

$4 = q \times 2 \times 10^{-2} \times 2 \times 10^5 \sin 30°$

$\Rightarrow$ q = 2 mC

6. $W = q\Delta V$

as ΔV is same in all conditions, work will be same.

7. $\frac{K\times(\Delta e)^2}{r^2}=\frac{Gm^2}{r^2}$

$\Delta e = m\sqrt{\frac{G}{K}} = 1.67\times10^{-27}\sqrt{\frac{6.67\times10^{-11}}{9\times10^9}}$ C

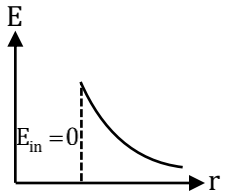
$= 1.436 \times 10^{-37}$ C

8. $S = ut + \frac{1}{2}at^2$

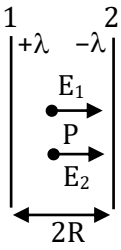
$$h = 0 + \frac{1}{2}\left(\frac{qE}{m}\right)t^2 \Rightarrow t = \sqrt{\frac{2hm}{qE}} \Rightarrow t \propto \sqrt{m}$$

as mass of proton is heavier than electron, hence electron will take less time.

9. For a metal sphere $E_{in} = 0$ and $\vec{E}_{out} = \frac{Kq}{r^2}\hat{r}$



10.



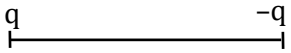
$\vec{E} = \vec{E}_1 + \vec{E}_2$

$E = E_1 + E_2$

$$E = \frac{\lambda}{2\pi\epsilon_0 R} + \frac{\lambda}{2\pi\epsilon_0 R}$$

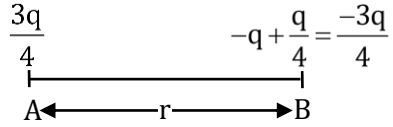
$$E = \frac{\lambda}{\pi\epsilon_0 R}\ N/C$$

11.



$F = \frac{-Kq^2}{r^2}$

25% charge from A is transferred to B



New force (F') = $\frac{K\left(\frac{3q}{4}\right)\left(\frac{-3q}{4}\right)}{r^2} = \frac{-9}{16}\frac{kq^2}{r^2} = \frac{9F}{16}$

12. $\phi = \frac{q_{in}}{\epsilon_0} = 0$

13. $V = \frac{kP\cos\theta}{r^2} = \frac{9\times10^9\times16\times10^{-9}(1/2)}{(0.6)^2} = 200V$

14. Potential is constant throughout the volume

$\therefore$ Electric field is zero.

15. $E = \frac{kQ}{r^2} = \frac{9\times10^9\times3.2\times10^{-7}}{(15\times10^{-2})^2}$

$E = 1.28 \times 10^5\ N/C$

16. Electric field in equatorial plane

$$\vec{E} = -\frac{k\vec{p}}{r^3}$$

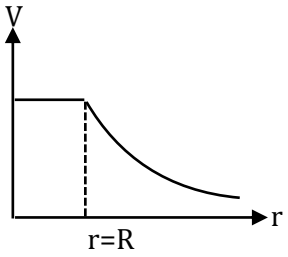
$$\vec{E} = -\frac{\vec{p}}{4\pi\epsilon_0 r^3}$$

17. $F = K\frac{e^2}{r^2} \Rightarrow a = K\frac{e^2}{mr^2}$

$$\Rightarrow a = 9\times10^9\frac{(1.6\times10^{-19})^2}{(1.6\times10^{-10})^2(9\times10^{-31})}$$

$\Rightarrow a = 10^{-29} \times 10^{51} = 10^{22}\ m/s^2$

18. $V_{in} = V_S = \frac{KQ}{R}$ and $V_{out} = \frac{KQ}{r}(r > R)$

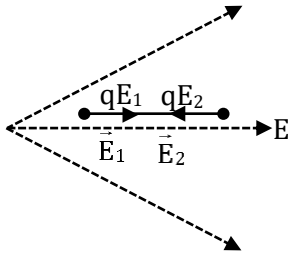


19. $|\vec{E}_1| > |\vec{E}_2|$

as field lines are closer at charge +q,

So, net force on the dipole acts towards right side.

Also, a system always moves to decrease its potential energy.



20. For a conducting sphere

$$E = \frac{\sigma}{\epsilon_0}$$

$$V = \frac{\sigma R}{\epsilon_0}$$

as both spheres have same potential after connecting with wire,

$V_1 = V_2$

$\sigma_1 R_1 = \sigma_2 R_2$

$$\Rightarrow \frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}$$

21. Volume conservation

$$\frac{4}{3}\pi R^3 = 27\left(\frac{4}{3}\pi r^3\right) \Rightarrow R = 3r \qquad ...(1)$$

$$\text{and } V = \frac{Kq}{r} \Rightarrow \frac{V_1}{V_2} = \left(\frac{q_1}{q_2}\right)\left(\frac{r_2}{r_1}\right)$$

$$\Rightarrow \frac{220}{V_2} = \left(\frac{q}{27q}\right)\left(\frac{3r}{r}\right)$$

$$\Rightarrow \frac{220}{V_2} = \frac{1}{9}$$

$\Rightarrow V_2 = 220 \times 9 = 1980$ Volt

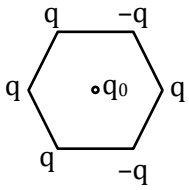
22. Electric field is always perpendicular to EPS.

23. It is electric dipole at large distance electric field intensity

$$E = \frac{KP}{R^3}\sqrt{1+3\cos^2\theta}$$

$$\therefore E \propto \frac{1}{R^3}$$

24.



Work = $U_f - U_i$

= 0 – 0

Work = 0

25. τ on a dipole = $\vec{p}\times\vec{E}$

$\tau = pE\sin\theta$

$4 = q \times \ell \times E \times \sin 30°$

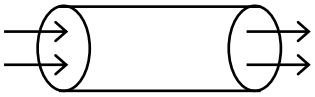
$$4 = q\times 2\times 10^{-2}\times 2\times 10^{5}\times\frac{1}{2}$$

$q = 2 \times 10^{-3}$

$q = 2$ mC

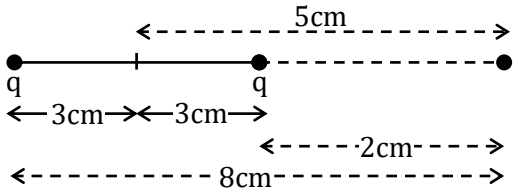
26. $\phi_{closed} = 0$

So $\phi_{in} = \phi_{out}$



Number of field lines entering is equal number of field lines leaving.

27.



$$v = \frac{Kq}{2\times 10^{-2}} - \frac{Kq}{8\times 10^{-2}}$$

$$= Kq\left[\frac{3}{8}\right]\times 10^2$$

28. $\phi = \frac{q_{inside}}{\varepsilon_0}$

only depends on charge enclosed by surface.

29. Total flux from cube = $\frac{q}{\varepsilon_0}$

$\therefore$ So flux of any one surface of cube

$$= \frac{q}{6\varepsilon_0} = \frac{Q\times 10^{-6}}{6\varepsilon_0}$$

30. $\therefore V = \frac{KQ}{R}$

$$E = \frac{KQ}{r^2}$$

$$E = \frac{VR}{r^2}$$

31. $V_{dipole} = \frac{kp\cos\theta}{r^2}$

at axis ($\theta = 0°$ or $180°$)

$$V_{Axis} = \pm\frac{kp}{r^2} = \pm\frac{9\times 10^9\times 4\times 10^{-6}}{2^2}$$

$$= \pm 9\times 10^3 V$$

A $\rightarrow$ Correct

R $\rightarrow$ False. $\quad \because V = \pm\frac{p}{4\pi\varepsilon_0 r^2}$

32. Shell is equipotential surface

So, $V_p = V_c$

$\therefore$ Potential Difference = 0

33. $\sigma = \frac{Q}{A} = \frac{6\times 10^{-6}}{6\times 5\times 5\times 10^{-4}}$

$$= \frac{1}{25}\times 10^{-2}$$

$= 4 \times 10^{-4}$

$= 0.4 \times 10^{-3}$ C/m^2

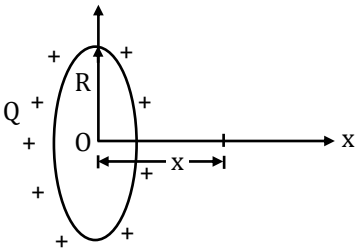
34. $V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$

$V = \frac{9\times10^9\times4\times10^{-7}}{9\times10^{-2}}$

$V = 4 \times 10^4$ Volt

Exercise-III (Analytical Questions)

1.



$E = \frac{KQx}{\left(R^2+x^2\right)^{3/2}}$, for (x, 0, 0)

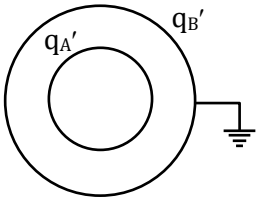
$E = 0$ & $V = \frac{KQ}{R}$ at centre/origin.

2. $|q| \propto$ No. of ELF

Neutral point of system of two unlike charges is outside the line joining the two charges near to & near the charge of smaller magnitude.

3. ∵ Both are unlike so neutral point exist outside & near the smaller charge.

4.



$2V = \frac{kq_A}{R}+\frac{kq_B}{2R}$

$\frac{3V}{2} = \frac{k(q_A+q_B)}{2R}$

on solving

$\frac{q_A}{q_B} = \frac{1}{2}$

$V_B = 0$

$\frac{kq'_A}{b}+\frac{kq'_B}{b} = 0$

$q'_B = -q_A'$

$\frac{|q'_A|}{|q'_B|} = 1$

5. Charge particle is in stable equilibrium if it is displaced along y-axis.

6. Think according to options.

7. For infinite non-conducting sheet ; $E = \frac{\sigma}{2\varepsilon_0}$

For infinite conducting sheet ; $E = \frac{\sigma}{\varepsilon_0}$

For non conducting sphere (at surface) ;

$E = \frac{\rho R}{3\varepsilon_0}$

For non conducting sphere at centre ; E = 0

8. Charge can't exist without mass, but mass can exist without charge. Charge is conserved but mass is not conserved.

9. $[F] = [MLT^{-2}]$

$\because \quad F = \frac{E}{q}$

$\therefore \quad E = \frac{MLT^{-2}}{AT} \Rightarrow E = [MLT^{-3}A^{-1}]$

ε_r is dimensionless

$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2} \Rightarrow \varepsilon_0 = \frac{[A^2T^2]}{[MLT^{-2}][L^2]}$

$= [M^{-1}L^{-3}A^2T^4]$

10. $V_{R/2} = \frac{kQ}{2R^3}\left[3R^2-\left(\frac{R}{2}\right)^2\right] = \frac{kQ}{2R^3}\left(\frac{11R^2}{4}\right)$

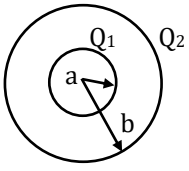
$V_{R/2} = \frac{11kQ}{8R} = \frac{11V}{8}$; $V_S = \frac{kQ}{R}$

$V_{2R} = \frac{kQ}{2R} = \frac{V}{2}$

$E_{R/2} = \frac{kQ}{R^3}\frac{R}{2} = \frac{kQ}{2R^2} = \frac{V}{2}$ (R = 1m)

$E_{2R} = \frac{kQ}{(2R)^2} = \frac{V}{4}$

11.



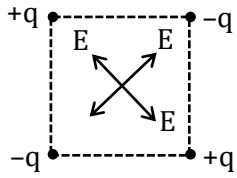
$\Delta V = KQ_1\left(\frac{1}{a}-\frac{1}{b}\right)$ does not depends on outer charge.

$E_{in} = \frac{KQ_1}{a^2}$ does not depends on outer charge.

12. Direction of force depends on nature of charge.

13. Potential depends on frame of reference while potential difference is a absolute.

14.



$E = 0, V = \frac{Kq}{x} + \frac{Kq}{x} - \frac{Kq}{x} - \frac{Kq}{x} = 0$

It is not necessary that if potential is zero then electric field is also zero.

15. Extra charge on aircraft body due to air friction is transferred to ground.

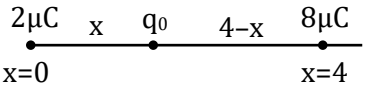
16. Electric field can not enter in conductor.

17. Charge of isolated body can not be changed.

18. Net flux is due to inside charges but electric field is due to all charges.

19. There is loss / gain of electrons so mass may decrease or increase.

22.

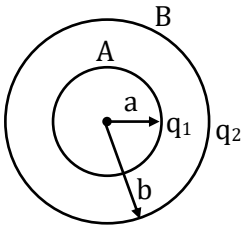


$\frac{k2q_0}{x^2} = \frac{k8q_0}{(4-x)^2}; \ \frac{1}{x} = \frac{2}{4-x}$

$4-x = 2x; \ 3x=4$

$x = \frac{4}{3} = \frac{\alpha}{\beta} \Rightarrow \alpha+\beta = 4+3 = 7$

24.



$V_A = \frac{q_2}{b} + \frac{q_1}{a}$

$V_B = \frac{q_1}{b} + \frac{q_2}{b}$

$V_A - V_B = q_1\left[\frac{1}{a} - \frac{1}{b}\right]$

25. $V = \frac{kQ}{r}$ & $E = \frac{kQ}{r^2}$

27. Centre of mass is taken but there is no concept of centre of charge.

AIM: FREE EDUCATION TO ALL APUL

Current Electricity

SOLUTIONS

Exercise-I (Conceptual Questions)

1. $I = \frac{dq}{dt} = \frac{10^6 \times 1.6 \times 10^{-19}}{1} = 1.6 \times 10^{-13}\,A$

2. $\because I = \frac{dQ}{dt} \Rightarrow dQ = Idt$

$Q = \int Idt = \int_2^3 (2t + 3t^2)dt$

on solving $Q = 24\ C$

3. $i = \frac{q}{t} = \frac{10^4 \times 1.6 \times 10^{-19}}{60} = 0.25 \times 10^{-16}\,A$

4. $i = neA\,v_d$

$v_d = \frac{i}{neA} = 1.56 \times 10^{-5}\ m/s$

5. $V_d = \frac{I}{neA} = 2.5 \times 10^{-3}\ m/s$

6. Ampere

7. If no. current passes through a conductor then average velocity of all electrons in conductor is zero for time.

8. $I = neAv_d$

$= \frac{N}{V}eAv_d = \frac{N}{A\ell}eAv_d$

$I = \frac{N}{\ell}ev_d$ solve it.

9. $i = 4.1 \times 10^{18} \times 1.6 \times 10^{-19}$

$= 0.66$ A towards right

10. $I = neAv_d$

$\Rightarrow v_d \propto \frac{1}{n}$

$\Rightarrow \frac{v_{d_1}}{v_{d_2}} = \frac{n_2}{n_1} = \frac{4n_1}{n_1} = 4:1$

11. $j = \frac{I}{A} = nev_d$

$\frac{4I}{\pi d^2} = nev$ ...(i)

$\frac{16I}{\pi d^2} = nev'$ ...(ii)

From equation (i) & (ii) $\frac{4I}{16I} = \frac{v}{v'} \Rightarrow v' = 4v$

12. $v_d = \frac{i}{Ane}$ As A ↑ so v_d↓ $\Rightarrow v_P > v_Q$

13. $q = \int i\,dt$ = area under i – t curve

$q_1 = 2C, q_2 = 2C, q_3 = 2C$

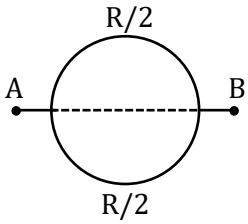
So, $q_1 : q_2 : q_3 = 1 : 1 : 1$

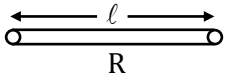
14. $R = \frac{\rho\ell}{A}$

$R_1 = \frac{\rho\ell}{A}$ $R_2 = \rho\left(\frac{4\ell}{A}\right)$ $R_3 = \rho\frac{\ell}{4A}$

$R_3 < R_1 < R_2$

15.





$\Rightarrow R_{AB} = \left(\frac{R/2}{2}\right) = \frac{R}{4}$

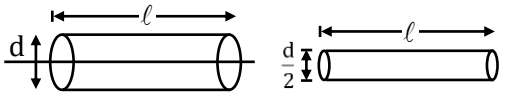
16. Material same $\Rightarrow \rho$ unchanged

$R = \frac{\rho\ell}{A} = \frac{\rho\ell}{\pi a^2} \Rightarrow R \propto \frac{\ell}{a^2}$

Therefore $R' \propto \frac{2\ell}{(2a)^2}$

$\Rightarrow R' = \frac{R}{2}$

17.



Initial volume = final volume

$\frac{\pi d^2 \ell}{4} = \frac{\pi d^2 L}{16}$ $\boxed{L = 4\ell}$ $A_f = \frac{A_i}{4}$

$R_i = \frac{\rho\ell}{A_i}$ $R_f = \rho\left(\frac{4\ell}{A_i}\right) = 16\,R_i$

18. T↑ ρ↑ σ↓

$\rho \times \sigma$ = constant

19. $V_{old} = A\ell$

$V = AL \qquad R = \frac{\rho L}{A}$

New

$L' = L + \frac{L}{5} = \frac{6}{5} L$

$V = A'L' = AL$

$A' = \frac{5A}{6}$

$R' = \frac{\rho L'}{A'} = \frac{36}{25}R$

% change $= \frac{R'-R}{R} \times 100 = 44\%$

20. Volume remains same $\Rightarrow AL = A'(2L)$

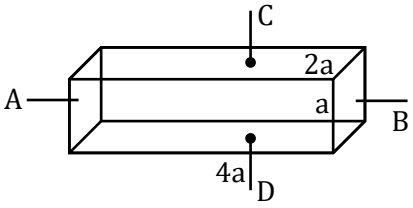
$A' = \frac{A}{2}$; $R_i = \frac{\rho L}{A}$

$R_f = \frac{\rho(2L)}{(A/2)} = 4R_i$

%change $= \frac{4R_i - R_i}{R_i} \times 100 = 300\%$

21. Increase and decrease respectively.

22.



$x > z > y$

$\left(R = \rho\frac{\ell}{A}\right)$, A = Area of crossection.

23. The value ρ does not depends on geometry but increases with increase in temperature

24. $R = R_0 (1 + \alpha\Delta T)$

$1 = R_0[1 + \alpha(27)]$

$2 = R_0[1 + \alpha(T)] \Rightarrow \frac{1}{2} = \frac{1+\alpha\times 27}{1+\alpha\times T}$ solve it.

25. For I – V slope $= \frac{I}{V} \propto \frac{1}{R} \propto \frac{1}{T}$ curve

Slope ↓ R ↑ T ↑

$\because R_2 > R_1 \ \therefore T_2 > T_1$

26. $\frac{2R}{2+R} = \frac{6}{5}$

$10R = 12 + 6R$; $R = 3\ \Omega$

27. $R = R_0 (1 + \propto\Delta t)$

$3R_0 = R_0(1 + 4 \times 10^{-3} T)$

$2 = 4 \times 10^{-3} T \quad \Rightarrow \quad T = \frac{1000}{2} = 500°C$

28. For conductor, $T\downarrow \ \rho\downarrow$

For semiconductor, $T\downarrow \ \rho\uparrow$

29. $R_1 = R_{01} (1+\alpha\Delta T)$

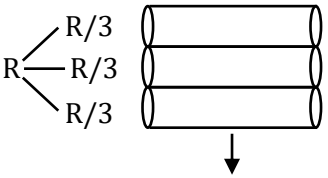
$R_2 = R_{02}(1 -\beta.\Delta T)$

$R_{series} = R_1 + R_2 = R_{01} + (R_{01}\alpha - R_{02}\beta)\Delta T + R_{02}$

Not depend on temperature

$R_{01}\alpha - R_{02}\beta = 0 \quad \Rightarrow \quad \frac{R_{01}}{R_{02}} = \frac{\beta}{\alpha}$

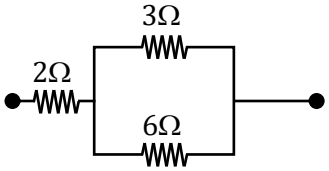
30.



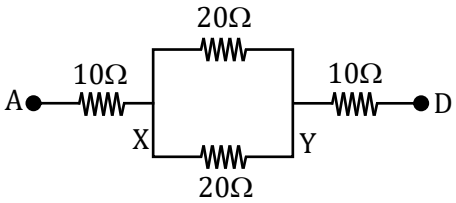
Parallel combination

$R_{eq} = \frac{R}{9}$

31.



32. Redraw circuit



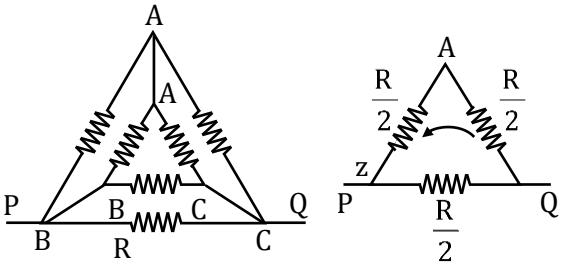
$R_{AD} = 30\ \Omega$

33. $\frac{(R+3).10}{(R+3)+10} + 3 = R$ (given)

solve it $(10R + 30) = (R - 3)(R + 13)$

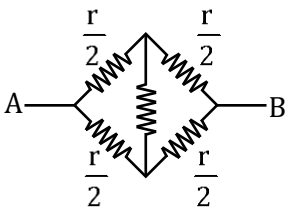
$10R + 30 = R^2 - 39 + 10R$ or $R = \sqrt{69}\ \Omega$

34.



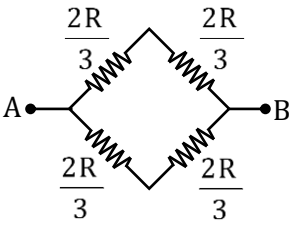
$R_{eq} = \frac{R}{3}$

35. After folding symmetry



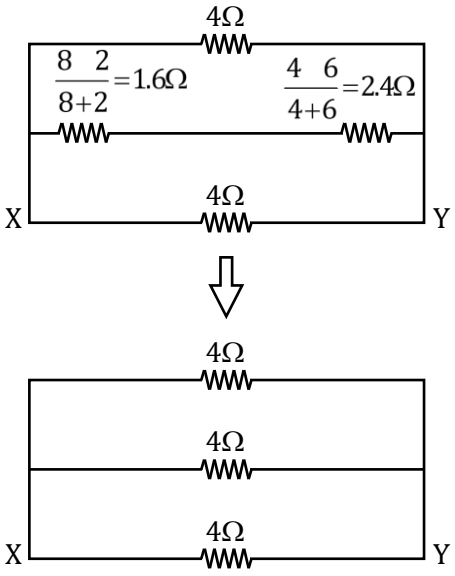
$R_{eq} = \frac{r}{2}$

36. It is balance Wheat Stone Bridge



$R_{AB} = \frac{2R}{3}\Omega$

37. Redrawing the circuit



$Req = \frac{4}{3}$

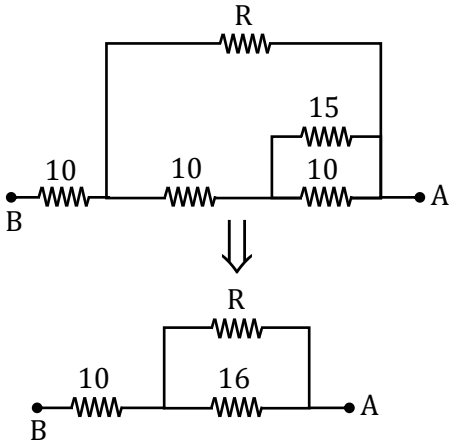
38. When they are connected in parallel

Then $R = \frac{r}{n} \Rightarrow r = nR$

When they are connected in series

Then $R_{eq.} = nr = n \times nR = n^2R$

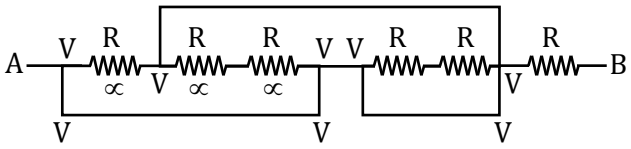
39.



$\frac{16R}{16+R} + 10 = 18$

40. $R_{AB} = 20\Omega$ Multiple W.S.B.

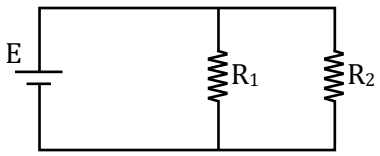
41.



$R_{eq} = R$

42. $R_{eq} = 6\Omega$

43.



$\frac{R_1R_2}{R_1+R_2} = 6\Omega \Rightarrow R_1$ could be $= 8\Omega$

44. $I_{in} = I_{out}$

$10 + 1 + 2 = I \Rightarrow \boxed{I = 13A}$

45. $V_x + 2 \times 1 - 3 \times 1 = V_y$

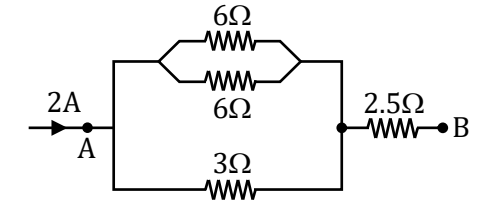
$V_x - V_y = +1$ volt

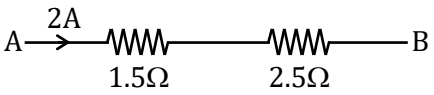
46. $I = \frac{V}{R_{eq}} = \frac{9}{\frac{9}{5}} = 5$ A

47. $i = 0.25 = \frac{12}{R+6} \Rightarrow R + 6 = 48 \Rightarrow R = 42\ \Omega$

48. $i = 2\ A = \frac{6}{2+R} \Rightarrow R = 1\Omega$

49.





$R_{eq} = 4\Omega$ $V_{eq} = 8V$

50. In loop AHDCBA

$-30\ I_1 - 45 - I_3 - 40\ I_3 = 0$

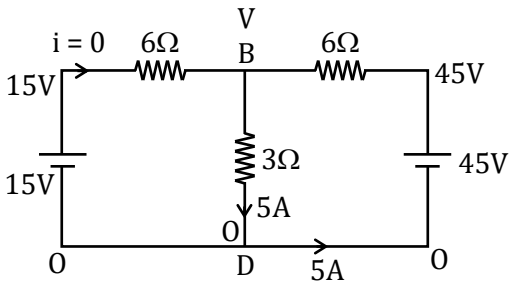
$-30\ I_1 - 41\ I_3 - 45 = 0$

In loop AHDCBA

$-30\ I_1 + 20\ I_2 + 1\ I_2 - 80 = 0$

$-30\ I_1 + 21\ I_2 - 80 = 0$

51.



$\frac{V-15}{6}+\frac{V}{3}+\frac{V-45}{6}$

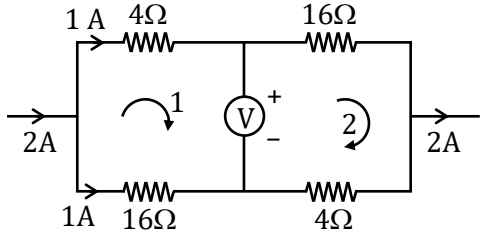
$V - 15 + 2V + V - 45 = 0$

$4V = 60$

$V = 15\ V$

$\therefore\ i_{BD} = \frac{V}{3}=\frac{15}{3}A = 5A$

52.



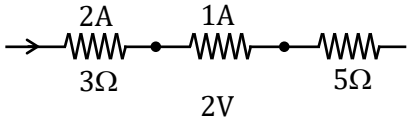
By KVL in loop (1)

$-4 \times 1 - V + 16 \times 1 = 0$

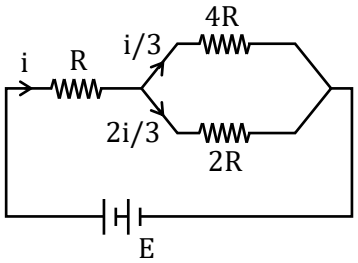
$V = 12$ Volts

53. $V_{JG}=\frac{32}{32+34}\times 60 = 20V$

54.

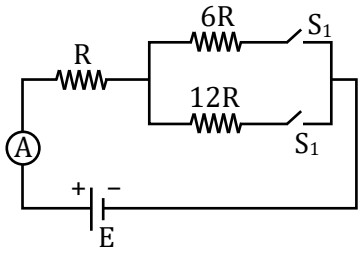


55. $i = \frac{3E}{7R}$



$V = (2R)\ \frac{2}{3}\left(\frac{3E}{7R}\right) = \frac{4}{7}E$

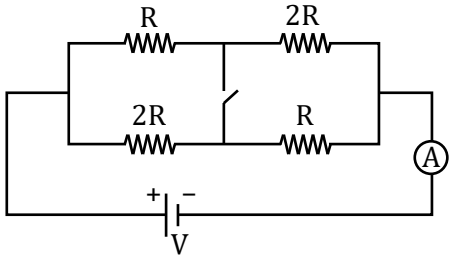
56.



$I_1 = \frac{E}{7R}$ $I_2 = \frac{E}{13R}$ $I_3 = \frac{E}{5R}$

$I_3 > I_1 > I_2$

57.

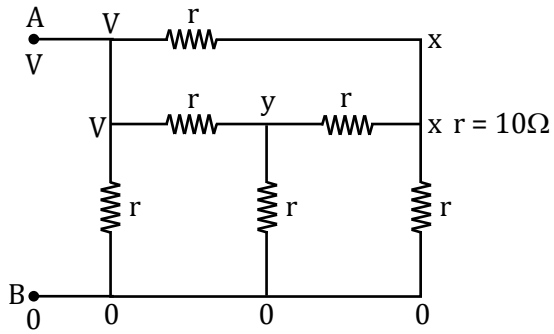


when key open when key closed

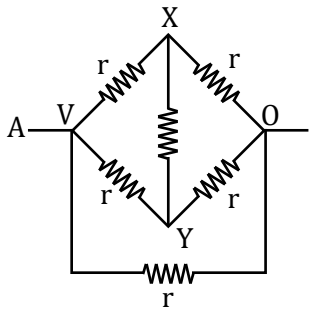
$R_{eq} = \frac{3R}{2}$ $R_{eq} = \frac{4R}{3}$

$i_0 = \frac{2V}{3R}$ $i_C = \frac{3V}{4R}$

58.



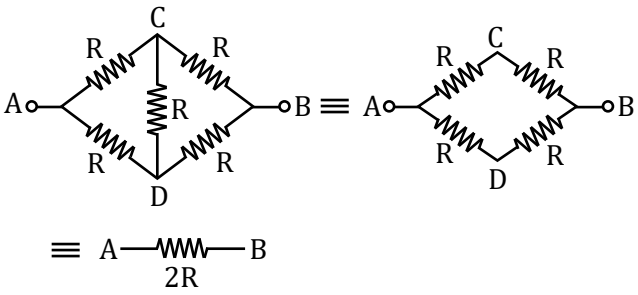
Balanced Wheat Stone Bridge



$R_{AB} = \frac{r}{2} = 5\Omega$

59. Given circuit can be reduced to

Balanced Wheat Stone Bridge

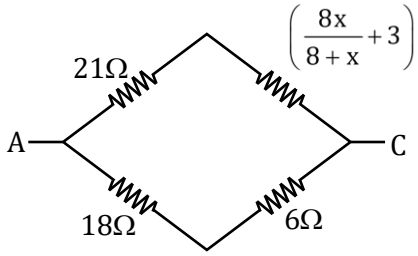


Required current = $\frac{V}{2R}$

60. Option (3) is correct.

61. Wheatstone bridge is in balance condition

62.

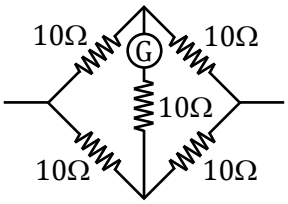


$\Rightarrow \frac{P}{Q} = \frac{R}{S}$

63. For balance wheat stone bridge

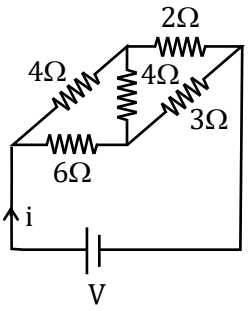
$\frac{16}{4} = \frac{x}{1/2}$; x = 2Ω.

64. R_{eq} = 10 Ω



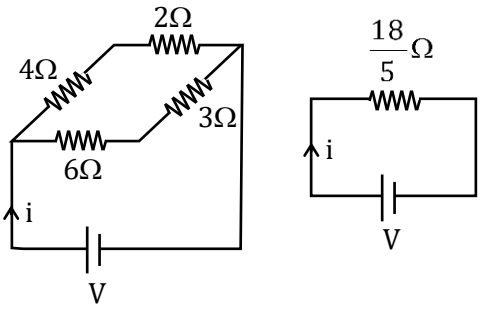
R_{eq} = 10Ω

65.



Here $\frac{4}{6} = \frac{2}{3} \Rightarrow$ Balanced Wheat Stone Bridge

Therefore given circuit can be reduced to



$\Rightarrow i = \frac{5V}{18}$

66. $V = E - Ir = 4 - \frac{10}{20} \times 8 = 0V \left(\because I = \frac{10}{20} = \frac{1}{2}A\right)$

67. When battery is connected to external resistance

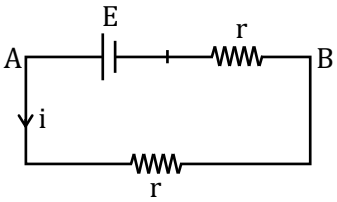
$\frac{4}{2+r} = 1 \Rightarrow r = 2\Omega$

When terminals of battery are connected directly

i = 4/2 = 2A

68. All

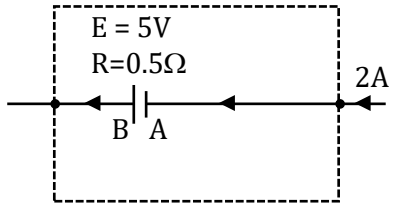
69.



$V_{AB} = 3 \quad \Rightarrow \quad i = 3/r$

$E - ir = 3 \quad \Rightarrow \quad E = 3 + ir = 3 + 3 = 6V$

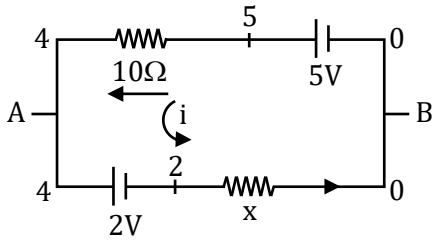
70.



$V_A + 5 - 2 \times 0.5 = V_B$

$10 + 5 - 1 = V_B \Rightarrow V_B = 14V$

71.



$i = \frac{5-4}{10} = 0.1A$

$x \times 0.1 = 2 \Rightarrow x = 20\Omega$

72. emf is basically work

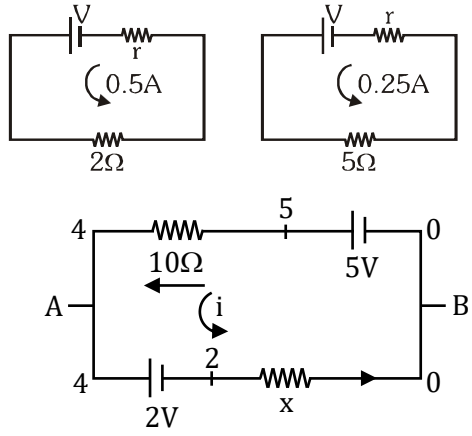
73. $\because I = \frac{E}{2+r}$, terminal potential $\frac{E}{2} = E - Ir$

$\Rightarrow \frac{E}{2} = E - \frac{E}{2+r}r$

$\Rightarrow 2 + r = 2r$

$\Rightarrow r = 2\Omega$

74.



$V=0.5(2+r)$ ...(i)

$V=0.25(5+r)$ ...(ii)

for (i) & (ii)

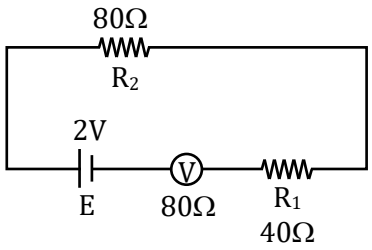
$0.5(2+r) = 0.25(5+r)$

$4 + 2r = 5 + r$

$r = 1$

$V = 0.5 \times 3 = 1.5V$

75.



$i = \frac{2}{200} = 0.01\ A$

$V = 0.8\ V$

76. $0.9 = \frac{V}{2+r}$

$0.3 = \frac{V}{7+r}$

$1.8 + 0.9r = 2.1 + 0.3r$

$0.6r = 0.3$

$r = \frac{1}{2}\Omega$

77. for maximum power $R_{internal} = R_{ext.}$

78. $R = \frac{v_0^2}{p_0}$; $P = \frac{V^2}{R} = \frac{V^2}{V_0^2}P_0 = \left(\frac{V}{V_0}\right)^2 P_0$

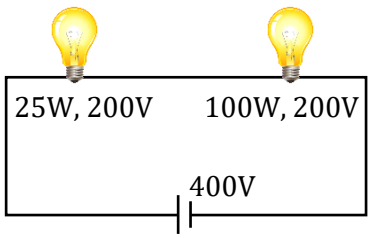
79. $P = \frac{V^2}{R_{eq.}} \Rightarrow R_{eq.} \downarrow P \uparrow$

80. $P=i^2R$ Current is same, so $P \propto R$.

In the first case it is 3r, in second case it is (2/3)r, in III case it is $\frac{r}{3}$ & in IV case the net resistance is $\frac{3r}{2}$

$R_{III} < R_{II} < R_{IV} < R_I \therefore P_{III} < P_{II} < P_{IV} < P_I$

81.

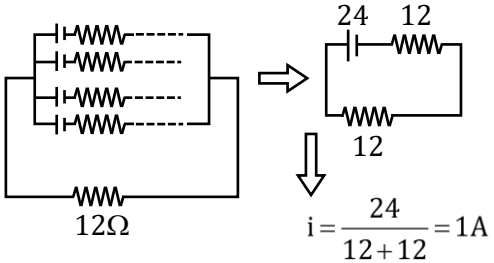


$R = \frac{V_{rated}^2}{P_{rated}}$; $R_1 = \frac{(200)^2}{25}$; $R_2 = \frac{(200)^2}{100}$

$R_1 > R_2$; $V_1 > V_2$

thats why 25 W bulb fuse first

82.



$i = \frac{24}{12+12} = 1A$

For $I_{max} = \frac{nr}{m} = R$ ...(1)

n = number of coloum

m = number of row

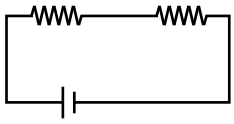
nm = 48 ...(2)

from eq (1) and (2)

n = 12

m = 4

83. $R_1 = \frac{V_1^2}{P}$ $R_2 = \frac{V_2^2}{P}$



$i = \frac{\varepsilon}{\left(\frac{V_1^2}{P} + \frac{V_2^2}{P}\right)}$; Brightness $\propto$ power = (i^2R)

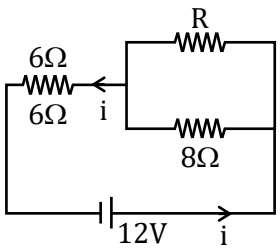
$P_i = \frac{\varepsilon^2 V_1^2}{(V_1^2 + V_2^2)}$

84. $P = \frac{V^2}{R_{eq}}$; $R_{eq} \downarrow P \uparrow$

85. For maximum power internal resistance should be equal to external resistance ($R_{ent} = R_{Int}$)

$R = \frac{r}{2}$

86.



$i^2 6 = 6 \quad \Rightarrow \quad i = \sqrt{\frac{6}{6}} = 1A$

$\left(6 + \frac{8R}{8+R}\right) = 6$

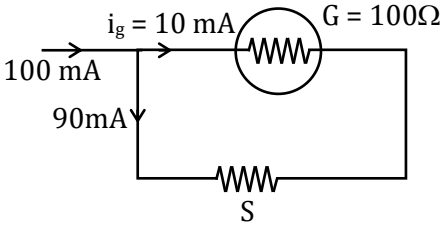
$\frac{8R}{8+R} = 6$; $4R = 24 + 3R \quad \Rightarrow R = 24\ \Omega$

87. max $\rightarrow$ $R = 1 \Rightarrow P = i^2R = 4W$

88. Net resistance of system will decrease.

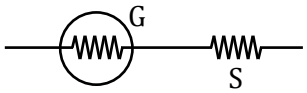
$\therefore$ Reading of A will increase & reading of V will decrease.

89.



$100 \times 10 = 90\ S \Rightarrow S = \frac{100}{9}\Omega$

90.



$V = G\, i_g$

$nV = (G + S)i_g \Rightarrow n = \left(\frac{G+S}{G}\right) \Rightarrow S = (n-1)\,G$

91. Initially

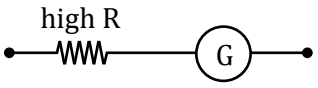
$\frac{R_1}{R_2} = \frac{x}{(1-x)} = \frac{10}{30} \Rightarrow x = 25$ cm

Finally after interchange

$\frac{y}{1-y} = \frac{30}{10} \Rightarrow y = 75$ cm

shifting = 50 cm

92. Voltmeter



93. If radius of wire is doubled then resistance becomes 1/4 times but ratio of AC : CB remains same.

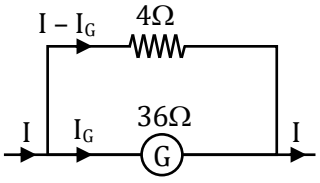
94. $V = I(R + R_A) \Rightarrow 12 = 0.1(R + 20)$

$\Rightarrow R = \frac{12}{0.1} - 20 \quad R = 100\ \Omega$

95. $R = (n-1)G$

Here $n = \frac{10}{1} = 10 \Rightarrow R = (10 - 1)G = 9G$

96.

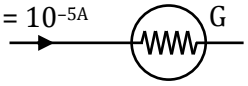


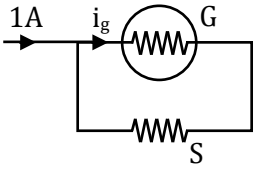
$V_G = V_R$

$\Rightarrow 4(I - I_G) = 36 I_G$

$\Rightarrow 10 I_G = I \Rightarrow \frac{I_G}{I} = \frac{1}{10}$

97. $i_g = 10^{-5}A$





$V = 10^{-3}$ V

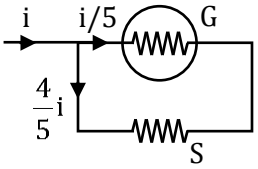
$Gi_g = S(1 - i_g)$

$10^{-3} = S(1 - 10^{-5})$

$S \approx 10^{-3}\ \Omega$

98. High resistance $\Rightarrow$ Good voltmeter

99.



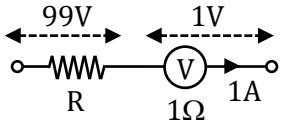
$\frac{i}{5}G = \frac{4i}{5}S;\ S = \frac{G}{4}$

100. $\frac{R_1 + 10}{R_2} = \frac{50}{50} = 1 \Rightarrow R_1 + 10 = R_2$ ... (i)

$\frac{R_1}{R_2} = \frac{40}{60} \Rightarrow 3R_1 = 2R_2$ ... (ii)

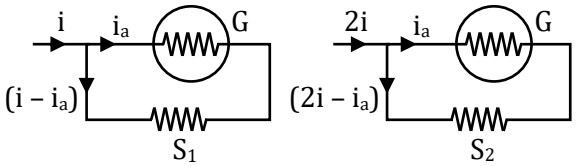
$R_1 + 10 = \frac{3}{2}R_1 \Rightarrow R_1 = 20\ \Omega$

101.



$99 = 1 \times R \Rightarrow R = 99\Omega$

102.



$Gi_a = S_1(i - i_a)$ ...(i)

$Gi_a = (2i - i_a)S_2$ ...(ii)

Equation (i) = (ii)

$S_1(i - i_a) = S_2(2i - i_a) \Rightarrow \frac{S_1}{S_2} = \left(\frac{2i - i_a}{i - i_a}\right)$

Exercise-II (Previous Year Questions)

1. Resistance = (0.5Ω/km) (150 km) = 75Ω

Total voltage drop = (8 V/km) (150 km) = 1200V

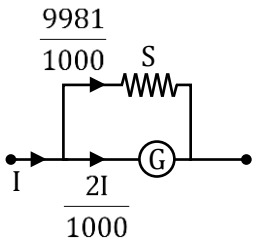
Power loss = $\frac{(\Delta V)^2}{R} = \frac{(1200)^2}{75}$ W

= 19200 W = 19.2 kW

2. $\frac{5}{R} = \frac{\ell_1}{100 - \ell_1}$ and $\frac{5}{R/2} = \frac{1.6\ell_1}{100 - 1.6\ell_1}$

$\Rightarrow$ R = 15Ω

3.



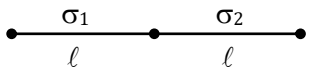
$\left(\frac{2I}{1000}\right)G = \left(\frac{998I}{1000}\right)S$

$\Rightarrow S = \frac{G}{499}$

Total resistance of Ammeter

$R = \frac{SG}{S+G} = \frac{\left(\frac{G}{499}\right)}{\left(\frac{G}{499}\right)+G} = \frac{G}{500}$

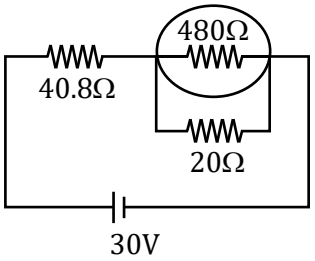
4.



$R_{eq} = R_1 + R_2$

$\Rightarrow \frac{2\ell}{\sigma_{eq}A} = \frac{\ell}{\sigma_1 A} + \frac{\ell}{\sigma_2 A} \Rightarrow \sigma_{eq} = \frac{2\sigma_1\sigma_2}{\sigma_1 + \sigma_2}$

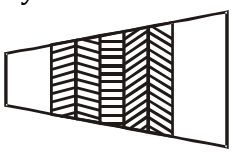
5.



$R_{eff} = 40.8 + \frac{480 \times 20}{480 + 20} = 40.8 + 19.2 = 60\ \Omega$

$I = \frac{V_{eff}}{R_{eff}} = 0.5\,A$

6. Metallic conductor can be considered as the combination of various conductors connected in series combination. In series combination the current always remains constant.



7. Effective resistance of B & C = $\frac{(1.5R)(3R)}{1.5R+3R}=R$

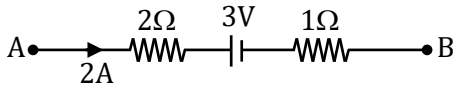
In series sequence $V \propto R$

so voltage across 'A' = voltage across B & C

Now B & C are parallel so $V_B = V_C$

$\Rightarrow V_A = V_B = V_C$

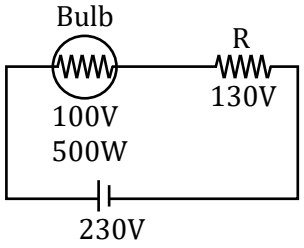
8.



$V_B = V_A - (2 \times 2) - 3 - (2 \times 1)$

$\Rightarrow V_A - V_B = 9V$

9.



Current through bulb = $\frac{P}{V}=\frac{500W}{100V}=5A$

Therefore $R=\frac{130V}{5A}=26\Omega$

10. $R=\frac{\rho\ell}{A}=\frac{\rho\ell^2}{\text{volume}}\Rightarrow R\propto \ell^2$

$\Rightarrow R_2 = n^2R_1$

11. $I=\frac{E}{nR+R}$ ... (1)

$10I=\frac{E}{\frac{R}{n}+R}=\frac{nE}{R+nR}$ ... (2)

From (1) & (2),

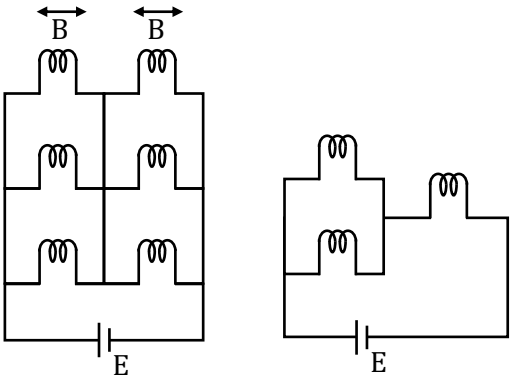
$n\frac{E}{R+nR}=10\left(\frac{E}{nR+R}\right)\Rightarrow n = 10$

12. $I=\frac{nE}{nr}=\frac{E}{r}=\text{constant}$

13. $\frac{\theta}{i}=\frac{50}{10^{-3}}=5000$, $\frac{\theta}{V}=20$

$\frac{\theta/i}{\theta/V}=\frac{5000}{10}=250\Rightarrow\frac{V}{i}=R=250\Omega$

14.



Case-I **Case-II**

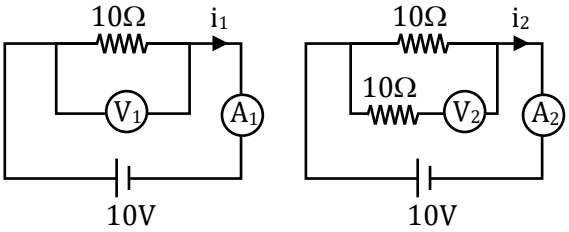
$R_{eq_1}=2R/3$ $R_{eq_2}=R/2+R=\frac{3R}{2}$

$P_{eq_1}=\frac{E^2}{2R/3}=\frac{3P}{2}$ $P_{eq_2}=\frac{E^2}{3R/2}=\frac{2P}{3}$

$\therefore P_{eq_1}: P_{eq_2} = 9:4$

15. Fuse is used for protection.

16.



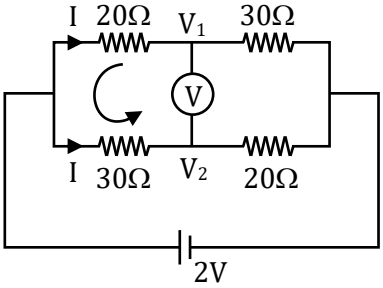
Circuit 1 Circuit 2

10Ω is in series with ideal voltmeter. Therefore it will not affect the circuit (Circuit-2)

$i_1=\frac{10}{10}=1A$ $i_2=\frac{10}{10}=1A$

$V_1 = 10V$ $V_2 = 10V$

17.



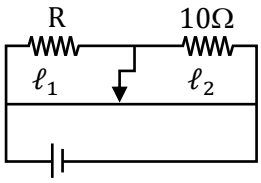
$(V_1 - V_2)=\frac{30}{25}-\frac{20}{25}=\frac{10}{25}=\frac{2}{5}=0.4\ V$

18. Interchanging cell and galvanometer do not effect balance condition.

19. $\mu=\frac{v_d}{E}=\frac{7.5\times10^{-4}}{3\times10^{-10}}=2.5\times10^6\ m^2/V\text{-}s$

20. For some metals like copper, resistivity is nearly proportional to temperature although a non linear region always exists at very low temperature.

21.



$\frac{R}{10}=\frac{\ell_1}{\ell_2} \Rightarrow \frac{R}{10}=\frac{3}{2} \Rightarrow R = 15\Omega$

Length of 15Ω resistance wire is 1.5 m

$\therefore$ length of 1Ω resistance wire $=\frac{1.5}{15}$ = 0.1 m

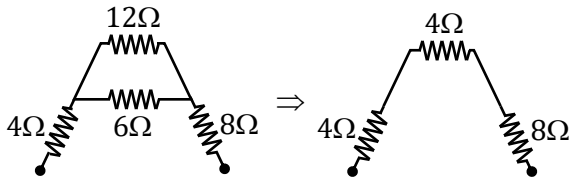
22. $I=\frac{2+4}{4+1+1}=\frac{6V}{6\Omega}=1\ A$

23. By KVL

$-I_2R_2 - E_2 + E_3 + I_3R_1 = 0$

$I_2R_2 + E_2 - E_3 - I_3R_1 = 0$

24.



$\Rightarrow R_{AB} = 16\Omega$

25. Resistance of conductor, $R=\frac{\rho\ell}{A} \Rightarrow A=\frac{\rho\ell}{R}$

$\Rightarrow \frac{A_1}{A_2}=\frac{\rho_1}{\rho_2}\times\frac{\ell_1}{\ell_2}\times\left(\frac{R_2}{R_1}\right)=1$

[$\because R_1 = R_2$, $\ell_1 = \ell_2$ and for same material $\rho_1 = \rho_2$]

26. (A) $v_d=\left(\frac{eE}{m}\right)\tau$

(B) $J = \sigma E = E/\rho \Rightarrow \rho = E/J$

(C) $\rho=\frac{E}{nev_d}$

$v_d=\frac{E}{ne\rho}$

$\frac{eE}{m}\tau=\frac{E}{ne\rho}$

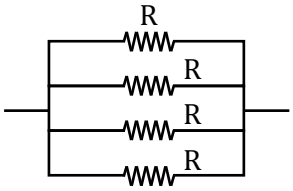
$\tau=\frac{m}{ne^2\rho}$

(D) $I = neAv_d$

$\frac{i}{A}=nev_d$

$J = nev_d$

27. $R_l=\frac{R}{4}=0.25\Omega$



$R = 1\ \Omega$

$R_{series} = 4R = 4(1) = 4\Omega$

28. $V = ir$

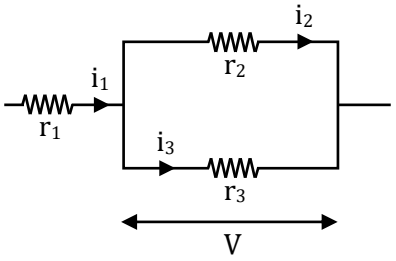
$i=\frac{V}{r}$

$i \propto \frac{1}{r}$ [V is same for r_2 & r_3]

$\frac{i_2}{i_3}=\frac{r_3}{r_2}$

$i_3=\frac{r_2}{r_2+r_3}i_1$

$\frac{i_3}{i_1}=\frac{r_2}{r_2+r_3}$



29. $V=\frac{1}{4\pi\in_0}\cdot\frac{Q}{R}$

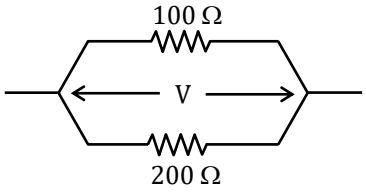
$\frac{1}{4\pi\in_0}=\text{constant}$

Q = same (Given)

$\therefore V \propto \frac{1}{R}$

$\therefore$ Potential is more on smaller sphere.

30.



As both resistors are in parallel combination so potential drop (V) across both are same.

$P=\frac{V^2}{R} \Rightarrow P \propto \frac{1}{R}$

$\frac{P_1}{P_2}=\frac{R_2}{R_1}=\frac{200}{100}=\frac{2}{1}=2:1$

31. For conductors α is (+)ve

For semiconductors & Insulators α is (–)ve

32. Radius of wire $= \frac{10^{-2}}{\sqrt{\pi}}$

Cross sectional area $A = \pi r^2 = 10^{-4}\ m^2$

$$j = \frac{i}{A} = \left(\frac{V}{R}\right).\frac{1}{A} = \frac{E\ell}{RA} \qquad R = \frac{\rho\ell}{A}$$

$$j = \frac{10\times 10}{10\times 10^{-4}} = 10^5\ A/m^2$$

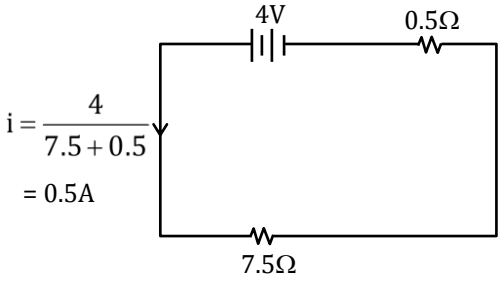
or

$$J = \sigma E \Rightarrow \frac{E}{\rho} = \frac{E\ell}{RA} = \frac{10\times 10\times \pi}{10\times 10^{-4}\times \pi}$$

$\Rightarrow 10^5\ A/m^2$

33. Resistance of P & Q should be approx. equal as it decreases error in experiment.

34.



$$i = \frac{4}{7.5+0.5}$$

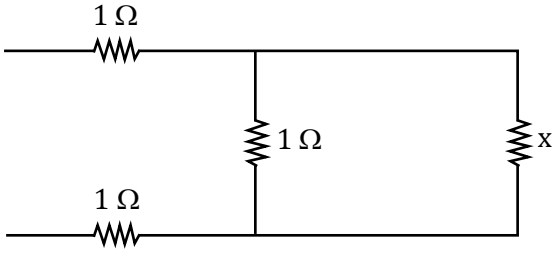
$= 0.5A$

TPD = 4 – 0.5 × 0.5

= 4 – 0.25

TPD = 3.75 volt

35. If effective resistance is x,



$$\Rightarrow x = 1 + \frac{x\times 1}{x+1} + 1$$

$$\Rightarrow (x-2) = \frac{x}{x+1}$$

$\Rightarrow x^2 - x - 2 = x$

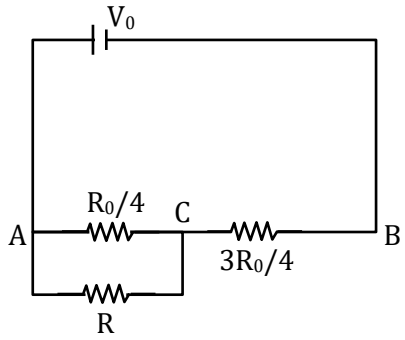
$\Rightarrow x^2 - 2x - 2 = 0$

So, $x = \frac{2\pm\sqrt{12}}{2} = 1 \pm \sqrt{3}\Omega$

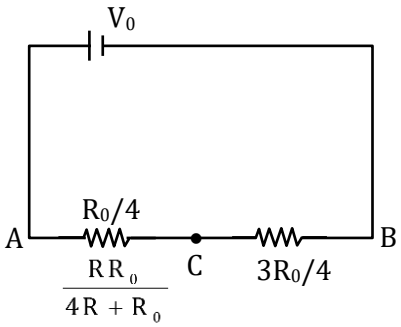
neglecting negative value, $x = 1 + \sqrt{3}\Omega$

36. Conductance = $\frac{1}{\text{Resistance}}$

37.



$\Rightarrow$

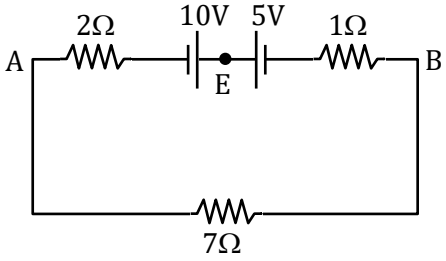


In series, potential divides in direct ratio of resistance,

So, $V_{AC} = \frac{R_{AC}}{R_{AC}+R_{CB}}V_0$

$$= \frac{\frac{RR_0}{4R+R_0}}{\frac{RR_0}{4R+R_0}+\frac{3R_0}{4}}\times V_0 = \frac{4RV_0}{16R+3R_0}$$

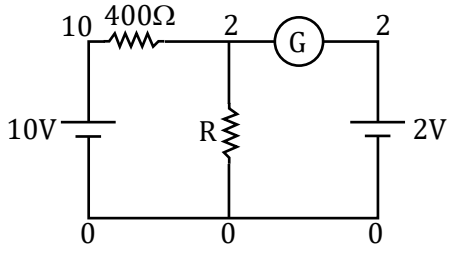
38. $i = \frac{10-5}{10} = \frac{5}{10}A$



= 0.5 A

from A to B through E.

39. For no reading galvanometer. Potential across it is same.



$i_{400\Omega} \Rightarrow \frac{10-2}{400} = \frac{8}{400} = \frac{1}{50} = i_R$

$i_R \Rightarrow \frac{V_R}{R} \Rightarrow \frac{2}{R} = \frac{1}{50} \Rightarrow R = 100\Omega$

40. $R_T = R_0\left[1+\alpha\left(T-T_0\right)\right]$

$6.8 = 2[1+\alpha(80-\alpha)]$

$\alpha = \frac{2.4}{80} = 0.03/°C = 3 \times 10^{-2}/°C$

41. $I_S = \frac{E}{10R}$ ... (1)

$I_P = \frac{E}{R/10} = \frac{10E}{R}$ ... (2)

$n = \frac{I_P}{I_S} = 100 \Rightarrow n = 100$

42. $R = \frac{\rho L}{A}$ If diameter becomes thrice then cross Section area will become 9 times so

$R \propto \frac{1}{A}$ Resistance will become $\frac{1}{9}$ times

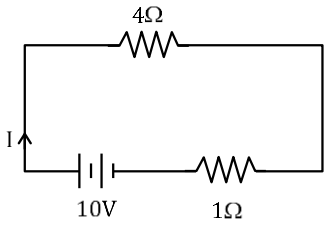
$R' = \frac{81\Omega}{9} = 9\Omega$

43. $I = neAV_d$

$V_d = \frac{I}{neA} = \frac{10}{10^{22} \times 1.6 \times 10^{-19} \times \pi \times 10^{-6}}$

$V_d = \frac{6.25}{\pi} \times 10^3\ ms^{-1}$

44.



$I = \frac{E}{R+r} = \frac{10}{4+1} = 2A$

$V_T = E - Ir$

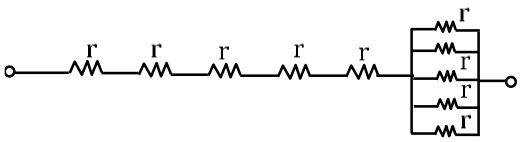
$= 10 - 2\ (1)$

$= 8V$

45. Wire resistance = 100Ω

Divided into 10 equal parts

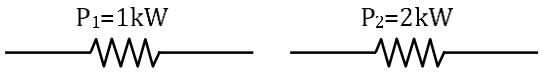
so each part resistance r = $\frac{100}{10} = 10\Omega$



Req. = $5(10) + \frac{10}{5}$

= 52 Ω

46.



In series $P_S = \frac{P_1P_2}{P_1+P} = \frac{1\times2}{1+2} = \frac{2}{3}kW$

In parallel $P_P = P_1 + P_2 = 1+2 = 3kW$

$\Rightarrow \frac{P_S}{P_P} = \frac{\frac{2}{3}}{3} = \frac{2}{9}$

47. Using $i = neAv_d$

$100 \times 10^{-3} = n \times \pi\left(\frac{d}{2}\right)^2 ev$ .....(1)

$200 \times 10^{-3} = n \times \pi\left[\frac{d/2}{2}\right]^2 \times e \times v'$ .....(2)

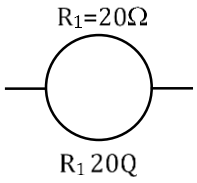
On Solving ; v' = 8v

48. $\ell \rightarrow 10\Omega$

$\because R \propto \ell^2$

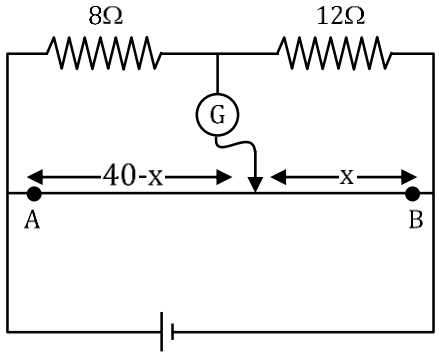
$\therefore$ for 2ℓ length R = 40Ω

Now



$R_{eff} = \frac{R_1}{2} = 10\Omega$

49.



8 (x) = 12(40 – x)

2x = 120 – 3x $\Rightarrow$ 5x = 120 $\Rightarrow$ x = 24 cm

Exercise-III (Analytical Questions)

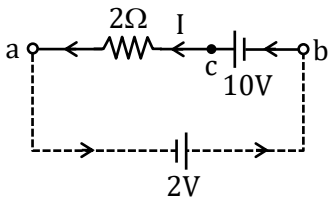
1. $\because V_1$ and V_2 are in series

$\therefore V \propto R$

$\therefore V_1 \neq V_2$ (as $R_1 \neq R_2$)

$V_1 + V_2 = V_3$ (as V is same in parallel)

2.



$I = \frac{12}{2} = 6A$

Apply KVL from a to c

$V_a + 6 \times 2 = V_c$

$V_a - V_c = -12$ Volt

3. $E = 10V$, $I_{max} = \frac{E}{r} = 2$ Amp.

$r = \frac{E}{2} \Rightarrow r = \frac{10}{2} = 5\Omega$

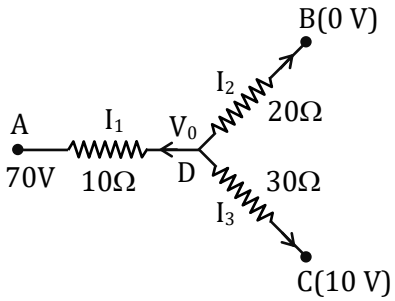
4. $\frac{V^2}{R}t$ = constant

$\therefore t \propto R$

In series $t_s = t_1 + t_2$

In parallel $t_p = \frac{t_1 t_2}{t_1 + t_2}$

5.



$I_1 + I_2 + I_3 = 0$

$\frac{V_0 - 70}{10} + \frac{V_0 - 0}{20} + \frac{V_0 - 10}{30} = 0$

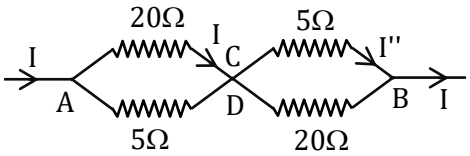
$\Rightarrow V_0 = 40$ volt

$I_1 = 3A$

$I_2 = 2A$

$I_3 = 1A$

6.

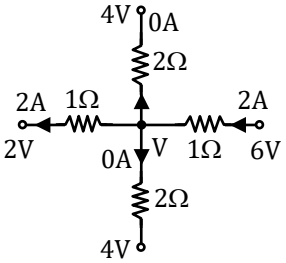


$I' = \frac{5}{25}I = \frac{I}{5}$ $\quad R_{eq} = 4 + 4 = 8\Omega$

$I'' = \frac{20}{25}I = \frac{4I}{5}$ $\quad$ Also = $V_C = V_D$

Current in DC = $\frac{4I}{5} - \frac{I}{5} = 3I/5$

7.



$\frac{V-2}{1} + \frac{V-6}{1} + \frac{V-4}{2} + \frac{V-4}{2} = 0$

$2V - 4 + 2V - 12 + V - 4 + V - 4 = 0$

$6V = 24$

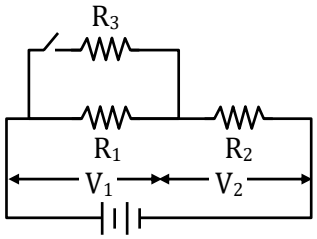
$V = 4$

2A, – 2A, 0A, 0A

8. i = constant

as area $\uparrow \Rightarrow R/\ell \downarrow$ & $E \downarrow \Rightarrow \frac{V}{L} \downarrow$ & $J = \frac{i}{A} \downarrow$

9.



After closing the switch $V_1 \downarrow$ and $V_2 \uparrow$

$\Rightarrow$ i through $R_1 \downarrow$ and i through $R_2 \uparrow$.

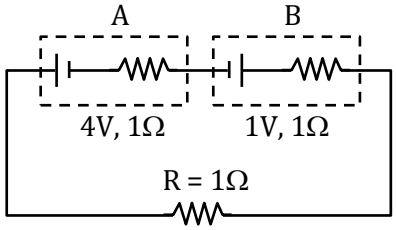
10. $V = \frac{\text{Work}}{q} = \frac{[ML^2T^{-2}]}{[AT]} = [ML^2T^{-3}A^{-1}]$

$R = \frac{V}{i} = [ML^2T^{-3}A^{-2}]$

$\rho = \frac{RA}{\ell} = [ML^3T^{-3}A^{-2}]$

$\sigma = \frac{1}{\rho} = [M^{-1}L^{-3}T^3A^2]$

11.



$i=\frac{4-1}{1+1+1}=1A$

Now TPD of A = $4 - 1 \times 1 = 3V$

TPD of B = $1 + 1 \times 1 = 2V$

Power consumed by R = $1^2 \times 1 = 1W$

P.D. of R = $1 \times 1 = 1V$

12. Increase in temperature causes atoms to vibrate more, which increases the number of collisions electrons suffer and decrease the drift velocity.

OR $\vec{v}_d=\mu E=\left(\frac{e\tau}{m}\right)\vec{E}$ & $\sigma=\frac{ne^2\tau}{m}$

$v_d \propto \tau$, $\sigma \propto \tau$ (temp. $\uparrow$; $\tau\downarrow$)

13. $S(i - i_0) = i_0G$

$\Rightarrow i=i_0\left(1+\frac{G}{S}\right)$

$\therefore$ As $S\uparrow$ $i\downarrow$

$\therefore$ Value of shunt is more in mili ammeter.

$R_{net}=\frac{GS}{S+G}=\frac{G}{1+\frac{G}{S}}$

$\therefore S\uparrow \Rightarrow R_{net}\uparrow$

$\Rightarrow R_{\text{mili ammeter}} > R_{\text{ammeter}}$

14. A current carrying wire is not charged.

15. In balanced condition, current through galvanometer is zero and does not affect main current.

16. For rating, $P=\frac{V^2}{R}$ $\therefore P\uparrow \Rightarrow R\downarrow$

17. * For maximum power R = r

* $P = I^2R \propto I^2$ (if R = constant)

18. At higher currents, joule's heating will be larger, increasing the temperature sharply. Since,

$R = R_0 (1 + \propto \Delta T)$, resistance varies and $V \propto I$ doesn't hold anymore.

19. For meter bridge experiment, unknown resistance $S=\frac{100-\ell}{\ell}R$

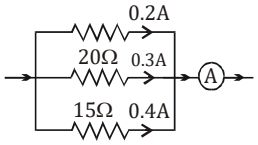
20. $\Delta KE = q(\Delta V)$

21. As i changes (in same resistor); n remain constant

by i = JA ; J become double

by i = $nAeV_d$; V_d become double

23.



$0.2R_1 = 20 \times 0.3 = 15 \times 0.4$

26. For (A) :

$\vec{E}=-\frac{\partial V}{\partial x}\hat{i}-\frac{\partial V}{\partial y}\hat{j}-\frac{\partial V}{\partial z}\hat{k}=\frac{1}{x^2}\hat{i}+\frac{1}{y^2}\hat{j}+\frac{1}{z^2}\hat{k}$

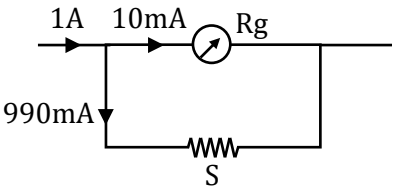
$\Rightarrow |\vec{E}|=|\hat{i}+\hat{j}+\hat{k}|=\sqrt{3}\ N/C$

For (B) :

$I=\frac{dQ}{dt}=4-4t$, current will be zero at t = 1s

Total heat produced

$\int_0^1 I^2Rdt=\int_0^1 (4-4t)^2(6)dt=(16\times6)\left[\frac{(1-t)^2}{-3}\right]_0^1=32J$



For(C) (10 mA) (Rg) = (990 mA) (s)

$\Rightarrow s=\frac{0.99}{99}=0.01\Omega$

For(D) $U=-\vec{p}.\vec{E}$ Here $\vec{p}=\pm qd\hat{i}$

$|\vec{P}| = (10 \times 10^{-3}) (8 \times 10^{-2}) = \pm (8 \times 10^{-4})\ \hat{i}$ C-m

$U = -pE\cos\theta$

$U = (8 \times 10^{-4}) (500)$

$U = 0.4$ J

AIM: FREE EDUCATION TO ALL APUL

Capacitor

SOLUTIONS

Exercise-I (Conceptual Questions)

1. $C \rightarrow$ Depends material & dimensions only

2. $Q = CV$

$C \rightarrow$ does not depend on Q and V

$Q \uparrow \qquad V \uparrow$

3. Charge on capacitor = Charge of positive plate or negative plate. Net charge on capacitor = 0

4. $C = 4\pi\varepsilon_0 R$ and $V = \frac{4}{3}\pi R^3$

$A = 4\pi R^2$

$R = \left(\frac{3V}{A}\right)$

$\therefore \quad C = 12\pi\varepsilon_0\left(\frac{V}{A}\right)$

5. Current

6. Energy Source → Battery, Capacitor

7. $C = 4\pi\varepsilon_0 R\varepsilon_r \quad \left[\because 2\pi R = 2 \Rightarrow R = \frac{1}{\pi}\right]$

$C = \frac{4\pi\varepsilon_0\varepsilon_r}{\pi}$

$C \approx 2800$ pF

8. Energy supplied by the battery is

$U = CV^2 = Q^2/C = QV$

$= (10^{-6})(300)$

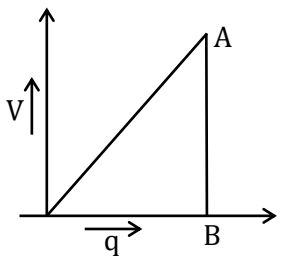
$= 3 \times 10^{-4}$J

9. It does not depend on the resistance

$\frac{1}{2}CV^2 = \frac{1}{2}\times 2\times 10^{-4}\times (200)^2 = 4J$

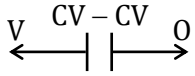
10. $V = \frac{Q}{C}$

11.



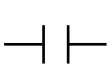
Area $= \frac{1}{2}$ qV = Energy stored in capacitor.

12.



Work $= CV^2 \Rightarrow U = \frac{1}{2}CV^2$

13.



$C = \frac{\in_0 A}{d}$

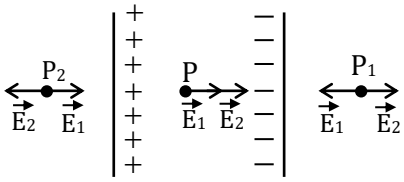
14. $C = \frac{\in_0 A}{d}$

$Q = CV = \frac{\in_0 AV}{d}$

$= \frac{8.85\times 10^{-12}\times 400\times 10^{-4}\times 200}{2\times 10^{-3}}$

$= 3.54 \times 10^{-8}$ C

15.



at P, $E = \frac{q}{2A\varepsilon_0}$

at P_1 & P_2 $\quad E = 0$

16. r = 20 mm for sphere

$A = \pi r^2 \qquad C' = 4\pi\in_0 R$

$C = \frac{\in_0 A}{d} \Rightarrow C = C'$

$\frac{\in_0 A}{d} = 4\pi\in_0 R \Rightarrow d = \frac{\in_0 r^2}{4\in_0 R} = 1 \times 10^{-4}$ m

$= 0.1$ mm

17. Use $U = \frac{1}{2}\in_0 E^2$

18. Q-same

$U' \propto \frac{1}{C} \qquad C' = KC$

$U' = \frac{CU_0}{C'} = \frac{U_0}{K}$

19. Q remains same.

$U = \frac{Q^2}{2C}$

$\therefore \quad C' = KC = 6C$

& $U' = \frac{U}{K} = \frac{U}{6}$

20. $C=\dfrac{\in_0 A}{d-t+\dfrac{t}{K}}$

21. $Q = CV$;
when a slab of dielectric medium is placed capacitance is increased
$Q \propto C$ so charge is also increased

22. Charge does not change. It remains constant.

23. $U=\dfrac{1}{2}CV^2$

$\because$ when a dielectric is inserted then $C\uparrow$ so $U\uparrow$

24. $U=\dfrac{Q^2}{2C}$ $\because$ $C\uparrow$ $\therefore$ $U\downarrow$

25. $U=\dfrac{Q^2}{2C}$

$\because$ $C=\dfrac{\in_0 A}{d}$

If $d\downarrow$ $C\uparrow$ $U\downarrow$

26. $E=\dfrac{V}{d}$

$\because$ $d\uparrow$ $E\downarrow$

$\because$ $Q = CV$ and $C=\dfrac{\in_0 A}{d}$

$\because$ $d\uparrow$ $C\downarrow$ $Q\downarrow$

27. Battery $\rightarrow$ removed
$Q \rightarrow$ constant

$V=\dfrac{Q}{C}$ if $d\uparrow C\downarrow V\uparrow$

28. No

29. $C=\dfrac{\in_0 A}{d}$; $C_1=\dfrac{\in_0 A}{2d}=\dfrac{C}{2}$

$C_2=\dfrac{5\in_0 A}{2d}=\dfrac{5}{2}C$

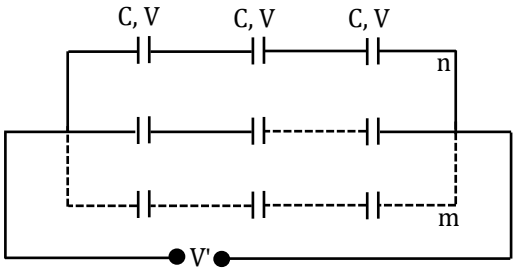
$C_{eq} = C_1 + C_2 = 3C$

$\dfrac{\Delta C}{C}\%=\dfrac{2C}{C}\times 100 = 200\%$

30. $C_{AB}=\dfrac{2}{3}+2=\dfrac{8}{3}\ \mu F$

31. $V_2 = V_3$
$Q_1 = Q_2 + Q_3$
$V = V_1 + V_2$

32. $C = 1\ \mu F$, $C' = 3\ \mu F$
$V = 500$ V, $V' = 2000$ V



Suppose m rows of given capacitors are connected in parallel and each row now contains n capacitors then potential difference across each capacitor $V=\dfrac{V'}{n}$ and equivalent capacitance of network $C'=\dfrac{mC}{n}$

on putting values,

$\Rightarrow \quad V=\dfrac{V'}{n} \quad \Rightarrow \quad 500=\dfrac{2000}{n}$

$n = 4 \quad \Rightarrow \quad C'=\dfrac{mC}{n}$

$3=\dfrac{m\times 1}{4} \quad \Rightarrow \quad m = 12$

$\therefore$ total capacitors = m × n = 48

33. Given circuit is balanced wheat stone bridge.
Hence effective capacity is 6μF.

34. $\dfrac{V_1}{V_2}=\dfrac{C_2}{C_1}=\dfrac{10}{1}$

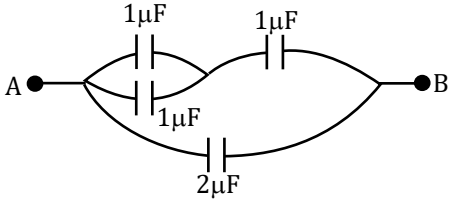
$V_1=\dfrac{500}{11}\times 10=\dfrac{5000}{11}$ volt

35. C and C are in parallel and in series with 2C.
Therefore resultant of these three will be

$$=\frac{(C+C)\times 2C}{C+C+2C}=C$$

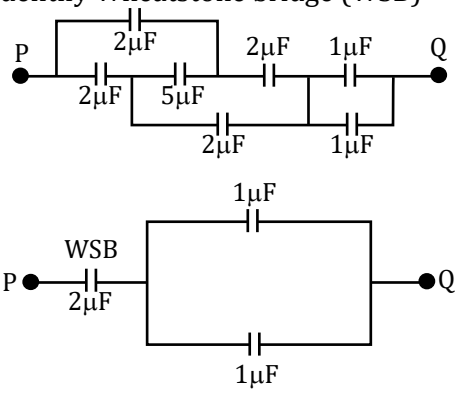
This equivalent system is in parallel with C.
It's equivalent capacitance = C + C = 2C
Now, 2C and 2C are in series and in parallel with 2C.
So, C_{net} = C + 2C = 3C

36.



Redraw the circuit $C_{AB}=\dfrac{8}{3}\ \mu F$

37. Identify Wheatstone bridge (WSB)



$C_{PQ} = 1\mu F$

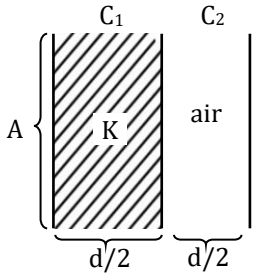
38. Let Initial charge = q
then from charge conservation
$Q_1 + Q_2 = Q'_1 + Q'_2$
$q + q = C_1V + C_2V = V[C_1+C_2]$

$$V = \frac{2q}{C_1+C_2} = \frac{2q}{4\pi\in_0(R_1+R_2)} = \frac{2kq}{(R_1+R_2)}$$

$$\Rightarrow q = \frac{V(R_1+R_2)}{2k}$$

39. As the potential of two spheres is same hence there will be no flow of charge.

40.



$$C = \frac{\in_0 A}{d} \Rightarrow C_1 = \frac{\in_0 KA}{d/2} = \frac{2\in_0 KA}{d}$$

$$C_2 = \frac{\in_0 A}{d/2} = 2\left(\frac{\in_0 A}{d}\right)$$

C_1 & C_2 are in series combination

Hence $\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{d}{2\in_0 A}\left(\frac{1}{K} + \frac{1}{1}\right)$

$$\frac{1}{C_s} = \frac{d}{2\in_0 A}\left(\frac{K+1}{K}\right)$$

$$C_s = \frac{2\in_0 A}{d}\left(\frac{K}{K+1}\right) \Rightarrow C_s = \frac{2CK}{K+1}$$

41. Q same $\rightarrow Q = C_1V_{AD} = C_1(V_1 - V_D)$

and $Q = C_2V_{DB} = C_2(V_D - V_2)$

$C_1(V_1 - V_D) = C_2(V_D - V_2)$

$C_1V_1 + C_2V_2 = (C_1 + C_2)V_D$

$$V_D = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

42. Since both capacitors are in series so charge on each capacitor will be same.
Applying KVL from A to B :-

$$V_A - V_B = \frac{Q}{C_1} - E + \frac{Q}{C_2}$$

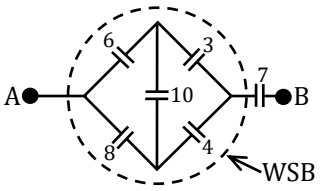
or $$V_A - V_B + E = Q\left[\frac{1}{C_1} + \frac{1}{C_2}\right]$$

$$5 + 10 = Q\left[\frac{3}{2}\times 10^6\right]$$

$$Q = \frac{30}{3} = 10\ \mu C$$

$$V_1 = \frac{Q}{C_1} = \frac{10\times 10^{-6}}{1\times 10^{-6}} = 10V$$

43. Redraw circuit



C_{net} of W.S.B.

$$= \frac{6\times 3}{6+3} + \frac{8\times 4}{8+4}$$

$$= 2 + \frac{32}{12} = \frac{14}{3}$$

This is in series with 7

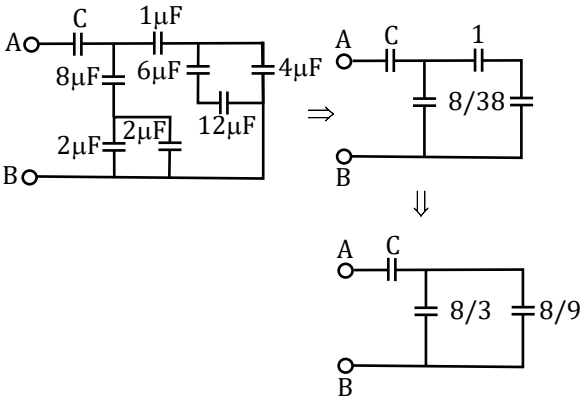
So, $$C_{eq.} = \frac{\frac{14}{3}\times 7}{\frac{14}{3}+7} = \frac{14}{5}$$

44. C_{net} of C_1 and C_2, $C_{net} = \frac{C_1C_2}{C_1+C_2}$ $\{C_1 < C_2\}$

So, C_{net} will be smaller than C_1

$C_{net} < C_1 < C_2$ so $\frac{C_1}{2} < C < C_1$

45.



$$C_{AB} = \frac{C\times\frac{32}{9}}{C+\frac{32}{9}} = 1 \Rightarrow C = \frac{32}{23}\mu F$$

46. When capacitor is fully charged. There will be no flow of current through capacitor.

So, $I = \frac{2.5}{2+0.5} = 1\ A$

P.D. across upper branch = $2\Omega \times 1A = 2V$

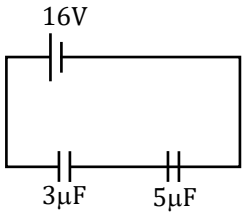
and same voltage on lower branch

$V_C + V_{2\Omega} = 2V$

{but voltage on $2\Omega = 0$ as no current}

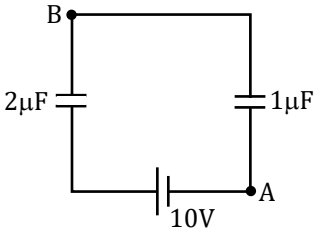
$V_C = 2V$, So $Q = 2V \times 5\mu F = 10\ \mu C$

47.



$V_3 = \frac{5}{3+5} \times 16 = 10V$

48. Given circuit can be reduced as



Charge on capacitors $= \left(\frac{2}{3}\right)(10)\mu C$

Now $V_B - V_A = \left(\frac{20}{3}\right)(1) = \frac{20}{3}$

$\Rightarrow V_B = V_A + \frac{20}{3} = 10 + \frac{20}{3} = \frac{50}{3} V$

49. C_2 & C_3 are in parallel so $V_2 = V_3$

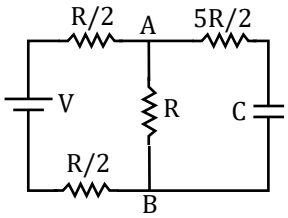
Let $C_1 = C$

$C_2 = C_3 = \frac{C_1}{2} = \frac{C}{2}$

Equivalent of C_2 & $C_3 = \frac{C}{2} + \frac{C}{2} = C$

So $V_1 = V_2 = V_3$

50. In steady state $V_C = V_{AB}$ = capacitor voltage = V/2



51. At $t \to \infty$ $V_{AB} = \frac{V}{2}, i_{AB} = \frac{V}{2R}$

52. Bulb light up during charging

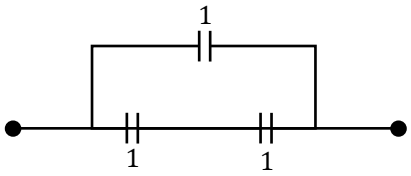
53. $C = \frac{\epsilon_0 A}{d} = 10\ \mu F$

$C_1 = \frac{k_1 \epsilon_0 A}{2d} = 10\ \mu F$

$C_2 = \frac{k_2 \epsilon_0 A}{2d} = 20\ \mu F$

$C_{eq} = C_1 + C_2 = 30\ \mu F$

54.



$C_{net} = 1.5\ \mu F$

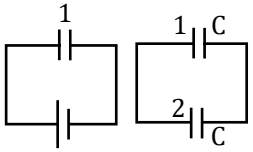
55. Whenever there is transfer of charge some energy dissipates in the form of heat hence potential energy of system decreases.

If $V_1 \neq V_2$ then charge transfer takes place

$\Rightarrow \frac{kQ_1}{R_1} \neq \frac{kQ_2}{R_2}$

$Q_1R_2 \neq Q_2R_1$

56.



energy stored $(U) = \frac{q^2}{2C}$

After connecting with another capacitor

$V_{Common} = \frac{q_1 + q_2}{C_1 + C_2} = \frac{q + 0}{C + C} = \frac{q}{2C}$

$\because$ Energy on each capacitor

$= \frac{1}{2} C V_{Common}^2 = \frac{1}{2} C \left(\frac{q}{2C}\right)^2 = \frac{1}{2} \frac{q^2}{4C} = \frac{U}{4}$

57. $\frac{1}{C_{eq}} = \frac{1}{3} + \frac{1}{10} + \frac{1}{15}$

$C_{eq} = 2\mu F$

$Q_{eq} = (2 \times 100)\ \mu C = 200 \mu C$

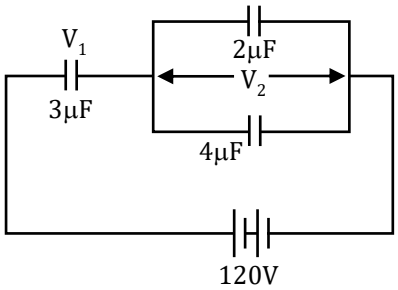
58. $V_A = \frac{C_B}{C_A + C_B} V_T = 6V \Rightarrow V_B = (10 - 6)V = 4V$

$\Rightarrow \frac{V_A}{V_B} = \frac{3}{2}$

59. $U_s = \frac{1}{2}kx^2 = 500\,J \qquad \left[k = \frac{F}{x}\right]$

$U_C = \frac{1}{2}CV^2 = 500\ J \Rightarrow \frac{U_s}{U_c} = \frac{1}{1}$

60.



$C_{eq} = \frac{6\times3}{6+3} = 2\mu F$

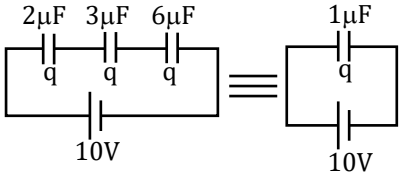
$\therefore\ Q = 120 \times 2\ \mu C = 240\mu C$

$V_1 = \frac{240}{3} = 80\ V$

$V_2 = 40\ V \Rightarrow Q_2 = 40 \times 2 = 80\ C$

$Q_3 = 40 \times 4 = 160\ C$

61.



$\frac{1}{C} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{6}{6} \Rightarrow C = 1\ \mu C$

$\Rightarrow q = 10^{-6} \times 10 = 10\ \mu C$

62. After connecting V become same

$\frac{E_a}{E_b} = \frac{b}{a}$

63. $C = \frac{4\pi\in_0 R_1R_2}{R_2 - R_1}$

64. $C = \frac{4\pi\in_0 b^2}{b-a}$

65. $\tau = RC$

66. Energy density $= \frac{1}{2}\in_0 \frac{V^2}{d^2}$

67. $V_{common} = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$ { $\because\ V_2 = 0$)

$\Rightarrow \quad V_{common} = \frac{C_1V}{C_1 + C_2}$

68. Potential same at both spheres $V_1 = V_2$

$\therefore\ \frac{kQ_1}{R_1} = \frac{kQ_2}{R_2} \Rightarrow \frac{Q_1}{Q_2} = \frac{R_1}{R_2}$

surface charge density

$\therefore\ \frac{\sigma_1}{\sigma_2} = \frac{Q_1}{4\pi R_1^2} \times \frac{4\pi R_2^2}{Q_2}$

$= \frac{Q_1}{Q_2} \times \left(\frac{R_2}{R_1}\right)^2 = \left(\frac{R_1}{R_2}\right)\left(\frac{R_2}{R_1}\right)^2$

$= \frac{R_2}{R_1} = \frac{20}{10} = \frac{2}{1}$

69. $C_{eq} = 4\mu F$ (Wheat stone bridge)

70. By charge conservation

$V_{com} = \frac{6\times12 - 3\times12}{9} = 4\ Volt$

71. $V_{com} = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$

$= \frac{0.2\times600 + 1\times0}{0.2+1}$

$= 100\ volt$

72. $U = \frac{q^2}{2C} = \frac{q^2d}{2\in_0 A}$

Volume = Ad

Energy Density $= \frac{U}{Volume} = \frac{q^2}{2\in_0 A^2}$

73. $W = \frac{1}{2}CV_2^2 - \frac{1}{2}CV_1^2 = \frac{1}{2}C(900-225)$

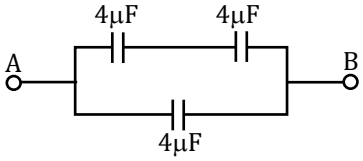
$C = \frac{2}{675}W$

Now in 2nd case

$\frac{1}{2}C\left[(60)^2 - (30)^2\right] = \frac{1}{2}C\left[3600 - 900\right]$

$= \frac{1}{2}.\frac{2}{675}W\left[2700\right] = 4W$

74.



$C_{AB} = (2 + 4)\ \mu F = 6\mu F$

75. $\frac{Q_1}{Q_2} = \frac{C_1V}{C_2V} = \frac{C_1V}{KC_1V}$ ($\because$ V = const.)

$\Rightarrow K = \frac{Q_2}{Q_1} = \frac{100}{40} = 2.5$

76. $C_{net} = \frac{C}{2}, V_{net} = 2V$

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77. $C_{eq} = 2\mu F$
$Q_{eq} = 4\mu C$
Q on $1\mu F = 2\mu C$
Q on $2\mu F$ each $= 2\mu C$

78. E remain same because V = constant

79. $P = \frac{W}{t} = \frac{\frac{1}{2}CV^2}{t} = \frac{\frac{1}{2}\times 40\times 10^{-6}\times (3000)^2}{2\times 10^{-3}}$
$= 9 \times 10^4 = 90$ kW

80. $C_S = \frac{C_1C_2}{C_1+C_2}$
$C_P = C_1 + C_2$

81. $V \propto \frac{Q}{R}$ So $Q \propto R$ at constant V
So $\frac{q_1}{q_2} = \frac{R_1}{R_2}$

82. $V_{com.} = \frac{Q}{C_1+C_2} = \frac{Q}{4\pi\in_0 (R_1+R_2)}$
$= \frac{150\times 10^{-6}\times 9\times 10^9}{(10+20)\times 10^{-2}} = 4.5\times 10^6 V$

83. $\frac{1}{2}C_1V_1^2 = \frac{1}{2}C_2V_2^2$ [Assume no energy lost]
because total energy is transferred (given).
$\therefore \frac{1}{2} \times 900 \times 10^{-6} \times 100^2 = \frac{1}{2} \times 100 \times 10^{-6} \times V^2$
$\therefore V^2 = 90000 \Rightarrow V = 300$ Volt

Exercise-II (Previous Year Questions)

1. Electric field, $E \propto \frac{1}{K}$ As $K_1 < K_2$ so $E_1 > E_2$

2. Once the capacitor is charged, its charge will be constant Q = CV. When dielectric slab is inserted
$C'_{New} = KC$
$E = \frac{Q^2}{2C} \Rightarrow E_{New} = \frac{1}{K} E_{initial}$
$V = \frac{Q}{C}$ so $V_{new} = \frac{1}{K} V$

3. $F = \frac{Q^2}{2\varepsilon_0 A}$
$\because$ $Q = CV$ and $C = \frac{\varepsilon_0 A}{d} \Rightarrow \varepsilon_0 A = Cd$
So $F = \frac{C^2V^2}{2Cd} = \frac{CV^2}{2d}$

4. Initial energy stored in capacitor $2\ \mu F$
$U_i = \frac{1}{2}2(V)^2 = V^2$
Final voltage after switch 2 is ON
$V_f = \frac{C_1V_1}{C_1+C_2} = \frac{2V}{10} = 0.2\ V$
Final energy in both the capacitors
$U_f = \frac{1}{2}(C_1+C_2)V_f^2 = \frac{1}{2}10\left(\frac{2V}{10}\right)^2 = 0.2\ V^2$
So energy dissipated $= \frac{V^2 - 0.2V^2}{V^2}\times 100 = 80\%$

5. $U_i = \frac{1}{2}CV^2$
$U_f = \frac{1}{2}[2C]\left[\frac{V}{2}\right]^2 = \frac{1}{2}U_i$
Decrease by a factor of 2

6. $F = QE = Q\frac{\sigma}{2\in_0} = \frac{Q^2}{2A\in_0}$
$\therefore$ Electrostatic force is independent of distance between plates

7. $U_{initial} = \frac{1}{2}CV^2$,
Loss $= \frac{C.C}{2(C+C)}(V-0)^2 = \frac{1}{4}CV^2$
% Loss $= \frac{\frac{1}{4}CV^2}{\frac{1}{2}CV^2}\times 100 = 50\%$

8.

(A)	(B)
R	2R
σ	σ

Total charge $= \sigma \times 4\pi R^2 + \sigma \times 4\pi (2R)^2$
$= 20\ \sigma\pi R^2$
$\frac{Q_A}{Q_B} = \frac{1}{2}$
$Q_A = \frac{20}{3}\sigma\pi R^2$ and $Q_B = \frac{40}{3}\sigma\pi R^2$
$\sigma_A = \frac{Q_A}{\text{area}} = \frac{20}{3}\frac{\sigma\pi R^2}{4\pi R^2} = \frac{5\sigma}{3}$
$\sigma_B = \frac{40\sigma\pi R^2}{4\pi(2R)^2} = \frac{5\sigma}{6}$

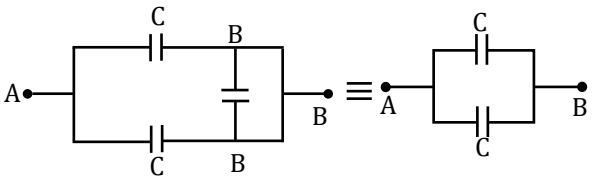
9. $C_m = \in_r C_o \Rightarrow \in_r = \frac{30}{6} = 5$

$\in = \in_0 \in_r = 8.85 \times 10^{-12} \times 5 = 0.44 \times 10^{-10}$

10. $C_O = \frac{\in_0 A}{d}$

$C_K = \frac{\in_0 A}{d - t + \frac{t}{k}} = \frac{\in_0 A}{d - \frac{d}{2} + \frac{d}{8}} = \frac{8}{5}\frac{\in_0 A}{d} = \frac{8}{5}C_O$

11.



(one capacitor gets shorted)

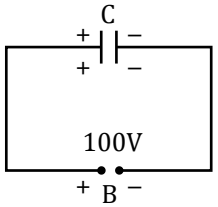
$\Rightarrow C_{eq} = C_1 + C_2$

$= C + C = 2C$

12. $E = \frac{1}{2}CV^2 = \frac{1}{2}\left(\frac{\varepsilon_0 A}{d}\right)(Ed)^2 = \frac{1}{2}\varepsilon_0 E^2 Ad$

13. Polar molecules have centres of positive and negative charges separated by some distance, so they have permanent dipole moment.

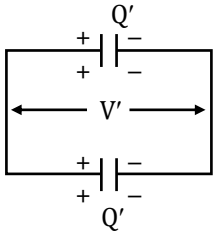
14.



$Q_i = CV$

$= 900 \times 10^{-12} \times 100$

$Q_i = 9 \times 10^{-8}\,C$



$Q_f = 2Q'$

From conservation of charge

$2Q = Q_f = Q_i$

$Q' = \frac{Q_i}{2} = \frac{9\times10^{-8}}{2}$

$Q' = 4.5\times10^{-8}C$

$E_{store} = 2\times\frac{1}{2}\frac{(Q')^2}{C}$

$= \frac{\left(4.5\times10^{-8}\right)^2}{900\times10^{-12}}$

$E_{store} = 2.25 \times 10^{-6}\,J$

15. $C = \frac{A\varepsilon_0}{d}$

$C' = \frac{A/2\,\varepsilon_0}{2d} = \frac{A\varepsilon_0}{4d} = \frac{C}{4}$

16. $\frac{C_1C_2}{C_1 + C_2} = 3$

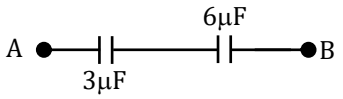
$C_1 + C_2 = 16$

$C_1\,C_2 = 48$

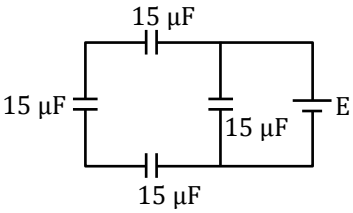
$C_1 = 12\mu F$

$C_2 = 4\mu F$

17. $C_{AB} = \frac{3\times6}{3+6} = 2\mu F$

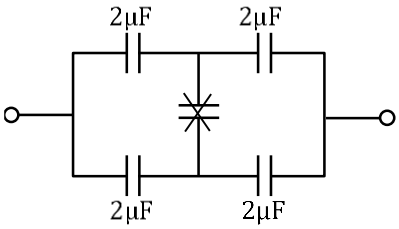


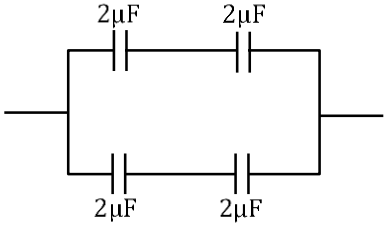
18.



$C_{eq} = 5 + 15 = 20\mu F$

19. Balanced wheat stone bridge





$C_{eq} = 2\mu F$

20. Battery connected so V = constant

Now $d\downarrow \Rightarrow C\uparrow$

A : $Q = CV \propto C \Rightarrow Q\uparrow$

B : $U = \frac{1}{2}CV^2 \propto C \Rightarrow U\uparrow$

C : $C = \frac{\epsilon_0 A}{d} \Rightarrow C\uparrow$

D : $\frac{Q}{V} = C \Rightarrow C\uparrow$

E : $(Q)(V) \propto C \Rightarrow QV\uparrow$

Therefore statements A, C and E are correct.

21. $U_{capacitor} = \frac{1}{2}CV^2$

$= \frac{1}{2} \times 12 \times 10^{-12} \times 50 \times 50$

$= 15 \times 10^{-9}$J

= 15 nJ

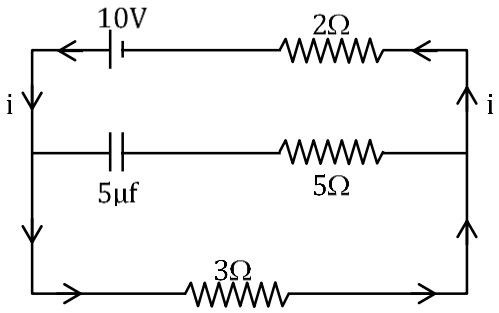
22. Capacitance depends only on

(1) Shape

(2) Size

(3) Medium

23.



There will be no current in middle branch containing capacitor.

$i = \frac{V}{R} = \frac{10}{2+3} = 2A$

Exercise-III (Analytical Questions)

1. ∵ battery is connected

∴ V = constant

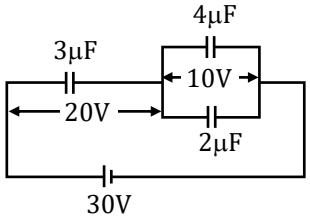
$U = \frac{1}{2}CV^2$, Q = CV

$U \propto C$, $Q \propto C$

4. $U = \frac{1}{2}\epsilon_0 E^2$ (Ad)

$U \propto E^2$

5.



$\because V \propto \frac{1}{C}$

$Q_{Battery} = C_{eff} V = 2 \times 20\ \mu C$

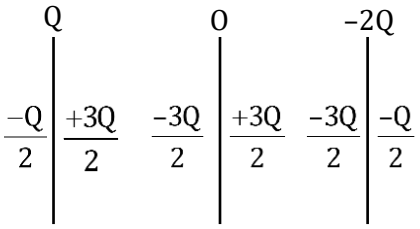
$Q_{battery} = 60\ \mu C$

$Q_{3\mu F} = 60\mu C$

$Q_{4\mu F} = 40\mu C$

$Q_{2\mu F} = 20\mu C$

6.



7. $U_{self} = \frac{KQ^2}{2R}$ Energy density $= \frac{1}{2}\varepsilon_0 E^2$

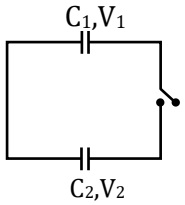
$F = \frac{Q^2}{2A\varepsilon_0}$ $W_{field} = U_i - U_f = \frac{kQ^2}{R} - 0$

$W_{field} = \frac{kQ^2}{R}$

8. Force between two point charges is independent of the presence of other charges.

9. $C = 4\pi\varepsilon_0 R \Rightarrow 1 = \frac{1}{9\times10^9} R \Rightarrow R = 9 \times 10^9$ m

10.



$V = \frac{C_1V_1 + C_2V_2}{C_2 + C_2}$

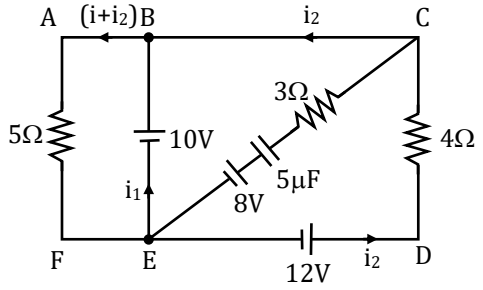
When two capacitors are connected then energy of system decreases. (but Not always)

11. A high capacity capacitor charged to high voltage has very high charges, that may discharge to ground through human body, when touched with bare hands.

12. $U=\frac{Q^2}{2C}$ and $C=\frac{\varepsilon_0 A}{d}$ $d\uparrow C\downarrow U\uparrow$

13. At facing surface same magnitude and opposite nature charges are there.

14.



When a steady state is reached, no current passes through the capacitor or the branch CE.

Considering the loop ABEFA,

$5 \times (i_1 + i_2) = 10$ or $i_1 + i_2 = 2A$ ...(i)

Considering the loop BCDEB

$4i_2 = 12 - 10 = 2 \Rightarrow i_2 = 0.5$ A

So $i_1 = 2 - 0.5 = 1.5$ A

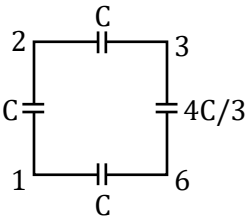
To find the charge on capacitors, we must know potential difference across the plates.

Consider the loop CEDC :

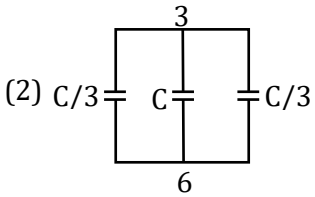
$-12 + 4i_2 + 3 \times 0 - V_C + 8 = 0 \Rightarrow V_C = -2V$.

So, charge on capacitor $Q = CV = 10\ \mu C$

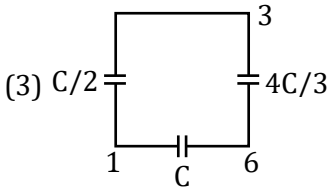
16. (1)



(2)



(3)

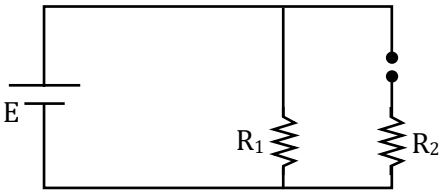


18. In steady state current through battery

$I=\frac{50}{23+3+17+7}=1A$

$\Rightarrow Q_{10\mu F} = CV = 10 \times 20 = 200\mu C$

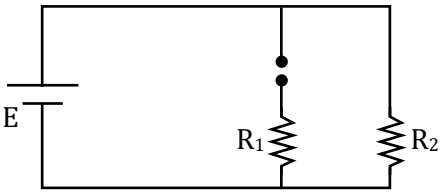
19.



At t = 0

$I_1 = \frac{E}{R_1} = \frac{25}{5} = 5A$

$I_2 = 0$



At $t = \infty$

$I_1 = 0$

$I_2 = \frac{E}{R_2} = \frac{25}{10} = 2.5$ A

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Magnetic Effect of Current and Magnetism

SOLUTIONS

Exercise-I (Conceptual Questions)

2. $B_{net} = B_1 + B_2 + B_3 + B_4$

$= 4B \quad (B_1 = B_2 = B_3 = B_4 = B)$

4. $B = \frac{\mu_0 I}{2\pi r} = \frac{\mu_0 10}{2\pi(1)}$

$= 2 \times 10^{-7} \times 10 = 2 \times 10^{-6} T = 2\mu T$

direction of $\vec{B} \Rightarrow i\vec{d\ell} \times \vec{r} = \hat{k} \times \hat{i} = +\hat{j}$

5. $B_x = \frac{B_0}{\left(1+\frac{x^2}{R^2}\right)^{3/2}}$

$x = \sqrt{3}R$

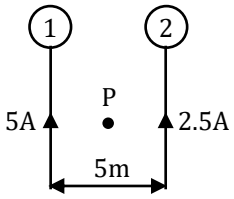
$\frac{B_x}{B_0} = \frac{1}{8} \Rightarrow \frac{B_0}{B_x} = \frac{8}{1}$

6. $B \propto N^2$ (for ℓ = constant)

9. $B = \frac{\mu_0 I}{2r} \Rightarrow I = \frac{2Br}{\mu_0}$

$I \propto Br \Rightarrow \frac{I_1}{I_2} = \frac{B_1 r_1}{B_2 r_2} = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$

10.



$B = \underset{\otimes}{B_1} - \underset{\odot}{B_2}$

$B = \frac{\mu_0 I_1}{2\pi d} - \frac{\mu_0 I_2}{2\pi d}$

$= \frac{\mu_0}{2\pi d}(I_1 - I_2)$

$= \frac{\mu_0}{2\pi(5/2)}(5 - 2.5) = \mu_0/2\pi \otimes$

13. $B = \frac{\mu_0 Ni}{2R}$

14. $B = \frac{\mu_0 i}{2\pi d} \Rightarrow d = \frac{\mu_0 i}{2\pi B}$

15. $B_x = \frac{B_0}{\left(1+\frac{x^2}{R^2}\right)^{3/2}}$

$\because x = R$

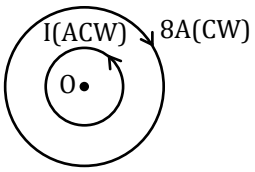
$\therefore B_x = \frac{B_0}{(1+1)^{3/2}} = \frac{B_0}{(2)^{3/2}} \Rightarrow \frac{B_x}{B_0} = \left(\frac{1}{2}\right)^{3/2}$

18. $B_x = \frac{\mu_0 NIR^2}{2(r^2+R^2)^{3/2}}$

If r >> R then $\frac{R^2}{r^2}$ <<< 1 (negligible)

$B_x \simeq \frac{\mu_0 NIR^2}{2(r^2)^{3/2}} \simeq \frac{\mu_0 NIR^2}{2r^3} \Rightarrow B_x \propto \frac{1}{r^3}$

19. $(B_0)_{net} = 0$

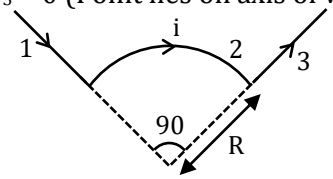


$\frac{\mu_0 I_1}{2r_1} = \frac{\mu_0 I_2}{2r_2}$ (I_1: $\otimes$, I_2: $\odot$)

$\frac{8}{0.1} = \frac{I}{0.08}$

$I = \frac{8 \times 0.08}{0.1} = 6.4A, ACW$

24. $B_1 = B_3 = 0$ (Point lies on axis of wire)



$B_2 = \frac{\mu_0 i}{2R} \times \left(\frac{90}{360}\right)$

$B_2 = \frac{\mu_0 i}{8R} \otimes$

27. Apply $B = \mu_0 ni$

$n = \frac{N}{\ell}$

According to the question,

$n' = \frac{N/2}{\ell/2} = \frac{N}{\ell}$

$\therefore$ B remains same

28. $B = \mu_0 \frac{N}{\ell} I$

$B' = \frac{\mu_0 (2N)(2I)}{\ell} = 4B$

29. $B = \mu_0 \frac{N}{2\pi R_m}.I$

$= 4\pi \times 10^{-7} \times 1000 \times \frac{1}{4\pi} = 10^{-4}$ (Wb/m²)

30. $B = \mu_0 \frac{N}{2\pi R_m}.I$ (given R_m = 10 cm)

31. $B = \mu_0 \left(\frac{nN}{L}\right) i$ nN = total turns

$B \propto D°$

32. $B = \mu_0 ni$

$\frac{B_2}{B_1} = \frac{n_2 i_2}{n_1 i_1}$; $n_2 = \frac{n_1}{2}$, $i_2 = 2i_1$

$= \frac{(n_1/2)(2i_i)}{n_1 i_1} = 1$

$B_2 = B_1 = B$

33. $B = \mu_0 ni$

$\frac{B_2}{B_1} = \frac{n_2\ i_2}{n_1\ i_1} = \frac{100 \times (i/3)}{200 \times i}$

$B_2 = \frac{1}{6} \times 6{\cdot}28 \times 10^{-2} = 1{\cdot}05 \times 10^{-2}$ Wb/m²

34. $r = \frac{P}{qB} \Rightarrow P = qBr$

$P \propto q$ (B, r → same)

$\frac{p_P}{p_\alpha} = \frac{q_P}{q_\alpha} = \frac{1}{2} = \frac{e}{2e} = \frac{1}{2}$

36. $f = \frac{qB}{2\pi m} \Rightarrow f \propto \frac{q}{m}$

37. $F = qvB \quad \Rightarrow F \propto q \quad$ (v, B → same)

38. $r = \frac{mv}{qB}$

$m_\beta = m_e \qquad m_\alpha = 4m_p$

$q_\beta = q_e \qquad q_\alpha = 2q_e$

$r_\beta < r_\alpha$, $\because C \propto \frac{1}{r}$ $(\therefore C_\beta > C_\alpha)$

39. $r = \frac{\sqrt{2mqV_{acc.}}}{qB}$

According to the question

$r \propto \sqrt{m}$

$\frac{r_1}{r_2} = \sqrt{\frac{m_1}{m_2}}$

$\frac{r_1^2}{r_2^2} = \frac{m_1}{m_2}$

42. $r = \frac{mv\sin\theta}{qB} = \frac{v\sin\theta}{(q/m)B}$

$= \frac{1}{10^8} \times \frac{3 \times 10^5}{0.3} \times \frac{1}{2} = 5 \times 10^{-3}$ m = 0.5 cm

43. Kinetic energy of charge particle is always constant, if it enters perpendicular in uniform magnetic field.

44. $r = \frac{\sqrt{2mE_k}}{qB}$

$r \propto \sqrt{m}$ (q, B, E_k → same)

or $m \propto r^2$

$\frac{m_1}{m_2} = \left(\frac{r_1}{r_2}\right)^2$

45. $f = \frac{qB}{2\pi m}$

46. $r = \frac{mv}{qB}$ (v, B →same)

$r \propto \frac{m}{q}$

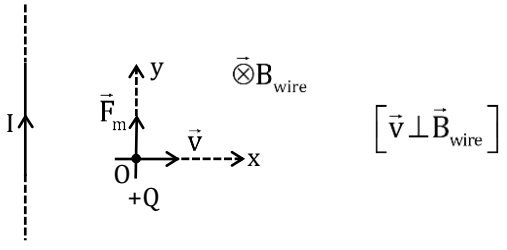
$\frac{r_A}{r_B} = \frac{m_A}{m_B} \times \frac{q_B}{q_A} = \left(\frac{24\text{a.m.u.}}{22\text{a.m.u.}}\right)\left(\frac{2e}{e}\right) = \frac{24}{11}$

50. $r_p : r_d : r_\alpha$

$r \propto \sqrt{\frac{m}{q}}$

$\sqrt{\frac{1}{1}} : \sqrt{\frac{2}{1}} : \sqrt{\frac{4}{2}} = 1 : \sqrt{2} : \sqrt{2}$

54. Use right hand palm rule to find direction of magnetic force.



$[\vec{v} \perp \vec{B}_{wire}]$

OR

$\vec{F}_m \to (\vec{v} \times \vec{B})$

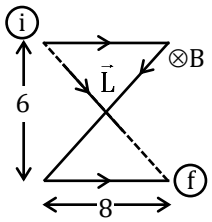
$\vec{F}_m \to (\hat{i} \times -\hat{k}) = +\hat{j} = y - axis$

(current carrying wire produces cross field therefore $-\hat{k}$ is taken in vector product)

$\vec{F}_m \to +\hat{j} = +y - axis$

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60. $F_m = BiL_{eff}$ $\left(\vec{L} \perp \vec{B}\right)$

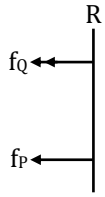


$L_{eff} = \sqrt{6^2 + 8^2} = 10$

$F_m = 10 \times 3 \times 10 \times 10^{-2}$

$= 3N$

62.



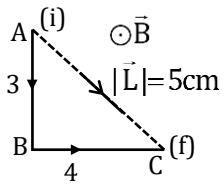
$f_{net} \Rightarrow$ (towards left)

like currents always attracts to each other

63. $f = \frac{\mu_0 I_1 I_2}{2\pi d} = \frac{4\pi \times 10^{-7} \times 10 \times 10}{2\pi \times 10 \times 10^{-2}} = 2 \times 10^{-4} N$

(Like currents $\rightarrow$ Attraction)

64. $\vec{F}_m = I\left(\vec{L} \times \vec{B}\right)$



$|\vec{F}_m| = BIL$, where $\vec{L} \perp \vec{B}$

$= 5 \times 10 \times \frac{5}{100} = 2.5N$

65. When direct current flows in spring, any two adjacent turns having the current in same direction so there is attractive magnetic force between them hence spring contracts.

69. $B = \frac{\mu_0 I}{2R}$ ...(1)

$M = I(\pi R^2)$ ...(2)

divide eqn. (2) by(1)

$\frac{M}{B} = \frac{I(\pi R^2)2R}{\mu_0 I}$

$M = \frac{2\pi B R^3}{\mu_0}$

71. $I = ef$

73. $B_0 = \frac{\mu_0 e\omega}{4\pi r} \Rightarrow \omega = \frac{4\pi r B_0}{\mu_0 e}$

77. $B = \frac{\mu_0 ef}{2r}$

78. $\vec{M} \| \vec{B} \Rightarrow \tau = 0$

81. $W_{90^\circ} = MB(1 - \cos 90^\circ)$

$= MB(1 - 0) = MB$

$W_{60^\circ} = MB(1 - \cos 60^\circ)$

$= MB\left(1 - \frac{1}{2}\right) = \frac{MB}{2}$

$W_{90^\circ} = n\, W_{60^\circ}$

$MB = n\left(\frac{MB}{2}\right) \quad \therefore n = 2$

112. Curie law $\chi_m \propto \frac{1}{T}$

113. A diamagnetic material in a magnetic field moves from stronger to the weaker part of the field.

Exercise-II (Previous Year Questions)

1. Net magnetic field, $B = \sqrt{B_1^2 + B_2^2}$

$= \sqrt{\left(\frac{\mu_0 I_1}{2\pi d}\right)^2 + \left(\frac{\mu_0 I_2}{2\pi d}\right)^2}$

$= \frac{\mu_0}{2\pi d}\sqrt{I_1^2 + I_2^2}$

2. Magnetic field due to a circular loop

$B = \frac{\mu_0 NI}{2r}$ Where N $\rightarrow$ no. of loops

$I = \frac{q}{T} = \frac{e}{1/n} = ne$

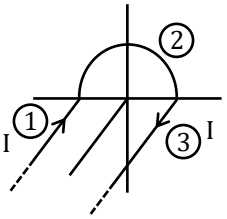
$B = \frac{\mu_0 ne}{2r}$

(Here N =1 as electron makes only one loop)

3. 'B' due to segment '1'

$\vec{B}_1 = \frac{\mu_0 I}{4\pi R}[\sin 90^\circ + \sin\theta](-\hat{k})$

$B_1 = \frac{\mu_0 I}{4\pi R}(-\hat{k}) = B_3$



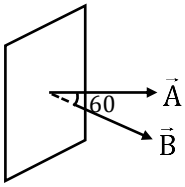
B due to segment '2'

$B_2 = \frac{\mu_0 I}{4R}$

so 'B' at center $\vec{B}_c = \vec{B}_1 + \vec{B}_2 + \vec{B}_3$

$\Rightarrow \vec{B}_C = \frac{-\mu_0 I}{4R}\left(\hat{i} + \frac{2\hat{k}}{\pi}\right) = \frac{-\mu_0 I}{4\pi R}(\pi\hat{i} + 2\hat{k})$

4.



$\vec{\tau} = \vec{M} \times \vec{B}$

$|\vec{\tau}| = MB\sin\theta = NIAB\sin\theta = 0.20\,Nm$

5. $R = \frac{mv}{q_B} = \frac{\sqrt{2mK}}{q_B}$

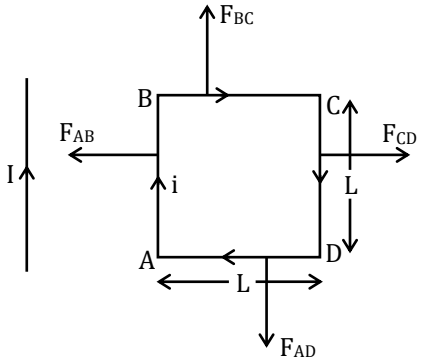
$\because \quad R_\alpha = R_p$

$\therefore \quad \frac{4m_\alpha k_a}{q_\alpha^2 B^2} = \frac{4m_p K_p}{q_p^2 B^2}$

$\Rightarrow \quad \frac{4m_p k_\alpha}{4e^2} = \frac{m_p(1MeV)}{e^2}$

$\Rightarrow \quad K_\alpha = 1MeV$

6.



$F_{AB} = i\ell B$ (Attractive)

$F_{AB} = i(L).\frac{\mu_0 I}{2\pi\left(\frac{L}{2}\right)}(\leftarrow) = \frac{\mu_0 iI}{\pi}(\leftarrow)$

F_{BC} ($\uparrow$) and F_{AD} ($\downarrow$) $\Rightarrow$cancels each other

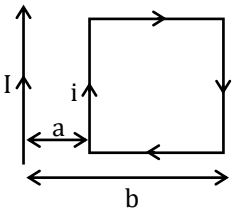
$F_{CD} = i\ell B$ (Repulsive)

$F_{CD} = i(L)\frac{\mu_0 I}{2\pi\left(\frac{3L}{2}\right)}(\rightarrow) = \frac{\mu_0 iI}{3\pi}(\rightarrow)$

$\Rightarrow F_{net} = \frac{\mu_0 iI}{\pi} - \frac{\mu_0 iI}{3\pi} = \frac{2\mu_0 iI}{3\pi}$

OR

Trick



$F_{net} = \frac{\mu_0 IiL}{2\pi}\left[\frac{1}{a} - \frac{1}{b}\right]$ (Attractive)

7. Magnetic susceptibility = χ

It is negative for diamagnetic materials only

8. For points inside the wire

$B = \frac{\mu_0 Ir}{2\pi R^2} \quad (r \le R)$

For points outside the wire

$B = \frac{\mu_0 I}{2\pi r} \quad (r \ge R)$

according to the question

$\frac{B}{B'} = \frac{\frac{\mu_0 I(a/2)}{2\pi a^2}}{\frac{\mu_0 I}{2\pi(2a)}} = 1:1$

9. Since $\ell = 2\pi R = n(2\pi r) \Rightarrow r = \frac{R}{n}$

For one turn $B = \frac{\mu_0 i}{2R}$ and

For n turn $B' = \frac{\mu_0 ni}{2r} \Rightarrow B' = \frac{\mu_0 n^2 i}{2R} = n^2 B$

10. $\tau = MB\sin 60°$ .....(1)

$W = MB(1 - \cos 60°)$ .....(2)

From (1) and (2)

$\frac{\tau}{W} = \frac{\sqrt{3}/2}{1/2} \Rightarrow \tau = W\sqrt{3}$

11. $f = \frac{eB}{2\pi m}$

$f = \frac{1.76\times10^{11}\times3.57\times10^{-2}}{2\times3.14}$ Hz

$f = 10^9$ Hz or 1 GHz

12. Work = $MB[\cos\theta_1 - \cos\theta_2]$

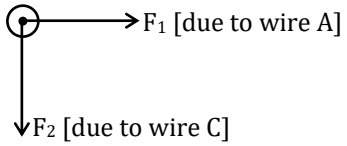
Work = $MB[\cos 0 - \cos 180°]$

$W = NiAB[1 - (-1)]$

$W \simeq 9.1\ \mu J$

13. $F = \frac{\mu_0 i_1 i_2}{2\pi d}$ = force per unit length

$F_1 = \frac{(\mu_0 i) i}{2\pi d} = \frac{\mu_0 i^2}{2\pi d} = F_2$

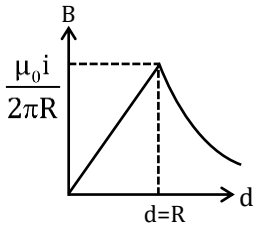


$F_{net} = \sqrt{F_1^2 + F_2^2} = \frac{\mu_0 i^2}{\sqrt{2}\pi d}$

14. When current source is switched on, magnetic field sets up between poles on electromagnet.

Diamagnetic material, due to its tendency to move from stronger to weaker field, is thus repelled out.

15. $B = \begin{cases} \frac{\mu_0 i d}{2\pi R^2} & : \ d \le R \\ \frac{\mu_0 i}{2\pi d} & : \ d > R \end{cases}$



16. $\frac{q_H}{q_\alpha} = \frac{1}{2}$

$r = \frac{mv}{qB}$

For same momenta, $r \propto \frac{1}{q}$

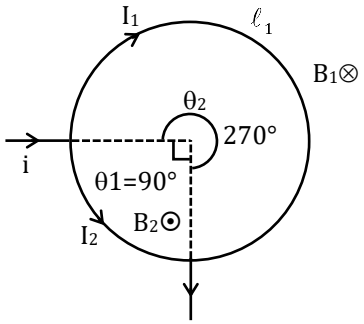
$\frac{r_H}{r_\alpha} = \frac{q_\alpha}{q_H} = \frac{2}{1}$

17. $B = \frac{\mu_0 N i}{2\pi r}$

$\frac{B_1}{B_2} = \frac{N_1}{N_2}\frac{r_2}{r_1} = \left(\frac{200}{100}\right)\left(\frac{20}{40}\right) = 1:1$

18. $\frac{R_1}{R_2} = \frac{\ell_1}{\ell_2} = \frac{I_2}{I_1} = \frac{\theta_1}{\theta_2}$

$\Rightarrow \quad \theta_1 I_1 = \theta_2 I_2$



$\left.\begin{aligned} B_1 &= \frac{\mu_0}{4\pi}\frac{I_1\theta_1}{r} \\ B_2 &= \frac{\mu_0}{4\pi}\frac{I_1\theta_1}{r} \end{aligned}\right\} \Rightarrow B_1 = B_2$

B_1 and B_2 are in opposite directions, hence resultant field at centre is zero.

19. $\mu_r = \chi_m + 1 = 599 + 1 = 600$

$\mu = \mu_o\mu_r = 4\pi \times 10^{-7} \times 600 = 2.4\pi \times 10^{-4} \ \frac{Tm}{A}$

20. $B = \mu_0 \frac{N}{\ell} I$

$= 4\pi \times 10^{-7} \times \frac{100}{(0.5)} \times 2.5$

$= 6.28 \times 10^{-4}$ T

21. $M = I(\pi r^2)$ Where, $r = \frac{L}{2\pi}$

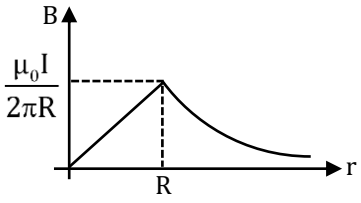
$\Rightarrow M = I(\pi)\left(\frac{L}{2\pi}\right)^2 = \frac{IL^2}{4\pi}$

22. Inside a current carrying cylindrical conductor,

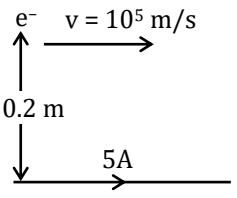
$B = \frac{\mu_0 I}{2\pi R^2} r \qquad \therefore B \propto r$

Outside the conductor,

$B = \frac{\mu_0 I}{2\pi r} \ \therefore B \propto \frac{1}{r}$



23.



F = qVB

$F = ev\left(\frac{\mu_0 i}{2\pi r}\right)$

$F = \frac{1.6\times10^{-19}\times10^{5}\times2\times10^{-7}\times5}{0.2}$

$F = 8 \times 10^{-20}$ Newton

24. $\vec{F} = q(\vec{v}\times\vec{B})$

$$4\hat{i} - 20\hat{j} + 12\hat{k} = 1\begin{vmatrix} i & j & k \\ 2 & 4 & 6 \\ B & B & B_0 \end{vmatrix}$$

Comparing

$\Rightarrow 4 = 4B_0 - 6B$
$-20 = -2B_0 + 6B$
$12 = 2B - 4B$

Solving
$B = -6$
$B_0 = -8$

$\therefore \quad \vec{B} = -6\hat{i} - 6\hat{j} - 8\hat{k}$

25. $M_1 = \left(\frac{\sqrt{3}}{4}a^2\right)I\times4 = \sqrt{3}\,Ia^2$ (no. of turns = 4)

$M_2 = a^2I \times 3 = 3Ia^2$ (no. of turns = 3)

32. Magnetic field exist in
Closed Loops (Monopoles do not exist)

$\boxed{\phi\vec{B}.d\vec{A} = 0}$

(Gauss law for magnetism)

33. $\vec{F} = I(\vec{\ell}\times\vec{B})$

$= I\left[(L\hat{i})\times(2\hat{i} + 3\hat{j} - 4\hat{k})\right]$

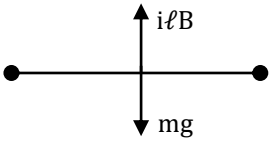
$= I(4L\hat{j} + 3L\hat{k})$

$|\vec{F}| = 5\,IL$

34. $B = \frac{\mu_0}{4\pi}\frac{I}{R}(\pi) - \frac{\mu_0}{4\pi}\frac{2I}{R}$

$= \frac{\mu_0 I}{4R}\left[1 - \frac{2}{\pi}\right]$ outward i.e away from page.

35.



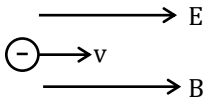
$mg = i\ell B$

$250 \times 10^{-3} \times 9.8 = i \times 2 \times 0.7$

$i = \frac{0.250\times9.8}{2\times0.7} = 0.250\times7$

$i = 1.75$ A

36. Speed of electron will decrease due to electric force magnetic force of electron is zero.



37. (A) Dia → II
(B) Ferro → III
(C) Para → (IV)
(D) Non magnetic → I

38. B = 0.049 T, $f = \frac{20}{5} = 4$Hz

$I = 9.8 \times 10^{-6}$ kg – m²

$M = x \times 10^{-5}$ A – m²

$f = \frac{1}{2\pi}\sqrt{\frac{MB}{I}}$

$M = \frac{f^2I(4\pi^2)}{B} = \frac{16\times4\pi^2\times98\times10^{-7}}{49\times10^{-3}}$

$x \times 10^{-5} = 128\pi^2 \times 10^{-4}$

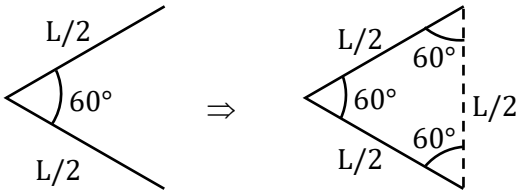
$x = 1280\,\pi^2$

39. $B = \frac{\mu_0 NI}{2R}$

$= \frac{4\pi\times10^{-7}\times100\times7}{2\times0.1}$

= 4.4 mT

40. Magnetic moment M = mL
where m is magnetic strength and L is length.
Now,

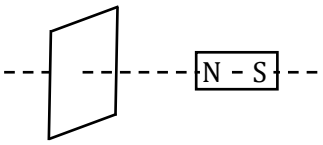


New magnetic moment,

$M' = m\times\frac{L}{2} = \frac{mL}{2}$

$M' = \frac{M}{2}$

41.



A force is needed to

(A) hold the sheet there if it is magnetic
(C) move the sheet away from the pole with uniform velocity if it is conducting.

42. U=–mBcosθ

$\because \theta = 90°$

U = 0

43. Diamagnetic material has

$-1 \le \chi < 0$

$0 < \mu_r < 1$

&

$\mu < \mu_0$

44. 10 Oscillations → 10s

⇒ 1 Oscillations → 1s

$$T = 2\pi\sqrt{\frac{I}{MB}}$$

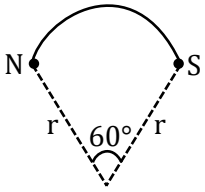
$$I = 2\pi\sqrt{\frac{10^{-6}}{\pi^2 \times 10^{-2} \times B}}$$

$$I = \frac{4\pi^2 \times 10^{-6}}{\pi^2 \times 10^{-2} B}$$

$B = 4 \times 10^{-4}$ T

B = 0.4 mT

45.



$M = m\ell$

M' = mr

$$M' = \frac{m3\ell}{\pi} \text{ and } \frac{\pi r}{3} = \ell$$

$$M = \frac{3M}{\pi}$$

Exercise-III (Analytical Questions)

1. Magnetic field due to coil of radius 5 cm

$$B_1 = \frac{N\mu_0 I}{2R} = \frac{50 \times 4\pi \times 10^{-7} \times 2}{2 \times (5 \times 10^{-2})} = 4\pi \times 10^{-4} \text{ T}$$

Magnetic field due to coil of radius 10 cm

$$B_2 = \frac{N\mu_0 I}{2R} = \frac{100 \times 4\pi \times 10^{-7} \times 2}{2 \times (10 \times 10^{-2})} = 4\pi \times 10^{-4} \text{ T}$$

$\therefore$ Current in same sense ; $B_{Net} = 8\pi \times 10^{-4}$ T

$\therefore$ Current in opposite sense ; $B_{Net} = 0$

2. Angle between $\vec{M}$ and $\vec{B}$ is 180°.

3. Current carrying wire is electrically neutral.

4. K.E. = qV ⇒ If V doubles, K.E. doubles

$$R = \frac{1}{B}\sqrt{\frac{2mV}{q}} \Rightarrow \text{If V doubles,}$$

R becomes $\sqrt{2}$ times

Time period $T = \frac{2\pi m}{qB} \Rightarrow$ Independent of speed

⇒ Angular velocity will not change.

5. $\vec{B}$ at a is inwards

$\vec{B}$ at f is zero (point on axis of wire)

$\vec{B}$ at c is outwards

$\vec{B}$ at g is outwards

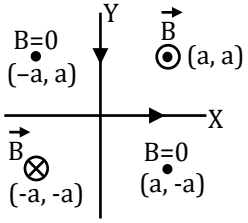
6. (a) θ = 180° (Q → angle between $\vec{M}$ & $\vec{B}$)

(b) θ = 90°

(c) θ = 0°

(d) θ is obtuse

7.



8. Wires carrying current in same direction attract each other.
Wires carrying current in opposite direction repel each other.

9. Use ACL and solve

for loop (1) $\oint \vec{B} \cdot \vec{d\ell} = \mu_0(-i)$

for loop (2) $\oint \vec{B} \cdot \vec{d\ell} = \mu_0 i$

for loop (3) $\oint B \cdot d\ell = 0$

for loop (4) $\oint B \cdot d\ell = 0$

10. $F_M = Bi\ell \sin\theta$

F_{AB} = Bia $\quad$ F_{CH} = Bia

$F_{BC} = 0$ $\quad$ $F_{CE} = Bi\sqrt{2}\,a$

11. Net force on a current carrying loop in a non uniform magnetic field may or may not be zero.
e.g. when a circular current loop is placed in a circular MF at a line of force then F = 0.

13. Magnetic field lines always form close curve.

14. There is moving charge in a current carrying element.

15. Current carrying wires (current = constant) produces magnetic field but not electric field. Charge particle beam produces both electric and magnetic field.

16. Ampere's law is valid for infinite wire of symmetrical current distribution.

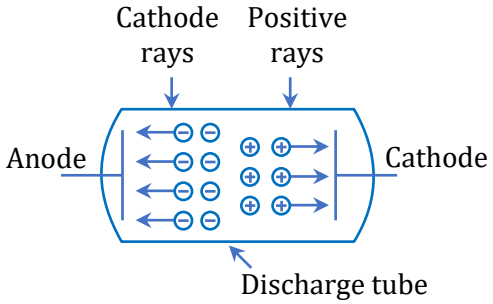
17. At curie point ferro start behaving as para.

19. (A) When V=0 or $\vec{V} \| \vec{E}$ $\rightarrow$ path will be straight When $\vec{V} \neq \vec{E}$ $\rightarrow$ path will be parabolic

(B) When $\vec{V} \| \vec{B} \rightarrow$ path will be straight

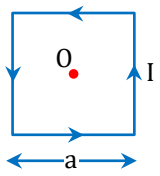
When $\vec{V} \perp \vec{B} \rightarrow$ path will be circular

21. Nature of both type of rays are unlike and they are moving in opposite direction in same transverse field so they are deflected in same direction according to



$\vec{F}_m = \pm q(\vec{v} \times \vec{B})$ or right hand palm rule.

22. Magnetic field at centre of square



$B = \frac{2\sqrt{2}\mu_0 I}{\pi a}$ ...(1)

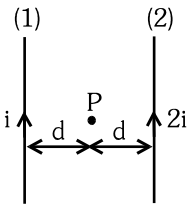
Magnetic moment

$M = Ia^2$ ...(2)

$\frac{M}{B} = \frac{Ia^2(\pi a)}{2\sqrt{2}\mu_0 I}$ [Divide equation (2) by (1)]

$M = \frac{\pi a^3 B}{2\sqrt{2}\mu_0}$

23. Magnetic field at mid point :-



$B = (B_2 - B_1)\odot$

$\therefore B = \left(\frac{\mu_0(2i)}{2\pi d} - \frac{\mu_0 i}{2\pi d}\right)\odot$

$= \frac{\mu_0 i}{2\pi d}\odot = B\odot$ .....(1)

After switching off 2i current, magnetic field remain due to wire (1) only :-

$B_1 = \frac{\mu_0 i}{2\pi d} = B\otimes$ [from (1)]

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Electromagnetic Induction

SOLUTIONS

Exercise-I (Conceptual Questions)

2. $\phi = BA\cos 0° = BA = 0.01\times10^3 = 10$ Wb

4. $e = A\dfrac{\Delta B}{\Delta t} = 2\left(\dfrac{4-1}{2}\right) = 3$ volt

7. $q = \dfrac{\Delta\phi}{R} = \dfrac{5.5\times10^{-4} - 5\times10^{-4}}{10} = 5\times10^{-6}C$

8. $q = \dfrac{\Delta\phi}{R} \Rightarrow q = \dfrac{NBA}{R}$

$B = \dfrac{qR}{NA} = \dfrac{32\times10^{-6}\times200}{100\times\pi(6\times10^{-3})^2} = 0.566T$

9. $\varepsilon = \dfrac{-\Delta\phi}{\Delta t} = -B\dfrac{(A_2 - A_1)}{\Delta t}$

$(2\pi r = 4\ell \Rightarrow r = \dfrac{2\ell}{\pi})$

$A_1 = A_{square} = (22\times10^{-2})^2$

$A_2 = A_{circular} = 3.14\times\left(\dfrac{2\times22}{3\cdot14}\times10^{-2}\right)^2$

$\varepsilon = -6\cdot6$ mV

10. $\varepsilon = \dfrac{\Delta\phi}{\Delta t}$ (1Wb = 10^8 Maxwell)

$= \dfrac{12\times10^{+3}\times10^{-8}}{0\cdot2} = 0\cdot6$ mV

11. $q = \dfrac{\Delta\phi}{R} = \dfrac{2NBA}{R}$

12. $\varepsilon = \dfrac{2NBA}{t}$

$= \dfrac{1000\times(0.4\times10^{-4})\times(500\times10^{-4})\times2}{0.1}$

$= 0.04V$

15. Speed of beam is constant so current is also constant. Due to this no current induced.

$(\because v = \text{constant} \Rightarrow I = \text{constant}\ \phi = \text{constant}$

$\Rightarrow \dfrac{d\phi}{dt} = 0 \Rightarrow \varepsilon = 0)$

16. Current in wire is constant so flux is also constant I = const $\Rightarrow$ B = const $\Rightarrow$ ϕ = constant $\Rightarrow$ NO EMI

20. $q = \dfrac{\Delta\phi}{R}$ $\quad\therefore\ q \propto (\Delta t)°$

21. $\phi_1 = NBA$ $\quad\phi_2 = -NBA$

$\Delta\phi = -2NBA$

$e = -\dfrac{d\phi}{dt} = -\left(\dfrac{-2NBA}{t}\right) = \dfrac{2NBA}{t}$

(Only magnitude)

$= \dfrac{2\times200\times0.8\times70\times10^{-4}}{0.1} = 22.4$ volt

25. Magnetic energy

$U_B = \dfrac{1}{2}LI^2 \propto I^2 \Rightarrow I \propto \sqrt{U_B}$

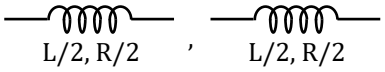
$\dfrac{I_2}{I_1} = \sqrt{\dfrac{U_2}{U_1}} = \sqrt{\dfrac{1}{4}} = \dfrac{1}{2} \Rightarrow I_2 = \dfrac{I_1}{2}$

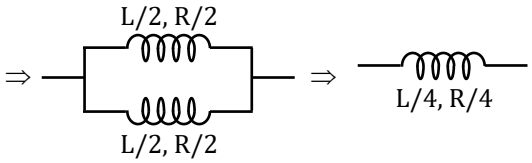
t is equal to Half life period therefore :

$t = 0\cdot693\ \dfrac{L}{R} = 0\cdot693 \times \dfrac{10}{2} = 3\cdot5$ sec.

26. $\lambda = \dfrac{L}{R} = 4$ sec. $\quad L = \mu_0 n^2 A\ell \Rightarrow L \propto \ell$

$R = \dfrac{\rho\ell}{A} \Rightarrow R \propto \ell$





$L_P = \dfrac{L_1L_2}{L_1+L_2} = \dfrac{L}{4}$, $R_P = \dfrac{R_1R_2}{R_1+R_2} = \dfrac{R}{4}$

$\lambda' = \dfrac{L/4}{R/4} = \dfrac{L}{R} = 4\text{sec}$

28. $N\phi = LI$

$L = \dfrac{N\phi}{I} = \dfrac{100\times5\times10^{-5}}{2} = 2.5\times10^{-3}$ H

$U = \dfrac{1}{2}LI^2 = \dfrac{1}{2}(2.5\times10^{-3})(2)^2 = 5\times10^{-3}$ J

OR

$U = (\phi_{total})I/2$

29. $L = \frac{\mu_0 N^2 \pi r^2}{\ell} = \mu_0 n^2 \ell \pi r^2$

$L' = \mu_0 n^2 \frac{\ell}{2} \pi (2r)^2 = 2L \Rightarrow$ two times

$\Rightarrow$ inductance increases by 100%.

31. $I = I_0\left(1 - e^{-\frac{R}{L}t}\right)$

$\frac{9I_0}{10} = I_0\left(1 - e^{-\frac{5}{5}t}\right) \Rightarrow e^{-t} = \frac{1}{10}$

After taking log both the sides

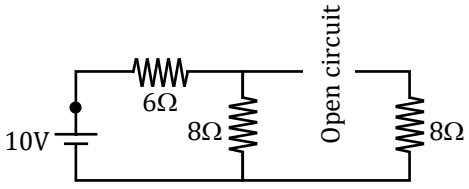
$t = \log_e 10 = 2.3$ sec.

33. $\varepsilon = L\frac{\Delta I}{\Delta t} = 60 \times 10^{-6} \times \frac{(1.5)-(1)}{0.1}$

$= 300 \times 10^{-6}$ V

$i = \frac{\varepsilon}{R} = \frac{300\times10^{-6}}{600\times10^{-6}} = 0.5$ Amp.

34. At t = 0



$I = \frac{V}{R_{net}} = \frac{10}{6+8} = \frac{10}{14} = \frac{5}{7}$A

35. In parallel combination

$\frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{3}{3} = 1$H

36. $\lambda = \frac{L}{R} = 2 \times 10^{-3}$ sec

$\frac{L}{R+90} = 0.5 \times 10^{-3}$ sec.

$(2 \times 10^{-3})\, R = (0.5 \times 10^{-3})(R + 90)$

$R = 30\Omega$

$L = 2 \times 10^{-3} \times 30 = 60$ mH

37. Magnetic energy

$U_B = \frac{1}{2}LI^2 = \frac{1}{2}(LI)I \qquad [\because \phi = LI] = \frac{1}{2}\phi I$

38. Self inductance of toroid is given by

$L = \frac{\mu_0 N^2 A}{2\pi R_m} = \frac{\mu_0 N^2 \pi r^2}{2\pi R_m} = \frac{\mu_0 N^2 r^2}{2R_m}$

41. $e = -\frac{L\Delta I}{\Delta t} \Rightarrow L = -\frac{e}{(\Delta I/\Delta t)}$

$L = -\frac{8}{(2/0.05)} = -0.2$H

$= 0.2$ H (only positive value)

42. $e = \frac{LdI}{dt} \Rightarrow L = e\frac{dt}{dI}$

$L = \frac{12}{48} \times \frac{60}{1} = 15$H

44. $U = \frac{1}{2}LI^2$, where $I_0 = \frac{E}{R}$

46. Self inductance of solenoid

$L = \frac{\mu_0 N^2 A}{\ell} \Rightarrow L \propto N^2$

47. $\lambda = \frac{L}{R} = \frac{40}{8} = 5$ second

48. $e = \frac{LdI}{dt} \Rightarrow L = \frac{e}{(dI/dt)} = \frac{5}{(1/10^{-3})}$

$= 5 \times 10^{-3} = 5$mH

49. Current becomes half of its peak value in half life time

$T = \frac{L}{R} \times 0{\cdot}693 = \frac{300 \times 10^{-3}}{2} \times 0.693 = 0.1$ sec.

50. $i = i_0(1 - e^{-t/\lambda})$

$i_0 = \frac{5V}{5\Omega} = 1$ amp.

$l = \frac{L}{R} = \frac{10}{5} = 2$ sec.

at t = 2 sec.

$i = 1(1 - e^{-2/2})$

$i = (1 - e^{-1})$

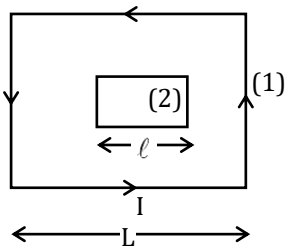
52. In both cases (a) & (b) M is constant

For (a) $\varepsilon_2 = M\frac{dI_1}{dt}$

$M = \frac{\varepsilon_2}{(dI_1/dt)} = \frac{2\times10^{-3}}{5} = 4 \times 10^{-4}$ H

For (b) $e_1 = M\frac{dI_2}{dt} = 4 \times 10^{-4} \times 2 = 8 \times 10^{-4}$ V

53. $M = \frac{N_2 B_1 A_2}{I_1}$



$= (1)\left(\frac{2\sqrt{2}\mu_0 I}{\pi L}\right)\frac{\ell^2}{I} = \frac{2\sqrt{2}\mu_0\ell^2}{\pi L}$

$M \propto \frac{\ell^2}{L}$

61. $\phi = MI \Rightarrow \frac{d\phi}{dt} = M\frac{dI}{dt} \Rightarrow M = \frac{d\phi}{dI} = \frac{2\times10^{-2}}{0.01} = 2H$

64. $M = K\sqrt{L_1L_2}$ here (K = 1)

so $M = \sqrt{L_1L_2}$

66. $\frac{V_S}{V_P} = \frac{N_S}{N_P} = \frac{1}{20} \Rightarrow V_s = \frac{1}{20}\times V_P$

$V_s = \frac{2500}{20}$ volt

67. Transfarmer cannot works on dc supply.

68. $e = -\frac{d\phi}{dt} = -\frac{d}{dt}(10t^2 - 50t + 250) = -(20t - 50)$

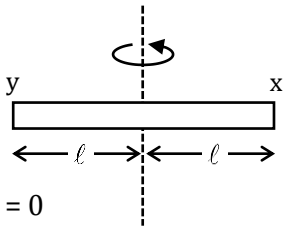
Put t = 3 sec

$e = -(20 \times 3 - 50) = -10V$

69. $M = \frac{\mu_0 N_1 N_2 A}{\ell}$

70. $e_{ox} = \frac{1}{2}B\omega\ell^2$

$e_{oy} = \frac{1}{2}B\omega\ell^2$

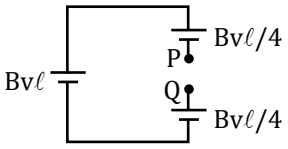


$e_{xy} = \frac{1}{2}B\omega\ell^2 - \frac{1}{2}B\omega\ell^2 = 0$

71. $e = \frac{1}{2}B\omega\ell^2 = \frac{1}{2}B(2\pi f)\ell^2$

$= \frac{1}{2}(6.28\times10^{-3})(2\times3.14\times50)(1)^2 = 1$ V

73. Using KVL



$V_P + \frac{Bv\ell}{4} - Bv\ell + \frac{Bv\ell}{4} = V_Q$

$V_P - V_Q = \frac{Bv\ell}{2}$

$V_P - V_Q$ is positive so $V_P > V_Q$

75. Effective length of the loop = 2R

perpendicular component of velocity to effective length vsinθ

$e = B|(\vec{\ell}\times\vec{v})|$

$e = B2R\,v\sin\theta$

76. For rotating conducting rod emf across its end is

$\varepsilon = \frac{1}{2}B\omega\ell^2$

So $\varepsilon_{OP} = \frac{1}{2}B\omega(4\ell)^2 = 8B\omega\ell^2$

$\therefore \quad \varepsilon_{OQ} = \frac{1}{2}B\omega(2\ell)^2 = 2B\omega\ell^2$

therefore $e_{PQ} = e_{OP} - e_{OQ}$

$\varepsilon_{PQ} = 6B\omega\ell^2$

77. An emf will induce in that close loop in such a way that it opposes the motion of rod.

79. $\Delta\phi = 0$ (due to motion inside a uniform magnetic field)

$\therefore \varepsilon = 0$

87. $\varepsilon_0 = NBA\omega$

91. Induced electric field in this loop

$E(2\pi R) = \pi R^2\frac{dB}{dt}$ (r = R, on the surface)

$E = \frac{R}{2}\frac{dB}{dt} = 2\times10^{-2}\times0.2 = 4\times10^{-3}$ v/m

Force applied by this electric field

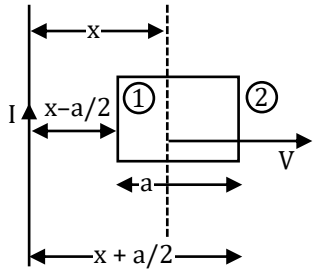
$F = qE = 2\times4\times10^{-3} = 8\times10^{-3}$ N

Exercise-II (Previous Year Questions)

1. $\eta = \frac{V_S I_S}{V_P I_P} \Rightarrow 0.9 = \frac{V_S(6)}{3\times10^3} \Rightarrow V_S = 450V$

As $V_P I_P = 3000$ so $I_P = \frac{3000}{200}A = 15A$

2.



emf Induced in side (1)

$\varepsilon_1 = B_1V\ell$

emf Induced is side (2)

$\varepsilon_2 = B_2V\ell$

emf in the frame

$= B_1V\ell - B_2V\ell$

$\varepsilon = V\ell\,[B_1 - B_2]$

$\Rightarrow \quad \varepsilon \propto B_1 - B_2 \quad$ Since $B \propto \dfrac{1}{r}$

So $\varepsilon \propto \left[\dfrac{1}{x-\dfrac{a}{2}} - \dfrac{1}{x+\dfrac{a}{2}}\right]$

$\Rightarrow \quad \varepsilon \propto \left[\dfrac{1}{(2x-a)} - \dfrac{1}{(2x+a)}\right]$

3. First current develops in direction of abcd but when electron moves away, then magnetic field inside loop decreases & current will change its direction.

4. Flux linked with each turn = 4×10^{-3} Wb

$\therefore$ Total flux linked

$= 1000[4\times10^{-3}]$ Wb = 4 Wb

$\phi_{total} = L\,i = 4 \Rightarrow L = 1H$

5. For Loop 1

$$\varepsilon_{ind} = -\frac{d\phi}{dt} = -A\left(\frac{dB}{dt}\right)\cos 0° = -\pi r^2\left(\frac{dB}{dt}\right)$$

For Loop 2, $\varepsilon_{ind} = 0$ as no flux linkage

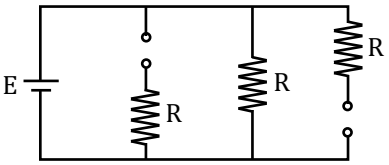
6. $$q = \left[\left(\frac{\Delta\phi}{\Delta t}\right)\cdot\frac{1}{R}\right]\Delta t$$

$$q = \left[\mu_0 nN\pi r^2\frac{\Delta i}{\Delta t}\right]\frac{1}{R}\Delta t$$

$$q = \left[\begin{matrix}4\pi\times10^{-7}\times2\times10^4\times\\ 100\times\pi\times(10^{-2})^2\times\left(\dfrac{4}{.05}\right)\end{matrix}\right]\frac{1}{10\pi^2}\times0.05$$

$q = 32\ \mu C$

7. Just after switch closed then



Here E = 18V, R = 9Ω

$I = E/R = \dfrac{18}{9} = 2A$

8. Magnetic potential energy $= \dfrac{1}{2}LI^2$

$$\Rightarrow 25\times10^{-3} = \frac{1}{2}L(60\times10^{-3})^2$$

$\Rightarrow L = 13.89$ H

9. Given

$N = 800$, $A = 0.05\ m^2$, $B = 5\times10^{-5}$ T

$\Delta t = 0.15$ s

As $e = -\dfrac{(\phi_f - \phi_i)}{\Delta t} = -\dfrac{(0 - NBA)}{\Delta t}$

$= \dfrac{800\times5\times10^{-5}\times5\times10^{-2}}{0.1} = 0.02$ V

10. Eddy current effect is not used in electric heater

11. $E = \dfrac{B\omega\ell^2}{2} = \dfrac{0.1(10)(0.5)^2}{2} = 0.125$ V

12. $\phi = 5t^2 + 3t + 60$

$|\varepsilon| = \left|\dfrac{d\phi}{dt}\right| = 10t + 3$

At t = 4 sec.

$|\varepsilon| = 40 + 3 = 43$ volt

13. $\varepsilon = \dfrac{1}{2}B\omega r^2$

$$\varepsilon = \frac{1}{2}\times(0.4\times10^{-4})\times\left(2\pi\left[\frac{120}{60}\right]\right)(1)^2$$

$\varepsilon = 0.8\pi \times 10^{-4} = 2.512 \times 10^{-4}$ V

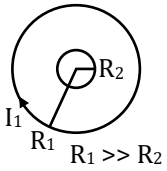
14. $220 \times i_p = 44$

$\Rightarrow i_p = \dfrac{44}{220} = \dfrac{1}{5} = 0.2A$

15. $M = \frac{\phi_{12}}{I_1} = \frac{B_1 A_2}{I_1} = \frac{\left(\frac{\mu_0 I_1}{2R_1}\right)(\pi R_2^2)}{I_1}$

$M = \frac{\mu_0 \pi R_2^2}{2R_1}$

$M \propto \frac{R_2^2}{R_1}$



18. $V_S I_S = V_P I_P$ (ideal Transformer)

$\Rightarrow P_{out} = P_{in}$

$\Rightarrow 60 = 220 \times I_P$

$I_P = \frac{60}{220} = 0.27A$

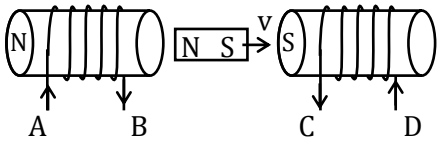
19. Energy = $\frac{1}{2}Li^2$

$= \frac{1}{2} 4 \times 10^{-6} \times 2^2$

$= 8 \times 10^{-6}$ J

energy = 8 μJ

20.



A to B & D to C

21. For ideal transformer

$\frac{V_S}{V_P} = \frac{N_S}{N_P} = 2:1$

22. $L = \mu_0 n^2 A\ell$

$\frac{L_A}{L_B} = \frac{\mu_r \mu_0 (n/2)^2 A(2\ell)}{\mu_r \mu_0 n^2 A\ell}$

$\frac{L_A}{L_B} = \frac{1}{4} \times 2 = \frac{1}{2}$

Exercise-III (Analytical Questions)

1. $e = (vB\ \ell_{eff})$

$\ell_{eff} = L/2$

$\Rightarrow e = \frac{1}{2}$ BLv [P is at higher potential]

2. (a) $\phi = LI$ = Henry – ampere

(b) Energy = $\frac{1}{2} LI^2$

$L \rightarrow \left(\frac{J}{A^2}\right)$

(c) emf = $-L\frac{dI}{dt}$

$L \rightarrow \frac{\text{volt} - \text{second}}{\text{ampere}}$

(d) S.I, unit of magnetic induction is Tesla.

3.

$q = 2t^2$	At $t = 1$ sec
$I = \frac{dq}{dt} = 4t$	$q = 2C$
$\frac{dI}{dt} = 4$	$I = 4A$
	$\frac{dI}{dt} = 4A/s$

(i) $V_a - L.\frac{dI}{dt} = V_b \Rightarrow V_a - V_b = L.\frac{dI}{dt} = 4V$

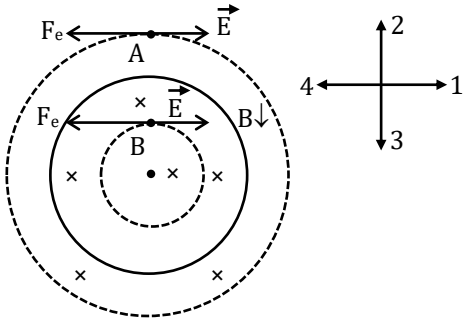
(ii) $V_b - \frac{q}{C} = V_c \Rightarrow V_b - V_c = \frac{q}{C} = 1V$

(iii) $V_c - IR = V_d \Rightarrow V_c - V_d = IR = 16V$

(iv) $V_a - L.\frac{dI}{dt} - \frac{q}{c} - 4I = V_d$

$\Rightarrow V_a - V_d = 21$ V

4.



5. $B = \frac{F}{i\ell} = \frac{MLT^{-2}}{AL}$

$L = \frac{\text{Energy}}{i^2} = [ML^2T^{-2}A^{-2}]$

$T = 2\pi\sqrt{LC} \Rightarrow LC = [T^2]$

$\phi = LI = [M^1L^2T^{-2}A^{-1}]$

6. At t = 0 $V_L = V_0 = 10V$

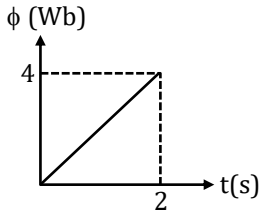
$V_R = V_0 = 0V$

At t = 1 $V_L = V_0 e^{-t/\tau}$ | $V_R = V_0 [1 - e^{-t/\tau}]$

$V_L = 10\, e^{-1}$ | $V_R = 10\left[1 - \frac{1}{e}\right]$

$V_L = \frac{10}{e}$

7.



$R = 2\Omega$

$\Delta\phi = 4$

$e_{in} = \frac{\Delta\phi}{\Delta t} = \frac{4}{2} = 2\,\text{volt}$

$i_{in} = \frac{\varepsilon_{in}}{R} = \frac{2}{2} = 1\,A$

$\Delta q = \frac{\Delta\phi}{R} = \frac{4}{2} = 2C$

$H = I^2Rt$

$H = (1)^2 \times 2 \times 2 = 4\,J$

8. D.C. voltage is steady with time.

9. According to lezs law induced current always opposes the change in flux.

10. Inductor tends to keep flux constant.

11. $M_{1,2} = M_{2,1}$

Independent of current.

12. $\oint \vec{E} \cdot \vec{d\ell} = -A\frac{dB}{dt}$

13. $\phi = \vec{B}.\vec{A}$

14. When flux is zero, at that instant rate of flux change is maximum.

AIM: FREE EDUCATION TO ALL APUL

Alternating Current

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Given that,

$H_{AC} = 3\ H_{DC}$

$I_{rms}^2\ Rt = 3\ (I^2Rt)$

$I_{rms}^2 = 3I^2$

$I_{rms} = \sqrt{3}I = \sqrt{3}\ 2$

$I_{rms} = 3.46$ A

2. $E = E_0 \cos \omega t$

Given, $E_0 = 10V$, $t = \frac{1}{600}$ sec., $f = 50Hz$

$E = 10 \cos (2\pi \times 50 \times \frac{1}{600})$

$E = 10 \cos \frac{\pi}{6} \rightarrow E = 5\sqrt{3}$ V

3. $\because 2\pi$ (phase difference) $\equiv$ T(Time difference)

1 rad (PD) $\equiv \frac{T}{2\pi}$ (Time difference)

So, $\frac{\pi}{4}$ (PD) $\equiv \frac{T}{2\pi} \times \frac{\pi}{4}$ (Time difference)

$= \frac{T}{8} \Rightarrow$ time difference $= \frac{1}{50\times 8}$ s

or time difference = 2.5 ms.

4. $I = 3 + 4 \sin \omega t$

(i) square $= 9 + 16 \sin^2 \omega t + 24 \sin \omega t$

(ii) mean value $= 9 + 16 \times \frac{1}{2} + 0 = 17$

(iii) Root $= \sqrt{17}$

(iv) Rms value $= \sqrt{17}$

OR

$I = a + b \sin \omega t$

$\therefore \boxed{I_{rms} = \sqrt{a^2 + \frac{b^2}{2}}} = \sqrt{9 + \frac{16}{2}} = \sqrt{9+8} = \sqrt{17}$

6. $i = i_1 \cos \omega t + i_2 \sin \omega t$

Let $i_1 = a \sin \phi$, and $i_2 = a \cos \phi$

than $i = a \sin\phi \cos \omega t + a \cos \phi \cos \omega t$

$i = a \sin (\phi + \omega t)$

[$\because \sin A \cos B + \sin B \cos A = \sin (A+B)$]

So $I_{rms} = \frac{a}{\sqrt{2}}$

$a = \sqrt{i_1^2 + i_2^2}$ $\left(\because a = \sqrt{i_1^2 + i_2^2}\right)$

OR

$i = i_1 \cos \omega t + i_2 \sin \omega t$

$I_{max} = \sqrt{i_1^2 + i_2^2} \Rightarrow I_{rms} = \frac{\sqrt{i_1^2 + i_2^2}}{\sqrt{2}} = \sqrt{\frac{i_1^2 + i_2^2}{2}}$

7. $V = 120 \sin (100\pi\ t) \cos (100\pi t)$

$\because 2 \sin\theta \cos\theta = \sin 2\theta$

$\therefore \sin (100\pi t) \cos (100\pi t) = \frac{\sin(2\times 100\pi t)}{2}$

$V = 60 \sin (200\pi\ t)$

compare from

$V = V_0 \sin \omega t$

$V_0 = 60$ volt,

$\omega = 200\pi$

$2\pi f = 200\pi$

$f = 100$ Hz

8. $V_{avg} = \frac{2V_0}{\pi} = \frac{2\sqrt{2}V_{rms}}{\pi} = \frac{2\sqrt{2}\times 220}{3.14} = 198$Volt

12. Peak voltage in primary $V_0 = \sqrt{2}\ V_{rms} = 100\sqrt{2}$

$\frac{V_s}{V_p} = \frac{N_2}{N_1} \Rightarrow V_s = \frac{N_s}{N_p} \times V_p$

$V_s = 5 \times 100\sqrt{2} = 500\sqrt{2}$Volt

15. $X_L = 2\pi fL$

$X_L \propto f$(graph straight line)

$X_C = \frac{1}{2\pi fC}$; $X_C \propto \frac{1}{C}$

(graph rectangular hyperbola)

16. $\tan \phi = \frac{X_L}{R}$

$\tan \phi = \frac{2\pi\times 50\times 0.21}{12}$

$\tan \phi \approx 5.5$

Now $\tan 60° = \sqrt{3}$ and $5.5 > \sqrt{3}$

So, angle will be greater 60°.

17. $V^2 = V_R^2 + V_C^2$

$(220)^2 = (110)^2 + V_C^2$

$V_C = 190$ Volt.

19. For DC. V = IR

$R = \frac{V}{I} = \frac{100}{25}$

R = 4Ω (Resistance of coil)

For AC. V = IZ

$Z = \frac{V}{I} = \frac{100}{20}$ $\quad Z = 5\ \Omega$

$Z^2 = R^2 + X^2 \rightarrow X^2 = Z^2 - R^2$

$= 5^2 - 4^2$

Reactance of coil X = 3Ω.

24. $I = \frac{V}{Z}$

$Z = \omega L = 2\pi \times 50 \times 2 = 200\pi$

$I_1 = \frac{V}{200\pi}$

on frequency 400 Hz

$Z = \omega L = 2\pi \times 400 \times 2$

$I_2 = \frac{V}{1600\pi}$; $\frac{I_1}{I_2} = \frac{V}{200\pi} \times \frac{1600\pi}{V} = 8$

$I_2 = \frac{I_1}{8}$ (lagging by 90° from V due to purely inductive circuit)

OR

$X_L \propto f$, f becomes 8 times, so X_L also becomes 8 times, so current becomes 1/8 times and lagging by 90° from V (pure inductive circuit)

$\therefore$

27. Use $X_c = \frac{1}{\omega C}$

31. $\tan\phi = \frac{X_L}{R}$

$\tan\phi = \frac{40\pi}{100}$ $\quad \begin{cases} X_L = \omega L \\ = 2\pi \times 50 \times 0.4 \\ = 40\pi\Omega \\ R = 100\Omega \end{cases}$

$\tan\phi = 0.4\pi$,

$\phi = \tan^{-1}(0.4\pi)$

32. $\tan\phi = \frac{X_C}{R} = \frac{1}{\omega RC}$

$= \frac{1}{2\pi \times 50 \times 100 \times 100 \times 10^{-6}} = \frac{1}{\pi}$

$\rightarrow \phi = \tan^{-1}\left(\frac{1}{\pi}\right)$

33. $V_L = L\frac{dI}{dt} = L\frac{d}{dt}(I_0 \sin\omega t)$

$= I_0\omega L\cos\omega t = I_0\omega L\sin(\omega t + \pi/2)$

35. Use $V^2 = V_R^2 + V_C^2$

39. $V_L = I \times X_L$

$I = \frac{0.6}{2\pi \times 60 \times 10^{-3}}$

40. $\because$ $X_L = X_C$

$f = \frac{1}{2\pi\sqrt{LC}} \rightarrow X_L = 2\pi fL$

$X_L = 2\pi L \times \frac{1}{2\pi\sqrt{LC}} = \sqrt{\frac{L}{C}}$

$X_L = \sqrt{\frac{10^{-3}}{10 \times 10^{-6}}} \Rightarrow X_L = 10\Omega.$

OR

At resonance, $\omega = \frac{1}{\sqrt{LC}}$

$\omega = 10^4$ Rad/sec.

$\therefore X_L = \omega L = 10^4 \times 10^{-3} = 10\ \Omega$

42. Impedance of series circuit

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

If X_L and X_C are interchanged then there is no effect on Z.

47. Power dissipation = VI cosϕ

$= V.\frac{V}{Z}.\frac{R}{Z} = \frac{V^2R}{Z^2}$

Now $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$Z = R\ (\because X_L = X_C)$

Power dissipations = $\frac{V^2R}{R^2} = \frac{V^2}{R}$

$V = \frac{V_0}{\sqrt{2}} = \frac{10}{\sqrt{2}} \Rightarrow R = 10\Omega$

Power dissipation = $\frac{10 \times 10}{\sqrt{2} \times \sqrt{2} \times 10}$ = 5 Watt.

OR

At resonance

$I_0 = \frac{10}{10} = 1\ A$

$I_{rms} = \frac{1}{\sqrt{2}} A$

$P_{av} = V_{rms}\ I_{rms} = \frac{10}{\sqrt{2}} \times \frac{1}{\sqrt{2}}$ = 5 Watt.

48. At resonance, maximum power impart to circuit.

$f = \frac{1}{2\pi\sqrt{LC}} \Rightarrow 50 = \frac{1}{2\pi\sqrt{10\times C}}$

$\rightarrow C = 10^{-6}$ F

49. $f = \frac{1}{2\pi\sqrt{LC}}$

Now $L' = L + \frac{L}{4} = \frac{5L}{4}$

$C' = C - \frac{C}{5} = \frac{4C}{5}$

$f' = \frac{1}{2\pi\sqrt{L'C'}} = \frac{1}{2\pi\sqrt{\frac{5L}{4}\times\frac{4C}{5}}}$

$f' = \frac{1}{2\pi\sqrt{LC}} = f$

51. R Changed $\Rightarrow$ Z Changed $\Rightarrow$ I Changed

$V_L = I X_L$ (Changed)

$V_C = I X_C$ (Changed)

V_L and V_C are changed by same amount

So Voltage across LC combination ($V_L - V_C$) remains same.

52. $\tan\phi = \frac{V_L - V_C}{V_R} = \frac{1}{1} = 45°$

$V_L > V_C$

So current lags by $\frac{\pi}{4}$

$\therefore I = I_0 \cos\left(\omega t - \frac{\pi}{4}\right)$

53. $Z = \sqrt{R^2+X^2}$ [X = reactance]

if $\omega\uparrow$ than $X_C \downarrow$, $Z\downarrow$

So circuit contain resistance and capacitance

54. $V_L = V_C$

$\therefore X_L = X_C$ (Resonance condition)

$i = \frac{V}{R} = \frac{100}{50} = 2A$

$V_R = iR = 2 \times 50 = 100$ V

56. $V^2 = V_R^2 + (V_L - V_C)^2$

59. $P_{loss} = VI\cos\phi$

$\cos\phi = \frac{P_{loss}}{VI} = \frac{600}{5\times160}$

$\cos\phi = 0.75$

64. $E = E_0 \sin\omega t$

$I = I_0 \sin(\omega t + \frac{\pi}{2}) \Rightarrow \phi = \frac{\pi}{2}$

$\cos\phi = 0$ so $P = 0$

65. Power factor $\cos\phi = \frac{R}{Z}$

For negligible power factor $\Rightarrow$ R Low

Z high (L high)

66. Wattless power = $V I \sin\phi$,

Wattless power

$$= \frac{100}{\sqrt{2}} \times \frac{100}{\sqrt{2}} \times \sin\frac{\pi}{6} \begin{cases} V = \frac{100}{\sqrt{2}} V \\ I = \frac{100}{\sqrt{2}} A \\ \phi = \frac{\pi}{6} \end{cases}$$

$= 2.5 \times 10^3$ Watt

69. Power factor = $\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2+X_L^2}}$

$$\cos\phi = \frac{1100}{\sqrt{(1100)^2 + \left(2\times\frac{22}{7}\times50\times\frac{7}{2}\right)^2}}$$

$$= \frac{1100}{\sqrt{(1100)^2+(1100)^2}}; \quad \cos\phi = \frac{1}{\sqrt{2}}$$

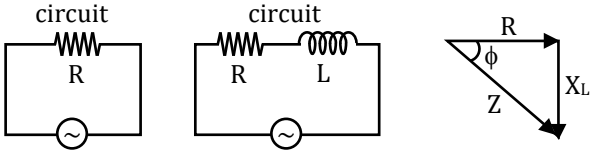
Exercise-II (Previous Year Questions)

1. When capacitor is filled with mica then capacitance C increases so X_C decreases

In case (b) $X_C \downarrow$ so voltage across capacitor decreases. so $V_a > V_b$

2.

Pure resistor circuit | Resistor & Inductor circuit | (Phasor diagram)



AC source Impedance = Z

$P' = V.I.\cos\phi$

$P = \frac{V^2}{R}$ $P' = V.\left[\frac{V}{Z}\right].\cos\phi$

$\Rightarrow V^2 = PR$ $P' = \frac{V^2}{Z}\cdot\frac{R}{Z}$

(From phasor diagram)

$P' = \frac{(PR)R}{Z^2}$

$P' = \left(\frac{R}{Z}\right)^2 P$

3. $\frac{1}{\omega C}=\frac{1}{340\times50\times10^{-6}}$ X_C = = 58.8 Ω

$X_L = \omega L = 340 \times 20 \times 10^{-3} = 6.8\ \Omega$

$Z = \sqrt{R^2+(X_C-X_L)^2}$

$= \sqrt{40^2+(58.8-6.8)^2} = \sqrt{4304}\ \Omega$

$P = i_{rms}^2 R = \left(\frac{V_{rms}}{Z}\right)^2 R$

$=\left(\frac{10/\sqrt{2}}{\sqrt{4304}}\right)^2 \times 40 = \frac{50\times40}{4304}$ = 0.47 W

So best answer (nearest answer) will be (1)

4. Input voltage, $V(t) = V_0 \sin\omega t$

For capacitor, $I(t) = \frac{dq}{dt} = C\frac{dV}{dt} = \omega C V_0 \cos\omega t$

$\Rightarrow$ Current I(t) leads voltage V (t) by 90°

Also, Capacitor does not consume any energy over a full cycle.

5. For better tuning, Q-factor must be high.

$\therefore\ Q = \frac{\omega_o L}{R} = \frac{1}{\sqrt{LC}}\left(\frac{L}{R}\right) = \frac{1}{R}\sqrt{\frac{L}{C}}$

R and C should be small and L should be high.

6. $\tan\phi = \frac{V_L - V_C}{V_R} = \frac{100-40}{80} = \frac{3}{4}$ or ϕ = 37°

Power factor = cosϕ = cos 37° = $\frac{4}{5}$ or 0.8

7. Here

$X_L = \omega L = (314)\ (20 \times 10^{-3}) = 6.280$

$X_C = \frac{1}{\omega C} = \frac{1}{314 \times 100 \times 10^{-6}} = 31.84\Omega$

$R = 50\Omega$

$Z = \sqrt{(X_C - X_L)^2 + R^2}$

$= \sqrt{(31.84 - 6.28)^2 + (50)^2}$ = 56Ω

$\Rightarrow P = \left(\frac{v_{rms}}{Z}\right)^2 R = \left(\frac{10}{\sqrt{2}\times56}\right)^2 \times 50$

$= 0.79 \text{watt}$

8. Changing polarity is termed as AC.

9. $Z = \frac{12}{0.2} = 60\ \Omega$ and $R = \frac{12}{0.4} = 30\ \Omega$

10. $I = \frac{V}{X_C} = \frac{V}{1/C\omega} = VC\omega$

$= 200 \times 40 \times 10^{-6} \times 2\pi \times 50$

= 2.5 A

11. When L removed $\tan\phi = \frac{X_C}{R}$

When L removed $\tan\phi = \frac{X_L}{R}$

$\frac{X_C}{R} = \frac{X_L}{R} \Rightarrow$ Resonance $\Rightarrow$ Z = R

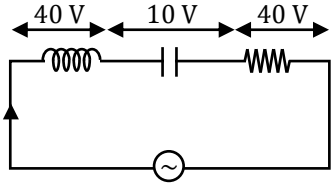
$\cos\phi = \frac{R}{Z} = \frac{R}{R} = 1$

12. Impedance, $z = \sqrt{R^2 + X_L^2}$

$X_L \uparrow$, $Z \uparrow$, $I \downarrow$

13. $I_0 = 10\sqrt{2}$ A

$I_{RMS} = \frac{I_0}{\sqrt{2}} = 10A$



$V_{RMS} = \sqrt{V_R^2 + (V_L - V_C)^2}$

$= \sqrt{(40)^2 + (40-10)^2}$

= 50 V

$Z = \frac{V_{RMS}}{I_{RMS}} = \frac{50\ V}{10\ V} = 5\Omega$

14. $Q = \frac{\omega}{\Delta\omega} = \frac{\omega L}{R} \Rightarrow \Delta\omega = R/L = \frac{50}{4} = 8\ \text{rad/sec}$

$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{5\times80\times10^{-6}}} = 50\ \text{rad/sec.}$

$\omega_{min} = \omega_0 - \frac{\Delta\omega}{2} = 46\ \text{rad/sec}$

$\omega_{max} = \omega_0 + \frac{\Delta\omega}{2} = 54\ \text{rad/sec}$

19. L = 10 × 10^{-3} H

C = 1 × 10^{-6} F

R = 100 Ω

At resonance $X_L = X_C$

$\omega L = \frac{1}{\omega C}$

$f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10\times10^{-3}\times10^{-6}}} = 1.59\ \text{KHz}$

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20. $i_C = i_D = \frac{V_0}{X_C}\sin\omega t$

$i_C = i_D = (V_0\omega C)\sin\omega t$

On decreasing frequency $i_C \downarrow$

21. $X_L = \frac{50}{L}\times 10^{-3}\times 2\pi\times 50 = 5\Omega$

$X_C = \frac{1}{2\pi\times 50\times\frac{10^3}{\pi}\times 10^{-6}} = 10\Omega$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$= \sqrt{(10)^2 + (5)^2}$

$= 5\sqrt{5}\,\Omega$

22. $\tan\phi = \frac{X_L}{R}$

$X_L = \omega l = 100\pi \times \frac{1}{\pi} = 100\Omega$

$R = 100\Omega$

$\tan\phi = \frac{100}{100} = 1$

$\phi = 45°$

23. $Z_1 = \sqrt{X_L^2 + R^2}$ $X_L = 0$ (D.C. circuit)

$Z_1 = 10\Omega$

$Z_2 = \sqrt{X_C^2 + R^2}$

$X_C = \frac{1}{2\pi\times 50\times\frac{10^3}{\pi}\times 10^{-6}} = 10\Omega$

$Z_2 = \sqrt{(10)^2 + (10)^2}$

$= 10\sqrt{2}$

$Z_1 < Z_2$

24. Power dissipated is maximum of purely resistive circuit.

25. $\omega = 2\pi\frac{\text{rad}}{\text{sec}}$

$E_{max} = NBA\omega$

$= 100 \times 0.05 \times 1 \times 2\pi$

$= 10 \times \pi$

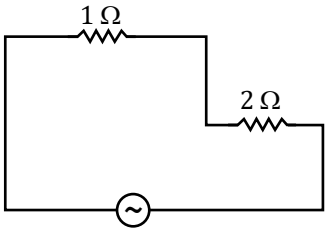
$= 31.4$ V

26. as frequency is very high

$X_C \approx 0$

$X_L \to \infty$

Effective circuit will be



Effective impedance of circuit will be = 3Ω

27. $I_{Peak} = \frac{V_{Peak}}{X_c} = V_{Peak}(\omega c)$

$= V_{Peak}(2\pi fc)$

$= (210\sqrt{2})\ (2\pi \times 50 \times 10 \times 10^{-6})$

$= 0.93$ Ampere

28. $V_L = i\,X_L$

$V_L = \omega L\, i_0 \cos\omega t$

29. $P = Vi$

$88 = 11000\, i$

$i = 8 \times 10^{-3}$ A

$i = 8$ mA

30. $Q = Q_0\, e^{\frac{-Rt}{2L}}$

putting $Q = 0.50\, Q_0$;

$0.50\, Q_0 = Q_0\, e^{\frac{-Rt}{2L}}$

$\Rightarrow e^{\frac{Rt}{2L}} = 2$

$\Rightarrow \frac{Rt}{2L} = \ln(2)$

$\Rightarrow t = \frac{2L}{R}(\ln 2) = \frac{2\times 12\times 10^{-3}}{1.5}.(0.693)$

$= 11.088\times 10^{-3} = 11.09$ ms

Exercise-III (Analytical Questions)

1. At frequency lower than resonant frequency since $X_C > X_L$

$\therefore$ Current leads voltage

$\to$ At resonance, impedance is minimum.

$\to$ At resonance, phase diff. between voltage across L and current in C is $\frac{\pi}{2}$

$\to$ $f = \frac{1}{2\pi\sqrt{LC}}$

2. $\tau = RC \Rightarrow R = \frac{\tau}{C}$

$[R] = \left[\frac{1}{fC}\right]$

$\tau = \frac{L}{R} \Rightarrow R = \frac{L}{\tau} = (f.L)$

$\therefore$ b & c are correct.

3. $V_1 = \sqrt{V_R^2 + V_L^2} = 100\text{Volt}$ ...(1)

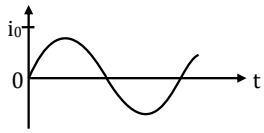
$V_C - V_L = 120$ Volt ...(2)

$130 = \sqrt{V_R^2 + (V_L - V_C)^2} \Rightarrow V_R = 50\text{Volt}$

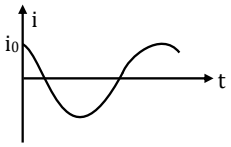
$V_L = 50\sqrt{3}$ Volt = 86.6 Volt & V_C = 206.6 Volt

$\cos\phi = \frac{V_R}{130} = \frac{5}{13}$

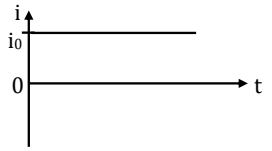
4.



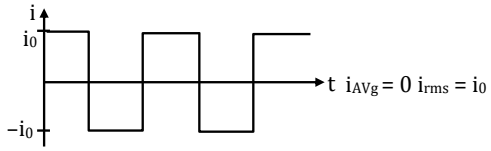
$i_{AVg} = 0$, $i_{rms} = \frac{i_0}{\sqrt{2}}$



$i_{avg} = 0$, $i_{rms} = \frac{i_0}{\sqrt{2}}$

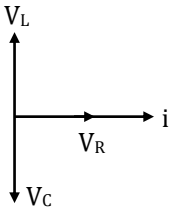


$i_{AVg} = i_{rms} = i_0$

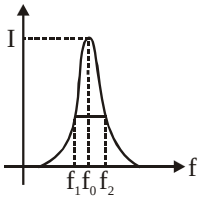


5. $Q_{factor} = \frac{1}{R}\sqrt{\frac{L}{C}}$

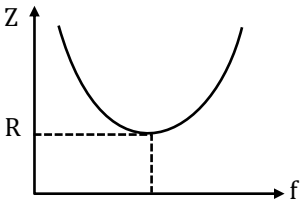
6.



7. $\boxed{f_0 = \sqrt{f_1 f_2}}$

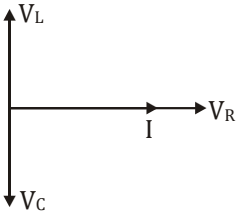


8. $z = \sqrt{R^2 + (x_L - x_C)^2}$



Impedance can be greater than or equal to resistance, but it can't be less than resistance of circuit.

9.



At resonance

$V_L = V_C$ & $Z = R$

there V_L & V_C are out of phase

10. For capacitor and inductor

$\phi = 90° \Rightarrow \cos\phi = 0 \Rightarrow P_{avg} = 0$

11. **For (A)** : Current and voltage are in phase

$\Rightarrow$ only R or RLC in resonance.

For (B): Current leads voltage by $\frac{\pi}{2} \Rightarrow$ only C

$\Rightarrow$ power factor = 0

For (C) : Current leads voltage $\Rightarrow$ RC or RLC with $X_C > X_L$

For (D) : Current leads voltage

$\Rightarrow$ RL or RLC with $X_L > X_C$

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Electromagnetic Waves

SOLUTIONS

Exercise-I (Conceptual Questions)

3. $\Psi(x, t) = 10^3 \sin\pi(3 \times 10^6 x - 9 \times 10^{14} t)$

$\Psi(x, t) = 10^3 \sin(3\pi \times 10^6 x - 9\pi \times 10^{14} t)$

$$\text{speed } v = \frac{\omega}{k} = \frac{9\pi\times10^{14}}{3\pi\times10^6} = 3 \times 10^8 \text{ m/s}$$

4. $\lambda = \frac{2\pi}{k}; \lambda = \frac{2\pi}{3\pi\times10^6} = \frac{2}{3}\times10^{-6} = 666 \text{ nm}$

Exercise-II (Previous Year Questions)

3. To generate electromagnetic waves we need accelerating charge particle.

4. $(i_d)_{max} = (i_c)_{max} = i_0 = \frac{E_0}{Z} = \frac{220\sqrt{2}}{\sqrt{100^2+100^2}} = 2.2A$

As we are asked amplitude of displacement current. So, need not worry about charge on capacitor.

5. $E_0 = cB_0$

$$E_{rms} = \frac{E_0}{\sqrt{2}}$$

$$\Rightarrow E_{rms}\sqrt{2} = cB_0$$

$$\Rightarrow B_0 = \frac{E_{rms}\sqrt{2}}{c} = \frac{6\times\sqrt{2}}{3\times10^8} = 2.83 \times 10^{-8} \text{ T}$$

6. $\hat{V} = \hat{E}\times\hat{B}, \quad \hat{i} = \hat{j}\times\hat{k}$

$\therefore \hat{B} = \hat{k}$, +z direction

7. $$v = \frac{1}{\sqrt{\mu\in}} = \frac{1}{\sqrt{\mu_r \in_r \mu_0 \in_0}} = \frac{c}{\sqrt{\mu_r\in_r}}$$

$$= \frac{3\times10^8}{\sqrt{1.44}} = \frac{30}{12}\times10^8 = 2.5\times10^8 \text{ ms}^{-1}$$

8. $$I = \frac{E}{At} \Rightarrow E = IAt = \frac{20}{10^{-4}}\times20\times10^{-4}\times60$$

$$= 24 \times 10^3 \text{ J}$$

9. In EMW, electric field and magnetic field have same energy density and same intensities.

10. Gamma rays have shortest wavelength among all options.

11. $\lambda = \frac{2\pi}{K} = \frac{2\pi}{\pi\times10^3} = 2\times10^{-2}\text{m}$

12. $q = CV$

$$\frac{dq}{dt} = C\frac{dv}{dt}$$

$I_d = C(V_0\,\omega\cos\omega t)$

$= V_0\omega C\cos\omega t$

13. $\vec{v}$ is parallel to $\vec{E}\times\vec{B}, \hat{v} = \hat{i}$ $\quad [\text{as}\,\hat{v} = \hat{i}]$

Option (1) $\vec{E}\times\vec{B} = \vec{0}$ $(\vec{E}\|\vec{B})$

Option (2) $\vec{E}\times\vec{B} = 2\hat{i}$ (parallel to $\vec{v}$)

Option (3) $\vec{E}\times\vec{B} = \vec{0}$ $(\vec{E}\perp\vec{B})$

Option (4) $\vec{E}\times\vec{B} = \vec{0}$ $(\vec{E}\|\vec{B})$

18. $C = \frac{E_0}{B_0}$

$$B_0 = \frac{E_0}{C}$$

$$= \frac{48}{3\times10^8}$$

$= 1.6 \times 10^{-7}$ T

19. $$\mu = \frac{C}{v} = \frac{\frac{1}{\sqrt{\mu_0\in_0}}}{\frac{1}{\sqrt{1.5\mu_0\times2\in_0}}} = \sqrt{3}$$

20. $Q = CV$

$$\frac{dQ}{dt} = C.\frac{dV}{dt} \Rightarrow \frac{dV}{dt} = \frac{I}{C} = \frac{2\times10^{-3}}{4\times10^{-6}}$$

$$= \frac{10^3}{2} = 500\frac{V}{s}$$

21. Displacement current is equal to conduction correct and flows in same direction.

22. EMW is emitted from charge performing nonuniform motion

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23. Gamma rays has smallest wavelength.

24. $\frac{E}{H}=\left(\frac{E}{\mu.H}\right)\mu=vu$

$= \frac{1\mu}{\sqrt{\mu\varepsilon}}$

$= \sqrt{\frac{\mu_r\mu_0}{\varepsilon_0\varepsilon_r}}=120\pi\times\frac{1}{2}$

$= 60\ \pi : 1$

Exercise-III (Analytical Questions)

2. $\vec{E} = 30 \sin (6 \times 10^8 t + kz)\,\hat{i}$

$E = E_0 \sin(\omega t + kz)$

Direction of propagation = –z axis

wave length $\lambda = \pi = 3.14$ m

Direction of M. field direction of $\vec{v}$ = direction of $\vec{E}\times\vec{B}$

$-\hat{k}=\hat{i}\times\hat{B}$

$\hat{B}=-\hat{j}$

$a \to s, b \to r, c \to p, d \to q$

3. Energy of x ray is higher than radio waves.

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Ray Optics and Optical Instruments

SOLUTIONS

Exercise-I (Conceptual Questions)

1. $\angle i = \angle r$

$\because \angle i = 0, \quad \therefore \angle r = 0$

2. $\delta = 180 - 2i$

$\delta = 180 - 2 \times 30°$

$\delta = 120°$

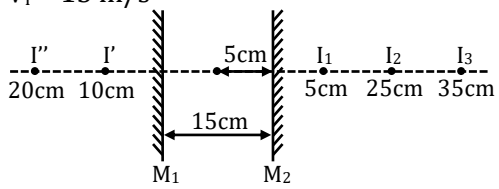
3. $n = \frac{360}{90} = 4$ (even number)

No. of images = 4 – 1 = 3

4. $V_I = V_0$, (with respect ot mirror)

$V_I = 15$ m/s

5.



then ans. will be 5 cm, 25 cm, 35 cm

6. $\delta = 360° - 2\theta$ (for two deviation)

$\delta = 360° - 2 \times 40°$

$\delta = 280°$

7. Required minimum length of the mirror

$= \frac{H}{2} = \frac{6}{2}$ =3 feet

8. Minimum length of plane mirror required

$= \frac{H}{2} = \frac{180}{2} = 90\text{cm}$

9. $n = \frac{360°}{\theta} = \frac{360°}{45} = 8$

n is even so number of images = (n – 1) = 7

10. $m = \frac{f}{f-u}$

m = – 2, → Real, Inverted

$-2 = \frac{-50}{-50-u}$; u = – 75 cm

11. R = 25 cm

f = 12.5 cm

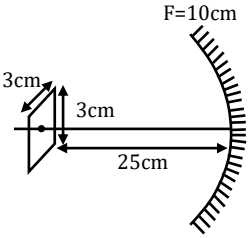
$m = \frac{f}{f-u} = \frac{12.5}{12.5-(-40)}$

m = 0.23

$m = \frac{\text{height of image}}{\text{height of object}}$

height of image = 0.23 × 7.5 = 1.78

12.



Side of square = 3cm

$A_0 = 3 \times 3 = 9\ \text{cm}^2$

$A_I = m^2 \times A_0$

$m = \frac{f}{f-u} = \frac{-10}{-10-(-25)} = \frac{-10}{15} = \frac{-2}{3}$

$A_I = \left(\frac{-2}{3}\right)^2 \times 9 = 4\ \text{cm}^2$

13. $m = \frac{-1}{4}$; $m = \frac{f}{f-u}$; $-\frac{1}{4} = \frac{-12}{-12-u}$

u = –60 cm

14. $m = \frac{f}{f-u}$; $m = \frac{-30}{-30-(-90)}$

$m = \frac{-30}{60}$; $m = -\frac{1}{2}$ → Image size is smaller than object

$\Rightarrow m = \frac{-v}{u}$; $\frac{-1}{2} = \frac{-v}{-90}$

– v = 45; v = – 45 cm

15. $m = \frac{h_i}{h_o} = \frac{-3}{1.5} = -2$ (Real image)

$m = \frac{f}{f-u} = -2 = \frac{f}{f-(-15)}$

–2f – 30 = f

f = –10 cm

16. Erect diminished image formed by convex mirror.

17. $V_I = -m^2 \times V_0$

$m = \frac{f}{f-u} = \frac{-24}{-24-(-60)} = \frac{-24}{36} = \frac{-2}{3}$

$V_I = -\left(\frac{-2}{3}\right)^2 \times 9$;

V_I = –4 cm/sec (toward the mirror)

18. $m = \frac{f}{f-u}$

Image to be real and three times m = –3

$-3 = \frac{f}{f-(-20)}$

f = –15 cm

20. $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$\because$ R = 22 cm

$\therefore$ f = 11 cm

Now, $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$\frac{1}{v}+\frac{1}{-14}=\frac{1}{11}$

v = +6.16 [Back side of mirror]

21. Focal length of spherical mirror does not depend on wavelength of light.

22. u = –10 cm, v= –20 cm

$m=-\frac{v}{u}\Rightarrow\frac{-(-20)}{-10}=-2$

$dv = -m^2 \times du$

$\Rightarrow \quad \Delta v = -4 \times 0.1$

$\Rightarrow \quad \Delta v = -0.4$ cm

–ve sign indicate object and image are moving opposite to each other

23. R = 2f; R = 2 × 20 = 40 cm

25. Frequency depend upon the source

26. $\sin\theta_c = \frac{1}{{}_R\mu_D}$; $\left[\because \mu \propto \frac{1}{\lambda}\right]$

$\sin\theta_C = \frac{\lambda_D}{\lambda_R}=\frac{3500\text{Å}}{7000\text{Å}}$; $\sin\theta_C = \frac{1}{2}$; $\theta_C = 30°$

27. Shifting = $t\left(1-\frac{1}{\mu}\right)=6\left(1-\frac{1}{1.5}\right)$

Shifting = 2 cm

28. ${}_{air}\mu_{glass} = \frac{v_{air}}{v_{glass}}$

$v_{air} = 2 \times 10^8 \times 1.5$ m/sec

$v_{air} = 3 \times 10^8$ m/s

${}_{air}\mu_{liquid} = \frac{v_{air}}{v_{liquid}}=\frac{3\times10^8}{2.5\times10^8} = 1.2$

29. $r=\frac{h}{\sqrt{\mu_w^2-1}}=\frac{4}{\sqrt{\left(\frac{5}{3}\right)^2-1}}=3\text{m}$

Diameter of disc = 6m

30. $\mu_R \sin i = \mu \sin\left(\frac{i}{2}\right)$

$1 \times 2\sin\left(\frac{i}{2}\right)\times\cos\left(\frac{i}{2}\right) = \mu \times \sin\left(\frac{i}{2}\right)$

$\cos\left(\frac{i}{2}\right)=\left(\frac{\mu}{2}\right)$

$i = 2\cos^{-1}\left(\frac{\mu}{2}\right)$

31. $\sin\theta_C = \frac{1}{\mu}$

VIBGYOR $\rightarrow \lambda \uparrow$

$\rightarrow \mu \downarrow$

$\rightarrow \theta_C \uparrow$

$\mu \propto \frac{1}{\lambda}$

32. Apparent depth = $\frac{\text{Actual depth}}{\mu}$

and $\mu \propto \frac{1}{\lambda}$

33. from one side

$(d_{app})_1 = \frac{(d_{ac})_1}{\mu}$

$(d_{ac})_1 = \mu(d_{app})_1 = 1.5\times5 \Rightarrow 7.5\text{cm}$

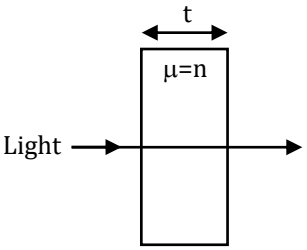
from second surface

$(d_{ac})_2 = \mu(d_{app})_2 = 1.5\times2 \Rightarrow 3.0\text{cm}$

Total $d_{ac} = (d_{ac})_1 + (d_{ac})_2 = 7.5+3.0 \Rightarrow 10.5\text{cm}$

34. Frequency doesn't change with change in medium

35.



$\mu=\frac{c}{v_m}\Rightarrow v_m=\frac{c}{\mu}=\frac{c}{n}$ ...(1)

$\because$ Velocity = $\frac{\text{displacement}}{\text{time}}=\frac{t}{\text{time}}=v_m$ ...(2)

(1) = (2)

$\frac{c}{n}=\frac{t}{\text{time}}\Rightarrow \text{time}=\frac{nt}{c}$

36. $\sin\theta_C = \frac{\mu_R}{\mu_D}$

$\theta_C = \sin^{-1}\left(\frac{\mu_R}{\mu_D}\right) = \sin^{-1}\left(\frac{\frac{4}{3}}{\frac{3}{2}}\right)$

$\theta_C = \sin^{-1}\left(\frac{8}{9}\right)$

37. $\sin\theta_C = \frac{1}{{}_a\mu_g} = \frac{2}{3}$; $\sin\theta_C = \frac{1}{{}_a\mu_\omega} = \frac{3}{4}$

$\Rightarrow C_w > C_g$

38. $\sin\theta_c = \frac{1}{\mu}$ & $\mu \propto \frac{1}{\lambda}$

42. If object will be invisible then R.I. of material and R.I. of medium should be same.

43. $\delta_m = (\mu_P - 1)\,A$

$\delta_m = (1.52 - 1) \times 4.5°$

$\delta_m = 0.52 \times 4.5°$

$\delta_m = 2.34°$

44. $\delta_m = 2i - A$

$e = i = \frac{3}{4} \times 60° \Rightarrow i = 45°$

$\delta_m = 2 \times 45° - 60°$

$\delta_m = 30°$

45. $\mu = \frac{\sin\left(\frac{\delta_m + A}{2}\right)}{\sin\frac{A}{2}}$; $\mu = \frac{\sin\left(\frac{30° + 60°}{2}\right)}{\sin\frac{60°}{2}}$;

$\frac{\sin 45°}{\sin 30°} = \frac{1}{\sqrt{2}} \times 2$; $\mu = \sqrt{2}$

46. $\sin\theta_C = \frac{1}{\mu}$; $\sin 45° = \frac{1}{\mu}$

$\mu = \sqrt{2}$

48. $\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\text{Sin}\frac{A}{2}}$

A is same for all prism

Here $\mu_1 < \mu_2 < \mu_3$

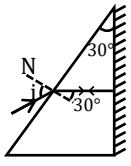
$\therefore \quad \delta_1 < \delta_2 < \delta_3$

49. Light should be incident $\perp$ on silvered surface

$\sin i = \mu \sin r$

$\sin i = \sqrt{2}\sin 30$

$i = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45°$



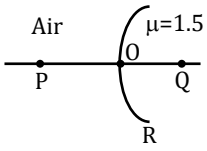
50. $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R} \Rightarrow \frac{1.5}{v} - \frac{1}{-15} = \frac{1.5 - 1}{30}$;

$v = -30$ cm

51. PO = OQ ; u = v

$\frac{1.5}{u} - \frac{1}{-u} = \frac{1.5 - 1}{R}$

$\frac{1.5 + 1}{u} = \frac{0.5}{R}$; $\frac{2.5}{u} = \frac{0.5}{R}$



$u = 5R$

52. $P = \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$; $P = \frac{100}{20} + \frac{100}{25} = +9D$

53. $P = P_1 - P_2 \Rightarrow P = +2.50 - 3.75$

$P = -1.25$

$f = \frac{-100}{1.25} = -80$ cm

54. $P_{net} = P_1 + P_2$; $P_{net} = 3 - 5$; $P_{net} = -2D$

$f = -\frac{100}{2} = -50$cm

$\frac{1}{v} - \frac{1}{-50} = \frac{1}{-50}$; $v = -25$ cm

55. $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$

$\Rightarrow P = \frac{1}{f} = \frac{100}{40} - \frac{100}{25} = 2.5 - 4$

$\Rightarrow P = -1.5$ D

56. $f = \sqrt{x_1 x_2}$

$10 = \sqrt{5x_2} \Rightarrow x_2 = 20$ cm

distance of image from lens = $x_2 + f = 30$ cm

57. $f = \frac{100}{5} = +20$ cm

$\frac{1}{20} = (1.5 - 1)\left(\frac{1}{R} - \frac{1}{-R}\right)$

$\frac{1}{20} = \frac{0.5 \times 2}{R}$; $R = 20$ cm



58. $f^2 = x_1 x_2$

$f^2 = 9 \times 25 \Rightarrow f = 15$ cm

59. $\frac{1}{f_{air}} = \{\mu_L - 1\}\left\{\frac{1}{R_1} - \frac{1}{R_2}\right\}$ ...(1)

$\frac{1}{f_m} = \left\{\frac{\mu_L - \mu_m}{\mu_m}\right\}\left\{\frac{1}{R_1} - \frac{1}{R_2}\right\}$ ...(2)

Equation $\frac{(1)}{(2)} = \frac{f_m}{f_{air}} = \frac{(\mu_L - 1)}{\left(\frac{\mu_L - \mu_m}{\mu_m}\right)}$

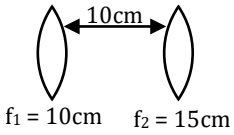
$$\frac{f_m}{8}=\frac{\left(\frac{3}{2}-1\right)}{\left\{\frac{\frac{3}{2}-\frac{4}{3}}{\frac{4}{3}}\right\}}\Rightarrow\frac{\frac{1}{2}}{\frac{1}{8}}=4$$

$f_m = 4 \times 8 = 32$ cm

60. $\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}-\frac{d}{f_1f_2}$

$\frac{1}{f}=\frac{1}{10}+\frac{1}{15}-\frac{10}{10\times15}$

$\frac{1}{f}=\frac{1}{10}+\frac{1}{15}-\frac{1}{15}=\frac{1}{10}$



$f = 10$ cm

61. Power = $\frac{1}{\text{Focus}}$

due to immersed in water f↑

$\frac{1}{f}=\left(\frac{\mu_L}{\mu_m}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

62. R.I ∝ no. of images

63. $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

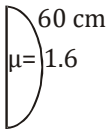
$\frac{1}{10}=(1.5-1)\left(\frac{1}{7.5}-\frac{1}{R_2}\right)$



$R_2 = -15$ cm

64. $\frac{1}{f}=(1.6-1)\left(\frac{1}{60}\right)$

$\frac{1}{f}=\frac{0.6}{60}\Rightarrow f = 100$ cm



65. $\frac{1}{f}=(\mu_L-1)\left\{\frac{1}{\infty}-\frac{1}{-R}\right\}$

$\frac{1}{f}=(\mu_L-1)\times\frac{1}{R}$



$\Rightarrow \quad \frac{1}{30}=(\mu-1)\frac{1}{10}$

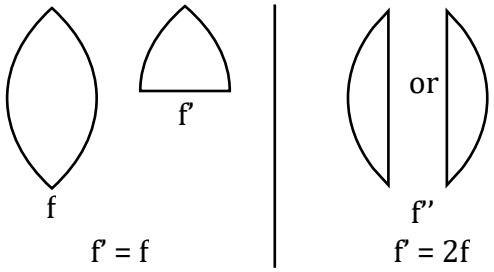
$\Rightarrow \quad \mu=\frac{1}{3}+1=\frac{4}{3}=1.33$

66. $P = P_1 + P_2 = \frac{100}{80}+\frac{100}{(-50)}$

$P = 1.25 - 2$

$P = -0.75$ D

68.



$\frac{1}{f}=(\mu-1)\left(\frac{2}{R}\right)$

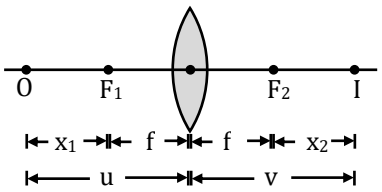
$f=\frac{R}{2(\mu-1)}$

$f'=\frac{R}{2(\mu-1)}$

$f''=\frac{R}{\mu-1}$

$\therefore$ f' = f & f'' = 2f

69.



Here, $x_1 = 25$ cm, $x_2 = 40$ cm

According to Newton's formula

$f^2 = x_1x_2 = (25)(40)$

$f=\sqrt{(25)(40)}=10\sqrt{10}\,\text{cm}=32\,\text{cm}$

Object distance, $u=-(f+x_1)=-(32+25)\text{cm} = -57\text{cm}$

Image distance, $v=(f+x_2)=(32+40)\text{cm} = 72\text{cm}$

Magnification, $m=\frac{v}{u}=\frac{72}{-57}=-1.3$

70. $m = 1+\frac{D}{f_e}$

$m = 1+\frac{25}{2.5}$; $m = 11$

71. $m = \frac{-f_0}{f_e}=-8$

$|f_0| + |f_e| = 45$

$8f_e + f_e = 45$

$9f_e = 45$

$f_e = 5$ cm

$f_0 = 40$ cm

72. $m_e \times m_0 = 35$

$m_e \times 7 = 35$

$m_e = 5$

73. $MP = \frac{-f_0}{f_e}\left(1+\frac{f_e}{D}\right)$

$MP = \frac{-100}{10}\left(1+\frac{10}{25}\right)$

$MP = -14$

74. $|f_0| + |f_e| = 105$ and $-\frac{f_0}{f_e} = -20$; $f_0 = 20\, f_e$

Now $20\, f_e + f_e = 105 \Rightarrow 21\, f_e = 105$

$f_e = 5$ cm and $f_0 = 100$ cm

75. $m = 1+\frac{D}{f_e}$; $m = 1+\frac{25}{5}$

$m = 6$

77. $MP = \frac{\beta}{\alpha} = -\frac{f_0}{f_e} = \frac{\beta}{0.5} = -\frac{100}{2}$

$\beta = -25°$

78. $M = -\frac{f_0}{f_e} = -4$

$f_0 = 4f_e$ ....(i)

$L = |f_0| + |f_e|$

$10 = |f_0| + |f_e|$ ....(ii)

from (i) & (ii) equations

$10 = |4f_e| + |f_e|$

$f_e = 2$ cm

$f_0 = 8$ cm i.e. L_4, L_1

79. Image is at D distance

$MP = \left(\frac{D}{f_e}+1\right) \Rightarrow 6 = \frac{25}{f_e}+1$

$5 = \frac{25}{f_e} \Rightarrow f_e = 5$ cm or $= 0.05$ m

80. $MP = -\frac{f_0}{f_e}$

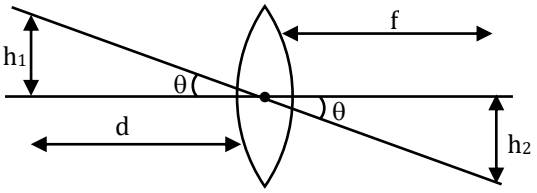
$-\frac{f_0}{20} = -10\, b$

$f_0 = 200$ cm

OR

The focal length of objective lens is greater than eye piece so answer should greater than 20.

81.



$\tan\theta = \frac{h_1}{d} = \frac{h_2}{f}$

$h_2 = \frac{h_1 \times f}{d} = \frac{50\times200\times10^{-2}}{2\times10^3} = 5\times10^{-2}\,m$

$h_2 = 5$ cm

Exercise-II (Previous Year Questions)

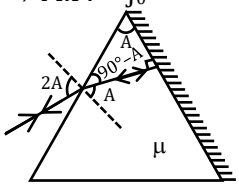
1. Magnifying power of Microscope $= \frac{LD}{f_0 f_e}$

$\Rightarrow$ M.P. $\propto \frac{1}{f_0}$

Magnifying power of Telescope $= \frac{f_0}{f_e}$

$\Rightarrow$ M.P. $\propto f_0$

2.



By Snell's law

(1) $\sin 2A = (\mu)\sin A \Rightarrow \mu = 2\cos A$

$2\sin A\cos A = \mu \sin A$

$2\cos A = \mu$

3. $\mu = \frac{\sin\left(\frac{\delta_m + A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

$\because \mu = \cot(A/2)$

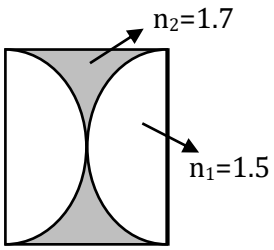
$\therefore \cot(A/2) = \frac{\sin\left(\frac{\delta_m + A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

$\Rightarrow \cos(A/2) = \sin\left(\frac{\delta_m + A}{2}\right)$

$\Rightarrow 90° - A/2 = \frac{\delta_m + A}{2}$

$\Rightarrow \delta_m = 180° - 2A$

4.



From lens maker's formula

$$\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

we have $\frac{1}{f_1}=(1.5-1)\left(\frac{1}{20}\right)=\frac{1}{40}$

$$\frac{1}{f_2}=(1.5-1)\left(\frac{1}{20}\right)=\frac{1}{40}$$

& $\frac{1}{f_3}=(1.7-1)\left(\frac{2}{-20}\right)=-\frac{7}{100}$

Now $\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}+\frac{1}{f_3}$

$\Rightarrow \frac{1}{f}=\frac{1}{40}+\frac{1}{40}-\frac{7}{100}$

$\Rightarrow$ f = – 50 cm

5. i = 45°; A = 60°; δ_m = 2i – A = 30°

$$\mu=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin A/2}=\frac{\sin 45^\circ}{\sin 30^\circ}=\frac{1}{\sqrt{2}}.\frac{2}{1}=\sqrt{2}$$

6. Using lens formula for objective lens

$\Rightarrow \frac{1}{v_0}-\frac{1}{u_0}=\frac{1}{f_0} \Rightarrow \frac{1}{v_0}=\frac{1}{f_0}+\frac{1}{u_0}$

$\Rightarrow \frac{1}{v_0}=\frac{1}{40}+\frac{1}{-200}=\frac{+5-1}{200}$

$\Rightarrow v_0$ = 50 cm

Tube length $\ell = |v_0| + f_e$ = 50 + 4 = 54 cm.

7. For spherical mirrors

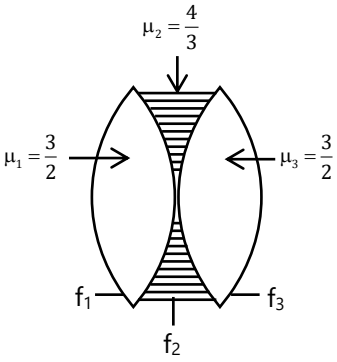
m = +ve $\Rightarrow$ virtual image

m = –ve $\Rightarrow$ real image

|m| > 1 $\Rightarrow$ magnified image

|m| < 1 $\Rightarrow$ diminished image

8.



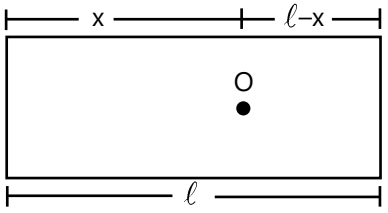
$$f_1=f_3=\frac{R}{2\left(\frac{3}{2}-1\right)}=R=f \text{ (given)}$$

$$f_2=\frac{-R}{2\left(\frac{4}{3}-1\right)}=\frac{-3}{2}R=-\frac{3}{2}f$$

$$\frac{1}{f_{eq}}=\frac{1}{f_1}+\frac{1}{f_2}+\frac{1}{f_3}=\frac{1}{f}+\left(-\frac{2}{3f}\right)+\frac{1}{f}$$

$$\Rightarrow \frac{1}{f_{eq}}=\frac{4}{3f} \Rightarrow f_{eq}=\frac{3f}{4}$$

9.



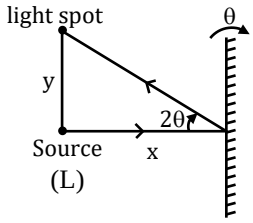
$\frac{x}{\mu}=5cm$ ... (i)

$\frac{\ell-x}{\mu}=3cm$ ... (ii)

From (i) and (ii)

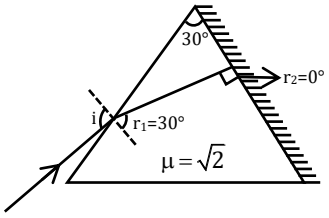
$\ell = (5+3)\mu = 12cm$

10.



$2\theta=\frac{y}{x}$; $\theta=\frac{y}{2x}$

11.

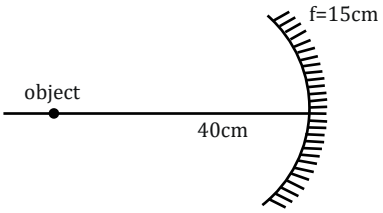


$r_2 = 0$, $r_1 = A = 30^\circ$

By using snell's law

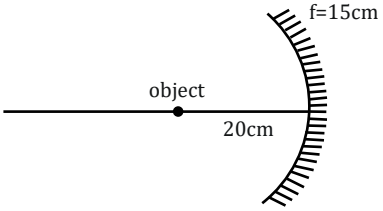
$\sin i = \sqrt{2}\sin 30° = \sqrt{2}\times\frac{1}{2} = \frac{1}{\sqrt{2}} \Rightarrow \angle i = 45°$

12. (i)



$v_1 = \frac{uf}{u-f} = \frac{(-40)(-15)}{-40+15} = \frac{600}{-25} = -24\text{cm}$

(ii)



$v_2 = \frac{uf}{u-f} = \frac{(-20)(-15)}{-20+15} = -60$ cm

Displacement of image = $v_2 - v_1 = -36$ cm

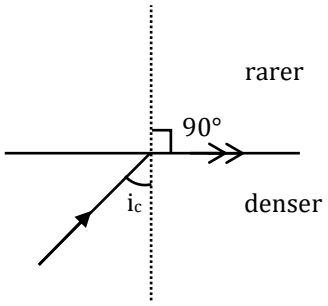
= 36 cm away from the mirror

13. Longest wavelength is of red colour.

14. $\frac{1}{F_1} = \frac{1}{f} + \frac{1}{f} \Rightarrow F_1 = f/2$

And $F_2 = f \Rightarrow \frac{F_1}{F_2} = \frac{1}{2}$

15. At critical angle



angle of refraction = 90°

16. Focal length does not change → Power does not change

17. $\frac{1}{f} = \left(\frac{\mu_2-\mu_1}{\mu_1}\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

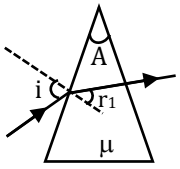
$\Rightarrow \frac{1}{25} = \left(\frac{1.5-1}{1}\right)\left(\frac{1}{R}-\frac{1}{(-2R)}\right)$

$\Rightarrow \frac{2}{25} = \frac{3}{2R} \Rightarrow R = \frac{75}{4} = 18.75$

Iˢᵗ surface radius = 18.75 cm

IIⁿᵈ surface radius = 37.50 cm

18.



Here $r_2 = 0$ so $r_1 = A$

Apply Snell's law $\sin i = \mu \sin r_1 = \mu \sin A$

for small angle

19. Spherometer is used to measure radius of curvature of the curved surface.

20. $P = \frac{100}{f} \Rightarrow f = \frac{100}{p} = \frac{100}{10} = 10\text{cm}$

$f = \frac{R}{2(\mu-1)}$ (for equiconvex lens)

$\Rightarrow 10 = \frac{10}{2(\mu-1)}$

$\Rightarrow (\mu-1) = \frac{1}{2} \Rightarrow \mu = \frac{1}{2}+1 = \frac{3}{2}$

21. By using mirror formula $\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$

We have $\frac{1}{-1.5\,f}+\frac{1}{v}=\frac{1}{-f} \Rightarrow \frac{1}{v} = -\frac{1}{f}+\frac{1}{1.5\,f}$

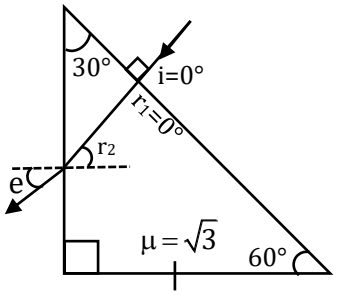
$\Rightarrow \frac{1}{v} = \frac{-1.5+1}{1.5\,f} = \frac{-0.5}{1.5\,f} \Rightarrow v = -3f$

22. $\sin\theta_C = \frac{1}{\mu}$

$\mu = \frac{1}{\sin\theta_C} = \frac{1}{\sin 45°} = \frac{1}{(1/\sqrt{2})} = \sqrt{2}$

$\mu = \frac{c}{v} \Rightarrow v = \frac{c}{\mu} = \frac{3\times10^8}{\sqrt{2}}$ m/s

23.



$r_1 + r_2 = A = 30°$

$r_2 = 30°$ ($r_1 = 0°$)

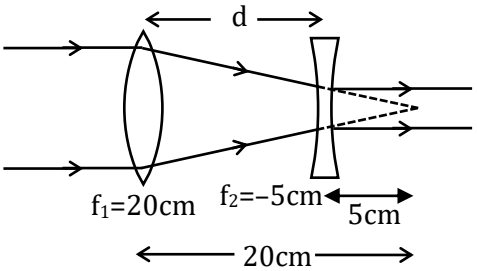
from Snell's law

$\sqrt{3}\sin r_2 = 1\times\sin e$

$\sqrt{3}\sin 30° = \sin e$

$e = 60°$

24.



$d = f_1 + f_2$

$= 20 - 5$

$= 15$ cm

28. Speed of light is air $V_1 = \frac{x}{t_1}$

speed of light is a medium $V_2 = \frac{10x}{t_2}$

$\sin\theta_c = \frac{V_2}{V_1} = \frac{10x}{t_2}\frac{t_1}{x}$

$\theta_c = \sin^{-1}\left(\frac{10t_1}{t_2}\right)$

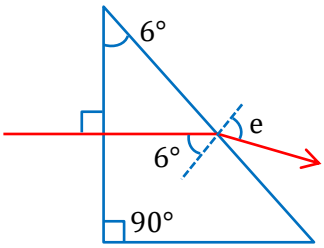
29. $\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}$

$\frac{1}{f_{eq}} = \frac{1}{f} - \frac{1}{f}$

$f_{eq} = \infty$

30. Since lens is made of three materials so three μ and hence three images.

31.



$A = r_1 + r_2 = 6°$

$\mu = 1.5\ \frac{\sin e}{\sin r_2} = \frac{\sin e}{\sin 6}$

$\text{sine} = 1.5 \times 6°$

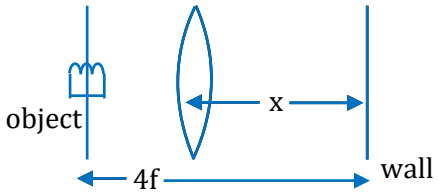
$e \simeq 9°$

32. $\text{Sin}\, i_c = \frac{1}{\mu} \propto \lambda$

$i_c \propto \lambda$

Yellow, orange, red emerge from air.

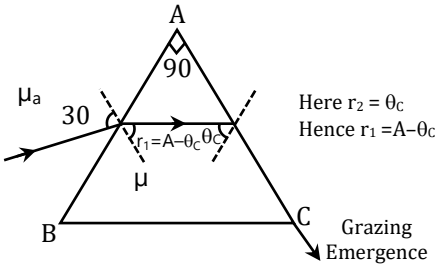
33.



$x = 2f$

$f = x/2$

34.



By snell's law

$\mu_a \sin 30 = \mu \sin r_1$

$\Rightarrow$ (1) $\sin 30 = \mu \sin(A - \theta_C)$

$\Rightarrow \sin 30 = \mu \sin(90 - \theta_C)$

$\Rightarrow \sin 30 = \mu \cos\theta_C$

$\Rightarrow \sin 30 = \mu\ \frac{\sqrt{\mu^2 - 1}}{\mu}$

$\Rightarrow \frac{1}{2} = \sqrt{\mu^2 - 1}$ $\quad \because \sin\theta_C = \frac{1}{\mu}$

$\Rightarrow \frac{1}{4} = \mu^2 - 1$ $\quad \cos\theta_C = \frac{\sqrt{\mu^2 - 1}}{\mu}$

$\Rightarrow \mu^2 = 5/4$

$\Rightarrow \mu = \frac{\sqrt{5}}{2}$

35. For Telescope –

Magnifying power

$\text{M.P.} = \frac{-fo}{fe} = \frac{-140}{5} = -28$

MP = –28

36. **Statement 1 is correct :-** Image is only formed by regular reflection not by irregular reflection.

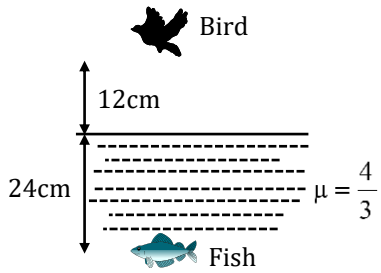
Statement 2 is correct :- The variety in colour of objects we see around us is due to the constituent colours of the light incident on them.

Exercise-III (Analytical Questions)

2. Simple microscope forms erect image while compound microscope form inverted image.

3. $M.P \propto \frac{1}{f}$ (for normal adjustment M.P = $\frac{D}{f}$)

4.



(A) $(d_{fb})_{surface} = \frac{24}{4/3} \times 1 = 18$ cm

(B) $(d_{bf})_{surface} = \frac{12}{1} \times \frac{4}{3} = 16$ cm

(C) $d_{fb} = 12 + \frac{24}{4/3} \times 1 = 12 + 18 = 30$ cm

(D) $d_{bf} = 24 + \frac{12}{1} \times \frac{4}{3} = 24 + 16 = 40$ cm

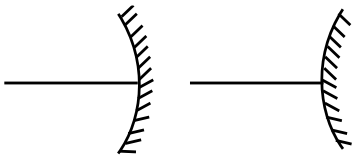
6. When mirror moves parallel to its plane, then image will not move.

7. $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ Both are same for thin lens.

8. On covering upper half of the lens its aperture decreases which results in decreasing of intensity but nature, position, shape and size of image remain same.

9. $M.P. = \frac{D}{f}$ and D may be different for different persons.

10. $v = \frac{uf}{u - f}$



(A) u = –f, f = +f $\quad v = \frac{-f \times f}{-f - f} = \frac{+f}{2} \Rightarrow m = -\frac{v}{u} = \frac{1}{2}$

(B) u = –f, f = –f $\quad v = \frac{-f \times -f}{-f - (-f)} = \infty \Rightarrow m = \infty$

(C) u = –2f, f = –f $\quad v = \frac{-2f \times -f}{-2f - (-f)} = -2f \Rightarrow m = -1$

(D) u = –2f, f = +f $\quad v = \frac{-2f \times f}{-2f - f} = \frac{+2}{3} f \Rightarrow m = \frac{1}{3}$

11. For $A > 2\theta_C$; the light ray never crosses the prism for any angle of incidence.

14. $e = (A + \delta) - i = (30° + 30°) - 60° = 0$

AIM: FREE EDUCATION TO ALL APUL

Wave Optics

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Polarization is not explained by Huygen's wave theory.

2. Huygen's principle is applicable to all type of waves.

3. Huygen's theory explained propagation of light in medium, reflection of light, refraction of light.

4. Huygen's theory of secondary waves can be used to find wavefront geometrically.

5. Huygen's theory not discussed about Newton's ring but his assumption of ether medium is not experimentally verified (failure of experimental verification)

6. Reflection & refraction both are explained by Newton's & Huygen's theory but interference explained the wave nature of light.

7. Wave nature of light is verified by interference.

8. Energy in the phenomenon of interference is conserved, only it gets redistributed.

9. $A_{Res} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\Delta\phi)}$

10. Transverse and longitudinal nature of light wave is not supported by the phenomenon of interference.

11. For distinct interference pattern to be observed

$\frac{I_1}{I_2} = 1$

$\therefore I_{max} = 4I$ & $I_{min} = 0$

12. $\frac{\Delta\phi}{2\pi} = \frac{\Delta x}{\lambda}$ $\{\Delta x = x\}$

$\boxed{\Delta\phi = \frac{2\pi x}{\lambda}}$

13. For constructive interference

$\Delta x = n\lambda$

$\Delta\phi = 2n\pi$

14. Two independent lamp can never be coherent.

$\therefore$ phase difference is not constant.

15. Coherent sources can be obtained by both division of wavefront & division of Amplitude.

16. $\Delta x = \frac{3\lambda}{2}$ is the condition of destructive interference $\therefore$ dark fringe is formed.

17. $I_A = I_1 + I_2 + 2\sqrt{I_1I_2}\cos\phi$

$I_A = I + 4I + 2\sqrt{I(4I)}\cos\frac{\pi}{2}$

$I_A = 5I$ ...(i) $\left\{\cos\frac{\pi}{2} = 0\right\}$

$I_B = I + 4I + 2\sqrt{I(4I)}\cos 2\pi$

$I_B = 9I$ ....(ii) $\{\cos 2\pi = 1\}$

$I_B - I_A = 9I - 5I = 4I$

18. $I_{min} \propto (a_1 - a_2)^2 \Rightarrow (a - 2a)^2 \Rightarrow a^2$

19. Constructive interference

$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2 \Rightarrow \left(\sqrt{I} + \sqrt{I}\right)^2 \Rightarrow 4I$

20. $I_R = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2}\cos\phi$

$= I + I + 2\sqrt{I \times I}\cos 120°$

$= I + I + 2I\left(-\frac{1}{2}\right)$ $\left\{\cos 120° = -\frac{1}{2}\right\}$

$I_R = I$

21. $\frac{I_1}{I_2} = \frac{25}{1}$

$\frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_1} + \sqrt{I_2}\right)^2}{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2} \Rightarrow \frac{\left(\sqrt{25} + \sqrt{1}\right)^2}{\left(\sqrt{25} - \sqrt{1}\right)^2}$

$\frac{I_{max}}{I_{min}} \Rightarrow \frac{(5+1)^2}{(5-1)^2} \Rightarrow \frac{36}{16} = \frac{9}{4}$

22. $I_{AV} = I_1 + I_2 = 2 + 3 = 5$

23. Two independent light source are incoherent.

24. Interference is shown by all types of wave.

25. For sustained interference the necessary condition is that two sources should have constant phase difference.

26. $\frac{I_{max}}{I_{min}}=\left(\frac{a_1+a_2}{a_1-a_2}\right)^2$

$=\left\{\frac{3+5}{3-5}\right\}^2=\left(\frac{8}{2}\right)^2=\left(\frac{4}{1}\right)^2=\frac{16}{1}$

27. $I_{max}=\left(\sqrt{I_1}+\sqrt{I_2}\right)^2=\left(\sqrt{I}+\sqrt{4I}\right)^2$

$\Rightarrow \quad \left(\sqrt{I}+2\sqrt{I}\right)^2=9I$

$I_{min}=\left(\sqrt{I}-\sqrt{4I}\right)^2\Rightarrow\left(\sqrt{I}-2\sqrt{I}\right)^2=I$

28. $I_{max}=\left(\sqrt{I_1}+\sqrt{I_2}\right)^2$

29. $y_1 = a\sin\omega t$ ...(1)

$y_2 = a\cos\omega t$

or

$y_2 = a\sin\left(\frac{\pi}{2}+\omega t\right)$ ...(2)

First wave lags $\frac{\pi}{2}$ from Second wave

or

Second wave leads $\frac{\pi}{2}$ from first wave

30. $A=\sqrt{a_1^2+a_2^2+2a_1a_2\cos\phi}$

$=\sqrt{a^2+a^2+2a^2\cos\frac{\pi}{3}}\Rightarrow a\sqrt{3}$

31. $\frac{I_{max}}{I_{min}}=\left(\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right)^2\Rightarrow\frac{25}{1}$

$\frac{5}{1}=\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}$

$\frac{\sqrt{I_1}}{\sqrt{I_2}}=\frac{3}{2}$ or $\frac{I_1}{I_2}=\frac{9}{4}$

32. Path difference of destructive interference

$=(2n+1)\frac{\lambda}{2}$

33. $\frac{I_{max}}{I_{min}}=\left(\frac{a_1+a_2}{a_1-a_2}\right)^2$

$\frac{36}{1}=\left(\frac{a_1+a_2}{a_1-a_2}\right)^2$

$\frac{6}{1}=\frac{a_1+a_2}{a_1-a_2}$ or $\frac{a_1}{a_2}=\frac{7}{5}$

34. $\Delta x_{slab} = (\mu-1)t$

35. Thickness of oil layer ≈ 0.1 μm

36. $\frac{I_1}{I_2}=\frac{9}{1}$ or $\left(\frac{a_1}{a_2}\right)^2=\frac{9}{1}\Rightarrow\frac{a_1}{a_2}=\frac{3}{1}$

$\frac{A_{max}}{A_{min}}=\left(\frac{a_1+a_2}{a_1-a_2}\right)=\left(\frac{3a_2+a_2}{3a_2-a_2}\right)$ = 2 : 1

37. For coherent sources $\Delta\phi$ should be constant.

38. Thin film of oil on water appear colourful due to the phenomenon interference.

39. $\frac{I_1}{I_2}=\frac{9}{1}$ or $\left(\frac{a_1}{a_2}\right)^2=\frac{9}{1}\Rightarrow\frac{a_1}{a_2}=\frac{3}{1}$

$\frac{I_{max}}{I_{min}}=\left(\frac{a_1+a_2}{a_1-a_2}\right)^2\Rightarrow\frac{4}{1}$

40. Soap bubble appear colourful due to the phenomenon of interference by division of amplitude.

41. Two coherent light sources emit light of the same frequency having constant phase difference.

42. If the amplitude of interferring waves are unequal then intensity is also not equal and fringe visibility is not 100%

$\therefore$ contrast in the fringes decreases.

43. Y.D.S.E. proves that light is made up of waves.

44. $\beta=\frac{\lambda D}{d}$, if $d\uparrow\Rightarrow\beta\downarrow$

45. In Y.D.S.E. both interference & diffraction occurs.

46. If the amplitude of interferring waves are unequal then intensity is also not equal and fringe visibility is not 100%

$\therefore$ contrast in the fringes decreases and dark fringe will become less dark and bright fringe will become less bright.

47. Nearest to the central bright fringe red colour appear.

48. Third Bright

$\Delta x = 3\lambda$

$\Delta\phi = 6\pi$

49. $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

$I_R = I_0 + I_0 + 2\sqrt{I_0 I_0}\cos\phi$

$\boxed{I_R = 2I_0(1+\cos\phi)}$

51. In presence of thin mica sheet fringe width remain same but fringe pattern shift (w' = w)

52. $\frac{I_{coherent}}{I_{noncoherent}} = \frac{4I_0}{2I_0} = \frac{2}{1}$

53. Width of the slit radio is 4 : 9

Intensity $\propto$ width of slit $\frac{I_1}{I_2} = \frac{4}{9}$

$$\frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2}{\left(\sqrt{I_1}-\sqrt{I_2}\right)^2} = \left(\frac{2+3}{2-3}\right)^2 = \frac{25}{1}$$

54. Spacing between successive maxima or successive minima is fringe width (β) = $\frac{\lambda D}{d}$

55. $\beta_{water} = \frac{\beta_{air}}{\mu}$

56. Interference pattern is not formed by incoherent source.

57. $y = \frac{(\mu-1)tD}{d}$

58. $\beta \propto \lambda$

$$\frac{\beta_\ell}{\beta_a} = \frac{\lambda_\ell}{\lambda_a} \qquad \left\{\begin{matrix}\lambda_\ell = \frac{\lambda_a}{\mu_\ell} \\ \frac{\lambda_\ell}{\lambda_a} = \frac{1}{\mu_\ell}\end{matrix}\right\}$$

$$\beta_\ell = \beta_a\left(\frac{1}{\mu_\ell}\right)$$

$$\beta_\ell = \frac{0.8\text{mm}}{1.6} = 0.5\text{mm}$$

59. $x = \frac{(\mu-1)tD}{d}$

$\Rightarrow \quad x = \frac{(1.5-1)\times 2.5\times 10^{-5}\times 1}{0.5\times 10^{-3}}$

$\Rightarrow \quad 2.5$ cm

60. In presence of thin mica sheet fringe width remain same but fringe pattern shift (w' = w)

61. $\lambda = 6000$ Å

C.F. is shifted to 4th bright fringe

$\mu = 1.5$ and $\Delta x = 4\lambda$

$\Rightarrow \Delta x = (\mu - 1)t = n\lambda$

$\Rightarrow (1.5 - 1)t = 4 \times 6000 \times 10^{-10}$

$t = 48000 \times 10^{-10} \Rightarrow 4.8 \times 10^{-6}$ m $= 4.8$ μm

62. $\because$ Central fringe at same position.

$\therefore$ Path difference (Δx) = 0

Now $(\mu_1 - 1)t_1 = (\mu_2 - 1)t_2$

$(1.5-1) \times 1.2 = (2.5-1)t_2$

$t_2 = \frac{0.5\times 1.2}{1.5} = 0.4\mu\text{m}$

63. $\beta = 0.2$ mm $= 0.2 \times 10^{-3}$ m

$\lambda' = \frac{11}{10}\lambda$ & $d' = \frac{11}{10}d$

$\beta = \frac{\lambda D}{d}$ & $\beta' = \frac{\lambda' D}{d'}$

$$\beta' = \frac{\frac{11}{10}\lambda D}{\frac{11}{10}d} = \frac{\lambda D}{d}$$

$\therefore \quad \beta' = \beta$

64. When thickness tends to zero

thin film appear dark in reflected system

$\Delta x = 2\mu t\cos r - \frac{\lambda}{2}$

65. Central fringe is white and all other fringes are coloured.

66. $\beta = \frac{\lambda D}{d}$ $\therefore \beta \propto \lambda$

67. $\beta = \frac{\lambda D}{d} = \frac{550\times 10^{-9}\times 1}{1.1\times 10^{-3}} = 500 \times 10^{-6} \Rightarrow 0.5$ mm

68. $\beta = \frac{\lambda D}{d} \Rightarrow \beta \propto \lambda$

λ is least for violet

$\Rightarrow \beta$ is least for violet, here blue for given options.

69. $\lambda_{Red} > \lambda_{Yellow}$ (Sodium lamp)

$\beta_{Red} > \beta_{Yellow}$

70. $(\mu-1)t = n\lambda \Rightarrow \quad n = \dfrac{(\mu-1)t}{\lambda}$

$\Rightarrow n = \dfrac{(1.5-1)\times 2\times 10^{-6}}{500\times 10^{-9}}$

n = 2 (Shift by 2 fringes upwards)

71. $\Delta x = (2n-1)\,\lambda/2$ & n = 3

$\therefore \quad \Delta x = \dfrac{5\lambda}{2}$

72. In presence of thin mica sheet fringe width remain same but fringe pattern shift (w' = w)

73. At same spot, so fringes coincide

$n_1\lambda_1 = n_2\lambda_2$

$\Rightarrow \quad 3 \times 700 = 5 \times \lambda_2$

$\Rightarrow \quad \lambda_2 = 420$ nm

74. $\beta = \dfrac{\lambda D}{d}$ or $\beta \propto \lambda$

$\therefore \quad \dfrac{\beta_2}{\beta_1} = \dfrac{\lambda_2}{\lambda_1} \quad \Rightarrow \quad \dfrac{\beta_2}{1\text{mm}} = \dfrac{6000\text{Å}}{5000\text{Å}}$

$\beta_2 = 1.2$ mm

75. $\beta = \dfrac{\lambda D}{d} = \dfrac{5\times 10^{-7}\times 2}{1\times 10^{-3}} = 1$ mm

76. $\because \quad \beta \propto \lambda, \quad \therefore \quad \dfrac{\beta_2}{\beta_1} = \dfrac{\lambda_2}{\lambda_1}$

$\dfrac{\beta_2}{0.7\text{mm}} = \dfrac{500\text{nm}}{700\text{nm}}$

$\beta_2 = 0.5$ mm

77. $\beta = \dfrac{\lambda D}{d}$ & $\beta' = \dfrac{\lambda D}{2d}$

$\therefore \quad \beta' = \dfrac{\beta}{2}$

78. Because of diffraction, waves bend around the corners of obstacle hence sound can travel from inside to outside of room.

79. Diffraction is performed by all types of waves.

80. Radio waves because its wavelength is more than the other options.

81. In Diffraction intensity and width of fringe decreases as we move away from CBF.

82. Width of CBF = $\dfrac{2D\lambda}{a}$

83. Light travels in the form of wave and it get diffracted by the tiny particles.

84. Width of CBF is double the width of other near by fringes.

85. In Diffraction intensity and width of fringe decreases as we move away from CBF.

86. Width of CBF = $\dfrac{2D\lambda}{a}$ and it is of maximum intensity.

87. Width of CBF = $\dfrac{2D\lambda}{a}$

88. Width of CBF = $\dfrac{2D\lambda}{a}$ and if 'a' decreases then width of CBF increases.

89. Width of CBF = $\dfrac{2D\lambda}{a}$, CBF is the region between both the first minima and angular position of first minima is $\sin\theta = \pm\dfrac{\lambda}{a}$

When $\lambda = a$

$\theta = \pm 90° \quad \therefore$ All are correct

90. Width of CBF = $\dfrac{2D\lambda}{a} \approx \dfrac{2f\lambda}{a}$

91. Path difference of nth order minima = $n\lambda$

92. n^{th} minima

$x = \dfrac{n\lambda D}{a} \qquad \{n = 1\}$

$x = \dfrac{\lambda D}{a} \quad \Rightarrow \quad a = \dfrac{\lambda D}{x} = \dfrac{5\times 10^{-7}\times 2}{5\times 10^{-3}}$

a = 0.2 mm

93. $w = \dfrac{2\lambda D}{d}$

$\Rightarrow \quad w = \dfrac{2\times 6\times 10^{-7}\times 2}{0.2\times 10^{-3}}$

$\Rightarrow \quad w = 12$ mm

94. $\sin\theta = \dfrac{n\lambda}{a} \qquad \{n = 1\}$

$\sin\theta = \dfrac{\lambda}{a} \quad \Rightarrow \quad \sin\theta = \dfrac{600\times 10^{-6}}{1.2\times 10^{-3}}$

$\sin\theta = \dfrac{1}{2}$

$\theta = 30°$ or $\dfrac{\pi}{6}$ radian

95. CBF is always bright

96. Angular width of CBF = $\frac{2\lambda}{d}$

97. Diffraction

98. Wave nature of light

99. Grimaldi

100. Angular width of CBF = $\frac{2\lambda}{a}$

it does not depend on D

101. Width of CBF decreases as λ decreases

102. $\lambda_{sound} > \lambda_{light}$

103. Plane

104. Width of CBF = $\frac{2D\lambda}{a}$

105. Polariser is used to produce polarised light

106. Transverse

107. Polarisation, because sound (longitudinal wave) cannot be polarised.

108. Ultrasonic waves (longitudinal)

109. Transverse nature of light is shown by polarisation.

110. $\because \quad \mu = \tan\theta_P$

$\therefore \quad \mu = \tan 60° \Rightarrow \mu = \sqrt{3}$

Now $\sin\theta_C = \frac{1}{\mu}$

$$\sin\theta_C = \frac{1}{\sqrt{3}} \Rightarrow \theta_C = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

111. $\because \quad \mu = \tan\theta_P$

$\therefore \quad n = \tan\theta_P$

$\theta_P = \tan^{-1}(n)$

112. 90°

113. Brewster's law

$\tan\theta_p = \mu$

114. At polarising angle, reflected beam is 100% polarised and reflected & refracted ray are perpendicular to each other.

115. $\because \quad \mu = \tan\theta_P$

$\therefore \quad \mu = \tan 60° = \sqrt{3}$

Now $\mu = \frac{c}{v}$

$$v = \frac{c}{\mu} = \frac{3\times10^8}{\sqrt{3}}$$

$$v = \sqrt{3}\times10^8 \text{ m/s}$$

116. After polarisation intensity becomes 50% of incidence.

117. Intensity of light is twice maximum and twice zero.

118. It reduces the light intensity to half on account of polarisation.

119. Varies between a maximum and zero

120. $\frac{I_0}{2}$ get transmitted and remaining $\frac{I_0}{2}$ does not get transmitted.

Exercise-II (Previous Year Questions)

1. For path difference λ, phase difference = 2π rad

For path difference $\frac{\lambda}{4}$, phase difference = $\frac{\pi}{2}$ rad

As $K = 4I_0$. So, intensity at given point where path difference is $\frac{\lambda}{4}$

$$K' = 4I_0\cos^2\left(\frac{\pi}{4}\right) = 2I_0 = \frac{K}{2}$$

2. Width of central bright fringe

$$= \frac{2\lambda D}{a}$$

$$= \frac{2\times600\times10^{-9}\times2}{1\times10^{-3}}\text{m}$$

$$= 2.4\times10^{-3}\text{ m}$$

$$= 2.4\text{ mm}$$

3. Angular fringe width of interference in double slit experiment = $\frac{\beta}{D} = \frac{\frac{\lambda D}{d}}{D} = \frac{\lambda}{d}$

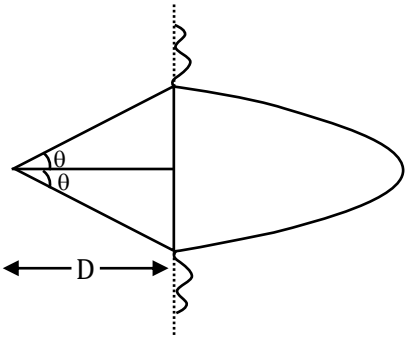
Angular width of central maxima in diffraction experiment = $\frac{2\lambda}{d'}$

According to the question $\frac{10\lambda}{d} = \frac{2\lambda}{d'}$

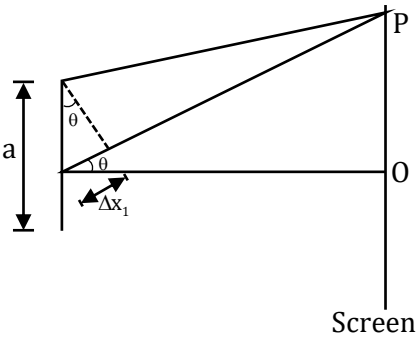
$\Rightarrow d' = 0.2\ d = 0.2$ mm

4. Linear width of central maxima

$$= D(2\theta) = 2D\theta = 2D\frac{\lambda}{a}$$



5.



For first minima at P, $\Delta x_1 = \lambda/2$

So phase difference $\Delta\phi_1 = \frac{\Delta x_1}{\lambda}\times 2\pi = \frac{\lambda/2}{\lambda}\times 2\pi$

$$\Delta\phi_1 = \frac{\lambda}{2\lambda}\times 2\pi = \pi \text{ radian}$$

6. $\frac{I_1}{I_2} = \frac{W_1}{W_2} = \frac{1}{25} \Rightarrow \frac{I_2}{I_1} = \frac{25}{1}$

$$\frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_2}+\sqrt{I_1}\right)^2}{\left(\sqrt{I_2}-\sqrt{I_1}\right)^2} = \left(\frac{\sqrt{\frac{I_2}{I_1}}+1}{\sqrt{\frac{I_2}{I_1}}-1}\right)^2$$

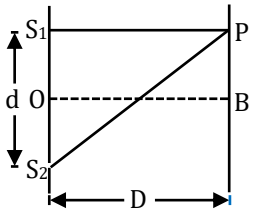
$$= \left(\frac{5+1}{5-1}\right)^2 = \left(\frac{6}{4}\right)^2 = \frac{9}{4}$$

7. For first minima, $\sin 30° = \frac{\lambda}{a} = \frac{1}{2}$

First secondary maxima will be at

$$\sin\theta = \frac{3\lambda}{2a} = \frac{3}{2}\left(\frac{1}{2}\right) \Rightarrow \theta = \sin^{-1}\left(\frac{3}{4}\right)$$

8. Path difference



$= S_2P - S_1P$

$$= \sqrt{D^2+d^2} - D$$

$$= D\left(1+\frac{1}{2}\frac{d^2}{D^2}\right) - D$$

$$= D\left[1+\frac{d^2}{2D^2}-1\right] = \frac{d^2}{2D}$$

$$\Delta x = \frac{d^2}{2\times 10d} = \frac{d}{20} = \frac{5\lambda}{20} = \frac{\lambda}{4}$$

$$\Delta\phi = \frac{2\pi}{\lambda}.\frac{\lambda}{4} = \frac{\pi}{2}$$

So, intensity at the desired point is

$$I = I_0\cos^2\frac{\phi}{2} = I_0\cos^2\frac{\pi}{4} = \frac{I_0}{2}$$

9. Let $\frac{I_1}{I_2} = \frac{n}{1}$

$$\frac{I_{max}-I_{min}}{I_{max}+I_{min}} = \frac{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2-\left(\sqrt{I_1}-\sqrt{I_2}\right)^2}{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2+\left(\sqrt{I_1}-\sqrt{I_2}\right)^2}$$

$$= \frac{4\sqrt{I_1I_2}}{2(I_1+I_2)}$$

Dividing numerator and denominator by I_2

$$\text{required ratio} = \frac{2\sqrt{I_1/I_2}}{\left(\frac{I_1}{I_2}+1\right)} = \frac{2\sqrt{n}}{n+1}$$

10. f = D = 60 cm

For first minima,

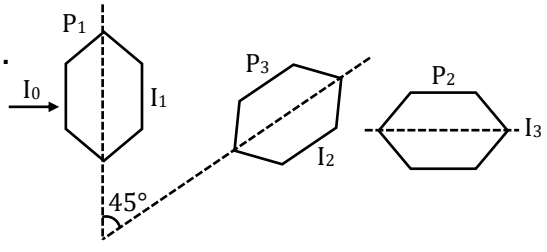
$$y = \frac{\lambda D}{a} = \frac{5\times 10^{-7}\times 60}{2\times 10^{-2}\times 10^{-2}} = \frac{5\times 10^{-3}\times 60}{2} = 0.15 \text{ cm}$$

11. $(y_8)_{\text{Bright, medium}} = (y_5)_{\text{Dark, air}}$

$$\frac{8\lambda_m D}{d} = \left(\frac{2(5)-1}{2}\right)\frac{\lambda D}{d}$$

$$\frac{5\lambda}{\mu}\frac{D}{d} = \frac{9}{2}\frac{\lambda D}{d} \Rightarrow \mu = \frac{16}{9} = 1.78$$

12.



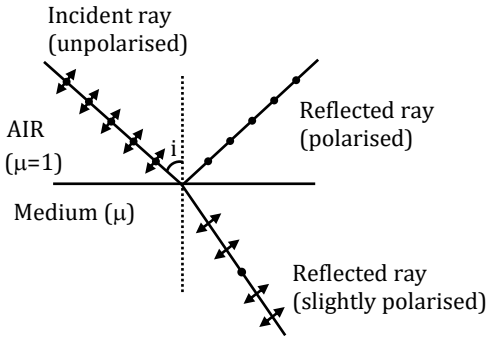
$$I_1 = \frac{I_0}{2}$$

$I_2 = \frac{I_0}{2}\cos^2 45° = \frac{I_0}{4}$

$I_3 = \frac{I_0}{4}\cos^2 45° = \frac{I_0}{8}$

13. When reflected and refracted rays are perpendicular, reflected light is polarised with electric field vector perpendicular to the plane of incidence.

tan i = μ, where i = Brewster's angle or polarisation angle



14. Angular width, $\theta = \frac{\lambda}{d} \Rightarrow \frac{\theta_1}{\theta_2} = \frac{d_2}{d_1}$

$\frac{0.20}{0.21} = \frac{d_2}{2} \Rightarrow d_2 = \frac{0.20}{0.21} \times 2 = 1.9\ \text{mm}$

15. For astronomical refracting telescope

Angular magnification $\left(\frac{f_0}{f_e}\right)$ is more for large focal length of objective lens

Angular resolution or Resolving power

$= \frac{d}{1.22\lambda}$

Resolving power is high for large diameter.

16. $\theta' = \theta/\mu \quad \therefore\ \theta' = \frac{0.2°}{4/3} = 0.15°$

17. Path difference for n^{th} minima = $(2n-1)\frac{\lambda}{2}$

For fifth minima (n=5) = $\frac{9\lambda}{2}$

18. Angular width $\propto \frac{\lambda}{d}$

$\Rightarrow \frac{\theta_0}{0.7\theta_0} = \frac{\frac{6000\,\text{Å}}{d}}{\frac{\lambda}{d}} \Rightarrow \lambda = 4200\ \text{Å}$

19. $\tan i_b = \frac{\mu_2}{\mu_1} = \frac{\mu_2}{1}$

$\because\ \mu_2 > 1\ \therefore \tan i_b > 1 \Rightarrow 90° > i_b > 45°$

20. Here $\beta = \frac{\lambda D}{d}$ & $\beta' = \frac{\lambda D'}{d'}$

Where D' = 2D, d' = $\frac{d}{2}$

$\Rightarrow \beta' = \frac{\lambda \times 2D}{d/2} = \frac{4\lambda D}{d}$

$\Rightarrow \beta' = 4\beta$

Fringe width becomes 4 times

21. For central maximum, the phase difference between the two waves will be zero.

22. (f_0 = focal length of objective)

(a = aperture of objective)

large aperture(a) of the objective lens provides better resolution $\therefore$ good quality of image is formed and also it gathers more light.

26. Angular width, $\theta_w = \frac{\lambda}{d}$

θ_w independent of D but depends on λ

27. At Brewster's angle reflected and refracted rays are perpendicular to each other. Reflected light is completely polarised and refracted light is partially polarised.

28. When white light is used, then path difference due to all the colours at centre will be zero. Hence at centre, central bright white fringe will be observed but surrounding fringes will be coloured.

29. Fringe separation (β) = $\frac{\lambda D}{d}$

$= \frac{6\times10^{-7}\times1}{1.5\times10^{-3}} = 4 \times 10^{-4}\ \text{m}$

30. Output from A = $\frac{I_0}{2}$

Output from B = $\frac{I_0}{2}\cos^2 60 = \frac{I_0}{8}$

Output from C = $\frac{I_0}{8}\cos^2 45 = \frac{I_0}{16}$

Exercise-III (Analytical Questions)

2. $\beta = \frac{\lambda D}{d}$, $\alpha = \frac{\lambda}{d}$

$\left(y_{n^{th}}\right)_{Bright} = \frac{n\lambda D}{d}$; $\left(y_{n^{th}}\right)_{Dark} = \frac{(2n-1)\lambda D}{2d}$

4. $I \propto w$ and $I \propto A^2$; So intensity of source are I & 4I

$I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2 = \left(\sqrt{I} + 2\sqrt{I}\right)^2 = 9I$

$I_{min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2 = \left(\sqrt{I} - 2\sqrt{I}\right)^2 = I$

$I_{max}/I_{min} = 9 : 1$

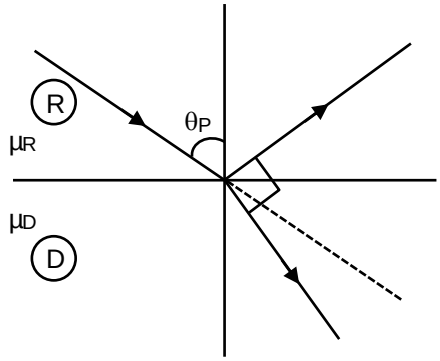
7. For higher order maxima in diffraction, the light waves from major part of aperture meet destructively and light from small part remain present. This remaining effective part of aperture decreases with increase in order of maximum.

10. $Z_f = \frac{a^2}{\lambda}$

11. For first maxima; $a\sin\theta = (2n + 1)\frac{\lambda}{2}$

$\boxed{\Delta x = \frac{3}{2}\lambda \ \& \ \phi = 3\pi}$ & $\boxed{n = 1}$

12.



$\theta_P = \tan^{-1}\left(\frac{\mu_D}{\mu_R}\right)$	**for D → R**
$i_C = \sin^{-1}\left(\frac{\mu_R}{\mu_D}\right)$	$\mu_D \sin\theta_P = \mu_R \sin r$
$\tan\theta_P = \frac{\mu_D}{\mu_R} = \frac{1}{\sin i_C}$	$\mu_D \sin\theta_P = \mu_R \sin(90-\theta_P)$
$\theta_P = \tan^{-1}(\text{cosec } i_C)$	$\tan\theta_P = \frac{\mu_R}{\mu_D}$
$i_C = \sin^{-1}(\cot\theta_P)$	$\tan\theta_P = \sin i_C$ $\theta_P = \tan^{-1}(\sin i_C)$ $i_C = \text{cosec}^{-1}(\cot\theta_P)$

13. Reflected ray has vibrations in the plane perpendicular to incident plane but incident light has no vibrations in that plane.

14. In the nicol prism, the e-ray and o-ray are produced by double refraction and the o-ray is turned in another direction by TIR.

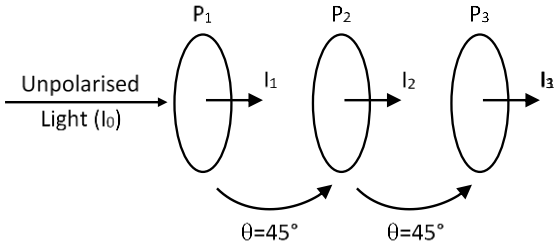
15. Reflected ray is completely polarised. Hence, polarising angle is 53°.

$\theta_P = \tan^{-1}(\mu)$

$\mu = \tan\theta_P = \tan 53° = \frac{4}{3}$

$r = 90° - \theta_P = 90° - 53° = 37°$

16.



$I_1 = \frac{I_0}{2}$, $I_2 = I_1\cos^2\theta$

$= \frac{I_0}{2}\cos^2 45° = \frac{I_0}{4}$

$I_3 = I_2\cos^2 45° = \frac{I_0}{4}\cos^2 45° = \frac{I_0}{8}$

$I_0 \xrightarrow{P_1} \frac{I_0}{2} \xrightarrow{P_2} \frac{I_0}{4} \xrightarrow{P_3} \frac{I_0}{8}$

Hence, intensity becomes 50% each time, by Malus law.

AIM: FREE EDUCATION TO ALL APUL

Modern Physics-1 [Dual Nature of Radiation and Matter]

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Maximum wavelength in visible light is 7800 Å. So energy of photon of this wavelength is

$$E = \frac{hc}{\lambda} = \frac{6.62\times10^{-34}\times3\times10^{8}}{7800\times10^{-10}\times1.6\times10^{-19}} = 1.6 \text{ eV}$$

2. Momentum of photon $p = \frac{h\nu}{c}$

$$= \frac{6.6\times10^{-34}\times1.5\times10^{13}}{3\times10^{8}} = 3.3\times10^{-29} \text{ kg m/s}$$

6. $W_0 = \frac{hc}{\lambda_0} = \frac{12400}{\lambda_0}$ eV–Å

or $\lambda_0 = \frac{12400}{4} = 3100$ Å $= 310$ nm

7. For element A, $\lambda_0 = \lambda$

$\therefore \quad W_A = \frac{hc}{\lambda_0} = \frac{hc}{\lambda} = 2.5$ eV .....(1)

For element B, $W_B = \frac{hc}{\lambda_B} = 5$ eV.....(2)

Dividing (1) & (2), we get

$$\frac{hc}{\lambda}\times\frac{\lambda_B}{hc} = \frac{2.5}{5} \quad \Rightarrow \quad \lambda_B = \frac{\lambda}{2}$$

8. $W_0 = \frac{hc}{\lambda_0} = \frac{12400}{\lambda_0}$ eV–Å $= \frac{12400}{6000}$

$= 2.06$ eV ≈ 2 eV

9. Minimum kinetic energy is always zero.

10. $K.E._{max} = eV_0 = 4$eV

$\therefore \quad V_0 = 4$ volt

11. Stopping potential does not depend on relative distance between the source and cell and saturation current $\propto \frac{1}{d^2}$.

12. $\lambda_0 = 200$ nm $= 2000$ Å

$\lambda = 100$ nm $= 1000$ Å

$$K.E._{max} = \left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right) 12400\text{eV} - \text{Å}$$

$$= 12400\left(\frac{1}{1000} - \frac{1}{2000}\right) \text{eV} = \frac{12400}{2000} = 6.2 \text{ eV}$$

17. $n_e \propto \frac{1}{d^2}$

19. $(KE)_{max} = h\nu - \phi_0 = 6\text{eV} - 4 \text{ eV} = 2 \text{ eV}$

21. $\nu_0 = \frac{\phi_0}{h} = \frac{3.3\times1.6\times10^{-19}}{6.6\times10^{-34}} = 8\times10^{14}$ Hz

25. The stopping potential depends on the substance of the cathode. Since it is same for A and B, the cathodes are made of the same substance. Since the current of A is greater than the current of B, the lights used are of different intensities. Light of A is more intense.

28. $(K.E.)_{max} = h\upsilon - \phi$

$= 4$ eV $- 2$ eV

K.E. $= 2$ eV; V $= 2$ volt

29. $\lambda = \frac{hc}{E} = \frac{h}{p}$

on decreasing λ, both p and E will increase

31. E $= 100$ eV

V $= 100$ volt

$\lambda = \frac{12.27}{\sqrt{V}}$ Å $= \frac{12.27}{\sqrt{100}} = 1.227$ Å

32. $\lambda = \frac{h}{\sqrt{2mqV}} \quad \therefore \quad \lambda \propto \frac{1}{\sqrt{mq}}$ ($\because$ V is constant)

$$\frac{\lambda_d}{\lambda_P} = \sqrt{\frac{m_P q_P}{m_d q_d}} = \sqrt{\frac{m_P q_P}{2m_P q_P}} = \frac{1}{\sqrt{2}}$$

33. V $= 40 - 20 = 20$ volt

$\lambda = \frac{12.27}{\sqrt{V}}$ Å $= \frac{12.27}{\sqrt{20}}$ Å $= 2.75$ Å

34. $\lambda = \frac{h}{mv} = \frac{6.62\times10^{-34}}{9\times10^{-31}\times6.6\times10^{3}} = 1.1\times10^{-7}$ m

35. Here $\lambda_2 = \lambda_1/2$

$$\lambda = \frac{h}{\sqrt{2mE}} \quad \Rightarrow \quad \lambda \propto \frac{1}{\sqrt{E}}$$

$$E \propto \frac{1}{\lambda^2} \quad \Rightarrow \quad \frac{E_1}{E_2} = \left(\frac{\lambda_2}{\lambda_1}\right)^2$$

$$E_2 = E_1 \times \left(\frac{\lambda_1}{\lambda_2}\right)^2 = E_1\left(\frac{\lambda_1}{\lambda_1/2}\right)^2 = 4E_1 = E_1 + 3E_1$$

energy added should be thrice initial energy.

36. $\lambda = \frac{h}{\sqrt{2mE}}$ Here E is same. So $\lambda \propto \frac{1}{\sqrt{m}}$

$m_\alpha > m_n > m_P > m_e \quad \therefore \quad \lambda_\alpha < \lambda_n < \lambda_P < \lambda_e$

37. $\lambda = \frac{12.27}{\sqrt{V}}$ Å $\Rightarrow \lambda = \frac{12.27}{\sqrt{50000}} = 0.055$ Å

38. For a neutron $m = 1.7 \times 10^{-27}$ kg

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.62\times10^{-34}}{\sqrt{2\times1.7\times10^{-27}\times E}} = \frac{0.284}{\sqrt{E(eV)}} \text{ Å}$$

$$= \frac{0.286}{\sqrt{3}} \simeq 0.16 \text{ Å} = 1.6 \times 10^{-11} \text{ m}$$

39. $\lambda = \frac{h}{\sqrt{2mqV}}$ Here V is same. So $\lambda \propto \frac{1}{\sqrt{mq}}$

Or $\frac{\lambda_P}{\lambda_\alpha} = \sqrt{\frac{m_\alpha q_\alpha}{m_P q_P}} = \sqrt{\frac{4m_P \times 2q_P}{m_P q_P}} = \frac{2\sqrt{2}}{1}$

41. Energy in ground state = 13.6 eV

Ist Method : $\lambda = \frac{h}{\sqrt{2mE}}$

$$= \frac{6.62\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times13.6\times1.6\times10^{-19}}}$$

$= 3.3 \times 10^{-10}$ m $= 3.3$ Å

IInd Method : $\lambda = \frac{12.27}{\sqrt{V}}$ Å

$\Rightarrow \lambda = \frac{12.27}{\sqrt{13.6}} = 3.3$ Å

47. By de Broglie formula

$$p = \frac{h}{\lambda} = \frac{6.6\times10^{-34}}{4400\times10^{-10}} = 1.5\times10^{-27} \text{ kgm/s}$$

For photon, velocity = $c = 3 \times 10^8$ m/s

$\therefore m = \frac{p}{c} = \frac{1.5\times10^{-27}}{3\times10^8}$ or $m = 5 \times 10^{-36}$ kg

50. $\because \quad 2\pi r = n\lambda$

Here $n = 1$

$\therefore \quad 2\pi r = \lambda$

52. K.E = 1 KeV

54. $2\pi r_n = n\lambda \Rightarrow 2\pi r_n = 2 \times 10^{-9}$m

55. $\lambda = \frac{h}{mv}$

$$\therefore \frac{\lambda_2}{\lambda_1} = \frac{m_1 v_1}{m_2 v_2} = \frac{2m_1}{m_1}\frac{2v_1}{v_1} = \frac{4}{1}$$

$\therefore \lambda_2 = 4\lambda_1$

New wavelength = 4 × original wavelength

$\therefore$ The wavelength increases by a factor more than 2.

56. For charged particle

$\lambda = \frac{h}{\sqrt{2mqV}}$

for proton and electron

$q \rightarrow$ same, $V \rightarrow$ same

So, $\lambda \propto \frac{1}{\sqrt{m}} \Rightarrow \frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}}$

57. de-Broglie wavelength $\lambda = \frac{h}{\sqrt{2mE}}$

where E is the kinetic energy

Since E is the same, $\therefore \lambda \propto \frac{1}{\sqrt{m}}$

Since $m_\alpha > m_p > m_e$, $\therefore \lambda_e > \lambda_p > \lambda_\alpha$.

58. For a particle, de Broglie wavelength $\lambda = \frac{h}{p}$

Kinetic energy $K = \frac{p^2}{2m}$

$\therefore \lambda = \frac{h}{\sqrt{2mK}}$

For an electron $\lambda = \frac{h}{\sqrt{2m_e K_e}}$

$$= \frac{(6.63\times10^{-34}\text{J})}{\sqrt{2\times(9.11\times10^{-31}\text{kg})\times(120\times1.6\times10^{-19}\text{J})}}$$

$= 112 \times 10^{-12}$ m $= 112$ pm [$\because$ 1 pm $= 10^{-12}$ m]

OR

K.E. = 120 eV V = 120 volt

For electron

$\lambda = \frac{12.27}{\sqrt{120}} = 112$ pm.

59. $\lambda = \frac{h}{\sqrt{2mk}}$

For protons $\lambda_p = \frac{h}{\sqrt{2m_p.k_p}}$ ....(i)

For α – particle $\lambda_\alpha = \frac{h}{\sqrt{2m_\alpha.k_\alpha}}$

$\lambda_\alpha = \frac{h}{\sqrt{2.(4m_p).k_p}}$ ($\because k_p = k_\alpha$) given

$\lambda_\alpha = \frac{h}{2\sqrt{2m_p k_p}}$ ....(ii)

Dividing (i) by (ii) we get $\frac{\lambda_p}{\lambda_\alpha} = \frac{\frac{h}{\sqrt{2m_pk_p}}}{\frac{h}{2\sqrt{2m_pk_p}}}$

$\frac{\lambda_p}{\lambda_\alpha} = 2$, $\lambda_p = 2\lambda_\alpha$

Exercise-II (Previous Year Questions)

1. Average force $F_{av} = \frac{\Delta p}{\Delta t} = \frac{2IA}{c}$

$= \frac{2\times25\times10^{4}\times15\times10^{-4}}{3\times10^{8}} = 2.50 \times 10^{-6}$ N

2. By using $h\nu = \phi_0 + K_{max}$

We have

$h\nu = \phi_0 + 0.5$ ..... (i)

and $1.2h\nu = \phi_0 + 0.8$ ..... (ii)

Therefore $\phi_0 = 1.0$ eV

3. $\lambda = \frac{h}{\sqrt{2mK}}$

$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{K_2}{K_1}} = \sqrt{\frac{16K}{K}} = \frac{4}{1}$

% change $= \frac{1-4}{4}\times100 = -75\%$

4. $eV_S = E - \phi \Rightarrow V_S = \frac{hc}{\lambda e} - \frac{hc}{\lambda_0 e}$

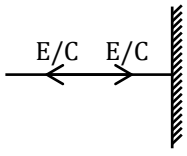
here $3V_0 = \frac{hc}{\lambda e} - \frac{hc}{\lambda_0 e}$ ...(1)

and $V_0 = \frac{hc}{2\lambda e} - \frac{hc}{\lambda_0 e}$ ...(2)

equation (1) – 3 × equation (2)

$\Rightarrow 0 = -\frac{hc}{2\lambda e} + \frac{2hc}{\lambda_0 e} \Rightarrow \lambda_0 = 4\lambda$

5.



Momentum of light $p = \frac{E}{C}$

So momentum transferred to the surface

$= p_f - p_i = \frac{2E}{C}$

6. $p = \frac{h}{\lambda} \Rightarrow p \propto \frac{1}{\lambda}$ (Rectangular hyperbola)

7. Energy of photon (E) $= \frac{12400}{5000} = 2.48$ eV

Work function (ϕ_0) = 2.28 eV

According to Einstein equation

$E = \phi_0 + (K.E.)_{max}$

$\Rightarrow$ 2.,48 = 2.28 + $(K.E.)_{max}$

$\Rightarrow$ $(K.E.)_{max}$ = 0.20 eV

For electron $\lambda = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda \approx 28$ Å

So $\lambda \geq 2.8 \times 10^{-9}$ m

8. $KE_1 = \frac{hc}{\lambda} - \phi$

$KE_2 = \frac{hc}{\lambda/2} - \phi = \frac{2hc}{\lambda} - \phi$

$KE_2 = 3KE_1$

$\Rightarrow \frac{2hc}{\lambda} - \phi = 3\left(\frac{hc}{\lambda} - \phi\right)$

$\Rightarrow 2\phi = \frac{hc}{\lambda}$

$\Rightarrow \phi = \frac{hc}{2\lambda}$

9. For electron $\lambda_e = \frac{h}{\sqrt{2mE}}$

for Photon E = pc

$\Rightarrow \lambda_{Ph} = \frac{hc}{E}$

$\Rightarrow \frac{\lambda_e}{\lambda_{Ph}} = \frac{h}{\sqrt{2mE}} \times \frac{E}{hc} = \left(\frac{E}{2m}\right)^{1/2}\frac{1}{c}$

10. $eV = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$ ...(i)

$\frac{eV}{4} = \frac{hc}{2\lambda} - \frac{hc}{\lambda_0}$ ...(ii)

From equation (i) and (ii)

$\Rightarrow 4 = \frac{\frac{1}{\lambda} - \frac{1}{\lambda_0}}{\frac{1}{2\lambda} - \frac{1}{\lambda_0}}$ On solving $\lambda_0 = 3\lambda$

11. $eV_s = \frac{1}{2}mv_{max}^2 = h\nu - \phi_0$

$2 = 5 - \phi_0 \Rightarrow \phi_0 = 3$ eV

In second case

$eV_s = 6 - 3 = 3$ eV $\Rightarrow V_s = 3$ V.

$\therefore$ $V_{AC} = -3$ V

12. Kinetic energy of thermal neutron with equilibrium is $\frac{3}{2}KT$

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2m\,K.E}} = \frac{h}{\sqrt{2m\left(\frac{3}{2}KT\right)}}$$

$$= \frac{h}{\sqrt{3mKT}}$$

13. $\lambda_0 = 3250$ Å

$\lambda = 2536$ Å

$$\frac{1}{2}mv^2 = hc\left[\frac{1}{\lambda} - \frac{1}{\lambda_0}\right]$$

$$v = \sqrt{\frac{2hc}{m}\left[\frac{1}{\lambda} - \frac{1}{\lambda_0}\right]}$$

$$= \sqrt{\frac{2\times 12400\times 1.6\times 10^{-19}}{9.1\times 10^{-31}}\left[\frac{714}{2536\times 3250}\right]}$$

$= 0.6 \times 10^6$ m/s $= 6 \times 10^5$ m/s

14. $\vec{a} = \frac{\vec{F}}{m} = \frac{q\vec{E}}{m} = \frac{(-e)(-E_0)\hat{i}}{m} = \frac{eE_0}{m}\hat{i}$

$$V = V_0 + \frac{eE_0}{m}t$$

$$\lambda = \frac{h}{mv} = \frac{h}{m\left[V_0 + \frac{eE_0 t}{m}\right]} = \frac{h}{mV_0\left[1 + \frac{eE_0 t}{mV_0}\right]}$$

$$= \frac{\lambda_0}{1 + \frac{eE_0 t}{mV_0}}$$

15. From Einstein's equation of PEE

$\frac{1}{2}mv_1^2 = 2h\nu_0 - h\nu_0 \Rightarrow \frac{1}{2}mv_1^2 = h\nu_0$ ...(1)

$\frac{1}{2}mv_2^2 = 5h\nu_0 - h\nu_0 \Rightarrow \frac{1}{2}mv_2^2 = 4h\nu_0$ ...(2)

(1) $\div$ (2)

$$\frac{v_1^2}{v_2^2} = \frac{1}{4} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

16. $\lambda = \sqrt{\frac{150}{V}}$ Å

$\lambda = \sqrt{\frac{150}{10^4}}$ Å $= 12.27 \times 10^{-12}$ m

17. $\phi = \frac{hc}{\lambda}$

$\Rightarrow 4$ eV $= \frac{1240}{\lambda}$

$\Rightarrow \lambda = \frac{1240}{4} = 310$ nm

18. $\lambda = \frac{h}{\sqrt{2mE_k}}$ ($\because$ E_k is same for both)

$$\Rightarrow \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{4m}{m}} = 2:1$$

19. $I = \frac{E}{At} \Rightarrow E = IAt = \frac{20}{10^{-4}} \times 20 \times 10^{-4} \times 60$

$= 24 \times 10^3$ J

20. $E = \frac{10^{-20}}{1.6\times 10^{-19}}$ eV $= 0.0625$ eV

21. $\lambda = 1.227 \times 10^{-2}$ nm $= 0.1227$ Å

$\lambda = \frac{12.27}{\sqrt{V}}$ Å $\Rightarrow 0.1227 = \frac{12.27}{\sqrt{V}}$ Å

$\Rightarrow \sqrt{V} = 10^2 \Rightarrow V = 10^4$ volt

22. $K_1 = 1.5\,h\nu_0 - \phi_0 = 0.5\,h\nu_0$

$K_2 = \frac{1.5}{2}h\nu_0 - h\nu_0 = -0.25\,h\nu_0$

$\because$ Kinetic energy can never be negative

So, no emission and i = 0

OR

In second case the incident frequency is halved

Incident frequency $= \frac{1.5}{2}\nu_0 = 0.75\,\nu_0$

Now the incident frequency is less than threshold frequency so no emission of electron take place therefore no current. (i = 0)

23. $\lambda = \frac{12.27}{\sqrt{V}}$ Å $= \frac{12.27}{\sqrt{144}} \times 10^{-10}$

$= 1.02 \times 10^{-10}$ m $= 102 \times 10^{-3}$ nm

24. The wave nature of electrons was experimentally verified by Davission and Germer.

25. $P=\frac{nhc}{\lambda}\Rightarrow n=\frac{P\lambda}{hc}$

$n=\frac{3.3\times10^{-3}\times600\times10^{-9}}{6.6\times10^{-34}\times3\times10^{8}}=10^{16}$

26. $\frac{hc}{\lambda}=K_{max}+\phi$ [given ϕ is negligible]

so, $\frac{hc}{\lambda}=K_{max}$

and $\lambda_d=\frac{h}{\sqrt{2m\,K_{max}}}\Rightarrow K_{max}=\frac{h^2}{2m\lambda_d^2}$

$\left(\frac{hc}{\lambda}\right)=\frac{h^2}{2m\lambda_d^2}\Rightarrow\lambda=\left(\frac{2mc}{h}\right)\lambda_d^2$

30. Given energy of photon E = 2.20 eV

Work function of Cs ϕ_0 = 2.14 eV, K ϕ_0 = 2.30 eV, Na ϕ_0 = 2.75 eV

We know that e^- emitts when $h\nu > \phi_0$

here it is clear that energy of photon is more than the work function of Cs [Caesium] only so Ans. only (Cs).

31. $\lambda_e=\frac{12.27}{\sqrt{V}}\text{Å}=\frac{12.27}{\sqrt{81}}\text{Å}=\frac{12.27}{9}\text{Å}$

$=1.36\text{ Å}\left(\because 1\text{Å}=\frac{1}{10}\text{nm}\right)$

$=0.136$ nm

32. Maximum kinetic energy of emitted electron is independent of intensity of radiation.

33. For a photon,

(i) Energy $E = h\nu \Rightarrow$ (statement A is correct)

(ii) All photons travel with speed of light (= c in free space)

$\Rightarrow$ statement B is correct

(iii) Momentum of a photon. $p=\frac{E}{c}=\frac{h\nu}{c}$

$\Rightarrow$ Statement C is correct.

(iv) In a photon-electron collision, total energy and total momentum are conserved.

$\Rightarrow$ statement D is also correct.

(v) Photons are massless and do not carry any charge.

$\Rightarrow$ statement E is incorrect.

Correct choice (2)

A, B, C, & D are correct.

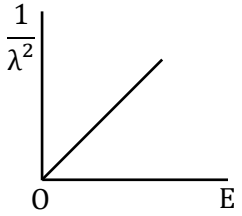
34. de-Broglie wavelength and energy relation of a free particle

$\lambda=\frac{h}{\sqrt{2mE}}$

$\lambda^2=\frac{h^2}{2mE}$

$\frac{1}{\lambda^2}=\frac{2m}{h^2}E$

Graph $\frac{1}{\lambda^2}$ v / s E



35. Ve = Vx

$V \propto$ Same

$\lambda=\frac{h}{\sqrt{2mqv}}$

$\lambda\propto\frac{1}{\sqrt{mq}}$

$\frac{\lambda_e}{\lambda_\alpha}=\sqrt{\frac{m_\alpha q_\alpha}{m_e q_e}}=\sqrt{\frac{4m_p\times2q_e}{m_e\times q_e}}$ $(q_e)=(q_p)$

$\frac{\lambda_e}{\lambda_\alpha}=\sqrt{8}\sqrt{\frac{m_p}{m_e}}$ $\begin{cases}m_p > m_e\\ \sqrt{8}=2.82\end{cases}$

$\lambda_e > \lambda_\alpha$

36. $\lambda=\frac{hc}{e}$

$K_{max}=\frac{hc}{\lambda}-\phi$

$K_{max}=\frac{ehc}{hc}-\phi$

$K_{max} = e - \phi$

Exercise-III (Analytical Questions)

1. Photo electric equation

$KE_{max} = E - \phi = h\nu - \phi$

$(4.14 \times 10^{-15}\ eV\text{–}s)\ (6 \times 10^{14}\ Hz) - 2.0\ eV$

$= 2.48 - 2.0 = 0.48\ eV \approx 0.5eV$

$= 0.5 \times 1.6 \times 10^{-19} = 8 \times 10^{-20}\ J$

Stopping potential

$eV_s = KE_{max} = 0.48eV$

$V_s = 0.48$ volt

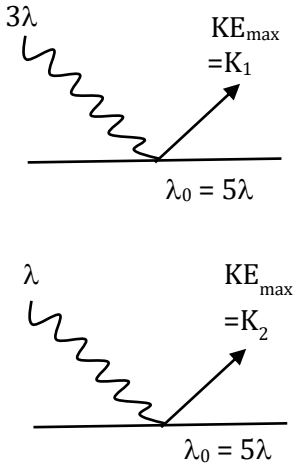
$$n = \frac{I\lambda}{hc} \times A = \frac{I}{h\nu} \times A$$

$$= \frac{60 \times 10^3 \times 6.6 \times 10^{-4}}{6.6 \times 10^{-34} \times 6 \times 10^{14}}$$

$n = 10^{20}$ per second

$i = ne = 10^{20} \times 1.6 \times 10^{-19} = 16\ A$

4.



$$K_1 = \frac{hc}{3\lambda} - \frac{hc}{5\lambda} = \frac{hc}{\lambda}\left[\frac{1}{3} - \frac{1}{5}\right] = \frac{hc}{\lambda}\left[\frac{2}{15}\right]$$

$$K_2 = \frac{hc}{\lambda} - \frac{hc}{5\lambda} = \frac{hc}{\lambda}\left[1 - \frac{1}{5}\right] = \frac{hc}{\lambda}\left[\frac{4}{5}\right]$$

(a) Ratio of KE

$$\frac{K_1}{K_2} = \frac{\frac{hc}{\lambda}\left[\frac{2}{15}\right]}{\frac{hc}{\lambda}\left[\frac{4}{5}\right]} = \frac{1}{6} = 1:6$$

(b) Ratio of speeds of incident photons

$$\frac{c_1}{c_2} = \frac{1}{1} = 1:1$$

Ratio of momentum of incident photons

$$\frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1} = \frac{\lambda}{3\lambda} = 1:3$$

(c) Ratio of fastest speed of photo electrons

$$\frac{v_1}{v_2} = \sqrt{\frac{K_1}{K_2}} = \sqrt{\frac{1}{6}} = 1:\sqrt{6}$$

6. Statement-I is true, statement-II is false.

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\lambda \propto \frac{1}{\sqrt{m}}\ (\because \text{KE = same})$$

7. (A) $\lambda = \frac{h}{mv}$ or $\lambda \propto \frac{1}{m}$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{m_\alpha}{m_p} = \frac{4}{1}\ (\because m_\alpha \approx 4m_p)$$

(B) $\lambda = \frac{h}{\sqrt{2mqV}}$ or $\lambda \propto \frac{1}{\sqrt{m.q}}$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\left(\frac{m_\alpha}{m_p}\right)\left(\frac{q_\alpha}{q_p}\right)} = \sqrt{\frac{4}{1} \times \frac{2}{1}} = \frac{2\sqrt{2}}{1}$$

$(\because q_\alpha = 2q_p)$

(C) $\lambda = \frac{h}{\sqrt{2mE_k}}$ or $\lambda \propto \frac{1}{\sqrt{m}}$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha}{m_p}} = \sqrt{\frac{4}{1}} = \frac{2}{1}$$

(D) $\lambda = \frac{h}{p}$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{1}{1}$$

8. Give $\lambda_p = 2\lambda_e$

So $\frac{p_p}{p_e} = \frac{\lambda_e}{\lambda_p} = \frac{1}{2}\ \left(\because \lambda = \frac{h}{p}\right)$

$$\frac{E_e}{E_p} = \frac{\frac{1}{2}m_e v^2}{\frac{hc}{\lambda}} = \frac{\frac{1}{2}(mv)v}{\left(\frac{h}{\lambda}\right)c}$$

$$= \frac{1}{2}\frac{p_e.v}{p_p c} = \frac{1}{2}\left(\frac{p_e}{p_p}\right)\frac{v}{c}$$

$$\frac{E_e}{E_p} = \frac{1}{2}\left(\frac{2}{1}\right)\left(\frac{1}{100}\right) = 10^{-2}$$

and $\frac{p_e}{m_e c} = \frac{m_e v_e}{m_e c} = \frac{v_e}{c} = \frac{1}{100} = 10^{-2}$

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Modern Physics-2 [Nuclear Physics & Radioactivity]

SOLUTIONS

Exercise-I (Conceptual Questions)

2. B.E. of A = 240 × 7.6 = 1824 MeV
B.E. of B = 100 × 8.1 = 810 MeV
B.E. of C = 140 × 8.1 = 1134 MeV
So Q = (810 + 1134) MeV – 1824 MeV
= 120 MeV

3. Energy released in 10 kg of Uranium
$E = mc^2 = 10 \times (3 \times 10^8)^2 = 9 \times 10^{17}$ J

4. Energy released in 2 mole of U^{235}
$= \frac{6.02\times10^{23}\times2\times235}{235}\times200$ MeV
$= 24 \times 10^{25}$ MeV

22. Released power 1 kW or 1×10^3 W
and per fission released energy 200 MeV
or $200 \times 1.6 \times 10^{-13}$ J
Number of fission = $\frac{10^3}{200\times1.6\times10^{-13}}$
per sec = 3.125×10^{13}

29. Binding energy = $[Zm_p + (A - Z)m_n] - M_n$
= [2 × 1.0073 + 2 × 1.0087] – 4.0015
= 4.032 – 4.0015 = 0.0305 u
or 0.0305 × 931 = 28.4 MeV

33. Nuclear density is of the order of 10^{17} kg/m^3.

36. Mass of nucleus is slightly less than sum of masses of its constituents. This mass difference is equivalent to binding energy.
$\Delta m = \frac{B}{c^2} = (Zm_P + NM_n) - M(N,Z)$
Hence M (N, Z) = $NM_n + ZM_P - B/c^2$

38. m = 1mg = 10^{-6} kg
$E = mc^2 = 10^{-6} \times (3 \times 10^8)^2 = 9 \times 10^{10}$ J

41. Isotones → number of neutrons are same(A-Z)

42. In option (1) charge is not conserved
i.e. (5 + 2 ≠ 7 + 1) ⇒ not possible
In (2) this reaction is possible only at very high temperature (or $^{23}_{11}Na$ is a very stable isotope and given nuclear reaction is very difficult)
In (3) & (4) $\bar{\nu}$ is emitted with β^- so option (4) is also wrong so correct answer is (3)

43. Energy released = binding energy of products – binding energy of reactants
= c – (a + b) = c – a – b

52. $N_\alpha = \frac{A_i - A_t}{4} = \frac{238-222}{4} = 4$
$N_\beta = Z_f - Z_i + 2N_\alpha = 83 - 90 + 2(4) = 1$

53. $N_\alpha = \frac{A_i - A_f}{4} = \frac{238-206}{4} = 8$
$N_\beta = Z_f - Z_i + 2N_\alpha = 82 - 92 + 2(8) = 6$

Exercise-II (Previous Year Questions)

1. BE of $_2He^4 = 4 \times 7.06 = 28.24$ MeV
BE of $_3Li^7 = 7 \times 5.60 = 39.20$ MeV
$_3Li^7 + {}_1H^1 \longrightarrow {}_2He^4 + {}_2He^4 + Q$
39.20 28.24 × 2
Q = 56.48 – 39.20 = 17.28 MeV

2. $R \propto A^{1/3}$
$\frac{R_{Al}}{R_{Te}} = \left(\frac{27}{125}\right)^{1/3}$
$\Rightarrow R_{Te} = \frac{5}{3} R_{Al}$

3. By COLM :
$p_f - p_i = 0$
$\Rightarrow p_{He} - p_{Th} = 0 \qquad \left[K.E. = \frac{p^2}{2m}\right]$
$\Rightarrow p_{He} = p_{Th}$
but $K \propto \frac{1}{m}$ and $m_{He} < m_{Th}$ So $K_{He} > K_{Th}$

4. $\alpha = {}^4_2He^{2+}$ = Helium Nucleus
2 protons and 2 neutrons

5. $^{235}_{92}U + {}^1_0n \rightarrow {}^{89}_{36}Kr + {}^{144}_{56}Ba + 3{}^1_0n + Q$

6. $E = mc^2$
$= 0.5 \times 10^{-3} \times 9 \times 10^{16}$
$= 4.5 \times 10^{13}$ J

7. $_ZX^A \xrightarrow{\gamma\text{-decay}} {}_ZZ^A$
Due to gamma emission, there is no change in mass number and atomic number.

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8. $X^{240} \rightarrow Y^{120} + Z^{120}$

given binding energy per nucleon of X, Y & Z are 7.6 MeV, 8.5 MeV & 8.5 MeV respectively.

Gain in binding energy is :-

Q = Binding Energy of products – Binding energy of reactants

= (120 × 8.5 × 2) – (240 × 7.6) MeV

= 216 MeV

9. ${}^{A}_{Z}X \xrightarrow{\beta^+} {}_{Z-1}B \xrightarrow{\alpha} {}_{Z-3}C \xrightarrow{\beta^-} {}_{Z-2}D$

β^+ decreases atomic number by 1

α decreases atomic number by 2

β^- increases atomic number by 1

12. ${}^{290}_{82}X \xrightarrow{\alpha} Y \xrightarrow{e^+} Z \xrightarrow{\beta^-} P \xrightarrow{e^-} {}_{81}Q^{286}$

13. Slow neutron can cause fission in ${}_{92}U^{235}$ than fast neutrons.

α-rays are Helium nuclei

β-rays are fast moving electron or positrons.

Exercise-III (Analytical Questions)

1. Assertion is true but reason is false.

4. Informative facts. At distance less than 0.7 fm, the nuclear force becomes repulsive.

5. (i) $R = R_0 A^{1/3}$

$$\frac{R_1}{R_2} = \left(\frac{2}{16}\right)^{1/3} = \frac{1}{2}$$

(ii) ρ = constant $\quad \frac{\rho_1}{\rho_2} = \frac{1}{1}$

(iii) $v \propto A$, $\frac{v_1}{v_2} = \frac{2}{16} = \frac{1}{8}$

(iv) $\frac{m_1}{m_2} = \frac{v_1}{v_2} = \frac{2}{16} = \frac{1}{8}$

7. larger is the binding energy per nucleon more is the stability of nucleus.

Modern Physics-3 [Atomic Structure]

SOLUTIONS

Exercise-I (Conceptual Questions)

1. For a Lyman series

$$\frac{1}{\lambda}=R\left[\frac{1}{1^2}-\frac{1}{n^2}\right] \quad ...(i)$$

where n = 2, 3, 4 .....

By putting n = 2, is equation (i), we obtain the longest wavelength of a Lyman series

$\frac{1}{\lambda}=R\left[\frac{1}{1^2}-\frac{1}{2^2}\right]$ or λ = 121.5 nm. This is the longest wavelength of the Lyman series.

Energy of photon $E=\frac{hc}{\lambda}$ or $E\propto\frac{1}{\lambda}$

For the longest wavelength, energy of the photon is the least.

2. For a Balmer series

$$\frac{1}{\lambda_B}=R\left[\frac{1}{2^2}-\frac{1}{n^2}\right] \quad ...(i)$$

where n = 3, 4, ...

By putting n = ∞ in equation (i), we obtain the series limit of the Balmer series. This is the shortest wavelength of the Balmer series. or

$\lambda_B=\frac{4}{R}$

For a Lyman series

$$\frac{1}{\lambda_L}=R\left[\frac{1}{1^2}-\frac{1}{n^2}\right] \quad ...(i)$$

where n = 2, 3, 4.....

By putting n = ∞ in equation (i), we obtain the series limit of the Lyman series. This is the shortest wavelength of the Lyman series.

or $\lambda_L=\frac{1}{R}$ ...(ii)

From (i) and (ii), we get $\frac{\lambda_B}{\lambda_L}=\frac{4}{1}$

3. KE = – TE = –(–13.6 eV) = 13.6 eV

8. According to the Bohr's theory of hydrogen atom

(i) The speed of the electron in the nth orbit

$V_n=2.2\times10^6\times\frac{Z}{n}\Rightarrow\boxed{V_n\alpha\frac{1}{n}}$

(ii) The energy of the electron in the nth orbit is

$E_n=\frac{-13.6Z^2}{n^2}\Rightarrow E_n\alpha\frac{1}{n^2}$

(iii) The radius of the electron in the nth orbit is

$r_n=\frac{0.53n^2}{Z}\text{Å}\Rightarrow r_n\alpha n^2$

9. $E_n=-\frac{13.6}{n^2}\text{eV}=-\frac{13.6}{(3)^2}\text{eV}=-1.51\text{ eV}$

Exercise-II (Previous Year Questions)

1. $\frac{1}{\lambda}=R\left(\frac{1}{1^2}-\frac{1}{n^2}\right)$

$\frac{912}{975}=\frac{1}{1}-\frac{1}{n^2}\Rightarrow n=4$

Number of spectral lines = $\frac{n(n-1)}{2}=6$

2. For H-like atoms

$v=\frac{Z}{n}\times2.188\times10^6$ m/s

here Z = 2, n = 3

$v=1.46\times10^6$ m/s

3. For Lyman series

$$\left(\frac{1}{\lambda_{max}}\right)_L=R(1)^2\left[\frac{1}{(1)^2}-\frac{1}{(2)^2}\right]$$

$$(\lambda_{max})_L=\frac{4}{3R}$$

For Balmer series

$$\left(\frac{1}{\lambda_{max}}\right)_B=R(1)^2\left[\frac{1}{(2)^2}-\frac{1}{(3)^2}\right]$$

$$(\lambda_{max})_B=\frac{36}{5R}$$

$$\frac{(\lambda_{max})_L}{(\lambda_{max})_B}=\frac{4}{3R}\times\frac{5R}{36}=\frac{5}{27}$$

4. At closest distance of approach, the kinetic energy of the particle will convert completely into electrostatic potential energy.

$\Rightarrow\frac{1}{2}mv^2=\frac{KZe^2}{d_{min}}\Rightarrow d_{min}\propto\frac{1}{m}$

5. Wave number, $\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_2^2}-\frac{1}{n_1^2}\right)$

$= 10^7 \times 1^2\left(\frac{1}{2^2}-\frac{1}{\infty^2}\right) = 0.25 \times 10^7\ m^{-1}$

6. Transition : $3 \to 2 \Rightarrow$ Wavelength λ.

Transition : $4 \to 3 \Rightarrow$ Wavelength $\lambda' = ?$

$$\frac{\frac{1}{\lambda}=RZ^2\left(\frac{1}{2^2}-\frac{1}{3^2}\right)}{\frac{1}{\lambda'}=RZ^2\left(\frac{1}{3^2}-\frac{1}{4^2}\right)} \Rightarrow \frac{\lambda'}{\lambda}=\frac{20}{7} \Rightarrow \lambda'=\frac{20\lambda}{7}$$

8. $E_{total} = -KE \Rightarrow \frac{KE}{E_{Total}} = \frac{1}{-1}$

9. TE = –3.4 eV

KE = –T.E PE = 2T.E

$\Rightarrow$ KE = +3.4 eV $\Rightarrow$ PE = –6.8 eV

10. $m_\mu = 207m_e$, $q_\mu = q_{e^-}$, $M_{nucleus} = 1836\ m_e$

Reduced mass

$$\mu = \frac{mM}{M+m} = \frac{207m_e \times 1836m_e}{207m_e + 1836m_e} = 186\,m_e$$

$\because\ r_1 = \frac{n^2h^2}{4\pi^2mkze^2} = 0.51$ Å

(Given in Question)

Radius of first orbit of new atom

$$r_1' = \frac{m_e r_1}{\mu} = \frac{m_e}{186m_e} \times 0.51\text{Å} = 2.56 \times 10^{-13}\ m$$

$$E_1' = \frac{\mu}{m}E_1 = \frac{186\ m_e}{m_e}(-13.6\ eV)$$

$= -2.5\ keV \approx -2.8 keV$

(according to given options)

11. Bohr model is applicable for only single electron species.

12. For hydrogen atom

$E = -\frac{13.6}{n^2}$ eV

15. Radius of nth orbit in Hydrogen Atom

$r_n = 0.53 \times \frac{n^2}{Z}$ Å

So, radius of third orbit

$$r_3 = 0.53 \times \frac{(3)^2}{(1)}\text{Å} = 4.77\text{Å}$$

16. $E_n = -13.6\frac{Z^2}{n^2}$

for ground state n = 1 and for second excited state n = 3

So $\frac{E_3}{E_1} = \left(\frac{n_1}{n_3}\right)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$

$E_3 = \frac{E_1}{9} = \frac{13.6\,eV}{9} = 1.51\,eV$

17. $\frac{1}{\lambda} = R\left(\frac{1}{(1)^2}-\frac{1}{n^2}\right)$ n = 2, 3, 4, ...

$\left(\frac{1}{\lambda_L}\right)_{max} = R\left(\frac{1}{(1)^2}-\frac{1}{(2)^2}\right)$ $\left(\because \frac{1}{R} \simeq 912\text{Å}\right)$

$\left(\lambda_L\right)_{max} = \frac{4}{3}\frac{L}{R}$

$\left(\lambda_L\right)_{max} = \frac{4}{3} \times 912\text{Å} = 4 \times 304\ \text{Å} = 1216\text{Å}$

$\left(\frac{1}{\lambda_L}\right)_{min} = R\left(\frac{1}{(1)^2}-\frac{1}{(\infty)^2}\right)$

$\left(\lambda_L\right)_{min} = \frac{1}{R} \simeq 912\text{Å}$

Range of λ is 912Å to 1216Å which lies in U.V. region.

18. Statement I is correct.

Statement II is incorrect because atom of radioactive elements are not stable.

19. $\Delta E = \frac{hc}{\lambda}$

$\Delta E \to$ less

$\lambda \to$ large

$E_A < E_B < E_C < E_D$

$\Rightarrow$ 656.3 > 486.1 > 434.1 > 410.2

III IV II I

$\Rightarrow$ A-III B-IV C-II D-I

20. Transition from $n_2 = 5, 6.....$ to $n_1 = 4$ is Brackett series.

21. Energy = $\frac{hc}{\lambda}$

(a) $E = \frac{hc}{\lambda_2} \Rightarrow \lambda_2 \propto \frac{1}{E}$

(b) $1.5E = \frac{hc}{\lambda_1} \Rightarrow \lambda_1 \propto \frac{1}{1.5E}$

(c) $ZE = \frac{hc}{\lambda_3} \Rightarrow \lambda_3 \propto \frac{1}{2E}$

$\lambda_2 > \lambda_1$, $\lambda_2 = 2\lambda_3$

Semiconductor Electronics

SOLUTIONS

Exercise-I (Conceptual Questions)

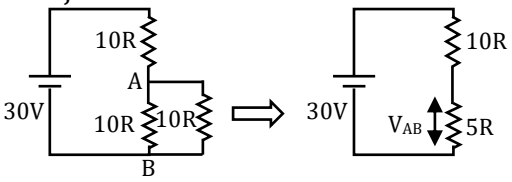
19. $n = AT^{3/2}e^{\frac{-\Delta Eg}{2kT}}$

35. When P-N junction is operated at 5mA

$(2 - 0.5) = 5 \times 10^{-3} \times R$

$R = 300\ \Omega$

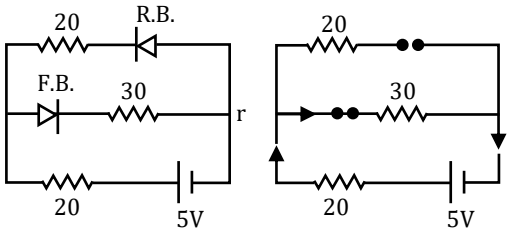
37. P-N junction in forward bias



$V_{AB} = IR' = \left(\frac{30}{15R}\right) \times 5R = 10V$

38. $I = \frac{3V - 1V}{100\Omega} = 20mA$

40.



$I = \frac{5}{50}A$

41. Electric field $= \frac{V}{d} = \frac{0.1}{10^{-6}} = 10^5$ (V/m)

49. To produce e–h pair

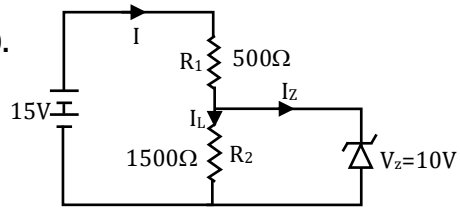
energy of incident photon $\geq \Delta E_g$

$\frac{hc}{\lambda} \geq \Delta E_g$

λ (In Å) $\leq \frac{12400}{\Delta E_g (\text{in eV})} \leq \frac{12400}{0.75} \leq 16500$ Å

So $\lambda_{max} = 16500$Å

50.



$10 = I_L R_L = I_L \times 1500$

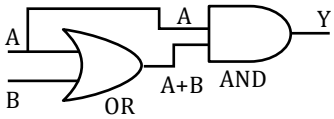
$I_L = 6.67$ mA

$I = \frac{V - V_z}{R_1} = \frac{15 - 10}{500} = \frac{5}{500} = 10mA$

$I = I_L + I_Z$

$I_Z = I - I_L = (10 - 6.67)mA = 3.33mA$

67.



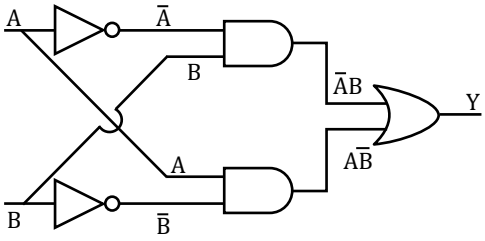
$Y = A.(A+B)$

$Y = A.A + AB$

$Y = A + AB = A(1+B)$

$Y = A$

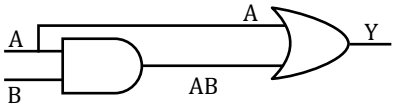
75.



$Y = \bar{A}B + A\bar{B}$

$= A \oplus B$ = XOR gate

76.



$Y = A + AB = A(1+B)$

$= A$

Exercise-II (Previous Year Questions)

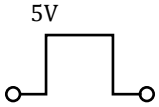
3. $y_1 = \bar{A}, y_2 = \bar{B}$,

$y = \overline{y_1 + y_2} = \overline{\bar{A} + \bar{B}}$ (using De-morgan's theorem)

$y = \bar{\bar{A}} \cdot \bar{\bar{B}} = A \cdot B$ AND gate

Hence this logic gate represents AND gate.

4. This is the circuit where P-N junction is acting as a Half-wave rectifier so the output will be



5. Potential difference on R = 3.5 – 0.5 = 3.0 volt

Current in circuit $i = \frac{V}{R} = \frac{3}{100} = 30mA$

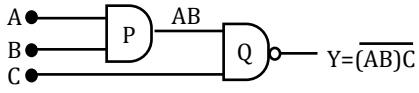
6. Since diode is in forward bias

$i = \frac{\Delta V}{R} = \frac{4 - (-6)}{1 \times 10^3} = \frac{10}{10^3} = 10^{-2}$ A

8. Current will not flow through D_1 as it is reverse biased. Current will flow through cell, R_1, D_2 and R_3.

$\therefore \quad i = \frac{10}{2+2} = 2.5\ A$

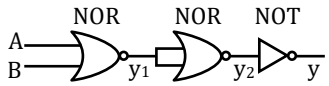
9.



for A = B = C = 0 ; y = 1

for A = B = C = 1 ; y = 0

10.

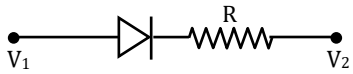


$y_1 = \overline{A + B}$

$y_2 = \overline{y_1 + y_1} = \overline{y_1} = \overline{\overline{A + B}} = A + B$

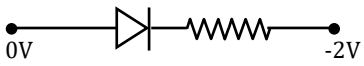
$y = \overline{y_2} = \overline{A + B}$ NOR GATE

11.



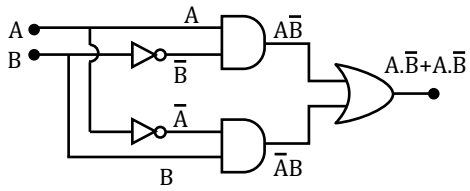
In forward bias $V_1 > V_2$

$\Rightarrow$ only



is in forward bias

12.



13. Due is heating, number of electron hold pairs will increase that affects the overall V – I characteristics of p-n junction

14.

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

$\therefore$ It is a NAND Gate

15. For P type

Holes are majority & trivalent atoms are the dopants.

16. $\lambda = \frac{1240\,nm}{1.9} = 652.6\ nm \simeq 654\ nm$

A	B	Y
0	0	1

17.

0	1	0	NOR gate
1	0	0	
1	1	0	

18. In reverse bias external battery attract majority charge carriers.

so width of the depletion region increase

19. In semiconductors and insulators, the number of charge carries per unit volume increases with an increase in temperature, so $\propto$ is negative for these.

20. $Y = \overline{\overline{A}+\overline{B}} = \overline{\overline{A}}.\overline{\overline{B}} = A.B$ = AND gate

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

21. In forward bias $V_P > V_N$

22. NAND gate and NOR gate are universal logic gates.

23. For N type semi-conductor intrinsic semiconductor doped by pentavalent impurity.

24. In N type semiconductor majority charge carriers are e and P type semiconductor majority charge carriers are holes.

$I = neAV_d = neA\ (\mu E)$

$\mu_e > \mu_h \quad \Rightarrow \quad I_e > I_h$

25. Reverse bias Zener diode is used a voltage regulator

for Ge Potential barrier $V_0 = 0.3$ V

Si Potential barrier $V_0 = 0.7$ V

26. $Y = A \cdot B + \overline{B \cdot C} = AB + \overline{B} + \overline{C}$

(i) 0 to t_1 A = 0, B = 0, C = 1

$Y = 0 \cdot 0 + \overline{0 \cdot 1} = 0 + 1 = 1$

(ii) t_1 to t_2 A = 1, B = 0, C = 1

$Y = 1 \cdot 0 + \overline{0 \cdot 1} = 0 + 1 = 1$

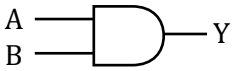
(iii) t_2 to t_3 A = 0, B = 1, C = 0

$Y = 0 \cdot 1 + \overline{1 \cdot 0} = 0 + 1 = 1$

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32. $Y = \overline{\bar{A}+\bar{B}} = \bar{\bar{A}}\cdot\bar{\bar{B}} = A.B$

AND gate

for AND gate 

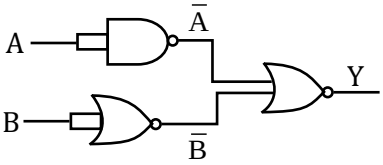
33. For p type semiconductor trivalent impurity added

34.

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

According to truth table, relation is inverse between Y and B.

Here, $Y = \bar{B}$

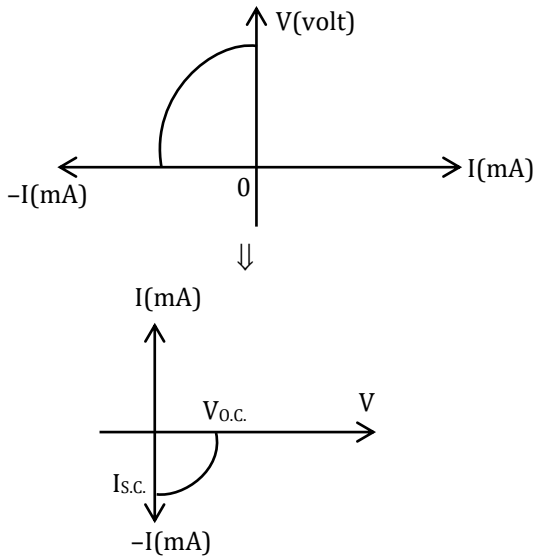
35. 

$\overline{\bar{A}+\bar{B}} \Rightarrow \bar{\bar{A}}\cdot\bar{\bar{B}} \Rightarrow A\cdot B$ (AND GATE)

36. A. Solar-cell, the I-V characteristics lie in the IV quadrant.

B. In reverse biased condition due to drift of minority charge carriers current flow in μA

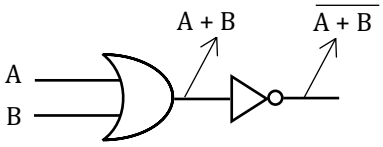
Answer should be (1) A is correct and (B) is incorrect

37. 

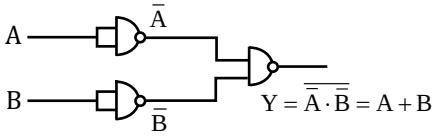
$V_{o.c.} \rightarrow$ Open circuit voltage

$I_{s.c.} \rightarrow$ short circuit voltage

Note : Graph of solar cell drawn in IV- Quadrant because it does not draw the current but supply same to the load.

38. 

$Y = \overline{A+B}$ (NOR gate)

39. 

Exercise-III (Analytical Questions)

1. Indium is a trivalent impurity.

4. When temp. increases in semiconductor then number of charge carriers increase.

6. In an unbiased P-N junction, majority charge carriers diffuse due to concentration difference.